Yair Shapira

Linear Algebra and Group Theory for Physicists and Engineers

Second Edition

Birkhäuser



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Preface

What's new in this edition? Every matrix has a Jordan form (Chap. 16). To show this, we'll design new vectors: generalized eigenvectors. In terms of these vectors, the matrix looks quite simple. On the main diagonal, we have the eigenvalues, one by one. On the superdiagonal just above it, we have 1's or 0's. All the rest is zero. Isn't this simple? It is also useful. For Hermitian matrix, for example, this gives the eigenbasis.

To design this, we also use a fundamental theorem in number theory: the Chinese remainder theorem. This will help design the Jordan decomposition (Chap. 17). This is a bit different from the Jordan form: it is given in terms of a new polynomial in the original matrix. To show how useful this is, we'll design the Jordan form in a special case of interest (Chap. 18).

This edition also contains more applications. It shows how to linearize the Einstein equations in general relativity (Chap. 19). In finite elements, you can then use the Newton iteration, to solve the nonlinear system of PDEs. There are some numerical results in electromagnetics (Tables 9.1–9.4).

Linear algebra and group theory are closely related. This book introduces both at the same time. Why? Because they go hand in hand. This is particularly good for (undergraduate) students in physics, chemistry, engineering, CS, and (applied) math. This is a new (interdisciplinary) approach: math is no longer isolated, but enveloped with practical applications in applied science and engineering.

The linear-algebra part introduces both vectors and matrices from scratch, with a lot of examples in two and three dimensions: 2×2 Lorentz matrices, and 3×3 rotation and inertia matrices. This prepares the reader for the group-theory stuff: 2×2 Moebius and Pauli matrices, 3×3 projective matrices, and more. This way, the reader gets ready for higher dimensions as well: big Fourier and Markov matrices, quantum-mechanical operators, and stiffness and mass matrices in (high-order) finite elements.

Once matrices are ready, they are used to mirror (or represent) useful groups. This is how linear algebra is used in group theory. This makes groups ever so concrete and accessible.

Why learn linear and modern algebra at the same time? Because they can complete and support each other. Indeed, in the applications, we also work the other way around: group theory paves the way to linear algebra, to uncover the electronic structure in the atom.

How to Use the Book in Academic Courses?

The book could be used as a textbook in three (undergraduate) math courses:

- Linear algebra for physicists and engineers (Chaps. 1–4 and 14–18)
- Group theory and its geometrical applications (Chaps. 5–7 and 14–15)
- Numerical analysis: finite elements and their applications (Chaps. 8–15 and 19)

Indeed, Part I introduces linear algebra, with applications in physics and CS. Part II, on the other hand, introduces group theory, with applications in projective geometry. Parts III–IV introduce high-order finite elements, in a regular mesh in 3-D. Part V assembles the stiffness and mass matrices in quantum chemistry. Part VI designs the Jordan form of a matrix, and the eigenbasis of a Hermitian matrix. Finally, Part VII linearizes the Einstein equations in numerical relativity.

The book is nearly self-contained: the only prerequisite is elementary calculus, which could be attended at the same time. There are plenty of examples and figures, to make the material more vivid and visual.

Each chapter ends with a lot of relevant exercises, with solutions or at least guidelines. This way, the reader gets to see how the theory develops step by step, exercise by exercise.

Roadmaps: How to Read the Book?

How to read the book? Here are a few options: (Figs. 1, 2, and 3):

- Physicists, chemists, and engineers could
  - Start from Chaps. 1–2 about linear algebra, with applications in geometrical mechanics
  - Skip to Chaps. 4–5, to use small matrices in special relativity and group theory
  - Use bigger matrices in quantum mechanics as well (Chaps. 7 and 14–15)
  - Proceed to the Jordan form (Chap. 16)
  - Conclude with advanced applications in general relativity (Chap. 19)
- Computer scientists, on the other hand, could
  - Start from Chaps. 1–2 about linear algebra
  - Proceed to Chap. 3, which uses a Markov matrix to design a search engine

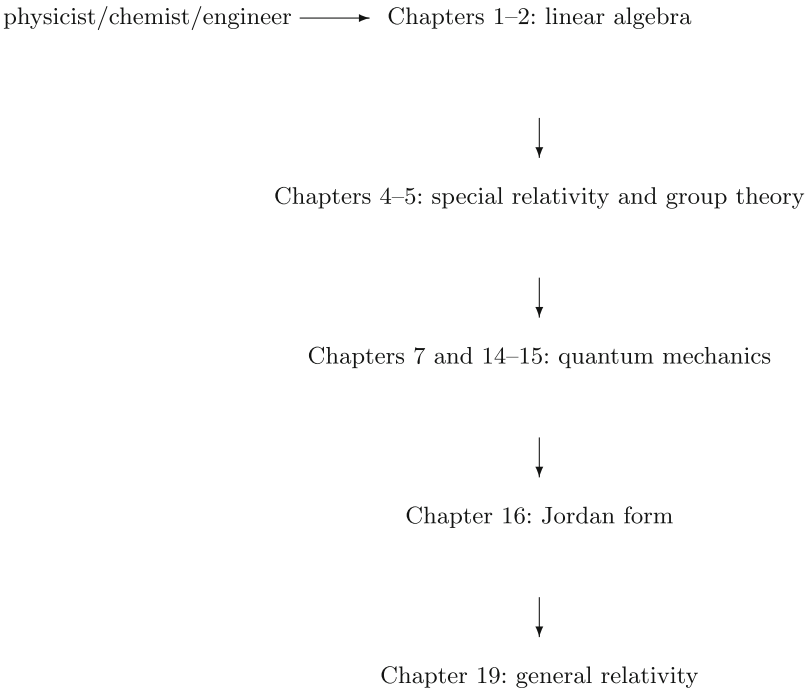


Fig. 1 How could a physicist/chemist/engineer read the book?

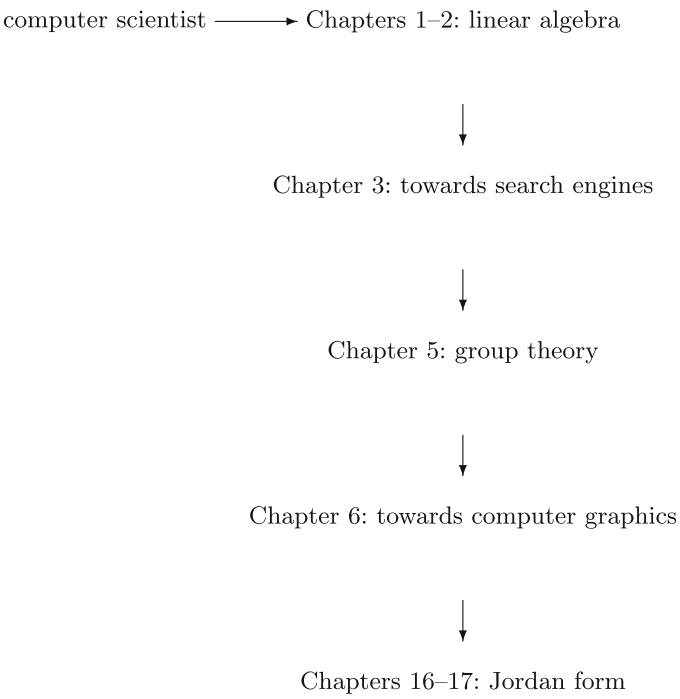


Fig. 2 How could a computer scientist read the book?

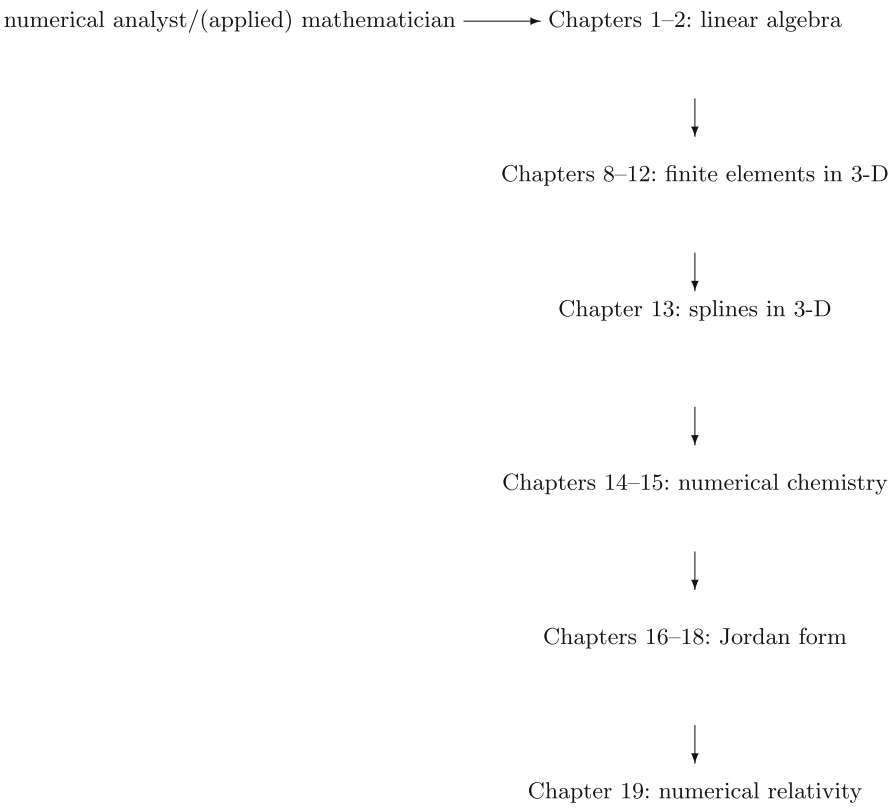


Fig. 3 How could a numerical analyst or a (applied) mathematician read the book?

  - Skip to Chaps. 5–6 about group theory and its applications in computer graphics
  - Conclude with Chaps. 16–17 about the Jordan form
- Numerical analysts and (applied) mathematicians, on the other hand, could
  - Start from Chaps. 1–2 about linear algebra
  - Skip to Chaps. 8–15 about finite elements and regular meshes, with applications in quantum chemistry and splines
  - Proceed to Chaps. 16–18 about the Jordan form
  - Conclude with Chap. 19 about numerical relativity

And a little remark: how to pronounce the title of the book? This is a bit tricky: write "physics," but say "phyZics." This is a phonetic law: use the easiest way to pronounce, no matter how the word is written. Likewise, write "tensor" and "isomorphism," but say "tenZor" and "iZomorphiZm," and so on.

Haifa, Israel Yair Shapira

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Part I
Introduction to Linear Algebra

We are already familiar with elementary algebraic objects: numbers (or scalars), along with the arithmetic operations between them. In this part, on the other hand, we look at more complicated structures: vector and matrix. Fortunately, it is also possible to define arithmetic (or algebraic) operations between them.

With these new operations, the vectors make a new linear space. The (nonsingular) matrices, on the other hand, make yet another important structure: a group. In a group, the associative law must hold. The commutative law, on the other hand, not necessarily holds.

The matrix is not just an algebraic object. It may also have a geometrical meaning: a mapping or transformation. This is most useful in many applications.

In special relativity, for example, the Lorentz transformation can be written as a small 2×2 matrix. In geometrical mechanics, on the other hand, 3×3 matrices are more useful. Finally, a yet bigger matrix is often used in stochastic analysis to model a Markov chain in a graph. This has an interesting application in modern search engines in the internet.

Chapter 1
Vectors and Matrices



What is a vector? It is a finite list of (real) numbers: scalars or components.

In a geometrical context, the components have yet another name: coordinates. In the two-dimensional Cartesian plane, for example, a vector contains two coordinates: the x- and y-coordinates. This is why the vector is often denoted by the pair (x, y). Geometrically, this can also be viewed as an arrow, leading from the origin $(0, 0)$ to the point $(x, y) \in \mathbb{R}^2$. Here, $\mathbb{R}$ is the real axis, $\mathbb{R}^2$ is the Cartesian plane, and "$\in$" means "belongs to."

In the three-dimensional Cartesian space, on the other hand, the vector also contains a third coordinate: the z-coordinate. This is why the vector is often denoted by the triplet $(x, y, z) \in \mathbb{R}^3$.

Still, vectors are more than just lists of numbers. They also have linear arithmetic operations: addition, multiplication by a scalar, and more. With these operations, the vectors form a complete linear space.

What is a matrix? It is a rectangular frame, full of numbers: scalars, or elements, ordered row by row in the matrix.

Unlike the vector, the matrix has a new arithmetic operation: multiplication. A matrix could multiply a vector or be applied to a vector. This is done from the left: first write the matrix, then the vector. Likewise, a matrix could multiply another matrix.

In geometrical terms, the matrix can also be viewed as a linear mapping (or transformation) from one vector space to another. To map a vector, the matrix should be applied to it. This produces the new image (or target) vector. With this new interpretation, the matrix is now more active: it acts upon a complete vector space. For example, the matrix could simply rotate the original vector (see exercises at the end of Chap. 2).

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1.1 Vectors in Two and Three Dimensions

1.1.1 Two-Dimensional Vectors

What is a vector? It is a finite list or sequence: a finite set of numbers, ordered one by one in a row. In this list, each number is also called a scalar or a component. The total number of components is often denoted by the natural number n.

In geometrical terms, on the other hand, the components are also viewed as coordinates. This way, the vector also takes a new interpretation: an n-dimensional vector, in a new n-dimensional linear space.

In the trivial case of $n = 1$, for example, the vector contains just one component or coordinate. In this degenerate case, the one-dimensional vector (x) mirrors the scalar x: both can be interpreted geometrically as the point x on the real axis.

In the more interesting case of $n = 2$, on the other hand, the two-dimensional vector is a pair of two numbers: x and y. Here, the first component x serves as the horizontal coordinate, whereas the second component y serves as the vertical coordinate.

This way, the original vector (x, y) takes its geometrical meaning as well: the new point (x, y) in the Cartesian plane. To illustrate the vector, draw an arrow from the origin $(0, 0)$ to the point $(x, y) \in \mathbb{R}^2$ (Fig. 1.1).

1.1.2 Adding Vectors

Consider two vectors: (x, y) and $(\hat{x}, \hat{y})$ that lie somewhere in the Cartesian plane. How to add them to each other? For this purpose, use the parallelogram rule (Fig. 1.2). After all, we already have three points: $(0, 0)$, (x, y), and $(\tilde{x}, \tilde{y})$. To make a parallelogram, we need just one more point. This new point will indeed be the required sum $(x, y) + (\hat{x}, \hat{y})$.

Unfortunately, this is still too geometrical. After all, we can never trust our own human eye or hand to draw this accurately. Instead, we better have a more algebraic method, independent of geometry.

As a matter of fact, the above geometrical rule also has an algebraic face: add component by component:

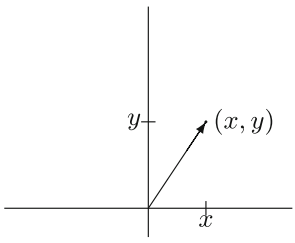


Fig. 1.1 The vector (x, y) is drawn as an arrow, issuing from the origin $(0, 0)$, and leading to the point (x, y) in the Cartesian plane

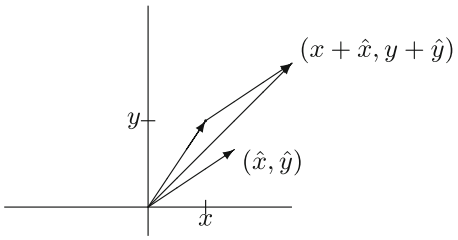


Fig. 1.2 How to add (x, y) to $(\hat{x}, \hat{y})$? Use them as sides in a new parallelogram, and let their sum be the diagonal of this parallelogram. This is the parallelogram rule

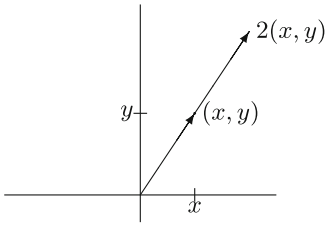


Fig. 1.3 How to multiply (or stretch) the original vector (x, y) by factor 2? Well, multiply coordinate by coordinate, to produce the new vector $2(x, y) = (2x, 2y)$, which is twice as long

$$(x, y) + (\hat{x}, \hat{y}) \equiv (x + \hat{x}, y + \hat{y}).$$

This way, in the sum, each individual coordinate is easy enough to calculate: it is just the sum of the corresponding coordinates in the original vectors. This algebraic formulation is much more practical: it is easy to implement on the computer and to extend to higher dimensions as well.

1.1.3 Scalar Times Vector

A vector can also be multiplied by a number (or scalar, or factor), either from the left (scalar times vector) or from the right (vector times scalar). What is the result? It is a new vector that could be either shorter or longer but must still point in the same direction as before. After all, the ratio between the coordinates remains the same. The only thing that has changed is the length (or magnitude) (Fig. 1.3).

Unfortunately, this is still too geometrical: it gives no practical algorithm. After all, we can never trust our human eye or hand to draw the new vector accurately. How to do this algebraically? Easy: multiply coordinate by coordinate. For example, to multiply the vector (x, y) by the scalar a from the left, define

$$a(x, y) \equiv (ax, ay).$$

Likewise, to multiply from the right, define

$$(x, y)a \equiv (xa, ya),$$

which is the same as before. In this sense, the multiplication is indeed commutative.

Later on, we will see a lot of examples with two-dimensional vectors. In Sect. 1.3.1, we will see that a complex number is actually a two-dimensional vector. Furthermore, in the exercises at the end of Chap. 2, we will see how to rotate a vector. Moreover, in Chaps. 4–5, we will see Lorentz and Moebius transformations. Here, however, we have no time for examples. To see how algebra and geometry go hand in hand, we better go ahead and extend the above to three spatial dimensions as well.

1.1.4 Three-Dimensional Vectors

So far, we have considered the two-dimensional case $n = 2$. In the three-dimensional case $n = 3$, on the other hand, we introduce one more dimension: the z-dimension. This way, a vector is now a triplet of three (rather than two) components or coordinates: (x, y, z).

In geometrical terms, the vector (x, y, z) represents a point in the three-dimensional Cartesian space, with the horizontal coordinates x and y, and the height coordinate z. This is why the vector is often illustrated as an arrow, issuing from the origin $(0, 0, 0)$, and leading to the point $(x, y, z) \in \mathbb{R}^3$ (Fig. 1.4).

As in Sect. 1.1.2, addition is still made coordinate by coordinate:

$$(x, y, z) + (\hat{x}, \hat{y}, \hat{z}) \equiv (x + \hat{x}, y + \hat{y}, z + \hat{z}).$$

Furthermore, as in Sect. 1.1.3, multiplication by a scalar is still made coordinate by coordinate as well. This could be done either from the left:

$$a(x, y, z) \equiv (ax, ay, az)$$

or from the right:

$$(x, y, z)a \equiv (xa, ya, za).$$

In both cases, the result is the same. In this sense, the commutative law indeed applies.

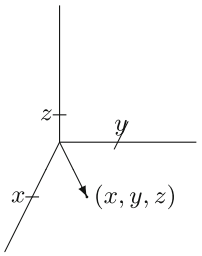


Fig. 1.4
The three-dimensional vector (x, y, z) is an arrow, issuing from the origin $(0, 0, 0)$, and leading to the point (x, y, z) in the Cartesian space

Later on, we will see a lot of examples. In Chap. 2, in particular, we will rotate a three-dimensional vector. Here, however, we have no time for this. After all, the complete algebraic picture is far more general. To realize this, we better proceed to a yet higher dimension, with no apparent geometrical meaning any more.

1.2 Vectors in Higher Dimensions

1.2.1 Multidimensional Vectors

So far, our vectors had a concrete geometrical meaning. For $n > 3$, on the other hand, they are no longer geometrical but only algebraic.

The n-dimensional vector is first of all a set: a finite list or sequence of n individual numbers (components):

$$v \equiv (v_1, v_2, v_3, \ldots, v_n) \in \mathbb{R}^n,$$

where $\mathbb{R}$ is the real axis, and $\mathbb{R}^n$ is the n-dimensional space. Still, the vector is more than that: it is also an algebraic object, with all sorts of arithmetic operations. To see this, consider yet another vector:

$$u \equiv (u_1, u_2, \ldots, u_n) \in \mathbb{R}^n.$$

This vector can now be added to v, component by component:

$$u + v \equiv (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n).$$

1.2.2 Associative Law

Fortunately, this operation is associative. Indeed, let $w \equiv (w_1, w_2, \ldots, w_n)$ be yet another vector in $\mathbb{R}^n$. Now, to sum these three vectors, ordering does not matter: either add the first two to each other and then add the third one as well or start from the second and third vectors and add them to each other, and then add the first as well:

$$\begin{aligned} u + (v + w) &= (u_1 + (v_1 + w_1), u_2 + (v_2 + w_2), \ldots, u_n + (v_n + w_n)) \\ &= ((u_1 + v_1) + w_1, (u_2 + v_2) + w_2, \ldots, uv_n + v_n) + w_n) \\ &= (u + v) + w. \end{aligned}$$

This is indeed the associative law for addition.

Still, the associative law applies not only to addition but also to multiplication by a scalar.

1.2.3 The Origin

In $\mathbb{R}^n$, there is one special vector: the zero vector (or the origin), with n zeroes:

$$\mathbf{0} \equiv (0, 0, 0, \ldots, 0) \quad (n \text{ zeroes}).$$

In what way is this vector special? Well, it is the only vector that can be added to any n-dimensional vector v, with no effect whatsoever:

$$\mathbf{0} + v = v + \mathbf{0} = v.$$

1.2.4 Multiplication and Its Laws

Now, how to multiply a vector by a scalar $a \in \mathbb{R}$, either from the left or from the right? Well, as before, this is done component by component:

$$av \equiv va \equiv (av_1, av_2, \ldots, av_n).$$

In this sense, multiplication is indeed commutative. Fortunately, it is associative as well. To see this, consider yet another scalar b:

$$b(av) = b(av_1, av_2, \ldots, av_n) = (bav_1, bav_2, \ldots, bav_n) = (ba)v.$$

1.2.5 Distributive Laws

Furthermore, the above arithmetic operations are also distributive in two senses. On the one hand, you can push the same vector v into parentheses:

$$\begin{aligned}(a + b)v &= ((a + b)v_1, (a + b)v_2, \ldots, (a + b)v_n) \\ &= (av_1 + bv_1, av_2 + bv_2, \ldots, av_n + bv_n) \\ &= av + bv.\end{aligned}$$

On the other hand, you can push the same scalar a into parentheses:

$$a(u + v) = (a(u_1 + v_1), a(u_2 + v_2), \ldots, a(u_n + v_n))$$

$$= (au_1 + av_1, au_2 + av_2, \ldots, au_n + av_n)$$
$$= au + av.$$

This completes the definition of the new vector space $\mathbb{R}^n$, and the linear algebraic operations in it.

1.3 Complex Numbers and Vectors

1.3.1 Complex Numbers

The two-dimensional vectors defined above may also help model complex numbers. As a matter of fact, complex numbers have just one more algebraic operation: multiplication. Let us explain this briefly.

The negative number -1 has no square root: there is no real number whose square is -1. Fortunately, it is still possible to introduce a new auxiliary (not real) number—the imaginary number i. This way, i is the only number whose square is -1:

$$i^2 = -1,$$

or

$$i \equiv \sqrt{-1}.$$

Because it lies outside the real axis, i may now span a new vertical axis, perpendicular to the original real axis (Fig. 1.5). This way, i marks a new direction: vertical. To "span" means to introduce a complete new axis, containing infinitely many points: multiples of i.

For this purpose, place i at $(0, 1)$, above the origin. This way, i spans the entire imaginary axis:

$$\{bi = (0, b) \mid -\infty < b < \infty\}.$$

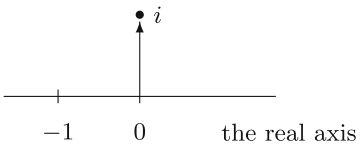


Fig. 1.5 The imaginary number i. The arrow leading from the origin to i makes a right angle with the real axis. In $i^2 = -1$, on the other hand, this angle doubles to make a flat angle with the positive part of the real axis

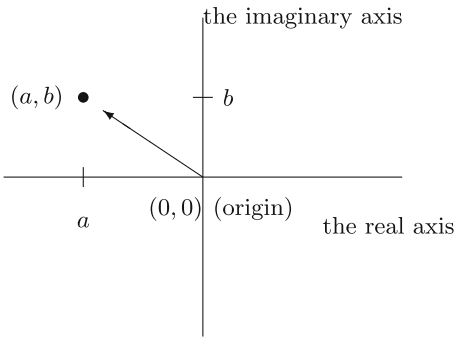


Fig. 1.6 The complex plane $\mathbb{C}$. The imaginary number $i \equiv \sqrt{-1}$ is at $(0, 1)$. A complex number $a + bi$ is at (a, b)

Here, b is some real number. In the above, the algebraic multiple bi also takes the geometrical form $(0, b)$: a new point on the vertical imaginary axis. In particular, if $b = 1$, then we obtain the original imaginary number $i = (0, 1)$ once again.

Together, the real and imaginary axes span the entire complex plane:

$$\mathbb{C} \equiv \{a + bi \equiv (a, b) \ \mid \ -\infty < a, b < \infty\}\,.$$

Here, a and b are some real numbers. The new complex number $a + bi$ also takes its geometrical place $(a, b) \in \mathbb{C}$ (Fig. 1.6).

So far, the complex plane is defined geometrically only. What is its algebraic meaning? Well, since complex numbers are two-dimensional vectors, we already know how to add them to each other and multiply them by a real scalar. How to multiply them by each other? Well, we already know that

$$i^2 = -1.$$

After all, this is how i has been defined in the first plane.

What is the geometrical meaning of this algebraic equation? Well, look at the negative number -1. What angle does it make with the positive part of the real axis? A flat angle of 180°. The imaginary number i, on the other hand, makes a right angle of 90° with the real axis (Fig. 1.5). Thus, multiplying i by itself means adding yet another right angle, to make a flat angle together.

Let us go ahead and extend this linearly to the entire complex plane. In other words, let us use the distributive and commutative laws to multiply $a + bi$ times $c + di$ (where a, b, c, and d are some real numbers):

$$(a+bi)(c+di) = a(c+di)+bi(c+di) = ac+adi+bci+bdi^2 = ac-bd+(ad+bc)i.$$

For example, look at the special case

$$c = a \quad \text{and} \quad d = -b.$$

This way, $c + di$ is the complex conjugate of $a + bi$, denoted by a small bar (or an asterisk) on top:

$$c + di \equiv a - bi \equiv (a + bi) \equiv (a + bi)^*.$$

In this case, the above product is a new real number: the squared absolute value of $a + bi$:

$$|a + bi|^2 \equiv (a + bi)(a - bi) = a^2 + b^2.$$

Thanks to this definition, the absolute value is the same as the length of the vector (a, b), obtained from Pythagoras theorem.

If this is nonzero, then we could divide by it:

$$(a + bi)\frac{a - bi}{a^2 + b^2} = 1.$$

So, we now also have the reciprocal (or inverse) of $a + bi$:

$$(a + bi)^{-1} = \frac{a - bi}{a^2 + b^2}.$$

1.3.2 Complex Vectors

In our n-dimensional vector, each individual component can now be a complex number. This yields a new vector space: $\mathbb{C}^n$. The only difference is that the components (and the scalar that may multiply them) can now be not only real but also complex.

Because the arithmetic operations are defined in the same way, the same associative and distributive laws still hold. In other words, the algebraic operations remain linear. For this reason, $\mathbb{C}^n$ could be viewed as a natural extension of the original vector space $\mathbb{R}^n$.

1.4 Rectangular Matrix

1.4.1 Matrices

A matrix is a frame, full of numbers. In an m by n (or $m \times n$) rectangular matrix, there are mn numbers: m rows and n columns (Fig. 1.7).

The original m by n matrix is often denoted by

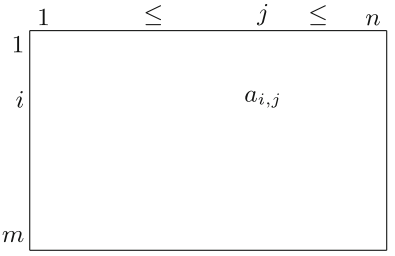


Fig. 1.7 A rectangular $m \times n$ matrix: there are m rows and n columns

$$A \equiv \left(a_{i,j}\right)_{1 \le i \le m,\ 1 \le j \le n}.$$

Here, the individual element $a_{i,j}$ is some number, placed in the ith row, and in the jth column. This way, A contains n columns, ordered one by one, left to right. Each column contains m individual elements (numbers), ordered top to bottom.

Let us look at each column as an individual object, which could serve as a new component in a new row vector. From this point of view, A is actually an n-dimensional vector, with n "components" in a row: each "component" is not just a scalar but a complete vector in its own right: an m-dimensional column vector, containing m numbers:

$$A \equiv \left(v^{(1)} \ \middle|\ v^{(2)} \ \middle|\ v^{(3)} \ \middle|\ \cdots \ \middle|\ v^{(n)}\right),$$

where $v^{(1)}, v^{(2)}, \ldots, v^{(n)}$ are column vectors in $\mathbb{R}^m$. This way, for $1 \le j \le n$, $v^{(j)}$ is the jth column in A:

$$v^{(j)} \equiv \begin{pmatrix} v_1^{(j)} \\ v_2^{(j)} \\ v_3^{(j)} \\ \vdots \\ v_m^{(j)} \end{pmatrix} \equiv \begin{pmatrix} a_{1,j} \\ a_{2,j} \\ a_{3,j} \\ \vdots \\ a_{m,j} \end{pmatrix}.$$

In this column vector, for $1 \le i \le m$, the ith component is the matrix element

$$v_i^{(j)} \equiv a_{i,j}.$$

For example, if $m = 3$ and $n = 4$, then A is a 3×4 matrix:

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \end{pmatrix}.$$

In this form, A could also be viewed as a list of three rows. Each row contains four numbers, ordered left to right. In total, A contains twelve elements.

1.4.2 Adding Matrices

Let

$$B \equiv \left(b_{i,j}\right)_{1\le i\le m,\ 1\le j\le n}$$

be yet another $m \times n$ matrix. This way, B can now be added to A, element by element:

$$A + B \equiv \left(a_{i,j} + b_{i,j}\right)_{1\le i\le m,\ 1\le j\le n}.$$

Alternatively, B could also be written in terms of its columns:

$$B \equiv \left(u^{(1)} \mid u^{(2)} \mid u^{(3)} \mid \cdots \mid u^{(n)}\right),$$

where $u^{(1)}, u^{(2)}, \ldots, u^{(n)}$ are column vectors in $\mathbb{R}^m$. This way, B could also be added column by column:

$$A + B \equiv \left(v^{(1)} + u^{(1)} \mid v^{(2)} + u^{(2)} \mid v^{(3)} + u^{(3)} \mid \cdots \mid v^{(n)} + u^{(n)}\right).$$

It is easy to see that this operation is associative:

$$(A + B) + C = A + (B + C),$$

where C is yet another $m \times n$ matrix.

1.4.3 Scalar Times Matrix

To multiply A by a real number $r \in \mathbb{R}$ (either from the left or from the right), just multiply element by element:

$$rA \equiv Ar \equiv \left(ra_{i,j}\right)_{1\le i\le m,\ 1\le j\le n}.$$

Clearly, this operation is associative as well: if $q \in \mathbb{R}$ is yet another scalar, then

$$q(rA) \equiv q\left(ra_{i,j}\right)_{1\le i\le m,\ 1\le j\le n} = \left(qra_{i,j}\right)_{1\le i\le m,\ 1\le j\le n} = (qr)A.$$

Furthermore, the above operations are also distributive in two senses. On the one hand, you can push the same matrix A into parentheses:

$$(r+q)A = rA + qA.$$

On the other hand, you can push the same scalar r into parentheses:

$$r(A+B) = rA + rB.$$

1.4.4 Matrix Times Vector

Recall that A has already been written column by column:

$$A = \left(v^{(1)} \mid v^{(2)} \mid \cdots \mid v^{(n)} \right).$$

This way, each column is in $\mathbb{R}^m$. Consider now a different column vector, not in $\mathbb{R}^m$ but rather in $\mathbb{R}^n$:

$$w = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}.$$

How many components are here? As many as the number of columns in A. This is just enough! We can now scan A column by column, multiply each column by the corresponding component from w, and sum up:

$$Aw \equiv w_1 v^{(1)} + w_2 v^{(2)} + \cdots + w_n v^{(n)} = \sum_{j=1}^{n} w_j v^{(j)}.$$

This is indeed A times w: a new linear combination of the columns of A, with coefficients taken from w.

Thus, w and Aw may have different dimensions: w is n-dimensional, whereas Aw is m-dimensional. Later on, we will use this property to interpret A geometrically, as a mapping:

$$A : \mathbb{R}^n \to \mathbb{R}^m.$$

This means that A maps each n-dimensional vector to an m-dimensional vector. In other words, A is a function: it takes an n-dimensional vector and returns an m-dimensional vector. Here, "$\to$" stands for mapping, not for a limit.

In Aw, what is the ith component? It is

$$(Aw)_i = \sum_{j=1}^{n} w_j v_i^{(j)} = \sum_{j=1}^{n} a_{i,j} w_j, \quad 1 \le i \le m.$$

To calculate this, scan the ith row in A element by element. Multiply each element by the corresponding component from w, and sum up.

This is the new matrix-times-vector operation. What are its algebraic properties? Well, first of all, it is associative for scalars: for any scalar $r \in \mathbb{R}$,

$$A(rw) = r(Aw) = (rA)w.$$

Furthermore, it is also distributive in two senses. On the one hand, you can push the same vector into parentheses:

$$(A + B)w = Aw + Bw,$$

where B has the same dimensions as A. On the other hand, you can push the same matrix into parentheses:

$$A(w + u) = Aw + Au,$$

where u has the same dimension as w.

In summary, $w \to Aw$ is indeed a linear transformation. This will be useful later. In Chap. 2, for example, we will see how a matrix could rotate a vector. Here, however, we have no time for examples. To have the complete algebraic picture, we better extend the above to a yet more complicated operation: matrix-times-matrix.

1.4.5 Matrix-Times-Matrix

The above can now be extended to define yet another kind of multiplication: matrix-times-matrix. For this purpose, however, we must be careful to pick a matrix B with proper dimensions: an $l \times m$ matrix, where l is some natural number as well.

Why must B have these dimensions? Because, this way, the number of columns in B is the same as the number of rows in A. This is a must: it will help multiply B times A soon.

Fortunately, A has already been written column by column. Now, apply B to each individual column:

$$BA \equiv \left(Bv^{(1)} \mid Bv^{(2)} \mid \cdots \mid Bv^{(n)} \right).$$

Why is this legitimate? Because A has m rows, and B has m columns. This way, the product BA is a new $l \times n$ matrix: it has as many rows as in B, and as many columns as in A.

In BA, what is the (i,k)th element? Well, for this purpose, focus on the kth column: $Bv^{(k)}$. In it, look at the ith component. It comes from the ith row in B: scan it element by element, multiply each element by the corresponding component in $v^{(k)}$, and sum up:

$$(BA)_{i,k} = \left(Bv^{(k)}\right)_i = \sum_{j=1}^{m} b_{i,j} v_j^{(k)} = \sum_{j=1}^{m} b_{i,j} a_{j,k}, \quad 1 \le i \le l,\ 1 \le k \le n.$$

Thus, at the same time, two things are scanned element by element: the ith row in B, and the kth column in A. This makes a loop of m steps. In each step, pick an element from the ith row in B, pick the corresponding element from the kth column in A, multiply, and sum up:

$$(BA)_{i,k} = \sum_{j=1}^{m} b_{i,j} a_{j,k}.$$

1.4.6 Distributive and Associative Laws

Fortunately, the new matrix-times-matrix operation is distributive in two senses. On the one hand, you can push the same matrix into parentheses from the right:

$$\left(B + \hat{B}\right) A = BA + \hat{B}A$$

(where $\hat{B}$ has the same dimensions as B). On the other hand, you can push the same matrix into parentheses from the left:

$$B\left(A + \hat{A}\right) = BA + B\hat{A}$$

(where $\hat{A}$ has the same dimensions as A).

Moreover, matrix-times-matrix is also an associative operation. To see this, let

$$C = \left(c_{i,j}\right)_{1 \le i \le k,\ 1 \le j \le l}$$

be a $k \times l$ matrix, where k is some natural number as well.

Why are the dimensions of C picked in this way? Well, this guarantees that the number of columns in C is the same as the number of rows in B (and also in BA).

This way, we can now apply C from the left and produce the new matrices $C(BA)$ and $(CB)A$.

Are they the same? To check on this, let us calculate the (s,t)th element, for some $1 \le s \le k$ and $1 \le t \le n$:

$$\begin{aligned}
(C(BA))_{s,t} &= \sum_{i=1}^{l} c_{s,i}(BA)_{i,t} \\
&= \sum_{i=1}^{l} c_{s,i} \sum_{j=1}^{m} b_{i,j} a_{j,t} \\
&= \sum_{i=1}^{l} \sum_{j=1}^{m} c_{s,i} b_{i,j} a_{j,t} \\
&= \sum_{j=1}^{m} \sum_{i=1}^{l} c_{s,i} b_{i,j} a_{j,t} \\
&= \sum_{j=1}^{m} \left(\sum_{i=1}^{l} c_{s,i} b_{i,j} \right) a_{j,t} \\
&= \sum_{j=1}^{m} (CB)_{s,j} a_{j,t} \\
&= ((CB)A)_{s,t}.
\end{aligned}$$

This can be done for every pair (s,t). Therefore,

$$C(BA) = (CB)A,$$

as asserted. This proves that matrix multiplication is not only distributive but also associative. In summary, it is a linear operation. This will be useful later.

1.4.7 The Transpose Matrix

Consider again our matrix A. Look at it the other way around: view the rows as columns, and the columns as rows. This yields a new $n \times m$ matrix—the transpose matrix A^t:

$$\left(A^t\right)_{j,i} \equiv a_{i,j}, \quad 1 \le i \le m,\ 1 \le j \le n.$$

For example, if A is the 3×4 matrix

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \end{pmatrix},$$

then A^t is the 4×3 matrix

$$A^t = \begin{pmatrix} a_{1,1} & a_{2,1} & a_{3,1} \\ a_{1,2} & a_{2,2} & a_{3,2} \\ a_{1,3} & a_{2,3} & a_{3,3} \\ a_{1,4} & a_{2,4} & a_{3,4} \end{pmatrix}.$$

From the above definition, it clearly follows that

$$(A^t)^t = A.$$

In Sect. 1.4.5, we have defined the product BA, where B is an $l \times m$ matrix. Why is this product well-defined? Because the number of rows in A is the same as the number of columns in B.

How to multiply the transpose matrices? Well, the number of rows in B^t is the same as the number of columns in A^t. So, we can construct a new $n \times l$ matrix: the product $A^t B^t$.

Is it really new? Not quite. After all, we have already seen it before, at least in its transpose form:

$$(BA)^t = A^t B^t.$$

To prove this, pick some $1 \leq i \leq l$ and $1 \leq k \leq n$. Consider the (k, i)th element in $(BA)^t$:

$$(BA)^t_{k,i} = (BA)_{i,k} = \sum_{j=1}^{m} b_{i,j} a_{j,k} = \sum_{j=1}^{m} A^t_{k,j} B^t_{j,i} = \left(A^t B^t\right)_{k,i},$$

as asserted.

1.5 Square Matrix

1.5.1 Symmetric Square Matrix

So far, we have assumed that A was rectangular: the number of rows was not necessarily the same as the number of columns. In this section, on the other hand, we focus on a square matrix of order $m = n$. Since A is square, it has a main

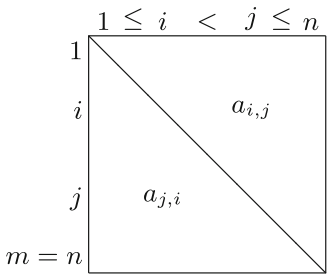


Fig. 1.8 In a square matrix A of order n, the main diagonal contains $a_{1,1}, a_{2,2}, \ldots, a_{n,n}$. If A is symmetric, then the lower triangular part mirrors the upper-triangular part: $a_{j,i} = a_{i,j}$

diagonal—from the upper-left corner to the lower-right corner:

$$a_{1,1},\ a_{2,2},\ a_{3,3},\ \ldots,\ a_{n,n}$$

(Fig. 1.8). This diagonal splits A into two triangular parts: its upper-right part and its lower-left part. If they mirror each other, then we say that A is symmetric.

This means that one could place a mirror on the main diagonal: the (i, j)th element is the same as the (j, i)th element:

$$a_{i,j} = a_{j,i}, \quad 1 \le i, j \le n.$$

In other words, A remains unchanged under interchanging the roles of rows and columns:

$$A = A^t.$$

1.5.2 The Identity Matrix

Here is an example of a symmetric matrix: the identity matrix of order n, denoted by I. On the main diagonal, it is 1:

$$I_{i,i} \equiv 1, \quad 1 \le i \le n.$$

Off the main diagonal, on the other hand, it is 0:

$$I_{i,j} \equiv 0, \quad 1 \le i \ne j \le n.$$

In summary,

$$I \equiv \begin{pmatrix} 1 & & 0 \\ & 1 & \\ & \ddots & \\ 0 & & 1 \end{pmatrix}.$$

Off the main diagonal, all elements vanish. No need to write all these zeroes. A blank space stands for a zero element.

Why is the identity matrix so special? Well, apply it to any n-dimensional vector v, and you would see no effect whatsoever:

$$Iv = v.$$

Likewise, apply I to any square matrix A of order n, and you would see no effect whatsoever:

$$IA = AI = A.$$

This is why I is also called the unit matrix.

1.5.3 The Inverse Matrix as a Mapping

The square matrix A may also have an inverse matrix: a new matrix A^{-1}, satisfying

$$A^{-1}A = I.$$

In this case, we say that A is nonsingular or invertible.

Thus, in the world of matrices, a nonsingular matrix plays the role of a nonzero number, and its inverse plays the role of the reciprocal. Later on, we will see this in a yet wider context: group theory.

Recall that A could also be viewed as a mapping of vectors:

$$v \to Av.$$

How to map Av back to v? Easy: just apply A^{-1} to it:

$$Av \to A^{-1}(Av) = \left(A^{-1}A\right)v = Iv = v.$$

This way, thanks to associativity, A^{-1} maps the vector Av back to v.

So, both A and A^{-1} could actually be viewed as mappings. This could be quite useful. Indeed, let us look at them the other way around: A^{-1} is now the original mapping that maps

$$v \to A^{-1}v,$$

and A is its mirror: the inverse mapping that maps $A^{-1}v$ back to v:

$$A^{-1}v \to A\left(A^{-1}v\right) = v.$$

In the language of matrices, this could be written simply as

$$AA^{-1} = I.$$

In summary, although matrices not necessarily commute, A and A^{-1} do commute with each other.

1.5.4 Inverse and Transpose

The inverse matrix could be quite hard to calculate. Still, once calculated, it is useful for many purposes.

What is the inverse of A^t? No need to calculate it! After all, A^{-1} is already available. To have the answer, just take its transpose. Indeed, from Sect. 1.4.7,

$$\left(A^{-1}\right)^t A^t = \left(AA^{-1}\right)^t = I^t = I.$$

In other words,

$$\left(A^t\right)^{-1} = \left(A^{-1}\right)^t,$$

as asserted. In summary, the inverse of the transpose is the transpose of the inverse. Thus, no parentheses are needed: one could simply write

$$A^{-t} \equiv \left(A^{-1}\right)^t = \left(A^t\right)^{-1}.$$

Finally, let A and B be two square matrices of order n. What is the inverse of the product? It is the product of the inverses, in the reverse order:

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Indeed, thanks to associativity,

$$\begin{aligned}(AB)\left(B^{-1}A^{-1}\right) &= A\left(B\left(B^{-1}A^{-1}\right)\right) = A\left(\left(BB^{-1}\right)A^{-1}\right)\\ &= A\left(IA^{-1}\right) = AA^{-1} = I.\end{aligned}$$

1.6 Complex Matrix and Its Hermitian Adjoint

1.6.1 The Hermitian Adjoint

Consider again a rectangular matrix. So far, we focused on a real matrix, with real elements in $\mathbb{R}$. Here, on the other hand, we consider a more general case: an $m \times n$ complex matrix, with complex elements in $\mathbb{C}$.

Fortunately, complex matrices still have the same algebraic operations: addition and multiplication. Furthermore, they still obey the same laws: distributive and associative. Instead of the transpose, we now have a better notion: the Hermitian adjoint. Let us see what it means.

Let $i \equiv \sqrt{-1}$ be the imaginary number. Recall that the complex number

$$c \equiv a + bi$$

has the complex conjugate

$$\bar{c} \equiv a - bi.$$

This helps define the Hermitian adjoint of a given complex matrix. This could be done in two stages:

- First, take the transpose.
- Then, for each individual element, take the complex conjugate.

This way, the original $m \times n$ matrix

$$A \equiv \left(a_{i,j}\right)_{1 \le i \le m,\ 1 \le j \le n}$$

makes a new $n \times m$ matrix, denoted by A^h (or A^*, or $A^\dagger$):

$$\left(A^\dagger\right)_{j,i} \equiv \left(A^*\right)_{j,i} \equiv \left(A^h\right)_{j,i} \equiv \bar{a}_{i,j}, \quad 1 \le i \le m,\ 1 \le j \le n.$$

In particular, if A happens to be real, then the complex conjugate is the same. In this case, the Hermitian adjoint is the same as transpose:

$$A^h = A^t.$$

In this sense, the Hermitian adjoint is a natural extension of the transpose. This is why it inherits a few attractive properties:

$$(A^h)^h = A,$$

and

$$(BA)^h = A^h B^h,$$

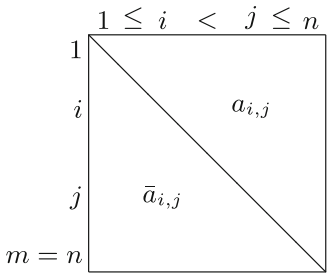


Fig. 1.9 In a Hermitian matrix, the lower triangular part mirrors the upper-triangular part, with a bar on top

where the number of columns in B is the same as the number of rows in A.

1.6.2 Hermitian (Self-Adjoint) Matrix

Consider now a square matrix A, of order $m = n$. We say that A is Hermitian (or self-adjoint) if it is the same as its Hermitian adjoint:

$$A = A^h.$$

In this case, the main diagonal acts like a mirror (Fig. 1.9). In fact, the lower triangular part mirrors the upper-triangular part, with a bar on top:

$$a_{j,i} = \left(A^h\right)_{j,i} = \bar{a}_{i,j}, \quad 1 \leq i, j \leq n.$$

In particular, look at an interesting case: $i = j$. This tells us that the main-diagonal must be real:

$$a_{i,i} = \bar{a}_{i,i}, \quad 1 \leq i \leq n.$$

1.7 Inner Product and Norm

1.7.1 Inner (Scalar) Product

Let

$$u \equiv \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \quad \text{and} \quad v \equiv \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

be two column vectors in $\mathbb{C}^n$. They can also be viewed as narrow $n \times 1$ "matrices," with just one column. This way, they also have their own Hermitian adjoint:

$$u^h = (\bar{u}_1, \bar{u}_2, \ldots, \bar{u}_n) \quad \text{and} \quad v^h = (\bar{v}_1, \bar{v}_2, \ldots, \bar{v}_n)\,.$$

These are $1 \times n$ "matrices," or n-dimensional row vectors.

Thus, u^h has n "columns," and v has n "rows." Fortunately, this is the same number. So, we can go ahead and multiply u^h times v. The result is a new 1×1 "matrix," or just a new (complex) scalar:

$$(u, v) \equiv u^h v = \sum_{j=1}^{n} \bar{u}_j v_j .$$

This is called the scalar (or inner) product of u and v.

1.7.2 Bilinearity

The inner product has an attractive property: bilinearity. To see this, let $c \in \mathbb{C}$ be some complex number. Now, in our inner product, we could "pull" c out (possibly with a bar on top):

$$(cu, v) = \bar{c}(u, v), \quad \text{and} \quad (u, cv) = c(u, v).$$

Furthermore, let w be yet another vector. Then, we also have a distributive law:

$$(u + w, v) = (u, v) + (w, v) \quad \text{and} \quad (u, v + w) = (u, v) + (u, w).$$

This means that the inner product is bilinear.

1.7.3 Skew-Symmetry

Moreover, the inner product is skew-symmetric: interchanging u and v just places a new bar on top. Indeed,

$$(v, u) = \sum_{j=1}^{n} \bar{v}_j u_j$$

is the complex conjugate of (u, v).

1.7.4 Norm

What is the inner product of v with itself? This is a real nonnegative number:

$$(v, v) = \sum_{j=1}^{n} \bar{v}_j v_j = \sum_{j=1}^{n} |v_j|^2 \geq 0.$$

Could this vanish? Only if v was the zero vector:

$$(v, v) = 0 \quad \Leftrightarrow \quad v = \mathbf{0}.$$

Thus, (v, v) has a square root. Let us use it to define the norm (or length, or magnitude) of v:

$$\|v\| \equiv +\sqrt{(v, v)}.$$

This way,

$$\|v\| \geq 0,$$

and

$$\|v\| = 0 \quad \Leftrightarrow \quad v = \mathbf{0},$$

as expected from a magnitude.

1.7.5 Normalization

Furthermore, every complex number $c \in \mathbb{C}$ could be "pulled" out:

$$\|cv\| = \sqrt{(cv, cv)} = \sqrt{\bar{c}c(v, v)} = \sqrt{|c|^2(v, v)} = |c|\sqrt{(v, v)} = |c| \cdot \|v\|.$$

This way, if v is a nonzero vector, then it could be normalized. Indeed, since $\|v\| > 0$, one could pick $c = 1/\|v\|$, to produce the normalized unit vector $v/\|v\|$: a new vector of norm 1, proportional to v.

1.7.6 Other Norms

$\|v\|$ is also called the l_2-norm and is also denoted by

$$\|v\| \equiv \|v\|_2,$$

to distinguish it from other norms: the l_1-norm

$$\|v\|_1 \equiv \sum_{i=1}^{n} |v_i|,$$

and the l_∞- norm (or the maximum norm):

$$\|v\|_\infty \equiv \max_{1 \le i \le n} |v_i|.$$

Like the l_2-norm, these norms also make sense: they are nonnegative and vanish if and only if v is the zero vector. Furthermore, every complex number $c \in \mathbb{C}$ could be pulled out:

$$\|cv\|_1 = |c| \cdot \|v\|_1, \quad \text{and} \quad \|cv\|_\infty = |c| \cdot \|v\|_\infty.$$

Still, we often use the l_2-norm. This is why we denote it simply by $\|v\|$.

1.7.7 Inner Product and the Hermitian Adjoint

Assume again that A is an $m \times n$ rectangular complex matrix. Let u and v be complex vectors of different dimensions: u is m-dimensional and v is n-dimensional. This way, Av is m-dimensional and has an inner product with u:

$$(u, Av) \equiv u^h Av.$$

Likewise, A^h is an $n \times m$ rectangular complex matrix. Thus, A^h could be applied to u, to produce $A^h u$: a new n-dimensional vector. As such, it has an inner product with v: $(A^h u, v)$. Is it the same?

To see this, recall that both u and v could also be viewed as narrow matrices. Thanks to associativity,

$$(u, Av) = u^h(Av) = (u^h A)v = (A^h u)^h v = (A^h u, v).$$

Let us use this in a special case: a square matrix.

1.7.8 Inner Product and a Hermitian Matrix

Assume now that A is square: $m = n$. Assume also that A is Hermitian:

$$A = A^h.$$

In this case, the above formula simplifies to read

$$(u, Av) = (Au, v),$$

for every two n-dimensional complex vectors u and v.

Next, let us work the other way around. Assume that we did not know that A was Hermitian. Instead we only knew that A was a square complex matrix of order $m = n$, satisfying

$$(u, Av) = (Au, v)$$

for every two n-dimensional complex vectors u and v. Would this imply that A was Hermitian?

To see this, pick u and v cleverly. Let i and j be some fixed natural numbers ($1 \leq i, j \leq n$). Now, let u and v be standard unit vectors, with just one nonzero component:

$$u_k = \begin{cases} 1 & \text{if } k = i \\ 0 & \text{if } k \neq i \end{cases} \quad (1 \leq k \leq n)$$

$$v_k = \begin{cases} 1 & \text{if } k = j \\ 0 & \text{if } k \neq j \end{cases} \quad (1 \leq k \leq n).$$

With this choice,

$$a_{i,j} = (u, Av) = (Au, v) = \bar{a}_{j,i},$$

implying that A is indeed Hermitian (Fig. 1.9).

1.8 Orthogonal and Unitary Matrix

1.8.1 Inner Product of Column Vectors

Assume again that A is an $m \times n$ rectangular complex matrix. For $1 \leq j \leq n$, let $v^{(j)}$ denote the jth column in A: an m-dimensional column vector. Now, for $1 \leq i, j \leq n$, what is the inner product of the ith and jth columns? Well, it is the same as the (i, j)th element in $A^h A$. Indeed,

$$(A^h A)_{i,j} = \sum_{k=1}^{m} \left(A^h\right)_{i,k} a_{k,j}$$

$$
\begin{aligned}
&= \sum_{k=1}^{m} \bar{a}_{k,i} a_{k,j} \\
&= \sum_{k=1}^{m} \bar{v}_k^{(i)} v_k^{(j)} \\
&= \left(v^{(i)}, v^{(j)} \right).
\end{aligned}
$$

This will be useful later.

1.8.2 Orthogonal and Orthonormal Column Vectors

Consider two complex vectors u and v, of the same dimension. If they have zero inner product:

$$(u, v) = 0,$$

then we say that they are *orthogonal* to each other. What is the geometrical meaning of this? Well, in 2-D or 3-D, they are perpendicular to each other. If they also have norm 1:

$$(u, v) = 0 \text{ and } \|u\| = \|v\| = 1,$$

then we also say that they are *orthonormal*. What is the geometrical meaning of this? Well, in 2-D, they span new axes and make a new coordinate system. Still, here we are more interested in higher dimensions, where linear algebra extends the limits of standard geometrical intuition.

Look again at our $m \times n$ complex matrix, and rewrite it column by column:

$$A \equiv \left(v^{(1)} \mid v^{(2)} \mid \cdots \mid v^{(n)} \right),$$

for some $m \geq n$. Assume that its columns are orthonormal:

$$\left(v^{(i)}, v^{(j)} \right) = 0, \quad 1 \leq i, j \leq n, \ i \neq j,$$

and

$$\left\| v^{(i)} \right\| = 1, \quad 1 \leq i \leq n.$$

In this case, what is the (i, j)th element in $A^h A$? Well, it is either zero or one:

$$(A^h A)_{i,j} = \left(v^{(i)}, v^{(j)}\right) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

(Sect. 1.8.1). In other words, $A^h A$ is the identity matrix of order n:

$$A^h A = I.$$

As a result, A preserves the inner product of any two n-dimensional vectors u and v:

$$(Au, Av) = (A^h Au, v = (Iu, v)) = (u, v).$$

In particular, by picking $u = v$, A preserves norm as well:

$$\|Av\|^2 = (Av, Av) = (v, v) = \|v\|^2.$$

Next, let us work the other way around. Assume that we did not know that A had orthogonal columns. Instead, we only knew that $A^h A$ was the identity matrix:

$$A^h A = I.$$

Does this imply that A has orthonormal columns? Yes, it does! Indeed, from Sect. 1.8.1,

$$\left(v^{(i)}, v^{(j)}\right) = (A^h A)_{i,j} = I_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j, \end{cases}$$

as asserted. In summary, A has orthonormal columns if and only if

$$A^h A = I$$

is the identity matrix of order $n \leq m$.

1.8.3 Projection Matrix and Its Null Space

So far, we studied the product $A^h A$. Next, let us multiply the other way around: AA^h. This could be a different matrix: after all, the commutative law does not work any more. Still, what can we say about AA^h?

Thanks to our assumptions ($m \geq n$, and A has orthonormal columns), we can now use the above equation and the associative law:

$$\left(AA^h\right) A = A\left(A^h A\right) = AI = A.$$

Thus, AA^h has no effect on the columns of A: it leaves them unchanged. In this respect, it behaves quite like the identity matrix.

Still, this is not the whole story. Since $m \geq n$, there may be a few nonzero vectors that are orthogonal to all columns of A. Let v be such a vector:

$$A^h v = \mathbf{0}.$$

What is the effect of AA^h on v? Thanks to associativity,

$$\left(AA^h\right) v = A\left(A^h v\right) = A\mathbf{0} = \mathbf{0}.$$

Thus, v is in the null space of AA^h. In summary, AA^h is nearly the identity matrix: on the columns of A, it acts like the identity matrix. On vectors that are orthogonal to the columns of A, on the other hand, it acts like the zero matrix.

In summary, AA^h is a special kind of matrix: taking its square does not change it at all. Indeed, thanks to associativity,

$$\left(AA^h\right)^2 = \left(AA^h\right)\left(AA^h\right) = A\left(A^h A\right)A^h = AIA^h = AA^h.$$

In other words, AA^h is a projection matrix: it is the same as its own square.

1.8.4 Unitary and Orthogonal Matrix

Assume now that A is a square matrix of order $m = n$. If A has orthonormal columns, then we say that A is unitary. In this case, thanks to the above discussion,

$$A^h A = AA^h = I$$

(the identity matrix of order $m = n$). This could also be written in terms of the inverse matrix:

$$A^h = A^{-1}, \quad \text{and} \quad A = \left(A^h\right)^{-1}.$$

Now, the inverse of the transpose is the transpose of the inverse. Therefore, one could drop these parentheses and simply write

$$A = A^{-h}.$$

If A is also real, then it is also called an orthogonal matrix. In this case,

$$A^t A = AA^t = I,$$

so

$$A^t = A^{-1}, \quad \text{and} \quad A = A^{-t}.$$

1.9 Eigenvalues and Eigenvectors

1.9.1 Eigenvectors and Their Eigenvalues

Let A be a square complex matrix of order n. An eigenvector of A is a nonzero vector $v \in \mathbb{C}^n$ satisfying

$$Av = \lambda v,$$

for some number $\lambda \in \mathbb{C}$, called the eigenvalue. In other words, applying A to v has the same effect as multiplying by λ. This is why λ is called the eigenvalue of A associated with the eigenvector v.

Note that v is not unique: it could be scaled. In fact, v could be multiplied by any nonzero number $c \in \mathbb{C}$, producing a "new" eigenvector: cv. Indeed, cv still satisfies the same equation:

$$A(cv) = cAv = c\lambda v = \lambda(cv).$$

Thus, the eigenvector associated with λ is not defined uniquely but only up to a (nonzero) scalar multiple.

What is the best way to pick c? Well, since $v \neq \mathbf{0}$, $\|v\| > 0$ (Sect. 1.7.4). Thus, best pick $c \equiv 1/\|v\|$. This would "normalize" v and produce a "new" unit eigenvector:

$$A\left(\frac{v}{\|v\|}\right) = \lambda\left(\frac{v}{\|v\|}\right), \quad \left\|\frac{v}{\|v\|}\right\| = 1.$$

This is the unit eigenvector proportional to v.

1.9.2 Singular Matrix and Its Null Space

What are the algebraic properties of the eigenvector v? First of all, it is nonzero:

$$v \neq \mathbf{0},$$

where $\mathbf{0}$ is the n-dimensional zero vector. Furthermore, v also satisfies

$$(A - \lambda I)\, v = \lambda v - \lambda v = \mathbf{0},$$

where I is the identity matrix of order n. Thus, the matrix $A - \lambda I$ maps v to $\mathbf{0}$. In other words, v is in the null space of $A - \lambda I$.

For this reason, $A - \lambda I$ must be singular (noninvertible). Indeed, by contradiction: if there were an inverse matrix $(A - \lambda I)^{-1}$, then apply it to the zero vector, to map it back to v. On the other hand, from the very definition of matrix-times-vector, this must be the zero vector:

$$v = (A - \lambda I)^{-1}\mathbf{0} = \mathbf{0},$$

in violation of the very definition of v as a nonzero eigenvector. Thus, $A - \lambda I$ must indeed be singular, as asserted. Let us use this to design an eigenvalue for A^h as well.

1.9.3 Eigenvalues of the Hermitian Adjoint

So far, we discussed an eigenvalue of A: λ (associated with the eigenvector v). What about A^h? It also has an eigenvalue: $\bar{\lambda}$.

Indeed, A, λ, and v have a joint algebraic property: $A - \lambda I$ is singular. What about its Hermitian adjoint

$$(A - \lambda I)^h = A^h - \bar{\lambda} I?$$

Is it singular as well? It sure is. (After all, it has the same determinant: 0. Still, let us prove this without using determinants.)

Indeed, by contradiction: if it were nonsingular (invertible), then there would be some nonzero vector u, mapped to v:

$$\left(A^h - \bar{\lambda} I\right) u = v.$$

This would lead to a contradiction:

$$0 = (\mathbf{0}, u) = ((A - \lambda I)\, v, u) = \left(v, \left(A^h - \bar{\lambda} I\right) u\right) = (v, v) > 0.$$

So, $A^h - \bar{\lambda} I$ must be singular as well. As such, it must map some nonzero vector $w \neq \mathbf{0}$ to the zero vector:

$$\left(A^h - \bar{\lambda} I\right) w = \mathbf{0},$$

or

$$A^h w = \bar{\lambda} w.$$

Thus, $\bar{\lambda}$ must be an eigenvalue of A^h. In summary, the complex conjugate of any eigenvalue of A is an eigenvalue of A^h.

1.9.4 Eigenvalues of a Hermitian Matrix

Assume now that A is also Hermitian:

$$A = A^h.$$

Thanks to Sect. 1.7.8,

$$\lambda(v, v) = (v, \lambda v) = (v, Av) = (Av, v) = (\lambda v, v) = \bar{\lambda}(v, v).$$

Now, since v is a nonzero vector, $(v, v) > 0$ (Sect. 1.7.4). Thus, we can divide by (v, v):

$$\lambda = \bar{\lambda},$$

so λ is real:

$$\lambda \in \mathbb{R}.$$

In summary, a Hermitian matrix has real eigenvalues only. Next, what can we say about its eigenvectors?

1.9.5 Eigenvectors of a Hermitian Matrix

Let u and v be two eigenvectors of the Hermitian matrix A:

$$Av = \lambda v \quad \text{and} \quad Au = \mu u,$$

where $\lambda \neq \mu$ are two distinct eigenvalues. What is the relation between u and v? Well, they must be orthogonal to each other. Indeed, thanks to Sects. 1.7.8 and 1.9.4,

$$\mu(u, v) = \bar{\mu}(u, v) = (\mu u, v) = (Au, v) = (u, Av) = (u, \lambda v) = \lambda(u, v).$$

Now, since $\lambda \neq \mu$, we must have

$$(u, v) = 0,$$

as asserted. Furthermore, we can normalize both u and v, to obtain the orthonormal eigenvectors $u/\|u\|$ and $v/\|v\|$. Let us see some interesting examples.

1.10 The Sine Transform

1.10.1 Discrete Sine Waves

An interesting example is the discrete sine wave. To obtain it, sample the sine function (Fig. 1.10).

For a fixed $1 \leq j \leq n$, consider the sine mode (or wave, or oscillation)

$$\sin(j\pi x).$$

What is the role of j? Well, j is the wave number: it tells us how fast the wave oscillates. In fact, as j increases, the above function oscillates more and more rapidly. For a small j, the function is smooth and oscillates just a little. For a greater j, on the other hand, the wave oscillates more rapidly: move x just a little, and you may already see an oscillation. In other words, if x measures the time, then the wave oscillates more frequently. This is why j is called the wave number, or the frequency.

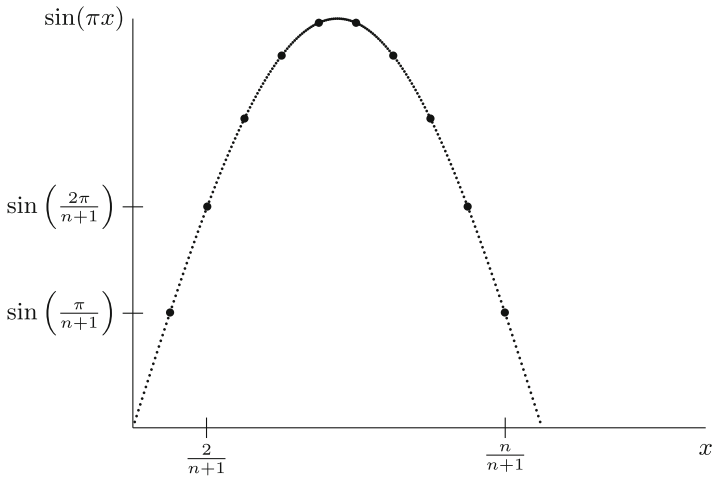


Fig. 1.10 The smoothest sine wave: $\sin(\pi x)$. To obtain the discrete sine mode, sample at n discrete points: $x = 1/(n+1)$, $x = 2/(n+1)$, $x = 3/(n+1)$, ..., $x = n/(n+1)$

The above sine mode is continuous: it is a function of every $x \in \mathbb{R}$. How to obtain the discrete sine mode? For this purpose, do two things:

- Discretize: sample the original sine mode at n equidistant points between 0 and 1:

$$x = \frac{1}{n+1}, \ \frac{2}{n+1}, \ \frac{3}{n+1}, \ \ldots, \ \frac{n}{n+1}.$$

  This produces the n-dimensional column vector

$$\begin{pmatrix} \sin\left(\frac{j\pi}{n+1}\right) \\ \sin\left(\frac{2j\pi}{n+1}\right) \\ \sin\left(\frac{3j\pi}{n+1}\right) \\ \vdots \\ \sin\left(\frac{nj\pi}{n+1}\right) \end{pmatrix}.$$

- Normalize: multiply by $\sqrt{2/(n+1)}$, to produce the new column vector

$$v^{(j)} \equiv \sqrt{\frac{2}{n+1}} \begin{pmatrix} \sin\left(\frac{j\pi}{n+1}\right) \\ \sin\left(\frac{2j\pi}{n+1}\right) \\ \sin\left(\frac{3j\pi}{n+1}\right) \\ \vdots \\ \sin\left(\frac{nj\pi}{n+1}\right) \end{pmatrix}.$$

This is the discrete sine mode. Thanks to the above normalization, it has norm 1:

$$\|v^{(j)}\| = 1$$

(see exercises below). Furthermore, as we will see next, the discrete sine modes are orthogonal to each other.

1.10.2 Orthogonality of the Discrete Sine Waves

Are the discrete sine modes orthogonal to each other? Yes! Indeed, they are the eigenvectors of a new symmetric matrix T. T is tridiagonal: it has nonzero elements on three diagonals only—the main diagonal, the diagonal just above it, and the diagonal just below it:

$$T \equiv \text{tridiag}(-1, 2, -1) \equiv \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

The rest of the elements, on the other hand, vanish. No need to write them explicitly—the blank spaces stand for zeroes.

As a real symmetric matrix, T is also Hermitian:

$$T^h = T^t = T.$$

What are its eigenvectors? The discrete sine modes! To see this, recall the formula:

$$\sin(\theta + \phi) + \sin(\theta - \phi) = 2\sin(\theta)\cos(\phi).$$

To study the ith component in $v^{(j)}$, set in this formula

$$\theta = \frac{ij\pi}{n+1} \text{ and } \phi = \frac{j\pi}{n+1}.$$

This gives

$$\sin\left(\frac{(i+1)j\pi}{n+1}\right) + \sin\left(\frac{(i-1)j\pi}{n+1}\right) = 2\sin\left(\frac{ij\pi}{n+1}\right)\cos\left(\frac{j\pi}{n+1}\right).$$

By doing this for all components $1 \le i \le n$, we have

$$Tv^{(j)} = \lambda_j v^{(j)},$$

where

$$\lambda_j \equiv 2 - 2\cos\left(\frac{j\pi}{n+1}\right) = 4\sin^2\left(\frac{j\pi}{2(n+1)}\right).$$

In summary, T is a Hermitian matrix, with n distinct eigenvalues. From Sect. 1.9.5, its eigenvectors are indeed orthogonal to each other.

Later on, we will see a more general theorem: a Hermitian matrix has a diagonal Jordan form, and an eigenbasis: a basis of (orthonormal) eigenvectors. Let us use them to expand a given vector.

1.10.3 The Sine Transform

Now, let us place the discrete sine modes in a new matrix, column by column:

$$W \equiv \left(v^{(1)} \mid v^{(2)} \mid \cdots \mid v^{(n)} \right).$$

By now, we already know that W is orthogonal. From Sect. 1.8.4, we therefore have

$$W^{-1} = W^h = W^t = W.$$

W is called the sine transform: it transforms each n-dimensional vector u to a new vector of the same norm:

$$\|Wu\| = \|u\|.$$

To uncover the original vector u, apply W once again:

$$u = W(Wu).$$

1.10.4 Diagonalization

Now, let us place the eigenvalues of T in a new diagonal matrix:

$$\Lambda \equiv \operatorname{diag}(\lambda_i)_{i=1}^n \equiv \operatorname{diag}(\lambda_1, \lambda 2, \ldots, \lambda_n) \equiv \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}.$$

Recall that the columns of W are the eigenvectors of T:

$$TW = W\Lambda,$$

or

$$T = W\Lambda W^{-1} = W\Lambda W.$$

This is called the diagonal form (or the diagonalization, or the spectral decomposition) of T. This is how the sine transform helps diagonalize the original matrix T.

1.10.5 Sine Decomposition

The sine transform W is a real, symmetric, and orthogonal matrix:

$$W^2 = W^t W = W^h W = W^{-1} W = I.$$

For this reason, every vector $u \in \mathbb{C}^n$ can be written uniquely as a linear combination of the columns of W, with coefficients that are the components of Wu:

$$u = Iu = W^2 u = W(Wu) = \sum_{j=1}^{n} (Wu)_j v^{(j)}.$$

This way, u is decomposed in terms of more and more frequent (or oscillatory) waves, each multiplied by the corresponding amplitude $(Wu)_j$.

For example, what is the smoothest part of u? It is now filtered out, in terms of the first discrete wave, times the first amplitude:

$$(Wu)_1 v^{(1)}.$$

What is the next (more oscillatory) part? It is

$$(Wu)_2 v^{(2)},$$

and so on, until the most oscillatory term

$$(Wu)_n v^{(n)}.$$

1.10.6 Multiscale Decomposition

The above can be viewed as a multiscale decomposition. The first discrete wave can be viewed as the coarsest scale, in which the original vector u is approximated rather poorly. The remainder (or the error) is

$$u - (Wu)_1 v^{(1)}.$$

This is approximated by the second discrete wave, on the next finer scale. This contributes a finer term to produce a better approximation of u. The up-to-date remainder is then

$$u - (Wu)_1 v^{(1)} - (Wu)_2 v^{(2)}.$$

This is approximated by the third discrete wave, on the next finer scale, and so on. In the end, the most oscillatory (finest) term is added as well to complete the entire multiscale decomposition. This is no longer an approximation: it is exactly u.

1.11 The Cosine Transform

1.11.1 Discrete Cosine Waves

Likewise, one could also define cosine waves. For a fixed $1 \le j \le n$, consider the function

$$\cos((j-1)\pi x).$$

To discretize, just sample the above cosine function at n equidistant points between 0 and 1:

$$x = \frac{1}{2n}, \frac{3}{2n}, \frac{5}{2n}, \ldots, \frac{2n-1}{2n}.$$

This produces a new $v^{(j)}$: the discrete cosine mode, whose components are now

$$v_i^{(j)} \equiv \cos\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right), \quad 1 \le i \le n$$

(Fig. 1.11). The normalization is left to the exercises below.

1.11.2 Orthogonality of the Discrete Cosine Waves

To show orthogonality, redefine T at its upper-left and lower-right corners:

$$T_{1,1} \equiv T_{n,n} \equiv 1$$

rather than 2. This way, T takes the new form

$$T \equiv \begin{pmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{pmatrix}.$$

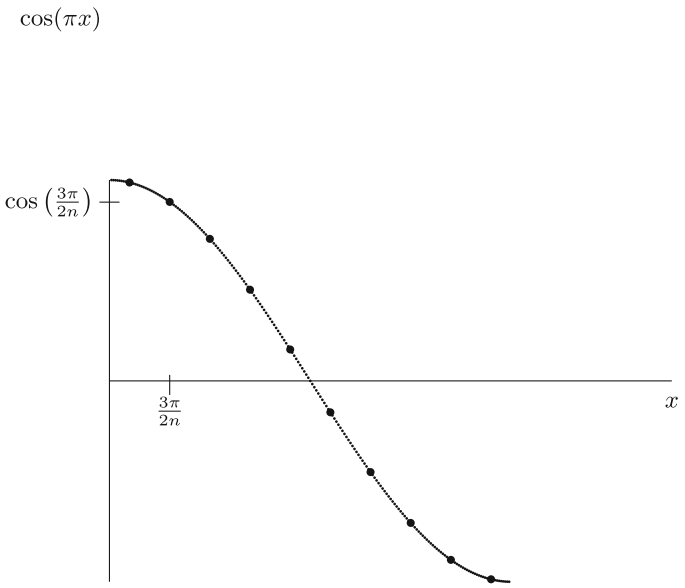


Fig. 1.11 The smoothest (nonconstant) cosine wave: $\cos(\pi x)$. Sample it at n discrete points: $x = 1/(2n)$, $x = 3/(2n)$, $x = 5/(2n)$, ..., $x = (2n-1)/(2n)$

Although the new matrix T is still real, tridiagonal, and symmetric, its eigenvectors are completely different from those in Sect. 1.10.2: they are now the discrete cosine waves. To see this, recall the formula:

$$\cos(\theta + \phi) + \cos(\theta - \phi) = 2\cos(\theta)\cos(\phi).$$

In particular, set

$$\theta = \left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n} \text{ and } \phi = (j-1)\frac{\pi}{n}.$$

With these θ and ϕ, the above formula takes the form:

$$\cos\left(\left(i + \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right) + \cos\left(\left(i - \frac{3}{2}\right)(j-1)\frac{\pi}{n}\right)$$
$$= 2\cos\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right)\cos\left((j-1)\frac{\pi}{n}\right).$$

Thus, the discrete cosine wave defined in Sect. 1.11.1 satisfies

$$Tv^{(j)} = \lambda_j v^{(j)},$$

with the new eigenvalue

$$\lambda_j = 2 - 2\cos\left((j-1)\frac{\pi}{n}\right) = 4\sin^2\left((j-1)\frac{\pi}{2n}\right).$$

Because T is symmetric, its eigenvectors are orthogonal to each other (Sect. 1.9.5). This proves orthogonality of the discrete cosine modes, as asserted.

1.11.3 The Cosine Transform

We are now ready to normalize:

$$v^{(j)} \leftarrow \frac{v^{(j)}}{\left\|v^{(j)}\right\|}$$

(see exercises below). Let us place these (normalized) cosine modes in a new matrix, column by column:

$$W \equiv \left(v^{(1)} \mid v^{(2)} \mid \cdots \mid v^{(n)}\right).$$

This new W, known as the cosine transform, is still real and orthogonal, but no longer symmetric:

$$W^{-1} = W^h = W^t \neq W.$$

1.11.4 Diagonalization

For this reason, our new T is diagonalized in a slightly different way:

$$T = W\Lambda W^{-1} = W\Lambda W^t \neq W\Lambda W,$$

where Λ is now a new diagonal matrix, containing the new eigenvalues:

$$\Lambda \equiv \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

1.11.5 Cosine Decomposition

Furthermore, every vector $u \in \mathbb{C}^n$ can now be decomposed uniquely in terms of discrete cosine waves as well:

$$u = Iu = (WW^t)u = W(W^t u) = \sum_{j=1}^{n} (W^t u)_j v^{(j)}.$$

This is called the cosine decomposition of u.

1.12 Positive (Semi)definite Matrix

1.12.1 Positive Semidefinite Matrix

Consider a Hermitian matrix A. Assume that, for every complex vector v,

$$(v, Av) \equiv v^h Av \geq 0.$$

We then say that A is positive semidefinite.

In particular, we could pick v to be an eigenvector of A, with the eigenvalue λ. This would give

$$\|v\|^2 \lambda = (v, v)\lambda = (v, \lambda v) = (v, Av) \geq 0.$$

Recall that v is a nonzero vector. As such, it has a nonzero norm. Therefore, we can divide by $\|v\|^2$:

$$\lambda \geq 0.$$

Thus, all eigenvalues of A must be nonnegative.

This also works the other way around: assume that we did not know that A was positive semidefinite. Instead, we only knew that A was Hermitian, with nonnegative eigenvalues only. Must A be positive semidefinite? Yes, it must. To see this, given a complex vector v, span it in terms of the (orthogonal) eigenvectors.

1.12.2 Positive Definite Matrix

Consider again a Hermitian matrix A. Assume that, for every nonzero vector $v \neq \mathbf{0}$, we even have a strict inequality:

$$(v, Av) \equiv v^h Av > 0.$$

We then say that A is not only positive semidefinite but also positive definite.

In particular, we could pick v to be an eigenvector of A, with the eigenvalue λ. This would give

$$\|v\|^2\lambda = (v, v)\lambda = (v, \lambda v) = (v, Av) > 0.$$

Now, divide by $\|v\|^2 > 0$:

$$\lambda > 0.$$

Therefore, all eigenvalues of A are positive. Thus, A must be nonsingular: 0 is not an eigenvalue.

This also works the other way around: assume that we did not know that A was positive definite. Instead, we only knew that A was Hermitian, with positive eigenvalues only. Must A be positive definite? Yes, it must. To see this, given a nonzero vector $v \neq \mathbf{0}$, span it in terms of the (orthogonal) eigenvectors.

1.13 Exercises: Generalized Eigenvalues

1.13.1 The Cauchy–Schwarz Inequality

1. Let u and v be n-dimensional (complex) vectors:

$$u \equiv (u_1, u_2, \ldots, u_n)^t \text{ and } v \equiv (v_1, v_2, \ldots, v_n)^t.$$

Prove the Cauchy–Schwarz inequality:

$$|(u, v)| \leq \|u\| \cdot \|u\|,$$

or

$$\left|\sum_{i=1}^{n} \bar{u}_i v_i\right|^2 = |(u, v)|^2 \leq (u, u)(v, v) = \left(\sum_{i=1}^{n} |u_i|^2\right)\left(\sum_{i=1}^{n} |v_i|^2\right).$$

Hint: pick two different indices between 1 and n: $0 \leq i \neq j \leq n$. Note that

$$\begin{aligned}
0 &\leq |u_i v_j - u_j v_i|^2 \\
&= \left(\bar{u}_i \bar{v}_j - \bar{u}_j \bar{v}_i\right)\left(u_i v_j - u_j v_i\right) \\
&= \bar{u}_i \bar{v}_j u_i v_j - \bar{u}_i \bar{v}_j u_j v_i - \bar{u}_j \bar{v}_i u_i v_j + \bar{u}_j \bar{v}_i u_j v_i \\
&= |u_i|^2 |v_j|^2 - (\bar{u}_i v_i)(u_j \bar{v}_j)) - (\bar{u}_j v_j)(u_i \bar{v}_i) + |u_j|^2 |v_i|^2,
\end{aligned}$$

so

$$(\bar{u}_i v_i)(u_j \bar{v}_j)) + (\bar{u}_j v_j)(u_i \bar{v}_i) \leq |u_i|^2 |v_j|^2 + |u_j|^2 |v_i|^2.$$

Do the same with the plus sign, to obtain

$$\left|(\bar{u}_i v_i)(u_j \bar{v}_j)) + (\bar{u}_j v_j)(u_i \bar{v}_i)\right| \leq |u_i|^2 |v_j|^2 + |u_j|^2 |v_i|^2.$$

Finally, sum this over $1 \leq i < j \leq n$.

2. Could the Cauchy–Schwarz inequality be an exact equality as well? Hint: only if u is proportional to v (or is a scalar multiple of v).

1.13.2 The Triangle Inequality

1. Conclude the triangle inequality:

$$\|u + v\| \leq \|u\| + \|v\|.$$

Hint:

$$\begin{aligned}
\|u + v\|^2 &= (u + v, u + v) \\
&= (u, u) + (u, v) + (v, u) + (v, v) \\
&= \|u\|^2 + (u, v) + (v, u) + \|v\|^2 \\
&\leq \|u\|^2 + 2|(u, v)| + \|v\|^2 \\
&\leq \|u\|^2 + 2\|u\| \cdot \|v\| + \|v\|^2 \\
&= (\|u\| + \|v\|)^2 .
\end{aligned}$$

1.13.3 Generalized Eigenvalues

1. Let A be a square complex matrix of order n. Let B be a Hermitian matrix of order n. Let v be an n-dimensional vector satisfying

$$Av = \lambda Bv \quad \text{and} \quad (v, Bv) \neq 0.$$

Then the (complex) scalar λ is called a generalized eigenvalue of A.

2. Could v be the zero vector?
3. Conclude that $A - \lambda B$ maps v to the zero vector.
4. Conclude that v is in the null space of $A - \lambda B$.
5. Is $A - \lambda B$ singular? Hint: otherwise,

$$v = (A - \lambda B)^{-1} \mathbf{0} = \mathbf{0},$$

in violation of $(v, Bv) \neq 0$.

6. Is $A^h - \bar{\lambda}B$ singular as well? Hint: otherwise, there would be a nonzero vector u mapped to v:

$$\left(A^h - \bar{\lambda}B\right) u = v,$$

leading to a contradiction:

$$0 = (\mathbf{0}, u) = ((A - \lambda B)\, v, u) = \left(v, \left(A^h - \bar{\lambda}B\right) u\right) = (v, v) > 0.$$

7. Prove this in yet another way. Hint:

$$\det\left(A^h - \bar{\lambda}B\right) = \det\left(A - \lambda B\right) = 0.$$

8. Conclude that $A^h - \bar{\lambda}B$ has a nontrivial null space.
9. Conclude that $\bar{\lambda}$ is a generalized eigenvalue of A^h.
10. Assume now that A is Hermitian as well. Must the generalized eigenvalue λ be real? Hint:

$$\lambda(v, Bv) = (v, \lambda Bv) = (v, Av) = (Av, v) = (\lambda Bv, v) = \bar{\lambda}(Bv, v) = \bar{\lambda}(v, Bv).$$

Finally, divide this by $(v, Bv) \neq 0$.

11. Let u and v be two such vectors:

$$Av = \lambda Bv \quad \text{and} \quad Au = \mu Bu,$$

where $\lambda \neq \mu$ are two distinct generalized eigenvalues. Show that

$$(u, Bv) = 0.$$

Hint:

$$\begin{aligned}
\mu(u, Bv) &= \mu(Bu, v) \\
&= \bar{\mu}(Bu, v) \\
&= (\mu Bu, v) \\
&= (Au, v) \\
&= (u, Av) \\
&= (u, \lambda Bv) \\
&= \lambda(u, Bv).
\end{aligned}$$

12. Assume now that B is not only Hermitian but also positive definite. Define the new "inner product:"

$$(\cdot, \cdot)_B \equiv (\cdot, B\cdot)\,.$$

13. Is this a legitimate inner product? Hint: show that it is indeed bilinear, skew-symmetric, and positive and vanishes only for the zero vector.
14. Look at the new matrix $B^{-1}A$. Show that it is "Hermitian" with respect to the new "inner product:"

$$\left(B^{-1}A\cdot, \cdot\right)_B = (A\cdot, \cdot) = (\cdot, A\cdot) = \left(\cdot, B^{-1}A\cdot\right)_B\,.$$

15. Show that the above u and v are eigenvectors of $B^{-1}A$.
16. Show that they are indeed orthogonal with respect to the new "inner product."
17. Normalize them with respect to the new "inner product."
18. For example, set $B \equiv I$. Is this familiar?

1.13.4 Root of Unity and Fourier Transform

1. Let n be some natural number. Define the complex number

$$w \equiv \exp\left(\frac{2\pi\sqrt{-1}}{n}\right).$$

Show that w can also be written as

$$w = \cos\left(\frac{2\pi}{n}\right) + \sqrt{-1}\sin\left(\frac{2\pi}{n}\right).$$

2. Show that w is the nth root of unity:

$$w^n = \exp^n\left(\frac{2\pi\sqrt{-1}}{n}\right) = \exp\left(\frac{2\pi\sqrt{-1}}{n}n\right) = \exp(2\pi\sqrt{-1}) = 1.$$

3. Look at Fig. 1.12. How many roots of unity are there in it? Hint: for each $1 \le j < n$,

$$\left(w^j\right)^n = w^{jn} = w^{nj} = \left(w^n\right)^j = 1^j = 1.$$

4. Use w and its powers to design a new $n \times n$ complex matrix:

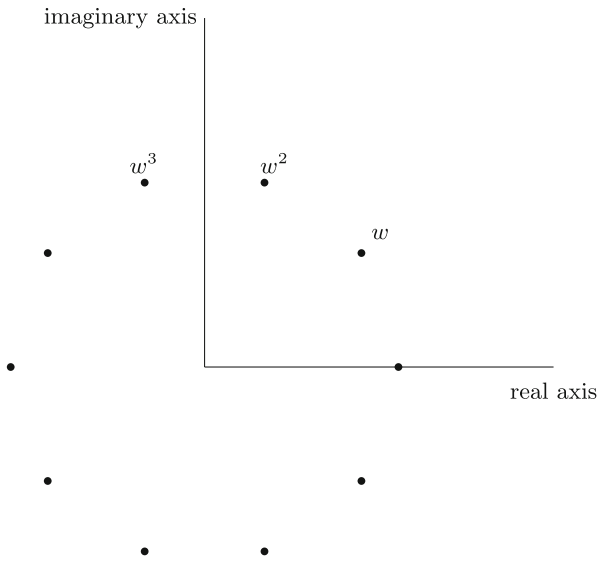


Fig. 1.12 The nth root of unity, and its powers in the complex plane: w, w^2, w^3, ..., $w^n = 1$

$$W \equiv n^{-1/2} \left(w^{(i-1)(j-1)} \right)_{1 \le i,j \le n}.$$

5. Let $1 \le j \le n$ be fixed. Show that the jth column in W could also be obtained from the exponent wave

$$\exp(2\pi\sqrt{-1}(j-1)x),$$

sampled at n equidistant points

$$x = 0, \ \frac{1}{n}, \ \frac{2}{n}, \ \frac{3}{n}, \ \ldots, \ \frac{n-1}{n},$$

and normalized by $n^{1/2}$.

6. Show that the Hermitian adjoint of W is the complex conjugate:

$$W^h = \bar{W}.$$

7. Show that the first column in W is the constant column vector

$$n^{-1/2}(1, 1, 1, \ldots, 1)^t.$$

8. Show that this is a unit vector of norm 1.
9. Show that this is indeed the first discrete cosine wave in Sect. 1.11.1 (in its normalized form).
10. Show that this eigenvector has the zero eigenvalue in Sect. 1.11.2.

11. Conclude that the tridiagonal matrix T defined in Sect. 1.11.2 is singular (not invertible).
12. Show that, in the new matrix W defined above, every column is a unit vector of norm 1 as well.
13. Show that, for $1 < j \leq n$, the jth column in W sums to zero:

$$\sum_{i=1}^{n} w^{(i-1)(j-1)} = \sum_{i=0}^{n-1} w^{i(j-1)} = \frac{1 - w^{n(j-1)}}{1 - w^{j-1}} = \frac{1-1}{1 - w^{j-1}} = 0.$$

14. Multiply the above equation by $w^{(j-1)/2}$, to obtain

$$\sum_{i=1}^{n} w^{(i-1/2)(j-1)} = 0$$

$(1 < j \leq n)$.
15. Look at the real part of this equation, to conclude that

$$\begin{aligned}
&\left(\sum_{i=1}^{n} \cos^2\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right)\right) - \left(\sum_{i=1}^{n} \sin^2\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right)\right) \\
&= \sum_{i=1}^{n} \left(\cos^2\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right) - \sin^2\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right)\right) \\
&= \sum_{i=1}^{n} \cos\left(\left(i - \frac{1}{2}\right)(j-1)\frac{2\pi}{n}\right) \\
&= 0
\end{aligned}$$

$(1 < j \leq n)$.
16. In the uniform grid in Sect. 1.11.1, sum the squares of sines and cosines:

$$\begin{aligned}
&\left(\sum_{i=1}^{n} \cos^2\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right)\right) + \left(\sum_{i=1}^{n} \sin^2\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right)\right) \\
&= \sum_{i=1}^{n} \left(\cos^2\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right) + \sin^2\left(\left(i - \frac{1}{2}\right)(j-1)\frac{\pi}{n}\right)\right) \\
&= \sum_{i=1}^{n} 1 \\
&= n
\end{aligned}$$

$(1 < j \leq n)$.

17. Add these two formulas to each other, to conclude that the discrete cosine waves in Sect. 1.11.1 have norm

$$\left\| v^{(j)} \right\| = \begin{cases} \sqrt{n} & \text{if } j = 1 \\ \sqrt{\frac{n}{2}} & \text{if } j > 1. \end{cases}$$

18. Show that the eigenvalues in Sect. 1.11.2 are distinct.
19. Conclude that the discrete cosine waves are indeed orthogonal to each other.
20. Rewrite the zero column sums in W

$$\sum_{i=1}^{n} w^{(i-1)(j-1)} = 0 \quad (1 < j \leq n)$$

in the simpler form

$$\sum_{i=0}^{n-1} w^{ij} = 0 \quad (1 \leq j < n),$$

or

$$\sum_{i=1}^{n-1} w^{ij} = -1 \quad (1 \leq j < n).$$

21. Look at the real part of this equation, to conclude that

$$\begin{aligned} -1 &= \sum_{i=1}^{n-1} \cos\left(\frac{2ij\pi}{n}\right) \\ &= \sum_{i=1}^{n-1} \left(\cos^2\left(\frac{ij\pi}{n}\right) - \sin^2\left(\frac{ij\pi}{n}\right) \right) \\ &= \left(\sum_{i=1}^{n-1} \cos^2\left(\frac{ij\pi}{n}\right) \right) - \left(\sum_{i=1}^{n-1} \sin^2\left(\frac{ij\pi}{n}\right) \right), \end{aligned}$$

$(1 \leq j < n)$.

22. Substitute $n + 1$ for n in the above equation, to read

$$\left(\sum_{i=1}^{n} \cos^2\left(\frac{ij\pi}{n+1}\right) \right) - \left(\sum_{i=1}^{n} \sin^2\left(\frac{ij\pi}{n+1}\right) \right) = -1$$

$(1 \leq j \leq n)$.

23. In the uniform grid in Sect. 1.10.1, sum the squares of sines and cosines:

$$\begin{aligned}
&\left(\sum_{i=1}^{n} \cos^2\left(\frac{ij\pi}{n+1}\right)\right) + \left(\sum_{i=1}^{n} \sin^2\left(\frac{ij\pi}{n+1}\right)\right) \\
&= \sum_{i=1}^{n} \left(\cos^2\left(\frac{ij\pi}{n+1}\right) + \sin^2\left(\frac{ij\pi}{n+1}\right)\right) \\
&= \sum_{i=1}^{n} 1 \\
&= n
\end{aligned}$$

$(1 \le j \le n)$.

24. Subtract these two formulas from each other, to confirm that the discrete sine waves introduced in Sect. 1.10.1 are indeed unit vectors of norm 1.
25. Show that the eigenvalues in Sect. 1.10.2 are distinct.
26. Conclude that the discrete sine waves are indeed orthonormal.
27. In Sect. 1.10.2, in the $n \times n$ matrix T, redefine the elements in the upper-right and lower-left corners as

$$T_{1,n} \equiv T_{n,1} \equiv -1$$

rather than the original definition

$$T_{1,n} \equiv T_{n,1} \equiv 0.$$

This way, T is no longer tridiagonal, but periodic:

$$T \equiv \begin{pmatrix} 2 & -1 & & & -1 \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ -1 & & & -1 & 2 \end{pmatrix},$$

or

$$T_{i,j} = \begin{cases} 2 & \text{if } i = j \\ -1 & \text{if } |i-j| = 1 \text{ or } |i-j| = n-1 \\ 0 & \text{otherwise} \end{cases}$$

$(1 \le i, j \le n)$. Show that this new T is no longer tridiagonal.

28. Show that, in this new T, the rows sum to zero.

29. Conclude that the first column of W (the constant n-dimensional unit vector) is an eigenvector of this new T, with the zero eigenvalue.
30. Conclude that this new T is singular.
31. More generally, show that the jth column of W ($1 \le j \le n$) is an eigenvector of this new T, with the new eigenvalue

$$\lambda_j = 2 - \left(w^{j-1} + w^{-(j-1)}\right) = 2 - 2\cos\left(\frac{(j-1)2\pi}{n}\right) = 4\sin^2\left(\frac{(j-1)\pi}{n}\right).$$

32. Use the above to form the matrix equation

$$TW = W\Lambda,$$

where Λ is now the new $n \times n$ diagonal matrix

$$\Lambda \equiv \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

33. So far, we have seen that T has complex eigenvectors: the columns of W. Does it have real eigenvectors as well?
34. Design them! Hint: follow the exercises below, one by one.
35. For this purpose, look at the jth column of W. Look at its real part. Is it an eigenvector of T in its own right? Hint: Yes! After all, T and λ_j are real.
36. What is its eigenvalue? Hint: λ_j.
37. Look again at the jth column of W. Look at its imaginary part. Is it an eigenvector of T in its own right? Hint: Yes! After all, T and λ_j are real.
38. What is its eigenvalue? Hint: λ_j.
39. Are the above eigenvalues different from each other? Hint: most of them are. Only for $1 \le j < n/2$ is the $(j+1)$st eigenvalue the same as the $(n-j+1)$st one:

$$\lambda_{n-j+1} = 4\sin^2\left(\frac{(n-j)\pi}{n}\right) = 4\sin^2\left(\pi - \frac{j\pi}{n}\right) = 4\sin^2\left(\frac{j\pi}{n}\right) = \lambda_{j+1}.$$

40. Conclude that the $(j+1)$st and the $(n-j+1)$st columns of W are eigenvectors of T, with the same eigenvalue.
41. Show that these column vectors are the complex conjugate of each other.
42. Conclude that their sum is twice their real part, which is an eigenvector of T as well, with the same eigenvalue: λ_{j+1}.
43. Conclude also that the difference between them is $2\sqrt{-1}$ times their imaginary part, which is an eigenvector of T as well, with the same eigenvalue: λ_{j+1}.
44. Show that those columns of W that correspond to different eigenvalues are indeed orthogonal to each other. Hint: T is symmetric (Sect. 1.9.5).
45. Show that the $(j+1)$st and the $(n-j+1)$st columns of W, although having the same eigenvalue λ_{j+1}, are orthogonal to each other as well:

$$\begin{aligned}\sum_{i=1}^{n} \bar{w}^{(n-j)(i-1)} w^{j(i-1)} &= \sum_{i=0}^{n-1} \bar{w}^{(n-j)i} w^{ji} \\ &= \sum_{i=0}^{n-1} w^{(j-n)i} w^{ji} \\ &= \sum_{i=0}^{n-1} w^{(j-n+j)i} \\ &= \sum_{i=0}^{n-1} w^{2ji} \\ &= \frac{1-w^{2jn}}{1-w^{2j}} \\ &= \frac{1-1}{1-w^{2j}} \\ &= 0\end{aligned}$$

$(1 \le j < n/2)$.

46. Use a similar calculation to verify directly that every two columns in W are indeed orthogonal to each other. Hint: for every $1 \le k \ne j \le n$,

$$\begin{aligned}\sum_{i=1}^{n} \bar{w}^{(k-1)(i-1)} w^{(j-1)(i-1)} &= \sum_{i=0}^{n-1} \bar{w}^{(k-1)i} w^{(j-1)i} \\ &= \sum_{i=0}^{n-1} w^{(1-k)i} w^{(j-1)i} \\ &= \sum_{i=0}^{n-1} w^{(j-k)i} \\ &= \frac{1-w^{(j-k)n}}{1-w^{j-k}} \\ &= \frac{1-1}{1-w^{j-k}} \\ &= 0.\end{aligned}$$

47. Show that the columns of W are also unit vectors of norm 1.
48. Conclude that the columns of W are orthonormal.

49. Conclude that W is a unitary matrix. (W is known as the discrete Fourier transform.)
50. Verify that W indeed satisfies

$$W^h W = \bar{W}^t W = \bar{W} W = I$$

and

$$W W^h = W \bar{W}^t = W \bar{W} = I,$$

where I is the $n \times n$ identity matrix.
51. Multiply the matrix equation

$$TW = W\Lambda$$

by W^{-1} from the right, to obtain our new T in its diagonal form:

$$T = W\Lambda\bar{W}.$$

52. Write an efficient algorithm to calculate Wu, for any given vector $u \in \mathbb{C}^n$. The solution can be found in Chapter 5 in [75].
53. Let $K \equiv (K_{i,j})_{1 \le i,j \le n}$ be the $n \times n$ matrix with 1s on the secondary diagonal (from the upper-right to the lower-left corner), and 0s elsewhere:

$$K_{i,j} = \begin{cases} 1 & \text{if } i + j = n + 1 \\ 0 & \text{otherwise.} \end{cases}$$

Show that K is both symmetric and orthogonal.
54. Conclude that

$$K^2 = K^t K = I.$$

55. Verify that $K^2 = I$ by a direct calculation.
56. Conclude that K is a projection matrix.

Chapter 2
Determinant and Vector Product and Their Applications in Geometrical Mechanics



How to use vectors and matrices? Well, we have already seen a few important applications: the sine, cosine, and Fourier transforms. Here, on the other hand, we use matrices and their determinant to introduce yet another practical operation: vector product in 3-D. This will help introduce angular momentum, and its conservation law.

2.1 The Determinant

2.1.1 Minors and the Determinant

For a real square matrix, the determinant is a real function: it maps the original matrix to a real number. For a complex matrix, on the other hand, the determinant is a complex function: it maps the original matrix to a new complex number: its determinant. To define the determinant, we must first define the minor.

Let A be a square matrix of order $n > 1$. Let $1 \leq i, j \leq n$ be two fixed indices. Define a slightly smaller $(n-1) \times (n-1)$ matrix: drop from A its ith row and jth column. The result is indeed a smaller matrix: just $n-1$ rows and $n-1$ columns. This is the (i, j)th minor of A, denoted by $A^{(i,j)}$.

Thanks to the minors, we can now go ahead and define the determinant recursively. If A is very small and contains one entry only, then its determinant is just this entry. If, on the other hand, A is bigger than that, then its determinant is a linear combination of the determinants of those minors obtained by dropping the first row:

$$\det(A) \equiv \begin{cases} a_{1,1} & \text{if } n = 1 \\ \sum_{j=1}^{n} (-1)^{j+1} a_{1,j} \det\left(A^{(1,j)}\right) & \text{if } n > 1. \end{cases}$$

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This kind of recursion could also be viewed as mathematical induction on $n = 1, 2, 3, \ldots$. Indeed, for $n = 1$, $\det(A)$ is just the only element in A: $\det(A) \equiv a_{1,1}$. For $n > 1$, on the other hand, the minors are smaller matrices of order $n - 1$, whose determinant has already been defined in the induction hypothesis, and can be used to define $\det(A)$ as well. This completes the induction step, and indeed the entire definition, as required. Later on, we will see yet another (equivalent) definition.

2.1.2 Examples

For example, let I be the identity matrix of order n. Then,

$$\det(I) = 1.$$

This could be proved easily by mathematical induction. After all, in the above formula, all minors drop, except for the $(1, 1)$st one: the $(n - 1) \times (n - 1)$ identity matrix.

For yet another example, let α be some scalar. Then,

$$\det(\alpha I) = \alpha^n.$$

This could be proved by mathematical induction as well. After all, in the above formula, all minors drop, except for the $(1, 1)$st one.

Another interesting example is the so-called switch matrix:

$$\det\left(\begin{pmatrix} & 1 & & & & \\ 1 & & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \end{pmatrix}\right) = -1.$$

Why is it called a switch matrix? Because, once applied to a vector, it interchanges its first and second components. Again, in the calculation of its determinant, most of its minors drop: only the $(1, 2)$nd minor survives—the $(n - 1) \times (n - 1)$ identity matrix, with a minus sign.

As a final example, look at the special case of $n = 2$. In this case, we have a small 2×2 matrix. Its determinant is

$$\det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = ad - bc.$$

Indeed, there are here two nonzero minors: the $(1, 1)$st minor is just the lower-right element d. To contribute to the determinant, it must be multiplied y a. The $(1, 2)$nd minor, on the other hand, is the lower-left element c. To contribute to the determinant, it must be multiplied by b, and pick a minus sign.

Later on, we will interpret this determinant geometrically: the area of the parallelogram that the columns (or rows) of the 2×2 matrix make in the Cartesian plane (Chap. 2, Sect. 2.3.3). This will be quite useful in special relativity later on (Chap. 4).

2.1.3 Algebraic Properties

Let us mention some algebraic properties of the determinant. Let A and B be square matrices of order n. The determinant of the product is the same as the product of the individual determinants:

$$\det(AB) = \det(A)\det(B).$$

(This will be proved in Chap. 14, Sect. 14.4.3.) This is quite useful. For example, what happens when two rows in A interchange? The determinant just picks a minus sign. For instance, to interchange the first and second rows, apply the above switch matrix:

$$\det\left(\begin{pmatrix} & 1 & & & & \\ 1 & & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \end{pmatrix} A\right) = \det\begin{pmatrix} & 1 & & & & \\ 1 & & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \end{pmatrix} \det(A) = -\det(A).$$

For this reason, if both rows are the same, then the determinant must vanish:

$$\det(A) = -\det(A) = 0.$$

In this case, A must be singular: it has no inverse. Fortunately, this is rather rare. More often, A has a nonzero determinant and is therefore nonsingular (invertible), as we will see below.

Another useful property is that the transpose has the same determinant:

$$\det\left(A^t\right) = \det(A).$$

(This will be proved in Chap. 14, Sect. 14.4.2.) Thus, to calculate the determinant, we could equally well work with columns rather than rows:

$$\det(A) = \begin{cases} a_{1,1} & \text{if } n = 1 \\ \sum_{i=1}^{n} (-1)^{i+1} a_{i,1} \det\left(A^{(i,1)}\right) & \text{if } n > 1. \end{cases}$$

This will be useful below.

Thanks to these two properties, we also have a third one: if Q is a real orthogonal matrix, then

$$(\det(Q))^2 = \det\left(Q^t\right) \det(Q) = \det\left(Q^t Q\right) = \det(I) = 1,$$

so

$$\det(Q) = \pm 1.$$

Later on in this chapter, we will pick the correct sign. Before going into this, let us use the determinant of a nonsingular matrix to design its inverse (Chap. 1, Sect. 1.5.3).

2.1.4 The Inverse Matrix in Its Explicit Form

If $\det(A) \neq 0$, then A is nonsingular: it has an inverse matrix. In this case, $\det(A)$ could be placed in the denominator and help define A^{-1} explicitly. More precisely, each individual element in A^{-1} is given in terms of the transpose minor:

$$\left(A^{-1}\right)_{i,j} = (-1)^{i+j} \frac{\det\left(A^{(j,i)}\right)}{\det(A)}, \quad 1 \leq i, j \leq n.$$

To check on this formula, let us use it to calculate a few elements in $A^{-1}A = I$. Let us start from the upper-left element:

$$\left(A^{-1}A\right)_{1,1} = \sum_{j=1}^{n} \left(A^{-1}\right)_{1,j} a_{j,1} = \frac{1}{\det(A)} \sum_{j=1}^{n} (-1)^{1+j} \det\left(A^{(j,1)}\right) a_{j,1} = \frac{\det(A)}{\det(A)} = 1$$

(Sect. 2.1.3). This is indeed as required.

Next, let us check an off-diagonal element as well. For this purpose, let us design a new matrix B as follows. B is nearly the same as A. Only its first column is different: it is the same as the second one. This way, both the first and second columns in B are the same as the second column in A. Thus, B must have a zero determinant (Sect. 2.1.3). Moreover, in B, most minors are the same as in A. So, we could work with B rather than A:

$$\begin{aligned}\left(A^{-1}A\right)_{1,2} &= \sum_{j=1}^{n}\left(A^{-1}\right)_{1,j} a_{j,2} \\ &= \frac{1}{\det(A)}\sum_{j=1}^{n}(-1)^{1+j}\det\left(A^{(j,1)}\right)a_{j,2} \\ &= \frac{1}{\det(A)}\sum_{j=1}^{n}(-1)^{1+j}\det\left(B^{(j,1)}\right)b_{j,1} \\ &= \frac{\det(B)}{\det(A)} \\ &= 0,\end{aligned}$$

as required. The explicit formula for A^{-1} is called Cramer's rule.

For example, consider the case $n = 2$. In this case, we have a small 2×2 matrix. Now, if its determinant is nonzero:

$$ad - bc \neq 0,$$

then its inverse is

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

(Check!) Next, we use Cramer's rule yet more efficiently.

2.1.5 Cramer's Rule

It is too expensive to calculate all minors, and their determinant. Fortunately, we often do not need the entire inverse matrix in its explicit form. Usually, we are given a specific vector

$$v \equiv (v_1,\ v_2,\ v_3,\ \ldots,\ v_n)^t,$$

and we only need to apply A^{-1} to it.

For example, let us calculate the first component of $A^{-1}v$. For this purpose, let us redefine B as a new matrix that is nearly the same as A. Only its first column is different: it is the same as v. This way, we now have

$$\left(A^{-1}v\right)_1 = \sum_{j=1}^{n}\left(A^{-1}\right)_{1,j} v_j = \frac{1}{\det(A)}\sum_{j=1}^{n}(-1)^{1+j}\det\left(A^{(j,1)}\right)v_j = \frac{\det(B)}{\det(A)}.$$

Now, there is nothing special about the first component. Likewise, let $1 \leq i \leq n$ be any fixed index. To calculate the ith component of $A^{-1}v$, use a similar approach: redefine B as a new matrix that is nearly the same as A. Only its ith column is now different: it is the same as v. With this new B, the ith component in $A^{-1}v$ is

$$\left(A^{-1}v\right)_i = \frac{\det(B)}{\det(A)}.$$

This formula will be useful in applied geometry: barycentric coordinates in 3-D (Chap. 9). In this chapter, on the other hand, we use the determinant for yet another geometrical purpose: to define the vector product in 3-D, with its physical applications.

2.2 Vector (Cross) Product

2.2.1 Standard Unit Vectors in 3-D

So far, the determinant was defined algebraically. Still, does it have a geometrical meaning as well? To see this, let us look at two vectors of the same dimension n.

Can they multiply each other, to form a new vector? In general, they cannot: we could calculate their inner product, but this would be just a scalar, not a vector (Chap. 1, Sect. 1.7.1). Still, there is one exception: the three-dimensional Cartesian space, obtained by setting $n = 3$. In this space, a vector product could indeed be defined.

For this purpose, define three standard unit vectors:

$$\mathbf{i} = (1, 0, 0)^t$$
$$\mathbf{j} = (0, 1, 0)^t$$
$$\mathbf{k} = (0, 0, 1)^t.$$

This way, $\mathbf{i}$ points in the positive x-direction, $\mathbf{j}$ points in the positive y-direction, and $\mathbf{k}$ points in the positive z-direction (Fig. 2.1).

These are orthonormal vectors: orthogonal unit vectors. In what way are they standard? Well, they align with the x-y-z axes in the three-dimensional space. In this sense, they actually make the standard coordinate system that spans the entire Cartesian space. Indeed, let

$$u \equiv (u_1, u_2, u_3)^t \quad \text{and} \quad v \equiv (v_1, v_2, v_3)^t$$

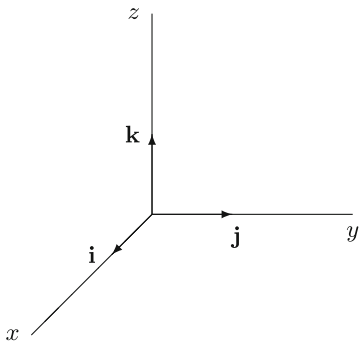


Fig. 2.1 The right-hand rule: the horizontal unit vectors, **i** and **j**, produce the vertical unit vector $\mathbf{i} \times \mathbf{j} = \mathbf{k}$

be some three-dimensional real vectors in $\mathbb{R}^3$. Thanks to the standard unit vectors, they could also be written as

$$u \equiv u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k} \quad \text{and} \quad v \equiv v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}.$$

This will be useful below.

2.2.2 *Inner Product—Orthogonal Projection*

In the above, we have considered two real three-dimensional vectors: u and v. Their inner product is the real scalar

$$(u, v) = u^h v = u^t v = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

This is actually a bilinear form: it takes two inputs (or arguments), u and v, to produce one new real number: their inner product.

What is the geometrical meaning of this? This is an orthogonal projection. Consider, for instance, the x-axis, spanned by the unit vector **i**. What is the inner product of v with **i**? This is

$$(v, \mathbf{i}) = v_1 \cdot 1 + v_2 \cdot 0 + v_3 \cdot 0 = v_1.$$

This is just the x-coordinate of v: the projection of v on the x-axis. In fact, if v makes angle η with the positive part of the x-axis, then

$$\cos(\eta) = \frac{v_1}{\|v\|} = \frac{(v, \mathbf{i})}{\|v\|}$$

(Fig. 2.2).

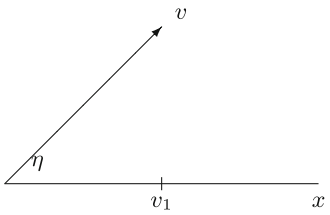


Fig. 2.2 The vector v makes angle η with the positive part of the x-axis: $\cos(\eta) = v_1/\|v\|$

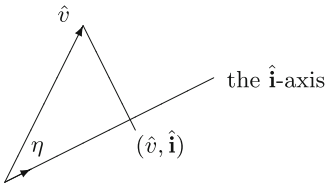


Fig. 2.3 The vector $\hat{v}$ makes angle η with the unit vector $\hat{\mathbf{i}}$. Once $\hat{v}$ projects onto the $\hat{\mathbf{i}}$-axis, we have $\cos(\eta) = (\hat{v}, \hat{\mathbf{i}})/\|\hat{v}\|$

Now, there is nothing special about the **i**-direction. Indeed, let us rotate both v and **i** by a fixed angle, to obtain the new vector $\hat{v}$, and the new unit vector $\hat{\mathbf{i}}$ (Fig. 2.3). This kind of rotation is an orthogonal transformation (see exercises below). As such, it preserves inner product (Chap. 1, Sect. 1.8.2). Furthermore, the angle η is still the same as before. Thus,

$$\cos(\eta) = \frac{(v, \mathbf{i})}{\|v\|} = \frac{\left(\hat{v}, \hat{\mathbf{i}}\right)}{\|\hat{v}\|}.$$

You could also look at things the other way around. After all, the axis system is picked arbitrarily. Why not pick the x-axis to align with the $\hat{\mathbf{i}}$-axis in Fig. 2.3? This way, we would have the same picture as in Fig. 2.2. Just rotate yourself, and look at it from a slightly different angle!

In either point of view, the inner product remains the same: the orthogonal projection of v (or $\hat{v}$) onto the unit vector **i** (or $\hat{\mathbf{i}}$). This is good: the inner product can now help calculate the cosine of the angle between the vectors. This angle will then give a geometrical meaning to the vector product defined below.

2.2.3 Vector (Cross) Product

What do we want from a vector product?

- It should take two inputs (or arguments): the original real three-dimensional vectors u and v.
- Likewise, its output should be a real three-dimensional vector, not just a scalar. In other words, once the symbol "$\times$" is placed in between u and v, $u \times v$ should be a new three-dimensional vector: their vector product.

- We already know that the inner product (u, v) is a bilinear form: it is linear in both u and v. Likewise, the vector product $u \times v$ should be a bilinear operator: linear in u, and linear in v as well.
- Interchanging the arguments should just change sign:

$$v \times u = -(u \times v).$$

- If both arguments are the same, then the vector product should vanish:

$$u \times u = \mathbf{0}$$

  (the origin).
- Take the triplet $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, and copy it periodically in a row:

$$\mathbf{i},\ \mathbf{j},\ \mathbf{k},\ \mathbf{i},\ \mathbf{j},\ \mathbf{k},\ \mathbf{i},\ \mathbf{j},\ \mathbf{k},\ \ldots.$$

  In this list, each pair should produce the next vector:

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}$$
$$\mathbf{j} \times \mathbf{k} = \mathbf{i}$$
$$\mathbf{k} \times \mathbf{i} = \mathbf{j}$$

  (Fig. 2.1). This is called the right-hand rule. Later on, we will motivate it geometrically as well.

How to define a good vector product, with all these properties? Fortunately, we can use the determinant. After all, thanks to its original definition, the determinant is just a linear combination of the items in the first row, which might be vectors in their own right:

$$\begin{aligned} u \times v &\equiv \det\left(\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}\right) \\ &= \mathbf{i}(u_2 v_3 - u_3 v_2) - \mathbf{j}(u_1 v_3 - u_3 v_1) + \mathbf{k}(u_1 v_2 - u_2 v_1) \\ &= (u_2 v_3 - u_3 v_2,\ u_3 v_1 - u_1 v_3,\ u_1 v_2 - u_2 v_1)^t. \end{aligned}$$

What is so good about this new definition? Well, let us see.

2.2.4 The Right-Hand Rule

In its new definition, does the vector product have the desirable properties listed above? Well, let us check:

- It is indeed bilinear: if w is yet another real three-dimensional vector, and α and β are some real numbers, then

$$(\alpha u + \beta w) \times v = \alpha(u \times v) + \beta(w \times v)$$
$$u \times (\alpha v + \beta w) = \alpha(u \times v) + \beta(u \times w).$$

  (Check!)
- Interchanging rows in a matrix changes the sign of its determinant. Therefore,

$$v \times u \equiv \det\left(\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ v_1 & v_2 & v_3 \\ u_1 & u_2 & u_3 \end{pmatrix}\right) = -\det\left(\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}\right) = -(u \times v).$$

  (Check!)
- As a result, a matrix with two identical rows must have a zero determinant. Therefore,

$$u \times u = \det\left(\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ u_1 & u_2 & u_3 \end{pmatrix}\right) = (0, 0, 0)^t = \mathbf{0}.$$

  (Check!)
- The right-hand rule indeed holds. Let us verify this for the standard unit vectors. (Later on, we will verify this for more general vectors as well.) For this purpose, take your right hand, with your thumb pointing in the positive x-direction, and your index finger pointing in the positive y-direction. Then, your middle finger will point in the positive z-direction:

$$\mathbf{i} \times \mathbf{j} \equiv \det\left(\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}\right) = \mathbf{k}.$$

  Now, let your thumb point in the positive y-direction, and your index finger in the positive z-direction. Then, your middle finger will point in the positive x-direction:

$$\mathbf{j} \times \mathbf{k} \equiv \det\left(\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right) = \mathbf{i}.$$

Finally, let your thumb point in the positive z-direction, and your index finger in the positive x-direction. Then, your middle finger will point in the positive y-direction:

$$\mathbf{k} \times \mathbf{i} \equiv \det\left(\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}\right) = \mathbf{j},$$

as required.

So far, we have considered standard unit vectors only. But what about more general vectors, like u and v above?

Well, let us focus on one component in $u \times v$. For instance, what is the z-coordinate? This is

$$(u \times v)_3 = \det\left(\begin{pmatrix} & & 1 \\ u_1 & u_2 & \\ v_1 & v_2 & \end{pmatrix}\right) = u_1 v_2 - u_2 v_1.$$

Is this positive? To check on this, let us look at the two-dimensional subvector $(u_1, u_2)^t \in \mathbb{R}^2$. Assume that it lies in the upper half of the Cartesian plane, where $u_2 > 0$. (Otherwise, just switch to $-u$) Likewise, look at the two-dimensional subvector $(v_1, v_2)^t \in \mathbb{R}^2$, and assume that it lies in the upper half of the Cartesian plane, where $v_2 > 0$ (Fig. 2.4).

This way, to check whether $u_1 v_2 - u_2 v_1$ is positive or not, one could divide it by $u_2 v_2 > 0$. When would this be positive? Only when

$$\operatorname{cotan}(\phi) = \frac{u_1}{u_2} > \frac{v_1}{v_2} = \operatorname{cotan}(\theta),$$

or

$$\phi < \theta,$$

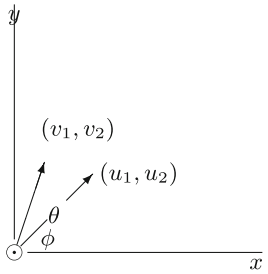


Fig. 2.4 The right-hand rule: take your right hand, and match your thumb to (u_1, u_2), and your index finger to (v_1, v_2). Then, your middle finger will point upwards, toward your own eyes, as indicated by the "$\odot$" at the origin

as in Fig. 2.4. (After all, the cotangent function is monotonically decreasing.) In this case, since its z-coordinate is positive, $u \times v$ points from the page outwards, toward your eyes. This is in agreement with the right-hand rule: take your right hand, and match your thumb to u, and your index finger to v. Then, your middle finger will point toward your own eyes, as required.

Finally, the vector product has yet another interesting property: $u \times v$ is orthogonal to both u and v. Indeed, the inner product of $u \times v$ with either u or v produces a matrix with two identical rows, with a zero determinant:

$$(u \times v)^t u = \det\left(\begin{pmatrix} u_1 & u_2 & u_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}\right) = 0$$

$$(u \times v)^t v = \det\left(\begin{pmatrix} v_1 & v_2 & v_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}\right) = 0.$$

This property will be useful in orthogonal transformations.

2.3 Orthogonalization

2.3.1 Invariance Under Orthogonal Transformation

The vector product is (nearly) invariant under orthogonal transformation: up to a sign, ordering does not matter: you could either apply the vector product and then transform or do things the other way around: first transform, and then apply the vector product (Fig. 2.5).

To see this, let Q be a 3×3 real orthogonal matrix. Let us show that, up to a sign, Q preserves vector product:

$$Q(u \times v) = \pm(Qu) \times (Qv).$$

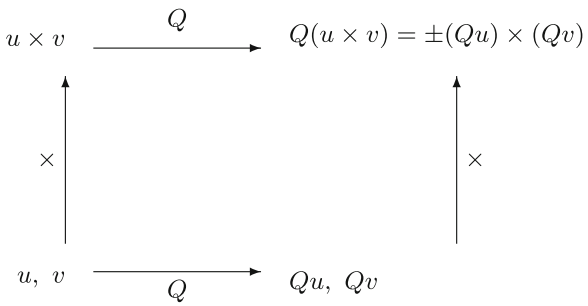


Fig. 2.5 Up to a sign, ordering does not matter. You could either apply the orthogonal transformation and then the vector product or work the other way around: apply the vector product first, and then the orthogonal transformation

(The proper sign will be specified later.)

Let us first show that $Q(u \times v)$ is proportional to $(Qu) \times (Qv)$, or orthogonal to both Qu and Qv. Fortunately, as discussed in Chap. 1, Sect. 1.8.2, Q preserves inner product, so

$$(Q(u \times v), Qu) = (u \times v, u) = 0$$
$$(Q(u \times v), Qv) = (u \times v, v) = 0.$$

This proves that $Q(u \times v)$ is indeed proportional to $(Qu) \times (Qv)$, as asserted. But what about their magnitude? Is it the same? To see this, just take their inner product:

$$\begin{aligned}
((Qu) \times (Qv), Q(u \times v)) &= \det \left(\begin{pmatrix} (Q(u \times v))^t \\ (Qu)^t \\ (Qv)^t \end{pmatrix} \right) \\
&= \det \left(\begin{pmatrix} (u \times v)^t Q^t \\ u^t Q^t \\ v^t Q^t \end{pmatrix} \right) \\
&= \det \left(\begin{pmatrix} (u \times v)^t \\ u^t \\ v^t \end{pmatrix} Q^t \right) \\
&= \det \left(\begin{pmatrix} (u \times v)^t \\ u^t \\ v^t \end{pmatrix} \right) \det \left(Q^t \right) \\
&= (u \times v, u \times v) \det (Q) \\
&= (Q(u \times v), Q(u \times v)) \det (Q) \\
&= \begin{cases} \|Q(u \times v)\|^2 & \text{if} \quad \det(Q) = 1 \\ -\|Q(u \times v)\|^2 & \text{if} \quad \det(Q) = -1. \end{cases}
\end{aligned}$$

So,

$$(Qu) \times (Qv) = \begin{cases} Q(u \times v) & \text{if} \quad \det(Q) = 1 \\ -Q(u \times v) & \text{if} \quad \det(Q) = -1. \end{cases}$$

In summary, ordering is (nearly) immaterial: making the vector product and then the orthogonal transformation is the same (up to a sign) as making the orthogonal transformation and then the vector product.

In particular, vector product is invariant under an orthogonal transformation with determinant 1 (rotation). This means that vector product is purely geometrical: it is independent of the coordinate system that happens to be used. For this reason, the vector product must have a pure geometrical interpretation, free of any tedious

algebraic detail. To see this, let us switch to a more convenient axis system, which is no longer absolute, but relative to the original vectors u and v.

2.3.2 Relative Axis System: Gram–Schmidt Process

Let us use the above properties to design a new coordinate system in $\mathbb{R}^3$. For this purpose, assume that u and v are linearly independent of each other: they are not a scalar multiple of each other. In this case, we can form a new 3×3 orthogonal matrix:

$$O \equiv \left(v^{(1)} \mid v^{(2)} \mid v^{(3)} \right).$$

What are these columns? Well, they should orthonormalize the original vectors u and v:

1. First, normalize u:

$$v^{(1)} \equiv \frac{u}{\|u\|}.$$

   This way, $v^{(1)}$ is the unit vector proportional to u.
2. Then, as in Sect. 2.2.2, project v on $v^{(1)}$, and subtract:

$$v^{(2)} \equiv v - \left(v^{(1)}, v \right) v^{(1)}.$$

   This is called a Gram–Schmidt process. Because u and v are linearly independent,

$$v^{(2)} \neq \mathbf{0}.$$

   By now, we have orthogonalized v with respect to u: $v^{(2)}$ is now orthogonal to $v^{(1)}$. Indeed, since $v^{(1)}$ is a unit vector,

$$\left(v^{(1)}, v^{(2)} \right) = \left(v^{(1)}, v - \left(v^{(1)}, v \right) v^{(1)} \right) = \left(v^{(1)}, v \right) - \left(v^{(1)}, v \right) = 0.$$

3. Next, normalize $v^{(2)}$ as well:

$$v^{(2)} \leftarrow \frac{v^{(2)}}{\|v^{(2)}\|}.$$

   This way, $v^{(1)}$ and $v^{(2)}$ are now orthonormal and span the same plane as the original vectors u and v.

4. Next, define a new vector, orthogonal to both u and v:

$$v^{(3)} \equiv v^{(1)} \times v^{(2)}.$$

5. Finally, normalize $v^{(3)}$ as well:

$$v^{(3)} \leftarrow \frac{v^{(3)}}{\|v^{(3)}\|}.$$

We will soon realize that this normalization is actually unnecessary: $v^{(3)}$ is already a unit vector.

In summary, O has real orthonormal columns. As such, O is an orthogonal matrix:

$$O^t O = O O^t = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus, the vector product is invariant under O. Indeed, in Sect. 2.3.1, the relevant sign is the plus sign, and O must have determinant 1:

$$v^{(1)} \times v^{(2)} = (O\mathbf{i}) \times (O\mathbf{j}) = O(\mathbf{i} \times \mathbf{j}) = O\mathbf{k} = v^{(3)}.$$

So, there was never any need to normalize $v^{(3)}$: it was a unit vector all along.

What are $v^{(1)}$, $v^{(2)}$, and $v^{(3)}$ geometrically? A new axis system! $v^{(1)}$ points in the positive direction of the new x-axis, $v^{(2)}$ points like the new y-axis, and $v^{(3)}$ points like the new z-axis.

Each three-dimensional vector can now be written in terms of this new coordinates. As an exercise, let us reconstruct u. Fortunately, u is orthogonal to both $v^{(2)}$ and $v^{(3)}$. Thus, the general expansion in Chap. 1, Sect. 1.11.5, reduces to

$$u = Iu = (OO^t)u = O(O^t u) = \sum_{j=1}^{3} (O^t u)_j v^{(j)} = (O^t u)_1 v^{(1)} = \left(v^{(1)}, u\right) v^{(1)} = \|u\| v^{(1)}.$$

What does this mean geometrically? It means that u is confined to the new x-axis, as required. In fact, in the new coordinates, u takes the simple form of

$$O^t u = \begin{pmatrix} \|u\| \\ 0 \\ 0 \end{pmatrix}.$$

v, on the other hand, is more complicated: it is expanded in two terms: its orthogonal projections on $v^{(1)}$, and on $v^{(2)}$. After all, v is orthogonal to $v^{(3)}$, but not necessarily to $v^{(1)}$, let alone to $v^{(2)}$:

$$\begin{aligned} v &= \sum_{j=1}^{3} (O^t v)_j v^{(j)} \\ &= (O^t v)_1 v^{(1)} + (O^t v)_2 v^{(2)} \\ &= \left(v^{(1)}, v\right) v^{(1)} + \left(v^{(2)}, v\right) v^{(2)} \\ &= \left(v^{(1)}, v\right) v^{(1)} + \left(v^{(2)}, v - \left(v^{(1)}, v\right) v^{(1)}\right) v^{(2)} \\ &= \left(v^{(1)}, v\right) v^{(1)} + \left\| v - \left(v^{(1)}, v\right) v^{(1)} \right\| v^{(2)}. \end{aligned}$$

What does this mean geometrically? It means that v is confined to the new x-y plane. In fact, in the new coordinates, v is actually two-dimensional: it has just two nonzero coordinates:

$$O^t v = \begin{pmatrix} \left(v^{(1)}, v\right) \\ \left\| v - \left(v^{(1)}, v\right) v^{(1)} \right\| \\ 0 \end{pmatrix}.$$

Thanks to this form, we can now redefine $u \times v$ geometrically rather than algebraically.

2.3.3 Angle Between Vectors

Like O, O^t is a rotation matrix as well: a real orthogonal matrix of determinant 1. Therefore, the vector product is invariant under O^t as well and can be calculated in the new coordinates, using the above form.

In the new coordinates, the original (algebraic) definition gets very simple. This is also apparent geometrically. After all, v contains only two nonzero coordinates: the $v^{(1)}$-coordinate (the new x-coordinate) and the $v^{(2)}$-coordinate (the new y-coordinate). In $u \times v$, only the latter coordinate is relevant. The former (which is proportional to u) drops, contributing nothing to $u \times v$:

$$\begin{aligned} u \times v &= \|u\| \cdot \left\| v - \left(v^{(1)}, v\right) v^{(1)} \right\| v^{(1)} \times v^{(2)} \\ &= \|u\| \cdot \|v\| \frac{\left\| v - \left(v^{(1)}, v\right) v^{(1)} \right\|}{\|v\|} v^{(3)} \\ &= \|u\| \cdot \|v\| \sin(\eta) v^{(3)}, \end{aligned}$$

where η is the angle between u and v in the u-v plane (the new x-y plane):

$$\cos(\eta) = \frac{(O^t v)_1}{\|v\|} = \frac{\left(v^{(1)}, v\right)}{\|v\|} = \frac{(u, v)}{\|u\| \cdot \|v\|}$$

(Fig. 2.3).

So, what is the norm $\|u \times v\|$? It is the area of the parallelogram that u and v make in their plane (the new x-y plane). This is a pure geometrical interpretation, independent of any coordinate system, and free of any algebraic detail. No wonder it is so useful in physics.

2.4 Linear and Angular Momentum

2.4.1 Linear Momentum

The vector product introduced above is particularly useful in geometrical physics. To see this, consider a particle of mass m, traveling in the three-dimensional Cartesian space. At time t, it is at position

$$r \equiv r(t) \equiv \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} \in \mathbb{R}^3.$$

Later on, in quantum mechanics, we will see that this is not so simple. Still, for the time being, let us accept this.

To obtain the velocity, differentiate with respect to time:

$$r' \equiv r'(t) \equiv \begin{pmatrix} x'(t) \\ y'(t) \\ z'(t) \end{pmatrix} \in \mathbb{R}^3.$$

To obtain the linear momentum, assume that the mass is constant, and multiply by it:

$$p \equiv p(t) \equiv mr'(t) \in \mathbb{R}^3.$$

Later on, in special relativity, we will redefine the linear momentum more carefully. Still, for the time being, let us accept this.

Finally, differentiate p, to obtain the force: the new vector $p'(t)$. In summary, at each particular time t, the linear momentum p describes the full motion, telling us how the particle travels. Still, we will soon see that only two directions are relevant: radial and nonradial.

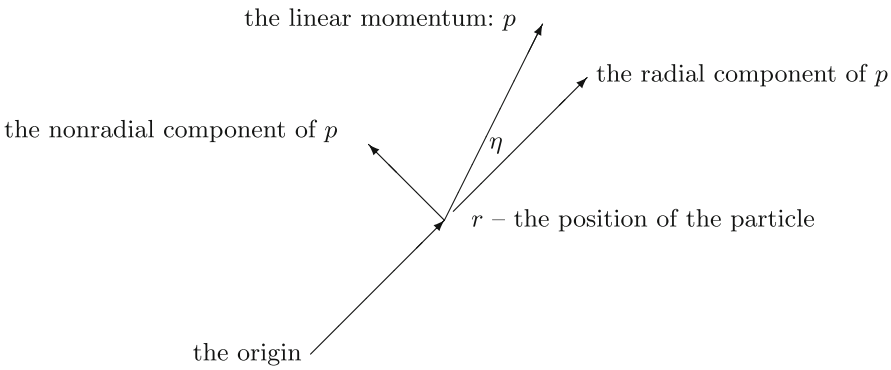


Fig. 2.6 The dynamic motion: at time t, the particle is at $r \equiv r(t) \in \mathbb{R}^3$, with linear momentum $p \equiv p(t) \in \mathbb{R}^3$, containing two parts: the radial part (proportional to r), and the nonradial part (perpendicular to r)

2.4.2 Radial Component: Orthogonal Projection

The radial component will tell us how fast $\|r\|$ grows. It will be the orthogonal projection of p onto the unit vector $r/\|r\|$:

$$\frac{(r,\, p)}{\|r\|^2} r = \left(\frac{r}{\|r\|},\, p\right) \frac{r}{\|r\|} = \cos(\eta)\|p\| \frac{r}{\|r\|},$$

where η is the angle between p and r (Figs. 2.2 and 2.6).

2.4.3 Angular Momentum

Still, the radial component does not tell us the whole story. After all, as time goes by, r may change not only magnitude but also direction. To model this motion, look at the nonradial component of p, telling us how fast the particle rotates around the origin.

Where does this rotation take place? Momentarily, it takes place in the r-p plane: the plane spanned by r and p. This is the plane perpendicular to the vector product $r \times p$.

This is why $r \times p$ will serve as the angular momentum, around which the particle rotates. How fast? To answer this, look at the nonradial component of p.

2.4.4 Angular Momentum and Its Norm

For this purpose, we need some preliminaries. First, what is the norm of $r \times p$? We already know what it is:

$$\|r \times p\| = \|r\| \cdot \|p\| \sin(\eta)$$

(Sect. 2.3.3).

2.4.5 Linear Momentum and Its Nonradial Component

To rotate around $r \times p$, the particle must make a little (infinitesimal) arc in the r-p plane. This is indeed the nonradial component of p: tangent to this arc, or perpendicular not only to $r \times p$ but also to r (Fig. 2.6). In summary, it must be proportional to

$$(r \times p) \times r.$$

What is the norm of this vector? Fortunately, we already know what it is:

$$\|(r \times p) \times r\| = \|r \times p\| \cdot \|r\| \sin\left(\frac{\pi}{2}\right) = \|r \times p\| \cdot \|r\| = \|p\| \sin(\eta) \|r\|^2.$$

Thus, to have the nonradial component, divide this by $\|r\|^2$:

$$\frac{r \times p}{\|r\|^2} \times r = \frac{r \times p}{\|r\|} \times \frac{r}{\|r\|}.$$

This way, the norm of this new vector is

$$\left\| \frac{r \times p}{\|r\|^2} \times r \right\| = \sin(\eta) \|p\|,$$

as required in Pythagoras' theorem.

2.4.6 Linear Momentum and Its Orthogonal Decomposition

In summary, we now have the complete orthogonal decomposition of the original linear momentum:

$$p = \frac{(r,\, p)}{\|r\|^2} r + \frac{r \times p}{\|r\|^2} \times r = \frac{(r,\, p)}{\|r\|} \cdot \frac{r}{\|r\|} + \frac{r \times p}{\|r\|} \times \frac{r}{\|r\|}.$$

What is so nice about this decomposition? Well, it is uniform: both terms are written in the same style. The only difference is that the former uses inner product, whereas the latter uses vector product.

Furthermore, both are orthogonal (perpendicular) to one another. After all, this is how they were designed in the first place: the former is proportional to r, whereas the latter is perpendicular to r. This is why they also satisfy Pythagoras' theorem: the sum of squares of their norms is

$$\begin{aligned}
\left\| \frac{(r,\, p)}{\|r\|} \cdot \frac{r}{\|r\|} \right\|^2 + \left\| \frac{r \times p}{\|r\|} \times \frac{r}{\|r\|} \right\|^2 &= \left(\frac{(r,\, p)}{\|r\|} \right)^2 + \left\| \frac{r \times p}{\|r\|} \right\|^2 \\
&= \cos^2(\eta) \|p\|^2 + \sin^2(\eta) \|p\|^2 \\
&= \left(\cos^2(\eta) + \sin^2(\eta) \right) \|p\|^2 \\
&= \|p\|^2.
\end{aligned}$$

2.5 Angular Velocity

2.5.1 Angular Velocity

What is the momentum? It is mass times velocity (assuming that mass is constant). This is true not only in one but also in three dimensions. So, to obtain the velocity vector, just divide by the mass of the particle:

$$v = \frac{p}{m} = \frac{r \times p}{m\|r\|^2} \times r + \frac{(r,\, p)}{m\|r\|^2} r.$$

This is just a special case of a more general (not necessarily orthogonal) decomposition:

$$v = u + w$$

(Fig. 2.7). Here,

$$u \equiv \boldsymbol{\omega} \times r,$$

and

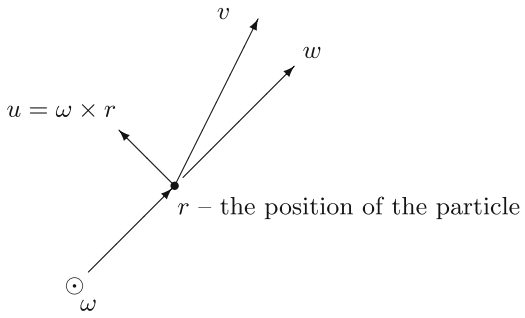


Fig. 2.7 Assume that the angular velocity $\boldsymbol{\omega}$ is perpendicular to r: it points from the page toward your eyes, as indicated by the '$\odot$' at the origin (Do not confuse $\boldsymbol{\omega}$ with the other vector w!)

$$\boldsymbol{\omega} \equiv \boldsymbol{\omega}(t) \equiv \begin{pmatrix} \boldsymbol{\omega}_1(t) \\ \boldsymbol{\omega}_2(t) \\ \boldsymbol{\omega}_3(t) \end{pmatrix} \in \mathbb{R}^3$$

is a new vector: the angular velocity. (Do not confuse $\boldsymbol{\omega}$ with the other vector w!) This way, the particle rotates around $\boldsymbol{\omega}$. The norm $\|\boldsymbol{\omega}\|$ tells us how fast: by what angle per second. By definition, u must be perpendicular to both $\boldsymbol{\omega}$ and r.

The angular velocity could be time-dependent: $\boldsymbol{\omega}$ could change in time, not only in magnitude but also in direction. For simplicity, however, we often look at some fixed time. This way, one could drop the argument "(t)" for short.

2.5.2 The Rotating Axis System

In general, $\boldsymbol{\omega}$ may point in any direction, not necessarily perpendicular to r or v. (See exercises below.) For simplicity, however, we often assume that $\boldsymbol{\omega}$ is perpendicular to r:

$$(\boldsymbol{\omega}, r) = 0.$$

Otherwise, just redefine the origin, and shift it along the $\boldsymbol{\omega}$-axis, until obtaining new orthogonal $\boldsymbol{\omega}$ and r. This way, $\boldsymbol{\omega}$-r-u make a new right-hand system, rotating around the $\boldsymbol{\omega}$-axis.

As a matter of fact, the rotating coordinates are more fundamental: they are closely related to the physical phenomenon, much more than the original Cartesian coordinates, which are more mathematical than physical. In terms of these new coordinates, the particle may "feel" a few new forces.

2.5.3 Velocity and Its Decomposition

As we have seen above, at time t, the particle rotates (momentarily) around the $\boldsymbol{\omega}$-axis (at angle $\|\boldsymbol{\omega}\|$ per second), making an infinitesimal arc. In our velocity

$$v = u + w,$$

the former term

$$u \equiv \boldsymbol{\omega} \times r$$

is tangent to this arc. The remainder

$$w \equiv v - u,$$

on the other hand, may be nontangential, and even perpendicular to the arc.

For the sake of better visualization, we often assume that $\boldsymbol{\omega}$ is perpendicular not only to r but also to w (Fig. 2.7). Still, this is not a must: in our final example, w will be nonradial and will contain a component parallel to $\boldsymbol{\omega}$. Still, in most of our discussion, $\boldsymbol{\omega}$ is perpendicular to both r and w (and v). To have this, it makes sense to define $\boldsymbol{\omega}$ to align with the angular momentum:

$$\boldsymbol{\omega} \equiv \frac{r \times p}{m\|r\|^2},$$

in agreement with the formula at the beginning of Sect. 2.5.1. This way, w is radial, so $\boldsymbol{\omega}$-w-u make the same right-hand system as $\boldsymbol{\omega}$-r-u: the rotating axis system (Sect. 2.5.2).

2.6 Real and Fictitious Forces

2.6.1 The Centrifugal Force

What is the centrifugal force? This is the force that the particle "feels" in its own ideal "world": the rotating coordinate system.

In general (even if $\boldsymbol{\omega}$ is not orthogonal to r), the centrifugal force is

$$-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r).$$

To illustrate, it is convenient to assume that $\boldsymbol{\omega}$ and r are orthogonal to each other:

$$(\boldsymbol{\omega}, r) = 0.$$

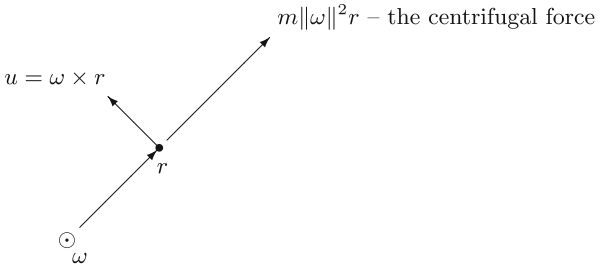


Fig. 2.8 The fictitious centrifugal force: $-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r)$. If $\boldsymbol{\omega}$ is perpendicular to r, then the centrifugal force is radial: $m\|\boldsymbol{\omega}\|^2 r$

This way,

$$\|\boldsymbol{\omega} \times r\| = \|\boldsymbol{\omega}\| \cdot \|r\|,$$

and the centrifugal force is radial:

$$-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r) = m\|\boldsymbol{\omega}\|^2 r$$

(Fig. 2.8). Still, this is "felt" in the rotating system only. In reality, on the other hand, there is no centrifugal force at all. Indeed, in the static system used in Figs. 2.6, 2.7, 2.8, the real force is just $p' = mv'$. So long as $v' = \mathbf{0}$, there is no force at all: Newton's first law holds, and the linear momentum is conserved. As a result, v can never change physically: what could change is just its writing style in rotating coordinates. This nonphysical "change" is due to the fictitious centrifugal "force."

Still, the rotating coordinates are quite legitimate, and we might want to work in them. In fact, if the particle stayed at the same rotating coordinates $(0, \|r\|, 0)$ all the time, then it would rotate physically round and round forever. For this purpose, however, a new counter-force must be introduced, to cancel the centrifugal force out.

2.6.2 The Centripetal Force

How to balance (or cancel) the centrifugal force? For this purpose, connect the particle to the origin by a wire. In rotating coordinates, this reacts to the centrifugal force. Indeed, thanks to Newton's third law, this supplies the required counter-force—the centripetal force:

$$m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r).$$

If $\boldsymbol{\omega}$ and r are perpendicular to each other, then this force is radial as well:

$$m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r) = -m\|\boldsymbol{\omega}\|^2 r$$

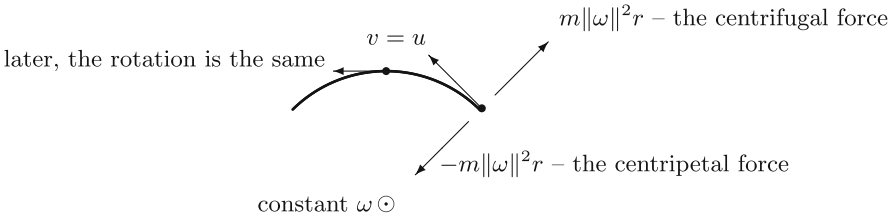


Fig. 2.9 Particle connected to the origin by a wire. This supplies the centripetal force, to balance the centrifugal force, and keep the particle at the same distance $\|r\|$ from the origin, rotating at the same angular velocity $\boldsymbol{\omega}$ all the time

(Fig. 2.9). This is indeed how the wire must pull the particle back toward the origin. After all, Newton's third law must work in just any coordinate system, rotating or not.

In the rotating system, the centripetal force balances the centrifugal force, leaving the particle at the same (rotating) coordinates $(0, \|r\|, 0)$ all the time. In the static coordinates, on the other hand, the centripetal force has a more physical job: to make u turn. In fact, the centripetal force pulls the particle just enough to keep it at r all the time, rotating at the constant angular velocity $\boldsymbol{\omega}$ forever. This way, the particle makes not only infinitesimal but also global arc around $\boldsymbol{\omega}$ (Fig. 2.9). In fact, this arc is as big as a complete circle of radius $\|r\|$. As a result, the velocity is tangent to the circle and has no nontangential part at all:

$$v = u \quad \text{and} \quad w = \mathbf{0}.$$

2.6.3 The Euler Force

In the above example, the particle rotates at the constant angular velocity $\boldsymbol{\omega}$. But what if $\boldsymbol{\omega}$ changed in time? In this case, $\boldsymbol{\omega}$ might have a nonzero time derivative:

$$\boldsymbol{\omega}' \equiv \boldsymbol{\omega}'(t) \equiv \begin{pmatrix} \boldsymbol{\omega}_1'(t) \\ \boldsymbol{\omega}_2'(t) \\ \boldsymbol{\omega}_3'(t) \end{pmatrix} \neq \mathbf{0}.$$

In the rotating coordinate system, this introduces the new Euler force:

$$-m\boldsymbol{\omega}' \times r.$$

What is the direction of this new force? Well, in general, $\boldsymbol{\omega}'$ could point in any direction. Indeed, as time goes by, the angular velocity may change not only magnitude but also direction, making the particle rotate in all sorts of new r-u planes.

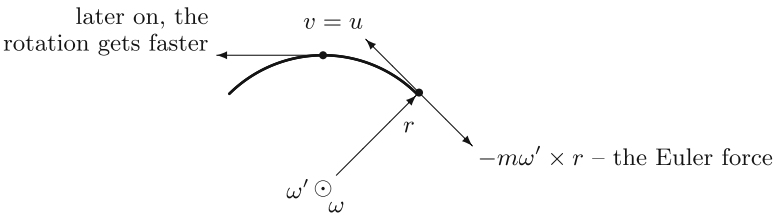


Fig. 2.10 Particle rotating counterclockwise, faster and faster. This way, $\boldsymbol{\omega}'$ points in the same direction as $\boldsymbol{\omega}$. This way, the Euler force pulls back the particle clockwise, opposing the original rotation, and slowing it down

Still, for simplicity, assume that $\boldsymbol{\omega}'$ keeps pointing in the same direction as the original $\boldsymbol{\omega}$ (Fig. 2.10). This way, $\boldsymbol{\omega}$ too keeps pointing in the same direction all the time: it only gets bigger and bigger in magnitude:

$$\|\boldsymbol{\omega}\|' \equiv \frac{d\|\boldsymbol{\omega}\|}{dt} > 0.$$

What happens physically? The particle rotates counterclockwise faster and faster. What could possibly supply the energy required for this? Well, assume that there is some angular accelerator that keeps increasing the frequency (number of cycles per second). This way, the particle keeps rotating counterclockwise in the same r-u plane, at a bigger and bigger angle $\|\boldsymbol{\omega}(t)\|$ per second.

In the rotating coordinates, the Euler force pulls the particle back clockwise, in an attempt to oppose this motion and slow it down. So, the accelerator must balance Euler force and supply a force in the amount of

$$m\|\boldsymbol{\omega}' \times r\| = m\|\boldsymbol{\omega}'\| \cdot \|r\|$$

counterclockwise. In the original static coordinate system, this accelerates the particle angularly, as required. In the rotating axis system, on the other hand, the particle remains at rest. It always keeps the same (rotating) coordinates: $(0, \|r\|, 0)$. This way, in its own subjective (rotating) "world," it keeps floating effortless, allowing the rotating axis system to carry it round and round, faster and faster.

2.6.4 The Earth and Its Rotation

The rotating coordinates are not only theoretical but also real and practical. It is time to use them.

In Figs. 2.9, 2.10, the particle makes a closed circle around the $\boldsymbol{\omega}$-axis. In fact, the velocity is tangent to this circle, with no nontangential part at all:

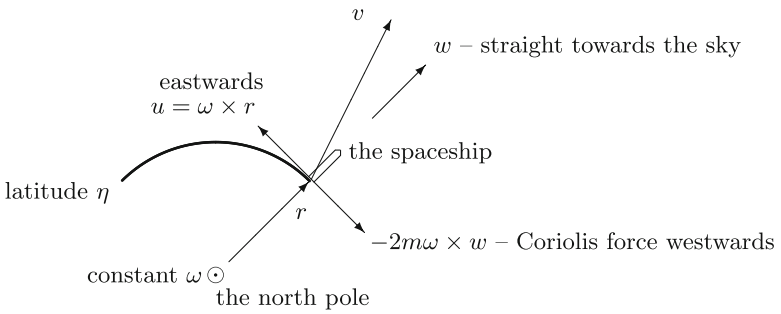


Fig. 2.11 A view on the Earth from above: a horizontal cross section of the northern hemisphere, at latitude $0 \le \eta < \pi/2$. The origin is not at the center of the Earth, but above it: at the center of the cross section. Thus, r is horizontal: it lies in the cross section. But w (the direction of the spaceship) is not: it makes angle η with the horizontal r-axis. The Coriolis force pulls the spaceship westwards

$$v = u \quad \text{and} \quad w = \mathbf{0}.$$

Consider now a more complicated case, in which v does have a nontangential part:

$$v = u + w, \quad \text{where} \quad w \neq \mathbf{0}.$$

As a matter of fact, we have already seen such a case (Fig. 2.7). Here, however, things are even more complicated: w is no longer radial but makes angle η with the horizontal r-axis, and angle $\pi/2 - \eta$ with the vertical $\boldsymbol{\omega}$-axis:

$$(\boldsymbol{\omega}, w) \ge 0$$

(Fig. 2.11). Consider, for example, a spaceship standing somewhere in the northern hemisphere of the Earth, at latitude $0 \le \eta < \pi/2$. Its nose points upwards: straight toward the sky.

The entire Earth rotates eastwards: this is why the Sun rises from the east. For this reason, in Fig. 2.11, the angular velocity $\boldsymbol{\omega}$ points northwards: from the page upwards, straight toward your eyes.

In reality, $\boldsymbol{\omega}$ is not quite constant: it changes direction, although very slowly. In fact, as the Earth rotates, $\boldsymbol{\omega}$ rotates too. Why? Because the Earth is not a perfect ball. For this reason, the north pole is not fixed: it loops clockwise. This is very slow: the loop takes 27,000 year to close. This is called precession.

Besides, the north pole makes yet another (small) loop clockwise. This loop is quicker: it takes 1.3 year to close. Still, it is very small: just ten meters in radius. Thus, both loops could be ignored.

In Fig. 2.11, we look at the Earth from above: say, from the North star. This way, we see a horizontal cross section of the Earth at latitude $0 \le \eta < \pi/2$. Recall that the origin can be picked arbitrarily. Here, we place it not at the center of the Earth

but above it: at the center of the cross section. This way, $\|r\|$ is not the radius of the Earth but the radius of the cross section. Indeed, r is horizontal: it lies in the cross section in its entirety.

Initially, at $t = 0$, the spaceship stands still on the boundary of the cross section. This way, r marks its location in the cross section. Furthermore, the body of the spaceship makes angle η with the horizontal r-axis.

At $t > 0$, on the other hand, the spaceship will fly away from the Earth. Fortunately, the cross section is part of an infinite horizontal plane. In this plane, r will mark the orthogonal projection (or "shadow") that the spaceship will make on the plane. This way, r will always be horizontal: it will remain in the same plane all the time.

Let us start with a simple case: $\eta = 0$. In this case, the cross section is surrounded by the equator, and its center coincides with the center of the Earth. Since r lies in the cross section, it coincides with the radius of the Earth. This way, r is perpendicular to the face of the Earth: it points straight into the sky. Fortunately, the centrifugal force in this direction is well-balanced by gravity, which supplies the centripetal force, as required.

This could answer an interesting question: why is not the Earth a perfect ball? Well, at the equator, some gravity is wasted to balance the centrifugal force. Thus, at the equator, gravity is a little weaker. This is probably how the Earth evolved in the first place: at the equator, it got a bit "fat." This is not quite a ball, but an oblate spheroid.

Consider now the more interesting case of $\eta > 0$. In this case, the cross section passes not through the center of the Earth but a little above it. But r is still horizontal, and shorter than before:

$$\|r\| = \cos(\eta) \text{ times the radius of the Earth.}$$

This way, r is no longer perpendicular to the face of the Earth but has a new component southwards.

What is the norm of this new component? Clearly, it is $\sin(\eta)\|r\|$. This produces a new centrifugal force in the amount of $m\|\boldsymbol{\omega}\|^2 \sin(\eta)\|r\|$ southwards, which is not balanced by gravity any more. After all, gravity pulls downwards, toward the ground, not northwards.

Recall that here we work in the rotating axis system: we stand on the face of the Earth, unaware of any rotation. Therefore, to us, the above force is real, and is truly felt. Likewise, the spaceship feels it as well, and its route could be affected, including the shadow r it makes on the (extended) cross section.

Fortunately, the above force can never affect $\boldsymbol{\omega}$, which produced it in the first place. Can it affect the "shadow" r? Not much: it has no time to act. In fact, for a small t, its effect on r is as small as t^2. (See exercises at the end of Chapter 1 in [76].) For this reason, it hardly affects the original motion, illustrated in Fig. 2.11 at the initial time of $t = 0$. Still, after a while, the effect may accumulate and grow and should be taken into account. To balance this, the nose of the spaceship should better point a little obliquely northwards, from the start.

2.6.5 Coriolis Force

Because the spaceship stands on the Earth, it also rotates eastwards at angle $\|\boldsymbol{\omega}\|$ per second. This produces one component of its velocity—the tangential part:

$$u \equiv \boldsymbol{\omega} \times r.$$

Here, however, we work in rotating axes: $\boldsymbol{\omega}$, r, and u. In rotating coordinates, there is no tangential motion at all. This is why we on the Earth are never aware of the Earth's rotation. For the spaceship too, the only relevant velocity is w: it remains valid in the rotating coordinates as well.

In fact, at time $t = 0$, the spaceship suddenly gets an initial velocity w: upwards, straight toward the sky. This is liftoff: the spaceship really feels it. Still, at the same time, it also feels a new force, which must not be ignored:

$$-2m\boldsymbol{\omega} \times w.$$

This is the Coriolis force (Fig. 2.11). What is its direction? Well, it must be perpendicular to the entire $\boldsymbol{\omega}$-w plane. Thanks to the right-hand rule, this must be westwards.

What is the norm of the Coriolis force? Well, this depends on the angle $\pi/2 - \eta$ between $\boldsymbol{\omega}$ and w:

$$2m \|\boldsymbol{\omega} \times w\| = 2m \|\boldsymbol{\omega}\| \cdot \|w\| \sin\left(\frac{\pi}{2} - \eta\right) = 2m \|\boldsymbol{\omega}\| \cdot \|w\| \cos(\eta).$$

Fortunately, this force does not have much time to act: for a small t, its effect on w is as small as t. (See exercises at the end of Chapter 1 in [76].) Thus, it hardly affects the motion, illustrated in Fig. 2.11 at the initial time of $t = 0$. Still, as time goes by, it may accumulate and must not be ignored. To balance it, the spaceship should better point a little obliquely eastwards, from the start.

2.7 Exercises: Inertia and Principal Axes

2.7.1 Rotation and Euler Angles

1. Let Q be an $n \times n$ orthogonal matrix. Does Q preserve inner product? Hint: for every two n-dimensional vectors u and v,

$$(Qu, Qv) = (u, Q^t Qv) = (u, Iv) = (u, v)$$

(where I is the $n \times n$ identity matrix).

2. Does Q preserve norm? Hint:

$$\|Qv\|^2 = (Qv, Qv) = (v, v) = \|v\|^2.$$

3. Let O and Q be two orthogonal matrices of the same order. What about their product? Is it orthogonal as well? Hint: thanks to associativity,

$$(OQ)^t(OQ) = \left(Q^t O^t\right)(OQ) = Q^t \left(O^t O\right) Q = Q^t I Q = Q^t Q = I.$$

4. Let $0 \leq \theta < 2\pi$ be some (fixed) angle. Let U be the following 2×2 matrix:

$$U \equiv U(\theta) \equiv \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

5. Show that U rotates the x-axis by angle θ counterclockwise. Hint: apply U to the standard unit vector that points rightwards:

$$U \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix}.$$

   The result is the $\tilde{x}$-unit vector in Fig. 2.12.
6. Show that U rotates the y-axis by angle θ counterclockwise as well. Hint: apply U to the standard unit vector $(0, 1)^t$.
7. Conclude that U rotates the entire x-y plane by angle θ counterclockwise. Hint: extend the above linearly. For this purpose, write a general two-dimensional vector as a linear combination of the standard unit vectors $(1, 0)^t$ and $(0, 1)^t$.
8. Show that the columns of U are orthogonal to each other.
9. Show that the columns of U are unit vectors of norm 1.
10. Conclude that the columns of U are orthonormal.
11. Conclude that the columns of U span new axes: the $\tilde{x}$- and $\tilde{y}$-axes in Fig. 2.12.

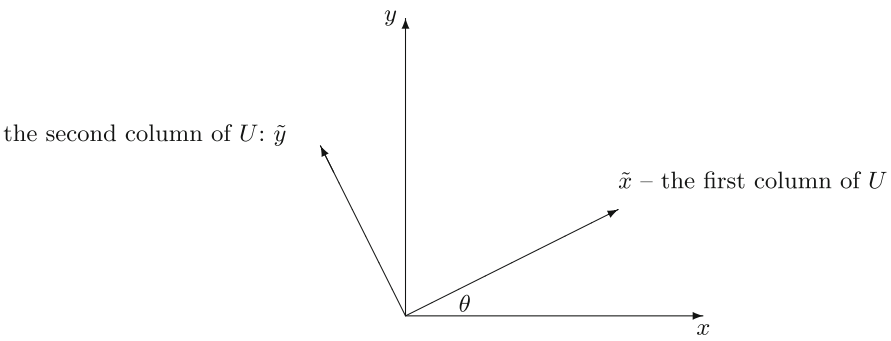


Fig. 2.12 The orthogonal matrix U rotates the entire x-y plane by angle θ counterclockwise, and maps it onto the new $\tilde{x}$-$\tilde{y}$ plane

12. Conclude also that U is orthogonal.
13. Verify that U indeed satisfies

$$U^t U = U U^t = I,$$

where I is the 2×2 identity matrix.
14. Conclude that U^t is orthogonal as well.
15. Verify that the columns of U^t (the rows of U) are indeed orthogonal to each other as well.
16. Verify that the columns of U^t (the rows of U) are indeed unit vectors of norm 1.
17. Recall that U has a geometrical meaning: once applied to a two-dimensional vector, it rotates it by angle θ counterclockwise. Hint: check this for the standard unit vectors $(1, 0)^t$ and $(0, 1)^t$. Then, extend this linearly.
18. Interpret U^t geometrically as the inverse rotation: once applied to a vector, it rotates it by angle θ clockwise.
19. Interpret the equation $U^t U = I$ geometrically. Hint: the composition of U^t on top of U is the identity mapping that changes nothing. Indeed, rotating clockwise cancels rotating counterclockwise.
20. Work the other way around: interpret the equation $U U^t = I$ geometrically as well.
21. Show that U has determinant 1:

$$\det(U) = \det\left(U^t\right) = 1.$$

22. Introduce a third spatial dimension: the z-axis. This makes the new x-y-z axis system in $\mathbb{R}^3$.
23. Consider yet another (right-hand) axis system: the $\tilde{x}$-$\tilde{y}$-$\tilde{z}$ axis system. (Assume that it shares the same origin.) How to map the original x-y-z system to the new $\tilde{x}$-$\tilde{y}$-$\tilde{z}$ system?
24. To do this, use three stages:
    - Rotate the entire $\tilde{x}$-$\tilde{y}$ plane by a suitable angle ψ clockwise, until the $\tilde{x}$-axis hits the x-y plane.
    - Then, rotate the entire x-y plane by a suitable angle ϕ counterclockwise, until the x-axis matches the up-to-date $\tilde{x}$-axis.
    - By now, the up-to-date x- and $\tilde{x}$-axes align with each other. So, all that is left to do is to rotate the up-to-date y-z plane by a suitable angle θ counterclockwise, until the y- and z-axes match the up-to-date $\tilde{y}$- and $\tilde{z}$-axes, respectively.
25. Conclude that, to map the original x-y-z axis system to the original $\tilde{x}$-$\tilde{y}$-$\tilde{z}$ axis system, one could use three stages:
    - Rotate the entire x-y plane by angle ϕ counterclockwise.
    - Then, rotate the up-to-date y-z plane by angle θ counterclockwise.
    - Finally, rotate the up-to-date x-y plane by angle ψ counterclockwise.

26. The angles ϕ, θ, and ψ are called Euler angles.
27. Show that this is a triple product of three orthogonal matrices.
28. Conclude that this triple product is orthogonal too.
29. What is its determinant? Hint: 1.
30. What matrix represents the first stage? Hint:

$$\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix}.$$

31. What matrix represents the second stage? Hint: note that this stage rotates the *up-to-date* y-z plane. Therefore, its matrix is

$$\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix}^{-1}.$$

32. What is the product of these two matrices? Hint:

$$\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix}.$$

33. What matrix represents the third stage? Hint: note that this stage rotates the *up-to-date* x-y plane. Therefore, its matrix is

$$\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \begin{pmatrix} U(\psi) & \\ & 1 \end{pmatrix} \left(\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \right)^{-1}.$$

34. What matrix represents the entire mapping? Hint: multiply the matrices of the three stages:

$$\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \begin{pmatrix} U(\psi) & \\ & 1 \end{pmatrix}.$$

35. What is its transpose? Hint:

$$\begin{pmatrix} U^t(\psi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U^t(\theta) \end{pmatrix} \begin{pmatrix} U^t(\phi) & \\ & 1 \end{pmatrix}.$$

36. How does this matrix map the original x-y-z system onto the new $\tilde{x}$-$\tilde{y}$-$\tilde{z}$ system? Hint: symbolically:

$$\begin{pmatrix} \tilde{x}\text{-axis} \\ \tilde{y}\text{-axis} \\ \tilde{z}\text{-axis} \end{pmatrix} = \begin{pmatrix} U^t(\psi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U^t(\theta) \end{pmatrix} \begin{pmatrix} U^t(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} x\text{-axis} \\ y\text{-axis} \\ z\text{-axis} \end{pmatrix}.$$

37. Indeed, take the transpose:

$$(x\text{-axis}, y\text{-axis}, z\text{-axis}) \begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \begin{pmatrix} U(\psi) & \\ & 1 \end{pmatrix} = (\tilde{x}\text{-axis}, \tilde{y}\text{-axis}, \tilde{z}\text{-axis}) .$$

38. Multiply this formula by $(1, 0, 0)^t$ from the right:

$$(x\text{-axis}, y\text{-axis}, z\text{-axis}) \begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \begin{pmatrix} U(\psi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = (\tilde{x}\text{-axis}, \tilde{y}\text{-axis}, \tilde{z}\text{-axis}) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \tilde{x}\text{-axis}.$$

39. How does the new $\tilde{x}$-axis look like? Hint: thanks to the above formula, it is spanned by the first column of the above matrix:

$$\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \begin{pmatrix} U(\psi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} .$$

40. How does the new $\tilde{y}$-axis look like? Hint: likewise, it is spanned by the second column:

$$\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \begin{pmatrix} U(\psi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} .$$

41. How does the new $\tilde{z}$-axis look like? Hint: likewise, it is spanned by the third column:

$$\begin{pmatrix} U(\phi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & U(\theta) \end{pmatrix} \begin{pmatrix} U(\psi) & \\ & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} .$$

42. Does the above matrix have determinant 1? Hint: this matrix is the product of three rotation matrices of determinant 1. Now, the determinant of a product is the product of determinants (Chap. 14, Sect. 14.4.3).
43. In terms of group theory, what does the above discussion tell us? Hint: the group of three-dimensional rotations is *generated* by two-dimensional rotations of individual planes.

2.7.2 Algebraic Right-Hand Rule

1. Consider a 3×3 matrix. Show that interchanging two rows in it changes the sign of the determinant.
2. Conclude that, if the matrix has two identical rows, then its determinant must vanish.
3. Let u, v, and w be real three-dimensional vectors in $\mathbb{R}^3$. Show that

$$u \times u = \mathbf{0}.$$

4. Show that

$$u \times v = -(v \times u).$$

5. Show that

$$(u, v \times w) = \det \left(\begin{pmatrix} u^t \\ v^t \\ w^t \end{pmatrix} \right).$$

6. Assume also that u, v, and w are linearly independent of each other: they do not belong to a plane that passes through the origin. Take the original triplet u, v, w, and copy it time and again in a row:

$$u,\ v,\ w,\ u,\ v,\ w,\ u,\ v,\ w,\ \ldots.$$

   In this list, start from some vector, and look ahead to the next two. Show that this produces the same determinant, no matter whether you started from u or v or w:

$$\det \left(\begin{pmatrix} u^t \\ v^t \\ w^t \end{pmatrix} \right) = \det \left(\begin{pmatrix} v^t \\ w^t \\ u^t \end{pmatrix} \right) = \det \left(\begin{pmatrix} w^t \\ u^t \\ v^t \end{pmatrix} \right).$$

   Hint: interchange the first and second rows, and then the second and third rows:

$$\det \left(\begin{pmatrix} u^t \\ v^t \\ w^t \end{pmatrix} \right) = -\det \left(\begin{pmatrix} v^t \\ u^t \\ w^t \end{pmatrix} \right) = \det \left(\begin{pmatrix} v^t \\ w^t \\ u^t \end{pmatrix} \right).$$

7. Conclude that

$$(u, v \times w) = (v, w \times u) = (w, u \times v).$$

8. Assume that u, v, and w satisfy the right-hand rule. Show that, in this case,

$$(u, v \times w) > 0.$$

Hint: recall that u, v, and $u \times v$ satisfy the right-hand rule (Fig. 2.4). So, to complete u and v into a right-hand system, one could add either w or $u \times v$. Thus, both w and $u \times v$ must lie in the same side of the u-v plane. As in Fig. 2.3, they must therefore have a positive inner product:

$$(w, u \times v) > 0.$$

9. Could this serve as a new (algebraic) version of the right-hand rule?

2.7.3 Linear Momentum and Its Conservation

1. Consider an aircraft, dropping a bomb. Look at the bomb on its own. Does it make a closed system? Hint: no—it is not isolated but attracted to the Earth.
2. Why? Hint: there is an outer force: gravity.
3. So, the bomb makes an open system. Why then is its total energy conserved? Hint: although gravity makes an outer force, its effect is still included in the system: the potential.
4. Now, look at the bomb and the Earth together. Do they make a closed system? Hint: yes, they attract each other.
5. In this system, what is the total momentum? Hint: zero. Indeed, while the bomb accelerates downwards, the entire Earth also accelerates upwards. Although this is tiny, the Earth has a very big mass. Thus, its momentum cancels the momentum of the bomb, leaving the total momentum at zero all the time.
6. Why? Hint: at each given time, the bomb is attracted down, and its momentum increases downwards. Thanks to Newton's third law, there is a reaction: the entire Earth is attracted up, and its momentum increases upwards.
7. Write this in a more general (differential) form. Hint: let the momentum p and the force F be vector functions, defined in a closed system $V \subset \mathbb{R}^3$. Since the total force vanishes, the total momentum is conserved:

$$\frac{d}{dt}\int\int\int_V p dx dy dz = \int\int\int_V \frac{dp}{dt} dx dy dz = \int\int\int_V F dx dy dz = \mathbf{0} \in \mathbb{R}^3.$$

2.7.4 Principal Axes

1. Recall that

$$r \equiv \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$$

is the position of the particle in 3-D.

2. Assume that r is fixed.
3. Show that rr^t is a 3×3 matrix.
4. Write it explicitly. Hint:

$$rr^t \equiv \begin{pmatrix} x \\ y \\ z \end{pmatrix} (x, y, z) = \begin{pmatrix} x^2 & xy & xz \\ yx & y^2 & yz \\ zx & zy & z^2 \end{pmatrix}.$$

5. Use this to show that rr^t is symmetric.
6. Show this in yet another way. Hint:

$$\left(rr^t\right)^t = \left(r^t\right)^t r^t = rr^t.$$

7. What are the eigenvalues and eigenvectors of rr^t? Hint: see below.
8. Show that r is an eigenvector of rr^t, with the eigenvalue $\|r\|^2$. Hint: thanks to associativity,

$$\left(rr^t\right) r = r \left(r^t r\right) = (r, r) r = \|r\|^2 r.$$

9. Let q be a vector perpendicular to r. Show that q is an eigenvector of rr^t, with the zero eigenvalue. Hint: thanks to associativity,

$$\left(rr^t\right) q = r \left(r^t q\right) = (r, q) r = 0r = \mathbf{0}.$$

10. Design two such q's: both perpendicular to r, and also perpendicular to each other. Hint: pick two linearly independent q's, both perpendicular to r. Apply to them the Gram–Schmidt process, to make them orthonormal.
11. Show that this could be done in many different ways.
12. Pick one particular pair of orthonormal q's.
13. Conclude that rr^t is positive semidefinite: its eigenvalues are greater than or equal to zero.
14. Conclude that rr^t has the following diagonal form:

$$rr^t = O \begin{pmatrix} \|r\|^2 & & \\ & 0 & \\ & & 0 \end{pmatrix} O^t,$$

where O is an orthogonal matrix.

15. What are the columns of O? Hint: $r/\|r\|$, and the above q's.

16. These columns are called *principal* axes. They span the entire Cartesian space, using new *principal* coordinates.
17. In terms of principal coordinates, where is the particle? Hint: its principal coordinates are: $(\|r\|, 0, 0)$.
18. Show that there are two principal axes that could be defined in many different ways. Still, pick arbitrarily one particular choice.

2.7.5 The Inertia Matrix

1. Let

$$I \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

be the 3×3 identity matrix.
2. Define a new 3×3 matrix:

$$A \equiv A(r) \equiv \|r\|^2 I - rr^t.$$

3. Later, A will be multiplied by the mass. This will produce mA: the inertia matrix of the particle. This will be useful later.
4. Show that A is symmetric.
5. Show that r is an eigenvector of A, with the zero eigenvalue.
6. Show that the above q's are eigenvectors of A, with the eigenvalue $\|r\|^2$.
7. Conclude that A is positive semidefinite: its eigenvalues are greater than or equal to zero.
8. These eigenvalues (times m) are called moments of inertia. They will be useful later.
9. Show that A has the following diagonal form:

$$A = O \begin{pmatrix} 0 & & \\ & \|r\|^2 & \\ & & \|r\|^2 \end{pmatrix} O^t,$$

where O is as above.
10. Recall that

$$\boldsymbol{\omega} \equiv \begin{pmatrix} \boldsymbol{\omega}_1 \\ \boldsymbol{\omega}_2 \\ \boldsymbol{\omega}_3 \end{pmatrix} \in \mathbb{R}^3$$

is the angular velocity.

11. To help visualize things better, we assumed so far that $\boldsymbol{\omega}$ was perpendicular to r.
12. This, however, is not a must: from now on, let us drop this assumption.
13. Show that, even if $\boldsymbol{\omega}$ is no longer perpendicular to r, $A\boldsymbol{\omega}$ still is:

$$(A\boldsymbol{\omega}, r) = 0.$$

Hint: in $A\boldsymbol{\omega}$, there is no radial component any more.

14. Show that

$$\|\boldsymbol{\omega} \times r\|^2 = \|\boldsymbol{\omega}\|^2 \|r\|^2 - (\boldsymbol{\omega}, r)^2.$$

Hint: see Sects. 2.2.2 and 2.3.3.

15. Conclude that

$$\|\boldsymbol{\omega} \times r\|^2 = \boldsymbol{\omega}^t A \boldsymbol{\omega}.$$

Hint: thanks to associativity,

$$\begin{aligned} \|\boldsymbol{\omega} \times r\|^2 &= \|\boldsymbol{\omega}\|^2 \|r\|^2 - (\boldsymbol{\omega}, r)^2 \\ &= \|r\|^2 \boldsymbol{\omega}^t \boldsymbol{\omega} - \left(\boldsymbol{\omega}^t r\right) \left(r^t \boldsymbol{\omega}\right) \\ &= \|r\|^2 \boldsymbol{\omega}^t I \boldsymbol{\omega} - \boldsymbol{\omega}^t \left(r r^t\right) \boldsymbol{\omega} \\ &= \boldsymbol{\omega}^t A \boldsymbol{\omega}. \end{aligned}$$

16. Show that the entire principal axis system could rotate around the $\boldsymbol{\omega}$-axis, carrying the "passive" particle at angle $\|\boldsymbol{\omega}\|$ per second.
17. What is the necessary condition for this? Hint: to have such a rotation, we must have no nontangential velocity: $w = \mathbf{0}$.
18. Give an example where this condition holds, and another example where it does not. Hint: in Fig. 2.11, for example, the entire Earth rotates around $\boldsymbol{\omega}$. The spaceship, on the other hand, spirals away from the Earth, toward the outer space. Indeed, it has two kinds of motion. On one hand, it rotates around $\boldsymbol{\omega}$, with the entire atmosphere. On the other hand, it also has a normal velocity w: toward the sky.

2.7.6 The Triple Vector Product

1. Let $\mathbf{v}$ and $\mathbf{w}$ be linearly independent vectors in $\mathbb{R}^3$. This means that $\mathbf{v}$ is not a scalar multiple of $\mathbf{w}$ but points in a different direction (Fig. 2.13). Let η be the angle between $\mathbf{v}$ and $\mathbf{w}$ ($0 < \eta < \pi$). Let $\mathbf{p}$ be yet another vector, perpendicular to $\mathbf{v}$ in the $\mathbf{v}$-$\mathbf{w}$ plane. Show that

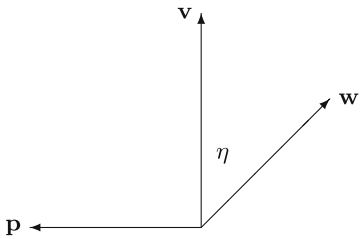


Fig. 2.13 The **v**-**w** plane: **p** is perpendicular to **v**, but not to **w**

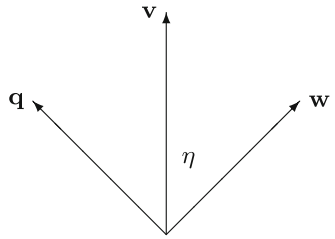


Fig. 2.14 The **v**-**w** plane: **q** is perpendicular to **w**, but not to **v**

$$\mathbf{p} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{p}, \mathbf{w})\mathbf{v}.$$

Hint: use the fact that

$$\cos\left(\frac{\pi}{2} + \eta\right) = -\sin(\eta).$$

2. Conclude that

$$\mathbf{p} \times (\mathbf{w} \times \mathbf{v}) = -(\mathbf{p}, \mathbf{w})\mathbf{v}.$$

3. In the latter formula, interchange the roles of **v** and **w**.
4. Conclude that

$$\mathbf{q} \times (\mathbf{v} \times \mathbf{w}) = -(\mathbf{q}, \mathbf{v})\mathbf{w},$$

where **q** is a new vector, perpendicular to **w** in the **v**-**w** plane (Fig. 2.14).

5. Add these formulas to each other:

$$(\mathbf{p} + \mathbf{q}) \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{p}, \mathbf{w})\mathbf{v} - (\mathbf{q}, \mathbf{v})\mathbf{w} = (\mathbf{p} + \mathbf{q}, \mathbf{w})\mathbf{v} - (\mathbf{p} + \mathbf{q}, \mathbf{v})\mathbf{w},$$

where **p** is perpendicular to **v** and **q** is perpendicular to **w** in the **v**-**w** plane.

6. Show that every vector **u** in the **v**-**w** plane could be written as $\mathbf{u} = \mathbf{p} + \mathbf{q}$, where **p** is perpendicular to **v** and **q** is perpendicular to **w** in the **v**-**w** plane. Hint: because **v** and **w** are linearly independent of each other, so are also **p** and **q**.
7. What does this mean geometrically? Hint: this is just the parallelogram rule.

8. Conclude that

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u}, \mathbf{w})\mathbf{v} - (\mathbf{u}, \mathbf{v})\mathbf{w},$$

   where $\mathbf{u}$ is any vector in the $\mathbf{v}$-$\mathbf{w}$ plane.
9. Extend this to a yet more general $\mathbf{u}$ that may exceed the $\mathbf{v}$-$\mathbf{w}$ plane. Hint: add to $\mathbf{u}$ a new component, perpendicular to the $\mathbf{v}$-$\mathbf{w}$ plane. This has no effect on either side of the above formula.

2.7.7 Linear Momentum: Orthogonal Decomposition

1. How to use this formula in practice? Give an example.
2. For instance, set $\mathbf{u} \equiv \mathbf{v} \equiv r$ (the position of the particle) and $\mathbf{w} \equiv p$ (the linear momentum). What do you get? Hint:

$$r \times (r \times p) = (r, p)r - (r, r)p,$$

   or

$$(r, r)p = -r \times (r \times p) + (r, p)r = (r \times p) \times r + (r, p)r.$$

3. Divide both sides by $\|r\|^2$.
4. Is this familiar? Hint: this is the orthogonal decomposition of the linear momentum p (Sect. 2.4.6).

2.7.8 The Centrifugal and Centripetal Forces

1. Use our triple-vector-product formula, in yet another practical example.
2. In particular, use this formula to write the centrifugal force in general, no matter whether $\boldsymbol{\omega}$ is perpendicular to r or not. (This way, the rotating axes not necessarily align with r any more.) Hint: set $\mathbf{u} = \mathbf{v} \equiv \boldsymbol{\omega}$ and $\mathbf{w} \equiv r$:

$$\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r) = (\boldsymbol{\omega}, r)\boldsymbol{\omega} - (\boldsymbol{\omega}, \boldsymbol{\omega})r.$$

   This way, the centrifugal force (divided by m) takes the form

$$\begin{aligned} -\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r) &= -(\boldsymbol{\omega}, r)\boldsymbol{\omega} + (\boldsymbol{\omega}, \boldsymbol{\omega})r \\ &= \|\boldsymbol{\omega}\|^2 \left(r - \frac{(\boldsymbol{\omega}, r)}{\|\boldsymbol{\omega}\|^2} \boldsymbol{\omega} \right) \\ &= \|\boldsymbol{\omega}\|^2 \hat{r}, \end{aligned}$$

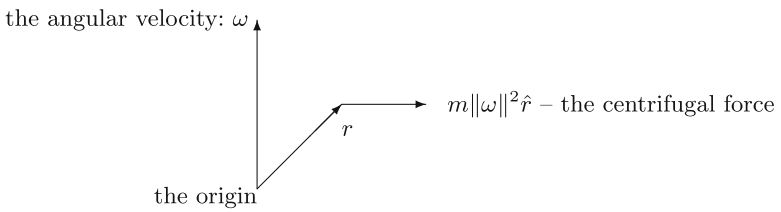


Fig. 2.15 A view from the side: r is not perpendicular to $\boldsymbol{\omega}$ any more. Instead, only $\hat{r}$ be that part of r that is perpendicular to $\boldsymbol{\omega}$. This way, the centrifugal force is $m\|\boldsymbol{\omega}\|^2\hat{r}$ rightwards

where $\hat{r}$ is the part of r that is perpendicular to $\boldsymbol{\omega}$ (Fig. 2.15).

3. Prove this in a more geometrical way. Hint: note that $\boldsymbol{\omega} \times r = \boldsymbol{\omega} \times \hat{r}$. Therefore,

$$-\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r) = -\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \hat{r}) = \|\boldsymbol{\omega}\|^2 \hat{r}.$$

4. Does the centrifugal force really exist? Hint: only in the rotating axis system! In the static system, on the other hand, it has no business to exist. Indeed, in Figs. 2.6, 2.7, the velocity is often constant, meaning equilibrium: no force at all.
5. Does the centripetal force exist? Hint: only if supplied by some source, such as gravity.
6. What is the role of the centripetal force? Hint: in the rotating axis system, it balances the centrifugal force and cancels it out. This way, there is no force at all, so the particle always has the same (rotating) coordinates and is carried effortless by the rotating axis system round and round forever. In the static system, on the other hand, the centripetal force has a more "active" role: to make u turn. This indeed makes the particle go round and round, as required.

2.7.9 The Inertia Matrix Times the Angular Velocity

1. Assume that the angular momentum is already available. How could it be used to define the angular velocity? Hint: at the end of Sect. 2.5.3, it is suggested to define the angular velocity proportional to the angular momentum:

$$\boldsymbol{\omega} \equiv \frac{r \times p}{m\|r\|^2}.$$

2. Why is this attractive? Hint: this makes $\boldsymbol{\omega}$ perpendicular to r.
3. Drop this assumption: $\boldsymbol{\omega}$ could now be perpendicular to r or not.
4. Work now the other way around: assume that the angular momentum is still unknown. Instead, assume that the angular velocity is available. Use it to uncover the angular momentum (assuming that w in Figs. 2.7 and 2.16 is

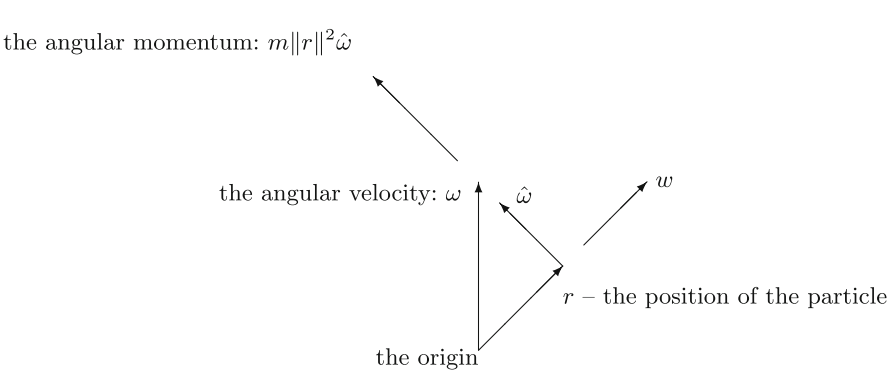


Fig. 2.16 $\hat{\boldsymbol{\omega}}$ is that part of $\boldsymbol{\omega}$ that is perpendicular to r. If w is radial, then the angular momentum is $m\|r\|^2\hat{\boldsymbol{\omega}}$, changing direction all the time, to point obliquely: inwards–upwards. The system is not closed: a horizontal centripetal force must be supplied from the outside. This is why the angular momentum is not conserved

radial):

$$\begin{aligned} r \times p &= m \cdot r \times v \\ &= m \cdot r \times (u + w) \\ &= m \cdot r \times u \\ &= m \cdot r \times (\boldsymbol{\omega} \times r) \\ &= m\left((r, r)\boldsymbol{\omega} - (r, \boldsymbol{\omega})r\right) \\ &= m\|r\|^2\left(\boldsymbol{\omega} - \frac{(r, \boldsymbol{\omega})}{\|r\|^2}r\right) \\ &= m\|r\|^2\hat{\boldsymbol{\omega}}, \end{aligned}$$

where $\hat{\boldsymbol{\omega}}$ is the part of $\boldsymbol{\omega}$ that is perpendicular to r. Hint: in our triple-vector-product formula, set $\mathbf{u} \equiv \mathbf{w} \equiv r$ and $\mathbf{v} \equiv \boldsymbol{\omega}$.

5. Prove this in a more geometrical way. Hint: note that $\boldsymbol{\omega} \times r = \hat{\boldsymbol{\omega}} \times r$. Therefore,

$$r \times (\boldsymbol{\omega} \times r) = r \times (\hat{\boldsymbol{\omega}} \times r) = \|r\|^2\hat{\boldsymbol{\omega}}.$$

6. Conclude that, if w in Figs. 2.7 and 2.16 is radial, then the angular momentum has a yet simpler form:

$$r \times p = mA\boldsymbol{\omega},$$

where $mA \equiv mA(r)$ is the inertia matrix of the particle.

7. Rewrite this in words. Hint: the angular momentum is the inertia matrix times the angular velocity.

8. What is the physical meaning of the inertia matrix? Hint: let us ask a yet deeper question: what is linear momentum? It is mass times velocity. More precisely, this is the *inertial* mass, resisting any change to the initial motion. Likewise, what is angular momentum? It is inertia matrix times angular velocity (if w is radial). Thus, the inertia matrix resists any change to the initial rotation.
9. In Fig. 2.11, how to make w radial? Hint: shift the origin downwards, until it meets the center of the Earth. This way, the new r is no longer horizontal, but oblique: it leads from the center of the Earth to the spaceship, at the same direction as w. As a result, w is now radial, as required. Thus, the angular momentum is not vertical but oblique: it is proportional to $\hat{\boldsymbol{\omega}}$, as in Fig. 2.16.
10. Assume now that we do not know yet that w is radial. Instead, we only know that the angular momentum is the inertia matrix times the angular velocity. Is this enough to prove that w is indeed radial? Hint: yes! See below.
11. Look at things the other way around. Assume now that the angular momentum is already available. How to define the angular velocity? Hint: design $\boldsymbol{\omega}$ in two stages: first, let it have some radial component. Then, define its nonradial component by

$$\hat{\boldsymbol{\omega}} \equiv \frac{r \times p}{m\|r\|^2}.$$

12. Why is this attractive? Hint: to see this, multiply this by $m\|r\|^2$:

$$r \times p = m\|r\|^2 \hat{\boldsymbol{\omega}}.$$

13. How to drop the hat? Hint: rewrite this as

$$r \times p = mA\boldsymbol{\omega}.$$

14. Rewrite this in words. Hint: the angular momentum is the inertia matrix times the angular velocity.
15. This way, must w be radial? Hint: thanks to the above definition of $\hat{\boldsymbol{\omega}}$,

$$\begin{aligned} r \times p &= m \cdot r \times v \\ &= m \cdot r \times (u + w) \\ &= m \cdot r \times u + m \cdot r \times w \\ &= m \cdot r \times (\boldsymbol{\omega} \times r) + m \cdot r \times w \\ &= m\|r\|^2 \hat{\boldsymbol{\omega}} + m \cdot r \times w \\ &= r \times p + m \cdot r \times w. \end{aligned}$$

Thus,

$$0 = m \cdot r \times w,$$

so

$$r \times w = 0.$$

This means that w must indeed be radial, as asserted.

16. Prove this in a more geometrical way. Hint: look again at the velocity $v = u + w$. Look also at its orthogonal decomposition at the beginning of Sect. 2.5.1. In it, the former term is now $u = \boldsymbol{\omega} \times r = \hat{\boldsymbol{\omega}} \times r$: the entire nonradial part. Therefore, the latter term must be w: the radial part.

2.7.10 Angular Momentum and Its Conservation

1. In Fig. 2.16, the angular momentum is not conserved. On the contrary: it changes direction all the time. Indeed, it is proportional to $\hat{\boldsymbol{\omega}}$, which changes direction to keep pointing obliquely (upwards and inwards). Why is not the angular momentum constant? Hint: only in a closed (isolated) system is the angular momentum conserved. The particle, however, is not isolated: there is an external force acting upon it—a horizontal centripetal force that makes it rotate around the vertical $\boldsymbol{\omega}$-axis.
2. To supply this centripetal force, introduce a new particle (of the same mass) at the new position

$$r^- \equiv \begin{pmatrix} -x \\ -y \\ z \end{pmatrix},$$

on the other side of the vertical $\boldsymbol{\omega}$-axis (Fig. 2.17).

3. What is its inertia matrix? Hint: $mA(r^-)$.
4. What are the eigenvectors of $A(r^-)$? Hint: one eigenvector is r^-. The two other eigenvectors are perpendicular to r^- and to one another.
5. Could the eigenvectors of $A(r^-)$ be the same as those of $A(r)$? Hint: only if r^- is proportional or perpendicular to r.
6. What are the principal axes of the second particle? Are they the same as those of the first particle?
7. In terms of its own principal coordinates, where is the second particle? Hint: it is always at the same principal coordinates: $(\|r\|, 0, 0)$.
8. Let $\hat{\boldsymbol{\omega}}^-$ be the part of $\boldsymbol{\omega}$ perpendicular to r^-. Show that the angular momentum of the second particle is

$$m\|r^-\|^2 \hat{\boldsymbol{\omega}}^- = mA(r^-)\boldsymbol{\omega}.$$

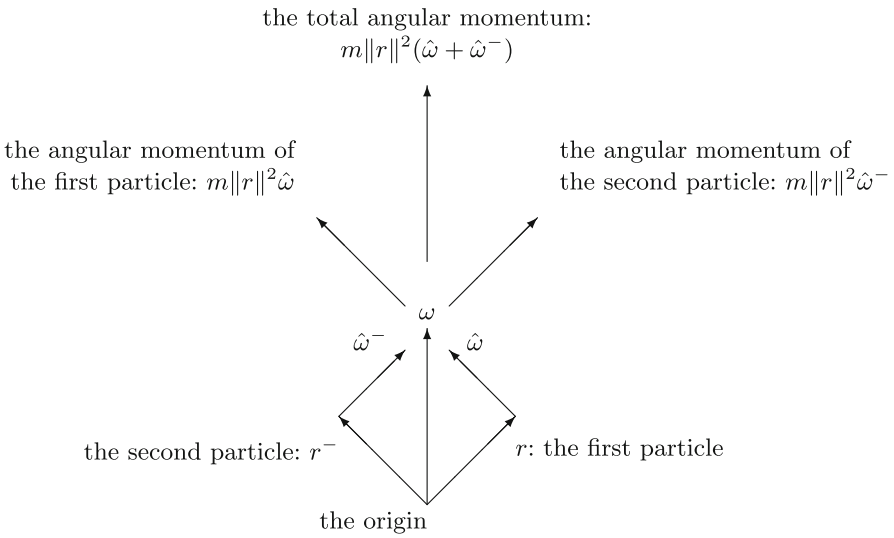


Fig. 2.17 Two particles at r and r^-: they attract one another, supplying horizontal centripetal force, to make them rotate together around the vertical $\boldsymbol{\omega}$-axis. The system is now closed: no external force. This is why the total angular momentum is now conserved: $\hat{\boldsymbol{\omega}}+\hat{\boldsymbol{\omega}}^-$ keeps pointing straight upwards all the time

9. Together, these particles make a closed system: for a suitable $\|r\|$, they attract each other just enough to supply the horizontal centripetal force required to make them rotate together around the vertical $\boldsymbol{\omega}$-axis.
10. This is called the two-body system.
11. Around what point does it rotate? Hint: around its center of mass: the midpoint in between the particles.
12. What is the total angular momentum of the two-body system? Hint: add up. (A conserved physical quantity can be added up.)
13. What is the direction of this vector?
14. Must it be vertical?
15. Show that the total angular momentum is now constant (conserved). Hint: the sum $\hat{\boldsymbol{\omega}}+\hat{\boldsymbol{\omega}}^-$ always points straight upwards, with no inward component any more.
16. Define the inertia matrix of the two-body system:

$$B \equiv B(r) \equiv mA(r) + mA(r^-).$$

17. Write the total angular momentum as

$$m\|r\|^2\left(\hat{\boldsymbol{\omega}}+\hat{\boldsymbol{\omega}}^-\right) = m\left(A(r)\boldsymbol{\omega} + A(r^-)\boldsymbol{\omega}\right) = B\boldsymbol{\omega}.$$

18. Rewrite this in words. Hint: the angular momentum is the inertia matrix times the angular velocity.
19. Without calculating B explicitly, show that it is symmetric.
20. Conclude that B has three orthogonal eigenvectors. Hint: see Chap. 1, Sect. 1.9.5.
21. Could these new eigenvectors be the same as those of $A(r)$? Hint: only if r^- is proportional or perpendicular to r.
22. Show that B is still positive semidefinite: its eigenvalues are greater than or equal to zero. Hint: given a three-dimensional vector q, span it in terms of eigenvectors of $A(r)$ (or $A(r^-)$). As a result,

$$(q, Bq) = m(q, A(r)q) + m(q, A(r^-)q) \geq 0.$$

23. Show that, if $r^- \neq -r$, then either $(q, A(r)q) > 0$ or $(q, A(r^-)q) > 0$.
24. Conclude that, in this case, $(q, Bq) > 0$ as well.
25. Conclude that, in this case, B is positive definite: its eigenvalues are strictly positive.
26. There are the moments of inertia of the two-body system.
27. Design an eigenvector of B that is perpendicular to both r and r^-. Hint: if $r^- \neq -r$, then take their vector product: $r \times r^-$.
28. What is its eigenvalue? Hint: $2m\|r\|^2$.
29. Still, this eigenvector depends on r. Design yet another eigenvector that does not depend on r, and remains the same all the time. Hint: take the vertical vector $(0, 0, 1)^t$.
30. What is its eigenvalue? Hint: $2m(\|r\|^2 - z^2) = 2m(x^2 + y^2)$.
31. Could this be negative or zero? Hint: no! If it were zero, then $r^- = r$, which is impossible.
32. Conclude that our vertical $\boldsymbol{\omega}$ remains an eigenvector of B all the time.
33. Conclude that the total angular momentum $B\boldsymbol{\omega}$ remains vertical all the time.
34. Conclude once again that the total angular momentum is indeed conserved.

2.7.11 Rigid Body

1. The above is just a special case of a more general setting: a closed system, containing many particles, rotating around a constant eigenvector $\boldsymbol{\omega}$ of the inertia matrix of the entire system. This way, the total angular momentum remains constant, as required.
2. How to use this in practice? Hint: the same is true even for infinitely many particles that make a new rigid body. In this case, use an integral rather than a sum:

$$B \equiv \int\int\int A(r)\rho(r)dxdydz.$$

This defines the new B: the total inertia matrix. Indeed, at $r = (x, y, z)^t$, ρ is the local density, so the local matrix $A(r)$ is multiplied by the local mass $\rho dxdydz$.

3. Is B still symmetric?
4. Conclude that B still has three orthogonal eigenvectors: the new principal axes.
5. Is B still positive semidefinite? Hint: the proof is as in the two-body system. Indeed, for a given three-dimensional vector q,

$$(q, Bq) = \int\int\int (q, A(r)q)\rho(r)dxdydz \geq 0.$$

6. If our rigid body rotates around a constant $\boldsymbol{\omega}$ (an eigenvector of B that does not depend on B), what happens to the angular momentum? Hint: it is constant: $B\boldsymbol{\omega}$.
7. In the rigid body, is the system closed? Hint: yes, no force acts on the rigid body from the outside.
8. Is the angular velocity constant? Hint: yes, we assume here that $\boldsymbol{\omega}$ is an eigenvector of B that does not depend on B. Therefore, $\boldsymbol{\omega}$ may remain constant, even if B changes during the rotation.
9. Is the angular momentum conserved? Hint: yes, $B\boldsymbol{\omega}$ remains constant as well.
10. What are the principal axes algebraically-geometrically? Hint: each principal axis is spanned by an eigenvector of B.
11. What are the principal axes physically? Hint: look at a principal axis spanned by an eigenvector of B that is independent of B. This principal axis marks the direction of constant angular velocity and momentum, around which the entire rigid body could rotate round and round forever.
12. Could B be singular (have a zero eigenvalue)? Hint: this could happen only in a very simple rigid body: a straight wire, in which all atoms lie on the same line. This way, B has a zero eigenvalue. The corresponding eigenvector points in the same direction as the wire itself. Around it, no rotation could take place, because the wire is too thin: it has no width at all.
13. In the wire, where should the origin be placed? Hint: best place it in the middle.
14. Does the wire have a more genuine principal axis? Hint: issue a vertical from the origin, in just any direction that is perpendicular to the wire. This could serve as a better principal axis, around which the entire wire could rotate round and round forever.
15. Consider yet another example of a rigid body: a thin disk. Where should the origin be placed? Hint: at its center.
16. What principal axis must the disk have? Hint: the vertical at its center. Around it, the disk could indeed rotate round and round forever.
17. What other principal axes could the disk have? Hint: every diameter could serve as a new principal axis, around which the entire disk could rotate round and round forever.
18. In Fig. 2.8, even if $\boldsymbol{\omega}$ is not perpendicular to r, show that the centrifugal force could be written simply as

$$-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times r) = mA(\boldsymbol{\omega})r.$$

2.7.12 The Percussion Point

1. Hold your baseball bat vertically in your hands, ready to hit a ball.
2. Assume that the bat has mass m and length d.
3. For simplicity, assume that the bat is uniform. This way, its center of mass is at the midpoint, at height $d/2$ (Fig. 2.18). Furthermore, its density is m/d kilogram per meter. (It is easy to extend this to a more standard, nonuniform bat.)
4. Let r be the vertical vector, leading from the bottom to the middle.
5. What is the norm of r? Hint: $\|r\| = d/2$.
6. What happens if you hit the ball at the midpoint r? Hint: this is not a good idea: this would push the entire bat leftwards into your hands, hurting you.
7. On the other hand, what happens if you hit the ball at the tip at $2r$? Hint: this is not a good idea either: this would rotate the bat slightly around the midpoint, pulling it away from your hands.
8. Where is the best place to hit the ball? Hint: somewhere in between, where no angular momentum is wasted on rotating the bat needlessly around the middle. This is the percussion point $d/2 < y < d$.
9. Where exactly is the percussion point y? Hint: see below.

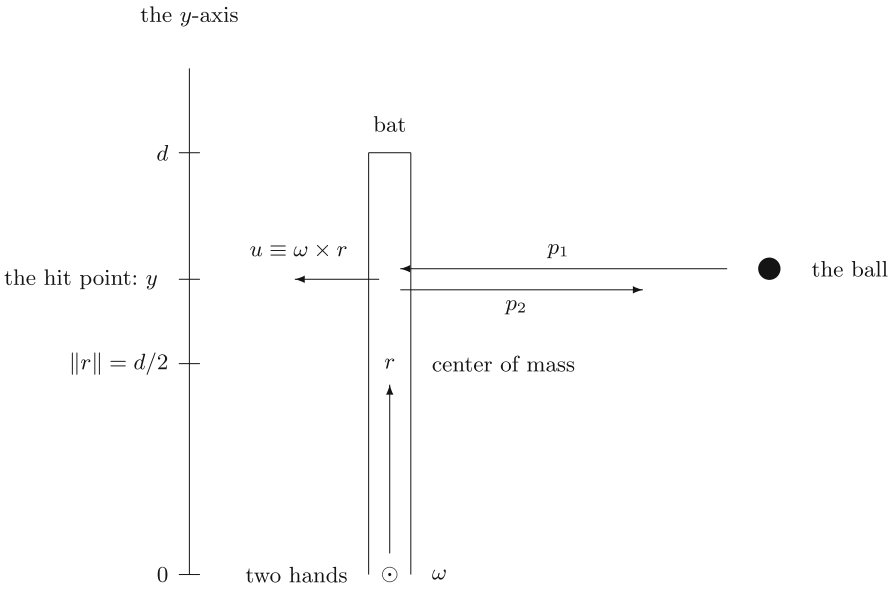


Fig. 2.18 A vertical baseball bat of height d. The ball meets it at height y, losing linear momentum in the amount of $p_1 - p_2$. This amount is transferred fully to the center of mass of the bat (at height $\|r\| = d/2$), producing a new velocity of $u \equiv \boldsymbol{\omega} \times r$

10. Let a ball come from the right, at a (horizontal) linear momentum p_1. Upon meeting the bat at height y, the ball returns rightwards, at a smaller linear momentum p_2. How much linear momentum did it lose? Hint: $p_1 - p_2$.
11. Was this really lost? Hint: no! Thanks to conservation of linear momentum, it wasn't lost, but transformed fully to the center of mass of the bat at $d/2$, pushing it at an initial linear momentum of

$$p_1 - p_2,$$

and an initial velocity of

$$\frac{p_1 - p_2}{m} = u = \boldsymbol{\omega} \times r$$

leftwards (Fig. 2.18).
12. What is $\boldsymbol{\omega}$? Hint: this is the angular velocity at which the bat rotates counterclockwise around its bottom. This way, $\boldsymbol{\omega}$ points from the page toward your eyes, as usual.
13. Conclude that $\boldsymbol{\omega}$-r-u is a right-hand system.
14. Use this in the above formula, and take the norm of both sides. Hint:

$$\frac{1}{m}\|p_1 - p_2\| = \|u\| = \|\boldsymbol{\omega}\| \cdot \|r\| = \|\boldsymbol{\omega}\|\frac{d}{2}.$$

15. Multiply both sides by m:

$$m\|\boldsymbol{\omega}\|\frac{d}{2} = \|p_1 - p_2\|.$$

This will be useful later.
16. Now, look at things the other way around. Just as the midpoint rotates around the bottom, the bottom also rotates counterclockwise around the midpoint, at the same angular velocity $\boldsymbol{\omega}$ (issuing now from the midpoint, as in Fig. 2.19).
17. What is the moment of inertia around the midpoint? Hint:

$$\begin{aligned}\frac{m}{d}\int_{-d/2}^{d/2} \tilde{y}^2 d\tilde{y} &= 2\frac{m}{d}\int_0^{d/2} \tilde{y}^2 d\tilde{y} \\ &= 2\frac{m}{d} \cdot \frac{1}{3}\left(\frac{d}{2}\right)^3 \\ &= 2\frac{m}{d} \cdot \frac{d^3}{24} \\ &= \frac{md^2}{12}.\end{aligned}$$

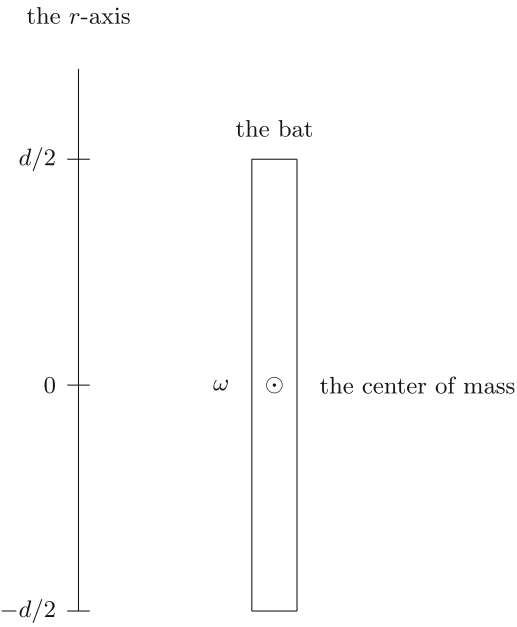


Fig. 2.19 The midpoint rotates around the bottom, but at the same time the bottom also rotates around the midpoint (at the same angular velocity $\boldsymbol{\omega}$, issuing now from the midpoint, not from the bottom). From the perspective of the midpoint, the bat rotates around it, at a new moment of inertia

18. How to use conservation of angular momentum? Hint: at hit time, the original angular momentum (of the ball around the midpoint) is transformed fully into a new angular momentum in the bat: the above moment of inertia times $\boldsymbol{\omega}$:

$$\left(y - \frac{d}{2}\right) \frac{r}{\|r\|} \times (p_1 - p_2) = \frac{md^2}{12}\boldsymbol{\omega}.$$

19. Take the norm of both sides:

$$\left(y - \frac{d}{2}\right) \|p_1 - p_2\| = \frac{md^2}{12}\|\boldsymbol{\omega}\|.$$

20. Recall that

$$\|p_1 - p_2\| = m\|\boldsymbol{\omega}\|\frac{d}{2}.$$

21. Plug this in:

$$\left(y - \frac{d}{2}\right) m\|\boldsymbol{\omega}\|\frac{d}{2} = \frac{md^2}{12}\|\boldsymbol{\omega}\|.$$

22. Divide both sides by $m\|\boldsymbol{\omega}\|d/2$:

$$y - \frac{d}{2} = \frac{d}{6}.$$

23. Conclude that

$$y = \frac{d}{2} + \frac{d}{6} = \frac{2}{3}d.$$

24. This is the percussion point: the best point to hit the ball.
25. Why is it best? Hint: at this point, thanks to conservation of angular momentum, there is no extra angular momentum around the midpoint, but only the exact amount required to rotate the bat around its bottom. This way, your hands will feel no strain at all. After all, rotating around the bottom is fine: only an extra rotation around the midpoint is bad, because it could pull the bat away from your hands. Fortunately, at the above y, there is none.

2.7.13 Bohr's Atom and Energy Levels

1. Look at two physical phenomena: gravity and electrostatics. What do they have in common? Hint: both make a force of $\|r\|^{-2}$, and a potential of $\|r\|^{-1}$ (times a constant, which we disregard).
2. Look at the solar system. The Sun is at the origin: $(0, 0, 0)$. Around it, there is a planet at r, orbiting the Sun.
3. What is the centrifugal force? Hint: it is proportional to $\|\boldsymbol{\omega}\|^2\|r\|$. (Forget about constants.)
4. What is the direction of the centrifugal force? Hint: radial, outwards.
5. What keeps the planet in orbit? In other words, what prevents the planet from escaping to the outer space? Hint: the centripetal force, which balances the centrifugal force (Fig. 2.9).
6. What is the direction of the centripetal force? Hint: radial, inwards.
7. What is its magnitude? Hint: the same as that of the centrifugal force.
8. What supplies the centripetal force? Hint: gravity.
9. What is the angular velocity? Hint: proportional to $\|r\|^{-3/2}$. This way, the centripetal force agrees with the gravitational force:

$$\|\boldsymbol{\omega}\|^2\|r\| \sim \left(\|r\|^{-3/2}\right)^2 \|r\| = \|r\|^{-2}.$$

10. What is the angular momentum? Hint:

$$\|r \times p\| \sim \|A(r)\boldsymbol{\omega}\| \sim \|r\|^2\|\boldsymbol{\omega}\| \sim \|r\|^2\|r\|^{-3/2} = \|r\|^{1/2}.$$

11. Is it conserved? Hint: yes—$\|r\|$ is constant in the orbit. Furthermore, $r \times p$ points in the same direction as $\boldsymbol{\omega}$ all the time: perpendicular to the plane of the orbit (Fig. 2.9).
12. What is the kinetic energy? Hint: kinetic energy is half momentum times velocity. In the context of circular motion, on the other hand, kinetic energy is half angular momentum times angular velocity:

$$\frac{1}{2}\|r \times p\| \cdot \|\boldsymbol{\omega}\| \sim \|r\|^{1/2}\|r\|^{-3/2} = \|r\|^{-1}.$$

13. Is it conserved? Hint: yes—$\|r\|$ is constant in the orbit.
14. Where did this energy come from? Hint: from the gravitational potential, which is proportional to $\|r\|^{-1}$ as well. (See below.)
15. Why does not the Earth spiral or fall into the Sun? Hint: because it has just enough kinetic energy to remain in orbit all the time.
16. Why is a Neptune-year much longer than an Earth-year? Hint: because the angular velocity decreases sharply with $\|r\|$:

$$\|\boldsymbol{\omega}\| \sim \|r\|^{-3/2}.$$

17. Next, look at the atom. In Bohr's model, the nucleus is at the origin, and the electrons orbit it.
18. To have an idea about the atom, use the same model as in the solar system. Hint: forget about constants, as above.
19. Later on, in quantum mechanics, we will see that angular momentum comes in discrete quanta: say, n quanta, where n is an integer number.
20. Assume that the electron is at r, orbiting the nucleus. What could the radius $\|r\|$ be? Hint: not every radius is allowed. Indeed, the angular momentum must be n (times a constant):

$$\|r \times p\| \sim \|r\|^{1/2} \sim n.$$

Thus, the radius must be n^2 (times a constant):

$$\|r\| \sim n^2.$$

21. What energy level is allowed? Hint: the energy level must be n^{-2} (times a constant):

$$-\|r\|^{-1} \sim -n^{-2}.$$

22. Where did the minus sign come from? Hint: the electron has a negative charge. This way, it is attracted to the positive charge of the proton in the nucleus. How to model this mathematically? Better insert a minus sign. This makes the potential increase monotonically away from the nucleus. This way, the minimal

potential is near the nucleus. To have a minimal potential, the electron must therefore travel inwards. This is indeed attraction.

23. Where did the electron get its kinetic energy from? Hint: assume that the electron was initially at infinity, where the potential is zero. Then, it started to spiral and fall toward the nucleus, gaining negative potential and positive kinetic energy, until reaching its current position: r. During its journey, it lost potential energy of

$$0 - \left(-\|r\|^{-1}\right) = \|r\|^{-1}.$$

Still, this wasn't lost for nothing: it was converted completely from potential to kinetic energy.

24. According to Maxwell's theory, while orbiting the nucleus, the electron should actually radiate an electromagnetic wave (light ray), and lose its entire energy very quickly. Why does not this happen? Hint: in terms of quantum mechanics, its position in its orbit is only nondeterministic: a periodic wave function, with an integer wave number. This is a standing wave that radiates nothing. This is how Debroglie proved stability in Bohr's model.
25. By now, the electron is at r, at radius $\|r\|$ from the nucleus. Could it get yet closer to the nucleus? In other words, could it "jump" inwards, from its allowed radius to an inner radius? Hint: only if the inner radius is allowed as well.
26. In the inner radius, how much potential energy would it have? Hint: less than before. Indeed, $-\|r\|^{-1}$ increases monotonically with $\|r\|$.
27. What happened to the potential energy that was lost? Hint: it transforms into light: the electron emits a photon.
28. On the other hand, could the electron get away from the nucleus, and "jump" outwards, to an outer radius? Hint: only if the outer radius is allowed as well.
29. In the outer radius, how much potential energy would it have? Hint: more than before. Indeed, $-\|r\|^{-1}$ increases monotonically with $\|r\|$.
30. What could possibly supply the extra energy required for this? Hint: a photon (light ray) comes from the Sun and hits the electron hard enough.
31. Could the photon hit the electron ever so hard, and knock it off the atom, sending it all the way to infinity? Hint: only if the light ray is of high frequency, with energy as high as n^{-2} (the old energy level). Only this could give the electron sufficient energy to escape to $\|r\| = \infty$ and have zero potential.
32. This is the photo-electric effect, discovered by Einstein. We will come back to this later.

Chapter 3
Markov Matrix and Its Spectrum: Toward Search Engines



So far, we have mostly used small matrices, with a clear geometrical meaning: 2×2 matrices transform the Cartesian plane, and 3×3 matrices transform the entire Cartesian space. What about yet bigger matrices? Fortunately, they may still have a geometrical meaning of their own. Indeed, in graph theory, they may help design a weighted graph, and model a stochastic flow in it. This makes a Markov chain, converging to a unique steady state. This has a practical application in modern search engines in the Internet [50].

3.1 Characteristic Polynomial and Spectrum

3.1.1 Null Space and Characteristic Polynomial

In this chapter, we will see how useful matrices are in graph theory. This will help design a practical ranking algorithm for search engines on the Internet. Before going into this, let us see some more background in linear algebra.

Let A be a square (real or complex) matrix of order n. In many cases, one might want to focus on the eigenvalues alone. After all, they tell us how A acts on important nonzero vectors: the eigenvectors. How to characterize the eigenvalues, without solving for the eigenvectors?

For this purpose, let I be the identity matrix of order n. Let λ be some (unknown) eigenvalue. Then $A - \lambda I$ maps the (unknown) eigenvector to zero. In other words, the eigenvector belongs to the null space of $A - \lambda I$ (Chap. 1, Sect. 1.9.2).

So, there is no need to know the eigenvector explicitly: it is sufficient to know that $A - \lambda I$ maps it to the zero vector. This means that $A - \lambda I$ has no inverse. After all, no matrix in the world could possibly map the zero vector back to the original nonzero eigenvector. Thus, $A - \lambda I$ must have zero determinant:

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Y. Shapira, *Linear Algebra and Group Theory for Physicists and Engineers*,
https://doi.org/10.1007/978-3-031-22422-5_3

$$\det(A - \lambda I) = 0.$$

In other words, λ solves the equation

$$\det(A - xI) = 0,$$

where x is the unknown variable. Indeed, if we set $x = \lambda$, then this equation is satisfied. For all other x's that are not eigenvalues, on the other hand, this equation is no longer true.

This equation is called the characteristic equation. It characterizes the eigenvalue λ in a simple algebraic way, as required. Even before λ is known, we already know something about it: it solves the characteristic equation. This gives us a practical algorithm: to discover the eigenvalue, just solve the characteristic equation. Then, solve for the eigenvector too, if necessary.

In the characteristic equation, what is the left-hand side? This is actually a polynomial in x:

$$p(x) \equiv \det(A - xI).$$

(See also Chap. 8 below.) This is called the characteristic polynomial. So, the original eigenvalue λ is a root of the characteristic polynomial: a special argument, for which the characteristic polynomial vanishes:

$$p(\lambda) = 0.$$

The eigenvalues are the only roots: any other x that is not an eigenvalue cannot possibly be a root. This is why it is called the characteristic polynomial: it characterizes the eigenvalues well.

How many roots are there? Well, there is at least one (complex) root, and at most n distinct roots, each with its own private eigenvector. Later, we will study the characteristic polynomial in some more detail (Chap. 14, Sects. 14.5.1–14.5.3).

3.1.2 Spectrum and Spectral Radius

Let us place the eigenvalues in a new set:

$$\text{spectrum}(A) \equiv \{x \in \mathbb{C} \mid \det(A - xI) = 0\}.$$

This is the spectrum of A: the set of eigenvalues.

How large could an eigenvalue be in magnitude? This is called the spectral radius:

$$\rho(A) \equiv \max_{\mu \in \text{spectrum}(A)} |\mu|.$$

Clearly, the spectral radius is a nonnegative real number. How large could it be? Well, it cannot exceed the maximal row-sum:

$$\rho(A) \leq \max_{1 \leq i \leq n} \sum_{j=1}^{n} |a_{i,j}|,$$

or the maximal column-sum:

$$\rho(A) \leq \max_{1 \leq j \leq n} \sum_{i=1}^{n} |a_{i,j}|.$$

This is proved in the exercises below.

Note that $\rho(A)$ is not necessarily an eigenvalue on its own right. After all, even if A is a real matrix, its eigenvalues (and eigenvectors) are not necessarily real: they could be complex as well. For this reason, $\rho(A)$ is not necessarily an eigenvalue: it is just the absolute value of a (complex) eigenvalue.

We already know that, if A is Hermitian, then its eigenvalues are real:

$$A = A^h \Rightarrow \text{spectrum}(A) \subset \mathbb{R}.$$

(Here, "$\subset$" means "contained in.") In this case, either $\rho(A)$ or $-\rho(A)$ must be an eigenvalue. Nevertheless, even in this case, the eigenvectors may still be complex.

Only if A is a real symmetric matrix must it have not only real eigenvalues but also real eigenvectors. Of course, the real eigenvector is defined up to a scalar multiple, so it could always be multiplied by a complex scalar, to produce a complex eigenvector as well. Still, to make your life easy, better design a real eigenvector, and stick to it. For this purpose, given a complex eigenvector, just look at its real part (or its imaginary part): this is indeed a real eigenvector, easy to use. (See exercises at the end of Chap. 1.)

3.2 Graph and Its Matrix

3.2.1 Weighted Graph

What is a graph? It is modeled in terms of two sets: N and E. N contains the nodes, and E contains the edges. Each edge is a pair of two nodes. Geometrically, the edge leads from one node to the other.

In a weighted graph, each edge is assigned a nonnegative number: its weight, or the amount it could carry. This way, the edge $(j, i) \in E$ (leading from node j to node i) has the weight $a_{i,j} \geq 0$. This lets the amount $a_{i,j}$ to flow from node j to node i. If, on the other hand, $(j, i) \notin E$, then no edge leads from node j to node i, and $a_{i,j} \equiv 0$.

3.2.2 Markov Matrix

Let us index the nodes by the index

$$i = 1, 2, 3, \ldots, |N|,$$

where $|N|$ is the total number of nodes. Consider the jth node ($1 \leq j \leq |N|$). How much weight flows from it to all nodes in N? Well, assume that this weight sums to 1:

$$\sum_{i=1}^{|N|} a_{i,j} = 1.$$

This way, $a_{i,j}$ can be viewed as the probability that a particle based at node j would pick edge (j, i) to move to node i. In particular, if $(j, j) \in E$, then there is a small circle: an edge leading from node j to itself. In this case, the particle could stay at node j. The probability for this is $a_{j,j} \geq 0$.

The weights (or probabilities) can now be placed in a new $|N| \times |N|$ matrix:

$$A \equiv \left(a_{i,j}\right)_{1 \leq i,j \leq |N|}.$$

This is the probability matrix, or the Markov matrix: its columns sum to 1. Let us use it to describe a discrete stochastic flow.

3.2.3 Example: Uniform Probability

Consider those edges issuing from node j:

$$\text{outgoing}(j) \equiv \{(j, i) \in E \mid i \in N\} \subset E.$$

This way, $|\text{outgoing}(j)|$ is the total number of those edges issuing from j. Consider a simple example, with a uniform probability: the particle has no preference—it is equally likely to move to any neighbor node:

$$a_{i,j} \equiv \begin{cases} \frac{1}{|\text{outgoing}(j)|} & \text{if } (j, i) \in E \\ 0 & \text{if } (j, i) \notin E. \end{cases}$$

Why is this a legitimate probability? Because the columns sum to 1:

$$\begin{aligned}\sum_{i=1}^{|N|} a_{i,j} &= \sum_{\{i\in N \mid (j,i)\in E\}} \frac{1}{|\text{outgoing}(j)|} \\ &= \sum_{(j,i)\in \text{outgoing}(j)} \frac{1}{|\text{outgoing}(j)|} \\ &= |\text{outgoing}(j)| \frac{1}{|\text{outgoing}(j)|} \\ &= 1.\end{aligned}$$

3.3 Flow and Mass

3.3.1 Stochastic Flow: From State to State

Let us use the weights (or probabilities) to form a new discrete flow, step by step. The flow is stochastic: we can never tell for sure what the result of a particular step is, but only how likely it is to happen.

So far, we've assigned weights to the edges. These weights are permanent: they are assigned once and for all and never change any more.

Assume now that each node contains a nonnegative mass. These masses are different from the above weights: they may change dynamically, step by step.

At the beginning, node j contains the initial mass $u_j \geq 0$. These masses can be placed in a new $|N|$-dimensional column vector:

$$u \equiv \left(u_1, u_2, u_3, \ldots, u_{|N|}\right)^t.$$

This is the initial state: the mass distribution among the nodes in N.

Next, what happens to the mass in the jth node? Well, in the first step, u_j may split into tiny bits, each flowing to a different neighbor node: each edge of the form (j, i) transfers the amount $a_{i,j}u_j$ from node j to node i. This way, the original mass is never lost, but only redistributed among the neighbor nodes:

$$\sum_{i=1}^{|N|} a_{i,j}u_j = u_j \sum_{i=1}^{|N|} a_{i,j} = u_j \cdot 1 = u_j.$$

This is indeed mass conservation. This will be discussed further below.

In the first step, node j may lose its original mass through outgoing edges. Fortunately, at the same time, it may also gain some new mass through incoming edges. In fact, through each incoming edge of the form $(i, j) \in E$, it gains the new mass of $a_{j,i}u_i$ Thus, at node j, the mass has changed from

$$u_j \to \sum_{i=1}^{|N|} a_{j,i} u_i = (Au)_j.$$

This is true for each and every node $j \in N$. In summary, the mass distribution has changed from the original state u to the new state Au:

$$u \to Au.$$

This completes the first step. The same procedure can now repeat in the next step as well, to change the state from

$$Au \to A(Au) = A^2 u,$$

and so on.

The process may then continue step by step forever. Thanks to mass conservation, the total mass never changes.

3.3.2 Mass Conservation

What is mass conservation? It means that the total mass remains unchanged. Indeed, since the columns of A sum to 1, the total mass after the step is the same as before:

$$\begin{aligned}
\sum_{i=1}^{|N|} (Au)_i &= \sum_{i=1}^{|N|} \sum_{j=1}^{|N|} a_{i,j} u_j \\
&= \sum_{j=1}^{|N|} \sum_{i=1}^{|N|} a_{i,j} u_j \\
&= \sum_{j=1}^{|N|} u_j \sum_{i=1}^{|N|} a_{i,j} \\
&= \sum_{j=1}^{|N|} u_j \left(\sum_{i=1}^{|N|} a_{i,j} \right) \\
&= \sum_{j=1}^{|N|} u_j \cdot 1 \\
&= \sum_{j=1}^{|N|} u_j.
\end{aligned}$$

The same is true in subsequent steps as well. By mathematical induction, mass is preserved throughout the entire process.

This is indeed an infinite process that may go on and on forever. Does it converge to a steady state? To answer this, we must study the spectrum of A.

3.4 The Steady State

3.4.1 The Spectrum of Markov Matrix

Fortunately, the spectral radius of A is as small as

$$\rho(A) \leq \max_{1 \leq j \leq n} \sum_{i=1}^{n} |a_{i,j}| = 1$$

(see exercises below). Is it exactly

$$\rho(A) = 1?$$

Well, to check on this, let us look at the transpose matrix A^t. We already know that its eigenvalues are the complex conjugate of those of A (Chap. 1, Sect. 1.9.3). Therefore, both have the same spectral radius:

$$\rho\left(A^t\right) = \rho(A) \leq 1.$$

What else do we know about A^t? Well, its rows sum to 1. Therefore, we already have one eigenvector: this is the constant $|N|$-dimensional vector

$$\mathbf{c} \equiv (1, 1, 1, \ldots, 1)^t,$$

satisfying

$$A^t \mathbf{c} = \mathbf{c}.$$

So, for A^t, 1 is indeed an eigenvalue. As a result,

$$\rho\left(A^t\right) = 1.$$

Is 1 an eigenvalue of A as well? It sure is. After all,

$$\bar{1} = 1$$

is an eigenvalue of A as well. Thus,

$$\rho(A) = 1$$

as well.

In summary, both A and A^t share a common eigenvalue: 1. Still, they do not necessarily share the same eigenvector. For A^t, this is the constant vector. For A, on the other hand, it could be completely different. Fortunately, we can still tell how it looks like. For this purpose, we need a new assumption.

3.4.2 Converging Markov Chain

Let us make a new assumption: our graph is well-connected—no node could drop from it. In other words, all nodes in N are important—no node is redundant. Every node is valuable—it may be used to receive some mass at some step. Dropping it may therefore spoil the entire flow.

Because all nodes may be used to take some mass, N has no invariant subset, from which no mass flows away. This means that the flow is global: there is no autonomous subgraph, in which the original mass circulates forever, never leaking to the rest of the graph.

In this case, we say that A is irreducible. This is a most desirable property: it guarantees that the infinite flow makes a Markov chain that converges to a unique steady state.

Why is this true? Well, we already know that A has eigenvalue 1. Thanks to the above assumption, we can now tell how the corresponding eigenvector looks like:

- A has a unique eigenvector v of eigenvalue 1:

$$Av = v$$

  (up to a scalar multiple).
- All its components are positive:

$$v_j > 0, \quad 1 \le j \le |N|.$$

- To have this property, v is defined up to multiplication by a positive number.
- 1 is maximal—all other eigenvalues are strictly smaller than 1 in magnitude:

$$\mu \in \text{ spectrum}(A) \Rightarrow |\mu| < 1.$$

This is indeed the Peron–Frobenius theory [93].

Thanks to these properties, the infinite flow makes a Markov chain that converges to a unique steady state. Even if the mass is initially concentrated in just one node, it will eventually get distributed globally among all nodes.

3.4.3 The Steady State

To design the steady state, let u be the initial state. Let us write

$$u = v + w,$$

where v is the unique eigenvector corresponding to 1 (Sect. 3.4.2), and the residual (or remainder, or error) w is a linear combination of the other (pseudo-) eigenvectors, associated with eigenvalues smaller than 1 in magnitude. This way, as the process progresses, we have

$$\|A^n w\| \to_{n\to\infty} 0,$$

so

$$A^n u = A^n(v + w) = A^n v + A^n w = v + A^n w \to_{n\to\infty} v.$$

Thus, the infinite flow converges to the steady state v.

We are not done yet. After all, v is not defined uniquely. In fact, although v contains positive components only, these components are defined up to multiplication by a positive constant. How to specify v uniquely?

Fortunately, thanks to mass conservation, we can tell the total mass in v:

$$\sum_{j=1}^{|N|} v_j = \sum_{j=1}^{|N|} u_j.$$

This determines v uniquely, as required. Moreover, this also shows that the error w must have zero total mass. After all, unlike u and v, w could contain not only positive but also negative masses, which cancel each other, and sum to zero.

3.4.4 Search Engine in the Internet

How to use a Markov chain in practice? Well, let us use it to model a communication network, such as the Internet. Each site is considered as an individual node. A link from one site to another makes an edge. This way, each site may contain a few links: its outgoing edges. For simplicity, assume that the probability to click on such a link is uniform, as in Sect. 3.2.3.

You, the surfer, play here the role of the particle. Initially, you are at the kth site, for some $1 \le k \le |N|$. While surfing, you may use a link to move to another site. Still, if this is a good site, and you are likely to stay there for long, then $a_{k,k}$ could be nearly 1.

In stochastic terms, you must eventually approach the steady state: v. How likely are you to enter the lth site ($1 \leq l \leq |N|$)? Just look at the positive component v_l, and see how large it is.

Still, you are not the only one. Like you, there are many other surfers. Thus, mass could stand here for the number of surfers at a particular site. The initial mass u tells us the initial distribution of surfers. The steady state v, on the other hand, tells us the "final" distribution: how likely are the surfers to enter a particular site eventually.

Now, suppose that you are interested in some keyword and want to search the web for those sites that contain it. Still, there are many correct answers: many sites may contain the same keyword. How should the search engine order (or rank) the answers? This is called the ranking problem.

For simplicity, assume that the search engine knows nothing about your surfing habits (which is highly unlikely these days). How should it rank the answers to your search? Well, it should start from the maximal component in v. If it indeed contains the keyword, then it should be ranked first. After all, this must be a popular site, in which you must be interested. Next, rank the second maximal component in v, and so on.

Still, the web is dynamic, not static. Every day, new sites are added, and old sites drop. For this reason, a good search engine should better update A often, and recalculate v often. This is indeed an eigenvector problem. (Later on in the book, we will present it in a more general form.) Fortunately, the positive constant that may multiply v is immaterial: it has no effect on the ranking.

3.5 Exercises: Gersgorin's Theorem

3.5.1 Gersgorin's Theorem

1. Let A be an $n \times n$ (complex) matrix. Let λ be an eigenvalue, associated with the eigenvector v:

$$Av = \lambda v, \quad v \neq \mathbf{0}.$$

Pick a specific i, for which v_i is a maximal component in v (in absolute value):

$$|v_j| \leq |v_i|, \quad 1 \leq j \leq n.$$

Show that

$$|v_i| > 0.$$

Hint: otherwise, $v = \mathbf{0}$, which is impossible for an eigenvector.

2. For the above i, show that the eigenvalue is not too far from the main-diagonal element:

$$\left|\lambda - a_{i,i}\right| \leq \sum_{1 \leq j \leq n,\ j \neq i} \left|a_{i,j}\right|.$$

This is Gersgorin's theorem. Hint:

$$\begin{aligned}
\left|\lambda - a_{i,i}\right| \cdot |v_i| &= \left|\left(\lambda - a_{i,i}\right) v_i\right| \\
&= \left| \sum_{1 \leq j \leq n,\ j \neq i} a_{i,j} v_j \right| \\
&\leq \sum_{1 \leq j \leq n,\ j \neq i} \left|a_{i,j} v_j\right| \\
&= \sum_{1 \leq j \leq n,\ j \neq i} \left|a_{i,j}\right| \cdot \left|v_j\right| \\
&\leq \sum_{1 \leq j \leq n,\ j \neq i} \left|a_{i,j}\right| \cdot |v_i| \\
&= |v_i| \sum_{1 \leq j \leq n,\ j \neq i} \left|a_{i,j}\right|.
\end{aligned}$$

Now, divide by $|v_i| > 0$.

3. For the above i, conclude that the eigenvalue is as small as the row-sum (in absolute value):

$$|\lambda| \leq \sum_{j=1}^{n} |a_{i,j}|.$$

Hint: use Gersgorin's theorem:

$$|\lambda| - |a_{i,i}| \leq \left|\lambda - a_{i,i}\right| \leq \sum_{1 \leq j \leq n,\ j \neq i} \left|a_{i,j}\right|.$$

4. Conclude that

$$\rho(A) \leq \max_{1 \leq i \leq n} \sum_{j=1}^{n} |a_{i,j}|.$$

Hint: the above could be done for each and every eigenvalue.

5. Do the same for the Hermitian adjoint. Conclude that

$$\rho(A) = \rho\left(A^h\right) \leq \max_{1 \leq j \leq n} \sum_{i=1}^{n} |a_{i,j}|.$$

6. Assume now that A is a Markov matrix. What does this mean? Hint: its elements are positive or zero, and its columns sum to 1.
7. Must A have a real spectrum? Hint: no—A might be nonsymmetric.
8. What about the transpose matrix, A^t? Must it be a Markov matrix as well? Hint: the rows of A not necessarily sum to 1.
9. Show that

$$\rho(A^t) \leq 1.$$

   Hint: use Gersgorin's theorem above.
10. Does this necessarily mean that

$$\rho(A^t) = 1?$$

    Hint: only if you could find at least one eigenvalue of modulus 1.
11. Design an eigenvector for A^t. Hint: the constant vector $\mathbf{c}$.
12. Show that

$$A^t \mathbf{c} = \mathbf{c}.$$

13. Conclude that

$$1 \in \text{ spectrum}(A^t).$$

14. Conclude that

$$\rho(A^t) = 1.$$

15. Conclude that

$$1 \in \text{ spectrum}(A)$$

    as well. Hint: see Chap. 1, Sect. 1.9.3. After all, the complex conjugate of 1 is 1 as well.
16. Conclude that

$$\rho(A) = 1$$

    as well.
17. Conclude that the spectrum of A lies in the closed unit circle in the complex plane:

$$\text{spectrum}(A) \subset \{z \in \mathbb{C} \mid |z| \leq 1\}.$$

18. Let v be the eigenvector of A satisfying

$$Av = v.$$

Prove that v indeed exists, and that $v \neq \mathbf{0}$. Hint: we have already seen that 1 is an eigenvalue of A.

19. Must v be the constant vector? Hint: the constant vector is an eigenvector of A^t, but not necessarily of A.
20. Assume that A is also irreducible (Sect. 3.4.2). What can you say now about v? Hint: up to a scalar multiple, v is unique and has positive components only.
21. What can you say now about the spectrum of A? Hint: 1 is the only eigenvalue of magnitude 1: all other eigenvalues are strictly smaller than 1 in absolute value:

$$\mu \in \text{spectrum}(A),\ \mu \neq 1 \ \Rightarrow\ |\mu| < 1.$$

22. Could A have an eigenvalue of the form

$$\exp(\theta\sqrt{-1}) = \cos(\theta) + \sin(\theta)\sqrt{-1}$$

for any $0 < \theta < 2\pi$? Hint: no—this complex number has magnitude 1, but is different from 1.

23. Conclude that the spectrum of A lies in the *open* unit circle, plus the point 1:

$$\text{spectrum}(A) \subset \{z \in \mathbb{C} \mid |z| < 1\} \cup \{1\}.$$

(Here, "$\cup$" means a union with the set that contains one element only: 1.)

24. Use the above properties to prove that the Markov chain indeed converges to a steady state. Hint: see Sect. 3.4.3.
25. Why is the steady state unique? Hint: mass conservation.
26. How could this help design a good search engine in the Internet? Hint: see Sect. 3.4.4.
27. How often should the search engine update A, and solve a new eigenvector problem, to update v as well?

Chapter 4
Special Relativity: Algebraic Point of View



Let us see how useful matrices are in practice. Indeed, a small 2×2 matrix is the key to a new theory in modern physics: special relativity.

How to model physics geometrically? For this task, Euclidean geometry is too poor: it only deals with a static shape in the two-dimensional plane. The real world, on the other hand, is three-dimensional. On top of this, we also need the (perpendicular) time axis, to help describe dynamics and motion.

Thanks to Newton, we indeed have this. We can now model a new force, acting on an object from the outside, to accelerate its initial speed, and make it faster.

This fits well in Plato's philosophy: to think about a new object, we need a new word in our language, to represent not only one concrete instance but also the "Godly" spirit behind all possible instances. Likewise, a Newtonian force acts from the outside, to give "life" to a static object.

Einstein, on the other hand, "threw" the time axis back into the very heart of geometry. This unites the time axis with the original space, producing a new four-dimensional manifold: spacetime.

This is more in the spirit of Aristotle's philosophy. A word takes its meaning not from the outside, but from the very inside: the deep nature of the concept it stands for.

Special relativity [20, 99] improves on Newtonian mechanics. It teaches us how to add velocities more accurately: nonlinearly! This gives us a new (relativistic) way to define energy and momentum. This is good physics: correct at all (inertial) reference frames, traveling at constant speed and direction with respect to each other.

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4.1 Adding Velocities (or Speeds)

4.1.1 How to Add Velocities?

In Newtonian mechanics, velocities are added linearly. (For us, velocity and speed are the same. We will use them both.) Consider a particle (or any object), traveling at constant speed v (at some constant direction). At the same time, another particle travels at the constant speed u at the opposite direction (Fig. 4.1). These speeds are not absolute: they are only relative to our lab. Indeed, we measure them here, from our lab.

But our lab is not an absolute reference frame. Indeed, it is not static, but dynamic: moves with the entire Earth, which orbits the Sun, which orbits the center of our galaxy: the Milky Way. Fortunately, this underlying motion is irrelevant. After all, we do not feel it at all. For us, the lab is as good as static.

Better yet, disregard the lab, and focus on the relation between the particles: how fast do they get away from one another? Assume that you stand on the second particle. How fast does the first particle get away from you? In this question, the lab plays no role any more.

In Galileo–Newton's theory, the velocities add up, so the answer is simply

$$u + v.$$

For small velocities, this indeed makes sense. But what happens when v is as large as the speed of light c? In this case, the sum $c + u$ would exceed the speed of light, which is impossible!

Fortunately, velocities are added nonlinearly: the true answer is not $u + v$ but

$$\frac{u + v}{1 + \frac{uv}{c^2}}$$

(Fig. 4.2). This makes a lot of sense, as we will see soon.

Indeed, when u and v are moderate (nonrelativistic), this is nearly the same as $u + v$. This is why Galileo–Newton's theory works well in practice. Still, it is inaccurate. At high (relativistic) velocities (as big as c, or a fraction of it), one must use the precise formula.

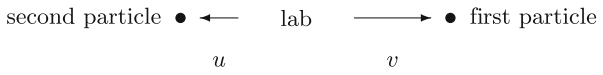


Fig. 4.1 In our lab, the first particle moves rightward at speed v, while the second particle moves leftward at speed u

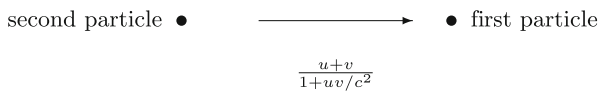


Fig. 4.2 From the second particle, the first particle gets away at speed $(u + v)/(1 + uv/c^2)$

4.1.2 Einstein's Law: Never Exceed the Speed of Light!

Thanks to this formula, we can now keep Einstein's law: never exceed the speed of light! Indeed, assume that both particles do not exceed the speed of light (with respect to the lab):

$$|u| \leq c \quad \text{and} \quad |v| \leq c.$$

Now, with respect to each other, could they ever exceed the speed of light? No! To see this, we must first scale the velocities right. For this purpose, just divide them by c, and define the new parameters

$$\beta_u \equiv \frac{u}{c} \ \ and \ \ \beta_v \equiv \frac{v}{c}.$$

In these new terms, Einstein's law tells us two things: first, c is the same in all reference frames (systems), provided that they are inertial: travel at a constant speed with respect to one another. Furthermore, no speed could ever exceed c:

$$|\beta_u| \leq 1 \ \ and \ \ |\beta_v| \leq 1.$$

What about our new formula? Does it obey this law? It sure does! To see this, assume first that both u and v are strictly smaller than the speed of light (in magnitude):

$$|u| < c \quad \text{and} \quad |v| < c,$$

so

$$|\beta_u| < 1 \ \ and \ \ |\beta_v| < 1.$$

In this case,

$$1 - \beta_u - \beta_v + \beta_u \beta_v = (1 - \beta_u)(1 - \beta_v) > 0,$$

so

$$\beta_u + \beta_v < 1 + \beta_u \beta_v,$$

or

$$\frac{\beta_u + \beta_v}{1 + \beta_u \beta_v} < 1.$$

Moreover, if both u and v change sign, then the above still holds. Therefore, we also have

$$\left| \frac{\beta_u + \beta_v}{1 + \beta_u \beta_v} \right| = \frac{|\beta_u + \beta_v|}{1 + \beta_u \beta_v} < 1.$$

By multiplying this by c, we obtain what we wanted:

$$\left| \frac{u + v}{1 + \frac{uv}{c^2}} \right| < c.$$

This is as required: with respect to one another, the particles never exceed the speed of light.

4.1.3 Particle as Fast as Light

So far, no particle was as fast as light. But what if one of them is? In this case, things get weird. To see this, look at

$$v = c \quad \text{and} \quad -c < u \leq c.$$

In this case,

$$\frac{u + c}{1 + \frac{uc}{c^2}} = \frac{u + c}{1 + \frac{u}{c}} = c.$$

What happens physically? The first particle travels rightward as fast as light. It cannot distinguish between the lab and the second particle: both are left behind at the same speed: c. Sounds strange? It is still true.

4.1.4 Singularity: Indistinguishable Particles

Still, there is a yet weirder case:

$$v = c \quad \text{and} \quad u = -c.$$

In this case, the second particle is as fast: it follows the first particle rightward, at the same speed: c. This way, our formula breaks down: it divides zero by zero.

What to do? Drop math, and use physics! After all, at the initial time $t = 0$, both particles are at the origin and have the same speed: c. This means that they are actually one and the same: indistinguishable. Later on, in quantum chemistry, we will discuss this more.

4.2 Systems and Their Time

4.2.1 Inertial Reference Frame

Both particles are measured from our lab. Thus, our lab makes a reference frame, to which our particles can refer. For this, our lab must have its own (Cartesian) coordinates: x, y, and z. We often assume that, at the initial time $t = 0$, both particles are at the origin: (0, 0, 0). From there, they start moving along the x-axis.

Because there is no external force, the particle never changes speed or direction. This is an inertial motion.

What if the motion is not in the x-direction, but in any other fixed direction? In this case, redefine your lab coordinates more cleverly: pick your new x-axis to match the direction of motion. This way, you actually get a one-dimensional motion: the new y and z coordinates are immaterial and can be dropped. As a matter of fact, we "live" in the x-t plane only.

In its own (private reference frame), the particle marks yet another origin. It stays there all the time, at rest. This is its self-reference frame (or self-system).

The entire self-system travels at a constant speed and direction (carrying the particle with it). This is why it is indeed inertial. Later on, we will see that it also has its own (private) time axis.

4.2.2 How to Measure Time?

In our lab, how to measure the time t? Well, a time unit is often measured in terms of length. For example, in 1 s, the long hand of the clock makes an angle of 6°: 1/60 of a complete circle.

This is a grandpa clock, with a circular scale. In special relativity, on the other hand, better use a linear clock. Do not use this at home: this is just a theoretical clock: a light ray. In each second, it advances one more light second. This tells us that one more second has passed.

Why is this a good clock? Because it scales the time by c. Instead of the original time variable t, measured in seconds, we now look at the length variable ct, measured in light seconds. After all, while t increases by 1 s, ct advances by one light second: 300,000 kilometer.

This way, instead of the original t-axis, we now use the new ct-axis. Why is this better? Because ct is a length variable, just like x. These axes are now more symmetric and have the same status. Together, they make the new x-ct plane.

4.2.3 The Self-system

The original coordinates x, y, and z tell us the position in the lab. Similarly, t tells us the time, as measured in the lab. Later on, we will see that these measurements are not absolute, but only relative: they make sense only in the lab, but not in other (inertial) reference frames, traveling away from the lab.

For example, the first particle has its own self-system, traveling with it at the same direction and the same speed: v. This system has new coordinates, with a prime on top: x' and t'. Do not get confused: this prime has nothing to do with differentiation. In fact, t' just tells us how much time passed since the initial time $t' = 0$ (measured in a tiny clock that the particle carries in its "pocket"). Likewise, x' tells us the distance from the origin $x' = 0$ (measured in a tiny ruler that the particle carries in its pocket). If we are to the right of the particle, then $x' > 0$. If, on the other hand, we are to the left of the particle, then $x' < 0$. The particle itself is at $x' = 0$, all the time. This way, in its self-system, the particle is static. As before, $y' \equiv y$ and $z' \equiv z$ can be dropped.

In the self-system, how to measure time? As before, better use not t' but ct'.

4.2.4 Synchronization

Assume that the systems are synchronized: initially, at $t' = t = 0$, the particle lies at the origin in both systems:

$$\begin{pmatrix} x \\ y \\ z \\ ct \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \\ ct' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

This is the initial event in spacetime. At any later time, on the other hand, the systems are no longer the same: they are different, in terms of both position and time: x' differs from x, and t' differs from t. Time is indeed relativistic: it depends on the system where it is measured.

4.3 Lorentz Group of Transformations (Matrices)

4.3.1 Space and Time: Same Status

In terms of the self-system, the particle is static. Indeed, its position is always at

$$x' = 0.$$

In terms of the lab, on the other hand, the particle is dynamic: it moves at speed v. Therefore, at time t, it is at

$$x = \frac{dx}{dt}t = vt = \frac{v}{c}ct = \beta_v ct,$$

where β_v is the scaled velocity, obtained from differentiation with respect to the scaled time ct:

$$\beta_v \equiv \frac{v}{c} \equiv \frac{dx}{d(ct)}.$$

Consider now a more general point at a fixed distance x' from the particle. If $x' > 0$, then this is to the right of the particle. If, on the other hand, $x' < 0$, then this is to the left of the particle. In either case, this remains static in the self-system: x' remains the same.

In the lab, on the other hand, the above point is dynamic: it moves at speed v. At the initial time of $t = t' = 0$, it starts from $x = x'$. At time $t > 0$, on the other hand, it advances to

$$x = x' + vt = x' + \beta_v ct.$$

In other words,

$$x' = x - \beta_v ct.$$

Still, this is not the whole story: x' and t' are inseparable. Together, they make a new pair, with the same speed of light: c. After all, the speed of light is a true physical quantity, independent of the coordinates that happen to be used.

In this new pair, x' and t' will indeed have the same status. This way, they will mirror x and t well. For this purpose, they should transform together, symmetrically.

4.3.2 Lorentz Transformation

How to transform the original x-ct lab coordinates to the new x'-ct' self-coordinates? This transformation must be insensitive to interchanging x and ct. After all, both measure distance: x measures the distance from the lab's origin, and ct measures the distance made by the light ray in the linear clock in the lab (Sect. 4.2.2).

Once the dummy coordinates $y' \equiv y$ and $z' \equiv z$ are dropped, life gets much easier: we get the new Lorentz transformation that transforms x and ct into x' and ct':

$$\begin{pmatrix} x \\ ct \end{pmatrix} \to \begin{pmatrix} x' \\ ct' \end{pmatrix} \equiv \gamma\,(\beta_v) \begin{pmatrix} 1 & -\beta_v \\ -\beta_v & 1 \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix},$$

where

$$\gamma \equiv \gamma\,(\beta_v) \equiv \frac{1}{\sqrt{1-\beta_v^2}} \geq 1$$

is picked to make sure that the determinant is 1.

This is the Lorentz transformation: it gives the self-coordinates in terms of the lab coordinates. This is carried out by a symmetric 2×2 matrix: the Lorentz matrix. This way, time and space indeed have equal status, as required.

4.3.3 Lorentz Matrix and the Infinity Point

The Lorentz transformation preserves area: thanks to the coefficient γ, the Lorentz matrix has determinant 1.

What happens when $|v| = c$? In this case, γ is no longer a number: it is the infinity point ∞. Still, it is assumed that

$$0 \cdot \infty = 0.$$

We will come back to this later.

Let us look at another extreme case: $v = 0$. In this case, Lorentz matrix is just the 2×2 identity matrix:

$$I \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

This makes sense: because $v = 0$, the self-system is the same as the lab, so the Lorentz transformation does nothing. This looks silly, but is not: it will help rewrite

the Lorentz matrix better. For this purpose, define the new 2×2 matrix

$$J \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Thanks to this matrix, the Lorentz matrix can now be written as

$$\gamma\left(\beta_v\right)\left(I - \beta_v J\right).$$

Let us go ahead and use this.

4.3.4 Interchanging Coordinates

Fortunately, J commutes with the Lorentz matrix:

$$J\gamma\left(\beta_v\right)\left(I - \beta_v J\right) = \gamma\left(\beta_v\right)\left(I - \beta_v J\right) J.$$

Therefore, the Lorentz transformation is insensitive to interchanging x and ct:

$$\begin{aligned}
\begin{pmatrix} ct \\ x \end{pmatrix} &= J \begin{pmatrix} x \\ ct \end{pmatrix} \\
&\rightarrow \gamma\left(\beta_v\right)\left(I - \beta_v J\right) J \begin{pmatrix} x \\ ct \end{pmatrix} \\
&= J\gamma\left(\beta_v\right)\left(I - \beta_v J\right) \begin{pmatrix} x \\ ct \end{pmatrix} \\
&= J \begin{pmatrix} x' \\ ct' \end{pmatrix} \\
&= \begin{pmatrix} ct' \\ x' \end{pmatrix}.
\end{aligned}$$

This makes sense: time and space have equal status.

4.3.5 Composite Transformation

Our new matrix J helps write the Lorentz matrix in a simple way: a linear combination of I and J. This leads to an important property: every two Lorentz matrices commute with each other. Thanks to this, we are now ready to prove the rule of adding velocities (Sect. 4.1.1).

To see this, let us design a composite Lorentz transformation. What is this physically? Let us look at our first particle, traveling rightward at speed v. The second particle, on the other hand, travels leftward at speed u, or rightward at speed $-u$. These velocities are with respect to our lab, which is assumed to be at rest.

Now, look at things the other way around: from the perspective of the second particle, the entire lab travels rightward at speed u. This way, the second particle is now assumed to be at rest. How does the first particle travel away from it?

To calculate this, we need to compose two motions: the first particle travels with respect to the lab, which travels away from the second particle. How to represent this algebraically? Easy: multiply two Lorentz matrices. Since

$$J^2 = I,$$

the product is

$$\begin{aligned}
\gamma(\beta_u)(I-\beta_u J)\gamma(\beta_v)(I-\beta_v J) &= \gamma(\beta_u)\gamma(\beta_v)(I-\beta_u J)(I-\beta_v J)\\
&= \gamma(\beta_u)\gamma(\beta_v)\left(I-\beta_u J-\beta_v J+\beta_u\beta_v J^2\right)\\
&= \gamma(\beta_u)\gamma(\beta_v)\left((1+\beta_u\beta_v)I-(\beta_u+\beta_v)J\right)\\
&= \gamma(\beta_u)\gamma(\beta_v)(1+\beta_u\beta_v)\left(I-\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}J\right)\\
&= \gamma\left(\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}\right)\left(I-\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}J\right)
\end{aligned}$$

(because the product must have determinant 1 as well). This composition describes the total motion of the first particle away from the second one. The total velocity is, therefore, not $u+v$ but

$$c\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}=\frac{u+v}{1+\frac{uv}{c^2}},$$

as illustrated in Fig. 4.2.

4.3.6 The Inverse Transformation

Let us look at a special case: $u=-v$. What is this physically? The particles coincide! In this case, the above composition takes the form of

$$\begin{aligned}
\gamma(\beta_{-v})(I-\beta_{-v}J)\gamma(\beta_v)(I-\beta_v J) &= \gamma\left(\frac{\beta_{-v}+\beta_v}{1+\beta_{-v}\beta_v}\right)\left(I-\frac{\beta_{-v}+\beta_v}{1+\beta_{-v}\beta_v}J\right)\\
&= \gamma(0)I\\
&= I.
\end{aligned}$$

This gives us the inverse Lorentz matrix:

$$\gamma\left(\beta_{-v}\right)\left(I-\beta_{-v}J\right)=\gamma\left(\beta_{v}\right)\left(I+\beta_{v}J\right).$$

What is this physically? This is a legitimate Lorentz matrix (of determinant 1), which transforms back to the lab. Indeed, with respect to the particles (which are now one and the same), the entire lab travels leftward at speed v (or rightward at speed $-v$). To transform back to the lab, the inverse matrix works the other way around: it views the particles as static, and the entire lab as traveling at speed $-v$ away from them. This is why the inverse matrix picks a minus sign and uses $-v$ instead of v.

4.3.7 Abelian Group of Lorentz Matrices

The inverse matrix could also be obtained from Cramer's formula (Chap. 2, Sect. 2.1.4):

$$\begin{pmatrix}1 & -\beta_{v}\\ -\beta_{v} & 1\end{pmatrix}^{-1}=\frac{1}{1-\beta_{v}^{2}}\begin{pmatrix}1 & \beta_{v}\\ \beta_{v} & 1\end{pmatrix}=\gamma^{2}\left(\beta_{v}\right)\begin{pmatrix}1 & \beta_{v}\\ \beta_{v} & 1\end{pmatrix}.$$

Upon dividing this by $\gamma(\beta_v)$, you get the same inverse matrix as before. In summary, the Lorentz matrices make an Abelian (commutative) group, mirrored by the open interval $(-1, 1)$:

$$\gamma\left(\beta_{v}\right)\left(I-\beta_{v}J\right)\leftrightarrow\beta_{v}\in(-1,1).$$

4.4 Proper Time in the Self-system

4.4.1 Proper Time: Invariant

In its self-system, the particle is static: it is always at $x' = 0$. In its "pocket," it carries a tiny clock. What time does it show in the self-system? This is the proper time of the particle:

$$s\equiv t'.$$

Why do we give it the new name s? To emphasize that it is invariant: it can be calculated not only from the self-system but also from any other (inertial) system, say our lab.

In the x'-t' self-system, the tiny clock is at

$$(x', t') = (0, s).$$

This makes the matrix

$$\begin{pmatrix} ct' & x' \\ x' & ct' \end{pmatrix} = \begin{pmatrix} cs & 0 \\ 0 & cs \end{pmatrix} = csI.$$

This matrix has determinant c^2s^2. To transform it back to the lab, let us apply the inverse Lorentz matrix to it. Let us do this column by column: start with the second column. The inverse Lorentz matrix transforms it back to the x-t lab coordinates:

$$\begin{pmatrix} x' \\ ct' \end{pmatrix} \to \gamma(\beta_v)(I + \beta_v J)\begin{pmatrix} x' \\ ct' \end{pmatrix} = \begin{pmatrix} x \\ ct \end{pmatrix}.$$

Fortunately, the inverse Lorentz matrix commutes with J (Sect. 4.3.4). Therefore, the first column transforms in the same way:

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = J\begin{pmatrix} x' \\ ct' \end{pmatrix} \to J\gamma(\beta_v)(I + \beta_v J)\begin{pmatrix} x' \\ ct' \end{pmatrix} = J\begin{pmatrix} x \\ ct \end{pmatrix} = \begin{pmatrix} ct \\ x \end{pmatrix}.$$

In summary, the entire matrix transforms to

$$\begin{pmatrix} ct & x \\ x & ct \end{pmatrix} = \gamma(\beta_v)(I + \beta_v J)\begin{pmatrix} ct' & x' \\ x' & ct' \end{pmatrix} = cs\gamma(\beta_v)(I + \beta_v J).$$

On both sides, calculate the determinant:

$$c^2t^2 - x^2 = c^2s^2.$$

4.4.2 Time Dilation

This is why s (the proper time of the particle) is invariant: it can be calculated not only from the self-system but also from the lab:

$$s \equiv \sqrt{t^2 - \frac{x^2}{c^2}}.$$

In these new terms, what is x? This is the location of the particle in the lab. On the other hand, what is t? This is the proper time of the lab: the time shown in a static clock in the lab. After all, this is how t was defined in the first place. Still, t has yet another meaning: the maximal proper time of any particle. Indeed, at any time t, the particle would be at $x = vt$, so its proper time would be

$$s \equiv \sqrt{t^2 - \frac{x^2}{c^2}} = \sqrt{t^2 - \beta_v^2 t^2} = \frac{t}{\gamma\left(\beta_v\right)} \leq t.$$

This is time dilation. If you want to see a quick time (many seconds passing quickly), then better look at your own (static) watch. If you looked at someone else's watch (moving away from you), then you would get a "penalty:" you would see a slower time, telling you that only a few seconds have passed. This is time dilation.

4.4.3 Length Contraction

In its self-system, the particle is always at $x' = 0$. In the lab, on the other hand, it is at x. How to find this point? Start from 0 in the lab, measure x rightward, and you arrive at x.

Thus, the location has no meaning on its own, but only relative to a reference point: 0. Only the distance between two locations is meaningful.

Consider, for example, a horizontal stick that travels at speed v with respect to our lab. In its self-system, the stick is at rest: one endpoint at x_1', and the other at x_2'. Thus, in its own system, its length is

$$\triangle x' \equiv x_2' - x_1'.$$

How does it look like from the lab? To see this, let us use the Lorentz transformation:

$$\begin{pmatrix} x_1' \\ ct_1' \end{pmatrix} = \gamma\left(\beta_v\right) \begin{pmatrix} 1 & -\beta_v \\ -\beta_v & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ ct_1 \end{pmatrix},$$

and

$$\begin{pmatrix} x_2' \\ ct_2' \end{pmatrix} = \gamma\left(\beta_v\right) \begin{pmatrix} 1 & -\beta_v \\ -\beta_v & 1 \end{pmatrix} \begin{pmatrix} x_2 \\ ct_2 \end{pmatrix}.$$

Now, subtract the former equation from the latter:

$$\begin{pmatrix} \triangle x' \\ c\triangle t' \end{pmatrix} = \gamma\left(\beta_v\right) \begin{pmatrix} 1 & -\beta_v \\ -\beta_v & 1 \end{pmatrix} \begin{pmatrix} \triangle x \\ c\triangle t \end{pmatrix}.$$

Now, assume that I sit in the lab. How could I measure the length of the moving stick? Unfortunately, I have no access to the self-system. Instead, I must use the x-t lab coordinates: at the same time $t_1 = t_2$, I need to look at the endpoints x_1 and x_2:

$$\begin{pmatrix} \triangle x' \\ c\triangle t' \end{pmatrix} = \gamma\left(\beta_v\right) \begin{pmatrix} 1 & -\beta_v \\ -\beta_v & 1 \end{pmatrix} \begin{pmatrix} \triangle x \\ 0 \end{pmatrix}$$
$$= \gamma\left(\beta_v\right) \begin{pmatrix} \triangle x \\ -\beta_v \triangle x \end{pmatrix}.$$

There are two scalar equations here. We only need the top one:

$$\triangle x' = \gamma\left(\beta_v\right) \triangle x,$$

or

$$\triangle x = \frac{\triangle x'}{\gamma\left(\beta_v\right)}.$$

Since $\gamma \geq 1$, $|\triangle x| \leq |\triangle x'|$. This is length contraction. From its own self-system, the stick looks longer than from any other system (including the lab). This is good for balance: it gives a counter-effect to time dilation (Chapter 11 in [76, 77]). Moreover, the above length $\triangle x$ (observed from the lab) decreases monotonically as $|v|$ increases. In the extreme case of $|v| = c$ and $\gamma = \infty$, for example, $\triangle x = 0$. This means that, from the lab, the stick seems like one point, passing as fast as light.

We have already seen this in Sect. 4.1.4. In fact, our stick passes so fast that its endpoints coincide: they shrink to just one point. This is also why, in such a fast stick, no change could ever be noticed. Its proper time gets stuck and does not tick at all!

4.4.4 Simultaneous Events

In the lab, both endpoints of the stick were measured at the same time: $t_1 = t_2$. These are simultaneous events. Still, in the x-t plane, they are not identical. After all, they take place in different locations: $x_1 \neq x_2$.

In the self-system, on the other hand, these events are no longer simultaneous. Indeed, in the equation in Sect. 4.4.3, look now at the bottom:

$$c\triangle t' = -\gamma\left(\beta_v\right)\beta_v \triangle x = -\beta_v \triangle x' \neq 0.$$

Thus, the events are simultaneous in the lab only. In every other (inertial) system, on the other hand, they are not.

4.5 Spacetime and Velocity

4.5.1 Doppler's Effect

In the above, the particle travels at speed v with respect to the lab. In its "pocket," it carries a tiny clock, to show its proper time. We, in the lab, can also read what this clock shows, with some time dilation. This is only theoretical: in practice, this information needs time to arrive.

In its own self-system, the tiny clock shows time t'. In the lab, on the other hand, this time transforms back to t. At time $t_1 > 0$ in the lab, for instance, the particle gets as far as $x_1 = vt_1$. At this time, a signal (as fast as light) issues from the particle, to carry the news all the way back. To arrive, it needs some more time: $x_1/c = vt_1/c$. (Here, we assume for simplicity that $v > 0$, as in Fig. 4.3.) Denote the arrival time by T_1. Later on, at time $t_2 > t_1$, the next signal will issue as well, to arrive at $T_2 > T_1$.

How to write $T_2 - T_1$ in terms of $t_2' - t_1'$? Thanks to time dilation (Sect. 4.4.2),

$$\begin{aligned}
\triangle T &\equiv T_2 - T_1 \\
&= t_2 + \frac{x_2}{c} - \left(t_1 + \frac{x_1}{c}\right) \\
&= t_2 + \frac{vt_2}{c} - \left(t_1 + \frac{vt_1}{c}\right) \\
&= t_2 - t_1 + \frac{v}{c}\,(t_2 - t_1) \\
&= (\triangle t)\,(1 + \beta_v) \\
&= (\triangle t')\gamma\,(\beta_v)\,(1 + \beta_v) \\
&= (\triangle t')\frac{1 + \beta_v}{\sqrt{1 - \beta_v^2}} \\
&= (\triangle t')\frac{1 + \beta_v}{\sqrt{1 - \beta_v}\sqrt{1 + \beta_v}}
\end{aligned}$$

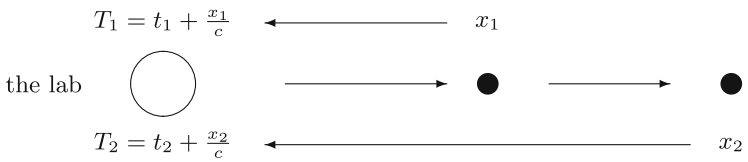


Fig. 4.3 The view from the lab: at time $t_1 > 0$, the particle arrives at $x_1 = vt_1$, and a signal issues back to the lab, to arrive at $T_1 = t_1 + x_1/c$. Later, at $t_2 > t_1$, the particle arrives at $x_2 = vt_2$, and another signal issues back to the lab, to arrive at $T_2 = t_2 + x_2/c$

$$= (\triangle t')\sqrt{\frac{1+\beta_v}{1-\beta_v}}$$
$$> \triangle t'$$

(since we assumed that $v > 0$). What is the practical effect of this? Well, in its pocket, the particle also carries a tiny camera, to make a movie. We, in the lab, will watch the movie in slow motion. Indeed, an original activity that takes $\triangle t'$ seconds will seem to us to take as many as $\triangle T$ seconds. This is Doppler's effect.

4.5.2 Velocity in Spacetime

How do things look like from the perspective of the second particle, traveling at speed $-u$ with respect to the lab? To describe its self-system, let us use now the x-t coordinates (with no prime). This will be our new spacetime, assumed to be at rest. With respect to it, the entire lab travels at speed u rightward. This is why the lab will now use the x'-t' coordinates (with a prime on top).

In the x'-t' lab coordinates, how does the first particle look like? It travels at speed v:

$$\frac{dx'}{dt'} = v,$$

or

$$\frac{dx'}{cdt'} = \frac{v}{c} = \beta_v.$$

This could also be written in terms of a two-dimensional column vector:

$$\begin{pmatrix} dx' \\ cdt' \end{pmatrix} = \begin{pmatrix} \beta_v \\ 1 \end{pmatrix}.$$

Indeed, just divide the top component by the bottom one, and get the same thing as before. (This is as in projective geometry in Chap. 6.)

But how do things look like from our new spacetime? To see this, we need to transform the above vector to our new spacetime: the x-t self-system of the second particle. For this purpose, apply an inverse Lorentz matrix:

$$\begin{pmatrix} dx \\ cdt \end{pmatrix} = \gamma\,(\beta_u) \begin{pmatrix} 1 & \beta_u \\ \beta_u & 1 \end{pmatrix} \begin{pmatrix} dx' \\ cdt' \end{pmatrix}$$

$$= \gamma(\beta_u) \begin{pmatrix} 1 & \beta_u \\ \beta_u & 1 \end{pmatrix} \begin{pmatrix} \beta_v \\ 1 \end{pmatrix}$$

$$= \gamma(\beta_u) \begin{pmatrix} \beta_v + \beta_u \\ \beta_u \beta_v + 1 \end{pmatrix}.$$

Again, divide the top component by the bottom one:

$$\frac{dx}{cdt} = \frac{\beta_v + \beta_u}{\beta_u \beta_v + 1},$$

or

$$\frac{dx}{dt} = c\frac{\beta_v + \beta_u}{\beta_u \beta_v + 1}, = \frac{u + v}{1 + \beta_u \beta_v}.$$

Isn't this familiar? This is our rule of adding velocities (Sect. 4.1.1). This is indeed how fast the first particle gets away from the second one.

4.5.3 Moebius Transformation

The Lorentz transformation transforms two-dimensional vector to two-dimensional vector. The Moebius transformation, on the other hand, works more directly: it transforms ratio to ratio: the old speed (of the first particle in the lab) to the new speed (as observed from the second particle):

$$\beta_v = \frac{dx'}{cdt'} \rightarrow \frac{dx}{cdt} = \frac{\beta_u + \beta_v}{1 + \beta_u \beta_v}.$$

Let us use the same approach to calculate the perpendicular velocity as well.

4.5.4 Perpendicular Velocity

So far, we modeled a one-dimensional motion only. How to model a two-dimensional motion, not only in the x' but also in the y'-direction? For this purpose, our lab system still uses primes: the x'-y'-t' system. In it, the first particle travels now at the new velocity $(v_{x'}, v_{y'})$ obliquely: $v_{x'}$ in the x'-direction, and $v_{y'}$ in the y'-direction (Fig. 4.4). (Do not get confused: these are not partial derivatives, but just velocity components.) The second particle, on the other hand, still travels at velocity $(-u, 0)$, in the x'-direction only. This is how things look like from the lab.

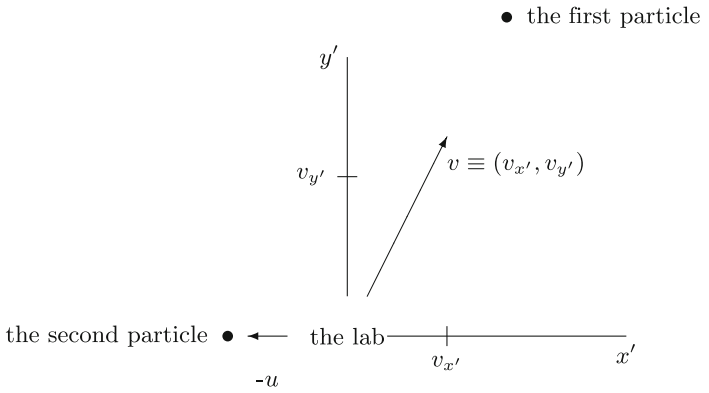


Fig. 4.4 The view from the lab: the first particle travels at velocity $(v_{x'}, v_{y'})$, and the second at velocity $(-u, 0)$

How do things look like from the second particle? This is now our new spacetime: the x-y-t self-system of the second particle (with no primes). To observe things from there, let us use the same approach as before, with a few changes.

In its own self-system, the second particle is at rest, and the entire lab travels at velocity $(u, 0)$ rightward. In these new terms, how does the first particle look like?

To see this, we must transform back to the self-system of the second particle. How to do this algebraically? Apply an extended 3×3 Lorentz matrix, which leaves the second component unchanged:

$$\begin{pmatrix} x' \\ y' \\ ct' \end{pmatrix} \to \begin{pmatrix} x \\ y \\ ct \end{pmatrix} = \begin{pmatrix} \gamma(\beta_u) & & \\ & 1 & \\ & & \gamma(\beta_u) \end{pmatrix} \begin{pmatrix} 1 & & \beta_u \\ & 1 & \\ \beta_u & & 1 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ ct' \end{pmatrix}.$$

(As usual, blank spaces stand for zero matrix elements.) Let us use this matrix to calculate the velocities:

$$\begin{aligned} \begin{pmatrix} dx \\ dy \\ cdt \end{pmatrix} &= \begin{pmatrix} \gamma(\beta_u) & & \\ & 1 & \\ & & \gamma(\beta_u) \end{pmatrix} \begin{pmatrix} 1 & & \beta_u \\ & 1 & \\ \beta_u & & 1 \end{pmatrix} \begin{pmatrix} dx' \\ dy' \\ cdt' \end{pmatrix} \\ &= \begin{pmatrix} \gamma(\beta_u) & & \\ & 1 & \\ & & \gamma(\beta_u) \end{pmatrix} \begin{pmatrix} 1 & & \beta_u \\ & 1 & \\ \beta_u & & 1 \end{pmatrix} \begin{pmatrix} \beta_{v_{x'}} \\ \beta_{v_{y'}} \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} \gamma(\beta_u) & & \\ & 1 & \\ & & \gamma(\beta_u) \end{pmatrix} \begin{pmatrix} \beta_{v_{x'}} + \beta_u \\ \beta_{v_{y'}} \\ \beta_u \beta_{v_{x'}} + 1 \end{pmatrix}. \end{aligned}$$

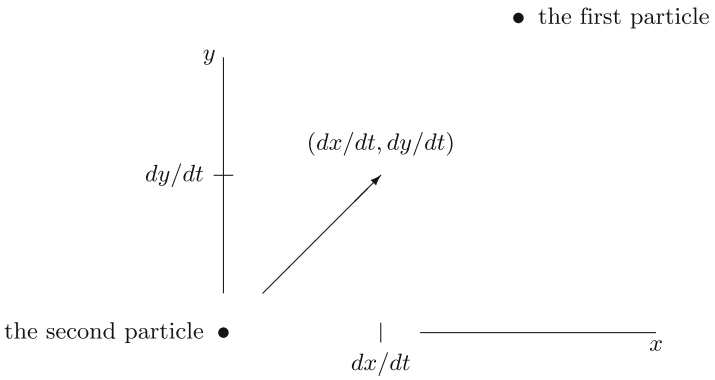


Fig. 4.5 The view from the second particle: the first particle gets away at a new velocity: $(dx/dt, dy/dt)$ in the x-y-t system

We are now ready to divide by dt, to obtain the new velocities dx/dt and dy/dt, as observed from the second particle. (This way, we actually eliminate the x'-y'-t' lab system and drop it completely.) As observed from the second particle, the x-velocity of the first particle is still

$$\frac{dx}{dt} = c\frac{\beta_u + \beta_{v_{x'}}}{1 + \beta_u \beta_{v_{x'}}} = \frac{u + v_{x'}}{1 + \beta_u \beta_{v_{x'}}}.$$

This is no surprise: it is as in our rule of adding velocities (Sects. 4.5.2–4.5.3). The y-velocity, on the other hand, is now

$$\frac{dy}{dt} = c\frac{\beta_{v_{y'}}}{\gamma\left(\beta_u\right)\left(1 + \beta_u \beta_{v_{x'}}\right)} = \frac{v_{y'}}{\gamma\left(\beta_u\right)\left(1 + \beta_u \beta_{v_{x'}}\right)}.$$

In magnitude, this is less than dy'/dt'. What does this mean physically? As observed from the second particle, the first particle follows a new arrow, with a more moderate slope (Fig. 4.5). To draw this arrow, we now have all we need: dx/dt and dy/dt are now both available in terms of three known parameters: u, $v_{x'}$, and $v_{y'}$.

4.6 Relativistic Momentum and its Conservation

4.6.1 Invariant Mass

Consider now a new particle, traveling rightward at a new speed: not v, but u (with respect to the lab). Assume that it has mass m. This mass is not relativistic, but invariant: once defined in one fixed system (the default is the self-system), it could

also be measured (or read, or calculated) in any other system and always give the same result: m. Still, to obtain m accurately, the act of measurement must not affect the velocity. In the lab, for example, the mass must be measured, while the particle is at motion, at the constant speed u.

4.6.2 Momentum: Old Definition

Initially, our particle travels calmly at speed u (Fig. 4.6). Suddenly, it explodes into two equal subparticles (without supplying any force or energy from the outside). How should they fly away (with respect to the original particle)? Well, thanks to symmetry, they fly sideways: one flies rightward (at the extra speed v), and the other leftward (at the extra speed v too).

In Newtonian mechanics, the momentum is defined as mass times velocity:

$$p \equiv mu$$

(Chap. 2, Sect. 2.4.1). This is the initial linear momentum in the horizontal x-direction. After the explosion, the momentum is still the same:

$$\frac{m}{2}(u+v)+\frac{m}{2}(u-v)=\frac{m}{2}(u+v+u-v)=\frac{m}{2}(u+u)=mu.$$

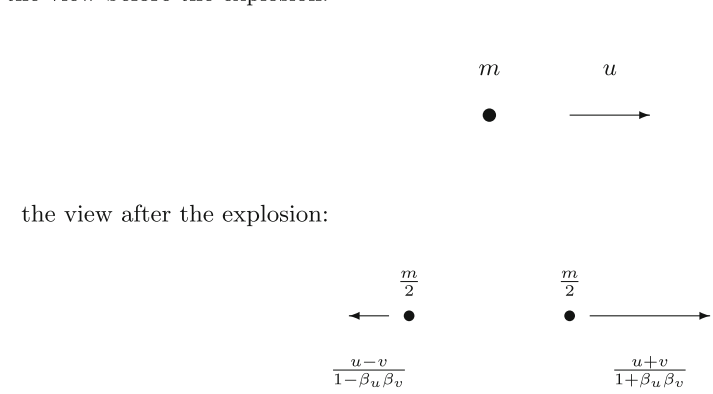


Fig. 4.6 After the original particle (top picture) explodes into two new subparticles (bottom picture), the total momentum is still $mu\gamma(\beta_u)$

4.6.3 Relativistic Momentum

Thanks to special relativity, however, we already know better: this is *not* the correct way to add velocities (Sect. 4.1.1). To fix this, let us redefine momentum in a more accurate way.

In the lab, the initial velocity is

$$u \equiv \frac{dx}{dt}.$$

To have a relativistic momentum, however, better differentiate with respect to the invariant time s. This helps define a new relativistic momentum p:

$$p \equiv m\frac{dx}{ds} = m\frac{dx}{dt}\cdot\frac{dt}{ds} = mu\gamma\left(\beta_u\right).$$

Thus, the relativistic momentum is a triple product: mass times velocity times γ. This defines p not absolutely but only relative to the lab. In the rest of this chapter, when talking about momentum, we mean relativistic momentum.

4.6.4 Rest Mass vs. Relativistic Mass

By now, we have used two invariant quantities: the proper time s and the mass m. m is also called the rest mass. Why? Because the default is to calculate it in the self-system, where the particle is at rest. Still, better call it just mass. As discussed above, the mass m is invariant: once defined in one system, it could also be measured from any other (inertial) system and give the same result: m.

To define the relativistic momentum, however, we used not m but the product

$$m\gamma\left(\beta_u\right).$$

This is also called relativistic mass. It is the product of two factors: mass times γ. To obtain the relativistic momentum, multiply also by u. This mirrors Newtonian mechanics. The relativistic mass plays the same role as the inertial mass in classical mechanics: it resists any force that may be applied from the outside in an attempt to change the momentum without changing m.

Still, if sufficient energy is available to overcome this resistance and apply such a force, then u could get very high, and even close to the speed of light (without changing m). This way, the relativistic mass gets high as well and resists the force even more. In this sense, the particle gets "heavier:" not in terms of mass, but only in terms of relativistic mass. This is why a particle of a positive mass $m > 0$ could never be as fast as light: its relativistic mass would get too big:

$$m\gamma\,(\beta_u) \to_{u\to c} \infty.$$

It would resist the outer force ever so strongly and win!

Still, rest mass is more elementary than relativistic mass. In the rest of this chapter, when talking about mass, we mean rest mass, not relativistic mass.

4.6.5 Moderate (Nonrelativistic) Velocity

What happens for a moderate (or nonrelativistic) velocity $|u| \ll c$? In this case, $\beta_u \ll 1$ and $\gamma(\beta_u) \sim 1$, so the relativistic momentum is nearly the same as the old Newtonian momentum:

$$mu\gamma(\beta_u) \sim mu.$$

For a large $|u|$, on the other hand, the new definition is a substantial improvement.

4.6.6 Closed System: Lose Mass—Gain Motion

Indeed, to make the explosion, some energy is needed, which must come from somewhere. Now, our system is closed and isolated: no force or energy could come from the outside. Where could the extra energy come from? It must come from mass: at the explosion, the subparticles must lose some of their original mass. Indeed, before the explosion, each (inner) subparticle had mass $m/2$. After the explosion, on the other hand, each subparticle has mass of only

$$\frac{m}{2\gamma\,(\beta_v)} < \frac{m}{2\gamma\,(0)} = \frac{m}{2}.$$

(We will come back to this later.) What is the physical meaning of this? Before the explosion, the subparticle was still inside the original particle and had velocity $v = 0$ with respect to the particle. This is why each subparticle had mass $m/2$. During the explosion, each subparticle lost a little bit of mass. Still, this served a good purpose: to supply the extra energy required to ignite the explosion.

4.6.7 The Momentum Matrix

Let us define the (so-called) momentum matrix: mass times Lorentz matrix. Later on, we will look at one matrix element: the relativistic momentum.

For this purpose, look at things from the lab, as in Fig. 4.6. For each subparticle, compose two Lorentz matrices, as in Sect. 4.3.5:

$$\begin{aligned}
&\text{mass} \cdot \text{ Lorentz matrix} \\
&= \frac{m}{2\gamma\left(\beta_v\right)}\gamma\left(\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}\right)\left(I+\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}J\right)+\frac{m}{2\gamma\left(\beta_v\right)}\gamma\left(\frac{\beta_u-\beta_v}{1-\beta_u\beta_v}\right)\left(I+\frac{\beta_u-\beta_v}{1-\beta_u\beta_v}J\right) \\
&= \frac{m}{2\gamma\left(\beta_v\right)}\gamma\left(\beta_v\right)\left(I+\beta_v J\right)\gamma\left(\beta_u\right)\left(I+\beta_u J\right)+\frac{m}{2\gamma\left(\beta_v\right)}\gamma\left(\beta_v\right)\left(I-\beta_v J\right)\gamma\left(\beta_u\right)\left(I+\beta_u J\right) \\
&= \frac{m}{2}\left(I+\beta_v J+I-\beta_v J\right)\gamma\left(\beta_u\right)\left(I+\beta_u J\right) \\
&= m\gamma\left(\beta_u\right)\left(I+\beta_u J\right).
\end{aligned}$$

Thus, after the explosion, the momentum matrix remains the same (with respect to the lab). This is indeed conservation of momentum, in its matrix form.

4.6.8 Momentum and its Conservation

The above is a matrix equation: it actually contains four scalar equations. Let us look at one of them, say the upper-right one:

$$\frac{m}{2\gamma\left(\beta_v\right)}\gamma\left(\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}\right)\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}+\frac{m}{2\gamma\left(\beta_v\right)}\gamma\left(\frac{\beta_u-\beta_v}{1-\beta_u\beta_v}\right)\frac{\beta_u-\beta_v}{1-\beta_u\beta_v}=m\gamma\left(\beta_u\right)\beta_u.$$

Better yet, multiply this by c:

$$\frac{m}{2\gamma\left(\beta_v\right)}\frac{u+v}{1+\beta_u\beta_v}\gamma\left(\frac{\beta_u+\beta_v}{1+\beta_u\beta_v}\right)+\frac{m}{2\gamma\left(\beta_v\right)}\frac{u-v}{1-\beta_u\beta_v}\gamma\left(\frac{\beta_u-\beta_v}{1-\beta_u\beta_v}\right)=mu\gamma\left(\beta_u\right).$$

This is indeed conservation of momentum: after the explosion, with respect to the lab, the new momentum (new mass times new velocity times γ) sums to the same as the original momentum (mass times velocity times γ in the beginning): $mu\gamma(\beta_u)$.

4.7 Relativistic Energy and its Conservation

4.7.1 Force: Derivative of Momentum

So far, we defined the relativistic momentum and saw that it is indeed conserved. What about relativistic energy? To define it, we need to differentiate γ with respect to β_v:

$$\begin{aligned}\gamma'(\beta_v) &\equiv \left(\left(1-\beta_v^2\right)^{-1/2}\right)' \\ &= -\frac{1}{2}\left(1-\beta_v^2\right)^{-3/2}(-2\beta_v) \\ &= \beta_v\left(1-\beta_v^2\right)^{-3/2}.\end{aligned}$$

Furthermore, let us differentiate $\gamma(\beta_v)$ as a composite function of v:

$$\begin{aligned}\frac{d}{dv}\gamma(\beta_v) &= \gamma'(\beta_v)\frac{d\beta_v}{dv} \\ &= \frac{1}{c}\gamma'(\beta_v) \\ &= \frac{1}{c}\beta_v\left(1-\beta_v^2\right)^{-3/2}.\end{aligned}$$

Moreover, let us differentiate the product $v\gamma(\beta_v)$:

$$\begin{aligned}\frac{d}{dv}(v\gamma(\beta_v)) &= \gamma(\beta_v) + v\frac{d}{dv}\gamma(\beta_v) \\ &= \gamma(\beta_v) + \frac{v}{c}\beta_v\left(1-\beta_v^2\right)^{-3/2} \\ &= \gamma(\beta_v) + \beta_v^2\left(1-\beta_v^2\right)^{-3/2} \\ &= \gamma^{-2}(\beta_v)\gamma^3(\beta_v) + \beta_v^2\left(1-\beta_v^2\right)^{-3/2} \\ &= \left(1-\beta_v^2\right)\left(1-\beta_v^2\right)^{-3/2} + \beta_v^2\left(1-\beta_v^2\right)^{-3/2} \\ &= \left(1-\beta_v^2\right)^{-3/2}.\end{aligned}$$

We are now ready to define relativistic energy. This definition will be accurate not only for small but also for large velocities.

Consider a particle of mass m that is initially at rest in the x-t lab system. Then, an external force F is applied to it from time 0 until time $q > 0$, to increase both its momentum and energy (without changing m).

4.7.2 Open System: Constant Mass

This is indeed an open system: neither closed nor isolated any more. Thanks to the external force, energy can now increase without changing m. This way, the particle

does not have to lose any mass any more: it may remain with the same mass m all the time.

To calculate the force, we need to differentiate the relativistic momentum (Sect. 4.6.3) with respect to time. This will help define the relativistic energy:

$$\begin{aligned}
\int_{x(0)}^{x(q)} F(x)dx &= \int_0^q F(x(t))dx(t) \\
&= \int_0^q F(x(t))\frac{dx(t)}{dt}dt \\
&= \int_0^q F(x(t))v(t)dt \\
&= m\int_0^q \frac{d}{dt}\left(v\gamma\left(\beta_v\right)\right)v(t)dt \\
&= m\int_0^q \frac{d}{dv}\left(v\gamma\left(\beta_v\right)\right)\frac{dv}{dt}v(t)dt \\
&= m\int_{v(0)}^{v(q)} \frac{d}{dv}\left(v\gamma\left(\beta_v\right)\right)vdv \\
&= m\int_{v(0)}^{v(q)} \left(1-\beta_v^2\right)^{-3/2}vdv \\
&= mc^2\int_{v(0)}^{v(q)} \left(1-\beta_v^2\right)^{-3/2}\frac{1}{c}\beta_v dv \\
&= mc^2\left(\gamma\left(\beta_{v(q)}\right)-\gamma\left(\beta_{v(0)}\right)\right) \\
&= mc^2\left(\gamma\left(\beta_{v(q)}\right)-1\right).
\end{aligned}$$

4.7.3 Relativistic Energy: Kinetic Plus Potential

This is indeed the new kinetic energy that the external force introduced into the particle from time 0 until time q. The potential (nuclear) energy stored in the particle, on the other hand, is not relativistic, but invariant. This is the latter term, subtracted in the latter right-hand side:

$$E_{\text{potential}} \equiv E(0) \equiv mc^2.$$

With this, we now have the total energy (as a smooth function of v):

$$E(v) \equiv E_{\text{potential}} + E_{\text{kinetic}}(v) = mc^2 + mc^2\left(\gamma\left(\beta_v\right)-1\right) = mc^2\gamma\left(\beta_v\right).$$

This is indeed the new relativistic energy: kinetic plus potential. This is stored in the momentum matrix

$$m\gamma\left(\beta_v\right)\left(I+\beta_v J\right).$$

Indeed, just look at the lower-right entry, and multiply by c^2. In the rest of this chapter, when talking about energy, we mean relativistic energy.

4.7.4 Moderate (Nonrelativistic) Velocity

The new definition is a natural extension of the old one. Indeed, what happens at a moderate (or nonrelativistic) velocity $|v| \ll c$? In this case, $\beta_v^2 \ll 1$. Thanks to the Taylor expansion around 0,

$$\gamma\left(\beta_v\right)=\left(1-\beta_v^2\right)^{-1/2}\sim 1+\frac{1}{2}\beta_v^2.$$

Thus, in this case, the new definition agrees with the old one:

$$E(v)=mc^2\gamma\left(\beta_v\right)\sim mc^2\left(1+\frac{1}{2}\beta_v^2\right)=mc^2+\frac{1}{2}mv^2.$$

Are these terms familiar? The former is the nuclear energy, and the latter is the good old Newtonian kinetic energy.

4.8 Mass and Energy: Closed vs. Open System

4.8.1 Why Is It Called Rest Mass?

Assume now that the system is closed: the particle takes no external force any more. In the beginning, it is at rest: $v=0$. This is when its mass measures m. This is why it is called rest mass.

Still, could the particle start moving, and take a nonzero velocity $v\neq 0$? For this, it must have more kinetic energy. Where does it come from? From its original mass. Only a static particle ($v=0$) can keep its entire original (maximal) mass m. This is why m is also called rest mass.

4.8.2 Mass is Invariant

But mass is invariant: it could be measured not only from the self-system but also from any other (inertial) system and give the same result: m. Only in the self-system is the particle at rest. In the other systems, on the other hand, it is not. Still, these systems are as legitimate and could be used (in theory) to measure m in the first place (provided that the measurement is made, while the particle is at motion).

Of course, in practice, it is much easier to measure m in the self-system, where the particle is at rest. This is why m is called either mass or rest mass.

4.8.3 Energy is Conserved—Mass Is Not

In Sect. 4.7.3, we assumed an open system: an external force is applied to the particle, to increase both its momentum and energy (without changing m). This is why its mass remains m all the time. This was necessary to define the relativistic energy in the first place.

In Fig. 4.6, on the other hand, the explosion takes place in a closed (isolated) system: no external force is allowed. For this reason, not only the total momentum but also the total energy remains unchanged. But mass does not.

During the explosion, where do the subparticles get their extra kinetic energy from? From their original mass. In fact, to ignite the explosion, some mass is lost.

4.8.4 Particle Starting to Move

As a matter of fact, this is true not only for an exploding particle but also for any other particle that starts to move. Indeed, at the beginning, the particle accelerates: it changes velocity from 0 to $v \neq 0$ (while preserving its total energy). Its new kinetic energy must come from somewhere: from its potential (or nuclear) energy. As a price, its mass must fall from $m(0) = m$ to

$$m(v) \equiv \frac{m(0)}{\gamma\left(\beta_v\right)} < m(0) = m.$$

This way, its total energy remains the same as at rest:

$$E(v) = m(v)c^2\gamma\left(\beta_v\right) = \frac{m(0)}{\gamma\left(\beta_v\right)}c^2\gamma\left(\beta_v\right) = m(0)c^2 = E(0).$$

This is indeed conservation of energy in a closed system.

4.8.5 Say Mass, Not Rest Mass

This is why $m(0)$ is also called rest mass. This is the original mass measured in the lab (when it was a self-system, before the particle started to move). This is the system that was chosen to define $m \equiv m(0)$ in the first place. At any other (inertial) reference frame, on the other hand, the particle travels, and its mass must be measured, while it is at motion. This must give the same result: m as well: mass is invariant.

But be careful: in the other reference frame, this mass has nothing to do with rest. It is measured at motion, not at rest. For this reason, better say mass, not rest mass.

4.8.6 Decreasing Mass in the Lab

The lab is the system we are really interested in. Why? Because our physical process takes place in the lab. We are not interested in any fictitious motion due to transforming to another system, but only in a real motion, coming from converting potential energy into kinetic. Once the particle starts to move in our lab, the new mass $m(v)$ gets smaller:

$$m(v) = \frac{m(0)}{\gamma\left(\beta_v\right)} < m(0) = \frac{m(0)}{\gamma(0)} = m.$$

4.8.7 Closed System: Energy Can Only Convert

Like the explosion studied before, the above decrease from $m(0)$ to $m(v)$ takes place in an isolated system. This is essential: since no external force is welcome, the total energy must remain constant. The mass $m(v)$, on the other hand, decreases as $|v|$ increases. Thus, energy is never lost but only converts from potential to kinetic. In this process, the total energy remains the same:

$$E = E_{\text{potential}}(v) + E_{\text{kinetic}}(v) = \frac{m(0)}{\gamma\left(\beta_v\right)}c^2 + \frac{m(0)}{\gamma\left(\beta_v\right)}c^2\left(\gamma\left(\beta_v\right) - 1\right) = m(0)c^2.$$

This is Einstein's famous formula. Thanks to the big factor c^2, even a tiny mass can still produce a lot of kinetic energy.

4.8.8 Open System

This is quite different from the situation in Sect. 4.7.3, where the system was open, letting the external force introduce more kinetic energy, without changing the mass. This was necessary to define the kinetic energy obtained from the work done by this force.

4.8.9 Mass in a Closed System

Mass, on the other hand, may change even in a closed system. In a particle that starts to move, for example, its mass $m(v)$ decreases as $|v|$ increases. If $|v|$ gets as large as $|v| = c$, then we have $\gamma = \infty$, so the particle has no mass at all any more: all its potential energy has already been exploited and converted into kinetic energy.

Still, even with no mass at all, the particle still has nonzero momentum. This is why momentum and energy are more fundamental than mass: they are relativistic, and transform together, as a pair.

4.9 Momentum–Energy and Their Transformation

4.9.1 New Mass

In their new definitions (Sects. 4.6.3–4.7.3), both energy and momentum are relativistic: they depend on the velocity, which changes from system to system. In the lab, in particular, the particle travels rightward at speed v, so its mass is $m(0)/\gamma(\beta_v)$. Still, who cares about its history? Who cares whether it was initially at rest or not? Better redefine m as

$$m \leftarrow \frac{m(0)}{\gamma\,(\beta_v)}.$$

This is the new mass, measured, while the particle is at motion in the lab. This new mass is now used to calculate both energy and momentum in the lab.

4.9.2 Spacetime

To describe the lab, let us use again the x'-t' coordinates (as in Sect. 4.5.2). (This prime has nothing to do with differentiation—it just reminds us that both x' and t' are measured in the lab.) Thus, in the lab, the momentum of the particle is primed

as well:

$$p' \equiv mv\gamma\left(\beta_v\right).$$

Likewise, in the lab, the total energy of the particle is primed as well:

$$E' \equiv mc^2\gamma\left(\beta_v\right).$$

Why are we using primes here? Because we are not really interested in the lab. We are more interested in a second particle, traveling leftward at velocity u with respect to the lab. To describe its self-system, we use the *unprimed* x-t coordinates again (as in Sect. 4.5.2). This will be our spacetime. In it, what are the (unprimed) energy E and momentum p of the first particle?

4.9.3 A Naive Approach

To transform to our new spacetime, we could take a naive approach:

- Use the rule of adding velocities to calculate the velocity of the first particle away from the second one. This would give us the velocity of the first particle in our new spacetime.
- Use this new velocity to calculate (from the original definition) the new momentum and energy in our new spacetime, as required.

Still, this is too complicated. How to avoid going back to the original definitions?

4.9.4 The Momentum–Energy Vector

For this purpose, note that p' and E'/c have a familiar ratio:

$$\frac{p'}{E'/c} = \frac{mv\gamma\left(\beta_v\right)}{mc\gamma\left(\beta_v\right)} = \frac{v}{c} = \beta_v.$$

So, let us put them in a new column vector, proportional to the column $(\beta_v, 1)^t$. In fact, this is just the second column in the momentum matrix (times c):

$$\begin{pmatrix} p' \\ E'/c \end{pmatrix} \equiv cm\gamma\left(\beta_v\right)\left(I + \beta_v J\right)\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

4.9.5 The Momentum Matrix in Spacetime

How do things look like from our new spacetime? Well, from the perspective of the second particle, the entire lab travels rightward at speed u. On top of this, the first particle travels in the lab at speed v. Thus, to transform the old momentum matrix to our new spacetime, we need to apply an inverse Lorentz matrix:

$$m\gamma\left(\beta_u\right)\left(I+\beta_u J\right)\gamma\left(\beta_v\right)\left(I+\beta_v J\right).$$

This is the new momentum matrix in spacetime.

4.9.6 Lorentz Transformation on Momentum–Energy

To have the energy and momentum of the first particle from the perspective of the second one, just look at the second column of this new momentum matrix, and multiply by c:

$$\begin{aligned}\begin{pmatrix} p \\ E/c \end{pmatrix} &= cm\gamma\left(\beta_u\right)\left(I+\beta_u J\right)\gamma\left(\beta_v\right)\left(I+\beta_v J\right)\begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \gamma\left(\beta_u\right)\left(I+\beta_u J\right)\begin{pmatrix} p' \\ E'/c \end{pmatrix}.\end{aligned}$$

So, we got what we wanted: to drop the primes from the old vector $(p', E'/c)^t$, just apply the inverse Lorentz transformation (Sect. 4.3.6). This is much more direct: you work with momentum and energy only, not time or space or velocity any more. This gives the energy and momentum of the first particle not only with respect to the lab but also with respect to the second particle, with no need to add the velocities u and v explicitly any more.

4.10 Energy and Mass

4.10.1 Invariant Nuclear Energy

So far, we placed the energy–momentum in the second column of the momentum matrix. Still, we could work the other way around and place them in the first column too:

$$\begin{pmatrix} E'/c & p' \\ p' & E'/c \end{pmatrix} = cm\gamma\left(\beta_v\right)\left(I+\beta_v J\right).$$

What is the determinant of this matrix? Since the Lorentz matrix has determinant 1,

$$\begin{aligned}\frac{E'^2}{c^2} - p'^2 &= \det\left(\begin{pmatrix} E'/c & p' \\ p' & E'/c \end{pmatrix}\right) \\ &= \det\left(cm\gamma\left(\beta_v\right)\left(I + \beta_v J\right)\right) \\ &= c^2 m^2 \det\left(\gamma\left(\beta_v\right)\left(I + \beta_v J\right)\right) \\ &= m^2 c^2.\end{aligned}$$

This is invariant—independent of system: once defined in the lab, it could also be calculated from any other system, using the same formula. Therefore, in the left-hand side, the primes could probably drop, with no effect. How to prove this? We already know: apply another Lorentz matrix (Sect. 4.9.6). Fortunately, this has no effect on the determinant:

$$\begin{aligned}\frac{E^2}{c^2} - p^2 &= \det\left(\begin{pmatrix} E/c & p \\ p & E/c \end{pmatrix}\right) \\ &= \det\left(\gamma\left(\beta_u\right)\left(I + \beta_u J\right)\begin{pmatrix} E'/c & p' \\ p' & E'/c \end{pmatrix}\right) \\ &= \det\left(\begin{pmatrix} E'/c & p' \\ p' & E'/c \end{pmatrix}\right) \\ &= m^2 c^2.\end{aligned}$$

Thus, the determinant is indeed invariant: it does not depend on u and does not change from system to system. To simplify, multiply by c^2:

$$E^2 - c^2 p^2 = m^2 c^4 = \left(mc^2\right)^2.$$

Is this familiar? This is indeed the squared nuclear energy stored in the particle. This is not relativistic, but invariant: once fixed in one system, it is the same at all systems. Indeed, you could put the primes back on, if you like.

What is so good about this formula? It is quite surprising: both momentum and energy are relativistic, not invariant. Still, they combine to make the potential nuclear energy: not relativistic, but invariant!

Is this familiar? We have already seen this before: space and time are both relativistic, not invariant. Still, they combine to make a new invariant quantity—proper time:

$$c^2 t^2 - x^2 = c^2 s^2$$

(Sect. 4.4.1).

4.10.2 Invariant Mass

Thus, the momentum and energy (both relativistic, not invariant) combine to form a new 2×2 matrix, with an invariant determinant. This shows once again that the mass m is indeed invariant: once fixed in one system, it remains the same in all systems and never changes under any Lorentz transformation. In our lab, m is measured, while the particle is at motion, at speed v (Sect. 4.9.1). In another (inertial) system, on the other hand, the particle may have a different speed. Still, its mass will measure the same: m (provided that the measurement is made, while the particle is at motion).

As a matter of fact, the above formula actually gives us a new method to calculate m:

- First, calculate the relativistic momentum and energy in your system.
- This will give you p and E (or p' and E').
- Fortunately, these primes do not matter at all: they have no effect on the (invariant) determinant.
- Calculate the determinant.
- Use this to obtain the same m, as required.

This will be useful below.

4.10.3 Einstein's Formula

In particular, why not calculate m in the self-system of the first particle itself? After all, in this system, there is no velocity or momentum or kinetic energy at all, so the above formula simplifies to read

$$E^2_{\text{potential}} = m^2 c^4,$$

or

$$E_{\text{potential}} = mc^2.$$

This is Einstein's formula back again.

4.11 Center of Mass

4.11.1 Collection of Subparticles

In the lab, look at our first particle again. Assume that its velocity v is still unknown. How to uncover it? Use the momentum p' and the energy E':

$$v = \frac{mv\gamma\left(\beta_v\right)}{mc^2\gamma\left(\beta_v\right)}c^2 = \frac{p'}{E'}c^2.$$

When is this useful? When the momentum and energy are available, but the velocity is not. This is quite practical: as relativistic quantities, p' and E' are more fundamental than v, which is often missing.

Throughout this chapter, the second particle could actually be theoretical and nonphysical, with no size or mass at all. After all, it only served as a reference point for the first particle.

The first particle, on the other hand, is much more real and physical. To emphasize this, let us break it into $k \geq 1$ subparticles, each with its own velocity v_i, momentum p'_i, and energy E'_i with respect to the lab ($1 \leq i \leq k$).

What are the total momentum and energy? As fundamental (and conserved) quantities, they sum up:

$$p' \equiv \sum_{i=1}^{k} p'_i \quad \text{and} \quad E' \equiv \sum_{i=1}^{k} E'_i.$$

The velocity of the entire collection, on the other hand, is not necessarily the sum of the v_i's. After all, the subparticles may have different masses, which are not always available. As a matter of fact, some subparticles may even have no mass at all (those that are as fast as light and have $|v_i| = c$). To define the total velocity properly, better use the fundamental (relativistic) quantities: momentum and energy.

4.11.2 Center of Mass

We are now ready to define the velocity of the entire collection:

$$v \equiv \frac{p'}{E'}c^2 = c^2\frac{\sum_{i=1}^{k} p'_i}{\sum_{i=1}^{k} E'_i}.$$

This new velocity describes the motion of no concrete physical object, but only an imaginary (theoretical) object: the center of mass of the entire collection. Where is this? To find out, let us use the second particle.

The above velocity is in the lab. Next, let us look at things from our new spacetime: the self-system of the second particle, which travels in the lab at velocity $-u$. In this spacetime, what are the new momentum and energy of the entire collection? We already know what they are:

$$\begin{pmatrix} p \\ E/c \end{pmatrix} = \gamma\left(\beta_u\right)\left(I + \beta_u J\right)\begin{pmatrix} p' \\ E'/c \end{pmatrix}$$

(Sect. 4.9.6). There are actually two scalar equations here. Look at the top one. Assume also that the second particle follows at the same speed:

$$u \equiv -v, \quad \text{so} \quad \beta_u = -\beta_v = -\frac{p'}{E'/c}.$$

This way, the top equation simplifies to read

$$p = 0.$$

Thus, with respect to the second particle, the entire collection has no momentum at all: it is at a complete rest. This is why the second particle marks the center of mass itself.

4.11.3 The Mass of the Collection

What is the mass m of the entire collection? Is it the sum of the individual masses? To find out, better work with more fundamental (relativistic, not invariant) quantities: momentum and energy.

In our spacetime (the self-system of the second particle, traveling in the lab at velocity $-u = v$), the collection has no momentum at all: $p = 0$. Therefore, the formula in Sect. 4.10.1 tells us that

$$E^2 = m^2c^4,$$

or

$$m \equiv \frac{E}{c^2}.$$

This should indeed serve as a good definition of the total mass of the entire collection.

What is the physical meaning of this? We are now in our spacetime: both p and E have no primes any more. This is also the self-system of the collection (or its center of mass). In it, the collection is at a complete rest, with no momentum or kinetic energy at all. Thus, the above formula actually defines its mass m (which is also its rest mass) in terms of its total energy E. Being relativistic and conserved, E could also be written as a sum:

$$m \equiv \frac{E}{c^2} = \frac{1}{c^2}\sum_{i=1}^{k} E_i,$$

where E_i is the energy of the ith subparticle in our spacetime (kinetic plus potential). Indeed, to sum up, E_i must be relativistic: not only potential but also kinetic. The total mass, on the other hand, is neither relativistic nor conserved: it is *not* the sum of the individual masses.

4.12 Oblique Force and Momentum

4.12.1 Oblique Momentum in x′-y′

So far, we worked in one spatial dimension: the x-axis. How to work in two? As in Sect. 4.5.4: assume that the lab is described by the x'-y'-t' coordinates. This way, in the lab, the entire collection travels obliquely, at the new velocity

$$v \equiv \begin{pmatrix} v_{x'} \\ v_{y'} \end{pmatrix}.$$

This means that its speed is $v_{x'}$ in the x'-direction, and $v_{y'}$ in the perpendicular y'-direction in the lab. This way, the momentum is oblique as well:

$$p' \equiv \begin{pmatrix} p'_{x'} \\ p'_{y'} \end{pmatrix},$$

proportional to v. This is the view from the lab (Fig. 4.7). From the second particle, on the other hand, things look different (Fig. 4.8).

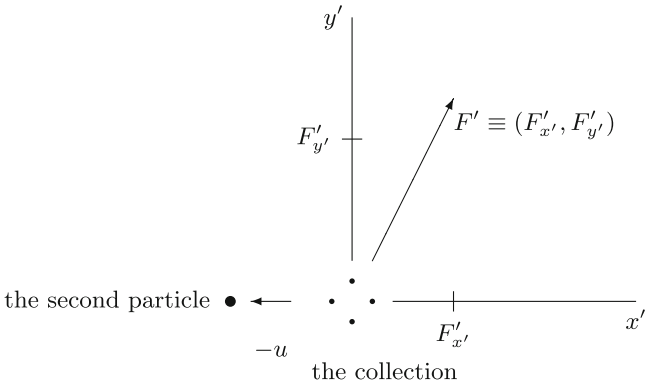


Fig. 4.7 The view from the lab: initially, at time $t' = 0$, the collection is still at rest at $x' = y' = 0$. At $t' = 0$, an oblique external force $F' = (F'_{x'}, F'_{y'})$ starts to act on it, to increase its momentum and kinetic energy (without changing its mass)

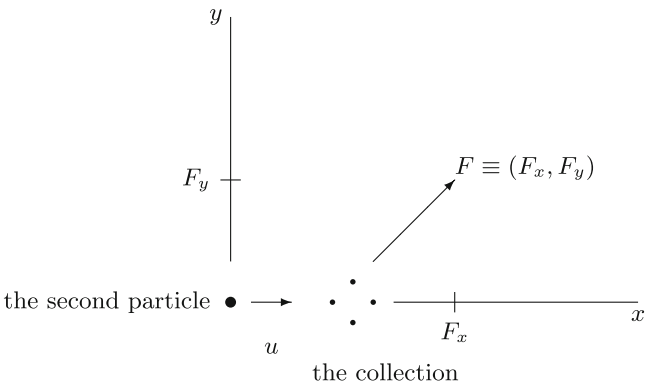


Fig. 4.8 From the second particle, on the other hand, the force looks different: the same in the x-direction, but weaker in the perpendicular y-direction

Do not get confused: there is no differentiation here. The primes mean no derivative but only remind us that we are in the lab. Likewise, the subscripts $_{x'}$ and $_{y'}$ mean no partial derivative, but only spatial coordinates in the lab.

Thus, in the formula in Sect. 4.10.1, p'^2 should be replaced by the inner product

$$\|p'\|^2 = (p', p') = p'^2_{x'} + p'^2_{y'}.$$

After all, in theory, we could always redefine x' to align with v and p' (see exercises below). Fortunately, there is no need to do this explicitly.

4.12.2 View from Spacetime

The second particle, on the other hand, travels horizontally, not obliquely: in the x'-direction only, at velocity $(-u, 0) \neq (0, 0)$ with respect to the lab. To transform from the lab to our new spacetime (the self-system of the second particle), we must now use an extended 3×3 Lorentz matrix:

$$\begin{pmatrix} p_x \\ p_y \\ E/c \end{pmatrix} = \begin{pmatrix} \gamma\,(\beta_u) & & \\ & 1 & \\ & & \gamma\,(\beta_u) \end{pmatrix} \begin{pmatrix} 1 & & \beta_u \\ & 1 & \\ \beta_u & & 1 \end{pmatrix} \begin{pmatrix} p'_{x'} \\ p'_{y'} \\ E'/c \end{pmatrix}.$$

Here, you are free again to pick u as you like: we no longer assume that $u = -v$. Thus, the second particle no longer coincides with the center of mass. This job is left to the lab itself.

4.12.3 The Lab: The New Self-system

Indeed, assume now that the lab system is initially the same as the self-system of the collection: at time $t' = 0$, the collection is at rest in the lab:

$$p' \equiv \begin{pmatrix} p'_{x'} \\ p'_{y'} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad E' = mc^2 > 0, \quad \text{and} \quad v \equiv \begin{pmatrix} v_{x'} \\ v_{y'} \end{pmatrix} \equiv \frac{p'}{E'}c^2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This is true at time $t' = 0$ only. At $t' > 0$, on the other hand, things may change, due to an external force.

4.13 Force in an Open System

4.13.1 Force in an Open Passive System

Unlike before, assume now that the lab is an open system: from time $t = t' = 0$ onward, an external force is applied to the entire collection, to accelerate it, and increase both its momentum and kinetic energy in the lab (with no change to its mass). In this sense, the lab is passive: the force acts directly on the static collection in it.

In the passive system, what is the force? This is the time derivative of the momentum (in the original x'-y'-t' coordinates):

$$F' \equiv \begin{pmatrix} F'_{x'} \\ F'_{y'} \end{pmatrix} \equiv \begin{pmatrix} \frac{dp'_{x'}}{dt'} \\ \\ \frac{dp'_{y'}}{dt'} \end{pmatrix}.$$

Unless said otherwise, we mean the force at the initial time $t' = 0$:

$$F' \equiv \begin{pmatrix} F'_{x'} \\ F'_{y'} \end{pmatrix} \equiv \begin{pmatrix} \frac{dp'_{x'}}{dt'}(0) \\ \\ \frac{dp'_{y'}}{dt'}(0) \end{pmatrix}.$$

(How to deal with a fixed time $t' > 0$? Easy: just shift it to zero, and rename it.) How does this force look like from the second particle? In other words, how to transform the force to our new spacetime (the x-y-t self-system of the second particle)?

4.13.2 What Is the Force in Spacetime?

Of course, we could take a naive approach: transform the momentum to our new spacetime, and differentiate it there with respect to t. Still, this too complicated. How to avoid this, and find F more directly?

4.13.3 Proper Time in the Lab

To differentiate the momentum with respect to time, use the same trick as in Sect. 4.5.4. For this purpose, differentiate t with respect to t'. This seems easy: after all, in the lab, t' is the proper time, isn't it? So, from the perspective of the second particle, t' should look slower (Sect. 4.4.2):

$$t' = \frac{t}{\gamma\left(\beta_u\right)},$$

shouldn't it? It should undergo time dilation, shouldn't it?

Unfortunately not. After all, t' could be proper only in an isolated lab, which welcomes no external force. In our open lab, on the other hand, t' is only *nearly* proper: only at $t' = 0$ (before the force had time to act) does t' behave like a proper time.

4.13.4 Nearly Proper Time in the Lab

To see this, transform the lab back to our spacetime, using the inverse Lorentz transformation:

$$ct = \gamma\left(\beta_u\right)\left(\beta_u x' + ct'\right),$$

or

$$t = \gamma\left(\beta_u\right)\left(\frac{\beta_u}{c}x' + t'\right).$$

Differentiate this with respect to t':

$$\frac{dt}{dt'} = \gamma\left(\beta_u\right)\left(\frac{\beta_u}{c}\cdot\frac{dx'}{dt'} + 1\right) = \gamma\left(\beta_u\right)\left(\frac{\beta_u}{c}v_{x'} + 1\right).$$

At $t' = 0$, in particular, $v_{x'} = 0$, so this simplifies to read

$$\frac{dt}{dt'} = \gamma\left(\beta_u\right).$$

Thus, at $t' = 0$, t' is indeed nearly proper: it behaves just like a proper time. Let us use this to look at the force from the second particle.

4.14 Perpendicular Force

4.14.1 Force: Time Derivative of Momentum

In our spacetime, the perpendicular force has no prime: F_y. How big is it? Thanks to the above 3×3 matrix, in the y-direction, the prime is easily dropped:

$$p_y = p'_{y'}.$$

Thus, the differentiation is simple:

$$\begin{aligned} F_y &\equiv \frac{dp_y}{dt} \\ &= \frac{dp'_{y'}}{dt} \\ &= \frac{dp'_{y'}}{\gamma\left(\beta_u\right) dt'} \\ &= \frac{1}{\gamma\left(\beta_u\right)} \cdot \frac{dp'_{y'}}{dt'} \\ &= \frac{1}{\gamma\left(\beta_u\right)} F'_{y'}. \end{aligned}$$

This is smaller than before (in magnitude).

4.14.2 Passive System—Strong Perpendicular Force

This is weaker! This is why the arrow in Fig. 4.8 has a moderate slope.

In summary, the passive system (our lab) feels the maximal perpendicular force. Any other system, on the other hand, would feel a weaker perpendicular force. Our spacetime, in particular, feels a perpendicular force $\gamma\left(\beta_u\right)$ times as weak.

4.15 Nonperpendicular Force

4.15.1 Force: Time Derivative of Momentum

What about the force in the x-direction? Does it also feel weaker? Let us use the same trick:

$$\begin{aligned} F_x &= \frac{dp_x}{dt} \\ &= \frac{d\left(\gamma\left(\beta_u\right)\left(p'_{x'} + \beta_u \frac{E'}{c}\right)\right)}{\gamma\left(\beta_u\right) dt'} \\ &= \frac{d\left(p'_{x'} + \beta_u \frac{E'}{c}\right)}{dt'} \\ &= \frac{dp'_{x'}}{dt'} + \beta_u \frac{dE'}{cdt'} \\ &= F'_{x'} + \frac{\beta_u}{c} \cdot \frac{dE'}{dt'}. \end{aligned}$$

4.15.2 Energy in an Open System

Look at the latter term. It contributes nothing! To see this, look at the equation

$$E'^2 = c^2\left(p', p'\right) + m^2c^4$$

that couples E' and p' in the lab. (Since the momentum is a two-dimensional vector, we use its inner product in 2-D.) This holds for every time $t' \geq 0$ (with the same m, but perhaps with a different E', p', or v). In a moment, we will come back to the initial time $t' = 0$.

4.15.3 Open System—Constant Mass

In this equation, the latter term remains constant all the time. After all, thanks to the external force, the potential energy (and the mass) remains unchanged all the time. So, upon differentiating both sides with respect to t', the latter term drops:

$$2E'\frac{dE'}{dt'} = 2c^2\left(p', \frac{dp'}{dt'}\right) = 2c^2\left(p', F'\right).$$

(This is still an inner product in 2-D.)

4.15.4 Nearly Constant Energy in the Lab

In particular, what happens at the initial time $t' = 0$? Well, at this time, the momentum was zero:

$$2E'\frac{dE'}{dt'} = 2c^2(0,0)F' = 0.$$

Since $E' > 0$, we must have

$$\frac{dE'}{dt'} = 0,$$

as required.

4.15.5 Nonperpendicular Force: Same at All Systems

In summary, at $t = t' = 0$,

$$F_x = F'_{x'} + \frac{\beta_u}{c} \cdot \frac{dE'}{dt'} = F'_{x'}.$$

Thus, unlike the perpendicular force, the nonperpendicular force is the same at all (inertial) systems.

4.15.6 The Photon Paradox

What is light? It has two faces. On one hand, it is a wave. On the other hand, it is also a particle: a massless photon, traveling at speed c with respect to any (inertial) reference frame.

The photon is in a kind of singularity. Indeed, from our perspective, it can have no size at all (due to length contraction, as in Sect. 4.4.3). Still, this is only from our own (subjective) point of view. The photon may disagree: in its own self-system, it is at rest, with positive mass and size, while the entire universe shrinks to a singular point, traveling as fast as light in the opposite direction. Who is right?

The answer is that the photon can never have an inertial self-system. After all, in an inertial system, the speed of light must be c, not zero. As a matter of fact, the

photon has no self-system at all. Indeed, as we will see in quantum mechanics, it has no certain position and momentum.

4.16 Exercises: Special Relativity in 3-D

4.16.1 Lorentz Matrix and its Determinant

1. Show that the determinant of a 2×2 matrix is the same as the area of the parallelogram made by its column vectors.
2. Show that the determinant of a 2×2 matrix is the same as the area of the parallelogram made by its row vectors.
3. Consider two 2×2 matrices. Show that the determinant of their product is the same as the product of their determinants. Hint: See Chap. 14, Sect. 14.4.3.
4. Show that the Lorentz matrix has determinant 1.
5. Consider a 2×2 matrix. Multiply it by a Lorentz matrix. What happened to the determinant? Hint: Nothing (thanks to the previous exercises).
6. Conclude that the Lorentz transformation preserves area in the two-dimensional Cartesian plane.
7. Calculate the inverse Lorentz matrix. Hint: Just change sign: replace v by $-v$ (Sect. 4.3.6).
8. What is the physical meaning of this? Hint: From the moving system, our lab seems to move in the opposite direction.
9. Conclude that the inverse Lorentz matrix has determinant 1 as well.
10. Conclude that the inverse transformation preserves area as well.
11. Use Cramer's formula (Chap. 2, Sect. 2.1.4) to calculate the inverse Lorentz matrix again.
12. Do you get the same inverse?

4.16.2 Motion in 3-D

1. Let

$$v \equiv \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \in \mathbb{R}^3$$

be some nonzero real vector in 3-D. Define the 3×3 matrix O_v, whose columns are v (normalized), a vector that is orthogonal to v (normalized as well), and their vector product:

$$O_v \equiv \left(\frac{v}{\|v\|} \;\middle|\; \frac{v^\perp}{\|v^\perp\|} \;\middle|\; \frac{v \times v^\perp}{\|v\| \cdot \|v^\perp\|} \right).$$

2. Show that O_v is an orthogonal matrix. Hint: See Chap. 1, Sect. 1.8.4.
3. Show that O_v has determinant 1. Hint: Use the Euler angles at the end of Chap. 2. Alternatively, see Chap. 14, Sect. 14.4.4.
4. Consider a particle that travels at velocity $v \in \mathbb{R}^3$ with respect to the lab.
5. More specifically, it travels at speed v_1 in x, speed v_2 in y, and speed v_3 in the z-spatial direction.
6. Interpret the motion better, in vector terms. Hint: The particle travels at direction $v/\|v\|$, at speed $\|v\|$.
7. Let (x', y', z', ct') denote the lab coordinates, and (x'', y'', z'', ct'') the self-coordinates of the particle.
8. We are now ready to define the more general Lorentz transformation

$$\begin{pmatrix} x' \\ y' \\ z' \\ ct' \end{pmatrix} \to \begin{pmatrix} x'' \\ y'' \\ z'' \\ ct'' \end{pmatrix} = L_v \begin{pmatrix} x' \\ y' \\ z' \\ ct' \end{pmatrix},$$

   where L_v is the following 4×4 Lorentz matrix:

$$L_v \equiv \begin{pmatrix} O_v & \\ & 1 \end{pmatrix} \begin{pmatrix} \gamma\left(\beta_{\|v\|}\right) & & & \\ & 1 & & \\ & & 1 & \\ & & & \gamma\left(\beta_{\|v\|}\right) \end{pmatrix} \begin{pmatrix} 1 & & & -\beta_{\|v\|} \\ & 1 & & \\ & & 1 & \\ -\beta_{\|v\|} & & & 1 \end{pmatrix} \begin{pmatrix} O_v^t & \\ & 1 \end{pmatrix}.$$

   (As usual, blank spaces stand for zero matrix elements.)
9. Show that this indeed transforms the lab system to the self-system of the particle.
10. Show that L_v has determinant 1.
11. Consider also a second particle, traveling at velocity $-u \in \mathbb{R}^3$ with respect to the lab.
12. Denote its self-coordinates by (x, y, z, ct) (with no primes at all). This will be our new spacetime.
13. With respect to this spacetime, how does the entire lab travel? Hint: It travels at velocity $u \in \mathbb{R}^3$.
14. How could this spacetime transform to the lab? Hint: The transformation is

$$\begin{pmatrix} x \\ y \\ z \\ ct \end{pmatrix} \to \begin{pmatrix} x' \\ y' \\ z' \\ ct' \end{pmatrix} = L_u \begin{pmatrix} x \\ y \\ z \\ ct \end{pmatrix}.$$

15. Consider the composite Lorentz transformation

$$\begin{pmatrix} x \\ y \\ z \\ ct \end{pmatrix} \to \begin{pmatrix} x'' \\ y'' \\ z'' \\ ct'' \end{pmatrix}$$

from our spacetime (the self-system of the second particle) to the self-system of the first particle.

16. Show that it is the matrix product $L_v L_u$:

$$\begin{pmatrix} x'' \\ y'' \\ z'' \\ ct'' \end{pmatrix} = L_v \begin{pmatrix} x' \\ y' \\ z' \\ ct' \end{pmatrix} = L_v L_u \begin{pmatrix} x \\ y \\ z \\ ct \end{pmatrix}.$$

17. Show that $L_v L_u$ has determinant 1 as well. Hint: See Chap. 14, Sect. 14.4.3.
18. Does L_u commute with L_v? Hint: Only if u is a scalar multiple of v.
19. Consider the inverse Lorentz transformation

$$\begin{pmatrix} x'' \\ y'' \\ z'' \\ ct'' \end{pmatrix} \to \begin{pmatrix} x \\ y \\ z \\ ct \end{pmatrix}$$

from the self-system of the first particle back to our spacetime (the self-system of the second particle).

20. Show that it is the inverse matrix

$$(L_v L_u)^{-1} = L_u^{-1} L_v^{-1} = L_{-u} L_{-v}.$$

21. Conclude that the last column in $L_{-u} L_{-v}$ describes the motion of the first particle away from the second one. Hint: See below.
22. Show that, in its own self-system, the first particle is at rest:

$$x'' = y'' = z'' = 0.$$

23. Conclude that t'' is its proper time. Hint: See Sect. 4.4.1.
24. Write the above equation in its vector form. Hint:

$$\begin{pmatrix} x'' \\ y'' \\ z'' \\ ct'' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ ct'' \end{pmatrix}.$$

25. Write the above equation in its differential form. Hint:

$$\begin{pmatrix} dx'' \\ dy'' \\ dz'' \\ cdt'' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

26. What is the meaning of this? Hint: Divide this by dt''. This gives zero velocity. After all, in its own self-system, the particle is at rest.
27. Transform this back to our spacetime. Hint:

$$\begin{pmatrix} dx \\ dy \\ dz \\ cdt \end{pmatrix} = L_{-u}L_{-v} \begin{pmatrix} dx'' \\ dy'' \\ dz'' \\ cdt'' \end{pmatrix} = L_{-u}L_{-v} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

28. How to do this efficiently? Hint: Up to a scalar multiple,

$$L_{-v} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} v \\ c \end{pmatrix}.$$

Therefore, we only need to calculate

$$\begin{pmatrix} dx \\ dy \\ dz \\ cdt \end{pmatrix} = L_{-u} \begin{pmatrix} v \\ c \end{pmatrix}.$$

29. Once this is solved, how to use it in practice? Hint: Divide the left-hand side by dt, and obtain the velocity in space:

$$\begin{pmatrix} dx/dt \\ dy/dt \\ dz/dt \end{pmatrix} \in \mathbb{R}^3.$$

30. What is the physical meaning of this? Hint: It tells us how the first particle gets away from the second one.
31. Consider a special case, in which u aligns with the x-axis:

$$u \equiv \begin{pmatrix} \|u\| \\ 0 \\ 0 \end{pmatrix}.$$

32. Show that, in this case, one could design

$$O_u \equiv I$$

(the 3×3 identity matrix).

33. Show that, in this case,

$$L_u \equiv \begin{pmatrix} \gamma\left(\beta_{\|u\|}\right) & & \\ & 1 & \\ & & 1 \\ & & \gamma\left(\beta_{\|u\|}\right) \end{pmatrix} \begin{pmatrix} 1 & & -\beta_{\|u\|} \\ & 1 & \\ & & 1 \\ -\beta_{\|u\|} & & 1 \end{pmatrix}.$$

34. Show that, in this case,

$$L_{-u} \equiv \begin{pmatrix} \gamma\left(\beta_{\|u\|}\right) & & \\ & 1 & \\ & & 1 \\ & & \gamma\left(\beta_{\|u\|}\right) \end{pmatrix} \begin{pmatrix} 1 & & \beta_{\|u\|} \\ & 1 & \\ & & 1 \\ \beta_{\|u\|} & & 1 \end{pmatrix}.$$

35. Show that this is not just a special case, but a most general case. Hint: For a general u, pick the x-axis to align with u in the first place.
36. Interpret L_{-u} as a projective mapping in the real projective space. Hint: See Chap. 6.
37. Interpret the above method as the three-dimensional version of the methods in Sects. 4.5.2–4.5.4.
38. Likewise, in Sect. 4.5.4, interpret the inverse Lorentz transformation back to spacetime as a projective mapping in the real projective plane.
39. What does this mapping do? Hint: It maps the original velocity $(dx'/dt', dy'/dt')$ of the first particle in the lab (Fig. 4.4) to the new velocity $(dx/dt, dy/dt)$ of the first particle away from the second one (Fig. 4.5).
40. In these figures, where is the time variable? Why is it missing? Hint: These are phase planes. They only show us the position that the particle reaches, but not when. Time remains implicit: it is just a parameter that "pushes" the particle along the arrow. To see this motion more vividly, make a movie that shows how the particle travels along the arrow.
41. In these figures, how did we get rid of time? Hint: We just divide by dt. This eliminates time, leaving just the velocity arrow in space.

Part II
Introduction to Group Theory

What have we done so far? Well, the vectors introduced above make a linear space. Indeed, the algebraic operations between them are linear. The (nonsingular) matrices, on the other hand, make a new mathematical structure: a group.

In a group, although the commutative law not necessarily holds, the associative law does hold. In what follows, we introduce group theory, including the first, second, and third isomorphism theorems, and their geometrical applications.

Matrices are particularly useful to represent all sorts of practical transformations in geometry and physics. In special relativity, for example, Lorentz transformations are written as 2×2 matrices. Here, we will put this in a much wider context: group representation. To show how useful this is, we will represent projective mappings as 3×3 matrices. This is particularly useful in computer graphics. Finally, we will also use matrices to introduce yet another important field: quantum mechanics.

Chapter 5
Groups and Isomorphism Theorems



What is the most elementary algebraic object? This could be the individual number. In the previous part, we also introduced more complicated algebraic structures: vectors and matrices.

Furthermore, elementary algebraic objects such as numbers, once used as input and output, form a yet more advanced mathematical object: a function. The polynomial, for example, is just a special kind of function, enjoying many algebraic operations: addition, multiplication, and composition.

Functions are indeed studied in a few major mathematical fields. In set theory, a set of functions is often studied just like any other set, and its cardinality is estimated. In algebra, on the other hand, functions are also viewed as algebraic objects that can be composed with each other. Finally, in calculus, functions are also considered as analytic objects that can be differentiated and integrated.

In this chapter, we consider a special kind of function: a mapping or transformation. Together, the transformations form a new mathematical structure: a group, with a lot of interesting properties.

To help study a mapping, we mirror it by a matrix. This way, algebraic operations are mirrored as well: composition of two mappings is mirrored by multiplication of two matrices. This is indeed group representation.

This point of view is most useful in the practical implementation. After all, a mapping could hardly be stored on the computer. A matrix, on the other hand, can. Furthermore, the representation could help understand the deep nature of the original mapping as an algebraic and geometrical object.

We have already seen an example of a useful transformation: in special relativity (Chap. 4), the Lorentz transformation has been represented as a 2×2 matrix. Here, on the other hand, we put this in a much wider context: group theory. In particular, we prove the first, second, and third isomorphism theorems, used later in projective geometry.

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5.1 Moebius Transformation and Matrix

5.1.1 Riemann Sphere—Extended Complex Plane

To make the discussion more concrete, we need a new geometrical concept: the infinity "point." As a matter of fact, this is not really a point. Still, it can be added to the complex plane, to form a complete "sphere."

The extended complex plane (or the Riemann sphere)

$$\mathbb{C} \cup \{\infty\}$$

is obtained from the original complex plane $\mathbb{C}$ by adding one more object: the infinity point ∞. This new "point" is not really a point, but a new artificial object, to help model a complex number with an arbitrarily large absolute value.

The infinity point is unique. In fact, in the complex plane, one could draw a ray issuing from the origin in just any angle. The complex number z could then "slide" along this ray and approach the same point: infinity. Later on, we will also meet more complicated spaces, with many infinity points.

5.1.2 Moebius Transformation and the Infinity Point

Thanks to the infinity point, we can now define the Moebius transformation. We have already met this transformation in the context of special relativity (Chap. 4, Sect. 4.5.2). Here, however, we introduce it in much more detail and depth and in a much wider context [4, 72].

A Moebius transformation (or mapping) from the extended complex plane onto itself is defined by

$$z \to \frac{az + b}{cz + d},$$

where a, b, c, and d are some fixed complex parameters. Here, "$\to$" stands for transformation, not for a limit.

Unfortunately, the definition is still incomplete. After all, the mapping is not yet defined at the infinity point $z = \infty$. Let us complete this gap in such a way that the mapping remains continuous:

$$\infty \to \frac{a}{c}.$$

This way, z could approach infinity on just any ray in the complex plane. After all, in every direction, the transformed values converge to a/c, as required.

Still, the definition is not yet complete. What happens at the pole $z = -d/c$, at which the denominator vanishes? Well, to preserve continuity, define

$$-\frac{d}{c} \to \infty.$$

This way, z could approach the pole from just any direction. In either case, the transformed values would approach a unique point: the infinity point.

There is one case in which these formulas coincide and agree with each other. This happens if $c = 0$. In this case, both formulas read

$$\infty \to \infty.$$

Still, this makes no problem: c could safely vanish. There is something else that must never vanish.

Indeed, to make sure that the mapping is invertible, the parameters a, b, c, and d must satisfy the condition

$$ad - bc \neq 0.$$

Otherwise, we would have $ad = bc$, which means that:

- Either $c \neq 0$, so the entire complex plane is mapped to

$$\frac{az+b}{cz+d} = \frac{1}{c} \cdot \frac{acz+bc}{cz+d} = \frac{1}{c} \cdot \frac{acz+ad}{cz+d} = \frac{a}{c}.$$

- Or $c = a = 0$, so the entire complex plane is mapped to b/d.
- Or $c = d = 0$, so the entire complex plane is mapped to ∞.

In either case, the transformation would be constant and not invertible. This is why the above condition is necessary to make sure that the original Moebius transformation is indeed nontrivial and invertible.

5.1.3 The Inverse Transformation

Fortunately, the condition $ad - bc \neq 0$ is not only necessary but also sufficient to guarantee that the original Moebius transformation is invertible. Indeed, assume that the original complex number z is mapped to the new complex number

$$u \equiv \frac{az+b}{cz+d}.$$

To have the inverse mapping in its explicit form, let us write z in terms of u. For this purpose, let us multiply the above equation by the denominator:

$$u(cz + d) = az + b.$$

Now, let us throw those terms that contain z to the left-hand side and the other terms to the right-hand side:

$$z(cu - a) = -du + b.$$

This implies that

$$z = \frac{-du + b}{cu - a} = \frac{du - b}{-cu + a}.$$

Thus, the required inverse transformation is

$$z \to \frac{dz - b}{-cz + a}.$$

To make sure that this transformation is indeed continuous, we must also define it properly at ∞ and at its pole, a/c:

$$\infty \to -\frac{d}{c}$$
$$\frac{a}{c} \to \infty.$$

These are just the reverse of the original definitions in Sect. 5.1.2. This way, the inverse transformation indeed maps the infinity point back to the pole of the original transformation, as required.

5.1.4 Moebius Transformation as a Matrix

The parameters in the original Moebius transformation are defined up to a scalar multiple. After all, for any nonzero complex number $q \neq 0$, the same transformation could also be defined by

$$z \to \frac{qaz + qb}{qcz + qd}.$$

Thus, the original Moebius transformation is associated with the 2×2 matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

or just any nonzero scalar multiple of it.

Similarly, the inverse mapping is associated with the matrix

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

or just any nonzero scalar multiple of it. As can be seen in Chap. 2, Sect. 2.1.4, this matrix is just a scalar multiple of the inverse of the original matrix.

Later on, we will see that this is not just an association, but much more: the matrix actually represents and mirrors the original transformation. In terms of a matrix, the original condition takes the form

$$\det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = ad - bc \neq 0.$$

Why does this condition make sense? Because it guarantees that the original matrix is nonsingular (invertible), as required.

Thus, the set of invertible Moebius transformations is mirrored by the set of (complex) nonsingular 2×2 matrices, defined up to a nonzero scalar multiple. Later on, we will refer to this as isomorphism or group representation [23, 28, 39, 40]. Indeed, it preserves the same algebraic structure: the composition of two Moebius transformations is mirrored by the product of the 2×2 matrices associated with them.

5.1.5 *Product of Moebius Transformations*

The product of the Moebius transformations m' and m is defined as their composition:

$$m'm \equiv m' \circ m.$$

This means that, for every $z \in \mathbb{C} \cup \{\infty\}$,

$$(m'm)(z) \equiv m'(m(z)).$$

Note that this algebraic operation is associative. Indeed, for every three Moebius transformations m, m', and m'',

$$\begin{aligned} ((m''m')m)(z) &= (m''m')(m(z)) \\ &= m''(m'(m(z))) \\ &= m''((m'm)(z)) \\ &= (m''(m'm))(z). \end{aligned}$$

Since this applies to each and every $z \in \mathbb{C} \cup \{\infty\}$, it can be written more concisely as

$$(m''m')m = m''(m'm),$$

which makes an associative law. This is no surprise: after all, as discussed below, this kind of composition is mirrored by matrix multiplication.

5.2 Matrix: A Function

5.2.1 Matrix as a Vector Function

As discussed above, each invertible Moebius transformation is associated with a nonsingular 2×2 complex matrix (defined up to a nonzero scalar multiple). Fortunately, matrix-times-matrix multiplication could be viewed as a composition as well.

For this purpose, consider a 2×2 matrix g. Let us interpret it as a special vector function, rather than just a matrix:

$$g : \mathbb{C}^2 \to \mathbb{C}^2,$$

with the explicit definition

$$g(v) \equiv gv, \quad (v \in \mathbb{C}^2).$$

Here, when g is followed by v, with no parentheses, then this is a matrix–vector product, as in Chap. 1, Sect. 1.4.4. This is then used to define the new vector function, which uses round parentheses.

Why is this new interpretation equivalent to the original one? Because it characterizes g uniquely! Indeed, given a matrix g, we have already seen how it is used to define a unique vector function. Conversely, given a vector function of the above form, we can easily reconstruct the unique matrix that defines it. For this purpose, just apply the vector function $g()$ to the standard unit vectors $(1, 0)^t$ and $(0, 1)^t$, to uncover the matrix g column by column:

$$g = \left(g^{(1)} \mid g^{(2)}\right), \text{ where } g^{(1)} = g\left(\begin{pmatrix}1\\0\end{pmatrix}\right) \text{ and } g^{(2)} = g\left(\begin{pmatrix}0\\1\end{pmatrix}\right).$$

5.2.2 Matrix Multiplication as Composition

Let g and g' be 2×2 matrices. With their new interpretation as vector functions, their product can now be viewed as a composition:

$$g'g \equiv g' \circ g.$$

This means that, for every two-dimensional vector $v \in \mathbb{C}^2$,

$$(g'g)(v) \equiv g'(g(v)).$$

Does this agree with the original definition—the matrix-times-matrix product in Chap. 1, Sect. 1.4.5? Well, let us look at the matrix $g'g$. Thanks to associativity, it defines the vector function

$$(g'g)(v) = (g'g)v = g'(gv) = g'(g(v)),$$

which is the same as the above composition.

Furthermore, we have already seen that this matrix is unique: $g'g$ is the only matrix that could be used to define the above composition.

So, matrices could mirror these special vector functions: matrix-times-matrix could mirror composition. How does this help to mirror Moebius transformations? To get to the bottom of this, we better use the principle of induction: study not only one special case, but also a much wider field: groups.

5.3 Group and its Properties

5.3.1 Group

The above g is just a special case. In general, g could be not only a 2×2 matrix but also any element in a group.

A group G is a set of elements or objects, with some algebraic operation between them. This operation is called multiplication or product. Thus, the group is closed under this kind of multiplication: for every two elements g and g' in G, their products gg' and $g'g$ (which are not necessarily the same) are legitimate elements in G as well.

This kind of multiplication might be rather strange and nonstandard. Fortunately, it must not be too nonstandard: although it does not have to be commutative, it must still be associative: for every three elements g, g', and g'' in G,

$$g(g'g'') = (gg')g''.$$

5.3.2 The Unit Element

Furthermore, it is also assumed that G contains a unit element I that satisfies

$$gI = Ig = g$$

for every element $g \in G$. (Do not confuse this with the identity matrix I, used often in linear algebra.) This is a very special element in G.

Indeed, this unit element is unique in G. Indeed, assume that $I' \in G$ was a unit element as well:

$$gI' = I'g = g$$

for every element $g \in G$. In particular, this would be true for $g = I$:

$$I'I = I.$$

Since I is the original unit element, we also have

$$I'I = I'.$$

In summary,

$$I' = I'I = I,$$

so I' is not really new: it is the same as the original unit element I.

5.3.3 Inverse Element

Finally, it is also assumed that every element $g \in G$ has an inverse element $g' \in G$ (dependent on g), for which

$$gg' = I.$$

Even if the commutative law does not hold in G in general, g' does commute with g:

$$g'g = I.$$

Indeed, thanks to the associative law,

$$g'g = g'(Ig) = g'((gg')g) = g'(g(g'g)) = (g'g)(g'g).$$

Now, let us multiplying this equation (from the right) by an inverse of $g'g$, denoted by $(g'g)'$. Thanks again to the associative law, we have

$$I = ((g'g)(g'g))(g'g)' = (g'g)\left((g'g)(g'g)'\right) = (g'g)I = g'g,$$

as asserted.

Furthermore, the inverse of g is unique. Indeed, assume that g had yet another inverse g'', satisfying

$$gg'' = I$$

as well. Thanks to the associative law, we would then have

$$g'' = Ig'' = (g'g)g'' = g'(gg'') = g'I = g'.$$

Thus, g'' is not really new: it is the same as g'. The unique inverse of g can now be denoted by g^{-1} rather than g'. Once we have this new notation, we can use g' once again to stand for a general element in G, independent of g.

Let us show that the inverse of the inverse is the original element itself:

$$\left(g^{-1}\right)^{-1} = g.$$

Indeed, we already know that g^{-1} is the inverse of g, not only from the right but also from the left:

$$g^{-1}g = I.$$

Fortunately, this equation could also be interpreted the other way around: g behaves as expected from an inverse to g^{-1}. But g^{-1} has only one inverse, so it must indeed be g:

$$\left(g^{-1}\right)^{-1} = g,$$

as asserted. Thus, the inverse operation is symmetric: not only g^{-1} is the inverse of g, but also g is the inverse of g^{-1}.

Finally, consider two general elements $g', g \in G$. How to invert their product $g'g$? Take the individual inverses, in the reverse order:

$$(g'g)^{-1} = g^{-1}g'^{-1}.$$

Indeed, look at the right-hand side. Does it behave as expected from the inverse of $g'g$? Well, thanks to the associative law,

$$(g'g)\left(g^{-1}g'^{-1}\right) = \left((g'g)g^{-1}\right)g'^{-1} = \left(g'\left(gg^{-1}\right)\right)g'^{-1} = (g'I)g'^{-1} = g'g'^{-1} = I.$$

The assertion follows now from the uniqueness property.

5.4 Mapping and Homomorphism

5.4.1 Mapping and its Origin

Recall some properties of sets and their mappings, useful in the present discussion. Let G and M be some sets (not necessarily groups). A mapping ξ from G to M is a function

$$\xi : G \to M$$

that maps each element $g \in G$ to an element of the form

$$\xi(g) \in M.$$

Consider now some element $m \in M$. What elements are mapped to it from G? There could be many. Let us place them in one subset, called the origin of m under ξ:

$$\xi^{-1}(m) \equiv \{g \in G \mid \xi(g) = m\} \subset G.$$

This is just a notation for a subset of G. It has nothing to do with inverse. In fact, ξ may have no inverse mapping at all. After all, $\xi^{-1}(m)$ might contain a few elements in G. On the other hand, $\xi^{-1}(m)$ might also be completely empty. In either case, no inverse mapping could possibly be defined.

We say that ξ is *onto* M if every element $m \in M$ has a nonempty origin, with at least one element in it:

$$\left|\xi^{-1}(m)\right| = |\{g \in G \mid \xi(g) = m\}| \geq 1, \quad m \in M.$$

Furthermore, we say that ξ is one-to-one if every element $m \in M$ has at most one element in its origin:

$$\left|\xi^{-1}(m)\right| = |\{g \in G \mid \xi(g) = m\}| \leq 1, \quad m \in M.$$

Clearly, we can now combine these properties: ξ is a one-to-one mapping from G *onto* M if every element $m \in M$ has exactly one element in its origin:

$$\left|\xi^{-1}(m)\right| = |\{g \in G \mid \xi(g) = m\}| = 1, \quad m \in M.$$

This element is quite special: it is the only one mapped to m. We can therefore define the inverse mapping: just map m back to this unique element. Only in this case can ξ^{-1} be interpreted as a legitimate mapping: the inverse mapping. Only in this case is ξ invertible.

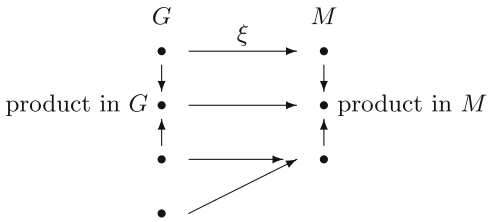


Fig. 5.1 The homomorphism ξ from the original group G onto the group M is not necessarily one-to-one. Still, it preserves (or mirrors) the algebraic operation, denoted by the vertical arrows

5.4.2 Homomorphism

A group is not just a set. It is much more: an algebraic structure. In fact, it has an algebraic operation: multiplication.

Indeed, the individual elements in the group are algebraic: they can multiply each other. Furthermore, they can often be mapped to yet another group.

This kind of mapping should better preserve the original multiplication. This way, it may indeed mirror the original algebraic structure and highlight its nature from a new angle.

To make this more concrete, let G and M be two groups. A mapping ξ from G *onto* M (not necessarily one-to-one) is called a homomorphism if it preserves multiplication: a product in G is mapped to the product in M (Fig. 5.1).

More precisely, for every two elements $g, g' \in G$, order does not matter: you could multiply in G and then map the product to M, or do things the other way around: first map to M, and then multiply in M. Either way, you would get the same result:

$$\xi(gg') = \xi(g)\xi(g').$$

Why is this attractive? Because, if ξ is *not* one-to-one, then M is "smaller" and easier to sort out. In some sense, it is "blind:" it does not distinguish between different elements in G that are of the same kind.

5.4.3 Mapping the Unit Element

Since the homomorphism preserves the algebraic operation, it must map the original unit element $I \in G$ to the unit element $i \in M$:

$$\xi(I) = i.$$

(Here, i has nothing to do with the notation $i = \sqrt{-1}$, used often in complex analysis. It is just a coincidence that both notations use the same letter i.) Indeed,

$$\xi(I) = \xi(II) = \xi(I)\xi(I).$$

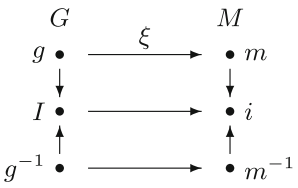


Fig. 5.2 Thanks to the homomorphism ξ, the inverse operation in the original group G is mirrored or preserved in the group M as well

Now, in M, $\xi(I)$ must have an inverse. Let us use it to multiply this equation (say from the right). Thanks to associativity, we then have

$$i = \xi(I)(\xi(I))^{-1} = (\xi(I)\xi(I))(\xi(I))^{-1} = \xi(I)\left(\xi(I)(\xi(I))^{-1}\right) = \xi(I)i = \xi(I),$$

as asserted.

5.4.4 Preserving the Inverse Operation

Thanks to the above properties, the homomorphism also preserves the inverse operation (Fig. 5.2): while g maps to $\xi(g)$, the inverse of g in G must map to the inverse of $\xi(g)$ in M. Indeed,

$$i = \xi(I) = \xi(gg^{-1}) = \xi(g)\xi(g^{-1}),$$

so

$$(\xi(g))^{-1} = \xi\left(g^{-1}\right)$$

in M, as asserted.

Still, recall that the homomorphism is not necessarily one-to-one. Thus, I may be not the only element that maps to i. The elements that map to i, including I, form a special subset: the kernel.

5.4.5 Kernel of a Mapping

The kernel of a mapping ξ (not necessarily a homomorphism) contains those elements that are mapped to the unit element i in M:

$$\xi^{-1}(i) \equiv \{g \in G \mid \xi(g) = i\}$$

(Fig. 5.3). Recall that this is just a notation for a subset of G. It means no inverse: after all, ξ is not necessarily invertible.

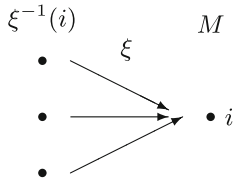


Fig. 5.3 The homomorphism ξ maps its entire kernel (on the left) to the unit element $i \in M$ (on the right)

If ξ is *onto* M, then the kernel must be nonempty (Sect. 5.4.1). For example, in the present context, in which ξ is a homomorphism, the kernel must contain at least one element: the unit element I (Sect. 5.4.3).

Unfortunately, even if the original homomorphism ξ maps G *onto* M (while preserving multiplication), something is still missing: ξ is not necessarily one-to-one, so it is not necessarily invertible. For example, the kernel may contain a few elements. This means that G and M do not exactly mirror each other any more. Fortunately, this could be fixed: ξ could still be modified to form an invertible mapping. For this purpose, we need a new concept: subgroup.

5.5 The Center and Kernel Subgroups

5.5.1 Subgroup

A subgroup is a subset that is a group in its own right. In other words, a subset $S \subset G$ is a subgroup if:

1. S is closed under multiplication:

$$s, s' \in S \Rightarrow ss' \in S.$$

2. S contains the unit element:

$$I \in S.$$

3. S is closed under the inverse operation:

$$s \in S \Rightarrow s^{-1} \in S.$$

Fortunately, S also inherits the associative law:

$$s, s', s'' \in S \Rightarrow s, s', s'' \in G \Rightarrow (ss')s'' = s(s's'').$$

Thus, S is indeed a legitimate group in its own right, as required.

As a matter of fact, to make sure that a subset S is also a subgroup, it is sufficient to check just two conditions:

1. S is closed under division:

$$s, s' \in S \Rightarrow ss'^{-1} \in S.$$

2. S contains the unit element:

$$I \in S.$$

Indeed, under these conditions, S is also closed under the inverse operation:

$$s \in S \Rightarrow s^{-1} = Is^{-1} \in S.$$

As a result, S is also closed under multiplication:

$$s, s' \in S \Rightarrow s, s^{-1} \in S \Rightarrow ss' = s\left(s'^{-1}\right)^{-1} \in S.$$

Thus, the original three conditions hold, so S is indeed a legitimate subgroup of G.

5.5.2 The Center Subgroup

Recall that our original group G is not necessarily commutative: it may contain "bad" elements that do not commute with each other. Still, it may also contain "good" elements that do commute with every element. Let us place them in a new subset: the center C:

$$C \equiv \{c \in G \mid cg = gc \text{ for every } g \in G\}.$$

By now, we only know that C is a subset. Is it also a legitimate subgroup? Let us check: is it closed under multiplication? Well, let c and c' be two elements in C. Thanks to the associative law inherited from G, the product cc' commutes with every element $g \in G$:

$$(cc')g = c(c'g) = c(gc') = (cg)c' = (gc)c' = g(cc').$$

Thus, cc' is in C as well, as required.

Next, is the unit element I in C? Well, it does commute with every element $g \in G$:

$$Ig = g = gI.$$

Finally, is C closed under the inverse operation? To check on this, pick some element $c \in C$. This way, c commutes with every $g \in G$. What about its inverse c^{-1}? Does it commute with g as well? Well, take the equation

$$cg = gc,$$

and multiply it by c^{-1} from the right. Thanks to associativity,

$$(cg)c^{-1} = (gc)c^{-1} = g(cc^{-1}) = gI = g.$$

Now, multiply this by c^{-1} from the left:

$$c^{-1}\left((cg)c^{-1}\right) = c^{-1}g.$$

Thanks again to associativity,

$$gc^{-1} = I\left(gc^{-1}\right) = \left(c^{-1}c\right)\left(gc^{-1}\right) = c^{-1}\left(c\left(gc^{-1}\right)\right) = c^{-1}\left((cg)c^{-1}\right) = c^{-1}g.$$

Thus, c^{-1} does commute with every $g \in G$, so it does belong to C as well, as required. This proves that the center C is not only a subset but also a legitimate subgroup of G.

5.5.3 *The Kernel Subgroup*

Thanks to our original homomorphism $\xi : G \to M$, G has yet another interesting subset: the kernel $\xi^{-1}(i)$ (Sect. 5.4.5). Is it a subgroup as well?

Well, let us check: is it closed under multiplication? Well, for every two elements $g, g' \in \xi^{-1}(i)$,

$$\xi(gg') = \xi(g)\xi(g') = ii = i,$$

so their product is in the kernel as well:

$$gg' \in \xi^{-1}(i).$$

Next, does the unit element $I \in G$ belong to $\xi^{-1}(i)$ as well? Yes, it sure does (Sect. 5.4.3).

Finally, is the kernel also closed under the inverse operation? Well, for each element $g \in \xi^{-1}(i)$, its inverse g^{-1} is in $\xi^{-1}(i)$ as well:

$$\xi(g^{-1}) = (\xi(g))^{-1} = i^{-1} = i$$

(Sect. 5.4.4). This proves that the kernel of ξ is not only a subset but also a legitimate subgroup of G.

To use the kernel fully, we need some background in set theory.

5.6 Equivalence Classes

5.6.1 Equivalence Relation in a Set

In a set G (not necessarily a group), what is a relation? Well, a relation is actually a subset of G^2: it may contain ordered pairs of the form (g, g'). We then say that g is related to g':

$$g \sim g'.$$

What is an equivalence relation? Well, this is a special kind of relation: it has three properties:

1. Reflexivity: Every element $g \in G$ is related to itself:

$$g \sim g.$$

2. Symmetry: For every two elements $g, g' \in G$, if g is related to g', then g' is related to g as well:

$$g \sim g' \Rightarrow g' \sim g.$$

3. Transitivity: For every three elements $g, g', g'' \in G$, if g is related to g' and g' is related to g'', then g is related to g'' as well:

$$g \sim g',\ g' \sim g'' \Rightarrow g \sim g''.$$

Let us use this to decompose the original set G.

5.6.2 Decomposition into Equivalence Classes

Thanks to the equivalence relation, we can now decompose the original set G (which may be a group or not) in terms of disjoint equivalence classes (Fig. 5.4). In this decomposition, each equivalence class contains those elements that are related (or equivalent) to each other. In particular, each element $g \in G$ belongs to one equivalence class only:

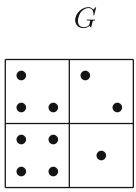


Fig. 5.4 The original set G is decomposed (or split) into disjoint lines, or equivalence classes

$$g \in \psi_g \equiv \left\{ g'' \in G \mid g'' \sim g \right\}.$$

Indeed, thanks to reflexivity, $g \sim g$, so $g \in \psi_g$. Now, could g belong to yet another equivalence class of the form $\psi_{g'}$, for any other $g' \in G$? Well, if it did, then this would mean that $g \sim g'$. Thanks to transitivity, we would then have

$$g'' \sim g \Rightarrow g'' \sim g',$$

so

$$\psi_g \subset \psi_{g'}.$$

On the other hand, thanks to symmetry, we would also have $g' \sim g$. Thanks again to transitivity, we would then have

$$g'' \sim g' \Rightarrow g'' \sim g,$$

so

$$\psi_{g'} \subset \psi_g.$$

In summary, we would have

$$\psi_{g'} = \psi_g,$$

so ψ_g is actually the only equivalence class containing g, as asserted.

5.6.3 Family of Equivalence Classes

Let us look at the family (or set) of these disjoint equivalence classes:

$$\left\{ \psi_g \mid g \in G \right\}.$$

In this family, it is assumed that there is no duplication: each equivalence class of the form ψ_g appears only once, with g being some representative picked arbitrarily

from it. Furthermore, in this family, each equivalence class is an individual element, not a subset. To pick its inner elements and obtain G one again, one must apply the union operation:

$$G = \cup_{g \in G} \psi_g.$$

This is the union of all the ψ_g's: it contains all their inner elements.

5.6.4 Equivalence Relation Induced by a Subgroup

So far, the discussion was rather theoretical. After all, we never specified what the equivalence relation was. Now, let us go back into business. Assume again that G is not just a set but actually a group, as before, with a subgroup $S \subset G$. This way, S may help define (or induce) a new relation: for every two elements g' and g in G, g' is related to g if their "ratio" is in S:

$$g' \sim g \text{ if } g'g^{-1} \in S.$$

In other words, there is an element $s \in S$ that can multiply g and produce g':

$$g' = sg.$$

In this case, $s = g'g^{-1}$ is unique.

Is this an equivalence relation? Well, let us check: is it reflexive? In other words, given a $g \in G$, is it related to itself? Fortunately, it is

$$gg^{-1} = I \in S,$$

as required.

Next, is it symmetric? Well, consider two elements $g', g \in G$. Assume that $g' \sim g$ or $g'g^{-1} \in S$. Since S is a subgroup, it also contains the inverse element:

$$gg'^{-1} = \left(g'g^{-1}\right)^{-1} \in S,$$

so $g \sim g'$ as well, as required.

Finally, is it transitive? Well, consider three elements $g, g', g'' \in G$. Recall that S is closed under multiplication: if $g'g^{-1} \in S$ and $g''g'^{-1} \in S$, then their product is in S as well. For this reason, thanks to associativity,

$$\begin{aligned} g''g^{-1} &= g''(Ig^{-1}) \\ &= g'' \left(\left(g'^{-1}g'\right)g^{-1}\right) \end{aligned}$$

$$
\begin{aligned}
&= g'' \left(g'^{-1} \left(g' g^{-1} \right) \right) \\
&= \left(g'' g'^{-1} \right) \left(g' g^{-1} \right) \\
&\in S
\end{aligned}
$$

as well, as required. In summary, this is indeed an equivalence relation in the original group G.

5.6.5 *Equivalence Classes Induced by a Subgroup*

With this new equivalence relation, how does an equivalence class look like? Well, consider a particular element $g \in G$. As discussed in Sect. 5.6.2, it is contained in one equivalence class only:

$$
\begin{aligned}
\psi_g &\equiv \left\{ g' \in G \;\middle|\; g' \sim g \right\} \\
&= \left\{ g' \in G \;\middle|\; g' g^{-1} \in S \right\} \\
&= \left\{ g' \in G \;\middle|\; g' g^{-1} = s \text{ for some } s \in S \right\} \\
&= \left\{ g' \in G \;\middle|\; g' = sg \text{ for some } s \in S \right\}.
\end{aligned}
$$

Thus, the equivalence class takes the special form

$$\psi_g = Sg \equiv \{ sg \;|\; s \in S \}.$$

5.7 The Factor Group

5.7.1 *The New Set G/S*

Let us place these equivalence classes as individual elements in a new set (or family):

$$\{ Sg \;|\; g \in G \}.$$

Unfortunately, in this family, there is some duplication: each equivalence class of the form Sg may appear many times. In fact, every two equivalent elements $g' \sim g$ introduce the same equivalence class $Sg' = Sg$ into the above family.

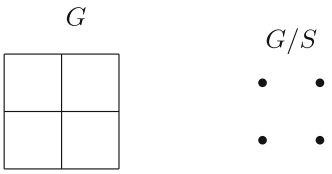


Fig. 5.5 Disjoint equivalence classes are considered as individual elements in G/S

To avoid this, one might want to drop all the duplicate copies of the form Sg'. This way, Sg appears only once, with g being some representative picked arbitrarily from it. The resulting family is called G/S (Fig. 5.5).

Why is this a suitable name? Because the original elements in G are regarded only up to multiplication (from the left) by just any element from S. This way, equivalent elements in G are united into one and the same element in G/S.

Thus, G/S is completely blind to any difference between equivalent elements $g' \sim g$ and can never distinguish between them. After all, in G/S, such elements coincide to form the same element $Sg' = Sg$.

5.7.2 Normal Subgroup

So far, G/S is just a set. To make it a group, we must define a proper multiplication between its elements. This is not always possible.

To guarantee that it is, we must also assume that S is normal in the sense that it "commutes" with every element $g \in G$:

$$Sg = gS \equiv \{gs \mid s \in S\}.$$

In other words, for every $s \in S$, there is an $s' \in S$ (dependent on both g and s) for which

$$gs' = sg.$$

In this case, $s' = g^{-1}sg$ is unique.

For example, S could be a subgroup of the center of G, defined in Sect. 5.5.2:

$$S \subset C \subset G.$$

In this case, $s' = s$ in the above equation.

What is so good about S being normal? Well, consider two elements $g, h \in G$. In G, their product is just gh. Now, let s and s' be two elements from S. What happens when g is replaced by the equivalent element sg, and h is replaced by the equivalent element $s'h$? Can we still calculate the product?

Well, since S is normal, there is an $s'' \in S$ such that

$$s''g = gs'.$$

Thanks to associativity, the product is now

$$(sg)(s'h) = s(g(s'h)) = s((gs')h) = s((s''g)h) = s(s''(gh)) = (ss'')(gh)).$$

Thus, the product gh has not changed much: it was just replaced by an equivalent element.

In summary, thanks to normality, multiplication is invariant under the equivalence relation. In other words, order does not matter: you could switch to equivalent elements and then multiply, or do things the other way around: first multiply, and then switch to the relevant equivalent element. Either way, you would get the same result. In terms of equivalence classes, this could be written as

$$(Sg)(Sh) = S(gh).$$

In G, this is an equality between two equivalence classes. In G/S, on the other hand, this is much more meaningful: an equality between two *elements*. Thus, this could be used as a new definition of a new kind of multiplication.

5.7.3 The Factor (Quotient) Group

In G/S, what are the individual elements? They are just equivalence classes from G. How to multiply them? We already know: consider two elements of the form

$$Sg,\ Sh\ \in G/S$$

(for some $g, h \in G$). What is their product in G/S? It is just

$$(Sg)(Sh) \equiv S(gh).$$

To be well defined, this product must not depend on the particular representatives g or h: replacing each of them by an equivalent element must not affect the result. Since S is normal, this is indeed the case.

Still, this is not good enough. To be a legitimate group, G/S must also be associative. To check on this, consider three elements of the form

$$Sg,\ Sh,\ Sk\ \in G/S$$

(for some $g, h, k \in G$). Since G is associative and S is normal,

$$(SgSh)Sk = S(gh)Sk = S((gh)k) = S(g(hk)) = SgS(hk) = Sg(ShSk),$$

so G/S is associative as well.

Still, this is not the end of it. To be a legitimate group, G/S must also contain a unit element. For this job, let us choose

$$S = SI \in G/S.$$

Indeed, for any element $Sg \in G/S$,

$$SgSI = S(gI) = Sg = S(Ig) = (SI)Sg.$$

Still, this is not the end. To be a group, G/S must also be closed under the inverse operation. Fortunately, for every element of the form $Sg \in G/S$, the inverse is just $Sg^{-1} \in G/S$. Indeed,

$$SgSg^{-1} = S\left(gg^{-1}\right) = SI = S.$$

This is the end of it: G/S is indeed a legitimate group. Let us use it to modify the original homomorphism ξ defined in Sect. 5.4.2, and make a new one-to-one isomorphism.

5.7.4 Is the Kernel Normal?

Consider again the homomorphism

$$\xi : G \to M.$$

As pointed out in Sect. 5.4.2, ξ is not necessarily one-to-one, so it is not necessarily invertible: there may be some element $m \in M$ with a few elements in its origin: $|\xi^{-1}(m)| > 1$. How to fix this?

Fortunately, ξ can still be modified to produce a new one-to-one homomorphism: a new isomorphism. To do this, consider the kernel $\xi^{-1}(i)$, defined in Sect. 5.4.5. As discussed in Sect. 5.5.3, this is a legitimate subgroup of G. Therefore, just like S above, it can be placed in the denominator of the new set

$$G/\xi^{-1}(i).$$

By now, this is a mere set. Is it also a legitimate group? In other words, is the kernel normal (commute with every element $g \in G$):

$$\xi^{-1}(i)g = g\xi^{-1}(i)?$$

To check on this, pick some x in the kernel. Then, look at the triple product $g^{-1}xg$. Is it in the kernel as well? It sure is

$$\xi\left(g^{-1}xg\right) = \xi\left(g^{-1}\right)\xi(x)\xi(g) = \xi\left(g^{-1}\right)i\xi(g) = \xi\left(g^{-1}\right)\xi(g) = \xi\left(g^{-1}g\right) = \xi(I) = i.$$

Thus, $G/\xi^{-1}(i)$ is not only a set but also a new group: the factor group (Sects. 5.7.2–5.7.3).

5.7.5 *Isomorphism on the Factor Group*

Fortunately, ξ does not distinguish between equivalent elements in G. For example, if g and g' are equivalent to each other, then there must be an element $s \in \xi^{-1}(i)$ for which

$$g' = sg$$

(Sect. 5.6.5). Therefore, we must have

$$\xi(g') = \xi(sg) = \xi(s)\xi(g) = i\xi(g) = \xi(g).$$

Thus, ξ maps the entire equivalence class to one and the same element in M. This observation can now be used to form a new one-to-one mapping from the factor group $G/\xi^{-1}(i)$ onto M. In this new mapping, the entire equivalence class will be mapped as a whole to its image: some element in M. This is indeed invertible: in the inverse mapping, this image element will be mapped back to the original equivalence class.

More precisely, the new isomorphism

$$\Xi : G/\xi^{-1}(i) \to M$$

is defined by

$$\Xi\left(\xi^{-1}(i)g\right) \equiv \xi(g), \quad g \in G$$

(Fig. 5.6). Why is this well defined? Because it does not depend on the particular representative g picked arbitrarily from the equivalence class. After all, one could

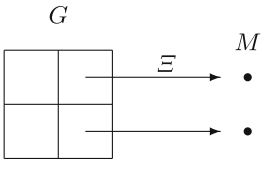


Fig. 5.6 The new mapping Ξ maps disjoint equivalence classes (or distinct elements in the factor group) to distinct elements in M

pick any equivalent element $g' \sim g$ and still have

$$\Xi\left(\xi^{-1}(i)g'\right) = \xi(g') = \xi(g),$$

as discussed above.

5.7.6 The Fundamental Theorem of Homomorphism

Like the original homomorphism, Ξ is *onto* M. Indeed, for each individual element $m \in M$, there is some element $g \in G$ for which $\xi(g) = m$. Therefore,

$$\Xi\left(\xi^{-1}(i)g\right) = \xi(g) = m,$$

as required.

Furthermore, Ξ is one-to-one. Indeed, consider two distinct elements $\xi^{-1}(i)g$ and $\xi^{-1}(i)g'$ in $G/\xi^{-1}(i)$. Clearly, $g' \not\sim g$, so $g'g^{-1} \notin \xi^{-1}(i)$, so

$$\xi(g')(\xi(g))^{-1} = \xi(g')\xi\left(g^{-1}\right) = \xi(g'g^{-1}) \neq i$$

(Sect. 5.4.4). As a result,

$$\Xi\left(\xi^{-1}(i)g\right) = \xi(g) \neq \xi(g') = \Xi\left(\xi^{-1}(i)g'\right),$$

as asserted. In summary, Ξ is indeed invertible.

Fortunately, Ξ also preserves algebraic operations (Fig. 5.7). Indeed, for every two elements $g, g' \in G$, we have

$$\begin{aligned}\Xi\left(\left(\xi^{-1}(i)g\right)\left(\xi^{-1}(i)g'\right)\right) &= \Xi\left(\xi^{-1}(i)(gg')\right)\\ &= \xi(gg')\\ &= \xi(g)\xi(g')\\ &= \Xi\left(\xi^{-1}(i)g\right)\Xi\left(\xi^{-1}(i)g'\right).\end{aligned}$$

In summary, the factor group $G/\xi^{-1}(i)$ is isomorphic to M:

$$G/\xi^{-1}(i) \simeq M.$$

In summary, these groups mirror each other and have the same algebraic structure.

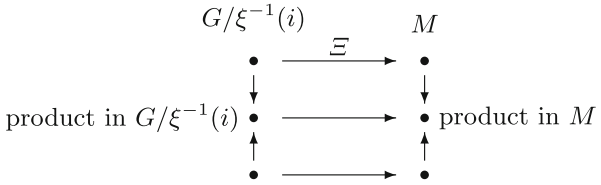


Fig. 5.7 The new isomorphism Ξ from the factor group onto M (the horizontal arrows) preserves (or mirrors) the algebraic operation (the vertical arrows)

This is the fundamental theorem of homomorphism, or the first isomorphism theorem. Later on, we will use it to prove two other important theorems: the second and third isomorphism theorems. Before doing this, however, we use it in our original application: Moebius transformations.

5.8 Geometrical Applications

5.8.1 Application in Moebius Transformations

Let us apply the above theory to the special case in which G is the group of 2×2 nonsingular complex matrices, and M is the group of invertible Moebius transformations (Sect. 5.1.4). In this case, the unit element $I \in G$ is the 2×2 identity matrix

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and the unit element $i \in M$ is the identity mapping

$$z \to z \text{ and } \infty \to \infty,$$

for which $a = d \neq 0$ and $b = c = 0$ in Sect. 5.1.2.

In this case, what is the center of G? Well, it contains the nonzero scalar multiples of the 2×2 identity matrix:

$$C \equiv \{c \in G \mid cg = gc \text{ for all } g \in G\} = \{zI \mid z \in \mathbb{C},\ z \neq 0\}$$

(see exercises below). From Sect. 5.5.2, C is indeed a subgroup of G.

Our job is to design a suitable homomorphism

$$\xi : G \to M,$$

with the kernel

$$\xi^{-1}(i) = C.$$

For this purpose, we need some geometrical preliminaries.

5.8.2 Two-Dimensional Vector Set

Let us use the above center subgroup $C \subset G$ to define an equivalence relation in the set

$$V \equiv \mathbb{C}^2 \setminus \{(0,0)\}.$$

Here, "\" means "minus" the set that contains the origin. Thus, V contains the nonzero two-dimensional complex vector. Although V is not a group but a mere set, it can still be decomposed in terms of disjoint equivalence classes, as in Sects. 5.6.1–5.6.3.

For every two vectors $v, v' \in V$, let

$$v' \sim v \text{ if } v' = cv \text{ for some } c \in C.$$

Since C is defined as in Sect. 5.8.1, this means that v' is just a nonzero scalar multiple of v. Still, in principle, the same could be done with other normal subgroups as well.

It is easy to see that this is an equivalence relation in the original set V. Indeed:

- For every $v \in V$, $v = Iv$. This shows reflexivity.
- Furthermore, for every $v, v' \in V$, if $v' = cv$ (for some $c \in C$), then $v = c^{-1}v'$. This shows symmetry.
- Finally, for every $v, v', v'' \in V$, if $v' = cv$ and $v'' = c'v'$ (for some $c, c' \in C$), then $v'' = c'(cv) = (c'c)v$. This shows transitivity.

This proves that the above relation is indeed a legitimate equivalence relation in V.

5.8.3 Geometrical Decomposition into Planes

Consider the nonzero two-dimensional complex vector

$$v \equiv \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \in V.$$

What is its equivalence class? Well, it takes the form

$$Cv \equiv \{cv \mid c \in C\} = \{zv \mid z \in \mathbb{C},\ z \neq 0\} = \left\{ z \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \mid z \in \mathbb{C},\ z \neq 0 \right\}.$$

In geometrical terms, this is just the oblique plane spanned by the vector $v = (c_1, c_2)^t$ (Fig. 5.8).

5.8.4 Family of Planes

Together, all such planes make the family

$$V/C \equiv \{Cv \mid v \in V\}.$$

To avoid duplication, each individual plane of the form Cv appears only once, with v being some representative picked arbitrarily from it. Note that, unlike in Sect. 5.7.3, here V/C is just a set, not a group. This is because the original set V is not a group in the first place.

In V, Cv is a subset: an oblique plane (Fig. 5.8). In the new set V/C, on the other hand, Cv is just an element. To obtain the original set V once again, one must therefore apply the union operation, to pick the inner vectors from each plane:

$$V = \cup_{Cv \in V/C} Cv$$

(Sect. 5.6.3).

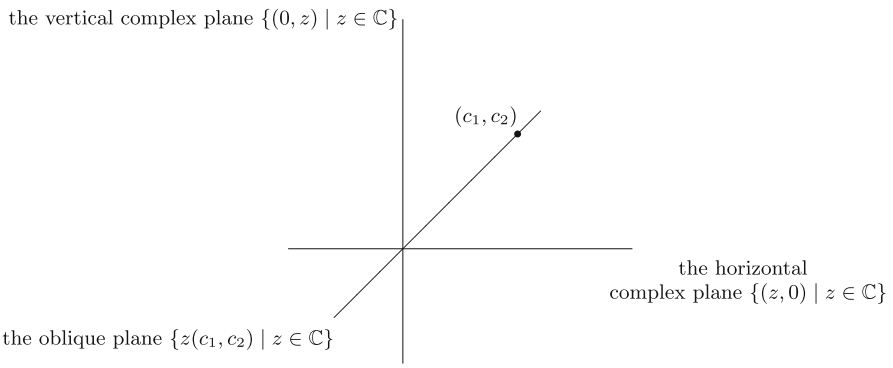


Fig. 5.8 A picture of $\mathbb{C}^2$: the two-dimensional complex vector $(c_1, c_2)^t$ spans an oblique plane—the equivalence class $C(c_1, c_2)^t$

5.8.5 Action of Factor Group

The original group G acts on the set V: each element $g \in G$ acts on each $v \in V$, transforming it into the new vector gv. Thanks to the above decomposition, this also applies to complete planes: the factor group G/C acts on v/C.

Indeed, an element of the form $Cg \in G/C$ acts not only on individual vectors of the form $v \in V$ but also on complete planes of the form Cv:

$$Cg(Cv) \equiv C(gv).$$

Why is this a legitimate definition? Because it is independent of the particular representative g or v. Indeed, since C is normal, replacing g by cg and v by $c'v$ (for some $c, c' \in C$) changes nothing:

$$C(cg)(C(c'v)) \equiv C(cgc'v) = C(cc'gv) = C(gv).$$

5.8.6 Composition of Functions

Thanks to the above action, each element of the form $Cg \in G/C$ can also be interpreted as a function

$$Cg : V/C \to V/C.$$

After all, the original algebraic operation in G/C, defined in Sect. 5.7.3 as

$$(Cg')(Cg) = C(g'g) \quad (g, g' \in G),$$

is mirrored well by function composition:

$$(Cg' \circ Cg)(Cv) = Cg'(Cg(Cv)) = Cg'(C(gv)) = C(g'gv) = C(g'g)(Cv),$$

for every $v \in V$.

5.8.7 Oblique Projection: Extended Cotangent

Let us define the oblique projection

$$P : V/C \to \mathbb{C} \cup \{\infty\}$$

by

$$P\left(C\begin{pmatrix}c_1\\c_2\end{pmatrix}\right) \equiv \begin{cases} c_1/c_2 & \text{if } c_2 \neq 0 \\ \infty & \text{if } c_2 = 0. \end{cases}$$

Fortunately, this definition is independent of the particular representative $(c_1, c_2)^t$. After all, for every nonzero complex number z, one may replace c_1 by zc_1 and c_2 by zc_2 and still have the same projection.

In geometrical terms, P can be viewed as an oblique projection on the horizontal plane

$$\{(z, 1) \mid z \in \mathbb{C}\}$$

(Fig. 5.9). This way, P actually extends the standard cotangent projection, to apply not only to real numbers but also to complex numbers.

The inverse mapping

$$P^{-1} : \mathbb{C} \cup \{\infty\} \to V/C$$

can now be defined simply by

$$P^{-1}(z) = \begin{cases} C\begin{pmatrix}z\\1\end{pmatrix} & \text{if } z \in \mathbb{C} \\ \\ C\begin{pmatrix}1\\0\end{pmatrix} & \text{if } z = \infty. \end{cases}$$

In what sense is this the inverse? Well, in two senses: on one hand,

$$P^{-1}P = CI = C$$

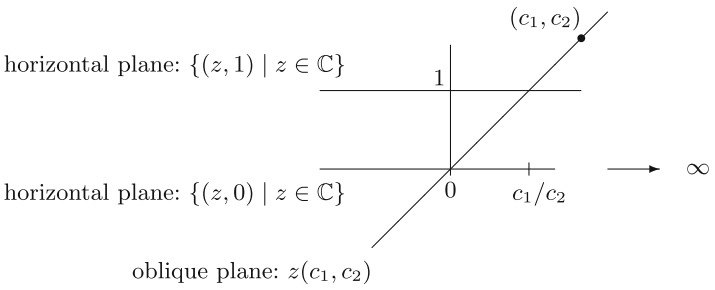


Fig. 5.9 The oblique projection P projects the oblique plane $C(c_1, c_2)^t$ to c_1/c_2. In particular, the horizontal complex plane $\{(z, 0) \mid z \in \mathbb{C}\}$ projects to ∞

is the unit element in G/C that leaves V/C unchanged: each oblique plane projects and then unprojects. On the other hand,

$$PP^{-1} = i$$

is the identity transformation in Sect. 5.8.1: each complex number unprojects and then projects back.

Let us use P and P^{-1} to associate the original Moebius transformation with the relevant 2×2 matrix.

5.8.8 *Homomorphism onto Moebius Transformations*

The association made in Sect. 5.1.4 takes now the form of a new homomorphism

$$\xi : G \to M,$$

from the group of 2×2 nonsingular complex matrices, onto the group of invertible Moebius transformations:

$$\xi(g) \equiv PCgP^{-1} \quad (g \in G).$$

Why is this a Moebius transformation? Well, look what happens to a complex number z: it transforms to

$$z \to PCgP^{-1}z,$$

or, in three stages,

$$z \to P^{-1}z \to CgP^{-1}z \to PCgP^{-1}z.$$

In other words, z first unprojects to an oblique plane:

$$z \to C\begin{pmatrix} z \\ 1 \end{pmatrix},$$

which is then multiplied from the left by

$$Cg \equiv C\begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and projects back:

$$C\begin{pmatrix} a & b \\ c & d \end{pmatrix} C\begin{pmatrix} z \\ 1 \end{pmatrix} = C\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} z \\ 1 \end{pmatrix}\right) = C\begin{pmatrix} az+b \\ cz+d \end{pmatrix} \to \frac{az+b}{cz+d},$$

as required.

Furthermore, what happens to the infinity point? Well, it first unprojects to a horizontal plane:

$$\infty \to C\begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

which then transforms by

$$C\begin{pmatrix} 1 \\ 0 \end{pmatrix} \to C\begin{pmatrix} a & b \\ c & d \end{pmatrix} C\begin{pmatrix} 1 \\ 0 \end{pmatrix} = C\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = C\begin{pmatrix} a \\ c \end{pmatrix},$$

which then projects back:

$$C\begin{pmatrix} a \\ c \end{pmatrix} \to \frac{a}{c},$$

as required.

Let us show that ξ is indeed a legitimate homomorphism. First, is it *onto* M? Well, as discussed in Sect. 5.1.4, every invertible Moebius transformation $m \in M$ has the explicit form

$$z \to \frac{az+b}{cz+d},$$

for some complex parameters a, b, c, and d, satisfying

$$ad - bc \neq 0.$$

As discussed above, this transformation could also be decomposed as

$$m = \xi(g) = PCgP^{-1},$$

where

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is a nonsingular matrix, with a nonzero determinant. So, ξ maps g to m, as required.

Finally, does ξ preserve algebraic operations? Well, thanks to the associativity of function composition (Sects. 5.8.5–5.8.6),

$$\begin{aligned}\xi(g')\xi(g) &= \left(PCg'P^{-1}\right)\left(PCgP^{-1}\right)\\ &= (PCg')\left(P^{-1}P\right)CgP^{-1}\\ &= P(Cg')(CI)(Cg)P^{-1}\\ &= P(Cg')(Cg)P^{-1}\\ &= PC(g'g)P^{-1}\\ &= \xi(g'g).\end{aligned}$$

This proves that ξ is indeed a legitimate homomorphism, as asserted.

5.8.9 The Kernel

To design a proper isomorphism as well, we must also have the kernel of ξ. Fortunately, this is just the center of G:

$$\xi^{-1}(i) = C = \{zI \mid z \in \mathbb{C},\ z \neq 0\}$$

(defined in Sect. 5.8.1). Let us prove this in two stages. First, let us show that

$$C \subset \xi^{-1}(i).$$

Indeed, for every $c \in C$, $\xi(c)$ is just the identity transformation that leaves every complex number unchanged:

$$\xi(c)(z) = PCcP^{-1}(z) = PCcC\begin{pmatrix} z \\ 1 \end{pmatrix} = PC\begin{pmatrix} z \\ 1 \end{pmatrix} = z,$$

and also leaves the infinity point unchanged:

$$\xi(c)(\infty) = PCcP^{-1}(\infty) = PCcC\begin{pmatrix} 1 \\ 0 \end{pmatrix} = PC\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \infty.$$

Thus, $\xi(c)$ is indeed the identity mapping, or the unit element in M:

$$\xi(c) = i \in M,$$

so

$$C \subset \xi^{-1}(i),$$

as asserted.

Conversely, let us also show that

$$\xi^{-1}(i) \subset C.$$

Indeed, if $\xi(g)$ is the identity transformation $z \to z$, then g must be a nonzero scalar multiple of the 2×2 identity matrix (see exercises below).

In summary,

$$\xi^{-1}(i) = C,$$

as asserted. This means that ξ does not distinguish between matrices that are a nonzero scalar multiple of each other. In view of the discussion in Sect. 5.1.4, ξ is indeed a good candidate to represent invertible Moebius transformations in terms of 2×2 nonsingular complex matrices.

5.8.10 Eigenvectors and Fixed Points

If $v \in V$ is an eigenvector of $g \in G$ with the eigenvalue $\lambda \in \mathbb{C}$:

$$gv = \lambda v,$$

then Cv contains eigenvectors only. After all, each element in Cv is a nonzero scalar multiple of v, so it must be an eigenvector as well, with the same eigenvalue λ. Furthermore, since g is nonsingular, λ must be nonzero.

In this case, Cv is a fixed point that remains unchanged under the action of Cg. Indeed, from the definitions in Sect. 5.8.5,

$$Cg(Cv) = C(gv) = C(\lambda v) = Cv.$$

Furthermore, in this case, PCv is a fixed point that remains unchanged under the Moebius transformation $\xi(g)$:

$$\xi(g)(PCv) = PCgP^{-1}PCv = PCgCv = PCv.$$

5.8.11 Isomorphism onto Moebius Transformations

Let us now use the fundamental theorem of homomorphism (Sects. 5.7.5–5.7.6) to design a new isomorphism

$$\Xi : G/C \to M.$$

Naturally, it is defined by

$$\Xi(Cg) \equiv \xi(g) = PCgP^{-1}, \quad Cg \in G/C.$$

This way, Ξ does not distinguish between matrices that are a nonzero scalar multiple of each other: it views them as one element in G/C and maps them as a whole to the same Moebius transformation.

This is indeed a proper group representation. To see this, look at things the other way around. A Moebius transformation is not easy to store on the computer. To do this, use Ξ^{-1}. In fact, each element $m \in M$ is mirrored (or represented) by the unique element

$$\xi^{-1}(m) = P^{-1}mP = C\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G/C.$$

After all, a matrix is easy to store on the computer. Furthermore, Ξ^{-1} also preserves the algebraic operation: it mirrors it by by matrix product, easy to calculate on the computer.

5.9 Application in Continued Fractions

5.9.1 Continued Fractions

Let us use the above theory to define a continued fraction. For $k = 1, 2, 3, \ldots$, consider the Moebius transformations

$$m_k(z) \equiv \frac{a_k}{z + b_k},$$

where a_k and b_k are some nonzero complex numbers (known as the coefficients). For $n = 1, 2, 3, \ldots$, consider the compositions

$$\begin{aligned} f_1 &\equiv m_1 \\ f_2 &\equiv m_1 \circ m_2 \\ f_3 &\equiv m_1 \circ m_2 \circ m_3 \\ f_n &\equiv m_1 \circ m_2 \circ m_3 \circ \cdots \circ m_n. \end{aligned}$$

As a matter of fact, this is just a mathematical induction:

$$f_n \equiv \begin{cases} m_1 & \text{if } n = 1 \\ f_{n-1} \circ m_n & \text{if } n > 1. \end{cases}$$

Now, let us apply these functions to $z = 0$:

$$f_1(0),\ f_2(0),\ f_3(0),\ \ldots.$$

These are the approximants. We say that they converge (in the wide sense) to the continued fraction f if

$$f_n(0) \to_{n\to\infty} f \in \mathbb{C} \cup \{\infty\}$$

[31, 32]. In particular, if $f \in \mathbb{C}$ is a concrete number, then the convergence is in the strict sense as well. If, on the other hand, $f = \infty$ is no number, then the convergence is in the wide sense only.

5.9.2 Algebraic Formulation

To study the convergence, let us use the factor group

$$G/C \simeq M,$$

with the isomorphism Ξ in Sect. 5.8.11. For this purpose, let us define the new matrices in Sect. 5.1.4:

$$g_k \equiv \begin{pmatrix} 0 & a_k \\ 1 & b_k \end{pmatrix}.$$

This way,

$$m_k = \Xi(Cg_k) = PCg_kP^{-1}.$$

For example, both sides of this equation could be applied to the complex number $0 \in \mathbb{C}$:

$$m_k(0) = PCg_kP^{-1}(0) = PCg_kC\begin{pmatrix} 0 \\ 1 \end{pmatrix} = PCg_k\begin{pmatrix} 0 \\ 1 \end{pmatrix} = PC\begin{pmatrix} a_k \\ b_k \end{pmatrix} = \frac{a_k}{b_k},$$

in agreement with the original definition of m_k.

5.9.3 The Approximants

Let us use the isomorphism Ξ to obtain the composition f_n as well:

$$\begin{aligned}\Xi(Cg_1g_2\cdots g_n) &= \Xi(Cg_1Cg_2\cdots Cg_n)\\ &= \Xi(Cg_1)\Xi(Cg_2)\cdots\Xi(Cg_n)\\ &= m_1\circ m_2\circ\cdots\circ m_n\\ &= f_n.\end{aligned}$$

Both sides of this equation could now be applied to $0\in\mathbb{C}$:

$$\begin{aligned}PCg_1g_2\cdots g_n\begin{pmatrix}0\\1\end{pmatrix} &= PCg_1g_2\cdots g_nP^{-1}(0)\\ &= \Xi(Cg_1g_2\cdots g_n)(0)\\ &= f_n(0).\end{aligned}$$

In other words, the approximant $f_n(0)$ is just the ratio between the upper-right and lower-right elements in the matrix product $g_1g_2\cdots g_n$:

$$f_n(0)=\frac{(g_1g_2\cdots g_n)_{1,2}}{(g_1g_2\cdots g_n)_{2,2}}.$$

This observation will be useful below.

5.9.4 Algebraic Convergence

Actually, this matrix product is defined by mathematical induction on $n = 1, 2, 3, \ldots$:

$$g_1g_2\cdots g_n\equiv\begin{cases}g_1 & \text{if } n=1\\ ((g_1g_2\cdots g_{n-1})\,g_n & \text{if } n>1.\end{cases}$$

Recall that g_n is a special 2×2 matrix: its first column is just the standard unit vector $(0, 1)^t$. For this reason, the above products also have a special property: the first column in $g_1g_2\cdots g_n$ is the same as the second column in $g_1g_2\cdots g_{n-1}$:

$$g_1g_2\cdots g_n\begin{pmatrix}1\\0\end{pmatrix}=(g_1g_2\cdots g_{n-1})\left(g_n\begin{pmatrix}1\\0\end{pmatrix}\right)=g_1g_2\cdots g_{n-1}\begin{pmatrix}0\\1\end{pmatrix}.$$

Thus, if convergence indeed takes place as $n\to\infty$, then both columns in $g_1g_2\cdots g_n$ must be nearly proportional to each other: the ratio between the upper and lower components must approach the same limit f.

Note that these ratios remain unchanged upon multiplying the original product $g_1g_2\cdots g_n$ by a nonsingular diagonal matrix from the right. Thus, the continued

fraction f exists if and only if there exist diagonal matrices $D_n \in G$ for which

$$g_1 g_2 \cdots g_n D_n \to_{n\to\infty} (v \mid v),$$

for some

$$v = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \in V.$$

What is the meaning of this convergence? Well, it is interpreted elementwise: there are actually two independent limit processes here—one for the upper-right element and another one for the lower-right element. Thanks to this convergence, the required continued fraction f can now be obtained by

$$f \equiv \lim_{n\to\infty} f_n(0) = \lim_{n\to\infty} PCg_1 g_2 \cdots g_n \begin{pmatrix} 0 \\ 1 \end{pmatrix} = PCv = \frac{c_1}{c_2} \in \mathbb{C} \cup \{\infty\}.$$

Thus, to guarantee convergence, one only needs to design suitable diagonal matrices $D_n \in G$, in such a way that $g_1 g_2 \cdots g_n D_n$ converge elementwise to a singular matrix of the form $(v \mid v) \notin G$, for some $v \in V$. This can also be written more concisely as

$$Cg_1 g_2 \cdots g_n \to_{n\to\infty} (Cv \mid Cv).$$

This means that G/C is not closed: elements from it could converge to a limit outside it [73]. In this case, if the second component in v is nonzero:

$$c_2 \neq 0,$$

then the convergence is also in the strict sense. Otherwise, it is in the wide sense only.

5.10 Isomorphism Theorems

5.10.1 The Second Isomorphism Theorem

By now, we are rather experienced in "playing" with groups. Let us use the fundamental theorem of homomorphism (Sect. 5.7.6) to prove another important theorem in group theory: the second isomorphism theorem, used later in projective geometry.

For this purpose, let G be a group. Let $T \subset G$ be a subgroup (normal or not). Let $S \subset G$ be a normal subgroup.

Look at their intersection: $T \cap S$. Is this a legitimate group? Well, it certainly contains the unit element. After all, $I \in T$, and $I \in S$. Now, is it closed under multiplication? Well, to check on this, let $s, s' \in T \cap S$. In this case, $ss' \in T$, and $ss' \in S$, as required. Finally, is it closed under the inverse operation? Well, to check on this, let $s \in T \cap S$. In this case, $s^{-1} \in T$, and $s^{-1} \in S$, as required. In summary, $T \cap S$ is indeed a legitimate subgroup of T.

Still, is it normal? Well, to check on this, let $s \in T \cap S$, and $t \in T$. Since $S \subset G$ is normal, there is an $s' \in S$ such that $st = ts'$. Fortunately, $s' = t^{-1}st \in T$, so $s' \in T \cap S$, as required. In summary, $T \cap S$ is indeed a normal subgroup of T. Later on, we will use this property to design the factor group.

Before doing this, let us define a new group. What is the product of T times S? Well, it contains those products of an element from T times an element from S:

$$TS \equiv \{ts \ \mid \ t \in T, \ s \in S\} \subset G.$$

Is this a legitimate group? Well, it certainly contains the unit element. After all, $I \in T$, and $I \in S$. Now, is it closed under multiplication? Well, to check on this, let $ts, t's' \in TS$. Since S is normal, $st' = t's''$, for some $s'' \in S$. Thus,

$$(ts)(t's') = t(st')s' = t(t's'')s' = (tt')(s''s') \in TS,$$

as required.

Finally, is it closed under the inverse operation? Well, to check on this, let $ts \in TS$. Now, since S is normal, $s^{-1}t^{-1} = t^{-1}s'''$, for some $s''' \in S$. Therefore,

$$(ts)(t^{-1}s''') = (ts)(s^{-1}t^{-1}) = t(ss^{-1})t^{-1} = tt^{-1} = I,$$

as required. So, TS is indeed a legitimate subgroup, although not necessarily normal.

If T was also normal, then TS would have been normal as well. Indeed, in this case, for each $g \in G$ and $ts \in TS$, there would be $t' \in T$ and $s' \in S$ for which

$$(ts)g = t(sg) = t(gs') = (tg)s' = (gt')s' = g(t's'),$$

as required. For our purpose, however, we do not need this, so T could be either normal or not.

S, on the other hand, is normal not only in G but also in TS. To see this, let $ts \in TS$, and $s' \in S$. Since $S \subset G$ is normal, and since $ts \in G$, there is an $s'' \in S$ such that $(ts)s' = s''(ts)$, as required.

The second isomorphism theorem says that

$$\frac{T}{T \cap S} \simeq \frac{TS}{S}.$$

To prove this, let us use the fundamental theorem of homomorphism. For this purpose, define the new homomorphism

$$\xi : T \to \frac{TS}{S} \quad \text{by} \quad \xi(t) \equiv St.$$

Is this a legitimate homomorphism? Well, is it *onto*? Fortunately, it certainly is: after all, every element $Sts \in TS/S$ can also be written as $Sts = Ss't = St$.

Furthermore, ξ certainly preserves the original algebraic operation in T. So, it is indeed a legitimate homomorphism.

Moreover, its kernel is $T \cap S$. So, we can now use the fundamental theorem of homomorphism to conclude that

$$\frac{T}{T \cap S} \simeq \frac{TS}{S},$$

as asserted.

5.10.2 The Third Isomorphism Theorem

Finally, let us use the fundamental theorem of homomorphism to prove yet another important theorem: the third isomorphism theorem. This will give us a new point of view: look at groups as simple fractions, and "canceled" a common factor out!

To see this, let G be a given group. Let $S, T \subset G$ be two normal subgroups. Assume also that $S \subset T$.

Note that S is normal not only in G but also in T. Indeed, let $t \in T$, and $s \in S$. Since $t \in G$, there is an $s' \in S$ such that $st = ts'$.

Now, consider $T/S \subset G/S$. Is it a legitimate subgroup? Well, it certainly contains the unit element: S. Still, is it closed under multiplication? To check on this, let $St, St' \in T/S$. Fortunately, $StSt' = S(tt')$ is in T/S as well, as required.

Furthermore, is it closed under the inverse operation? Fortunately, it is: the inverse element St^{-1} is in T/S as well.

So, $T/S \subset G/S$ is a legitimate subgroup. Is it normal? Well, to check on this, let $g \in G$, and $t \in T$. Since $T \subset G$ is normal, there is a $t' \in T$ such that

$$StSg = S(tg) = S(gt') = SgSt',$$

as required.

The third isomorphism theorem tells us that, as in simple fractions, S could be "canceled out:"

$$G/T \simeq \frac{G/S}{T/S}.$$

To prove this, let us use the fundamental theorem of homomorphism. For this purpose, define the new homomorphism

$$\xi : G \to G/S \quad \text{by} \quad \xi(g) \equiv Sg, \quad g \in G.$$

On top of this, define yet another homomorphism:

$$\xi' : G/S \to \frac{G/S}{T/S} \quad \text{by} \quad \xi'(Sg) \equiv (T/S)Sg, \quad Sg \in G/S.$$

Now, consider the composite homomorphism

$$\xi'\xi : G \to \frac{G/S}{T/S}.$$

What is its kernel? Well, it must include T. Indeed, in T/S, a typical element is of the form St (for some $t \in T$). Therefore,

$$\xi'\xi(t) = \xi'(St) = (T/S)St = T/S,$$

which is just the unit element in $(G/S)/(T/S)$.

By now, we have seen that the kernel of $\xi'\xi$ must include T. Must it also be included in T? Yes, it must! Indeed, if $g \in G \setminus T$, then

$$\xi'\xi(g) = \xi'(Sg) = (T/S)Sg \neq T/S,$$

because $Sg \notin T/S$.

Thanks to the fundamental theorem of homomorphism, we therefore have

$$G/T \simeq \frac{G/S}{T/S},$$

as asserted. In the next chapter, we will use the isomorphism theorems in projective geometry.

5.11 Exercises

1. Recall that G is the group of 2×2 nonsingular complex matrices. Let A and B be two matrices in G, denoted by

$$A \equiv \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} \quad \text{and} \quad B \equiv \begin{pmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{pmatrix}.$$

Assume that B is also a diagonal matrix:

$$b_{1,2} = b_{2,1} = 0.$$

Show that the upper-right element in the product AB is

$$(AB)_{1,2} = b_{2,2}a_{1,2}.$$

2. Show that the upper-right element in the product BA is

$$(BA)_{1,2} = b_{1,1}a_{1,2}.$$

3. Assume also that A and B commute with each other:

$$AB = BA.$$

Conclude that

$$b_{2,2}a_{1,2} = (AB)_{1,2} = (BA)_{1,2} = b_{1,1}a_{1,2}.$$

4. Assume also that B is nonconstant:

$$b_{1,1} \neq b_{2,2}.$$

Conclude that A must be lower triangular:

$$a_{1,2} = 0.$$

5. Similarly, show that the lower-left element in the product AB is

$$(AB)_{2,1} = b_{1,1}a_{2,1}.$$

6. Similarly, show that the lower-left element in the product BA is

$$(BA)_{2,1} = b_{2,2}a_{2,1}.$$

7. Conclude that, if A and B commute with each other, then

$$b_{1,1}a_{2,1}(AB)_{2,1} = (BA)_{2,1} = b_{2,2}a_{2,1}.$$

8. Conclude that, if B is also nonconstant, then A must be upper triangular:

$$a_{2,1} = 0.$$

9. Conclude that, if A commutes with the nonconstant diagonal matrix B, then A must be diagonal as well:

$$a_{1,2} = a_{2,1} = 0.$$

10. Conclude that, if A commutes with every matrix $B \in G$, then A must be diagonal.
11. Conclude that the center of G may contain diagonal matrices only.
12. Show that the diagonal matrices make a subgroup in G.
13. Conclude that the center must be a subgroup of that subgroup.
14. Assume now that A is diagonal:

$$a_{1,2} = a_{2,1} = 0,$$

and B is not necessarily diagonal. Show that the upper-right element in the product AB is

$$(AB)_{1,2} = a_{1,1}b_{1,2}.$$

15. Similarly, show that the upper-right element in the product BA is

$$(BA)_{1,2} = a_{2,2}b_{1,2}.$$

16. Conclude that, if A commutes with B, then

$$a_{1,1}b_{1,2} = (AB)_{1,2} = (BA)_{1,2} = a_{2,2}b_{1,2}.$$

17. Conclude that, if B is not lower triangular:

$$b_{1,2} \neq 0,$$

then A must be constant:

$$a_{1,1} = a_{2,2}.$$

18. Conclude that, if A commutes with every matrix $B \in G$, then A must be a constant diagonal matrix.
19. Conclude that the center of G may contain constant diagonal matrices only.
20. Conclude that the center of G may contain only nonzero scalar multiples of the identity matrix $I \in G$.
21. Show that the center of G contains all nonzero scalar multiples of $I \in G$.
22. Show that this is indeed a subgroup of G.
23. Recall that M is the group of invertible Moebius transformations, with composition as the algebraic operation. Show that the identity mapping

$$i(z) \equiv \frac{az + b}{cz + d} \equiv z$$

is indeed the unique unit element in M:

$$im = mi = m, \quad m \in M.$$

(Recall that i has nothing to do with the imaginary number $\sqrt{-1}$, denoted often by the same letter i.)

24. Show that, in the above formulation of i, in the numerator, we must have $a \neq 0$.
25. Show also that, in the above formulation of i, in the numerator, we must have $b = 0$. Hint:

$$i\left(\frac{-b}{a}\right) = \frac{a\frac{-b}{a} + b}{c\frac{-b}{a} + d} = \frac{-b + b}{\frac{ad-bc}{a}} = \frac{a(-b + b)}{ad - bc} = 0.$$

Now, since i is the identity mapping, we must also have $-b/a = 0$, or $b = 0$.

26. Show also that, in the above formulation of i, in the denominator, we must have $c = 0$. Hint:

$$i\left(\frac{-d}{c}\right) = \frac{a\frac{-d}{c} + b}{c\frac{-d}{c} + d} = \frac{-\frac{ad-bc}{c}}{-d + d} = \frac{ad - bc}{c(d - d)} = \infty.$$

Now, since i is the identity mapping, we must also have $-d/c = \infty$, or $d = 0$.

27. Conclude that, in the above formulation of i, we must also have $d = a$. Hint: Thanks to the previous exercises, $i(1) = a/d$. Now, since i is the identity mapping, we must also have $a/d = 1$, or $d = a$.
28. Recall that ξ is defined by

$$\xi(g) \equiv PCgP^{-1} \quad (g \in G)$$

(Sect. 5.8.8). Show that ξ maps G onto M.

29. Show that ξ preserves algebraic operations:

$$\xi(g)\xi(g') = \xi(gg'), \quad g, g' \in G.$$

30. Conclude that ξ is indeed a homomorphism.
31. From the above exercises about the identity mapping $i \in M$ and its algebraic formulation, conclude that the kernel of ξ, $\xi^{-1}(i)$, contains only nonzero scalar multiples of the 2×2 identity matrix $I \in G$.
32. Furthermore, show that $\xi^{-1}(i)$ contains all nonzero scalar multiples of $I \in G$.
33. Conclude that $\xi^{-1}(i)$ is the same as the center of G, calculated above.
34. Show that the upper-triangular matrices make a subgroup in Go.
35. Similarly, show that the lower-triangular matrices make a subgroup in Go.
36. Let n be some natural number. Consider the set G_n, containing the nonsingular $n \times n$ complex matrices. Show that I_n, the identity matrix of order n, is the unit element in G_n.

37. Show that G_n is indeed a group.
38. Show that the diagonal matrices make a subgroup in G_n.
39. Show that the center of G_n is

$$C_n \equiv \{zI_n \mid z \in \mathbb{C},\ z \neq 0\}.$$

40. Verify that C_n is indeed a group in its own right.
41. Conclude that C_n is indeed a legitimate subgroup of G_n.
42. Show that G_n acts on the set $\mathbb{C}^n \setminus \{\mathbf{0}\}$ (the n-dimensional space, without the origin).
43. Conclude that G_n can be interpreted as a group of vector functions defined on $\mathbb{C}^n \setminus \{\mathbf{0}\}$, with composition playing the role of the algebraic operation.
44. Show that the factor group G_n/C_n is indeed a group.
45. Show that G_n/C_n acts on $(\mathbb{C}^n \setminus \{\mathbf{0}\})/C_n$.
46. Conclude that G_n/C_n can be interpreted as a group of functions defined in $(\mathbb{C}^n \setminus \{\mathbf{0}\})/C_n$, with composition playing the role of the algebraic operation.
47. Show that the upper-triangular matrices make a subgroup in G_n.
48. Similarly, show that the lower-triangular matrices make a subgroup in G_n.
49. Use the discussions in Sects. 5.9.1–5.9.4 to study the convergence of periodic continued fractions. Let j be a fixed natural number (the period). Assume that the coefficients in the original continued fraction are periodic:

$$a_{k+j} = a_k \text{ and } b_{k+j} = b_k, \quad k \geq 1.$$

Find algebraic conditions on the eigenvalues of the matrix product

$$g_1 g_2 \cdots g_j$$

that guarantee convergence to the continued fraction f. The solution can be found in [73].

Chapter 6
Projective Geometry with Applications in Computer Graphics



What is a geometrical object? It is something that we humans could imagine and visualize. Still, in Euclidean geometry, a geometrical object is never defined explicitly, but only implicitly, in terms of relations, axioms, and logic ([24] and Chapter 6 in [74]).

For example, a point may lie on a line, and a line may pass through a point. Still, a line is not just a collection of points. It is much more than that: an independent object, which may contain a given point, or not.

This way, Euclidean geometry uses no geometrical intuition, but only logic. After all, logic is far more reliable than the human eye.

Still, logic does not give us sufficient order or method. For this purpose, linear algebra is far better suited. How to use it in geometry?

The answer is in analytic geometry: the missing link between geometry and algebra. For this purpose, we introduce a new axis system. This way, a line is no longer an independent object, but a set of points that satisfy a linear equation.

This way, points are the only low-level bricks. Lines, angles, and circles, on the other hand, are high-level objects, built of points. Since these points satisfy an algebraic equation, it is much easier to prove theorems.

In projective geometry, on the other hand, we move another step forward: we use not only analytic geometry but also group representation and topology [17, 101]. For this purpose, we use the isomorphism theorems proved above. This way, we have a complete symmetry between points and lines, viewed as algebraic (nongeometrical) objects: a line may now be interpreted as a point, and a point may be interpreted as a line.

In the projective plane, the original axioms in Euclidean geometry take a much more symmetric form. Just as every two distinct points make a unique line, every two distinct lines meet at a unique point. (In particular, two parallel lines meet at an infinity point.) After all, as pure algebraic objects, points and lines mirror each other, so their roles may interchange.

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Similarly, in the projective space, there is a complete symmetry between points and planes. Just as every three independent points make a unique plane, every three independent planes meet at a unique point. As a result, the roles of point and plane may interchange: a plane may be viewed as a single point, whereas a point may be viewed as a complete plane. After all, both can be interpreted as pure algebraic objects, free of any geometrical intuition.

In this chapter, we use group theory to introduce the field of projective geometry. In particular, we use matrix–vector and matrix–matrix multiplication to form a group of mappings of the original projective plane or space.

To introduce group theory, we focused on an individual transformation or on the matrix that represents it. Here, on the other hand, we focus on the original geometrical object (points, lines, and vectors), rather than the mapping that acts upon it. This approach is particularly useful in computer graphics [65, 95].

6.1 Circles and Spheres

6.1.1 Degenerate "Circle"

We start with some preliminary definitions, which will be useful later. In particular, we define circles, spheres, and hyperspheres in higher dimensions. For this purpose, we must start from a degenerate zero-dimensional "circle."

Consider the one-dimensional real axis. Consider two points on it: -1 and 1. They are antipodal (or opposite) points: placed symmetrically at opposite sides of 0. Together, they make a new set:

$$S^0 \equiv \{-1, 1\}.$$

Note that this notation has nothing to do with the subgroup S in Chap. 5.

In what sense is S^0 a "circle?" Well, it contains those real numbers of absolute value 1. On the real axis, there are just two such points: -1 and 1. In this sense, the diameter of S^0 is just the line segment leading from -1 to 1.

Now, let us also introduce an orthogonal axis: the y-axis. This forms the two-dimensional Cartesian plane. In this plane, the original one-dimensional real axis is embedded into the horizontal x-axis (Fig. 1.6). In particular, the original antipodal points -1 and 1 embed into a new pair of antipodal points: $(-1, 0)$ and $(1, 0)$, on the new x-axis. This is the embedded S^0.

The original diameter also embeds just the new line segment leading from $(-1, 0)$ to $(1, 0)$ on the new x-axis. Next, we will use this embedded diameter to produce a more genuine circle.

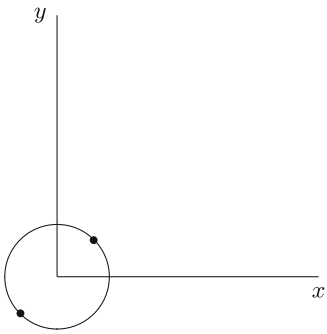


Fig. 6.1 Antipodal points on the unit circle

6.1.2 Antipodal Points in the Unit Circle

In the Cartesian plane, the embedded diameter can now rotate counterclockwise, making a larger and larger angle θ with the positive part of the x-axis. For each angle $0 < \theta < \pi$, this makes a new pair of antipodal points, placed symmetrically at opposite sides of $(0, 0)$:

$$(\cos(\theta), \sin(\theta)) \text{ from above, and } (-\cos(\theta), -\sin(\theta)) \text{ from below.}$$

(Fig. 6.1). This way, the original point $(1, 0)$ draws the upper semicircle. Its antipodal counterpart $(-1, 0)$, on the other hand, draws the lower semicircle. Together, they draw the entire unit circle:

$$S^1 \equiv \left\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\right\}.$$

6.1.3 More Circles

The unit circle S^1 can now shift to just any place in the Cartesian plane, to form a new circle.

In analytic geometry, a circle is characterized by two parameters: a point $O \equiv (x_o, y_o)$ to mark its center, and a positive number $r > 0$ to stand for its radius (Fig. 6.2). From Pythagoras' theorem, the circle contains those points of distance r from O:

$$\left\{(x, y) \in \mathbb{R}^2 \mid (x - x_o)^2 + (y - y_o)^2 = r^2\right\}.$$

This is indeed a link to algebra: the circle is no longer a low-level abstract object as in Euclidean geometry, but rather a set of points that satisfy an algebraic equation.

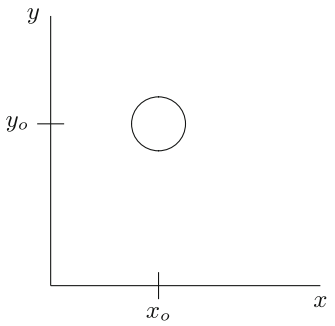


Fig. 6.2 A circle centered at $O \equiv (x_o, y_o)$

6.1.4 Antipodal Points in the Unit Sphere

What is a diameter in S^1? It is a line segment connecting two antipodal points:

$$[(-\cos(\theta), -\sin(\theta)), (\cos(\theta), \sin(\theta))].$$

Now, let us introduce yet another axis: the vertical z-axis, to form the three-dimensional Cartesian space. This way, the original unit circle embeds right into the horizontal x-y plane, for which $z = 0$:

$$S^1 \times \{0\} = \left\{(x, y, 0) \in \mathbb{R}^3 \mid x^2 + y^2 = 1\right\}.$$

In particular, the above diameter also embeds into

$$[(-\cos(\theta), -\sin(\theta), 0), (\cos(\theta), \sin(\theta), 0)].$$

Let us go ahead and rotate it at angle ϕ, upward into the new z-dimension. For each angle $0 < \phi < \pi$, this makes a new pair of antipodal points:

$$(\cos(\theta)\cos(\phi), \sin(\theta)\cos(\phi), \sin(\phi))$$

(which draws the upper semicircle) and

$$(-\cos(\theta)\cos(\phi), -\sin(\theta)\cos(\phi), -\sin(\phi))$$

(which draws the lower semicircle).

This can be done for each and every pair of antipodal points in the embedded S^1, characterized by some θ. Once this is done for all $0 \leq \theta < \pi$, the upper semicircles make the upper hemisphere, and the lower semicircles make the lower hemisphere. Together, we have the entire unit sphere:

$$S^2 \equiv \left\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\right\}.$$

6.1.5 General Multidimensional Hypersphere

The above procedure may repeat for higher and higher dimensions as well. By mathematical induction on $n = 1, 2, 3, \ldots$, we obtain the general hypersphere:

$$S^{n-1} \equiv \left\{ (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n \;\mid\; v_1^2 + v_2^2 + \cdots + v_n^2 = 1 \right\}.$$

Note that this is just a notation: the superscript $n - 1$ is not a power, although it has something to do with power: it reflects the fact that the above procedure has been iterated $n - 1$ times, to form $S^1, S^2, \ldots, S^{n-1}$.

For instance, by setting $n = 4$, we obtain the hypersphere

$$S^3 \equiv \left\{ (x, y, z, w) \in \mathbb{R}^4 \;\mid\; x^2 + y^2 + z^2 + w^2 = 1 \right\}.$$

6.1.6 Complex Coordinates

To visualize the above hypersphere geometrically, let us introduce the new complex coordinates c_1 and c_2. The original real coordinates x and y will then serve as real and imaginary parts in c_1. The third and fourth real coordinates, z and w, on the other hand, will serve as real and imaginary parts in c_2.

The original hypersphere S^3 can now be defined in terms of the new complex coordinates:

$$S^3 \equiv \left\{ (c_1, c_2) \in \mathbb{C}^2 \;\mid\; |c_1|^2 + |c_2|^2 = 1 \right\}.$$

To illustrate, let us define $r \equiv |c_1|$: the radius of some circle in the c_1-plane, around the origin $x = y = 0$ (Fig. 6.3). The complementary circle of radius $|c_2| = \sqrt{1 - r^2}$, on the other hand, can then be drawn in the c_2-plane, around the origin $z = w = 0$ (Fig. 6.4).

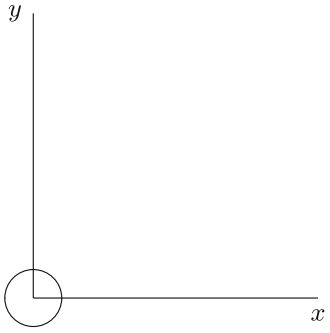


Fig. 6.3 The first complex coordinate $c_1 \equiv x + y\sqrt{-1}$. The circle contains complex numbers c_1 with $|c_1|^2 = x^2 + y^2 = r^2$, for a fixed radius $r \geq 0$

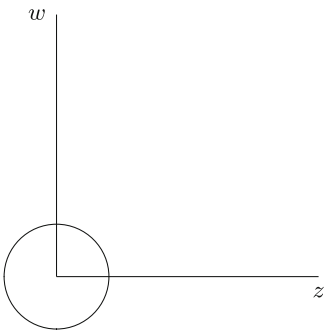


Fig. 6.4 The second complex coordinate $c_2 \equiv z + w\sqrt{-1}$. The circle contains complex numbers c_2 with $|c_2|^2 = z^2 + w^2 = 1 - r^2$, where $r \leq 1$ is the radius of the former circle: the c_1-circle above

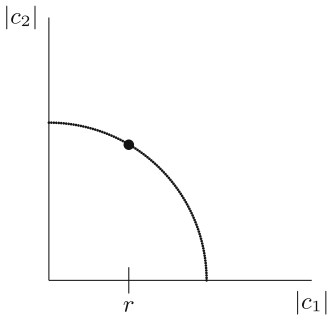


Fig. 6.5 The new $|c_1|$-$|c_2|$-plane, where $c_1 \equiv x + y\sqrt{-1}$ and $c_2 \equiv z + w\sqrt{-1}$ are formed from the original $(x, y, z, w) \in \mathbb{R}^4$. The arc contains those points for which $|c_1|^2 + |c_2|^2 = 1$, including those points in the c_1- and c_2-circles above

Now, let us pick one point from the former c_1-circle, and another point from the latter c_2-circle (Fig. 6.5). Together, they form a new four-dimensional point $(c_1, c_2) = (x, y, z, w) \in S^3$:

$$x^2 + y^2 + z^2 + w^2 = |c_1|^2 + |c_2|^2 = r^2 + 1 - r^2 = 1.$$

How does the c_1-circle in Fig. 6.3 relate geometrically to the c_2-circle in Fig. 6.4? This is illustrated in the two-dimensional $(|c_1|, |c_2|)$-plane (Fig. 6.5). This completes the missing link between c_1 and c_2: one just needs to pick a point from the arc

$$|c_1|^2 + |c_2|^2 = 1.$$

6.2 The Complex Projective Plane

6.2.1 The Complex Projective Plane

We start from the easy case: the complex projective plane. Actually, we have already met it in Chap. 5, Sect. 5.8.7. Here, however, we put it in a wider geometrical context.

Recall the set of nonzero two-dimensional complex vectors:

$$V \equiv \mathbb{C}^2 \setminus \{(0,0)\}$$

(Chap. 5, Sects. 5.8.2–5.8.4). Recall that it splits into disjoint planes of the form

$$C \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \equiv \left\{ z \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \mid z \in \mathbb{C},\ z \neq 0 \right\},$$

where c_1 and c_2 are some complex parameters that do not both vanish at the same time (Fig. 5.8).

How to visualize such a complex plane like? Well, for instance, by setting $c_2 = 0$, we have the horizontal complex plane in Fig. 5.9. By setting $c_1 = 0$, on the other hand, we have the vertical complex plane

$$\{(0, z) \mid z \in \mathbb{C},\ z \neq 0\}.$$

Thus, the complex projective plane is just the family of such planes, each considered as an individual element:

$$V/C \equiv \{Cv \mid v \in V\}.$$

In this set, there is no duplication: each element of the form Cv appears only once, with v being some representative picked arbitrarily from it.

This is no group: there is no algebraic operation. After all, V was never a group in the first place. So, there is no point to talk about homomorphism. Still, there is a point to talk about homeomorphism, to visualize how V/C looks like topologically, or how continuous it may be.

6.2.2 Topological Homeomorphism onto the Sphere

In Chap. 5, Sect. 5.8.7, we have already seen that the complex projective plane is topologically homeomorphic to the extended complex plane: both have the same continuity properties. Furthermore, the extended complex plane is topologically homeomorphic to the sphere. Thus, in summary, the complex projective plane is topologically homeomorphic to the sphere:

$$V/C \simeq \mathbb{C} \cup \{\infty\} \simeq S^2.$$

Here, "$\simeq$" means topological homeomorphism (an invertible mapping that preserves continuity), not algebraic isomorphism. After all, these are just sets, not groups, so there is no algebraic operation to preserve. Below, we will extend this to higher dimensions as well.

6.2.3 The Center and its Subgroups

In Sect. 6.1.5, we have defined the general hypersphere $S^{n-1} \subset \mathbb{R}^n$. Note that this notation has nothing to do with the subgroup S in Chap. 5.

Now, let us use G: the group of 2×2 nonsingular complex matrices (Chap. 5, Sect. 5.8.1). In G, the unit element is just the 2×2 identity matrix I. Furthermore, the center $C \subset G$ contains the nonzero scalar multiples of I. (See exercises at the end of Chap. 5.)

Let us write C as the product of two subgroups. For this purpose, let $H \subset C$ contain the positive multiples of I:

$$H \equiv \{rI \mid r \in \mathbb{R},\ r > 0\}.$$

It is easy to see that H is indeed a group in its own right.

Now, let us define yet another subgroup $U \subset C$:

$$U \equiv \{zI \mid z \in \mathbb{C},\ |z| = 1\}.$$

Using the unit circle S^1 (Sect. 6.1.2), this can be written more concisely as

$$U = S^1 I.$$

It is easy to see that U is indeed a group in its own right.

6.2.4 Group Product

What is the product of U and H? Well, it contains those products of an element from U with an element from H:

$$UH \equiv \{uh \mid u \in U,\ h \in H\}$$

(Chap. 5, Sect. 5.10.1).

The original group G is not commutative: two matrices not necessarily commute with each other. Fortunately, its center C is. For this reason,

$$uh = hu, \quad uh \in UH.$$

This implies that

$$UH = HU.$$

Moreover, it also implies that UH is indeed a group in its own right.

6.2.5 The Center—a Group Product

So, UH is a subgroup of G. Is it a subgroup of C as well? Well, to check on this, let us pick an element from UH:

$$wIrI = (wr)I \in C \quad (w \in \mathbb{C},\ |w| = 1,\ r \in \mathbb{R},\ r > 0).$$

This shows that

$$UH \subset C.$$

Conversely, is C a subgroup of UH? Well, to check on this, let us pick an element from C: a nonzero complex multiple of I. Fortunately, each nonzero complex number $z \in \mathbb{C}$ has the polar decomposition

$$z = |z| \exp(\theta\sqrt{-1}),$$

where

$$\theta \equiv \arg(z)$$

is the angle that z makes with the positive part of the real axis. (See exercises at the end of Chap. 8.) Thus,

$$zI = \exp(\theta\sqrt{-1})I|z|I \in UH \quad (z \in \mathbb{C},\ z \neq 0),$$

as required. This implies that

$$C \subset UH.$$

In summary, we have

$$C = UH.$$

Let us use this decomposition to visualize the complex projective plane geometrically.

6.2.6 How to Divide by a Product?

Thanks to the above factorization, each individual vector $v \in V$ spans the complex plane

$$Cv = (UH)v = U(Hv),$$

where Hv is just a "ray:"

$$Hv \equiv \{hv \mid h \in H\} = \{rv \mid r \in \mathbb{R},\ r > 0\},$$

and $U(Hv)$ is not just one ray but a complete fan of rays, making a complete complex plane:

$$\begin{aligned} U(Hv) &\equiv \{u(hv) \mid h \in H,\ u \in U\} \\ &= \left\{\exp(\theta\sqrt{-1})rv \mid r, \theta \in \mathbb{R},\ r > 0,\ 0 \le \theta < 2\pi\right\}. \end{aligned}$$

Together, all such planes make the complex projective plane:

$$V/C = \frac{V}{UH} \equiv \{(UH)v \mid v \in V\} = \{U(Hv) \mid Hv \in V/H\} = \frac{V/H}{U}.$$

In the above, there is no duplication: each equivalence class in V/H is represented, say, by a unique unit vector in S^3:

$$Hv = H\left(\frac{v}{\|v\|}\right).$$

Thus, V/H is mirrored by the hypersphere S^3 in Sect. 6.1.5. Since U is also mirrored by S^1, we have

$$V/C \simeq S^3/S^1.$$

Here, "$\simeq$" stands for topological homeomorphism (an invertible mapping that preserves continuity), and S^3/S^1 contains equivalence circles in the hypersphere S^3. Let us see how this looks like geometrically.

6.2.7 How to Divide by a Circle?

How to "divide" by S^1? Well, shrink each equivalence circle into a single point that lies in it. In Chap. 5, Sect. 5.8.7, this is done algebraically: divide by the second complex coordinate, c_2. Here, on the other hand, this is done geometrically.

Fortunately, the equivalence circle has already shrunk with respect to $|c_2|$. All that is left to do is to shrink it with respect to the angle $\arg(c_2)$ as well. For this purpose, the circle in Fig. 6.4 has to shrink to just one point on it. This way, the second complex coordinate, c_2, reduces to the nonnegative real coordinate $|c_2|$. This produces the top hemisphere in the three-dimensional $(x, y, |c_2|)$-space:

$$\left\{(x, y, |c_2|) \in \mathbb{R}^3 \;\middle|\; |c_2| \geq 0,\ x^2 + y^2 + |c_2|^2 = 1\right\}.$$

What do we have at the bottom of this hemisphere? Well, this is the equator:

$$x^2 + y^2 = 1, \quad \text{or} \quad c_2 = 0.$$

In the arc in Fig. 6.5, this is the lower endpoint: $r = |c_1| = 1$.

At the equator, we can no longer divide by $c_2 = 0$. Instead, we must divide by c_1. More precisely, it is only left to divide by $\arg(c_1)$. This shrinks the entire equator into the single point $C(1, 0)^t$: the infinity point in the complex projective plane. This "closes" the hemisphere from below, at the unique infinity point that contains the entire (shrunk) equator.

What is this topologically? We already know it well: this is just the sphere in Sect. 6.1.4! In summary, we have the topological homeomorphism

$$V/C \simeq S^3/S^1 \simeq S^2,$$

in agreement with Sect. 6.2.2.

6.2.8 *Second and Third Isomorphism Theorems*

In the above, we dealt with sets, not groups. This is why "$\simeq$" meant just topological homeomorphism, not algebraic isomorphism. After all, there is no algebraic operation to preserve. In this section, on the other hand, we deal with groups once again. In this context, "$\simeq$" means not only topological homeomorphism but also algebraic isomorphism.

To understand its inner structure, we better reconstruct the original factor group G/C more patiently in two stages. First, define G/H, whose elements are of the form

$$Hg \equiv \{rg \;|\; r \in \mathbb{R},\ r > 0\}.$$

What is Hg? Well, it has two possible interpretations. In G/H, it is an individual element. In G, on the other hand, it is a subset: an equivalence class (induced by the subgroup $H \subset G$), which may contain many elements. Fortunately, these interpretations mirror each other.

What is the center of G/H? Well, it is just

$$C/H \subset G/H$$

(see exercises below). Its elements are of the form

$$H\exp(\theta\sqrt{-1}),\quad \theta\in\mathbb{R},\ 0\le\theta<2\pi.$$

Now, let us go ahead and divide by this center, to have the factor group of factor groups: $(G/H)/(C/H)$. What is a typical element in it? Well, take the factor group C/H in the denominator, and use it to multiply a representative from the factor group G/H in the numerator. This produces an individual element in $(G/H)/(C/H)$, which can also be viewed as a subset of G/H: a complete equivalence class in G/H, induced by C/H:

$$\begin{aligned}(C/H)Hg &= \left\{H\exp(\theta\sqrt{-1})Hg \ \mid\ \theta\in\mathbb{R},\ 0\le\theta<2\pi\right\}\\ &= \left\{H\left(\exp(\theta\sqrt{-1})g\right) \ \mid\ \theta\in\mathbb{R},\ 0\le\theta<2\pi\right\}\\ &\subset G/H\end{aligned}$$

(Chap. 5, Sects. 5.7.3 and 5.8.5).

So, we also have a mirroring between an individual element in $(G/H)/(C/H)$ and a complete equivalence class in G:

$$(C/H)Hg\leftrightarrow\left\{\exp(\theta\sqrt{-1})rg \ \mid\ r,\theta\in\mathbb{R},\ r>0,\ 0\le\theta<2\pi\right\}=Cg\in G/C.$$

This makes a new isomorphism:

$$\frac{G/H}{C/H}\simeq G/C.$$

But we already know this formula well: this is just the third isomorphism theorem (Chap. 5, Sect. 5.10.2).

Furthermore, in our case, we know quite well how C/H looks like:

$$C/H\simeq U.$$

As a matter of fact, this is just a special case of the second isomorphism theorem. Indeed, in Chap. 5, Sect. 5.10.1, just substitute $T\leftarrow U$ and $S\leftarrow H$, and note that they have just one element in common: the unit element.

Combining these results, we can write the above less formally:

$$G/C\simeq\frac{G/H}{U}.$$

Thus, we got what we wanted. To divide by C, one could use two stages: first, divide by H; then, divide by U as well. After all, C factorizes as $C=UH$. This way, a typical element $Cg\in G/C$ is mirrored by $U(Hg)$.

How does Cg act on the plane $Cv\in V/C$? Well, this action can now factorize as well:

$$Cg(Cv) = U(Hg)U(Hv) = U(HgHv) = U(H(gv)) = (UH)gv = Cgv,$$

as required. This is a rather informal writing style. After all, as a subgroup of G, U could act on an individual element in V or G, but not in V/H or G/H. On the latter, what should act is C/H, not U. Still, as we have seen above, this is essentially the same.

All these algebraic games are very nice, but give little geometrical intuition. For this purpose, it is sometimes better to drop complex numbers altogether and stick to good old real numbers. Let us start from the simplest case.

6.3 The Real Projective Line

6.3.1 The Real Projective Line

What is the real projective line? First, redefine V to contain real vectors only:

$$V \equiv \mathbb{R}^2 \setminus \{(0,0)\}.$$

Furthermore, redefine G to contain real matrices only. This way, its center C contains only real multiples of I:

$$C \equiv \{xI \mid x \in \mathbb{R},\ x \neq 0\} = (\mathbb{R} \setminus \{0\})\, I.$$

This way, we have

$$V/C = \frac{\mathbb{R}^2 \setminus \{(0,0)\}}{(\mathbb{R} \setminus \{0\})\, I}.$$

This is the real projective line. Why line? Because, in the Cartesian plane, it can be modeled by the horizontal line $y \equiv 1$: each (oblique) line of the form $Cv \in V/C$ meets this horizontal line at one point exactly. This is indeed the oblique cotangent projection (Fig. 5.9).

There is just one exception: the x-axis does not meet the above horizontal line at all, so it must map to ∞. Fortunately, there is a more uniform way to model the real projective line geometrically. For this purpose, we must use some algebra once again.

What are the subgroups of the new center C? Well, once confined to real numbers only, U is redefined to contain two elements only: I and $-I$:

$$U \equiv \{xI \mid x \in \mathbb{R},\ |x| = 1\} = \pm I = S^0 I$$

(Sect. 6.1.1). The second subgroup H, on the other hand, remains the same as before. Thus, each ray of the form $Hv \in V/H$ is spanned by a unique unit vector in S^1:

$$Hv = H\left(\frac{v}{\|v\|}\right).$$

This can be done for every $v \in V$. Together, we have a fan of rays, each represented by a unique point, at which it meets the circle:

$$V/H = \left\{ H\begin{pmatrix} x \\ y \end{pmatrix} \mid x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\} \simeq S^1$$

(Sect. 6.1.2). Hereafter, "$\simeq$" means topological homeomorphism only, not algebraic isomorphism. After all, on the left, we have just a set, not a group, so there is no algebraic operation to preserve.

In summary, the real projective line takes the form

$$V/C = \frac{V}{UH} = \frac{V/H}{U} \simeq S^1/S^0.$$

This is the divided circle.

6.3.2 The Divided Circle

As discussed above, the real projective line is associated with the divided circle:

$$V/C \simeq S^1/S^0.$$

How to visualize this geometrically? Well, this is illustrated in Fig. 6.6. Each line of the form Cv (for some $v = (x, y)^t \in V$) meets the unit circle at two antipodal points: $v/\|v\|$ and $-v/\|v\|$. Fortunately, in the divided circle, they are just one and the same point.

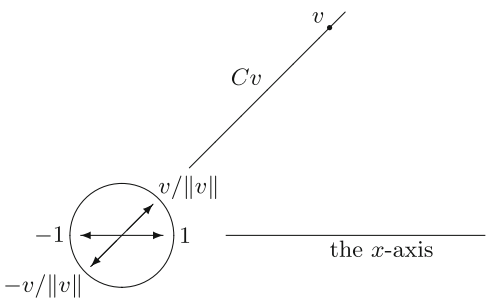


Fig. 6.6 The line Cv meets the unit circle at two antipodal points: $\pm v/\|v\|$. Fortunately, in the divided circle, they coincide with each other. For example, the horizontal x-axis, $C(1, 0)^t$, is represented by the pair $(\pm 1, 0)$

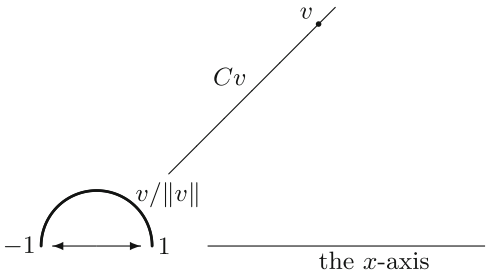


Fig. 6.7 The top semicircle is enough: each line of the form Cv is represented by the unique point $v/\|v\|$. There is just one exception: the horizontal x-axis $C(1,0)^t$ is still represented by the pair $(\pm 1, 0)$. In the divided circle, these points are considered as one and the same. Topologically, this "closes" the semicircle from below, producing a circle

What happens in the horizontal line $v = (1,0)^t$? Well, in this case, Cv is just the x-axis: the infinity object in the real projective line. Indeed, in the cotangent projection in Fig. 5.9, this line maps to ∞. Fortunately, in the divided circle, this line is mirrored well by the pair $(\pm 1, 0)$ (Fig. 6.6).

How to visualize the divided circle geometrically? Well, take the original unit circle in Fig. 6.1, and consider each pair of antipodal points as just one point. This is like taking just the upper semicircle (Fig. 6.7). The lower semicircle, on the other hand, could drop. After all, each point on it is no longer necessary: it is already mirrored by its upper counterpart.

Or is it? Well, there is just one exception: the points $(\pm 1, 0)$ are both needed and should not drop. Instead, they should unite into just one point. Topologically, this "closes" the semicircle at the bottom, producing a closed circle:

$$V/C \simeq S^1/S^0 \simeq S^1.$$

We are now ready to move on to higher dimensions as well.

6.4 The Real Projective Plane

6.4.1 The Real Projective Plane

Let us move on to a yet higher dimension. For this purpose, redefine V as a three-dimensional vector set:

$$V \equiv \mathbb{R}^3 \setminus \{(0,0,0)\}.$$

This way, a vector in V is specified by three real degrees of freedom: its first, second, and third coordinates. The projective plane V/C, on the other hand, gives away one

degree of freedom: the unspecified scalar multiple. This is why V/C depends on two degrees of freedom only and is referred to as a plane.

Furthermore, G is also redefined as the group of 3×3 nonsingular real matrices. The unit element in G is now the 3×3 identity matrix:

$$I \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The subgroups C, U, and H are also redefined to use this new I:

$$\begin{aligned} C &\equiv \{xI \mid x \in \mathbb{R},\ x \neq 0\} = (\mathbb{R} \setminus \{0\})\, I \\ U &\equiv \{xI \mid x \in \mathbb{R},\ |x| = 1\} = \pm I = S^0 I \\ H &\equiv \{xI \mid x \in \mathbb{R},\ x > 0\}. \end{aligned}$$

With these new definitions, V/C is called the real projective plane. Why plane? Because, in the Cartesian space, it can be modeled by the horizontal plane $z \equiv 1$: each oblique line of the form $Cv \in V/C$ meets this plane at one point exactly.

The real projective plane has an important advantage: it has not just one infinity point, but many infinity points from all directions.

6.4.2 Oblique Projection

To get a better idea about V/C, project it onto the horizontal plane

$$\{z \equiv 1\} = \{(x, y, 1) \mid x, y \in \mathbb{R}\}.$$

(Note that, unlike in the complex case, here z stands for a real coordinate.) More precisely, each line of the form $C(x, y, z)^t$ (with $z \neq 0$) projects onto the unique point at which it meets the above horizontal plane:

$$C \begin{pmatrix} x \\ y \\ z \end{pmatrix} \to \left(\frac{x}{z}, \frac{y}{z}, 1 \right).$$

In Fig. 6.8, this horizontal plane is viewed from an eye or a "camera" placed at the origin $(0, 0, 0)$, faced upward. Through this camera, one could see the semispace

$$\left\{ (x, y, z) \in \mathbb{R}^3 \mid z > 0 \right\}.$$

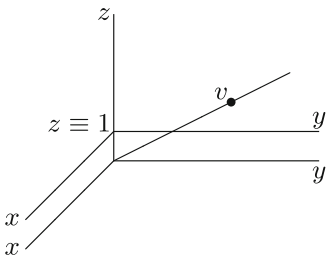


Fig. 6.8 Oblique projection onto the horizontal plane $z \equiv 1$. Each line of the form $C(x, y, z)^t$ $(z \neq 0)$ projects onto $(x/z, y/z, 1)$

More precisely, because one could only see a two-dimensional image, one sees the oblique projection onto the horizontal plane

$$\left\{(x, y, z) \in \mathbb{R}^3 \;\middle|\; z = 1\right\}.$$

And what about $z = 0$? Well, in this case, the projection must be radial:

$$C\begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \to \pm\frac{1}{\sqrt{x^2 + y^2}}(x, y, 0).$$

In summary, the entire projection is defined by

$$C\begin{pmatrix} x \\ y \\ z \end{pmatrix} \to \begin{cases} \left(\frac{x}{z}, \frac{y}{z}, 1\right) & \text{if } z \neq 0 \\ \pm\frac{(x,y,0)}{\sqrt{x^2+y^2}} & \text{if } z = 0. \end{cases}$$

Next, let us introduce a more uniform projection.

6.4.3 Radial Projection

Alternatively, one might want to use a more uniform approach: always use a radial projection, regardless of whether $z = 0$ or not:

$$C\begin{pmatrix} x \\ y \\ z \end{pmatrix} \to \pm\frac{1}{\sqrt{x^2 + y^2 + z^2}}(x, y, z)$$

(Fig. 6.9). Let us see what we obtain.

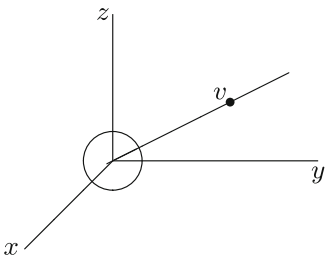


Fig. 6.9 Radial projection: each line of the form Cv projects onto the pair of antipodal unit vectors $\pm v/\|v\|$ in the sphere S^2

6.4.4 The Divided Sphere

Fortunately, we already have the decomposition

$$C = UH.$$

Now, each ray of the form $Hv \in V/H$ can also be represented by the unique vector $v/\|v\|$ in the unit sphere S^2. Thus,

$$V/C = \frac{V}{UH} = \frac{V/H}{U} \simeq S^2/S^0.$$

This is the divided sphere: the family of pairs of antipodal points in the original unit sphere. In this family, each pair of antipodal points is viewed as an individual object in its own right.

6.4.5 Infinity Points

Let us consider once again the oblique projection in Sect. 6.4.2. It is not quite uniform: it distinguishes between ordinary "points" and infinity "points," where a "point" means a complete line of the form $Cv \in V/C$.

What is an infinity point in the real projective plane? It is just a horizontal line of the form $C(x, y, 0)^t$, for some real numbers x and y that do not both vanish at the same time (Fig. 6.10). Unfortunately, the zero z-coordinate can no longer be used to divide, or normalize, or project to the horizontal plane

$$\{z \equiv 1\} = \{(x, y, 1) \mid x, y \in \mathbb{R}\}$$

as in Fig. 6.8.

Fortunately, the unit sphere S^2 is much more symmetric than the above plane. Therefore, the horizontal line $C(x, y, 0)^t$ can still project to S^2, as in Fig. 6.10. In both the oblique and the radial projections, this makes a pair of antipodal infinity points of the form

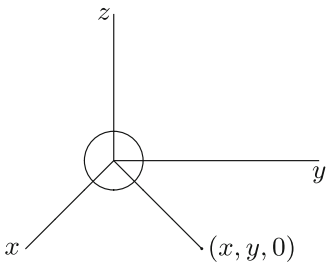


Fig. 6.10 What is an infinity point in the real projective plane? It is a horizontal line of the form $C(x, y, 0)^t$, projected onto the antipodal points $\pm(x, y, 0)/\sqrt{x^2 + y^2}$ in the infinity circle

$$\pm\frac{(x, y, 0)}{\sqrt{x^2 + y^2}}.$$

Together, they make the infinity circle.

6.4.6 The Infinity Circle

Together, these infinity points make the infinity circle:

$$\left\{(x, y, 0) \;\middle|\; x, y \in \mathbb{R},\ x^2 + y^2 = 1\right\} = S^2 \cap \{(x, y, 0) \mid x, y \in \mathbb{R}\} \simeq S^1.$$

This circle is just the equator in the original unit sphere. Fortunately, unlike in the complex projective plane in Sect. 6.2.7, it no longer shrinks into a single infinity point. On the contrary: it contains many useful infinity points, in all horizontal directions.

6.4.7 Lines as Level Sets

So far, we have used a vector of the form

$$v \equiv \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \in V$$

to stand for a particular point in V. Fortunately, this is not the only option: v could also make a complete plane in V.

Indeed, each real linear function

$$f : V \to \mathbb{R}$$

could also be defined in terms of an "inner product" with a fixed vector $v \in V$:

$$f(x, y, z) \equiv f_v(x, y, z) \equiv (x, y, z)v = xv_1 + yv_2 + zv_3.$$

(This is not quite an inner product, because we do not take the complex conjugate.) The vector v is also called the gradient of f, denoted by

$$\nabla f_v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = v.$$

Because f_v is linear, it must indeed have a constant gradient (Chap. 8, Sect. 8.12.2).

Now, let r be a fixed real number. What is the rth level set of f_v? Well, it is the "origin" of r under f_v: it contains those vectors $(x, y, z)^t \in V$ that f_v maps to r:

$$l_{v,r} \equiv f_v^{-1}(r) = \left\{(x, y, z)^t \in V \mid f_v(x, y, z) = r\right\}.$$

This notation has nothing to do with inverse. In fact, f_v may have no inverse at all. After all, a level set may contain a few points (Chap. 5, Sect. 5.4.1).

In particular, what is the zero level set? Well, it contains those vectors that are orthogonal to v. Together, they make a complete plane, orthogonal to v:

$$l_{v,0} \equiv f_v^{-1}(0) = \left\{(x, y, z)^t \in V \mid f_v(x, y, z) = (x, y, z)v = 0\right\}.$$

In a linear function as above, the gradient is a constant vector. In a more general function, on the other hand, the gradient may change from point to point, and the level set may be curved. Fortunately, at each point in it, the gradient is still normal (perpendicular) to it in a new sense: normal to the plane tangent to it.

Note that, if some vector is in $l_{v,0}$, then so is every nonzero scalar multiple of it. In other words, $l_{v,0}$ is invariant under C:

$$Cl_{v,0} = l_{v,0}.$$

This is how things look like in V. In V/C, on the other hand, this makes a complete line. To see this, let the plane $l_{v,0}$ cut the horizontal plane

$$\{z \equiv 1\} = \{(x, y, 1) \mid x, y \in \mathbb{R}\}.$$

This produces a line: the "shadow" of $l_{v,0}$ on this horizontal plane.

Note also that, for every $c \in C$,

$$l_{cv,0} = l_{v,0}.$$

Thus, $l_{v,0}$ could be defined not only by the original vector $v \in V$ but also by every nonzero scalar multiple of v.

Thus, in V/C, $l_{v,0}$ could be defined in terms of the entire line $Cv \in V/C$ rather than the concrete vector $v \in V$. Geometrically, this (oblique) line is represented by a unique point: the point at which it meets the horizontal plane $z \equiv 1$. This is the start of duality: in the real projective plane, a point is also a line, and a line is also a point.

6.5 Infinity Points and Line

6.5.1 Infinity Points and their Projection

In particular, $l_{v,0}$ contains the point

$$(-v_2, v_1, 0)$$

and every nonzero scalar multiple of it. As in Fig. 6.10, this point can project radially onto a pair of antipodal points in the infinity circle:

$$\pm \frac{1}{\sqrt{v_1^2 + v_2^2}}(-v_2, v_1, 0).$$

As discussed above, $l_{v,0}$ also makes a line: its shadow on (or intersection with) the horizontal plane $z \equiv 1$. Thus, in the real projective plane, the original plane $l_{v,0} \subset V$ is interpreted geometrically as a new extended line: the shadow on the horizontal plane $z \equiv 1$, plus its "endpoints:" the above pair of antipodal points on the infinity circle.

By doing this for every $v \in V$, the entire real projective plane is represented geometrically as a fan of infinite lines on the horizontal plane $z \equiv 1$, each extended by a pair of "endpoints." In summary, the entire real projective plane has projected onto the horizontal plane $z \equiv 1$, surrounded by the infinity circle. This is in agreement with the original oblique projection in Sect. 6.4.2.

6.5.2 Riemannian Geometry

In the radial projection in Sect. 6.4.3, on the other hand, the entire zero level set $l_{v,0}$ projects radially only, to cut the original unit sphere at a great circle, centered at the origin (0, 0, 0). Fortunately, in Riemannian geometry, such a circle is considered as a line. This way, lines are no longer linear in the usual sense, but rather circular. Furthermore, each point on a great circle coincides with its antipodal counterpart on the other side.

This way, the divided sphere S^2/S^0S mirrors the horizontal plane $z \equiv 1$. Each line on the latter can now extend to a complete plane that passes through the origin and cuts the sphere at a great circle. By doing this for two lines that cross each other at a unique point, we obtain two great circles that meet each other at two antipodal points, considered as one. Moreover, by doing this for two parallel lines that "meet" each other at an infinity point, we obtain two great circles that meet each other at two antipodal points on the equator.

6.5.3 A Joint Infinity Point

For example, consider the new vector

$$v' \equiv \begin{pmatrix} v_1 \\ v_2 \\ v_3' \end{pmatrix} \in V$$

that differs from the original vector v in the z-coordinate only:

$$v_3' \neq v_3.$$

Clearly, the oblique projections of the zero level sets $l_{v,0}$ and $l_{v',0}$ make two parallel lines on the horizontal plane $z \equiv 1$. Where do they "meet" each other? To find out, we must employ the radial projection, to obtain two great circles, which meet each other at two antipodal points on the equator:

$$\pm \frac{1}{\sqrt{v_1^2 + v_2^2}}(-v_2, v_1, 0),$$

which are considered as one.

Assume now that we are given two lines in the real projective plane. Where do they meet? To find out, let us introduce an easy algebraic method. This will show once again that the joint point is indeed unique.

6.5.4 Two Lines Share a Unique Point

Unlike in Euclidean or analytic geometry, here, in projective geometry, every two distinct lines meet each other at a unique point. This is true not only in Riemannian geometry, but also in the original oblique projection (Sect. 6.4.2).

What is this joint point? To find out, consider two independent vectors $v, v' \in V$, which are not a scalar multiple of each other. The corresponding zero level sets,

$l_{v,0}$ and $l_{v',0}$, make two distinct planes in V. In the real projective plane V/C, on the other hand, they are considered as two distinct lines. After all, in the horizontal plane $z \equiv 1$, they cut two lines: their shadow.

What is their intersection in V? Fortunately, this is available in terms of vector product:

$$l_{v,0} \cap l_{v',0} = C(v \times v').$$

After all, $v \times v'$ is orthogonal to both v and v' (Chap. 2, Sect. 2.2.4).

6.5.5 *Parallel Lines Do Meet*

Let us study the z-coordinate in $v \times v'$:

$$(v \times v')_3 = v_1 v_2' - v_2 v_1'.$$

This is zero if and only if (v_1', v_2') is a scalar multiple of (v_1, v_2), as in the example in Sect. 6.5.3. In this case, $l_{v,0}$ and $l_{v',0}$ cut not only the horizontal plane $z \equiv 1$ (at two parallel lines) but also the infinity circle, at two infinity points:

$$\pm \frac{1}{\sqrt{v_1^2 + v_2^2}}(-v_2, v_1, 0).$$

Thus, two parallel lines on the horizontal plane $z \equiv 1$ do meet each other at two antipodal infinity points, making one and the same point on the equator in the divided sphere.

Thus, in both Riemannian geometry and the real projective plane, there are no parallel lines any more: every two distinct lines meet each other at a unique point. Is this true even when one of the lines is the infinity line?

6.5.6 *The Infinity Line*

How does the infinity line look like? Well, we have already seen that an infinity point has a zero z-coordinate. What vector is orthogonal to all such points? This is just the standard unit vector

$$e \equiv \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \in V.$$

Indeed, every infinity point must lie in the plane orthogonal to e—the horizontal plane $z \equiv 0$:

$$l_{e,0} = \left\{(x, y, z)^t \in V \mid f_e(x, y, z) = (x, y, z)e = z = 0\right\} = \left\{(x, y, 0)^t \in V\right\}.$$

So, in the real projective plane V/C, the infinity line is just the horizontal plane $z \equiv 0$. After all, once projected onto the unit sphere as in Fig. 6.10, it makes the entire infinity circle.

The infinity line also meets every other line at a unique infinity point. Indeed, every vector $v \in V$ that is not a scalar multiple of e must have a nonzero component $v_1 \neq 0$ or $v_2 \neq 0$. Therefore, the zero level sets of v and e intersect each other at the line

$$l_{v,0} \cap l_{e,0} = C(v \times e) = C \begin{pmatrix} -v_2 \\ v_1 \\ 0 \end{pmatrix}.$$

Once projected on the unit sphere, this line makes the infinity point

$$\pm \frac{1}{\sqrt{v_1^2 + v_2^2}}(-v_2, v_1, 0).$$

This is indeed the unique joint point of the original lines $l_{e,0}$ and $l_{v,0}$ in the real projective plane.

We are now ready to see that, in projective geometry, there is a complete symmetry between points and lines: just as every two distinct lines meet each other at a unique point, every two distinct points make a unique line. This is proved algebraically: there is no longer any need to assume a specific axiom for this, as is done in Euclidean geometry.

6.5.7 *Duality: Two Points Make a Unique Line*

In projective geometry, a vector $v \in V$ may have two different interpretations: either as the point Cv, or as the line $l_{v,0}$. Let us use this duality to form a complete symmetry between points and lines.

Indeed, as discussed in Sect. 6.5.4, the vector product $v \times v'$ produces the unique joint point of the distinct lines $l_{v,0}$ and $l_{v',0}$. Fortunately, this also works the other way around: once v and v' are interpreted as the points Cv and Cv' in V/C, their vector product makes the unique line that passes through both of them.

Indeed, since both v and v' are orthogonal to $v \times v'$, both belong to the zero level set of $f_{v \times v'}$:

$$v,\ v' \in l_{v \times v',0},$$

or

$$Cv,\ Cv' \subset l_{v \times v',0}.$$

Thus, in projective geometry, vector product can be applied to two distinct objects of the same kind, to form a new object (of a new kind) that lies in both of them. If the original objects are interpreted as points, then the new object is the line that passes through them. If, on the other hand, the original objects are interpreted as lines, then the new object is the point they share.

6.6 Conics and Envelopes

6.6.1 Conic as a Level Set

So far, we have only studied the sets V and V/C. Now, let us also study the groups G and G/C that act upon them.

A conic (ellipsoid, hyperboloid, or paraboloid) in V is defined by some symmetric matrix $g \in G$ [57]. For this purpose, consider the quadratic function

$$q_g : V \to \mathbb{R},$$

defined by

$$q_g(v) \equiv v^t g v.$$

Here, we assume that g is indefinite, so q_g may return either positive or negative or zero value.

For each real number $r \in \mathbb{R}$, the rth level set of q_g (the origin of r under q_g) is denoted by

$$m_{g,r} \equiv q_g^{-1}(r) = \left\{ v \in V \ \mid \ q_g(v) = r \right\}.$$

This notation has nothing to do with inverse. In fact, q_g may have no inverse at all. After all, the level set may contain a few vectors (Chap. 5, Sect. 5.4.1).

In particular, the zero level set of q_g is

$$m_{g,0} = q_g^{-1}(0) = \left\{ v \in V \ \mid \ q_g(v) = 0 \right\}.$$

This zero level set is called a conic in V.

6.6.2 New Axis System

As a real symmetric matrix, g has real eigenvalues and real orthonormal eigenvectors. (See Chap. 1, Sects. 1.9.4 and 1.10.4, and exercises therein.) These eigenvectors form a new (real) axis system in V, which may differ from the standard x-y-z system. In the new axis system, g is in its diagonal form, with its (real) eigenvalues on the main diagonal. This is indeed the axis system in which the original conic visualizes best.

Thanks to the algebraic properties of the original matrix g, we have a rather good geometrical picture. Thanks to symmetry, we have the new axis system. Furthermore, thanks to indefiniteness, the eigenvalues are not of the same sign, so the zero level set $m_{g,0}$ is nonempty. For this reason, in terms of the new axis system, the original conic must be a hyperboloid.

Fortunately, for every $c \in C$,

$$m_{cg,0} = m_{g,0}.$$

Thus, $m_{g,0}$ can be defined not only by the original matrix $g \in G$ but also by the element Cg in the factor group G/C.

6.6.3 The Projected Conic

Clearly, if $v \in m_{g,0}$, then every nonzero scalar multiple of v is in $m_{g,0}$ as well: $Cv \subset m_{g,0}$. For this reason, $m_{g,0}$ is invariant under C:

$$Cm_{g,0} = m_{g,0}.$$

Thus, $m_{g,0}$ can be interpreted not only as a conic in V, but also as a lower-dimensional conic in V/C. Once projected on the horizontal plane $z \equiv 1$ as in Fig. 6.8, the original conic indeed produces a one-dimensional conic: ellipse, hyperbola, or parabola.

To make this more concrete, assume that a camera is placed at the origin $(0, 0, 0)$, faced upward. Through the camera, one could only see the upper part of the original conic, with $z > 0$. More precisely, one only sees a two-dimensional image: the horizontal plane $z \equiv 1$, with the conic's shadow in it: a curve of the form

$$\left\{ \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \in V \;\middle|\; (x, y, 1)g \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0 \right\}.$$

This is the projected one-dimensional conic in the horizontal plane $z \equiv 1$.

6.6.4 Ellipse, Hyperbola, or Parabola

How does the projected conic look like? Well, this depends on the leading quadratic terms: x^2, xy, and y^2 in the original function q_g. The coefficients of these terms can be found in the minor $g^{(3,3)}$: the 2×2 upper-left block in g (Chap. 2, Sect. 2.1.1).

Like g, $g^{(3,3)}$ is a real symmetric matrix, with a diagonal form: its real eigenvectors make a new two-dimensional axis system, which may differ from the standard x-y system. In this new axis system, the projected conic may indeed visualize best. In fact, if $g^{(3,3)}$ has a positive determinant:

$$\det\left(g^{(3,3)}\right) > 0,$$

then its (real) eigenvalues must have the same sign, so the projected conic must be an ellipse. If, on the other hand,

$$\det\left(g^{(3,3)}\right) < 0,$$

then the eigenvalues must have different signs, so the projected conic must be a hyperbola (in terms of the new two-dimensional axis system). Finally, if the determinant vanishes:

$$\det\left(g^{(3,3)}\right) = 0,$$

then one of the eigenvalues must vanish, so the projected conic must be a parabola (in terms of the new axis system) in the horizontal plane $z \equiv 1$.

6.6.5 Tangent Planes

Let v be some vector in the original conic $m_{g,0}$. Let us apply g to v, to produce the new vector $gv \in V$. As discussed in Sect. 6.4.7, gv defines the plane $l_{gv,0}$ that is orthogonal to gv. Furthermore, because $l_{gv,0}$ is invariant under C, it can be interpreted not only as a plane in V but also as a line in the real projective plane V/C. Moreover, it can project to a yet more concrete line: its shadow on the horizontal plane $z \equiv 1$.

Let us now return to the original conic $m_{g,0} \subset V$. Fortunately, the plane $l_{gv,0}$ is tangent to it at v. Indeed, since $v \in m_{g,0}$, $v^t gv = 0$, so $v \in l_{gv,0}$ as well. Furthermore, both the conic and the plane have the same normal vector at v. After all, both are level sets of functions with proportional gradients at v:

$$\nabla q_g(v) = 2gv = 2\nabla f_{gv}$$

(Sect. 6.4.7, and Chap. 8, Sect. 8.12.2). Thus, the mapping

$$v \rightarrow gv$$

maps the original point $v \in m_{g,0}$ to a vector that is normal (or perpendicular, or orthogonal) to both the original conic and the tangent plane at v.

Fortunately, once projected onto the horizontal plane $z \equiv 1$, the tangent plane also produces the line (or shadow) that is tangent to the projected conic at the projected v.

6.6.6 *Envelope*

The new vector gv studied above has yet another attractive property: it belongs to the zero level set of the quadratic function associated with the inverse matrix g^{-1}:

$$q_{g^{-1}}(gv) = (gv)^t g^{-1} gv = (gv)^t v = v^t g^t v = v^t gv = q_g(v) = 0,$$

or

$$gv \in m_{g^{-1},0}.$$

This can be written more compactly as

$$gm_{g,0} \subset m_{g^{-1},0}.$$

Now, let us substitute g^{-1} for g:

$$g^{-1} m_{g^{-1},0} \subset m_{g,0}.$$

By applying g to both sides, we have

$$m_{g^{-1},0} \subset gm_{g,0}.$$

In summary,

$$gm_{g,0} = m_{g^{-1},0}.$$

In the dual interpretation in Sect. 6.5.7, the original tangent plane $l_{gv,0}$ is viewed as a mere point: $Cgv \in m_{g^{-1},0}$. In this interpretation, the new conic $m_{g^{-1},0}$ makes a new envelope: a family of planes, all tangent to the original conic.

6.6.7 The Inverse Mapping

What is the envelope of the new conic $m_{g^{-1},0}$? To find out, we just need to use g^{-1}:

$$g^{-1} m_{g^{-1},0} = m_{g,0},$$

which is just the original conic once again.

More specifically, consider an individual vector of the form $gv \in m_{g^{-1},0}$. The inverse mapping

$$gv \to g^{-1} gv = v$$

maps it to the vector v that is normal to the new conic at gv:

$$\nabla q_{g^{-1}}(gv) = 2g^{-1} gv = 2v = 2\nabla f_v.$$

6.7 Duality: Conic–Envelope

6.7.1 Conic and its Envelope

Thus, the original roles have interchanged: v is no longer interpreted as a mere point in the original conic, but rather as a complete plane: $l_{v,0}$, tangent to the new conic at gv. The original tangent plane $l_{gv,0}$, on the other hand, is now interpreted as a mere point: gv, in the new conic. This is a geometrical observation. Algebraically, it has already been written most compactly as

$$g^{-1} m_{g^{-1},0} = m_{g,0}, .$$

In summary, projective geometry supports two kinds of duality. In the elementary level, a line takes the role of a point, whereas a point is viewed as a complete line (Sect. 6.5.7). In the higher level, on the other hand, the original conic is viewed as an envelope, whereas the original envelope is viewed as a conic.

6.7.2 Hyperboloid and its Projection

Consider, for example, the special case

$$g = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

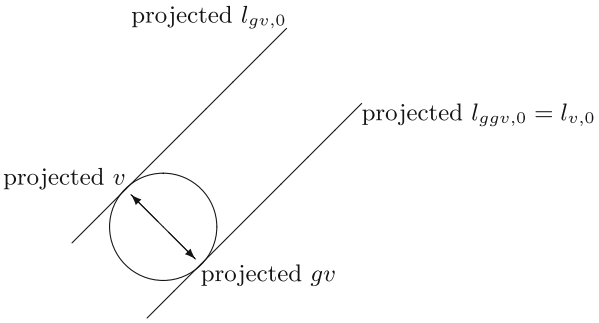


Fig. 6.11 The hyperboloid projects onto a circle in the horizontal plane $z \equiv 1$. Each plane tangent to the original hyperboloid projects to a line tangent to the circle and perpendicular to its radius

This way, the original conic is the hyperboloid

$$m_{g,0} = \left\{ v \in V \mid v^t g v = 0 \right\} = \left\{ (x, y, z)^t \in V \mid z^2 = x^2 + y^2 \right\}.$$

In this simple example,

$$g^2 = I,$$

or

$$g^{-1} = g,$$

so the new conic is the same.

Now, let us pick some vector v in the conic, say

$$v \equiv \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \equiv \begin{pmatrix} -1 \\ 1 \\ \sqrt{2} \end{pmatrix} \in m_{g,0}$$

(Fig. 6.11). The tangent plane at v is perpendicular to

$$gv = \begin{pmatrix} -v_1 \\ -v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ \sqrt{2} \end{pmatrix}.$$

More explicitly, the tangent plane at v is

$$l_{gv,0} = \left\{ (x, y, z)^t \in V \mid x - y = -z\sqrt{2} \right\}.$$

In particular, v itself belongs not only to the conic but also to this plane, as required.

To have a better geometrical understanding, let us project obliquely, to make a shadow on the horizontal plane $z \equiv 1$. The projected conic is the circle

$$\left\{(x, y, 1)^t \in V \mid x^2 + y^2 = 1\right\}.$$

Furthermore, the projected $l_{gv,0}$ is the line

$$\left\{(x, y, 1)^t \in V \mid x - y = -\sqrt{2}\right\}.$$

As can be seen in the upper-left part of Fig. 6.11, this line is indeed tangent to the circle at the projected v:

$$\frac{v}{v_3} = \frac{v}{\sqrt{2}} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 1 \end{pmatrix}.$$

So far, gv has been used only to form the tangent plane $l_{gv,0}$. Thanks to duality, gv can also be viewed as a mere point on the new conic, which is just the same hyperboloid:

$$m_{g^{-1},0} = m_{g,0}.$$

More explicitly, the vector

$$gv = \begin{pmatrix} 1 \\ -1 \\ \sqrt{2} \end{pmatrix}$$

projects onto the vector

$$\frac{gv}{(gv)_3} = \frac{gv}{\sqrt{2}} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 1 \end{pmatrix}.$$

The tangent plane at gv, which is just

$$l_{ggv,0} = l_{v,0} = \left\{(x, y, z)^t \in V \mid x - y = z\sqrt{2}\right\},$$

projects onto the tangent line

$$\left\{(x, y, 1)^t \in V \mid x - y = \sqrt{2}\right\},$$

as in the lower-right part of Fig. 6.11.

Thanks to duality, $l_{v,0}$ can also be interpreted as a mere point: the original vector $v \in V$. This is indeed duality: the tangent to the tangent is just the original vector itself, and the envelope of the envelope is just the original conic itself.

There is nothing special about the above choice of v: every two points v and gv on the original hyperboloid project onto two antipodal points:

$$\pm\left(\frac{v_1}{v_3}, \frac{v_2}{v_3}\right).$$

Furthermore, $l_{gv,0}$ and $l_{v,0}$ project onto two parallel lines that share the same normal vector:

$$\frac{1}{\sqrt{v_1^2 + v_2^2}} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

(Sect. 6.5.5). For this reason, the projected $l_{v,0}$ and the projected $l_{gv,0}$ are both perpendicular to the projected v and gv. This is indeed as expected from a circle in Euclidean geometry: the tangent should be perpendicular to the radius ([24] and Chapter 6 in [74]).

In summary, duality is relevant not only in the original real projective plane but also in the horizontal plane $z \equiv 1$: just as the projected $l_{gv,0}$ is tangent to the circle at the projected v, the projected $l_{v,0}$ is tangent to the circle at the projected gv, on the other side.

6.7.3 *Projective Mappings*

A projective mapping (or transformation) acts in the real projective plane V/C. This way, it can model a three-dimensional motion. Once projected onto the horizontal plane $z \equiv 1$, the original three-dimensional trajectory produces a two-dimensional shadow, easy to illustrate and visualize geometrically.

Let $g \in G$ be a real nonsingular 3×3 matrix. Clearly, g can be interpreted as a linear mapping:

$$v \to gv, \quad v \in \mathbb{R}^3.$$

As discussed in Chap. 5, Sect. 5.8.5, $Cg \in G/C$ can also act on the real projective plane V/C:

$$Cv \to Cg(Cv) \equiv C(gv), \quad Cv \in V/C.$$

In other words, if v is a representative from the equivalence class $Cv \in V/C$, and g is a representative from the equivalence class $Cg \in G/C$, then gv may represent the new equivalence class $Cg(Cv) \in V/C$.

This is how the projective mapping looks like in V/C. How does it look like in the horizontal plane $z \equiv 1$? Well, it can break into three stages: first unproject, then apply Cg, and then project. Together, this makes $PCgP^{-1}$:

$$v \to PCgP^{-1}v \equiv PCgCv \equiv PC(gv) \equiv \begin{cases} gv/(gv)_3 & \text{if} \quad (gv)_3 \neq 0 \\ \pm gv/\|gv\| & \text{if} \quad (gv)_3 = 0. \end{cases}$$

Let us look at a few examples.

6.8 Applications in Computer Graphics

6.8.1 Translation

Translation is an important example of a projective mapping that is often used in computer graphics. Let α and β be some real parameters. Consider the matrix

$$g \equiv \begin{pmatrix} 1 & 0 & \alpha \\ 0 & 1 & \beta \\ 0 & 0 & 1 \end{pmatrix}.$$

In the horizontal plane $z \equiv 1$, in particular, g translates by $(\alpha, \beta)^t$:

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \to g \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x + \alpha \\ y + \beta \\ 1 \end{pmatrix}.$$

Thus, the horizontal plane $z \equiv 1$ remains invariant.

This kind of translation, however, is too simple and naive to simulate or visualize a real three-dimensional motion. In computer graphics, one might want to simulate the original motion well, before projecting to two dimensions.

6.8.2 Motion in a Curved Trajectory

For this purpose, consider a planar object in the original space V, with the unit normal vector $\mathbf{n}$. Suppose that it moves along a given curve or trajectory $t \subset V$. How to simulate or visualize this motion best?

The original three-dimensional motion can be approximated by a composition of many tiny linear translations, each advances the object by a small step in the direction tangent to $t \subset V$. Let us focus on the first step: the next steps can model in the same way.

Initially, the object is placed at the beginning of t. At this point, let $\mathbf{p}$ be the unit vector tangent (or parallel) to t.

6.8.3 The Translation Matrix

Define the translation matrix g by

$$g \equiv I + \gamma \cdot \mathbf{p} \cdot \mathbf{n}^t,$$

where γ is a real parameter to be specified later, and $\mathbf{n}^t$ is the row vector transpose to $\mathbf{n}$ [89].

Consider, for example, the simple case in which both the planar object and the tangent vector $\mathbf{p}$ lie in the horizontal plane $z \equiv 1$:

$$\mathbf{n} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

and

$$\mathbf{p} = \begin{pmatrix} \alpha \\ \beta \\ 0 \end{pmatrix}.$$

In this case, the motion is actually two-dimensional:

$$g = I + \gamma \begin{pmatrix} \alpha \\ \beta \\ 0 \end{pmatrix} (0, 0, 1) = \begin{pmatrix} 1 & 0 & \gamma \cdot \alpha \\ 0 & 1 & \gamma \cdot \beta \\ 0 & 0 & 1 \end{pmatrix}.$$

With $\gamma = 1$, this is just the same as in Sect. 6.8.1. More general (genuinely three-dimensional) motion, on the other hand, requires a more general translation matrix.

6.8.4 General Translation of a Planar Object

What does a general translation do? Well, it translates the entire planar object in the direction pointed at by $\mathbf{p}$. To see this, let r be the (real) inner product that $\mathbf{n}$ makes

with the initial point in the trajectory. Initially, the planar object lies in its entirety in the plane (or level set, or shifted zero level set)

$$l_{\mathbf{n},r} = \left\{ v \in V \;\middle|\; \mathbf{n}^t v = r \right\} = r \cdot \mathbf{n} + l_{\mathbf{n},0}.$$

Then, each point v in the planar object translates by the same amount:

$$v \to gv = v + \gamma \cdot \mathbf{p} \cdot \mathbf{n}^t v = v + \gamma \cdot r \cdot \mathbf{p},$$

as required. This completes the first step in the discrete path that approximates the original motion.

6.8.5 Unavailable Tangent

In practice, however, the tangent $\mathbf{p}$ is not always available. Fortunately, it can still be approximated by the difference between two given points on the trajectory—the next point minus this point:

$$\mathbf{p} \doteq u_2 - u_1.$$

For instance, using the parameter

$$\gamma \equiv \frac{1}{\mathbf{n}^t u_1} = \frac{1}{r},$$

we have a new translation matrix that never uses $\mathbf{p}$:

$$g \equiv I + \frac{1}{r}(u_2 - u_1)\mathbf{n}^t.$$

This g indeed translates u_1 to u_2:

$$u_1 \to gu_1 = u_1 + \frac{1}{r}(u_2 - u_1)\mathbf{n}^t u_1 = u_1 + (u_2 - u_1) = u_2,$$

as required. The object is then ready to advance from u_2 to the next point on the trajectory.

In the next step, on the other hand, up-to-date values of u_1, u_2, r, and γ should be used, to design a new translation matrix g, and advance the object to the next point on the trajectory, and so on.

Finally, to visualize the discrete path well, project it on the horizontal plane $z \equiv 1$, as above. This may give a complete animation movie of the original three-dimensional motion.

6.8.6 Rotation

How to visualize the motion of the Moon in the solar system? It contains two inner rotations: the Moon around the Earth, and the Earth around the Sun. The latter may take place in the horizontal plane $z \equiv 1$. For instance, the Sun could be at $(0, 0, 1)$, and the Earth could start from $(1, 0, 1)$. In this case, the Earth's motion is governed by the matrix

$$g(\theta) \equiv \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This way, the Earth's orbit is

$$\left\{ g(\theta) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \;\middle|\; 0 < \theta \leq 2\pi \right\} = \left\{ \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 1 \end{pmatrix} \;\middle|\; 0 < \theta \leq 2\pi \right\} \simeq S^1.$$

The rotation of the Moon around the Earth, on the other hand, is not necessarily confined to any horizontal plane. On the contrary: it may take place in an oblique plane, with the normal vector

$$\mathbf{n} \equiv \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix},$$

with some nonzero real components n_1, n_2, and n_3. Let us define two more (real) orthonormal vectors:

$$\mathbf{m} \equiv \frac{1}{\sqrt{n_1^2 + n_2^2}} \begin{pmatrix} -n_2 \\ n_1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{k} \equiv \mathbf{n} \times \mathbf{m}.$$

This way, $\mathbf{n}$, $\mathbf{m}$, and $\mathbf{k}$ form a new axis system in $\mathbb{R}^3$ (Chap. 2, Sects. 2.2.4–2.3.2). Let us use them as columns in the new 3×3 (real) orthogonal matrix

$$O \equiv (\mathbf{n} \mid \mathbf{m} \mid \mathbf{k}).$$

We are now ready to define the matrix that rotates the Moon by angle θ in the oblique $\mathbf{m}$-$\mathbf{k}$ plane:

$$\hat{g}(\theta) \equiv O \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix} O^t.$$

This way, if the Moon is initially at $\mathbf{m}$ (relative to the Earth), then it will later be at

$$\begin{aligned} \hat{g}(\theta)\mathbf{m} &= O \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix} O^t \mathbf{m} \\ &= O \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\ &= O \begin{pmatrix} 0 \\ \cos(\theta) \\ \sin(\theta) \end{pmatrix} \\ &= \cos(\theta)\mathbf{m} + \sin(\theta)\mathbf{k} \end{aligned}$$

(relative to the Earth). For this reason, if the Earth was at the origin $(0, 0, 0)$, and the Moon was initially at $\mathbf{m}$, then the Moon's orbit would be

$$\left\{ \hat{g}(\theta)\mathbf{m} \;\middle|\; 0 < \theta \leq 2\pi \right\} = \{\cos(\theta)\mathbf{m} + \sin(\theta)\mathbf{k} \;|\; 0 < \theta \leq 2\pi\} \simeq S^1.$$

The Earth, however, is not static, but dynamic: it orbits the Sun at the same time. Therefore, the true route of the Moon in the solar system is the sum of these two routes: the Earth around the Sun, plus the Moon around the Earth, at a frequency twelve times as high:

$$\left\{ g(\theta) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \hat{g}(12\theta)\mathbf{m} \;\middle|\; 0 < \theta \leq 2\pi \right\}.$$

To visualize, let us use the discrete angles

$$0 < \theta_1 < \theta_2 < \cdots < \theta_N = 2\pi,$$

for some large natural number N. These N distinct angles produce the discrete path

$$\left\{ g(\theta_i) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \hat{g}(12\theta_i)\mathbf{m} \;\middle|\; 1 \leq i \leq N \right\}.$$

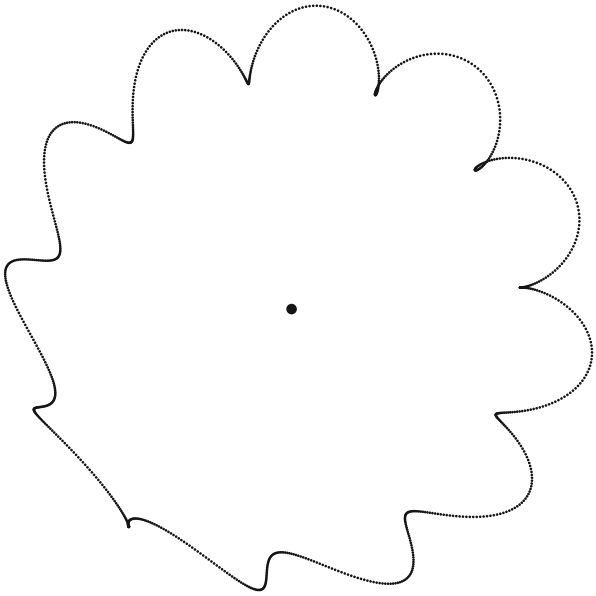


Fig. 6.12 The route of the Moon in the solar system, projected on the horizontal plane $z \equiv 1$. It is assumed that the Moon rotates around the Earth in an oblique plane, whose normal vector is $\mathbf{n} = (1, 1, 1)^t/\sqrt{3}$

This discrete path can now project onto the horizontal plane $z \equiv 1$, by just dividing each vector by its third component. This may produce a two-dimensional animation movie to visualize the original three-dimensional motion of the Moon in the solar system (Fig. 6.12).

6.8.7 Relation to the Complex Projective Plane

How does the real projective plane relate to the complex projective plane in Sect. 6.2.1? Well, recall that the latter was first reduced to a hemisphere. For this purpose, we divided by $\arg(c_2)$ (Sect. 6.2.7). At the equator at the bottom of the hemisphere, however, it is impossible to divide by $c_2 = 0$. Instead, we must divide by $\arg(c_1)$, shrinking the entire equator into just one infinity point, thus losing a lot of valuable information about the original direction of each individual infinity point on this equator.

Fortunately, the real projective plane improves on this. The equator no longer shrinks, so the original hemisphere no longer reduces to a standard sphere. This way, each pair of antipodal infinity points still points in the original direction, storing this valuable information for future use.

6.9 The Real Projective Space

6.9.1 The Real Projective Space

Let us now go ahead to a yet higher dimension: redefine V as

$$V \equiv \mathbb{R}^4 \setminus \{(0,0,0,0)\}.$$

Furthermore, redefine G as the group of 4×4 real nonsingular matrices. In this group, the unit element is the 4×4 identity matrix

$$I \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This new I is now used to redefine the subgroups C, U, and H:

$$\begin{aligned} C &\equiv \{xI \mid x \in \mathbb{R},\ x \neq 0\} = (\mathbb{R} \setminus \{0\})\, I \\ U &\equiv \{xI \mid x \in \mathbb{R},\ |x| = 1\} = \pm I = S^0 I \\ H &\equiv \{xI \mid x \in \mathbb{R},\ x > 0\}. \end{aligned}$$

We are now ready to project.

6.9.2 Oblique Projection

For each vector

$$v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} \in V.$$

In V, Cv is a complete equivalence class: a three-dimensional hyperplane in V. Still, it also has another face: an individual element (or "point") in the real projective space V/C. Let us go ahead and project it "obliquely."

If $v_4 \neq 0$, then Cv could indeed project on the hyperplane

$$\{(x, y, z, 1) \in V\} \subset V$$

simply by dividing by v_4:

$$v \to \frac{1}{v_4} v = \begin{pmatrix} v_1/v_4 \\ v_2/v_4 \\ v_3/v_4 \\ 1 \end{pmatrix}.$$

If, on the other hand, $v_4 = 0$, then v represents an infinity point that must project radially by

$$v \to \pm \frac{1}{\|v\|} v.$$

This completes our "oblique" projection.

6.9.3 Radial Projection

Alternatively, one could also use a more uniform approach: always project radially, regardless of whether v_1 vanishes or not:

$$v \to \pm \frac{1}{\|v\|} v.$$

Fortunately, in S^3/S^0, these antipodal points are considered as one and the same.

As in Sect. 6.4.1, this new projection could also be written algebraically as

$$V/C = \frac{V}{UH} = \frac{V/H}{U} \simeq S^3/S^0,$$

where "$\simeq$" stands for topological homeomorphism.

6.10 Duality: Point–Plane

6.10.1 Points and Planes

As discussed above, the point $v \in V$ represents the point Cv in the real projective space V/C. Furthermore, as in Sect. 6.5.7, v also has a dual interpretation: the three-dimensional hyperplane orthogonal to v:

$$\{(x, y, z, w) \in V \mid xv_1 + yv_2 + zv_3 + wv_4 = 0\}.$$

Fortunately, this hyperplane is invariant under C. Therefore, it might get rid of one redundant degree of freedom and be viewed as a two-dimensional plane in V/C.

In summary, in the real projective space, Cv may take two possible meanings: either the point Cv, or the plane orthogonal to v.

We have already seen duality in the context of the real projective plane: we have used vector product (defined in Chap. 2, Sect. 2.2.3) to show that two distinct lines meet at a unique point, and two distinct points make a unique line (Sects. 6.5.4 and 6.5.7).

To extend this, we must first extend vector product to four spatial dimensions as well. This will help establish duality in the real projective space as well: every three independent points make a unique plane, and every three independent planes meet at a unique point.

In a more uniform language, in V/C, three independent objects of one kind make a unique object of another kind. Thus, the three original objects could be interpreted in terms of either kind: there is a complete symmetry between both kinds.

6.10.2 The Extended Vector Product

To define the required vector product in four spatial dimensions as well, consider a row made of four column vectors—the standard unit vectors in V:

$$E \equiv \left(\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right).$$

This row will serve as the first row in the 4×4 matrix used in the vector product.

The so-called triple vector product can now be defined as a vector function of the form

$$\times : \left(\mathbb{R}^4\right)^3 \to \mathbb{R}^4.$$

What does it do? Well, it takes the column vectors $u, v, w \in \mathbb{R}^4$ and places them as rows in a new 4×4 matrix (whose first row is E). Finally, it returns the determinant:

$$\times (u, v, w) \equiv \det \begin{pmatrix} E \\ u^t \\ v^t \\ w^t \end{pmatrix}.$$

This is a new vector in $\mathbb{R}^4$, as required. Indeed, thanks to the original definition of a determinant (Chap. 2, Sect. 2.1.1), this is just a linear combination of the items in the first row—the standard unit vectors in $\mathbb{R}^4$. Let us use it in the real projective space.

6.10.3 Three Points Make a Unique Plane

Fortunately, a matrix with two identical rows has a zero determinant (Chap. 2, Sect. 2.2.4). This is why the new triple vector product is so attractive: it produces a new vector, orthogonal to u, v, and w:

$$(\times(u, v, w), u) = \det\left(\begin{pmatrix} u^t \\ u^t \\ v^t \\ w^t \end{pmatrix}\right) = 0$$

$$(\times(u, v, w), v) = \det\left(\begin{pmatrix} v^t \\ u^t \\ v^t \\ w^t \end{pmatrix}\right) = 0$$

$$(\times(u, v, w), w) = \det\left(\begin{pmatrix} w^t \\ u^t \\ v^t \\ w^t \end{pmatrix}\right) = 0.$$

For this reason, if u, v, and w are linearly independent vectors that represent three independent points in V/C, then their triple vector product represents the required "plane" in V/C: the unique plane that contains all three points—Cu, Cv, and Cw:

$$Cu, Cv, Cw \in \{C\mathbf{v} \in V/C \mid (\times(u, v, w), \mathbf{v}) = 0\} \subset V/C.$$

Let us now look at things the other way around.

6.10.4 Three Planes Share a Unique Point

In the dual interpretation, on the other hand, u is no longer a mere vector in V, but rather a complete hyperplane in V: the hyperplane orthogonal to u. Likewise, v is now viewed as the hyperplane orthogonal to v, and w is now viewed as the hyperplane orthogonal to w. What point do they share? This is just

$$\times(u, v, w).$$

After all, in V, this point belongs to all three hyperplanes. Therefore, in V/C,

$$C\,(\times(u, v, w))$$

is indeed the unique point shared by all three planes, as required.

Conics can now be defined in the spirit in Sect. 6.6.1. Tangent hyperplanes can also be defined in the spirit in Sect. 6.6.5. The details are left as an exercise.

6.11 Exercises

1. In Sect. 6.2.8, show that C/H is a legitimate subgroup of G/H. Hint: See Chap. 5, Sect. 5.10.2.
2. Furthermore, show that C/H is normal. Hint: See Chap. 5, Sect. 5.10.2.
3. In Sect. 6.2.8, what is the center of G/H?
4. Show that this center must include C/H. Hint: Each element $Hc \in C/H$ commutes with every element $Hg \in G/H$:

$$HcHg = H(cg) = H(gc) = HgHc.$$

5. Show that C/H must include the center of G/H. Hint: in the exercises at the end of Chap. 5, assume that the matrices A and B commute up to a scalar multiple. Fortunately, this scalar must be 1. After all, when either A or B is diagonal, both AB and BA have the same diagonal.
6. Conclude that C/H is indeed the center of G/H.
7. How can the formula $U \simeq C/H$ (end of Sect. 6.2.8) be deduced from the second isomorphism theorem in Chap. 5, Sect. 5.10.1? Hint: Assume that T and S have just one joint element: the unit element. Then, substitute $T \leftarrow U$ and $S \leftarrow H$.
8. Show that the set of real nonsingular 3×3 matrices is indeed a group.
9. Show that I, the identity matrix of order 3, is indeed the unit element in this group.
10. Show that the center of this group is the set of real nonzero scalar multiples of I. Hint: See exercises at the end of Chap. 5.
11. Show that U, defined in Sect. 6.4.1, is indeed a group in its own right.
12. Conclude that U is indeed a subgroup of the above center.
13. Show that H, defined in Sect. 6.4.1, is indeed a group in its own right.
14. Conclude that H is indeed a subgroup of the above center.
15. Conclude that UH is indeed a group in its own right.
16. Conclude that UH is a subgroup of the above center.
17. Show that UH is exactly the same as the above center.
18. Show that a "point" in the real projective plane could be viewed an object in $\mathbb{R}^3$: a real nonzero three-dimensional vector, defined up to a nonzero scalar multiple.
19. Show that a "line" in the real projective plane could be interpreted in terms of its normal vector, which is a real nonzero three-dimensional vector. This way, the "line" is still in $\mathbb{R}^3$: it contains those three-dimensional vectors orthogonal to that normal vector.

20. Show that multiplying that normal vector by a nonzero real scalar would still produce the same "line" as above.
21. Show that the above "line" is invariant under nonzero scalar multiplication.
22. Conclude that the above "line" is invariant under the above center.
23. Conclude that the above "line" is indeed a legitimate line in the real projective plane.
24. Conclude that the line can be rightly called a projective line.
25. Show that, in the real projective plane, two distinct points make a unique line. Hint: Use the vector product of the original three-dimensional vectors as a normal to the required line.
26. Show that, in the real projective plane, two distinct lines share a unique point (possibly an infinity point). Hint: Take the vector product of the original normal vectors.
27. Give an algebraic condition to guarantee that two such projective lines are "parallel" to each other or meet each other at a unique infinity point on the infinity circle. Hint: The original normal vectors must have a vector product with a vanishing z-coordinate. To guarantee this, from each normal vector, drop the third component. The resulting two-dimensional subvectors should be proportional to each other.
28. Show that the infinity line meets every other projective line at a unique infinity point on the infinity circle. Hint: Its normal vector is $(0, 0, 1)^t$, so the above condition indeed holds.
29. Show that a "point" in the real projective space could be viewed as an object in $\mathbb{R}^4$: a real nonzero four-dimensional vector, defined up to a nonzero scalar multiple.
30. Show that a "plane" in the real projective space could be interpreted as an object in $\mathbb{R}^4$, in terms of its four-dimensional normal vector. This way, the "plane" contains those four-dimensional vectors that are orthogonal to that normal vector.
31. Show that multiplying that normal vector by a nonzero real scalar still produces the same "plane" as above.
32. Show that the above "plane" is invariant under nonzero scalar multiplication.
33. Conclude that the above "plane" is indeed a legitimate plane in the real projective space.
34. Conclude that the above plane can be rightly called a projective plane.
35. Show that, in the real projective space, three independent points make a unique plane. Hint: Use the triple vector product (Sect. 6.10.2) of the original linearly independent four-dimensional vectors, to produce the required four-dimensional normal vector.
36. Show that, in the real projective space, three independent planes share a unique point. Hint: Take the triple vector product of the original linearly independent four-dimensional normal vectors. The resulting four-dimensional vector should be interpreted up to a nonzero scalar multiple.
37. Extend the definition of conics in Sect. 6.6.1 to the real projective space in Sect. 6.9.1 as well.

38. Define tangent planes in the real projective space, analogous to tangent lines in Sect. 6.6.5.
39. For $n = 1, 2, 3, \ldots$, define the $2n$-dimensional complex projective space.
40. Show that it is topologically homeomorphic to S^{2n+1}/S^1.
41. Show that this is just the top half of S^{2n}, with a rather strange "equator" at the bottom: not S^{2n-1} but rather S^{2n-1}/S^1—the infinity hyperplane, which is a lower-dimensional complex projective space in its own right, defined inductively.
42. Extend the duality established in Sects. 6.5.4–6.5.7 to the complex projective space as well. (Be sure to use the complex conjugate in your new kind of vector products.)
43. Produce an animation movie of a planar object traveling along a curved trajectory in the three-dimensional Cartesian space (Sect. 6.8.1).
44. Produce an animation movie of the Earth and the Moon, traveling in the solar system (Sect. 6.8.6).
45. Show that the set of real nonsingular 4×4 matrices is indeed a group.
46. Show that I, the identity matrix of order 4, is indeed the unit element in this group.
47. Show that the center of this group is the set of real nonzero scalar multiples of I. Hint: See exercises at the end of Chap. 5.
48. Show that U, defined in Sect. 6.9.1, is indeed a group in its own right.
49. Conclude that U is indeed a subgroup of this center.
50. Show that H, defined in Sect. 6.9.1, is indeed a group in its own right.
51. Conclude that H is indeed a subgroup of this center.
52. Conclude that UH is indeed a group in its own right.
53. Conclude that UH is a subgroup of the above center.
54. Show that UH is exactly the same as the above center.
55. Show that, in the method in Chap. 4, Sect. 4.5.4, the inverse Lorentz transformation back to the x-y-t self-system of the second particle is actually interpreted as a projective mapping in the real projective plane (Sects. 6.4.1 and 6.7.3). In this mapping, the original velocity $(dx'/dt', dy'/dt')$ of the first particle in the lab (Fig. 4.4) transforms to the new velocity $(dx/dt, dy/dt)$ of the first particle away from the second one (Fig. 4.5). Because we divide by t or t', the time variable is eliminated and is only used implicitly to advance the particle in the direction pointed at by the velocity vector.

Chapter 7
Quantum Mechanics: Algebraic Point of View



Matrices have two algebraic operations. Thanks to addition, they make a new linear space. Thanks to multiplication, they also form a group: the nonsingular matrices.

In this group, the commutative law does not hold any more. Indeed, multiplying from the left is not the same as multiplying from the right. How different could it be? To measure this, we need a new algebraic operation: the commutator.

Thanks to the commutator, we can now introduce yet another important field: quantum mechanics. Indeed, thanks to the above algebraic tools, this can be done in a straightforward and transparent way. For this purpose, we redefine momentum and energy in their stochastic (probabilistic) face.

In Chap. 2, we have already introduced angular momentum in classical mechanics. Here, on the other hand, we redefine angular momentum and highlight it from an algebraic point of view. This may help study a few elementary particles such as electrons and photons, with their new property: spin. This property is not well-understood physically. Fortunately, thanks to groups and matrices, it can still be modeled and understood mathematically.

What is the physical meaning of this? Well, in a very small scale, classical physics does not work any more. On the contrary: quantum mechanics teaches us new laws of nature. For example, ordering does matter: measuring position before momentum is not the same as measuring momentum before position. How to model this algebraically? We need a group with no commutative law: a non-Abelian group.

Fortunately, we already have such a group: the nonsingular matrices. Indeed, two matrices often have a nonzero commutator.

Quantum mechanics is not easy to grasp logically or intuitively. At best, it could be understood formally or mathematically, but not physically. Still, it helps make useful calculations and predictions in practice.

After learning about it, ask yourself: do I understand quantum mechanics? If you do, then something must be wrong. If, on the other hand, you are still puzzled, then you are on the right track.

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7.1 Nondeterminism

7.1.1 Relativistic Observation

In Newtonian mechanics, we often consider a particle, or just any physical object. Such an object must lie somewhere in space: this is its location or position.

At each individual time, the object may have a specific position. This is deterministic: we can measure the position and tell it for sure. This way, we can also calculate how fast it changes: the (momentary) velocity. Thanks to the velocity, we can also calculate the (momentary) momentum and kinetic energy. This tells us all we need to know: there is no doubt.

In Chap. 4, on the other hand, there is a new point of view: things are not absolute any more, but only relative. Indeed, the position that I measure in my lab may differ from the position that you measure in your own (inertial) system, traveling at a constant speed away from me. As a matter of fact, position is meaningless on its own, unless it has some reference point: the origin. What is meaningful is only the difference between two positions.

The same is true not only for distance but also for time. Indeed, time is not absolute, but only relativistic: the time that I see in my clock may differ from the time that you see in your own clock, traveling at a constant speed away from me.

Do not worry: no clock is bugged. Time is relativistic: it depends on the perspective it is measured from. In this sense, time is just an observation. To tell it, one must make a measurement or an experiment. This measurement is relativistic—it depends on perspective. This is why the time in my clock here on the Earth may differ from the time in some other clock, on board of a spaceship.

7.1.2 Determinism

Special relativity teaches us to be a little more humble: do not trust your eyes—they do not tell you the absolute truth. Not only time is relativistic, but also position, momentum, and energy.

Together, time and position make a new pair of relativistic observations. Likewise, momentum and energy make yet another pair, transformable by a Lorentz matrix. This is why momentum is more fundamental than velocity. Fortunately, this is still deterministic: in your own system, you still know what you see, with no doubt. In quantum mechanics, on the other hand, things are yet worse: you cannot be sure about anything any more.

7.1.3 Nondeterminism and Observables

In quantum mechanics, nothing is certain any more. Why? Because a subatomic particle is too small to occupy any specific point. If it had, then it would repel itself (if it had an electric charge) or attract itself and shrink (due to the strong nuclear force). Thus, the position is no longer an observation, but only an observable: a random variable. The particle *could* be at a certain point at some probability: a number that tells us how likely this is. Perhaps it is there, and perhaps not. We will never know, unless we conduct an experiment, and change the original state forever.

Indeed, to know the position for sure, you must take a measurement. But better not: the particle is often so small that this would require a complicated experiment, losing a lot of valuable information about other physical quantities (like momentum). This also works the other way around: better not measure the momentum, or you would lose all information about position forever.

Thanks to our advanced algebraic tools, we can now model even a weird state like this. To model an observable, use a matrix. This way, there is no need to look or observe as yet: this could wait until later. In the meantime, we can still "play" with our matrices algebraically and design more and more interesting observables.

7.2 State: Wave Function

7.2.1 Physical State

In Newtonian mechanics, we often consider a particle, traveling along the x-axis. At time t, it arrives at $x(t)$. Once differentiated, this gives the speed $x'(t)$ and the momentum $mx'(t)$ in the x-spatial direction (where m is the mass). This could be done at each and every time t.

In quantum mechanics, on the other hand, there is no determinism any more, but a lot of doubts. In "true" physics, the particle is nowhere (or everywhere). The physical state only tells us where it *might* be. Perhaps it is there, and perhaps not.

The physical state is no longer a function like $x(t)$, but a new random variable, whose probabilities are stored in a new (nonzero) n-dimensional (complex) vector:

$$v \in \mathbb{C}^n.$$

This vector contains every information that nature tells us about the particle, including where it might be at time t. For this purpose, v is not static, but dynamic: it may change in time:

$$v \equiv v(t) \in \mathbb{C}^n.$$

7.2.2 The Diagonal Position Matrix

Where might the particle be? For this purpose, we have a new $n \times n$ diagonal matrix: the position matrix X. On its main diagonal, you can find possible positions that the particle may (or may not) occupy. This is also called matrix mechanics.

Consider, for example, some element on the main diagonal: $X_{k,k}$ (for some $1 \leq k \leq n$). How likely is the particle to be at position $x = X_{k,k}$? The probability for this is stored in v: it is $|v_k|^2$.

7.2.3 Normalization

Clearly, the probabilities must sum to 1. For this purpose, we must assume that v has already been normalized to have norm 1:

$$\|v\|^2 = (v, v) = v^h v = \bar{v}^t v = \sum_{j=1}^{n} \bar{v}_j v_j = \sum_{j=1}^{n} |v_j|^2 = 1.$$

7.2.4 State and Its Overall Phase

Thus, the norm of v is actually immaterial: for all intents and purposes, one can actually assume that v is defined up to a scalar multiple only:

$$v \in \left(\mathbb{C}^n \setminus \{\mathbf{0}\}\right) / \left(\mathbb{C} \setminus \{0\}\right).$$

(See Chap. 5, Sects. 5.8.3–5.8.5. For $n = 2$, for instance, this is illustrated in Figs. 6.3, 6.4, 6.5.) This way, v is insensitive to the overall phase: it could precess, with no physical effect whatsoever. This will be discussed later. When is the overall phase relevant? Only in adding two states. Their phases may lead to interference (either constructive or destructive). Still, nature welcomes no addition: once v has been accepted as a legitimate state in nature, no other state could ever be added to it. As time goes by, v could only undergo a unitary transformation, in which each individual eigenvector precesses at its own frequency (or energy). This will be governed by the Schrodinger equation.

7.2.5 Dynamics: Schrodinger Picture

Unfortunately, n may be too small. After all, the particle could get farther and farther away from the origin and reach infinitely many positions. To allow this, X must be an infinite matrix, with an infinite order.

For example, a particle could move from number to number along the real axis. To model this, X must be as big as

$$X \equiv \begin{pmatrix} \ddots & & & & & & & \\ & -3 & & & & & & \\ & & -2 & & & & & \\ & & & -1 & & & & \\ & & & & 0 & & & \\ & & & & & 1 & & \\ & & & & & & 2 & \\ & & & & & & & 3 \\ & & & & & & & & \ddots \end{pmatrix}.$$

If the particle starts from 0 and makes exactly 5 steps (either rightward or leftward), then X must have order 11 (or more). If, on the other hand, the particle makes 7 steps, then X is of order 15 (or more) and so on.

To model an unlimited number of steps, on the other hand, X must be an infinite matrix. On its main diagonal, it must have all possible positions that the particle could reach: all integer numbers. Better yet, on its main diagonal, X should actually contain not only integer but also real numbers. This way, X makes a new operator.

This kind of dynamics is called Schrodinger's picture: X remains constant at all times, and $v \equiv v(t)$ changes in time. This is easier than Heisenberg's picture, which works the other way around: v remains constant, and $X \equiv X(t)$ changes in time.

For simplicity, however, we avoid an infinite dimension. Instead, we stick to our finite dimension n and our original n-dimensional vector and $n \times n$ matrix.

7.2.6 Wave Function and Phase

How does the state v look like? It may look like a discrete sine or cosine wave (Figs. 1.10, 1.11). This is why v is often called a wave function. Still, v is not necessarily real: it may be complex as well. For example, it may be a discrete Fourier mode (Fig. 7.1).

In general, each component v_k is a complex number, with its own polar decomposition: magnitude times exponent:

$$v_k = |v_k| \exp(i\theta_k),$$

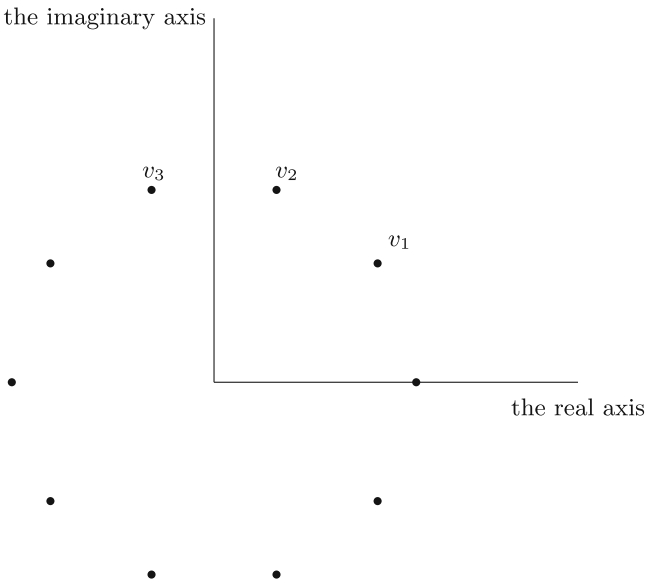


Fig. 7.1 Here, v looks like a discrete Fourier mode. Its components are distributed evenly on the unit circle in the complex plane. Once normalized, the probability is uniform: $|v_1|^2 = |v_2|^2 = \cdots = |v_n|^2 = 1/n$. Indeed, these sum to 1

where $i \equiv \sqrt{-1}$, and θ_k is the phase: the angle that v_k makes with the positive part of the real axis.

How likely is the particle to be at $X_{k,k}$? The probability for this is the square of the magnitude: $|v_k|^2$. The exponent, on the other hand, is a complex number of magnitude 1. As such, it has no effect on this probability. Still, it may store valuable information about other observables.

The original component v_k is often called the (complex) amplitude. Once all components are written in their polar decomposition, v takes a more familiar face: a wave function. This way, the original particle also has a new mathematical face: a wave. As such, it enjoys an interesting phenomenon: interference.

7.2.7 Phase and Interference

Two electrons cannot be in exactly the same state at the same time. This is Pauli's exclusion principle. (See exercises below.) Two photons, on the other hand, can. In this case, they have the same wave function.

Two wave functions may sum up and produce a new wave function. Since they are vectors, they add up linearly, component by component.

To add two corresponding components, their phases are most important. If they match, then they enhance each other. This is constructive interference. If, on the other hand, they do not match, then they can even cancel (annihilate) each other. This is destructive interference.

7.3 Observables: Which Is First?

7.3.1 Measurement: The State Is Gone

Let us go back to our original particle. It only has a nondeterministic position, where it *might* be. But where is it in fact?

Do not ask! Because, to find out, you must carry out an experiment. At probability $|v_k|^2$, you would find the particle at $x = X_{k,k}$ $(1 \leq k \leq n)$.

In this case, things have changed: you now know for sure that the particle is at $x = X_{k,k}$ (for some fixed k). There is no doubt any more: the probability for this is now as high as $|v_k|^2 = 1$. The probability to find the particle elsewhere, on the other hand, is now as low as $|v_j|^2 = 0$ $(j \neq k)$. So, v has changed forever. The valuable information it contained (about other physical quantities) is now gone.

Instead of the original v, you now have a new (boring) v: a standard unit vector:

$$v = e^{(k)}.$$

Only the kth component is 1, and the rest vanish. There is still one good thing about it: this is an eigenvector of X, with eigenvalue $X_{k,k}$:

$$Xe^{(k)} = X_{k,k}e^{(k)}.$$

Let us use this to rewrite our original v as a sum of standard unit vectors:

$$v = \sum_{j=1}^{n} v_j e^{(j)} = \sum_{j=1}^{n} \left(e^{(j)}, v\right) e^{(j)}.$$

Let us do the same for yet another interesting observable: momentum.

7.3.2 The Momentum Matrix and Its Eigenvalues

Let us study the linear momentum p of the particle (in the x-direction). For this, we are given yet another $n \times n$ Hermitian matrix: P:

$$P^h \equiv \bar{P}^t = P.$$

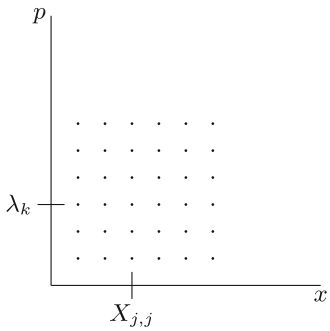


Fig. 7.2 The position–momentum grid. First, normalize v. Now, to be at $x = X_{j,j}$, the particle has probability $|v_j|^2$. To have momentum $p = \lambda_k$, the particle has probability $|(u^{(k)}, v)|^2$, where $u^{(k)}$ is an (normalized) eigenvector

Due to nondeterminism, the original state v does not tell us the momentum for sure, but only with a little doubt. For example, let λ_k be an (real) eigenvalue of P. How likely is the momentum to be $p = \lambda_k$? The probability for this is

$$\left|\left(u^{(k)}, v\right)\right|^2,$$

where v is our (normalized) state, and $u^{(k)}$ is the (normalized) eigenvector of P, corresponding to λ_k:

$$Pu^{(k)} = \lambda_k u^{(k)}.$$

In Fig. 7.2, this is illustrated in an $n \times n$ grid of points like $(X_{j,j}, \lambda_k)$ $(1 \leq j, k \leq n)$. Better yet, this should be an infinite two-dimensional plane. After all, in reality, position and momentum could take just any value. To model this, n should be infinite, and the inner product should be an infinite sum (or even an integral, as in Chap. 15, Sect. 15.1.1). For the time being, we stick to our finite dimension n and our n-dimensional vectors and $n \times n$ matrices.

7.3.3 Ordering Matters!

Why are matrices so good at modeling physical observables? Because they do not commute. This models quantum mechanics well: ordering matters.

Although you might be curious to know the exact position of the particle, better restrain yourself, and not look! Because, if you looked, then you would spoil v for good, losing all valuable probabilities of the form $|(u^{(k)}, v)|^2$ that gave you an idea about momentum. This also works the other way around. Better not measure momentum, or you would damage v, and lose the probabilities $|v_j|^2$ of the original position.

This means that ordering matters: measuring x and then p is not the same as measuring p and then x. This is modeled well by matrices: applying X before P is

not the same as applying P before X:

$$XP \neq PX.$$

7.3.4 Commutator

This is why matrices are so suitable to model physical observables. Indeed, they often do not commute with each other. In our case, X and P indeed have a nonzero commutator:

$$[X, P] \equiv XP - PX \neq (0)$$

(the $n \times n$ zero matrix).

Recall that X and P are both Hermitian. Therefore, their commutator $[X, P]$ is an anti-Hermitian matrix:

$$[X, P]^h = (XP - PX)^h = P^h X^h - X^h P^h = PX - XP = [P, X] = -[X, P].$$

This will be useful later.

7.3.5 Planck Constant

To have this commutator in its explicit form, we have a new law of nature:

$$[X, P] = i\bar{h}I,$$

where $i = \sqrt{-1}$, I is the $n \times n$ identity matrix, and $\bar{h}$ is the Planck constant (universal, positive, and very small).

Why must $[X, P]$ take this form? Well, it should better be a constant matrix, independent of the particle. After all, the same effect should appear even in vacuum, with no particle at all. This is due to zero-point (vacuum) energy at the ground state.

Furthermore, thanks to the imaginary number i, $[X, P]$ is indeed anti-Hermitian, as required. Finally, thanks to the small constant $\bar{h}$, $[X, P]$ is very small and has no effect in macroscale. This is why it was ignored in geometrical mechanics and special relativity (Chaps. 2 and 4).

7.4 Observable and Its Expectation

7.4.1 Observable (Measurable)

The above matrices are called observables (or measurables, or experiments). Why? Because they let us observe. For example, by applying X to v, we get to know the expectation of our random variable (the position) at v.

The actual observation, on the other hand, should better wait until later. After all, it requires an experiment, which may spoil the original state, including important probabilities of other observables. Before looking, let us "play" with our matrices a little.

7.4.2 Hermitian and Anti-Hermitian Parts

Consider an $n \times n$ matrix A (not necessarily Hermitian). Write it as the sum of two matrices:

$$A = \frac{A + A^h}{2} + \frac{A - A^h}{2}.$$

The former term is the Hermitian part:

$$\left(\frac{A + A^h}{2}\right)^h = \frac{A^h + A}{2} = \frac{A + A^h}{2}.$$

The latter is the anti-Hermitian part:

$$\left(\frac{A - A^h}{2}\right)^h = \frac{A^h - A}{2} = -\frac{A - A^h}{2}.$$

7.4.3 Symmetrization

An observable should be Hermitian. This way, its eigenvectors are orthogonal to each other. Once normalized, they make a new orthonormal basis, which can be used to decompose just any vector. Furthermore, a Hermitian matrix has real eigenvalues (Chap. 1, Sects. 1.9.4–1.9.5). This makes sense: in the entire nature around us, every observed value is real, with no imaginary part.

Fortunately, the anti-Hermitian part of A is often as small as $\bar{h}$ and can be dropped. This is how A is symmetrized: replaced by its Hermitian part. Fortunately,

this can wait until later. Meanwhile, we can still stick to our original A, Hermitian or not.

7.4.4 Observation

So far, we have seen two observables: the position X and the momentum P. In special relativity, the time t could also be viewed as an observation (Chap. 4). After all, to have the time, you must observe: either look at your own clock, to see your proper time, or at someone else's clock, to see a different observation: a slower time.

Of course, in special relativity, the scale is so big that everything is deterministic. For all intents and purposes, the observations commute with each other: ordering does not really matter.

In a very small scale, on the other hand, random effects can no longer be ignored. The "true" physical state is no longer deterministic. It only gives us the probability to observe something, not the actual observation. This is "true" nature: just probability. Of course, we humans will never get to see this "truth." After all, we must make a decision: what to measure first, and what to measure later. Each choice may give us different results: ordering does matter. Thus, the true (original) nature will remain a mystery.

7.4.5 Random Variable

Thus, an observable is more mathematical than physical. It makes a new random variable: we cannot tell for sure what its value is, but only what it *might* be. For example, its value could be λ: some eigenvalue of the observable. The probability for this is $|(u, v)|^2$, where v is the (normalized) state, and u is the (normalized) eigenvector, associated with λ.

An observable such as X or P must be Hermitian. How to have its expectation (or average) at state v? Easy: apply it to v, and take the inner product with v. (Below, we will see why.) As a matter of fact, this could be done even for a more general matrix A (not necessarily Hermitian). This gives us a new complex number:

$$(v, Av) = \left(v, \frac{A + A^h}{2} v\right) + \left(v, \frac{A - A^h}{2} v\right).$$

The former term is real, and the latter is pure imaginary. Thus, in terms of absolute value, each term is as small as the entire sum:

$$|(v, Av)| = \left|\left(v, \frac{A + A^h}{2} v\right) + \left(v, \frac{A - A^h}{2} v\right)\right| \geq \left|\left(v, \frac{A - A^h}{2} v\right)\right|.$$

This will be useful later.

7.4.6 Observable and Its Expectation

To write the expectation of X at state v, let us expand v in terms of standard unit vectors:

$$\begin{aligned}(v, Xv) &= \left(\sum_{j=1}^{n} v_j e^{(j)}, X \sum_{j=1}^{n} v_j e^{(j)}\right) \\ &= \left(\sum_{j=1}^{n} v_j e^{(j)}, \sum_{j=1}^{n} X_{j,j} v_j e^{(j)}\right) \\ &= \sum_{j=1}^{n} X_{j,j} \left|v_j\right|^2 .\end{aligned}$$

This is the expectation: multiply probability times value and sum over all possible values. Similarly, to write the expectation of P (at v), expand v in terms of (normalized) eigenvectors of P:

$$\begin{aligned}(v, Pv) &= \left(\sum_{k=1}^{n} \left(u^{(k)}, v\right) u^{(k)}, P \sum_{k=1}^{n} \left(u^{(k)}, v\right) u^{(k)}\right) \\ &= \left(\sum_{k=1}^{n} \left(u^{(k)}, v\right) u^{(k)}, \sum_{k=1}^{n} \lambda_k \left(u^{(k)}, v\right) u^{(k)}\right) \\ &= \sum_{k=1}^{n} \lambda_k \left|\left(u^{(k)}, v\right)\right|^2\end{aligned}$$

(thanks to orthonormality). Again, this is the correct expectation: sum of probability times value. Since both X and P are Hermitian, their expectations are real, as required.

7.5 Heisenberg's Uncertainty Principle

7.5.1 Variance

The original random variable might take all sorts of possible values. How likely are they to spread out around the average? To get some idea about this, we also introduce the variance (at state v). Again, this could be done even for our general matrix A:

$$\|(A - (v, Av)\, I)\, v\|^2 .$$

To estimate the variance of X and P, let us use the covariance.

7.5.2 Covariance

At state v, consider the product of the variances of X and P. How to estimate this from below? Thanks to the Cauchy–Schwarz inequality (end of Chap. 1),

$$\|(X - (v, Xv)\, I)\, v\| \cdot \|(P - (v, Pv)\, I)\, v\| \geq |((X - (v, Xv)\, I)\, v, (P - (v, Pv)\, I)\, v)| .$$

What do we have on the right-hand side? This is the covariance of X and P (at state v). To estimate it from below, recall that although X and P are both Hermitian, their product XP is not. Fortunately, we still have the estimate in Sect. 7.4.5:

$$\begin{aligned}
\|(X - (v, Xv)\, I)\, v\| \cdot \|(P - (v, Pv)\, I)\, v\| &\geq |((X - (v, Xv)\, I)\, v, (P - (v, Pv)\, I)\, v)| \\
&= |(v, (X - (v, Xv)\, I)\, (P - (v, Pv)\, I)\, v)| \\
&\geq \frac{1}{2}\, |(v, [X - (v, Xv)\, I, P - (v, Pv)\, I]\, v)| \\
&= \frac{1}{2}\, |(v, [X, P]\, v)| \\
&= \frac{1}{2}\, \left|\left(v, i\bar{h} I v\right)\right| \\
&= \frac{\bar{h}}{2}\, |(v, v)| \\
&= \frac{\bar{h}}{2}
\end{aligned}$$

(assuming that v has norm 1).

7.5.3 Heisenberg's Uncertainty Principle

Finally, take the square of both sides. This gives us a lower bound for the variance of X times the variance of P (at v):

$$\|(X - (v, Xv)\, I)\, v\|^2\, \|(P - (v, Pv)\, I)\, v\|^2 \geq \frac{\bar{h}^2}{4}.$$

This is Heisenberg's uncertainty principle. It says that you cannot enjoy both worlds. If you measured the precise position, then the variance of X becomes zero. Unfortunately, there is a price to pay: the variance of P gets infinite. As a result, there is no hope to measure the original momentum any more.

This also works the other way around: if you measured the precise momentum, then the variance of P would vanish. In this case, the variance of X gets infinite, so you can never measure the original position any more. Thus, better avoid the actual observation. Instead, study the original observables and their algebraic properties.

7.6 Wave: Debroglie Relation

7.6.1 Infinite Matrix (or Operator)

So far, the position x could take only a few discrete values. These are the diagonal elements in the position matrix X. Next, let us extend X to an infinite matrix (or an operator). On its (infinite) main diagonal, place all numbers: not only integer but also real. This is more realistic: x could now take just any real position in $\mathbb{R}$.

7.6.2 Momentum: Another Operator

In differential geometry, x is often replaced by a more general concept: the differential operator $\partial/\partial x$. This is relevant not only in a linear plane but also in a curved surface and even in a differentiable manifold like spacetime.

To follow this, let us introduce a new momentum operator:

$$P \equiv -i\bar{h}\frac{\partial}{\partial x}.$$

What is the physical reasoning behind this? Well, what happens when P is applied to the wave function? This differentiates the wave function with respect to position. For instance, consider a fixed position x. If the momentum there is positive, then the above definition tells us that the wave function should have a slope upward: the

particle is likely to be ahead of x, more than behind. This makes sense physically as well: the particle probably gets a "kick" in the x-direction.

The above definition makes sense not only analytically but also algebraically. Indeed, thanks to it, P does not commute with X. On the contrary: they have a nonzero commutator, as required in quantum mechanics.

7.6.3 The Commutator

Indeed, the commutator of X and P is a new operator. How does it act on a real (differentiable) function $f : \mathbb{R} \to \mathbb{R}$? Like this:

$$\begin{aligned}[X, P]f &= (XP - PX)f \\ &= -(PX - XP)f \\ &= i\bar{h}((xf)' - xf') \\ &= i\bar{h}(f + xf' - xf') \\ &= i\bar{h}f.\end{aligned}$$

Thus, the commutator is $i\bar{h}$ times the identity operator, as required.

7.6.4 Wave: An Eigenfunction

What are the eigenfunctions of P? These are waves of frequency ω and wave number k, traveling at speed ω/k. Indeed,

$$P\exp(i(kx - \omega t)) = -i\bar{h} \cdot ik\exp(i(kx - \omega t)) = \bar{h}k\exp(i(kx - \omega t)).$$

7.6.5 Duality: Particle—Matter or Wave?

This is indeed duality: in quantum mechanics, a particle may behave not only as matter but also as a wave. Its momentum could be just any eigenvalue of P.

7.6.6 Debroglie's Relation: Momentum–Wave Number

This is indeed Debroglie's relation: momentum is proportional to the wave number:

$$p = \bar{h}k.$$

This is nondeterministic: the probability to pick a concrete k is stored in the state.

7.7 Planck and Schrodinger Equations

7.7.1 Hamiltonian: Energy Operator

In special relativity, we have already seen that momentum mirrors space, and energy mirrors time (Chap. 4, Sect. 4.6.3). The energy operator, the Hamiltonian H, should thus differentiate in time:

$$H \equiv i\hbar h \frac{\partial}{\partial t}.$$

This is indeed Schrodinger's equation.

7.7.2 Time–Energy Uncertainty

Thanks to the Schrodinger equation, time–energy mirror space–momentum. Therefore, they also have their own uncertainty principle: the better you know the energy, the worse you know what time it is, and vice versa. Furthermore, thanks to the Schrodinger equation, we immediately have the eigenfunctions of H: our good old waves.

7.7.3 Planck Relation: Frequency–Energy

Indeed, let us apply H to our wave:

$$H \exp(i(kx - \omega t)) = i\bar{h}(-i\omega) \exp(i(kx - \omega t)) = \bar{h}\omega \exp(i(kx - \omega t)).$$

This is indeed Planck's relation: energy is proportional to frequency:

$$E = \bar{h}\omega.$$

This was used by Einstein to explain the photo-electric effect: only a photon with sufficiently high frequency has enough energy to knock out an electron.

7.7.4 No Potential: Momentum Is Conserved Too

Thanks to the above design, P and H share the same eigenfunctions: our waves. In other words, P and H commute with one another. Why? Because there is no force or potential at all. This way, both momentum and (kinetic) energy are conserved. Later on, we will see a more complicated example, with force and potential: the harmonic oscillator. There, the momentum will not commute with the Hamiltonian any more. Why? Because the Hamiltonian will contain a new potential term: X^2. This does not commute with P. Thus, both momentum and kinetic energy will not be conserved any more.

7.7.5 Stability in Bohr's Atom

In our original wave, let us separate variables and decompose:

$$\exp(i(kx - \omega t) = \exp(-i\omega t)\exp(ikx).$$

Now, assume that k is integer. Assume also that x is now confined to a circle, say the orbit of the electron around the nucleus in the atom. x is now interpreted as angle, and p as angular momentum. Fortunately, P still commutes with H. After all, in polar coordinates, the centrifugal and centripetal forces cancel one another, so there is no force or potential at all. This is why P and H still share the same eigenfunctions: standing waves, which are functions of x, precessing at frequency ω in time. (This is just a nonphysical phase, with no effect on the magnitude.)

The state, however, is not just one standing wave, but a superposition (sum) of many, each with its own frequency, wave number, and coefficient, telling us how likely the electron is to have certain momentum and (kinetic) energy. In such a superposition, the standing waves may interfere with each other and change dynamically in terms of position (but not in terms of energy or momentum, which remain conserved, keeping their original probabilities). This way, the model remains stable: each standing wave represents an electron that radiates no energy and loses no mass in its orbit. This is how Debroglie proved stability in Bohr's atom.

So far, we have discussed infinite matrices (or operators). Next, let us go back to our original $n \times n$ matrix and study its algebraic properties.

7.8 Eigenvalues

7.8.1 Shifting an Eigenvalue

Thanks to linear algebra, we already have an important tool: eigenvectors and their eigenvalues. Let us use these to study the commutator in general. For this purpose, let C and T be $n \times n$ matrices (Hermitian or not). Assume also that they do not commute but have a nonzero commutator, proportional to T:

$$[C, T] \equiv CT - TC = \alpha T,$$

for some (complex) parameter $\alpha \neq 0$. Let u be an eigenvector of C, with eigenvalue λ:

$$Cu = \lambda u.$$

How to find more eigenvectors? Just apply T to u. Indeed, if $Tu \neq \mathbf{0}$, then this is an eigenvector as well:

$$CTu = (TC + [C, T])\, u = (TC + \alpha T)\, u = (\lambda + \alpha)\, Tu.$$

Thus, we have "shifted" λ by α, obtaining a new eigenvalue of C: $\lambda + \alpha$ (so long as $Tu \neq \mathbf{0}$).

Moreover, we can now apply T time and again (so long as we do not hit the zero vector). This way, we obtain more and more eigenvalues of C:

$$\lambda,\ \lambda + \alpha,\ \lambda + 2\alpha,\ \lambda + 3\alpha,\ \ldots.$$

Since n is finite, this cannot go on forever. For infinite matrices, on the other hand, it can.

7.8.2 Shifting an Eigenvalue of a Product

Let us look at a special case. Let A and B be $n \times n$ matrices (Hermitian or not). Assume that they do not commute but have a nonzero commutator, proportional to the identity matrix:

$$[A, B] \equiv AB - BA = \beta I,$$

for some (complex) parameter $\beta \neq 0$.

Now, let us look at the product BA. What is its commutator with B? We already know what it is:

$$[BA, B] \equiv BAB - BBA = B(AB - BA) = B[A, B] = \beta B.$$

We can now use Sect. 7.8.1 to shift an eigenvalue of BA. For this purpose, let u be an eigenvector of BA with the eigenvalue λ:

$$BAu = \lambda u.$$

If $Bu \neq \mathbf{0}$, then this is an eigenvector of BA as well:

$$BA(Bu) = (\lambda + \beta)Bu.$$

Likewise, one could also shift in the opposite direction. For this purpose, look again at the product BA. What is its commutator with A? It is

$$[BA, A] = BAA - ABA = (BA - AB)A = -[A, B]A = -\beta A.$$

We can now use Sect. 7.8.1 once again and design a new eigenvector of BA: take the original eigenvector u and apply A to it:

$$BA(Au) = (\lambda - \beta)Au.$$

This way, if $Au \neq \mathbf{0}$, then this is indeed an eigenvector of BA as well, with the new eigenvalue $\lambda - \beta$. Moreover, we can go on and on (so long as we do not hit the zero vector). This way, we obtain more and more eigenvalues of BA:

$$\lambda,\ \lambda \pm \beta,\ \lambda \pm 2\beta,\ \lambda \pm 3\beta,\ \ldots.$$

Since n is finite, this cannot go on forever. For infinite matrices, on the other hand, it can.

7.8.3 A Number Operator

Consider now a special case, in which

$$B \equiv A^h \quad \text{and} \quad \beta \equiv 1.$$

This way, the assumption in the beginning of Sect. 7.8.2 takes the form

$$[A, A^h] = I.$$

Furthermore, the product studied above is now

$$BA = A^h A.$$

As a Hermitian matrix, A^hA has real eigenvalues only. Furthermore, thanks to the above discussion, we can now lower or raise an eigenvalue of A^hA: from λ to

$$\lambda + 1,\ \lambda + 2,\ \lambda + 3,\ \ldots$$

(until hitting the zero vector), and also to

$$\lambda - 1,\ \lambda - 2,\ \lambda - 3,\ \ldots$$

(until hitting the zero vector, which will be very soon).

A^hA is an important matrix: it is called a number operator. Why? Because its eigenvalues are $0, 1, 2, 3, 4, \ldots$.

7.8.4 Eigenvalue—Expectation

Indeed, let u be an eigenvector of A^hA, with the eigenvalue λ:

$$A^hAu = \lambda u.$$

With some effort, both λ and u could be calculated. But there is no need. Indeed, since A^hA is Hermitian, λ must be real (Chap. 1, Sect. 1.9.4). Could λ be negative? No! Indeed, look at the expectation of A^hA at u:

$$\lambda(u, u) = \left(u, A^hAu\right) = (Au, Au) \geq 0.$$

Next, let us use u to design more eigenvectors.

7.8.5 Down the Ladder

Recall that we are considering now a special case: in Sect. 7.8.2, set $B \equiv A^h$, and $\beta \equiv 1$. This way, the above eigenvector u can be used to design a new eigenvector: Au. Indeed, if $Au \neq \mathbf{0}$, then this is an eigenvector of A^hA in its own right, with a lower eigenvalue: $\lambda - 1$. Thus, A serves as a ladder operator: by applying A to u, we go down the ladder, to a yet smaller eigenvalue.

What is the norm of Au? To find out, look again at the expectation of A^hA at u:

$$\|Au\|^2 = (Au, Au) = \left(u, A^hAu\right) = \lambda(u, u) = \lambda\|u\|^2.$$

In other words,

$$\|Au\| = \sqrt{\lambda}\|u\|.$$

Later, we will use this to normalize the eigenvectors.

7.8.6 Null Space

This lowering process cannot go on forever, or we would eventually hit a negative eigenvalue, which is forbidden (Sect. 7.8.4). It must terminate upon hitting some eigenvector w in the null space of $A^h A$:

$$A^h A w = \mathbf{0}.$$

At this point, lowering must stop. Thus, λ must have been a nonnegative integer.

What is so good about w? Well, at w, $A^h A$ must have zero expectation:

$$(Aw, Aw) = \left(w, A^h A w\right) = (w, \mathbf{0}) = 0.$$

Thus,

$$Aw = \mathbf{0}.$$

So, w is in the null space of A too (Chap. 1, Sect. 1.9.2). Starting from w, we can now work the other way around: raise eigenvalues back again.

7.8.7 Up the Ladder

For this purpose, use Sect. 7.8.2 once again (with $B \equiv A^h$ and $\beta \equiv 1$, as before). This time, look at $A^h u$: a new eigenvector of $A^h A$, with a bigger eigenvalue: $\lambda + 1$. Here, A^h serves as a new ladder operator, to "climb" up.

What is the norm of $A^h u$? Since $[A, A^h] = I$,

$$\begin{aligned}
\|A^h u\|^2 &= \left(A^h u, A^h u\right) \\
&= \left(u, A A^h u\right) \\
&= \left(u, \left(A^h A + [A, A^h]\right) u\right) \\
&= \left(u, \left(A^h A + I\right) u\right) \\
&= (u, (\lambda + 1)\, u)
\end{aligned}$$

$$= (\lambda + 1)\,(u, u)\,.$$

In other words,

$$\|A^h u\| = \sqrt{\lambda + 1}\|u\|.$$

Later on, we will use this to normalize these eigenvectors, as required.

7.9 Hamiltonian

7.9.1 Harmonic Oscillator

So far, we studied the number operator $A^h A$ from an algebraic point of view: its eigenvalues and eigenvectors. Still, what is its physical meaning? To see this, let us model an important physical phenomenon: a harmonic oscillator (or a spring).

For this purpose, let us use our position and momentum matrices: X and P. Let us use them to define a new matrix—the Hamiltonian:

$$H \equiv \frac{m\omega^2}{2}\left(X^2 + \frac{1}{m^2\omega^2}P^2\right),$$

where m is the mass, and ω is the frequency. (These are given real parameters.) Do not confuse ω with the vector w in Sect. 7.8.6, or with the angular velocity in geometrical mechanics. H is the Hamiltonian observable: the total energy of the harmonic oscillator: potential plus kinetic. Like X and P, H is nondeterministic: the total energy *might* be an eigenvalue of H. How to calculate the probability for this? Take the corresponding eigenvector, normalize it, calculate its inner product with the (normalized) state v, take the absolute value, and square it up.

7.9.2 Concrete Number Operator

How do the eigenvalues of H look like? To see this, define A (in Sect. 7.8.3) more concretely:

$$A \equiv= \sqrt{\frac{m\omega}{2\bar{h}}}\left(X + \frac{i}{m\omega}P\right).$$

This way, its Hermitian adjoint is

$$A^h = \sqrt{\frac{m\omega}{2\bar{h}}}\left(X - \frac{i}{m\omega}P\right).$$

Thus, the commutator is

$$\left[A, A^h\right] = \frac{m\omega}{2\bar{h}}\left[X + \frac{i}{m\omega}P, X - \frac{i}{m\omega}P\right] = -\frac{m\omega}{2\bar{h}} \cdot \frac{i}{m\omega}\left([X, P] - [P, X]\right) = I,$$

as required in Sect. 7.8.3. Thus, we can now lower and raise eigenvalues. To construct normalized eigenvectors of $A^h A$, start from some w in the null space of A, normalize it, apply A^h time and again, and normalize:

$$w,\ A^h w,\ \frac{1}{\sqrt{2}}\left(A^h\right)^2 w,\ \frac{1}{\sqrt{3!}}\left(A^h\right)^3 w,\ \ldots\ \frac{1}{\sqrt{k!}}\left(A^h\right)^k w,\ \ldots.$$

These are the orthonormal eigenvectors of $A^h A$. Thanks to them, we can now span any vector (as in Chap. 1, Sect. 1.9.5).

7.9.3 Energy Levels

Mathematically, we already know a lot about our concrete number operator, and its eigenvalues and eigenvectors. Still, what is its physical meaning? To see this, let us calculate it explicitly:

$$\begin{aligned} A^h A &= \frac{m\omega}{2\bar{h}}\left(X - \frac{i}{m\omega}P\right)\left(X + \frac{i}{m\omega}P\right) \\ &= \frac{m\omega}{2\bar{h}}\left(X^2 + \frac{1}{m^2\omega^2}P^2 + \frac{i}{m\omega}[X, P]\right) \\ &= \frac{m\omega}{2\bar{h}}\left(X^2 + \frac{1}{m^2\omega^2}P^2\right) + \frac{i}{2\bar{h}}i\bar{h}I \\ &= \frac{1}{\bar{h}\omega}H - \frac{1}{2}I, \end{aligned}$$

or

$$H = \bar{h}\omega\left(A^h A + \frac{1}{2}I\right).$$

What does this mean physically? Well, in the harmonic oscillator, the total energy (potential plus kinetic) cannot be just any real number: it is confined to specific energy levels. These are the only values allowed:

$$\frac{\bar{h}\omega}{2},\ \frac{3\bar{h}\omega}{2},\ \frac{5\bar{h}\omega}{2},\ \frac{7\bar{h}\omega}{2},\ \ldots.$$

This is indeed quantum mechanics: energy is no longer continuous but comes in discrete levels. Position and time, on the other hand, remain continuous: at each individual time, the particle may still lie in just any position.

Furthermore, the eigenvectors of H are the same as those of $A^h A$ (designed above). Thanks to conservation of energy, each of them makes a constant state, with no dynamics at all: if your initial state is an eigenvector, then it must remain so forever. After all, it must keep the same energy level (its eigenvalue). For this reason, the wave function must be a standing function that never travels. (At most, it can only precess at a constant frequency, with no physical effect.) Let us see how it looks like.

7.9.4 Ground State (Zero-Point Energy)

First, let us look at w: the eigenvector that lies in the null space of A (Fig. 7.3):

$$Aw = \mathbf{0}.$$

What is its physical meaning? It represents a strange case: no frequency, and no motion at all.

This is the minimal energy level. Indeed, w is also in the null space of $A^h A$:

$$A^h A w = \mathbf{0}.$$

Therefore, w is an eigenvector of H as well:

$$Hw = \bar{h}\omega \left(A^h A + \frac{1}{2} I \right) w = \frac{\bar{h}\omega}{2} w.$$

This is why w is called the ground state: even with no particle at all, there is still some tiny energy: the zero-point (or vacuum) energy.

7.9.5 Gaussian Distribution

How does w look like? It makes a Gaussian distribution, with zero expectation. In Fig. 7.3, we illustrate the components w_k as a function of x. To be at $x = X_{k,k}$, the particle has probability $|w_k|^2$ (provided that $\|w\| = 1$).

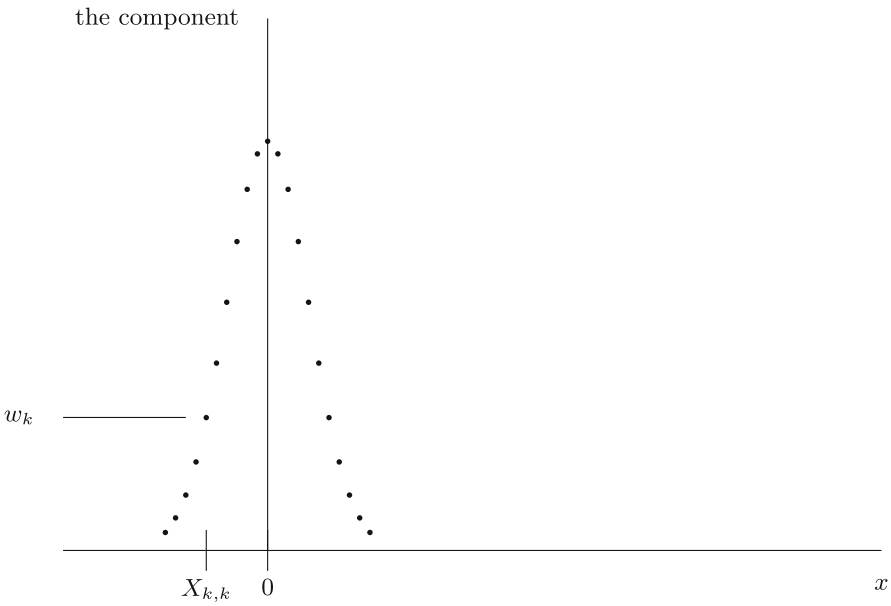


Fig. 7.3 The Gaussian distribution: w is in the null space of A (and $A^h A$), with zero expectation. To be at $x = X_{k,k}$, the particle has probability $|w_k|^2$. This is why it is likely to be at $x = 0$ (the expectation)

7.10 Coherent State

7.10.1 Energy Levels and Their Superposition

In Schrodinger's picture, the matrices X, P, and H are fixed (Sect. 7.2.5). The dynamics is in the state, which may change in time, along with the physical information it carries: the probabilities encapsulated in it.

How can such a moving state look like? Let us try a vector we already know: an (normalized) eigenvector of H:

$$\frac{1}{\sqrt{k!}} \left(A^h\right)^k w, \quad k \geq 0.$$

But this is no good: there is no dynamics here at all! After all, can this state ever change? No! Indeed, the total energy must remain the same eigenvalue of H. So, this state can only multiply by the (complex) number $\exp(i\phi t)$. But this introduces no dynamics at all. After all, this has no effect on the probabilities. Besides, the state is defined up to a scalar multiple only. Thus, this is still a standing function that travels nowhere.

A standing wave is rather rare: most waves do travel in time. To see this, look at a general state v and expand it in terms of the orthonormal eigenvectors of H:

$$v = \sum_{k \geq 0} \left(\frac{1}{\sqrt{k!}} \left(A^h \right)^k w, v \right) \frac{1}{\sqrt{k!}} \left(A^h \right)^k w.$$

7.10.2 Energy Levels and Their Precession

In this expansion, look at the kth coefficient. How does it change in time? It is multiplied by $\exp(-i\omega(k + 1/2)t)$:

$$v(t) = \sum_{k \geq 0} \exp\left(-i\omega \left(k + \frac{1}{2} \right) t \right) \left(\frac{1}{\sqrt{k!}} \left(A^h \right)^k w, v(0) \right) \frac{1}{\sqrt{k!}} \left(A^h \right)^k w.$$

This new exponent has no effect on the probability to be in the kth energy level, which remains

$$\left| \left(\frac{1}{\sqrt{k!}} \left(A^h \right)^k w, v \right) \right|^2 = \frac{1}{k!} \left| \left(\left(A^h \right)^k w, v \right) \right|^2$$

(assuming that $\|v(0)\| = 1$). Still, it does have an impact on the probability to be at a certain position and to have a certain momentum. Why? Because, to calculate these probabilities, different energy levels sum up and interfere with one another. Since each energy level precesses at a different frequency, this may produce either constructive or destructive interference.

7.10.3 Coherent State

To see the dynamics, assume that v is a coherent state: an eigenvector of A, not of $A^h A$:

$$Av = \lambda v,$$

where λ is now not necessarily real (because A is not Hermitian).

How does v look like? It makes a Gaussian distribution, shifted by the complex number λ. This is illustrated in Fig. 7.4: to be at position $x = X_{k,k}$, the particle has probability $|v_k|^2$. Remember: v is *not* an eigenvector of H, so it is nondeterministic in terms of energy too.

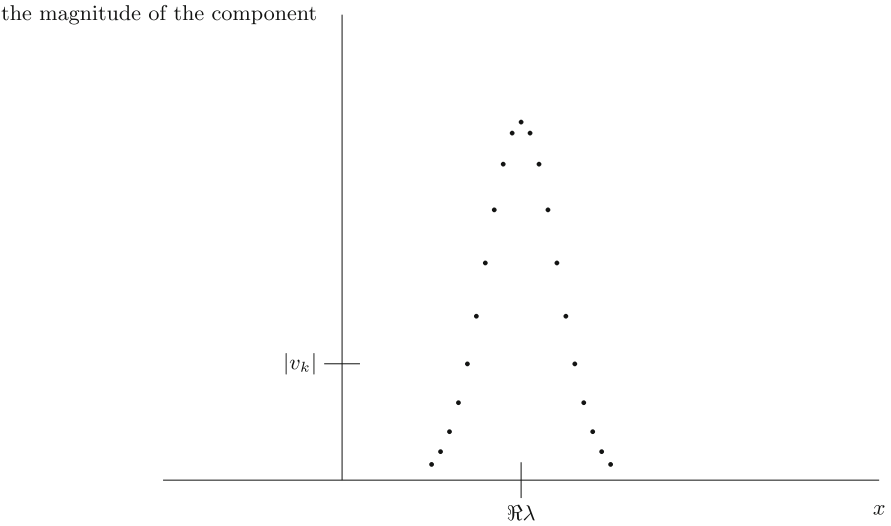


Fig. 7.4 The coherent state looks like a Gaussian distribution, shifted by λ. $|v_k|^2$ is the probability to be at $x = X_{k,k}$

7.10.4 Probability to Have Certain Energy

In a coherent state, we just saw the probability to be at position $x = X_{k,k}$. This is illustrated in Fig. 7.4. Still, there is yet another interesting probability: to have a certain amount of total energy.

Remember that not every amount is allowed, but only our discrete energy levels. Fortunately, the probability to be at the kth energy level is already available in Sect. 7.10.2. In a coherent state, it is even simpler:

$$\begin{aligned}\frac{1}{k!}\left|\left(\left(A^h\right)^k w, v\right)\right|^2 &= \frac{1}{k!}\left|\left(w, A^k v\right)\right|^2 \\ &= \frac{1}{k!}\left|\left(w, \lambda^k v\right)\right|^2 \\ &= \frac{1}{k!}|\lambda|^{2k}\,|(w, v)|^2 \\ &= |(w, v)|^2\,\frac{|\lambda|^{2k}}{k!}.\end{aligned}$$

Here, we have the coefficient $|(w, v)|^2$. How to calculate it? For this, note that the probabilities must sum to 1. For infinite matrices, we therefore have

$$1 = |(w, v)|^2 \sum_{k\geq 0} \frac{|\lambda|^{2k}}{k!} = |(w, v)|^2 \exp\left(|\lambda|^2\right).$$

As a result,

$$|(w, v)|^2 = \exp\left(-|\lambda|^2\right).$$

7.10.5 Poisson Distribution

This is the Poisson distribution (Fig. 7.5). Unlike the Gaussian distribution, it tells us the probability to have a certain energy (not position). In fact, the probability to have energy $\bar{h}\omega(k + 1/2)$ is

$$\exp\left(-|\lambda|^2\right) \frac{|\lambda|^{2k}}{k!},$$

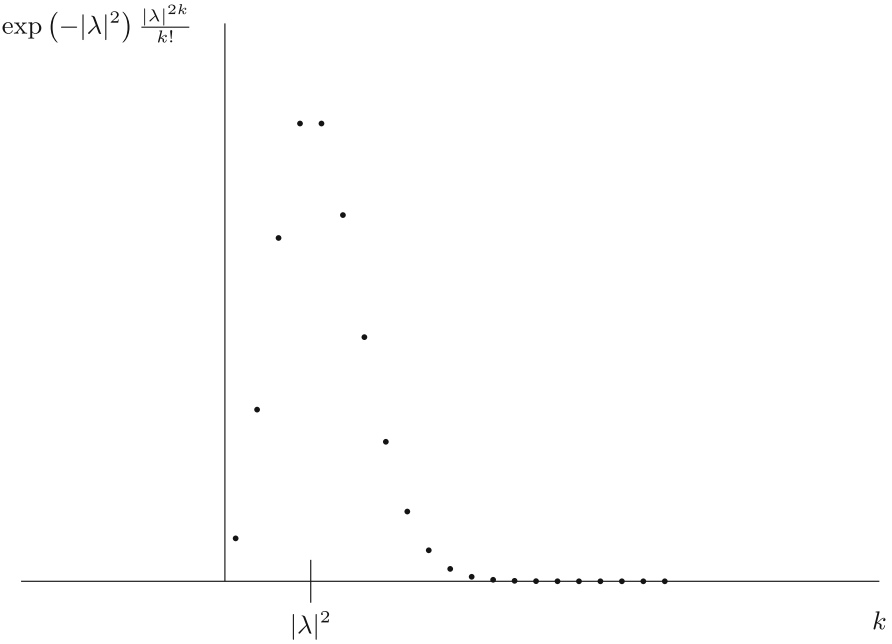


Fig. 7.5 The Poisson distribution. To have energy $\bar{h}\omega(k + 1/2)$, the probability is $\exp(-|\lambda|^2)|\lambda|^{2k}/k!$, where $Av = \lambda v$. The maximum probability is at $k \doteq |\lambda|^2$

where

$$Av = \lambda v.$$

7.10.6 Conservation of Energy

In summary, our coherent state is completely different from the ground state, or any other eigenvector of H. Indeed, energy is no longer known for sure. On the contrary: energy is nondeterministic: a wave function, expanded as a sum (superposition). In this sum, each term stands for some energy level. The probability to be in this energy level remains the same all the time. This is indeed conservation of energy in its nondeterministic face.

7.11 Particle in 3-D

7.11.1 The Discrete 2-D Grid

So far, we have seen three important observables: position, momentum, and (total) energy. Let us design a new observable: angular momentum. For this purpose, we need more dimensions.

Assume now that $n = m^2$, for some natural number m. This way, our original state v can be viewed not only as a vector but also as a grid function. For this purpose, v must be indexed by two indices, i and j:

$$v \equiv \left(v_{i,j}\right)_{1 \le i,j \le m} \in \mathbb{C}^n \cong \mathbb{C}^{m \times m}$$

(Fig. 7.6). Let us redefine our observables in the new $m \times m$ grid. To represent x-position, for example, our new position matrix should act in each horizontal row on its own: it should not mix different rows in the grid.

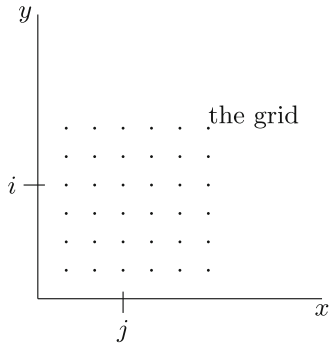


Fig. 7.6 Two-dimensional grid: m horizontal rows, each of them has m points. Since $n = m^2$, our state v makes a (complex) grid function. On it, both X and P act horizontally, on each row alone

The state v is now interpreted as a grid function, with two indices. How likely is the particle to be at $(X_{j,j}, X_{i,i})$? The probability for this is just $|v_{i,j}|^2$ (see exercises below).

7.11.2 Position and Momentum

Let x and p denote the x-position and x-momentum in one horizontal row in the grid. Let X and P be the corresponding $m \times m$ matrices, acting on this row only. How to extend them to the entire grid as well? For this purpose, let I be the $m \times m$ identity matrix. Let us use X, P, and I to define extended $n \times n$ block-diagonal matrices that act in the x-direction only, x-row by x-row.

7.11.3 Tensor Product

This is the tensor product:

$$X \otimes I \equiv \begin{pmatrix} X & & & \\ & X & & \\ & & \ddots & \\ & & & X \end{pmatrix} \quad \text{and} \quad P \otimes I \equiv \begin{pmatrix} P & & & \\ & P & & \\ & & \ddots & \\ & & & P \end{pmatrix}.$$

These are indeed block-diagonal matrices: the off-diagonal blocks vanish. This is how the new symbol "$\otimes$" produces a bigger matrix: the tensor product of two smaller matrices. (Clearly, the bigger matrix is Hermitian too.)

Likewise, to act in the y-direction (y-column by y-column in the grid), define new $n \times n$ (Hermitian) matrices:

$$I \otimes X \equiv \begin{pmatrix} X_{1,1}I & & & \\ & X_{2,2}I & & \\ & & \ddots & \\ & & & X_{m,m}I \end{pmatrix} \quad \text{and} \quad I \otimes P \equiv \begin{pmatrix} P_{1,1}I & \cdots & \cdots & P_{1,m}I \\ \vdots & \ddots & \cdots & \vdots \\ \vdots & \cdots & \ddots & \vdots \\ P_{m,1}I & \cdots & \cdots & P_{m,m}I \end{pmatrix}.$$

7.11.4 Commutativity

Thus, $X \otimes I$ is completely different from $I \otimes X$: the former acts on the horizontal x-rows (one by one), whereas the latter acts on the vertical y-columns in the grid. Because they act in different directions, these matrices commute.

Furthermore, although X and P do not commute, $X \otimes I$ and $I \otimes P$ do:

$$(X \otimes I)(I \otimes P) = X \otimes P = \begin{pmatrix} P_{1,1}X & \cdots & P_{1,m}X \\ \vdots & \ddots & \vdots \\ P_{m,1}X & \cdots & P_{m,m}X \end{pmatrix} = (I \otimes P)(X \otimes I).$$

Likewise, $P \otimes I$ and $I \otimes X$ do commute with one another:

$$(P \otimes I)(I \otimes X) = P \otimes X = (I \otimes X)(P \otimes I).$$

Let us extend this to a yet higher dimension.

7.11.5 3-D Grid

So far, we have only considered scalar position and momentum: x and p. Let us move on to a three-dimensional position:

$$r \equiv \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

(Chap. 2, Sect. 2.4.1). This way, r encapsulates three degrees of freedom: the x-, y-, and z-positions. Likewise, p is now a three-dimensional vector: the linear momentum in the x-, y-, and z-directions. In quantum mechanics, each component should be mirrored by a matrix.

For this purpose, assume now that $n = m^3$, for some natural number m. This way, our original state v can now be interpreted not only as a vector but also as a grid function:

$$v \equiv \left(v_{i,j,k}\right)_{1 \le i,j,k \le m} \in \mathbb{C}^n \cong \mathbb{C}^{m \times m \times m}.$$

Our state v is now interpreted as a grid function, with three indices. How likely is the particle to be at $(X_{k,k}, X_{j,j}, X_{i,i})$? The probability for this is just $|v_{i,j,k}|^2$ (see exercises below).

7.11.6 Bigger Tensor Product

Let us take our original $m \times m$ matrices X and P and place them in suitable tensor products. This way, we obtain extended $n \times n$ matrices, to help observe position and momentum in the x-, y-, and z-directions:

$$\begin{aligned} R_x &\equiv X \otimes I \otimes I \\ R_y &\equiv I \otimes X \otimes I \\ R_z &\equiv I \otimes I \otimes X \\ P_x &\equiv P \otimes I \otimes I \\ P_y &\equiv I \otimes P \otimes I \\ P_z &\equiv I \otimes I \otimes P. \end{aligned}$$

Do these new matrices commute with each other? Well, it depends: if they act in different directions, then they do (Sect. 7.11.4). For example,

$$\big[R_x, P_y\big] = (0).$$

If, on the other hand, they act in the same direction, then they do not. For example,

$$[R_x, P_x] = i\bar{h} I \otimes I \otimes I.$$

This will be useful below.

7.12 Angular Momentum

7.12.1 Angular Momentum Component

Thanks to these new matrices, we can now define a new kind of observable: angular momentum component. This mirrors the original (deterministic) angular momentum:

$$\begin{aligned} L_x &\equiv R_y P_z - R_z P_y \\ L_y &\equiv R_z P_x - R_x P_z \\ L_z &\equiv R_x P_y - R_y P_x. \end{aligned}$$

Indeed, as in the classical case, these definitions are cyclic: the coordinates x, y, and z shift cyclically.

Are these legitimate observables? Well, are they Hermitian? Thanks to commutativity (Sects. 7.11.4–7.11.6), they sure are. For instance,

$$L_x^h = \big(R_y P_z - R_z P_y\big)^h = P_z^h R_y^h - P_y^h R_z^h = P_z R_y - P_y R_z = R_y P_z - R_z P_y = L_x.$$

This mirrors the x-component in the vector product $r \times p$ (Chap. 2, Sect. 2.4.3). Here, however, we have an algebraic advantage: two matrices can combine to form the third one. This shows once again how clever it was to use matrices to model physical observables.

7.12.2 Using the Commutator

The above matrices do not commute with each other. Thanks to Sects. 7.3.4 and 7.11.4–7.11.6, they have a nonzero commutator:

$$\begin{aligned}
\big[L_x, L_y\big] &= \big[R_y P_z - R_z P_y, R_z P_x - R_x P_z\big] \\
&= \big[R_y P_z, R_z P_x\big] - \big[R_y P_z, R_x P_z\big] - \big[R_z P_y, R_z P_x\big] + \big[R_z P_y, R_x P_z\big] \\
&= R_y P_x \,[P_z, R_z] - R_y R_x \,[P_z, P_z] - P_y P_x \,[R_z, R_z] + P_y R_x \,[R_z, P_z] \\
&= -i\bar{h} R_y P_x - (0) - (0) + i\bar{h} P_y R_x \\
&= i\bar{h} L_z.
\end{aligned}$$

This extends cyclically to the other components as well:

$$\begin{aligned}
\big[L_x, L_y\big] &= i\bar{h} L_z \\
\big[L_y, L_z\big] &= i\bar{h} L_x \\
[L_z, L_x] &= i\bar{h} L_y.
\end{aligned}$$

This will be used next.

7.12.3 Up the Ladder

Let us use this to raise an eigenvalue. For example, let u be an eigenvector of L_z, with eigenvalue λ:

$$L_z u = \lambda u.$$

(Since L_z is Hermitian, λ must be real.) How to raise λ? Define a new matrix:

$$T \equiv L_x + iL_y.$$

This way, T does not commute with L_z. They have a nonzero commutator:

$$\begin{aligned}[L_z, T] &= \left[L_z, L_x + iL_y\right] \\ &= [L_z, L_x] + i\left[L_z, L_y\right] \\ &= i\bar{h}L_y + \bar{h}L_x \\ &= \bar{h}T.\end{aligned}$$

Thanks to Sect. 7.8.1, we can now raise λ: if $Tu \neq \mathbf{0}$, then it is an eigenvector of L_z as well, with the new eigenvalue $\lambda + \bar{h}$. This way, T serves as a new ladder operator, to help "climb" up. This can be done more and more, producing bigger and bigger eigenvalues, until hitting the zero vector, and reaching the maximal eigenvalue of L_z. This is quantum mechanics for you: angular momentum is no longer continuous but can take a few discrete values only.

7.12.4 Down the Ladder

This also works the other way around: not only to raise but also to lower λ. For this purpose, redefine T as

$$T \equiv L_x - iL_y.$$

This way, T still does not commute with L_z. They have a nonzero commutator again:

$$\begin{aligned}[L_z, T] &= \left[L_z, L_x - iL_y\right] \\ &= [L_z, L_x] - i\left[L_z, L_y\right] \\ &= i\bar{h}L_y - \bar{h}L_x \\ &= -\bar{h}T.\end{aligned}$$

Thanks to Sect. 7.8.1, we can now lower λ: if $Tu \neq \mathbf{0}$, then it is an eigenvector of L_z as well, with a new eigenvalue: $\lambda - \bar{h}$. This way, our new T serves as a new ladder operator, to help go down the ladder. Furthermore, this can be done more and more, until hitting the zero vector, and reaching the minimal eigenvalue of L_z.

So far, we have studied the eigenvalues and eigenvectors of L_z. The same can also be done for L_x and L_y. Let us place our three matrices in a new rectangular matrix.

7.12.5 Angular Momentum

Let us place these matrices as blocks in a new rectangular $3n \times n$ matrix:

$$L \equiv \begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix}.$$

This is the nondeterministic angular momentum. It mirrors the deterministic angular momentum $r \times p$ (Chap. 2, Sect. 2.4.3). In the following exercises, we will see an interesting example: spin.

7.13 Toward the Path Integral

7.13.1 What Is an Electron?

What is an electron? Is it a point? No! After all, a point is too small to contain any matter. The electron needs some more room: a big vector, with infinitely many coordinates. Each coordinate (or component) refers to one point in space and tells us whether the electron could be there or not. For this, each coordinate is a complex number: magnitude times phase. The magnitude tells us how likely the electron is to be there. The phase, on the other hand, will help us sum up later on. Together, all such functions make a complete linear space, with linear operations. This is what we wanted: we can now use the logic of linear algebra. This answers Einstein's doubts: this logic makes perfect sense, even more than the naive pointwise electron.

7.13.2 Dynamics

And what about dynamics? How does the electron move? After all, it is not a point any more, but a complete function! How does the function evolve and develop in time? Well, where is the function defined? It is defined on some surface in spacetime. Now, consider another surface. How does the function change from one surface to another? Linearly! At each point, it is just multiplied by a new phase. How to calculate this phase? Draw a path from the first surface to the second. Along the path,

integrate the Lagrangian. This is your new frequency. Now, sum up, over all paths that can be drawn in this way. Physically, this is interference: some contributions will annihilate each other (destructive interference), and some will accumulate on top of each other (constructive interference). In the end, we will get the new function, on the new surface in spacetime, as required. Thanks to linear algebra, we got what we wanted: how the electron transforms from one function on one surface to a new function on another surface in spacetime.

7.13.3 Reversibility

This is reversible: in theory, this also works the other way around—back in time. This is stable, as nature really is.

More precisely, the functions actually make not only a linear space but also a little more: a complete Hilbert space, which also supports integration (along the paths) and differentiation (to write the Schrodinger equation). In some simple cases, this equation can be solved in a closed form. This tells us where the electron could be and how likely this is. In general, however, the equation cannot be solved analytically, but only numerically, on finite elements.

7.13.4 Toward Spin

Still, this is not the end of it. As a matter of fact, things are a little more complicated. Each electron actually comes in two possible flavors: spin-up or spin-down. Therefore, it actually needs two functions: one for spin-up, and one for spin-down. But these are just details. This is a good approach to do science: pick a suitable mathematical model, to help describe nature, and even explain mysterious phenomena such as interference (as in the double-slit experiment).

7.14 Exercises: Spin

7.14.1 Eigenvalues and Eigenvectors

1. What is an observable? Hint: An $n \times n$ Hermitian matrix (or operator).
2. What can you say about its eigenvalues? Hint: They are real (Chap. 1, Sect. 1.9.4).
3. Consider two eigenvectors, corresponding to two distinct eigenvalues. What can you say about them? Hint: They are orthogonal to each other (Chap. 1, Sect. 1.9.5).

4. Can you make them orthonormal? Hint: Just normalize them to have norm 1.
5. Consider now two (linearly independent) eigenvectors that share the same eigenvalue. Can you make them orthogonal to each other? Hint: Pick one of them, and modify it: subtract its orthogonal projection on the other one (obtained from their inner product). This is the Gram–Schmidt process.
6. Normalize them to have norm 1.
7. How many (linearly independent) eigenvectors does an observable have? Hint: n.
8. How many (distinct) eigenvalues can an observable have? Hint: At least one, and up to n.
9. A degenerate observable has less than n (distinct) eigenvalues. What can you say about its eigenvectors? Hint: There are two (linearly independent) eigenvectors that share the same eigenvalue.
10. Could the position matrix X be degenerate? Hint: X must have distinct main-diagonal elements, to stand for distinct possible positions.

7.14.2 Hamiltonian and Energy Levels

1. Consider the Hamiltonian of the harmonic oscillator (Sect. 7.9.1). Is it Hermitian? Hint: X and P are Hermitian and so are also X^2 and P^2.
2. Is it a legitimate observable?
3. Can it have an eigenvalue with a nonzero imaginary part? Hint: No. A Hermitian matrix can have real eigenvalues only.
4. Can it have a negative eigenvalue? Hint: No. The number operator must have a nonnegative expectation (Sects. 7.8.3–7.8.4).
5. What is the minimal eigenvalue of the Hamiltonian?
6. Can it be zero? Hint: It must be bigger than the minimal eigenvalue of the number operator, which is zero.
7. What is the physical meaning of this eigenvalue? Hint: This is the minimal energy of the harmonic oscillator.
8. May the harmonic oscillator have no energy at all? Hint: No! Its minimal energy is positive.

7.14.3 The Ground State and Its Conservation

1. In the Hamiltonian of the harmonic oscillator, look again at the minimal eigenvalue. What is the corresponding eigenvector? Hint: The ground state: the state of minimal energy.
2. How does it look like? Hint: A Gaussian distribution (Fig. 7.3).

3. At the ground state, what is the probability to have a particular amount of energy? Hint: At probability 1, it has the minimal energy. It cannot have any other energy level. This is deterministic.
4. Can the ground state change dynamically in time? Hint: No! Energy must remain at its minimum.
5. Consider now yet another state: some other eigenvector of the Hamiltonian. Can it change dynamically in time? Hint: No! Energy must remain the same eigenvalue of the Hamiltonian.
6. How much energy may the harmonic oscillator have? Hint: The allowed energy levels are the eigenvalues of the Hamiltonian: $\bar{h}\omega(k + 1/2)$ $(k \geq 0)$.
7. To model the Hamiltonian well, what must the dimension n be? Hint: n should better be infinite. This way, one can start from the ground state and raise eigenvalues time and again, designing infinitely many new states, with more and more energy.
8. This way, is there a maximal energy? Hint: The above process never terminates. Indeed, the zero vector is never reached (Sect. 7.8.7).

7.14.4 Coherent State and Its Dynamics

1. In a coherent state, what is the probability to have a particular amount of energy? Hint: See Fig. 7.5.
2. Is this deterministic? Hint: No! Each energy level has its own (nonzero) probability.
3. Is this probability constant in time? Hint: Yes! Each energy level only precesses, with no effect on the probability.
4. Is the coherent state constant in time? Hint: No! Each energy level precesses dynamically (at its own frequency).
5. Does this affect the probability to have a particular amount of energy? Hint: No! Each energy level keeps the same absolute value and the same probability.
6. Could this affect any observable that commutes with the Hamiltonian? Hint: No! It has the same eigenvectors, which only precess, with no effect on the probabilities.
7. What is the physical meaning of this? Hint: This observable behaves like the total energy: it has its own conservation law.
8. Still, could this kind of precession affect any other observable that does not commute with the Hamiltonian? Hint: Yes, due to interference (either constructive or destructive).
9. Give an example. Hint: Position.

7.14.5 Entanglement

1. Consider now a particle in a two-dimensional $m \times m$ grid. How does the state v look like? Hint: v is now a grid function: a complex function $v_{i,j}$, defined for every $1 \leq i, j \leq m$.
2. v has norm 1. What does this mean? Hint: Its sum of squares is 1:

$$\sum_{i=1}^{m} \sum_{j=1}^{m} |v_{i,j}|^2 = 1.$$

3. How likely is the particle to be at $(X_{j,j}, X_{i,i})$? Hint: The probability for this is $|v_{i,j}|^2$.
4. How likely is the y-position to be $X_{i,i}$? Hint: Sum the above probabilities over the ith row in the grid:

$$\sum_{k=1}^{m} |v_{i,k}|^2.$$

5. Assuming that $y = X_{i,i}$, how likely is the x-position to be $X_{j,j}$? Hint: The probability for this is

$$\frac{|v_{i,j}|^2}{\sum_{k=1}^{m} |v_{i,k}|^2}.$$

   Indeed, look at the ith row on its own.
6. From the above formulas, how likely is the particle to be at $(X_{j,j}, X_{i,i})$? Hint: Multiply the above probabilities by each other:

$$\frac{|v_{i,j}|^2}{\sum_{k=1}^{m} |v_{i,k}|^2} \sum_{k=1}^{m} |v_{i,k}|^2 = |v_{i,j}|^2.$$

7. Is this familiar?
8. In our grid, could $v_{i,j}$ be factored as a product like $v_{i,j} = u_i w_j$, where u is defined in one column only, and w is defined in one row only? Hint: This is a very special case: disentanglement. In general, on the other hand, such a factorization is impossible. Indeed, in our grid, look at two different columns: the state could look completely different. Likewise, look at two different rows: they may have a completely different state.
9. This is called entanglement: the y-position depends on the x-position: it interacts with it and is *entangled* to it.
10. How about the other way around: does the x-position depend on the y-position?

11. Is the entanglement relation symmetric (as in Chap. 5, Sect. 5.6.1)? Hint: Yes: if v cannot be factored in the above form, then it cannot.
12. Is the disentanglement relation symmetric? Hint: Yes: if v can be factored in the above form, then it can.
13. Is the entanglement relation reflexive? Hint: Yes. Indeed, look at just one isolated row. In it, look at x-position only. Let x have some fixed value: $x = X_{j,j}$ (for some fixed j). This is determinism: only $v_j = 1$, and the other components of v vanish. But this depends on j: if it changed, then v would change as well.
14. Is the disentanglement relation reflexive? Hint: No.
15. Is the entanglement relation transitive? Hint: No! For example, in a three-dimensional grid, assume that the wave function could be factored as $v_{i,j,k} = u_{i,j} w_{i,k}$. In this case, the x-position is entangled to the z-position, which is entangled to the y-position. Still, the x-position is disentangled from the y-position.
16. Is the entanglement relation an equivalence relation? Hint: No: it is not transitive.
17. Is the disentanglement relation transitive? Hint: No! For example, in a three-dimensional grid, assume that the wave function could be factored as $v_{i,j,k} = u_i w_{j,k}$. In this case, the x-position is disentangled from the z-position, which is disentangled from the y-position. Still, the x-position is entangled to the y-position.
18. Is the disentanglement relation an equivalence relation? Hint: No: it is neither reflexive nor transitive.

7.14.6 Angular Momentum and Its Eigenvalues

1. Consider now a particle in a three-dimensional $m \times m \times m$ grid. How does the state v look like? Hint: v is now a grid function: a complex function $v_{i,j,k}$, defined for every $1 \le i, j, k \le m$.
2. v has norm 1. What does this mean? Hint: Its sum of squares is 1:

$$\sum_{i=1}^{m} \sum_{j=1}^{m} \sum_{k=1}^{m} |v_{i,j,k}|^2 = 1.$$

3. How likely is the particle to be at $(X_{k,k}, X_{j,j}, X_{i,i})$? Hint: The probability for this is $|v_{i,j,k}|^2$.
4. How likely is the z-position to be $X_{i,i}$? Hint: Sum the above probabilities over $j, k = 1, 2, 3, \ldots, m$:

$$\sum_{j=1}^{m}\sum_{k=1}^{m}|v_{i,j,k}|^2.$$

5. Consider now an angular momentum component like L_z (Sect. 7.12.1). Is it Hermitian?
6. Is it a legitimate observable?
7. Are its eigenvalues real? Hint: See Chap. 1, Sect. 1.9.4.
8. Are its eigenvectors orthogonal to each other? Hint: See Chap. 1, Sect. 1.9.5.
9. Make them orthonormal. Hint: See exercises above.
10. Consider some eigenvalue of L_z. Consider an n-dimensional state $v \in \mathbb{C}^n$. How likely is the z-angular momentum to be the same as this eigenvalue? Hint: Normalize v to have norm 1. Then, take its inner product with the relevant (orthonormal) eigenvector. Finally, take the absolute value, and square it up.
11. Must zero be an eigenvalue of L_z? Hint: Yes—look at the constant eigenvector, or any other grid function that is invariant under interchanging the x- and y-coordinates: $x \leftrightarrow y$ (Sect. 7.11.5).
12. Conclude that L_z must have a nontrivial null space.
13. Given a positive eigenvalue of L_z, show that its negative counterpart must be an eigenvalue as well. Hint: Interpret the eigenvector as a grid function. Interchange the x- and y-coordinates: $x \leftrightarrow y$. This makes a new eigenvector, with the negative eigenvalue.
14. Show that L_z has discrete eigenvalues of the form

$$0,\ \pm\bar{h},\ \pm 2\bar{h},\ \pm 3\bar{h},\ \cdots$$

(a finite list). Hint: See Sects. 7.12.3–7.12.4.

15. Show that L_z may (in theory) have a few more eigenvalues of the form

$$\pm\frac{\bar{h}}{2},\ \pm\frac{3\bar{h}}{2},\ \pm\frac{5\bar{h}}{2},\ \pm\frac{7\bar{h}}{2},\ \cdots$$

(a finite list). Hint: Use symmetry considerations to make sure that, in this (finite) list, the minimal and maximal eigenvalues have the same absolute value.

7.14.7 Spin-One

1. Define new 3×3 matrices:

$$S_x \equiv i\bar{h}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$S_y \equiv i\bar{h}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

$$S_z \equiv i\bar{h}\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where $i \equiv \sqrt{-1}$ is the imaginary number.

2. These are called spin matrices.
3. Show that these definitions are cyclic under the shift $x \to y \to z \to x$.
4. Are these matrices Hermitian?
5. Are they legitimate observables?
6. Calculate their commutator, and show that it is cyclic as well:

$$[S_x, S_y] = i\bar{h}S_z$$
$$[S_y, S_z] = i\bar{h}S_x$$
$$[S_z, S_x] = i\bar{h}S_y.$$

7. Conclude that these matrices mirror the angular momentum components in Sect. 7.12.2.
8. Focus, for instance, on S_z. What are its eigenvectors and eigenvalues?
9. Show that $(1, i, 0)^t$ is an eigenvector of S_z, with eigenvalue $\bar{h}$.
10. What is its geometrical meaning?
11. Interpret this eigenvector to point in the positive z-direction. Hint: Its y-component is at phase 90° ahead of its x-component (Fig. 7.7). Now, follow the right-hand rule: place your right hand with your thumb pointing in the positive x-direction and your index finger pointing in the positive y-direction. This way, your middle finger will point in the positive z-direction (Fig. 7.7).
12. Show that $(1, -i, 0)^t$ is an eigenvector as well, with eigenvalue $-\bar{h}$.
13. What is its geometrical meaning?
14. Interpret this eigenvector to point in the negative z-direction (Fig. 7.8).
15. Normalize these eigenvectors to have norm 1. Hint: Divide by $\sqrt{2}$.
16. Show that $(0, 0, 1)^t$ is an eigenvector as well, with eigenvalue 0.
17. Conclude that this eigenvector is in the null space of S_z.
18. Show that these eigenvectors are orthogonal to each other. Hint: Calculate their inner product. (Do not forget the complex conjugate!)
19. Conclude that they are not only orthogonal but also orthonormal.
20. Is this as expected from a Hermitian matrix? Hint: Yes—a Hermitian matrix must have real eigenvalues and orthonormal eigenvectors.
21. Is this also as expected from Sects. 7.12.3–7.12.4? Hint: Yes—an eigenvalue can be raised or lowered by $\bar{h}$.

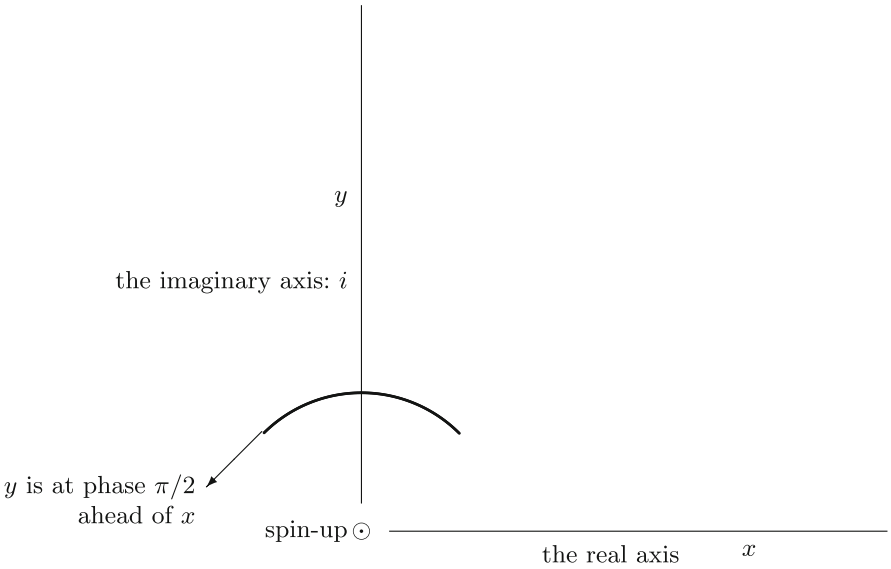


Fig. 7.7 How likely is the boson to spin up? Look at the eigenvector $(1, i, 0)^t/\sqrt{2}$. Thanks to the right-hand rule, it points from the page toward your eye, as indicated by the "$\odot$" at the origin. Now, calculate its inner product with the (normalized) state v. Finally, take the absolute value, and square it up. This is the probability to have spin-up

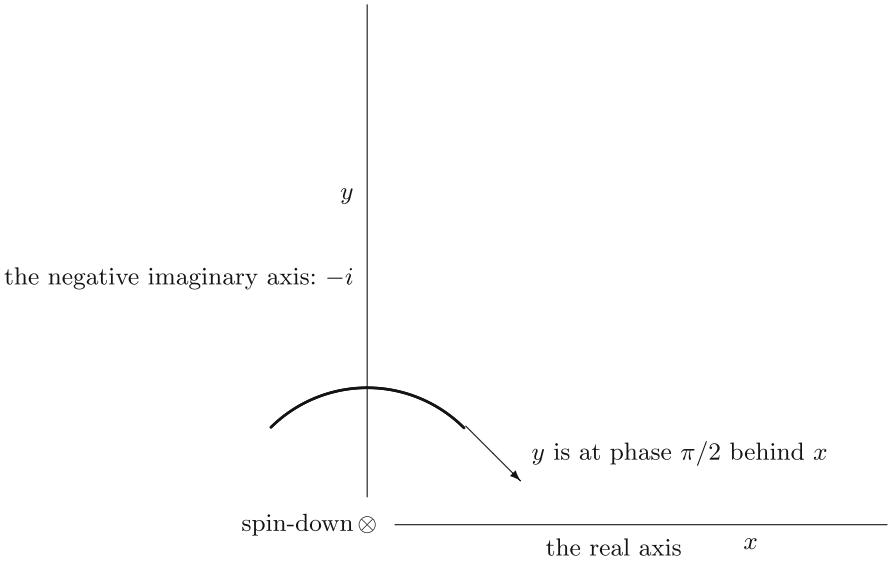


Fig. 7.8 How likely is the boson to have spin-down? Look at the eigenvector $(1, -i, 0)^t/\sqrt{2}$. Thanks to the right-hand rule, it points deep into the page, as indicated by the "$\otimes$" at the origin. Now, calculate its inner product with the (normalized) state v. Finally, take the absolute value, and square it up. This is the probability to have spin-down

22. Consider a new physical system: a boson. This is an elementary particle, with a new property: spin. This is a degenerate kind of angular momentum: it has no value, but only direction. As such, it is more mathematical than physical.
23. Interpret S_z as a new observable, telling us the spin around the z-axis. This is a degenerate kind of angular momentum: a new random variable, with only three possible values. Indeed, the boson "spins" around the z-axis either counterclockwise (spin-up) or clockwise (spin-down), or neither.
24. How likely is the boson to spin up? Hint: Take the state $v \in \mathbb{C}^3$, normalize it to have norm 1, calculate its inner product with $(1, i, 0)^t/\sqrt{2}$, take the absolute value, and square it up. This is the probability.
25. How likely is the boson to spin down? Hint: Do the same with $(1, -i, 0)^t/\sqrt{2}$.
26. How likely is the boson to have spin-zero? Hint: Do the same with $(0, 0, 1)^t$. The result is $|v_3|^2$.
27. Repeat the above exercises for S_x as well. This makes a new observable: spin-right (or left), in the x-direction.
28. Repeat the above exercises for S_y as well. This makes a new observable: spin-in (or out), in the y-direction.

7.14.8 Spin-One-Half and Pauli Matrices

1. The above spin is also called spin-one because the maximal eigenvalue is 1 (times $\bar{h}$).
2. Consider now a new physical system: a fermion. (For example, an electron or a proton or a neutron.)
3. This is a simpler particle, with a simpler spin, called spin-one-half, because its maximal observation is $1/2$ (times $\bar{h}$).
4. For this purpose, consider a new state, of a lower dimension: $v \in \mathbb{C}^2$.
5. Define the 2×2 Pauli matrices:

$$\sigma_z \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\sigma_x \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_y \equiv \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

6. Are these matrices Hermitian?
7. Are they legitimate observables?
8. Multiply the Pauli matrices by $\bar{h}/2$:

$$\tilde{S}_z \equiv \frac{\bar{h}}{2}\sigma_z$$
$$\tilde{S}_x \equiv \frac{\bar{h}}{2}\sigma_x$$
$$\tilde{S}_y \equiv \frac{\bar{h}}{2}\sigma_y.$$

9. These are also called spin matrices.
10. Calculate the commutator of these matrices, and show that it is cyclic:

$$\left[\tilde{S}_z, \tilde{S}_x\right] = i\bar{h}\tilde{S}_y$$
$$\left[\tilde{S}_x, \tilde{S}_y\right] = i\bar{h}\tilde{S}_z$$
$$\left[\tilde{S}_y, \tilde{S}_z\right] = i\bar{h}\tilde{S}_x.$$

11. Conclude that these matrices mirror the angular momentum components in Sect. 7.12.2.
12. What are the eigenvectors of $\tilde{S}_z$? Hint: The standard unit vectors $(1, 0)^t$ and $(0, 1)^t$.
13. Are they orthonormal?
14. What are their eigenvalues?
15. What are the eigenvectors of $\tilde{S}_x$? Hint: $(1, 1)^t$ and $(1, -1)^t$.
16. Are they orthogonal to each other?
17. Normalize them to have norm 1.
18. What are their eigenvalues? Hint: $\pm\bar{h}/2$.
19. Is this as expected? Hint: Yes—a Hermitian matrix must have real eigenvalues and orthonormal eigenvectors.
20. Is this as in Sects. 7.12.3–7.12.4? Hint: Yes—an eigenvalue can be raised or lowered by $\bar{h}$.
21. What are the eigenvectors of $\tilde{S}_y$? Hint: $(1, i)^t$ and $(1, -i)^t$.
22. Are they orthogonal to each other? Hint: Calculate their inner product. (Do not forget the complex conjugate.)
23. Normalize them to have norm 1. Hint: Divide by $\sqrt{2}$.
24. What are their eigenvalues?
25. As in spin-one above, interpret these eigenvectors to indicate spin. Hint: See Figs. 7.7, 7.8.
26. Show that these matrices have the same determinant:

$$\det\left(\tilde{S}_z\right) = \det\left(\tilde{S}_x\right) = \det\left(\tilde{S}_y\right) = -\frac{\bar{h}^2}{4}.$$

27. Show that these matrices have the same square:

$$\tilde{S}_z^2 = \tilde{S}_x^2 = \tilde{S}_y^2 = \frac{\bar{h}^2}{4} I,$$

where I is the 2×2 identity matrix.

28. Does this agree with the eigenvalues calculated above? Hint: Take an eigenvector, and apply the matrix to it twice.
29. Consider now both spin and position at the same time. To tell us about both, how should the state look like? Hint: It should be a $2n$-dimensional vector:

$$v \in \mathbb{C}^{2n} :$$

a complex grid function, defined on a new $2 \times n$ grid.

30. Consider now a new physical system: two fermions (say, two electrons). Could they have exactly the same state in $\mathbb{C}^{2n}$? Hint: No! This is Pauli's exclusion principle.

7.14.9 Polarization

1. The Pauli matrices could help observe not only spin-one-half but also a new physical property, in a new physical system: a photon.
2. The photon is a boson. As such, it has a state in $\mathbb{C}^3$, to help model its spin-one.
3. This spin also gives rise to a state in $\mathbb{C}^2$, to help model yet another physical property: polarization.
4. Indeed, the photon is not only a particle but also a light ray, or an electromagnetic wave. As such, it travels in some direction: say upward, in the positive z-direction. At the same time, it also oscillates in the horizontal x-y plane.
5. To help observe this, the Pauli matrices can also serve as new observables. For example, $\tilde{S}_z$ tells us how likely the photon is to oscillate in the x- or y-direction. In a (normalized) state $v \equiv (v_1, v_2)^t$, the probability to oscillate in the x-direction is $|v_1|^2$, whereas in the y-direction it is $|v_2|^2$.
6. How is this related to the eigenvectors of $\tilde{S}_z$? Hint: v_1 is the inner product of v with $(1, 0)^t$, and v_2 is the inner product with $(0, 1)^t$.
7. At the same time, $\tilde{S}_x$ tells us how likely the photon is to oscillate obliquely, at angle 45° in between the x- and y-axes. To calculate the probability for this, take the eigenvector $(1, 1)^t/\sqrt{2}$, calculate its inner product with v, take the absolute value, and square it up.
8. Write the result as a formula in terms of v_1 and v_2.
9. Finally, $\tilde{S}_y$ tells us how likely the photon is to make circles in the x-y plane (Figs. 7.7, 7.8). To calculate the probability to make circles counterclockwise, take the eigenvector $(1, i)^t/\sqrt{2}$, calculate its inner product with v (do not forget the complex conjugate), take the absolute value, and square it up.
10. Write the result as a formula in terms of v_1 and v_2.

11. To calculate the probability to make circles clockwise, on the other hand, do the same with the orthonormal eigenvector $(1, -i)^t/\sqrt{2}$.

7.14.10 Conjugation

1. Look again at the Pauli matrix

$$\sigma_y \equiv \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

   where $i = \sqrt{-1}$.
2. Is it unitary?
3. Is it Hermitian?
4. What is its inverse? Hint: Since it is unitary and Hermitian,

$$\sigma_y^{-1} = \sigma_y^h = \sigma_y.$$

5. Use σ_y to conjugate: multiply by σ_y from the left, and by σ_y^{-1} (which is actually σ_y) from the right.
6. For example, use σ_y to conjugate σ_y itself:

$$\sigma_y \to \sigma_y^{-1}\sigma_y\sigma_y = \sigma_y.$$

7. What is the effect? Hint: No effect at all: this changes nothing.
8. What is the effect on the two other Pauli matrices? Hint:

$$\sigma_x \to \sigma_y^{-1}\sigma_x\sigma_y = \sigma_y\sigma_x\sigma_y = i\sigma_y\sigma_z = i^2\sigma_x = -\sigma_x$$
$$\sigma_z \to \sigma_y^{-1}\sigma_z\sigma_y = \sigma_y\sigma_z\sigma_y = i\sigma_x\sigma_y = i^2\sigma_z = -\sigma_z.$$

9. What is the effect? Hint: Picking a minus sign.
10. Write the effect more uniformly, for all three Pauli matrices at the same time. Hint: Note that σ_x and σ_z are real, whereas σ_y is imaginary. Therefore, the effect is picking a minus sign and also placing a bar on top:

$$\sigma_x \to -\bar{\sigma}_x$$
$$\sigma_y \to -\bar{\sigma}_y$$
$$\sigma_z \to -\bar{\sigma}_z.$$

11. Is this a meaningful physical effect? Hint: No! This is just a conjugation: looking at the original spin-one-half from a different algebraic point of view.

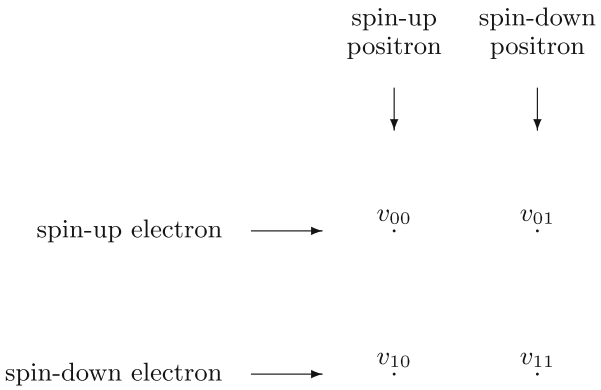


Fig. 7.9 The electron–positron system makes a 2×2 grid. On it, the state $v \in \mathbb{C}^4$ is a complex grid function. For example, spin-up electron and spin-down positron have a complex amplitude v_{01}, and probability $|v_{01}|^2$

12. Could this be used to model an anti-electron (positron)? Hint: Picking a minus sign and a bar on top can model an anti-quark in quantum chromodynamics. Unfortunately, it can never model a positron in our case. After all, it is completely nonphysical: just an algebraic conjugation.
13. How to model an electron and a positron at the same time? Hint: For this purpose, we need two more dimensions: not only $\mathbb{C}^2$, but $\mathbb{C}^4$: draw a new 2×2 grid (Fig. 7.9).
14. In this grid, what do the indices stand for? Hint: The first index stands for an electron: negative charge and positive energy. The second index, on the other hand, stands for a positron (a missing electron): positive charge and negative energy.
15. What do the columns stand for? Hint: The first column stands for spin-up positron, whereas the second stands for spin-down positron.
16. In this new system, what is the state v? Hint: $v \in \mathbb{C}^4$ is a complex grid function.
17. Let A and B be 2×2 matrices. What is the tensor product $A \otimes B$? Hint: See Fig. 7.6 and the text that follows it.
18. How to apply $A \otimes B$ to v? Hint: Look at v as a 2×2 matrix. Apply B from the left. This way, B acts on both columns, one by one, as required. Then, apply A^t from the right. This way, A acts on both rows, one by one, as required.
19. This can also be used in Dirac matrices below.

7.14.11 Dirac Matrices Anti-commute

1. Let

$$I \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

be the 2×2 identity matrix.

2. Show that the Pauli matrices have square I:

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I.$$

3. Show that the Pauli matrices anti-commute with each other:

$$\sigma_x\sigma_y + \sigma_y\sigma_x = \sigma_y\sigma_z + \sigma_z\sigma_y = \sigma_z\sigma_x + \sigma_x\sigma_z = (0)$$

(the 2×2 zero matrix).

4. Use the tensor product (Sect. 7.11.3) to define four new 4×4 matrices—the Dirac matrices:

$$\alpha_1 \equiv \sigma_x \otimes \sigma_z = \begin{pmatrix} \sigma_x & \\ & -\sigma_x \end{pmatrix}$$

$$\alpha_2 \equiv \sigma_y \otimes \sigma_z = \begin{pmatrix} \sigma_y & \\ & -\sigma_y \end{pmatrix}$$

$$\alpha_3 \equiv \sigma_z \otimes \sigma_z = \begin{pmatrix} \sigma_z & \\ & -\sigma_z \end{pmatrix}$$

$$\beta \equiv I \otimes \sigma_x = \begin{pmatrix} & I \\ I & \end{pmatrix}.$$

5. What is their square? Hint:

$$\alpha_1^2 = (\sigma_x \otimes \sigma_z)^2 = \sigma_x^2 \otimes \sigma_z^2 = I \otimes I$$

(the 4×4 identity matrix). Likewise,

$$\alpha_2^2 = \alpha_3^2 = \beta^2 = I \otimes I.$$

6. Show that they anti-commute with each other. Hint: For example,

$$\begin{aligned} \alpha_1\alpha_2 &= (\sigma_x \otimes \sigma_z)\left(\sigma_y \otimes \sigma_z\right) \\ &= \left(\sigma_x\sigma_y\right) \otimes \sigma_z^2 \\ &= -\left(\sigma_y\sigma_x\right) \otimes \sigma_z^2 \\ &= -\left(\sigma_y \otimes \sigma_z\right)(\sigma_x \otimes \sigma_z) \\ &= -\alpha_2\alpha_1, \end{aligned}$$

and so on. Furthermore,

$$\begin{aligned}\alpha_1\beta &= (\sigma_x \otimes \sigma_z)\,(I \times \sigma_x)\\ &= (\sigma_x I) \otimes (\sigma_z \sigma_x)\\ &= -\,(I\sigma_x) \otimes (\sigma_x \sigma_z)\\ &= -\,(I \otimes \sigma_x)\,(\sigma_x \otimes \sigma_z)\\ &= -\beta\alpha_1,\end{aligned}$$

and so on.

7.14.12 Dirac Matrices in Particle Physics

1. In particle physics, use an upper index to define four new 4×4 matrices:

$$\begin{aligned}\gamma^0 &\equiv \beta\\ \gamma^1 &\equiv \beta\alpha_1 = (I \otimes \sigma_x)\,(\sigma_x \otimes \sigma_z)\\ &= (I\sigma_x) \otimes (\sigma_x\sigma_z) = \sigma_x \otimes \begin{pmatrix} & -1\\ 1 & \end{pmatrix} = \begin{pmatrix} & -\sigma_x\\ \sigma_x & \end{pmatrix}\\ \gamma^2 &\equiv \beta\alpha_2 = (I \otimes \sigma_x)\,\big(\sigma_y \otimes \sigma_z\big)\\ &= \big(I\sigma_y\big) \otimes (\sigma_x\sigma_z) = \sigma_y \otimes \begin{pmatrix} & -1\\ 1 & \end{pmatrix} = \begin{pmatrix} & -\sigma_y\\ \sigma_y & \end{pmatrix}\\ \gamma^3 &\equiv \beta\alpha_3 = (I \otimes \sigma_x)\,(\sigma_z \otimes \sigma_z)\\ &= (I\sigma_z) \otimes (\sigma_x\sigma_z) = \sigma_z \otimes \begin{pmatrix} & -1\\ 1 & \end{pmatrix} = \begin{pmatrix} & -\sigma_z\\ \sigma_z & \end{pmatrix}.\end{aligned}$$

   Hint: On the left-hand side, the upper numbers are not powers, but only upper indices.
2. The new 4×4 matrices γ^0, γ^1, γ^2, and γ^3 are called Dirac matrices too.
3. What is γ^4? Hint: There is no such thing as γ^4.
4. What is γ^5? Hint: γ^5 is defined as the product of the above Dirac matrices:

$$\gamma^5 \equiv \gamma^0\gamma^1\gamma^2\gamma^3.$$

5. How does γ^5 look like? Hint: Thanks to anti-commutativity,

$$\begin{aligned}
\gamma^5 &\equiv \gamma^0\gamma^1\gamma^2\gamma^3 \\
&= \beta^2\alpha_1\beta\alpha_2\beta\alpha_3 \\
&= (-1)^3\beta^4\alpha_1\alpha_2\alpha_3 \\
&= -\alpha_1\alpha_2\alpha_3 \\
&= -\left(\sigma_x\sigma_y\sigma_z\right)\otimes\sigma_z^3 \\
&= -i\,I\otimes\sigma_z \\
&= i\begin{pmatrix} -I & \\ & I \end{pmatrix}.
\end{aligned}$$

Part III
Polynomials and Basis Functions

The polynomial is an algebraic object as well. Indeed, two polynomials can be added to or multiplied by each other. Still, the polynomial is also an analytic object: it can be differentiated and integrated, as in calculus. These two aspects can now combine to design a special kind of function: basis function, useful in many practical applications.

To study the polynomial, we can use tools from linear algebra. Indeed, a vector could be used to model a polynomial and store its coefficients. Still, the polynomial has more algebraic operations: multiplication and composition. This way, the polynomials make a new mathematical structure: a ring.

In turn, polynomials of a certain degree make a new vector space. Basis functions can then be designed carefully to extend this space further. This way, they can help design a smooth spline in three spatial dimensions. This is the key to the finite-element method.

Basis functions could be viewed as a special kind of vectors. Indeed, they span a new linear space, with all sorts of interesting properties. Later on, we will use them in a geometrical application: designing a new spline to approximate a given function on a mesh of tetrahedra. Furthermore, we will also use them in the finite-element method to help solve complex models in quantum mechanics and general relativity.

Chapter 8
Polynomials and Their Gradient



The polynomial is a special kind of function, easy to deal with. We start with a polynomial of one independent variable: x. This is an algebraic object: it supports addition, multiplication, and composition. Still, it is also a function that returns a value. To calculate it, we introduce the efficient Horner algorithm.

Next, we move on to a more complicated case: a polynomial of two (or even three) independent variables: x and y (and even z). Geometrically, we are now in a higher dimension: not only the one-dimensional line $\mathbb{R}$ but also the two-dimensional plane $\mathbb{R}^2$ (and even the three-dimensional space $\mathbb{R}^3$). This gives us new features: one can now differentiate in infinitely many spatial directions. This produces partial (and directional) derivatives, useful in applied science and engineering.

8.1 Polynomials and Their Arithmetic Operations

8.1.1 Polynomial of One Variable

What is a polynomial? It is a real function

$$p : \mathbb{R} \to \mathbb{R},$$

defined by

$$p(x) \equiv a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \equiv \sum_{i=0}^{n} a_i x^i,$$

where $n \geq 0$ is the degree of the polynomial, and $a_0, a_1, a_2, \ldots, a_n$ are the coefficients. Usually, we assume that $a_n \neq 0$. (Otherwise, a_n could drop anyway.)

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Y. Shapira, *Linear Algebra and Group Theory for Physicists and Engineers*,
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How to define a concrete polynomial of degree n? Specify its coefficients: $a_0, a_1, a_2, \ldots, a_n$. Thus, the polynomial is mirrored by the $(n+1)$-dimensional vector

$$(a_0, a_1, a_2, \ldots, a_n) \in \mathbb{R}^{n+1}.$$

8.1.2 Real vs. Complex Polynomial

So far, we looked at a real polynomial. A complex polynomial, on the other hand, is different in one aspect only: the coefficients $a_0, a_1, a_2, \ldots, a_n$ (and the independent variable x) could now be not only real but also complex. This way, the polynomial is now a complex function:

$$p : \mathbb{C} \to \mathbb{C}.$$

In what follows, we focus on a real polynomial. Still, a complex polynomial could be treated in the same way.

Polynomials are algebraic objects: they support arithmetic operations like addition, multiplication, and composition. Later on, we will see that they are also analytical objects: they can be differentiated and integrated.

We start with polynomials of just one independent variable: x. Later on, we will extend this to more complicated polynomials of two (and even three) independent variables: x and y (and even z). These too will be not only algebraic but also analytical, with addition, multiplication, differentiation, and integration.

8.1.3 Addition

By now, we already have an $(n+1)$-dimensional vector to mirror the polynomial p. This could help add or subtract. Indeed, let q be another polynomial (of degree m)::

$$q(x) \equiv \sum_{i=0}^{m} b_i x^i.$$

Without loss of generality, assume that $m \leq n$. (Otherwise, interchange the roles of p and q.) Define $n - m$ fictitious zero coefficients:

$$b_{m+1} \equiv b_{m+2} \equiv \cdots \equiv b_n \equiv 0.$$

This way, both p and q have the same number of coefficients: $n+1$. We are now ready to add them, term by term. This defines the new polynomial $p + q$:

$$(p+q)(x) \equiv p(x) + q(x) = \sum_{i=0}^{n} (a_i + b_i) x^i.$$

This is also mirrored in the underlying vectors of coefficients: they are added term by term: to produce a new vector of new coefficients. This works in subtraction as well:

$$(p-q)(x) \equiv p(x) - q(x) = \sum_{i=0}^{n} (a_i - b_i) x^i.$$

8.1.4 Scalar Multiplication

How to multiply p by a given scalar c? Term by term, of course! This produces the new polynomial cp:

$$(cp)(x) \equiv c \cdot p(x) = c \sum_{i=0}^{n} a_i x^i = \sum_{i=0}^{n} (ca_i) x^i.$$

As before, this is mirrored by scalar-times-vector of coefficients. Thus, polynomials are well-mirrored by vectors in both addition and scalar multiplication.

8.1.5 Multiplying Polynomials: Convolution

A polynomial of degree n is mirrored by the $(n+1)$-dimensional vector of its coefficients. Thanks to this, arithmetic operations like addition, subtraction, and scalar multiplication are carried out as in a linear space. Still, the polynomial is more than that: it also supports a new algebraic operation: multiplication.

To see this, consider two given polynomials: p and q. Let us multiply them by one another. In other words, define the new function pq:

$$(pq)(x) \equiv p(x)q(x).$$

After all, polynomials are just functions, which can be multiplied by one another. The result is a new function: the product pq. Still, is pq a *polynomial* in its own right? To see this, we must produce its new vector of coefficients.

By now, pq is only a new function. This means that, if you give me an x, then I can calculate pq at x. How? Easy:

- Calculate $p(x)$.
- Then calculate $q(x)$.

- Then multiply.

Still, is pq more than that? Does pq make a new polynomial, with new coefficients?

Fortunately, the product of

$$p(x) \equiv \sum_{i=0}^{n} a_i x^i \quad \text{and} \quad q(x) \equiv \sum_{j=0}^{m} b_j x^j$$

could be written as

$$(pq)(x) \equiv p(x)q(x) = \sum_{i=0}^{n} a_i x^i \sum_{j=0}^{m} b_j x^j = \sum_{i=0}^{n} \sum_{j=0}^{m} a_i b_j x^{i+j}.$$

This is not good enough. After all, this double sum scans the $(n+1) \times (m+1)$ grid

$$\{(i, j) \mid 0 \le i \le n,\ 0 \le j \le m\}$$

(Fig. 8.1). In this rectangular grid, each point of the form (i, j) contributes a term of the form $a_i b_j x^{i+j}$. This is not quite what we wanted. A legitimate polynomial must have a simpler form: not a double sum that uses two indices, but a standard sum that uses just one index: k.

For this purpose, scan the grid diagonal by diagonal, as in Fig. 8.1. After all, on the kth diagonal, each point of the form (i, j) satisfies $i + j = k$, and contributes $a_i b_j x^{i+j} = a_i b_{k-i} x^k$.

Thus, the grid is scanned diagonal by diagonal, not row by row. The diagonals are indexed by the new index $k = i + j = 0, 1, 2, \ldots, n + m$. This designs a new (outer) sum over k.

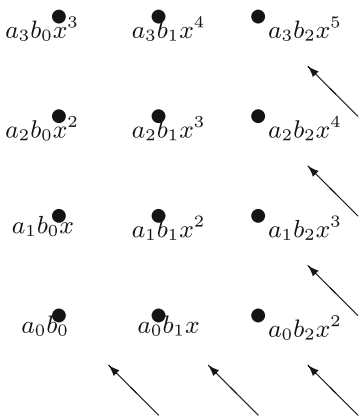


Fig. 8.1 How to multiply $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ by $q(x) = b_0 + b_1x + b_2x^2$? Sum the terms diagonal by diagonal, where the kth diagonal ($0 \le k \le 5$) contains terms with x^k only

In the inner sum, on the other hand, we scan one individual diagonal: the kth diagonal. For this purpose, we need an inner index. Fortunately, we already have a good index: i. It must not exceed the original grid:

$$(pq)(x) = \sum_{i=0}^{n} \sum_{j=0}^{m} a_i b_j x^{i+j} = \sum_{k=0}^{n+m} \sum_{i=\max(0,k-m)}^{\min(k,n)} a_i b_{k-i} x^k.$$

This is what we wanted. In this form, pq is a legitimate polynomial. Indeed, it has a new vector of coefficients:

$$(c_0, c_1, c_2, c_3, \ldots, c_{n+m}) \equiv (c_k)_{k=0}^{n+m},$$

where c_k is the inner sum:

$$c_k \equiv \sum_{i=\max(0,k-m)}^{\min(k,n)} a_i b_{k-i}.$$

This is also called convolution (see exercises below).

8.1.6 Example: Scalar Multiplication

For example, consider a simple case, in which q has degree $m = 0$:

$$q(x) \equiv b_0.$$

In this case, c_k is just

$$c_k = \sum_{i=k}^{k} a_i b_{k-i} = a_k b_0$$

($0 \le k \le n$). Thus, in this case,

$$(pq)(x) = \sum_{k=0}^{n} c_k x^k = \sum_{k=0}^{n} b_0 a_k x^k = b_0 \sum_{i=0}^{n} a_i x^i = b_0 \cdot p(x),$$

in agreement with Sect. 8.1.4.

8.2 Polynomial and Its Value

8.2.1 Value at a Given Point

What is a polynomial? It is just a special kind of function that maps x to $p(x)$. Indeed, for every given x, it returns a new value: $p(x)$. How to calculate $p(x)$ efficiently?

8.2.2 The Naive Method

The naive algorithm uses three stages.

- First, use recursion to calculate the powers of x:

$$x^i = x \cdot x^{i-1} \quad (i = 2, 3, 4, \ldots, n).$$

  This costs just $n - 1$ scalar-times-scalar multiplications.
- Then, multiply each power of x by the corresponding coefficient. This produces the terms

$$a_1 x, a_2 x^2, a_3 x^3, \ldots, a_n x^n.$$

  This costs n scalar-times-scalar multiplications.
- Finally, sum up:

$$a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = p(x).$$

  This costs n scalar-plus-scalar additions.

The total cost (the complexity) is, thus, $2n-1$ multiplications and n additions. Could this be improved? Fortunately, it could. For this purpose, introduce new parentheses, and take a common factor out of them.

8.2.3 Using the Distributive Law

Here is a simple problem: let a, b, and c be some given numbers. How to calculate

$$ab + ac.$$

efficiently? Well, the naive method uses three arithmetic operations:

- Calculate ab.

- Calculate ac.
- Add up: $ab + ac$.

Could this improve? Yes, it can: use the distributive law. For this purpose, introduce new parentheses, and take the common factor a out of them:

$$ab + ac = a(b + c).$$

This can be calculated in just two arithmetic operations:

- Add $b + c$.
- Multiply $a(b + c)$.

Let us use this idea in our original polynomial as well.

8.2.4 Recursion: Horner's Algorithm

Let us use the above to help calculate

$$p(x) = \sum_{i=0}^{n} a_i x^i$$

recursively. For this purpose, write p in the form

$$p(x) = a_0 + x p_1(x),$$

where $p_1(x)$ is a new polynomial of a lower degree:

$$p_1(x) = a_1 + a_2 x + a_3 x^2 + \cdots + a_n x^{n-1} = \sum_{i=0}^{n-1} a_{i+1} x^i.$$

To calculate $p_1(x)$, use the same algorithm recursively. This is indeed Horner's algorithm.

8.2.5 Complexity: Mathematical Induction

Is this efficient? Yes: it requires just n scalar-times-scalar multiplications, and n scalar-plus-scalar additions. To see this, use mathematical induction on n. For $n = 0$, p is constant:

$$p(x) \equiv a_0,$$

so there is nothing to calculate. Now, for $n = 1, 2, 3, \ldots$, assume that we already know how to calculate $p_1(x)$, in $n-1$ multiplications and $n-1$ additions. (This is indeed the induction hypothesis.) For this reason, to calculate

$$p(x) = a_0 + xp_1(x),$$

one needs just one more multiplication and one more addition, which makes a total of n multiplications and n additions, as asserted.

8.3 Composition

8.3.1 Mathematical Induction

So far, we have used Horner's algorithm to calculate the value of a polynomial. But this is not the only purpose: Horner's algorithm could also be used to calculate a composition like

$$(p \circ q)(x) \equiv p(q(x)).$$

It is easy enough to calculate the value of $p \circ q$ at a given x:

- Calculate $q(x)$.
- Use it as an argument in p, to calculate $p(q(x))$.

Still, here we want more: to have the entire vector of coefficients of the new polynomial $p \circ q$. To design this, use mathematical induction on the degree of p: n. For $n = 0$, p is constant: $p(x) \equiv a_0$. Therefore,

$$(p \circ q)(x) = p(q(x)) = a_0.$$

Thus, the vector of coefficients is quite short: it contains just one coefficient:

$$(a_0)\,.$$

Let us move on to a more general case.

8.3.2 The Induction Step

Now, for $n = 1, 2, 3, \ldots$, assume that the induction hypothesis holds: for every polynomial p_1 of degree $n-1$ (including the specific polynomial p_1 defined in Sect. 8.2.4), we already have the entire vector of coefficients of $p_1 \circ q$. Fortunately,

we already know how to multiply polynomials (Sect. 8.1.5). So, we can go ahead and multiply q times $p_1 \circ q$, to obtain the entire vector of coefficients of the product

$$q \cdot (p_1 \circ q).$$

Finally, add a_0 to the first coefficient, to obtain the entire vector of coefficients of

$$p \circ q = a_0 + q \cdot (p_1 \circ q),$$

as required.

8.3.3 Recursion: A New Horner Algorithm

This completes the induction step. This completes the inductive (or recursive) Horner algorithm for composing two polynomials with one another.

8.4 Natural Number as a Polynomial

8.4.1 Decimal Polynomial

So far, the coefficients were not yet specified. Let us go ahead and specify them in practice. For example, pick a natural number k, and write it as a a polynomial. For this purpose, note that

$$10^n \leq k < 10^{n+1},$$

for some nonnegative integer n. Therefore, k could be written as a decimal number, containing $n + 1$ digits:

$$a_n a_{n-1} a_{n-2} \cdots a_1 a_0.$$

This is actually a polynomial, with powers of 10:

$$k = a_0 + a_1 \cdot 10 + a_2 \cdot 10^2 + \cdots + a_n \cdot 10^n = \sum_{i=0}^{n} a_i \cdot 10^i = p(10).$$

This is a special kind of polynomial: a decimal polynomial, with coefficients that are digits between 0 and 9. To obtain k, evaluate this polynomial at $x = 10$.

8.4.2 Binary Polynomial

So far, we used base 10. But this is not a must: we could equally well use any other basis, say base 2. After all, k must also lie in between

$$2^m \leq k < 2^{m+1},$$

for some nonnegative integer m. Thus, k could also be written in terms of $m+1$ binary digits (0 or 1):

$$b_m b_{m-1} b_{m-2} \cdots b_1 b_0.$$

This is actually a binary polynomial, with powers of 2 rather than 10:

$$k = b_0 + b_1 \cdot 2 + b_2 \cdot 2^2 + \cdots + b_m \cdot 2^m = \sum_{j=0}^{m} b_j 2^j = q(2).$$

To have k, evaluate this polynomial at $x = 2$.

This could help store very long natural numbers in cryptography. Let us see yet another interesting application.

8.5 Monomial and Its Value

8.5.1 Monomial

So far, we wrote k in its binary form:

$$k = q(2),$$

where q is a binary polynomial, with coefficients that are binary digits (bits): 0 or 1. What is this good for? Let us see an interesting application: calculate an individual monomial, or a power of x.

Thanks to Horner's algorithm, we already know how to calculate the value of a given polynomial at a given point x. Still, is this always the best method? What about a very short polynomial that contains one term only? Is there a better method that exploits this structure?

8.5.2 A Naive Method

This is a new problem: to calculate a monomial, or a power: x^k. How to do this? Well, the naive approach is recursive (or sequential): for $i = 2, 3, 4, \ldots, k$, calculate

$$x^i = x \cdot x^{i-1}.$$

This is a bit expensive: it requires $k-1$ multiplications. Still, it also gives us a bonus: not only one but also many new monomials: x^2, x^3, x^4, ..., x^k. But what if all we need is x^k alone? Is there a more efficient method?

8.5.3 Horner Algorithm: Implicit Form

Fortunately, there is: Horner's algorithm, applied to the binary polynomial that represents k. Indeed, thanks to Horner's algorithm, k could be written as

$$k = q(2) = b_0 + 2q_1(2),$$

where q_1 is a binary polynomial of degree $m-1$ (or less). Fortunately, there is no need to have q or q_1 explicitly.

8.5.4 Mathematical Induction

Thanks to this (implicit) representation of k, we are now ready to calculate x^k efficiently: in just $2m$ multiplications. Indeed, by mathematical induction on m: for $m = 0$, there is no work at all. Indeed, in this case, $k \equiv b_0$ is either 0 or 1, so $x^k = x^0 = 1$ or $x^k = x^1 = x$.

8.5.5 The Induction Step

Next, assume that we already know how to deal with shorter polynomials of degree $m-1$ or less (including the above q_1). This applies not only to x but also x^2. So, we already know how to calculate

$$\left(x^2\right)^{q_1(2)}$$

in just $2(m-1)$ multiplications. (This is the induction hypothesis.) Let us use this to calculate our original monomial:

$$x^k = x^{q(2)} = x^{b_0+2q_1(2)} = x^{b_0} x^{2q_1(2)} = x^{b_0} \left(x^2\right)^{q_1(2)} = \begin{cases} x \cdot \left(x^2\right)^{q_1(2)} & \text{if } b_0 = 1 \\ \left(x^2\right)^{q_1(2)} & \text{if } b_0 = 0. \end{cases}$$

8.5.6 Complexity: Total Cost

To calculate this right-hand side, we may need two more multiplications:

- Calculate $x^2 = x \cdot x$. Later on, this will help calculate $(x^2)^{q_1(2)}$ recursively.
- Finally, multiply $(x^2)^{q_1(2)}$ by x (if $b_0 = 1$).

Thus, the total cost is as low as

$$2 + 2(m-1) = 2m$$

multiplications, as asserted.

8.5.7 Recursion Formula

To simplify this recursion, let us get rid of q and q_1 altogether. For this purpose, note that b_0 is just the unit binary digit in

$$k = q(2) = b_0 + 2q_1(2).$$

Thus, if k is even, then

$$b_0 = 0 \quad \text{and} \quad q_1(2) = \frac{k}{2}.$$

If, on the other hand, k is odd, then

$$b_0 = 1 \quad \text{and} \quad q_1(2) = \frac{k-1}{2}.$$

Thus, the above recursion could be written more explicitly as

$$x^k = \begin{cases} x \cdot (x^2)^{(k-1)/2} & \text{if } k \text{ is odd} \\ (x^2)^{k/2} & \text{if } k \text{ is even.} \end{cases}$$

This is useful not only to calculate x^k: later on, we will mirror this to obtain q^k, where q is a polynomial in its own right.

8.6 Differentiation

8.6.1 Derivative of a Polynomial

So far, we have looked at the polynomial as an algebraic object, with addition and multiplication. Next, let us view it as a function, which takes an input x to return the output $p(x)$. As such, it can be differentiated and integrated.

For this purpose, consider again the polynomial

$$p(x) = \sum_{i=0}^{n} a_i x^i.$$

What is its derivative? Well, it depends: if $n = 0$, then the derivative is zero. If, on the other hand, $n > 0$, then the derivative is a shorter polynomial:

$$p'(x) \equiv \begin{cases} 0 & \text{if } n = 0 \\ \sum_{i=1}^{n} a_i i x^{i-1} = \sum_{i=0}^{n-1} (i+1) a_{i+1} x^i & \text{if } n > 0. \end{cases}$$

8.6.2 Second Derivative

p' is a polynomial in its own right. As such, it could be differentiated as well, to produce the second derivative of p:

$$p''(x) = \frac{d^2 p}{dx^2}(x) = (p'(x))'.$$

8.6.3 High-Order Derivatives

p'' is a polynomial in its own right. As such, it could be differentiated as well, to produce the third derivative of p, and so on. In general, for $i = 0, 1, 2, \ldots$, the ith derivative of p is defined recursively by

$$p^{(i)} \equiv \begin{cases} p & \text{if } \ i = 0 \\ \left(p^{(i-1)}\right)' & \text{if } \ i > 0. \end{cases}$$

In particular, the zeroth derivative is just the function itself:

$$p^{(0)} \equiv p,$$

and its derivative is

$$p^{(1)} \equiv p'.$$

Furthermore, the nth derivative of p is just a constant:

$$p^{(n)} = a_n n!.$$

Thus, higher derivatives vanish:

$$p^{(i)} = 0, \quad i > n.$$

8.7 Integration

8.7.1 Indefinite Integral

Let us move on to yet another analytic operation: integration. For this purpose, consider again our original polynomial

$$p(x) = \sum_{i=0}^{n} a_i x^i.$$

What is its antiderivative (or indefinite integral)? This is a new polynomial:

$$P(x) \equiv \sum_{i=0}^{n} \frac{a_i}{i+1} x^{i+1} = \sum_{i=1}^{n+1} \frac{a_{i-1}}{i} x^i.$$

To prove this, just differentiate:

$$P'(x) = p(x).$$

8.7.2 Definite Integral over an Interval

The indefinite integral can now be used in a new (geometrical) task: to calculate the area under the graph of p. For simplicity, assume that p is positive. This way, our area is bounded from below by the x-axis, and from above by the graph of p. To bound the area from the left and right too, issue two parallel verticals from the x-axis upwards: at $x = a$ on the left, and at $x = b$ on the right, where $a < b$ are given parameters.

How to calculate this area? This is the *definite* integral:

$$\int_a^b p(x)dx = P(b) - P(a).$$

So far, we assumed that p was positive. Otherwise, do some preparation work: split the original interval $[a, b]$ into a few subintervals. In each subinterval, make sure that p is either positive (so the subarea could be calculated as above) or negative (so the subarea picks a minus sign). Finally, sum the subareas up, to obtain the total area, as required.

8.7.3 Examples

Here are a few examples. If $p(x) = x$ (a linear monomial), then $P(x) = x^2/2$, so

$$\int_a^b x dx = \frac{b^2 - a^2}{2} = (b - a)\frac{a + b}{2}.$$

This is the length of the interval times the average of p at the endpoints. This is the trapezoidal (or trapezoid, or trapezium) rule.

If, on the other hand, $p(x) = x^2$ (a quadratic monomial), then $P(x) = x^3/3$, so

$$\int_a^b x^2 dx = \frac{b^3 - a^3}{3}.$$

Finally, if $p(x) \equiv 1$ (the constant function), then $P(x) = x$, so

$$\int_a^b dx = b - a.$$

This is just the length of the original interval.

8.7.4 Definite Integral over the Unit Interval

Consider again our general polynomial p. Now, set $a \equiv 0$, and $b \equiv 1$. This is the unit interval $[0, 1]$. What is the definite integral over it? Thanks to the above formula, the result is

$$\int_0^1 p(x)dx = P(1) - P(0) = P(1) = \sum_{i=1}^{n+1} \frac{a_{i-1}}{i}.$$

8.8 Sparse Polynomials

8.8.1 Sparse Polynomial

So far, we considered a dense polynomial, with many nonzero terms. To calculate its value, we used Horner's algorithm (Sect. 8.2.4). But what about a sparse polynomial, with only a few nonzero terms? In this case, Horner's algorithm should better be modified.

How could a sparse polynomial look like? Well, it could contain just one nonzero term. This is actually a monomial:

$$p(x) = x^k.$$

To calculate this, Horner's algorithm does a poor job. Indeed, it is the same as the naive algorithm in Sect. 8.5.2, which requires as many as $k - 1$ multiplications. Fortunately, we already have a better approach:

$$x^k = \begin{cases} 1 & \text{if } k = 0 \\ x & \text{if } k = 1 \\ x \cdot (x^2)^{(k-1)/2} & \text{if } k > 1 \text{ and } k \text{ is odd} \\ (x^2)^{k/2} & \text{if } k > 1 \text{ and } k \text{ is even.} \end{cases}$$

Could Horner's algorithm exploit this?

8.8.2 Sparse Polynomial: Explicit Form

For this purpose, write the sparse polynomial explicitly:

$$p(x) = a_0 + a_k x^k + a_l x^l + \cdots + a_n x^n.$$

In this form, we specify three terms only:

- The first term a_0 is either zero or nonzero.
- The next nonzero term is $a_k x^k$, where $a_k \neq 0$ is the next nonzero coefficient ($k \geq 1$).
- The next nonzero term is $a_l x^l$, where $a_l \neq 0$ is the next nonzero coefficient ($l > k$), and so on.

8.8.3 Sparse Polynomial: Recursive Form

In the original Horner algorithm, we pulled x out of parentheses. Here, on the other hand, we pull not only x but also x^k:

$$p(x) = a_0 + x^k p_1(x),$$

where the new polynomial p_1 is shorter:

$$p_1(x) \equiv a_k + a_l x^{l-k} + \cdots + a_n x^{n-k}.$$

8.8.4 Improved Horner Algorithm

Since p_1 is shorter, its value could be calculated recursively. This leads to the improved Horner algorithm:

$$p(x) = \begin{cases} a_0 + x^k p_1(x) & \text{if } a_0 \neq 0 \\ x^k p_1(x) & \text{if } a_0 = 0, \end{cases}$$

where

$$x^k = \begin{cases} x & \text{if } k = 1 \\ x \cdot (x^2)^{(k-1)/2} & \text{if } k > 1 \text{ and } k \text{ is odd} \\ (x^2)^{k/2} & \text{if } k > 1 \text{ and } k \text{ is even.} \end{cases}$$

This is a natural extension of the original algorithm. Indeed, if p is dense, then this is the same as the good old version. Let us mirror this in more complicated tasks.

8.8.5 Power of a Polynomial

So far, our task was easy: to calculate the value of a polynomial. Next, let us move on to a more complicated task: designing the entire vector of coefficients.

Let q be a given polynomial (dense or sparse). Assume that we already have its complete vector of coefficients. How to design the vector of coefficients of q^k as well? Easy: just mirror x^k:

$$q^k = \begin{cases} 1 & \text{if } k = 0 \\ q & \text{if } k = 1 \\ q \cdot (q^2)^{(k-1)/2} & \text{if } k > 1 \text{ and } k \text{ is odd} \\ (q^2)^{k/2} & \text{if } k > 1 \text{ and } k \text{ is even.} \end{cases}$$

Here, q^2 means the vector of coefficients of q^2, calculated as in Sect. 8.1.5.

8.8.6 Composition

Assume that p is still sparse and has the same form as before:

$$p(x) = a_0 + x^k p_1(x).$$

How to compose $p \circ q$, and have all coefficients at the same time? Just mirror the improved Horner's algorithm:

$$p \circ q = a_0 + q^k \cdot (p_1 \circ q).$$

Since p_1 is shorter than p, $p_1 \circ q$ could be obtained recursively (including all coefficients). This should be multiplied by q^k, to produce $q^k \cdot (p_1 \circ q)$ (including all coefficients). Finally, if $a_0 \neq 0$, then add a_0 to the first coefficient, as required.

8.9 Polynomial of Two Variables

8.9.1 Polynomial of Two Independent Variables

So far, we have considered a polynomial of one independent variable: x. Next, let us consider a polynomial of two independent variables: x and y.

What is this? Well, a real polynomial of two variables is a function of the form

$$p : \mathbb{R}^2 \to \mathbb{R}$$

that can be written as

$$p(x, y) = \sum_{i=0}^{n} a_i(x) y^i,$$

where x and y are real arguments, and $a_i(x)$ $(0 \leq i \leq n)$ is a real polynomial in one independent variable: x only.

Likewise, a complex polynomial in two independent variables is a function

$$p : \mathbb{C}^2 \to \mathbb{C}$$

with the same structure as above, except that x and y can now be not only real but also complex numbers, and the polynomials $a_i(x)$ can now be complex polynomials of one variable.

8.9.2 Arithmetic Operations

How to carry out arithmetic operations between polynomials of two variables? The same as before (Sects. 8.1.3–8.1.5). The only difference is that the a_i's are no longer scalars but polynomials (in x) in their own right. Fortunately, we already know how to add or multiply them by each other.

In summary, consider two polynomials of two variables:

$$p(x, y) = \sum_{i=0}^{n} a_i(x) y^i \quad \text{and} \quad q(x, y) = \sum_{j=0}^{m} b_j(x) y^j$$

(for some natural numbers $m \leq n$). How to add them to each other? Well, if $m < n$, then define a few dummy zero polynomials:

$$b_{m+1} \equiv b_{m+2} \equiv \cdots \equiv b_n \equiv 0.$$

We are now ready to add:

$$(p + q)(x, y) = p(x, y) + q(x, y) = \sum_{i=0}^{n} (a_i + b_i)(x) y^i,$$

where

$$(a_i + b_i)(x) = a_i(x) + b_i(x)$$

is just the sum of polynomials of one variable,

Furthermore, how to multiply p times q? Like this:

$$\begin{aligned}(pq)(x, y) &= p(x, y) q(x, y) \\ &= \left(\sum_{i=0}^{n} a_i(x) y^i \right) \left(\sum_{j=0}^{m} b_j(x) y^j \right) \\ &= \sum_{i=0}^{n} \left(\sum_{j=0}^{m} (a_i b_j)(x) y^{i+j} \right).\end{aligned}$$

This is just the sum of n polynomials of two variables, which we already know how to do.

8.10 Differentiation and Integration

8.10.1 Partial Derivatives

So far, we have looked at the polynomial

$$p(x, y) = \sum_{i=0}^{n} a_i(x) y^i$$

as an algebraic object, with two arithmetic operations: addition and multiplication. Fortunately, it can also be viewed as an analytic object, with a new analytic operation: partial differentiation.

For this purpose, let us view y as a fixed parameter, and differentiate $p(x, y)$ as a function of x only. The result is called the partial derivative of p with respect to x:

$$p_x(x, y) \equiv \sum_{i=0}^{n} a_i'(x) y^i,$$

where $a_i'(x)$ is the derivative of $a_i(x)$.

Now, let us work the other way around: view x as a fixed parameter, and differentiate p as a function of y only. This is the partial derivative of p with respect to y:

$$p_y(x, y) \equiv \begin{cases} 0 & \text{if } n = 0 \\ \sum_{i=1}^{n} a_i(x) i y^{i-1} = \sum_{i=0}^{n-1} (i+1) a_{i+1}(x) y^i & \text{if } n > 0. \end{cases}$$

Note that both partial derivatives are polynomials of two variables in their own right. Together, they make a pair, or a two-dimensional vector: the gradient.

8.10.2 The Gradient

Let p_x serve as the first component, and p_y as the second component in a new two-dimensional vector. This makes the gradient of p at the point (x, y):

$$\nabla p(x, y) \equiv \begin{pmatrix} p_x(x, y) \\ p_y(x, y) \end{pmatrix}.$$

Thus, the gradient of p is actually a vector function that not only takes but also returns a two-dimensional vector:

$$\nabla p : \mathbb{R}^2 \to \mathbb{R}^2.$$

8.10.3 Integral over the Unit Triangle

As an analytic object, $p(x, y)$ can be not only differentiated but also integrated. This could be viewed as an extension of the fundamental theorem of calculus (Sect. 8.7.2).

Where is the integration carried out? For this purpose, consider the so-called unit triangle

$$t \equiv \{(x, y) \mid 0 \leq x, y, x + y \leq 1\}$$

(Fig. 8.2). This way, the unit triangle sits on its base: the unit interval [0, 1]. From this interval, issue many verticals upwards, in the y-direction. To integrate on t, just integrate on each and every individual vertical.

Our aim is to calculate the volume under the surface that p makes in the Cartesian space: the two-dimensional surface (or manifold) $z = p(x, y)$ in the x-y-z space. For simplicity, assume that p is positive, so this surface lies above the x-y plane. What is the volume underneath it? More precisely, what is the volume between the surface and the horizontal x-y plane below it?

Still, to calculate a volume, we must be yet more precise. Our three-dimensional region must be bounded not only from above and below but also from all other sides. For this purpose, from the unit triangle t, issue three "walls" upwards, in the z-direction. This way, we got what we wanted: a closed three-dimensional region. What is its volume? It is just the integral of p over t:

$$\int \int_t p(x, y) dx dy.$$

Note that this volume may well be negative, if p is mostly negative. For simplicity, however, we assume that p is positive.

To calculate this integral, let $0 \leq x < 1$ be a fixed parameter, as in Fig. 8.3. Furthermore, let $P(x, y)$ be the indefinite integral of $p(x, y)$ with respect to y:

$$P(x, y) \equiv \sum_{i=0}^{n} \frac{a_i(x)}{i+1} y^{i+1} = \sum_{i=1}^{n+1} \frac{a_{i-1}(x)}{i} y^i.$$

This way, $P(x, y)$ is characterized by the property that its partial derivative with respect to y is the original polynomial $p(x, y)$:

$$P_y(x, y) = p(x, y).$$

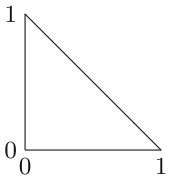


Fig. 8.2 The unit triangle t

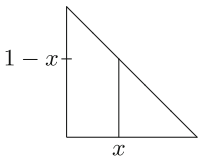


Fig. 8.3 Integration over the unit triangle: for each fixed x, integrate over the vertical $0 \leq y \leq 1 - x$

Fortunately, we have already split t into many verticals, issuing from its base upwards, in the y-direction. To integrate over t, just integrate on each and every vertical:

$$\int\int_t p(x,y)dxdy = \int_0^1 \left(\int_0^{1-x} p(x,y)dy\right)dx = \int_0^1 P(x,1-x)dx.$$

Thanks to the fundamental theorem of calculus, we already know how to calculate this. This is indeed the volume of our three-dimensional region, as required.

8.10.4 Second Partial Derivatives

Let us now return to differentiation. We have already differentiated p with respect to x and y, to produce the partial derivatives p_x and p_y. Fortunately, these are polynomials of two variables in their own right. As such, they can be differentiated as well, to produce the second partial derivatives of p. For example, the mixed partial derivative of p is

$$p_{xy}(x,y) \equiv (p_x(x,y))_y .$$

From Sect. 8.10.1, partial differentiation (or derivation) is insensitive to the order in which it is carried out:

$$p_{xy}(x,y) = \sum_{i=0}^{n-1}(i+1)a'_{i+1}(x)y^i = p_{yx}(x,y).$$

This is also called the $(1,1)$st partial derivative of p. After all, $x^1 = x$ and $y^1 = y$, so

$$p_{xy}(x,y) = p_{x^1y^1}(x,y).$$

With this notation, the $(0,0)$th partial derivative of p is nothing but p itself:

$$p_{x^0y^0}(x,y) = p(x,y).$$

The process may now continue yet more: differentiate a second partial derivative, and obtain a new partial derivative of order three. For example, the $(2,1)$st partial derivative of p is

$$p_{x^2y^1}(x,y) \equiv p_{xxy}(x,y) \equiv (p_{xx}(x,y))_y .$$

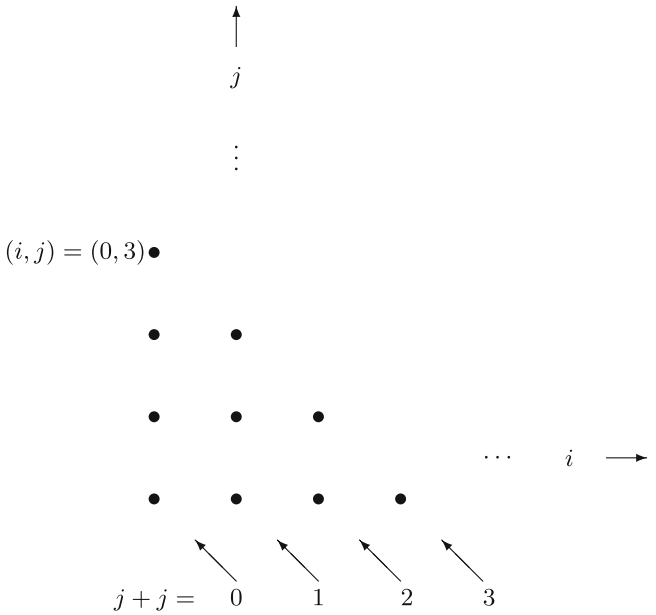


Fig. 8.4 To define the (i, j)th partial derivative, march diagonal by diagonal: use mathematical induction on $i + j = 0, 1, 2, 3, \ldots$

In general, the (i, j)th partial derivative of p is defined diagonal by diagonal, using mathematical induction on its order: $i + j = 0, 1, 2, 3, \ldots$ (Fig. 8.4):

$$p_{x^i y^j} \equiv \begin{cases} p & \text{if } i = j = 0 \\ \left(p_{x^{i-1} y^j}\right)_x & \text{if } i > 0 \\ \left(p_{x^i y^{j-1}}\right)_y & \text{if } j > 0. \end{cases}$$

Fortunately, if both $i > 0$ and $j > 0$, then these formulas agree with each other. Indeed, in the same mathematical induction, we could also prove that reordering does not matter:

$$\left(p_{x^i y^{j-1}}\right)_y = \left(p_{x^{i-1} y^{j-1}}\right)_{xy} = \left(p_{x^{i-1} y^{j-1}}\right)_{yx} = \left(p_{x^{i-1} y^j}\right)_x .$$

This completes the induction step, and indeed the entire definition, as required.

To count the partial derivatives, let us use some results from discrete math. How many distinct partial derivatives of order up to (and including) m are there? Well, from Chapter 8 in [76], the answer is

$$\binom{m+2}{2} = \frac{(m+2)!}{2! \cdot m!} = \frac{(m+1)(m+2)}{2}.$$

How many distinct partial derivatives of order m exactly are there? The answer to this is

$$\binom{m+2-1}{2-1} = \binom{m+1}{1} = m+1.$$

Indeed, here they are:

$$p_{x^0y^m},\ p_{x^1y^{m-1}},\ p_{x^2y^{m-2}},\ \ldots,\ p_{x^my^0}.$$

8.10.5 Degree

The original form

$$p(x,y) = \sum_{i=0}^{n} a_i(x)y^i$$

is somewhat incomplete: the degree is not necessarily n. To uncover the degree, a yet more explicit form is needed.

For this purpose, one must write each polynomial $a_i(x)$ more explicitly:

$$a_i(x) = \sum_{j\geq 0} a_{i,j}x^j,$$

where the $a_{i,j}$'s are some scalars. This way, p can now be written as

$$p(x,y) = \sum_{i\geq 0} a_i(x)y^i = \sum_{i\geq 0}\sum_{j\geq 0} a_{i,j}x^j y^i.$$

The degree of p is the maximal sum $i+j$ for which there is here a nontrivial monomial of the form $a_{i,j}x^j y^i$ (with $a_{i,j}\neq 0$). Note that, unlike in a polynomial of one variable, here the degree may be greater than n.

In a polynomial $p(x,y)$ of degree m, what is the maximal number of distinct monomials? Well, this is the same as the total number of distinct pairs of the form (i,j), with $i+j\leq m$. From Chapter 8 in [76], this number is just

$$\binom{m+2}{2} = \frac{(m+2)!}{m!\cdot 2!} = \frac{(m+1)(m+2)}{2}.$$

8.11 Polynomial of Three Variables

8.11.1 Polynomial of Three Independent Variables

A polynomial of three independent variables is defined by

$$p(x, y, z) = \sum_{i=0}^{n} a_i(x, y)z^i,$$

where the coefficients a_i are now polynomials of two independent variables: x and y.

How to add, subtract, or multiply polynomials of three variables? Fortunately, this is done in the same way as in Sects. 8.1.3–8.1.5. There is just one change: the a_i's are now polynomials of two variables in their own right. Fortunately, we already know how to "play" with them algebraically.

8.12 Differentiation and Integration

8.12.1 Partial Derivatives

Let us view both y and z as fixed parameters, and differentiate $p(x, y, z)$ as a function of x only. This produces the partial derivative with respect to x:

$$p_x(x, y, z) \equiv \sum_{i=0}^{n} (a_i)_x(x, y)z^i.$$

In this sum, the coefficients a_i are differentiated with respect to x as well.

Similarly, let us view both x and z as fixed parameters, and differentiate $p(x, y, z)$ as a function of y only. This produces the partial derivative with respect to y:

$$p_y(x, y, z) \equiv \sum_{i=0}^{n} (a_i)_y(x, y)z^i.$$

Finally, let us view both x and y as fixed parameters, and differentiate $p(x, y, z)$ as a function of z only. This produces the partial derivative with respect to z:

$$p_z(x, y, z) \equiv \begin{cases} 0 & \text{if } n = 0 \\ \sum_{i=1}^{n} a_i(x, y) i z^{i-1} = \sum_{i=0}^{n-1} (i+1) a_{i+1}(x, y) z^i & \text{if } n > 0. \end{cases}$$

Together, these partial derivatives make a new three-dimensional vector: the gradient.

8.12.2 The Gradient

Once placed in a three-dimensional vector, these partial derivatives form the gradient of p:

$$\nabla p(x, y, z) \equiv \begin{pmatrix} p_x(x, y, z) \\ p_y(x, y, z) \\ p_z(x, y, z) \end{pmatrix}.$$

Often, the gradient is nonconstant: it may change from point to point. Only when p is linear is its gradient constant.

Thus, the gradient of p is actually a vector function (or field):

$$\nabla p : \mathbb{R}^3 \to \mathbb{R}^3.$$

8.12.3 Vector Field (or Function)

A vector field could actually be even more general than that. For this purpose, consider three real functions (not necessarily polynomials):

$$f \equiv f(x, y, z)$$
$$g \equiv g(x, y, z)$$
$$h \equiv h(x, y, z).$$

Let us place them in a new three-dimensional vector. This makes a new vector function:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \to \begin{pmatrix} f \\ g \\ h \end{pmatrix} \equiv \begin{pmatrix} f(x, y, z) \\ g(x, y, z) \\ h(x, y, z) \end{pmatrix}.$$

This is also called a vector field. In what follows, we consider a differentiable vector field, in which f, g, and h are differentiable functions.

8.12.4 The Jacobian

So far, we have defined the gradient of the polynomial p. This is a column vector. The transpose gradient, on the other hand, is a row vector:

$$\nabla^t p(x, y, z) \equiv \big(p_x(x, y, z),\ p_y(x, y, z),\ p_z(x, y, z)\big)\,.$$

Let us now apply the transpose gradient to a vector field, row by row:

$$\nabla^t \begin{pmatrix} f \\ g \\ h \end{pmatrix} \equiv \begin{pmatrix} \nabla^t f \\ \nabla^t g \\ \nabla^t h \end{pmatrix} = \begin{pmatrix} f_x & f_y & f_z \\ g_x & g_y & g_z \\ h_x & h_y & h_z \end{pmatrix}.$$

This 3×3 matrix is the Jacobian of the original vector field.

In summary, the original mapping

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \to \begin{pmatrix} f \\ g \\ h \end{pmatrix}$$

has the Jacobian matrix

$$\frac{\partial(f, g, h)}{\partial(x, y, z)} \equiv \nabla^t \begin{pmatrix} f \\ g \\ h \end{pmatrix} = \begin{pmatrix} f_x & f_y & f_z \\ g_x & g_y & g_z \\ h_x & h_y & h_z \end{pmatrix}.$$

(The dependence on the spatial variables x, y, and z is often omitted for short.) The Jacobian matrix will be particularly useful in integration.

8.12.5 Integral over the Unit Tetrahedron

The unit tetrahedron T is a three-dimensional region, with four corners (or vertices):

$$(0, 0, 0),\ (1, 0, 0),\ (0, 1, 0),\quad \text{and}\ \ (0, 0, 1)$$

(Fig. 8.5). Furthermore, T is bounded by four triangles (faces or sides). In particular, T sits on its base: the unit triangle t in Fig. 8.2. In terms of analytic geometry, T is defined by

$$T \equiv \{(x, y, z)\ \mid\ 0 \le x, y, z, x + y + z \le 1\}\,.$$

The original polynomial

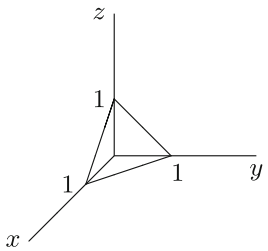


Fig. 8.5 The unit tetrahedron T

$$p(x, y, z) \equiv \sum_{i=0}^{n} a_i(x, y) z^i$$

can now be integrated in T. For this purpose, from each point on the base of T, just issue a vertical upwards, in the z-direction, until it hits the upper face of T. To integrate in T, just integrate on each and every individual vertical.

To carry out this plan, let $P(x, y, z)$ be the indefinite integral of $p(x, y, z)$ with respect to z:

$$P(x, y, z) \equiv \sum_{i=0}^{n} \frac{a_i(x, y)}{i+1} z^{i+1} = \sum_{i=1}^{n+1} \frac{a_{i-1}(x, y)}{i} z^i.$$

With this new definition, we are now ready to integrate:

$$\begin{aligned} \int\int\int_T p(x, y, z) dx dy dz &= \int\int_t \left(\int_0^{1-x-y} p(x, y, z) dz \right) dx dy \\ &= \int\int_t (P(x, y, 1-x-y) - P(x, y, 0)) \, dx dy \\ &= \int\int_t P(x, y, 1-x-y) dx dy. \end{aligned}$$

Fortunately, this is just a two-dimensional integration in t, which we already know how to do (Sect. 8.10.3).

8.13 Normal and Tangential Derivatives

8.13.1 Directional Derivative

Let us now return to the subject of differentiation. So far, we have differentiated $p(x, y, z)$ in a Cartesian direction: x, y, or z. This produces the three partial

derivatives. Let us now generalize this and differentiate p in just any spatial direction as well.

For this purpose, let $\mathbf{n}$ be a fixed three-dimensional vector in $\mathbb{R}^3$:

$$\mathbf{n} \equiv \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = (n_1, n_2, n_3)^t \in \mathbb{R}^3.$$

Assume also that $\mathbf{n}$ is a unit vector:

$$\|\mathbf{n}\|_2 \equiv \sqrt{n_1^2 + n_2^2 + n_3^2} = 1.$$

Define $p_n(x, y, z)$ as the directional derivative of $p(x, y, z)$ in the direction pointed at by $\mathbf{n}$. This is the inner product of $\mathbf{n}$ with the gradient of p at (x, y, z):

$$\begin{aligned} p_n(x, y, z) &\equiv (\mathbf{n}, \nabla p(x, y, z)) = \mathbf{n}^t \nabla p(x, y, z) \\ &= n_1 p_x(x, y, z) + n_2 p_y(x, y, z) + n_3 p_z(x, y, z). \end{aligned}$$

For short, we often omit the dependence on the specific point (x, y, z):

$$p_n \equiv (\mathbf{n}, \nabla p) = \mathbf{n}^t \nabla p = n_1 p_x + n_2 p_y + n_3 p_z.$$

This still depends on (x, y, z) implicitly and may still change from point to point. Only when p is linear is the directional derivative constant.

Next, let us look at an interesting special case.

8.13.2 Normal Derivative

Assume now that

$$\mathbf{n} \equiv (n_1, n_2, n_3)^t$$

is normal (or orthogonal, or perpendicular) to a particular plane in $\mathbb{R}^3$. In other words, $\mathbf{n}$ makes a zero inner product with the difference between any two points that lie on the plane. In the case, the above directional derivative is also called normal derivative.

As a matter of fact, $\mathbf{n}$ could be normal to a mere line in $\mathbb{R}^3$. For example, consider the line

$$\{(x, y, 0) \mid x + y = 1\} \subset \mathbb{R}^3.$$

(This line contains one of the edges in the unit tetrahedron in Fig. 8.5.) Consider two distinct points on this line:

$$(x, 1-x, 0) \text{ and } (\hat{x}, 1-\hat{x}, 0),$$

where $0 \le x, \hat{x} \le 1$, and $x \ne \hat{x}$. The difference between these two points is just

$$(x, 1-x, 0) - (\hat{x}, 1-\hat{x}, 0) = (x-\hat{x}, 1-x-(1-\hat{x}), 0) = (x-\hat{x}, \hat{x}-x, 0).$$

This difference is orthogonal to two constant vectors:

$$(1, 1, 0)^t \text{ and } (0, 0, 1)^t.$$

Thus, to be normal to the above line, $\mathbf{n}$ could be either

$$\mathbf{n} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = (0, 0, 1)^t \text{ or } \mathbf{n} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}(1, 1, 0)^t$$

or just any (normalized) linear combination of these two vectors.

8.13.3 Differential Operator

It is sometimes convenient to use differential operators: $\partial/\partial x$ means partial differentiation with respect to x, $\partial/\partial y$ means partial differentiation with respect to y, and $\partial/\partial z$ means partial differentiation with respect to z. With these new notations, the operator of normal differentiation can be written as

$$\frac{\partial}{\partial n} \equiv n_1 \frac{\partial}{\partial x} + n_2 \frac{\partial}{\partial y} + n_3 \frac{\partial}{\partial z}.$$

For example, if

$$\mathbf{n} = \frac{1}{\sqrt{2}}(1, 1, 0)^t,$$

then the operator of normal differentiation takes the form

$$\frac{\partial}{\partial n} = \frac{1}{\sqrt{2}} \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right).$$

For yet another example, consider the plane

$$\{(x, y, z) \mid x + y + z = 1\} \subset \mathbb{R}^3.$$

(This plane contains the upper face of the unit tetrahedron in Fig. 8.5.) The normal vector to this plane is

$$\mathbf{n} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Thus, in this case, the operator of normal differentiation is just

$$\frac{\partial}{\partial n} = \frac{1}{\sqrt{3}} \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right).$$

8.13.4 High-Order Normal Derivatives

Because the normal derivative of a polynomial is a polynomial in its own right, it has a normal derivative as well. The normal derivative of the normal derivative is called the second normal derivative.

This can be extended to a yet higher order. Indeed, by mathematical induction on $i = 1, 2, 3, \ldots$, the $(i + 1)$st normal derivative is just the normal derivative of the ith normal derivative.

8.13.5 Tangential Derivative

So far, we have assumed that $\mathbf{n}$ was normal to a given line or plane in the three-dimensional Cartesian space. Assume now that $\mathbf{n}$ is no longer normal but rather parallel to the line or the plane. This way, $\mathbf{n}$ is orthogonal to every vector that is normal to the original line or plane. In fact, if $\mathbf{n}$ was shifted to issue from the original line or plane rather than from the origin, then it would be contained in that line or plane and would indeed be tangent to it as well.

The directional derivative in the direction pointed at by $\mathbf{n}$ is then called the tangential derivative. Furthermore, we also have a yet higher order: the tangential derivative of the tangential derivative is called the second tangential derivative, or the tangential derivative of order 2.

Again, this is just mathematical induction: the zeroth tangential derivative is the original function itself. Now, for $i = 0, 1, 2, \ldots$, the $(i + 1)$st tangential derivative (or the tangential derivative of order $i + 1$) is defined as the tangential derivative of the ith tangential derivative. This will be useful later in the book.

8.14 High-Order Partial Derivatives

8.14.1 High-Order Partial Derivatives

For polynomials of two variables, high-order partial derivatives have already been defined in Sect. 8.10.4. For polynomials of three variables, on the other hand, high-order partial derivatives have also been used implicitly in Sect. 8.13.4. Here, however, we define them more explicitly, including mixed partial derivatives.

For this purpose, recall that the partial derivative of a polynomial of three variables is a polynomial of three variables in its own right. As such, it can be differentiated once again. For example, p_x can be differentiated with respect to z, to produce

$$p_{xz}(x, y, z) \equiv (p_x(x, y, z))_z.$$

From Sect. 8.12.1, the order in which the partial differentiation takes place is immaterial:

$$p_{xz}(x, y, z) = \sum_{i=0}^{n-1} (i + 1)\,(a_{i+1})_x\,(x, y) z^i = p_{zx}(x, y, z).$$

Furthermore, let us use differential operators (Sect. 8.13.3) to define the (i, j, k)th partial derivative:

$$p_{x^i y^j z^k}(x, y, z) = \left(\left(\frac{\partial}{\partial x} \right)^i \left(\frac{\partial}{\partial y} \right)^j \left(\frac{\partial}{\partial z} \right)^k p \right) (x, y, z).$$

For example, the $(2, 1, 0)$th partial derivative of p is just

$$p_{x^2 y^1 z^0}(x, y, z) = p_{xxy}(x, y, z).$$

In particular, the $(0, 0, 0)$th partial derivative of p is just p itself:

$$p_{x^0 y^0 z^0}(x, y, z) \equiv p(x, y, z).$$

The order of the (i, j, k)th partial derivative is the sum $i + j + k$. With this terminology, the (i, j, k)th partial derivative could have been defined more explicitly by mathematical induction on the order $i + j + k = 0, 1, 2, 3, \ldots$:

$$p_{x^i y^j z^k} \equiv \begin{cases} p & \text{if } i = j = k = 0 \\ \left(p_{x^{i-1} y^j z^k}\right)_x & \text{if } i > 0 \\ \left(p_{x^i y^{j-1} z^k}\right)_y & \text{if } j > 0 \\ \left(p_{x^i y^j z^{k-1}}\right)_z & \text{if } k > 0. \end{cases}$$

As discussed in Sect. 8.10.4, the same mathematical induction could also have been used to prove that these definitions always agree with each other.

To count the partial derivatives, let us use some results from discrete math. Thanks to Chapter 8 in [76], the total number of partial derivatives of order up to (and including) m is

$$\binom{m+3}{3} = \frac{(m+3)!}{3! \cdot m!} = \frac{(m+1)(m+2)(m+3)}{6}.$$

Furthermore, the total number of partial derivatives of order m exactly is

$$\binom{m+3-1}{3-1} = \binom{m+2}{2} = \frac{(m+2)!}{2! \cdot m!} = \frac{(m+1)(m+2)}{2}.$$

8.14.2 The Hessian

The second partial derivatives defined above can now be placed in a new 3×3 matrix. This is the Hessian.

For this purpose, recall that the transpose gradient of p is the row vector containing the partial derivatives of p:

$$\nabla^t p \equiv (\nabla p)^t = (p_x, p_y, p_z)$$

(Sect. 8.12.4). The dependence on the spatial variables x, y, and z is omitted here for short.

Now, to each individual component in this row vector, apply the gradient operator ∇:

$$\nabla\nabla^t p = (\nabla p_x \mid \nabla p_y \mid \nabla p_z) = \begin{pmatrix} p_{xx} & p_{yx} & p_{zx} \\ p_{xy} & p_{yy} & p_{zy} \\ p_{xz} & p_{yz} & p_{zz} \end{pmatrix}.$$

This is the Hessian of p: the 3×3 matrix that contains the second partial derivatives of p.

Fortunately, a mixed partial derivative is insensitive to the order in which the differentiation is carried out:

$$\begin{aligned} p_{xy} &= p_{yx} \\ p_{xz} &= p_{zx} \\ p_{yz} &= p_{zy}. \end{aligned}$$

Thus, the Hessian is a symmetric matrix, equal to its transpose:

$$\nabla\nabla^t p = \begin{pmatrix} p_{xx} & p_{yx} & p_{zx} \\ p_{xy} & p_{yy} & p_{zy} \\ p_{xz} & p_{yz} & p_{zz} \end{pmatrix} = \begin{pmatrix} p_{xx} & p_{xy} & p_{xz} \\ p_{yx} & p_{yy} & p_{yz} \\ p_{zx} & p_{zy} & p_{zz} \end{pmatrix} = \nabla^t \nabla p.$$

In other words, the Hessian is the Jacobian of the gradient.

8.14.3 Degree

As discussed in Sect. 8.10.5, the polynomials of two variables $a_i(x, y)$ could be written more explicitly:

$$a_i(x, y) = \sum_{j\geq 0} \sum_{k\geq 0} a_{i,j,k} x^k y^j,$$

where the $a_{i,j,k}$'s are some scalars. Using this formulation, the original polynomial of three variables could be written as

$$p(x, y, z) = \sum_{i=0}^{n} \sum_{j\geq 0} \sum_{k\geq 0} a_{i,j,k} x^k y^j z^i.$$

Thus, the degree of p could be much greater than n: it is the maximal sum $i + j + k$ for which there is a nontrivial monomial of the form $a_{i,j,k} x^k y^j z^i$ (with $a_{i,j,k} \neq 0$).

How many monomials are there? Well, thanks to Chapter 8 in [76], a polynomial of degree m may contain at most

$$\binom{m+3}{3} = \frac{(m+3)!}{m! \cdot 3!} = \frac{(m+1)(m+2)(m+3)}{6}$$

distinct monomials.

8.15 Exercises: Convolution

8.15.1 Convolution and Polynomials

1. Let

$$u \equiv (u_0, u_1, u_2, \ldots, u_n) \equiv (u_i)_{0\leq i\leq n}$$

be an $(n+1)$-dimensional vector. Likewise, let

$$v \equiv (v_0, v_1, v_2, \ldots, v_m) \equiv (v_i)_{0 \le i \le m}$$

be an $(m+1)$-dimensional vector. Complete both u and v into an $(n+m+1)$-dimensional vector by adding dummy zero components:

$$u_{n+1} \equiv u_{n+2} \equiv \cdots \equiv u_{n+m} \equiv 0,$$

and

$$v_{m+1} \equiv v_{m+2} \equiv \cdots \equiv v_{n+m} \equiv 0.$$

2. The convolution of u and v (denoted by $u * v$) is a new $(n+m+1)$-dimensional vector, with new components:

$$(u * v)_k \equiv \sum_{i=0}^{k} u_i v_{k-i}, \quad 0 \le k \le n+m.$$

3. Show that convolution is commutative:

$$u * v = v * u.$$

Hint: introduce a new index $j \equiv k - i$, and sum over it.

4. Show that convolution is also associative. Hint: see below.
5. Show that convolution is also distributive.
6. Use u as a vector of coefficients in a new polynomial p:

$$p(x) \equiv \sum_{i=0}^{n} u_i x^i = \sum_{i=0}^{n+m} u_i x^i.$$

7. Likewise, use v as a vector of coefficients in a new polynomial q:

$$q(x) \equiv \sum_{i=0}^{m} v_i x^i = \sum_{i=0}^{n+m} v_i x^i.$$

8. Consider the product pq. Is it a legitimate polynomial as well?
9. If so, what is its vector of coefficients? Hint: the convolution vector $u * v$:

$$(pq)(x) = \sum_{k=0}^{n+m} (u * v)_k x^k.$$

10. These are isomorphic algebras: polynomials (with their product) mirror vectors (with their convolution).
11. Thus, both algebras have the same algebraic properties.
12. In particular, both are commutative, associative, and distributive.
13. For polynomials, this is easy to prove. Use this to prove once again that convolution is indeed commutative, associative, and distributive as well.

8.15.2 Polar Decomposition

1. The infinite Taylor expansion of the exponent function around zero is

$$\exp(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

2. Truncate this series after $k + 1$ terms. This approximates $\exp(x)$ by a new polynomial of degree k:

$$\exp(x) \doteq \sum_{n=0}^{k} \frac{x^n}{n!}.$$

3. For a moderate $|x|$, this is indeed a good approximation.
4. For a big $|x|$, on the other hand, increase k until the tail is sufficiently small in absolute value.
5. To calculate this, design a new version of Horner's algorithm. Hint: avoid dividing by $n!$. The solution can be found in Chapter 1 in [75].
6. Likewise, the infinite Taylor expansion of the sine function around zero is

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

7. Truncate it as above.
8. Likewise, the infinite Taylor expansion of the cosine function around zero is

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}.$$

9. Truncate it as above.

10. Conclude that

$$\exp(i\theta) = \cos(\theta) + i\,\sin(\theta),$$

where $i \equiv \sqrt{-1}$ is the imaginary number. This is the polar decomposition of the complex number $\exp(i\theta)$, where $0 \leq \theta < 2\pi$ is the angle that it makes with the positive part of the real axis. Hint: add the above expansions, term by term.

Chapter 9
Basis Functions: Barycentric Coordinates in 3-D



Thanks to the above background, we are now ready to design a special kind of function: basis function (or B-spline). This will be the key to the finite-element method, with advanced applications in modern physics and chemistry.

We start from a simple case: just one tetrahedron. In it, the basis function is defined as a polynomial. To each adjacent tetrahedron, the basis function is then extended and defined as a different polynomial. This way, the basis function is piecewise-polynomial.

Still, at the interface between two adjacent tetrahedra, these polynomials must agree smoothly with each other. In other words, at a face shared by two adjacent tetrahedra, the basis function must be continuous. Moreover, across an edge shared by two adjacent tetrahedra, the basis function must be differentiable: have a continuous gradient. These properties will be used later to design a rather smooth spline to approximate a given function in 3-D. This is the key to the finite-element method, used in advanced applications in physics and chemistry later on.

9.1 Tetrahedron and its Mapping

9.1.1 General Tetrahedron

So far, we have considered a special tetrahedron: the unit tetrahedron T, vertexed at four special corners:

$$(0,0,0),\ (1,0,0),\ (0,1,0),\quad \text{and}\quad (0,0,1)$$

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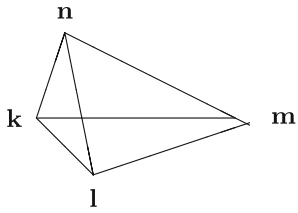


Fig. 9.1 A general tetrahedron t, vertexed at $\mathbf{k}$, $\mathbf{l}$, $\mathbf{m}$, and $\mathbf{n}$

(Fig. 8.5). Consider now a more general tetrahedron t, with more general corners (or vertices): $\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n} \in \mathbb{R}^3$. Each corner is a point (or a three-dimensional column vector) in $\mathbb{R}^3$. This way, our new tetrahedron t is denoted by

$$t \equiv (\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n}) \subset \mathbb{R}^3$$

(Fig. 9.1). Here, the order of the corners is determined arbitrarily in advance.

Let us now map T onto t. For this purpose, consider the vectors leading from $\mathbf{k}$ to the three other corners in t. Together, these column vectors form a new 3×3 matrix:

$$S_t \equiv (\mathbf{l} - \mathbf{k} \mid \mathbf{m} - \mathbf{k} \mid \mathbf{n} - \mathbf{k}) .$$

Later on, we will introduce the notion of regularity: t must be nondegenerate. In other words, its corners must not lie on the same plane. This way, S_t is nonsingular—it has a nonzero determinant:

$$\det (S_t) \neq 0.$$

More than that: t is rather thick, and not too thin. This means that S_t is far from being singular: its determinant is far away from zero.

To map T onto t, define the new mapping

$$E_t \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} \right) \equiv \mathbf{k} + S_t \begin{pmatrix} x \\ y \\ z \end{pmatrix} .$$

This way, the corners of T map to the corresponding corners of t:

$$E_t \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right) = \mathbf{k}$$

$$E_t \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right) = \mathbf{l}$$

$$E_t\left(\begin{pmatrix}0\\1\\0\end{pmatrix}\right) = \mathbf{m}$$

$$E_t\left(\begin{pmatrix}0\\0\\1\end{pmatrix}\right) = \mathbf{n}.$$

Furthermore, in the terminology in Chap. 8, Sect. 8.12.4, S_t is the Jacobian of E_t.

Clearly, the inverse mapping maps t back onto T:

$$E_t^{-1}\left(\begin{pmatrix}x\\y\\z\end{pmatrix}\right) = S_t^{-1}\left(\begin{pmatrix}x\\y\\z\end{pmatrix} - \mathbf{k}\right).$$

For this reason, S_t^{-1} is the Jacobian of E_t^{-1}.

9.1.2 Integral Over a Tetrahedron

Let F be an integrable function in t. How to integrate in t? This could be rather difficult. After all, t is a general tetrahedron. Better integrate in T—the unit tetrahedron. Still, F was never defined in T. What is defined in T is the composite function $F \circ E_t$. Let us go ahead and integrate it in T:

$$\int\int\int_t F(x,y,z)dxdydz = |\det(S_t)| \int\int\int_T (F \circ E_t)(x,y,z)dxdydz.$$

Let us consider a common example that arises often in practice. Assume that F is a product of two integrable functions:

$$F(x,y,z) = f(x,y,z)g(x,y,z), \quad (x,y,z) \in t.$$

To mirror these functions, define new composite functions in T:

$$\tilde{f} = f \circ E_t \text{ and } \tilde{g} = g \circ E_t.$$

What is so good about $\tilde{f}$ and $\tilde{g}$? Well, they are defined in the unit tetrahedron T, not the complicated tetrahedron t. As such, they are easier to differentiate and integrate.

It is sometimes useful to look at things the other way around:

$$f = \tilde{f} \circ E_t^{-1} \text{ and } g = \tilde{g} \circ E_t^{-1}$$

in t. Now, with $F \equiv fg$, the above formula takes the form

$$\begin{aligned}\int\int\int_t fg dx dy dz &= |\det(S_t)| \int\int\int_T (f \circ E_t)(g \circ E_t) dx dy dz \\ &= |\det(S_t)| \int\int\int_T \tilde{f}\tilde{g} dx dy dz.\end{aligned}$$

This will be useful below.

9.1.3 The Chain Rule

The chain rule tells us how to differentiate a composite function [16]. Thanks to it, instead of differentiating f directly, which is difficult, we can now differentiate $\tilde{f}$, which is easier.

Still, f and its gradient are evaluated in t. $\tilde{f}$ and its gradient, on the other hand, are evaluated in T, not t. In fact, f actually splits into two parts: first, use E_t^{-1} to map from t to T. Then, apply $\tilde{f}$.

Thus, differentiating $\tilde{f}$ is only half the story: it takes care of the latter part only. We must not forget the former: we must also differentiate E_t^{-1}, or multiply by its Jacobian:

$$\nabla^t f(x, y, z) = \nabla^t \tilde{f}\left(E_t^{-1}(x, y, z)\right) S_t^{-1}, \quad (x, y, z) \in t.$$

This is indeed the chain rule.

To have a shorter formula, better drop the dependence on x, y, and z:

$$\nabla^t f = \left(\left(\nabla^t \tilde{f}\right) \circ E_t^{-1}\right) S_t^{-1}$$

in t.

The transpose gradient gives us a row vector. To have a column vector, just transpose the above formula. This time, we use g rather than f:

$$\nabla g(x, y, z) = S_t^{-t} \nabla \tilde{g}\left(E_t^{-1}(x, y, z)\right), \quad (x, y, z) \in t.$$

Again, better drop the dependence on x, y, and z:

$$\nabla g = S_t^{-t}(\nabla \tilde{g}) \circ E_t^{-1}$$

in t. We can now take the inner product of these two gradients and integrate in t:

$$\int\int\int_t \left(f_x g_x + f_y g_y + f_z g_z\right) dxdydz = \int\int\int_t \nabla^t f \nabla g dxdydz$$
$$= |\det(S_t)| \int\int\int_T \nabla^t \tilde{f} S_t^{-1} S_t^{-t} \nabla \tilde{g} dxdydz.$$

This formula will be most useful later. We are now ready to design special polynomials in t.

9.1.4 Degrees of Freedom

Consider an unspecified polynomial $p(x, y, z)$ of degree m. How to specify it? Well, one way is to specify its coefficients $a_{i,j,k}$ $(0 \leq i + j + k \leq m)$. How many coefficients are there to specify?

We already know the answer:

$$\binom{m+3}{3} = \frac{(m+3)!}{m! \cdot 3!} = \frac{(m+1)(m+2)(m+3)}{6}$$

(Chap. 8, Sect. 8.14.3). For $m = 5$, for example, there is a need to specify

$$\binom{5+3}{3} = \frac{6 \cdot 7 \cdot 8}{6} = 56$$

coefficients of the form $a_{i,j,k}$ $(0 \leq i + j + k \leq 5)$. After all, even a zero coefficient must be specified as such.

This is a rather explicit way to specify a polynomial. Is there a more implicit way? After all, one could also specify any 56 independent pieces of information, or degrees of freedom.

For this purpose, a suitable piece of information is not necessarily a coefficient: it could also be the value of p (or any partial or directional derivative of p) at any point in the Cartesian space.

To specify a polynomial p of degree five, for example, let us look at the original tetrahedron t and pick degrees of freedom symmetrically in it. At each corner of t, specify the partial derivatives of order 0, 1, and 2.

How many such partial derivatives are there? From Chap. 8, Sect. 8.14.1, the answer is

$$\binom{2+3}{3} = \frac{4 \cdot 5}{2} = 10.$$

So, we have just specified a total of 40 degrees of freedom: ten at each corner. These, however, are not enough: to characterize p uniquely, we must specify 16 more. For

this purpose, look at the edges of t. In each edge, look at the midpoint, and pick two nontangential derivatives, in a direction not parallel to the edge.

For example, consider the edge $(\mathbf{k}, \mathbf{l})$, leasing from corner $\mathbf{k}$ to corner $\mathbf{l}$. Look at its midpoint: $(\mathbf{k} + \mathbf{l})/2$.

Now, look at the difference between the corners: $\mathbf{l} - \mathbf{k}$. Without loss of generality, assume that it is "nearly" in the z-direction. This means that, its maximal coordinate is the third coordinate:

$$|(\mathbf{l} - \mathbf{k})_3| \geq \max\left(|(\mathbf{l} - \mathbf{k})_1|, |(\mathbf{l} - \mathbf{k})_2|\right).$$

In this case, at the midpoint $(\mathbf{k}+\mathbf{l})/2$, do not pick the z-partial derivative: it is nearly tangent and has no chance to be normal. Instead, better pick the x- and y-partial derivatives, and specify

$$p_x\left(\frac{\mathbf{k}+\mathbf{l}}{2}\right) \quad \text{and} \quad p_y\left(\frac{\mathbf{k}+\mathbf{l}}{2}\right).$$

What is so good about this method? Well, it is purely geometrical: it depends on the edge only, not on the tetrahedron to whom it belongs. Later on, we will need to do the same not only in t but also in every neighbor tetrahedron that shares $(\mathbf{k}, \mathbf{l})$ as its joint edge. The same procedure will be carried out there consistently: it will come up with the same nontangential derivative. This will help extend our basis function to the entire mesh later on.

In our original tetrahedron t, there are six edges. Thus, the above specifies 12 more degrees of freedom.

So far, we have specified 52 degrees of freedom. Still, these are not enough: we need four more. What to pick? Well, at each side, look at the midpoint, and pick a nontangential derivative there.

For example, look at the face $\triangle(\mathbf{k}, \mathbf{l}, \mathbf{m})$, vertexed at $\mathbf{k}$, $\mathbf{l}$, and $\mathbf{m}$. Consider its normal vector: the vector product $(\mathbf{l} - \mathbf{k}) \times (\mathbf{m} - \mathbf{k})$. Without loss of generality, assume that it is "nearly" in the z-direction. In other words, its z-coordinate is maximal:

$$\left|((\mathbf{l} - \mathbf{k}) \times (\mathbf{m} - \mathbf{k}))_3\right| \geq \max\left(|((\mathbf{l} - \mathbf{k}) \times (\mathbf{m} - \mathbf{k}))_1|, |((\mathbf{l} - \mathbf{k}) \times (\mathbf{m} - \mathbf{k}))_2|\right).$$

In this case, at the midpoint $(\mathbf{k}+\mathbf{l}+\mathbf{m})/3$, do not pick the x- or y-partial derivative: they are nearly tangent and have no chance to be normal. Instead, better pick the z-partial derivative, and specify

$$p_z\left(\frac{\mathbf{k}+\mathbf{l}+\mathbf{m}}{3}\right).$$

Since there are four sides, this specifies four more degrees of freedom. This makes a total of 56 degrees of freedom, as required.

Still, are they independent of each other? In other words, do they specify p uniquely? To answer this, we need some more geometry and algebra.

9.2 Barycentric Coordinates in 3-D

9.2.1 Barycentric Coordinates in 3-D

Our original tetrahedron t could also be represented in a different way. For this purpose, let $\mathbf{d} = (x, y, z)^t$ be some point in t. Usually, $\mathbf{d}$ is in the interior of t. Still, this is not a must: $\mathbf{d}$ could also lie on the boundary of t: its faces, edges, or even corners. Anyway, $\mathbf{d}$ is a convex combination of the corners:

$$\mathbf{d} = \lambda_{\mathbf{k}}\mathbf{k} + \lambda_{\mathbf{l}}\mathbf{l} + \lambda_{\mathbf{m}}\mathbf{m} + \lambda_{\mathbf{n}}\mathbf{n},$$

where $\lambda_{\mathbf{k}}$, $\lambda_{\mathbf{l}}$, $\lambda_{\mathbf{m}}$, and $\lambda_{\mathbf{n}}$ are nonnegative real numbers that sum to 1:

$$\lambda_{\mathbf{k}} + \lambda_{\mathbf{l}} + \lambda_{\mathbf{m}} + \lambda_{\mathbf{n}} = 1.$$

The coefficients $\lambda_{\mathbf{k}}$, $\lambda_{\mathbf{l}}$, $\lambda_{\mathbf{m}}$, and $\lambda_{\mathbf{n}}$ are the barycentric coordinates of $\mathbf{d}$ [14, 49, 62, 69]. Together, they make a new four-dimensional vector:

$$\boldsymbol{\lambda} \equiv (\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}}, \lambda_{\mathbf{n}})^t.$$

Note that all the above vectors are column vectors, although not quite of the same dimension: $\mathbf{k}$, $\mathbf{l}$, $\mathbf{m}$, $\mathbf{n}$, and $\mathbf{d}$ are three-dimensional, whereas $\boldsymbol{\lambda}$ is four-dimensional. Thus, the above convex combination can also be written as a four-dimensional system:

$$\begin{pmatrix} \mathbf{d} \\ 1 \end{pmatrix} = \left(\begin{array}{c|c|c|c} \mathbf{k} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{array}\right) \boldsymbol{\lambda}.$$

This way, $\mathbf{d}$ is written in terms of $\boldsymbol{\lambda}$. In fact, this is just a projective mapping: the "oblique" real projective space

$$\{\boldsymbol{\lambda} \mid \lambda_{\mathbf{k}} + \lambda_{\mathbf{l}} + \lambda_{\mathbf{m}} + \lambda_{\mathbf{n}} = 1\}$$

is mapped to the "horizontal" real projective space in Chap. 6, Sects. 6.9.1–6.9.2.

9.2.2 The Inverse Mapping

Fortunately, this mapping could also be inverted, to give $\boldsymbol{\lambda}$ in terms of $\mathbf{d}$. Indeed, the above 4×4 matrix is nonsingular: it has a nonzero determinant. To see this, let us multiply it by a new matrix U—a 4×4 upper-triangular matrix:

$$U \equiv \begin{pmatrix} 1 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Clearly,

$$\det(U) = 1.$$

Therefore,

$$\begin{aligned} \det\left(\begin{pmatrix} \mathbf{k} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix}\right) &= \det\left(\begin{pmatrix} \mathbf{k} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix}\right) \det(U) \\ &= \det\left(\begin{pmatrix} \mathbf{k} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix} U\right) \\ &= \det\left(\begin{pmatrix} \mathbf{k} & \mathbf{l}-\mathbf{k} & \mathbf{m}-\mathbf{k} & \mathbf{n}-\mathbf{k} \\ 1 & 0 & 0 & 0 \end{pmatrix}\right) \\ &= \det\left(\begin{pmatrix} \mathbf{k} & S_t \\ 1\ 0 & 0\ 0 \end{pmatrix}\right) \\ &= -\det(S_t) \\ &\neq 0. \end{aligned}$$

9.2.3 Geometrical Interpretation

From Cramer's rule (Chap. 2, Sect. 2.1.5), we can now have the barycentric coordinates in their explicit form. The first one, for instance, is

$$\lambda_{\mathbf{k}}(\mathbf{d}) = \frac{\det\left(\begin{pmatrix} \mathbf{d} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix}\right)}{\det\left(\begin{pmatrix} \mathbf{k} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix}\right)}$$

$$
\begin{aligned}
&= \frac{\det\left(\begin{pmatrix} \mathbf{d} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix} U\right)}{\det\left(\begin{pmatrix} \mathbf{k} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix} U\right)} \\
&= \frac{\det\left(\begin{pmatrix} \mathbf{d} & \mathbf{l}-\mathbf{d} & \mathbf{m}-\mathbf{d} & \mathbf{n}-\mathbf{d} \\ 1 & 0 & 0 & 0 \end{pmatrix}\right)}{\det\left(\begin{pmatrix} \mathbf{k} & \mathbf{l}-\mathbf{k} & \mathbf{m}-\mathbf{k} & \mathbf{n}-\mathbf{k} \\ 1 & 0 & 0 & 0 \end{pmatrix}\right)} \\
&= \frac{\det\left(\begin{pmatrix} \mathbf{d} & S_{(\mathbf{d},\mathbf{l},\mathbf{m},\mathbf{n})} \\ 1\,0 & 0 \quad 0 \end{pmatrix}\right)}{\det\left(\begin{pmatrix} \mathbf{k} & S_t \\ 1\,0\,0 & 0 \end{pmatrix}\right)} \\
&= \frac{-\det\left(S_{(\mathbf{d},\mathbf{l},\mathbf{m},\mathbf{n})}\right)}{-\det(S_t)} \\
&= \frac{\det\left(S_{(\mathbf{d},\mathbf{l},\mathbf{m},\mathbf{n})}\right)}{\det(S_t)}.
\end{aligned}
$$

This gives $\lambda_{\mathbf{k}}$ an interesting geometrical meaning. To see this, just draw four inner edges, leading from $\mathbf{d}$ to the corners of t. This splits t into four disjoint subtetrahedra, each vertexed at $\mathbf{d}$ and three corners of t:

$$t = (\mathbf{d}, \mathbf{l}, \mathbf{m}, \mathbf{n}) \cup (\mathbf{k}, \mathbf{d}, \mathbf{m}, \mathbf{n}) \cup (\mathbf{k}, \mathbf{l}, \mathbf{d}, \mathbf{n}) \cup (\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{d}).$$

Now, look at the first subtetrahedron, vertexed at $\mathbf{d}$, $\mathbf{l}$, $\mathbf{m}$, and $\mathbf{n}$. Calculate its volume, and divide it by the volume of t. This gives the first barycentric coordinate: $\lambda_{\mathbf{k}}$.

In summary, $\lambda_{\mathbf{k}}$ is the relative volume of $(\mathbf{d}, \mathbf{l}, \mathbf{m}, \mathbf{n})$ (the subtetrahedron that lies across from $\mathbf{k}$ in t). Similar formulas can also be written for the three other barycentric coordinates:

$$
\begin{aligned}
\lambda_{\mathbf{l}}(\mathbf{d}) &= \frac{\det\left(S_{(\mathbf{k},\mathbf{d},\mathbf{m},\mathbf{n})}\right)}{\det(S_t)} \\
\lambda_{\mathbf{m}}(\mathbf{d}) &= \frac{\det\left(S_{(\mathbf{k},\mathbf{l},\mathbf{d},\mathbf{n})}\right)}{\det(S_t)} \\
\lambda_{\mathbf{n}}(\mathbf{d}) &= \frac{\det\left(S_{(\mathbf{k},\mathbf{l},\mathbf{m},\mathbf{d})}\right)}{\det(S_t)}.
\end{aligned}
$$

Why do the barycentric coordinate sum to 1? We can now interpret this not only algebraically but also geometrically: the four subtetrahedra sum to the original tetrahedron t.

What happens if $\mathbf{d}$ is a corner? In this case, its barycentric coordinates are either 0 or 1:

$$\lambda_{\mathbf{i}}(\mathbf{j}) = \begin{cases} 1 & \text{if } \mathbf{i} = \mathbf{j} \\ 0 & \text{if } \mathbf{i} \neq \mathbf{j} \end{cases} \qquad \mathbf{i}, \mathbf{j} \in \{\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n}\}.$$

After all, in this special case, one subtetrahedron is t, whereas the three others shrink to nothing. This nice observation will be useful later.

9.2.4 The Chain Rule and Leibniz Rule

As discussed above, we now have $\boldsymbol{\lambda}$ in terms of $\mathbf{d}$:

$$\boldsymbol{\lambda} = \begin{pmatrix} \mathbf{k} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{d} \\ 1 \end{pmatrix}.$$

Thus, $\boldsymbol{\lambda} \equiv \boldsymbol{\lambda}(\mathbf{d})$ is a function of $\mathbf{d}$. More specifically, the individual barycentric coordinates are functions of $\mathbf{d}$ as well:

$$\lambda_{\mathbf{k}} \equiv \lambda_{\mathbf{k}}(\mathbf{d}),\ \lambda_{\mathbf{l}} \equiv \lambda_{\mathbf{l}}(\mathbf{d}),\ \lambda_{\mathbf{m}} \equiv \lambda_{\mathbf{m}}(\mathbf{d}), \quad \text{and} \quad \lambda_{\mathbf{n}} \equiv \lambda_{\mathbf{n}}(\mathbf{d}).$$

We can now differentiate these functions with respect to x, y, and z. For this purpose, look at the above inverse matrix:

$$\begin{pmatrix} \mathbf{k} & \mathbf{l} & \mathbf{m} & \mathbf{n} \\ 1 & 1 & 1 & 1 \end{pmatrix}^{-1}.$$

Look at its first, second, and third columns. Together, they make a 4×3 rectangular matrix: the Jacobian of $\boldsymbol{\lambda}$ with respect to $\mathbf{d}$:

$$\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} = \frac{\partial\,(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}}, \lambda_{\mathbf{n}})}{\partial (x, y, z)}.$$

How to obtain the individual elements in this matrix? Just use Cramer's formula (Chap. 2, Sect. 2.1.4). Assume that this has already been done. This way, $\partial \boldsymbol{\lambda} / \partial \mathbf{d}$ can now be used in the chain rule. This will be quite useful: it will help differentiate a composite function of $\boldsymbol{\lambda} \equiv \boldsymbol{\lambda}(\mathbf{d})$.

To see this, let

$$f(\boldsymbol{\lambda}) \quad \text{and} \quad g(\boldsymbol{\lambda})$$

be two differentiable functions. This means that they have a gradient with respect to $\boldsymbol{\lambda}$:

$$\nabla_{\boldsymbol{\lambda}} f \equiv \begin{pmatrix} \partial f/\partial \lambda_{\mathbf{k}} \\ \partial f/\partial \lambda_{\mathbf{l}} \\ \partial f/\partial \lambda_{\mathbf{m}} \\ \partial f/\partial \lambda_{\mathbf{n}} \end{pmatrix} \quad \text{and} \quad \nabla_{\boldsymbol{\lambda}} g \equiv \begin{pmatrix} \partial g/\partial \lambda_{\mathbf{k}} \\ \partial g/\partial \lambda_{\mathbf{l}} \\ \partial g/\partial \lambda_{\mathbf{m}} \\ \partial g/\partial \lambda_{\mathbf{n}} \end{pmatrix}.$$

Later on, we would like to differentiate a product like fg with respect to λ. This is done as in Leibniz rule:

$$\nabla_{\boldsymbol{\lambda}}(fg) = (\nabla_{\boldsymbol{\lambda}} f)\, g + f \nabla_{\boldsymbol{\lambda}} g = g \nabla_{\boldsymbol{\lambda}} f + f \nabla_{\boldsymbol{\lambda}} g.$$

This is a column vector. To have a row vector, on the other hand, just take the transpose:

$$\nabla^t_{\boldsymbol{\lambda}}(fg) = g \nabla^t_{\boldsymbol{\lambda}} f + f \nabla^t_{\boldsymbol{\lambda}} g.$$

Now, let us go one step further. To both sides of this equation, apply $\nabla_{\boldsymbol{\lambda}}$ from the left:

$$\begin{aligned} \nabla_{\boldsymbol{\lambda}} \nabla^t_{\boldsymbol{\lambda}}(fg) &= \nabla_{\boldsymbol{\lambda}} \left(g \nabla^t_{\boldsymbol{\lambda}} f + f \nabla^t_{\boldsymbol{\lambda}} g \right) \\ &= \nabla_{\boldsymbol{\lambda}} g \nabla^t_{\boldsymbol{\lambda}} f + g \nabla_{\boldsymbol{\lambda}} \nabla^t_{\boldsymbol{\lambda}} f + \nabla_{\boldsymbol{\lambda}} f \nabla^t_{\boldsymbol{\lambda}} g + f \nabla_{\boldsymbol{\lambda}} \nabla^t_{\boldsymbol{\lambda}} g. \end{aligned}$$

This will be useful later.

So far, f and g have been treated as functions of $\boldsymbol{\lambda}$. Still, this is not the whole story: because $\boldsymbol{\lambda} \equiv \boldsymbol{\lambda}(\mathbf{d})$, f and g are actually composite functions of x, y, and z. As such, they can be differentiated with respect to x, y, and z, to form the gradient. Fortunately, we have the chain rule to help do this:

$$\nabla^t f = \nabla^t_{\boldsymbol{\lambda}} f \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \quad \text{and} \quad \nabla g = \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right)^t \nabla_{\boldsymbol{\lambda}} g.$$

Thus, the Hessian of f (with respect to x, y, and z) is

$$\nabla \nabla^t f = \nabla \nabla^t_{\boldsymbol{\lambda}} f \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) = \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right)^t \nabla_{\boldsymbol{\lambda}} \nabla^t_{\boldsymbol{\lambda}} f \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right).$$

9.2.5 Integration in Barycentric Coordinates

Now, look again at the oblique real projective space, introduced at the end of Sect. 9.2.1. It contains those $\boldsymbol{\lambda}$'s satisfying

$$\lambda_{\mathbf{n}} = 1 - \lambda_{\mathbf{k}} - \lambda_{\mathbf{l}} - \lambda_{\mathbf{m}}.$$

This is easy to integrate on. For instance, consider a practical application: as in Sect. 9.1.2, take the inner product of the gradient of f with the gradient of g and integrate in t:

$$\begin{aligned}
&\int\int\int_t \left(f_x g_x + f_y g_y + f_z g_z\right) dxdydz \\
&= \int\int\int_t \nabla^t f \nabla g dxdydz \\
&= |\det(S_t)| \int_0^1 d\lambda_{\mathbf{k}} \int_0^{1-\lambda_{\mathbf{k}}} d\lambda_{\mathbf{l}} \int_0^{1-\lambda_{\mathbf{k}}-\lambda_{\mathbf{l}}} d\lambda_{\mathbf{m}} \nabla^t_{\boldsymbol{\lambda}} f \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right)^t \nabla_{\boldsymbol{\lambda}} g,
\end{aligned}$$

where the gradients on the right, $\nabla^t_{\boldsymbol{\lambda}} f$ and $\nabla_{\boldsymbol{\lambda}} g$, are evaluated at four-dimensional points of the form

$$(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}}, 1 - \lambda_{\mathbf{k}} - \lambda_{\mathbf{l}} - \lambda_{\mathbf{m}}) .$$

Thus, this is just an integral over the unit tetrahedron T:

$$(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}}) \in T.$$

Fortunately, we already know how to do this (Chap. 8, Sect. 8.12.5). This will be useful later. For this purpose, our functions f and g will have to be continuous not only in t but also outside it. Let us see how such a function should be extended smoothly to a neighbor tetrahedron as well.

9.3 Independent Degrees of Freedom

9.3.1 Continuity Across an Edge

In Sect. 9.1.4, we have already introduced 56 independent degrees of freedom. In what sense are they independent? In the following sense:

A polynomial of degree five (or less) with 56 vanishing degrees of freedom must be identically zero.

Let us go ahead and prove this.

For this purpose, consider a polynomial $p(x, y, z)$ of degree five (or less). Consider the edge

$$(\mathbf{k}, \mathbf{l}) \subset t,$$

leading from corner $\mathbf{k}$ to corner $\mathbf{l}$ in t. In this edge, $\lambda_{\mathbf{m}} = \lambda_{\mathbf{n}} = 0$. Thus, unless p vanishes throughout the entire edge, p must not contain the factor $\lambda_{\mathbf{m}}$ or $\lambda_{\mathbf{n}}$.

For a start, assume that p has 20 vanishing degrees of freedom: ten vanishing partial derivatives (of order 0, 1, and 2) at $\mathbf{k}$, and the same at $\mathbf{l}$. So, how does p look like in the edge? Well, along the edge, look at p, and at its first and second tangential derivatives. Both vanish at both endpoints: $\mathbf{k}$ and $\mathbf{l}$. Thus, once restricted to the edge, p must contain cubic factors of the form $\lambda_{\mathbf{k}}^3$ and $\lambda_{\mathbf{l}}^3$:

$$p\,|_{(\mathbf{k},\mathbf{l})} = \lambda_{\mathbf{k}}^3 \lambda_{\mathbf{l}}^3 \cdots = \lambda_{\mathbf{k}}^3 \,(1-\lambda_{\mathbf{k}})^3 \cdots$$

(times an unknown factor, possibly zero). But p is of degree five only, and must not contain such a big factor. Therefore, p must vanish throughout the entire edge:

$$p\,|_{(\mathbf{k},\mathbf{l})} \equiv 0.$$

As a bonus, p also has a zero tangential derivative along the entire edge. Furthermore, in the face $\triangle(\mathbf{k},\mathbf{l},\mathbf{m})$, p must contain (at least a linear) factor $\lambda_{\mathbf{m}}$:

$$p\,|_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} = \lambda_{\mathbf{m}} \cdots$$

(times an unknown factor, possibly zero). Later on, we will use this to extend p smoothly outside t as well.

9.3.2 Smoothness Across an Edge

Furthermore, under some more conditions, the above factor is not only linear but also quadratic:

$$p\,|_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} = \lambda_{\mathbf{m}}^2 \cdots$$

(times an unknown factor, possibly zero). To have this, look at the gradient of p: ∇p. How does it look like in the edge $(\mathbf{k},\mathbf{l})$? Well, at the endpoints, we already know that ∇p vanishes, together with its own partial derivatives. So, at both $\mathbf{k}$ and $\mathbf{l}$, ∇p has a zero tangential derivative. Thus, once restricted to the edge, ∇p must contain quadratic factors of the form $\lambda_{\mathbf{k}}^2$ and $\lambda_{\mathbf{l}}^2$:

$$\nabla p\,|_{(\mathbf{k},\mathbf{l})} = \beta \lambda_{\mathbf{k}}^2 \lambda_{\mathbf{l}}^2 = \beta \lambda_{\mathbf{k}}^2 \,(1-\lambda_{\mathbf{k}})^2 \,.$$

Here, since ∇p is a polynomial of degree four (or less), β must be a constant three-dimensional column vector.

Assume now that p has two more vanishing degrees of freedom: at the midpoint $(\mathbf{k}+\mathbf{l})/2$, it also has two vanishing nontangential derivatives. Thus, thanks to the bonus at the end of Sect. 9.3.1, its gradient vanishes there:

$$\nabla p\left(\frac{\mathbf{k}+\mathbf{l}}{2}\right) = (0,0,0)^t.$$

Therefore, we must have $\beta = (0,0,0)^t$, so ∇p vanishes throughout the entire edge:

$$\nabla p \mid_{(\mathbf{k},\mathbf{l})} \equiv (0,0,0)^t.$$

Therefore, in the face $\triangle(\mathbf{k},\mathbf{l},\mathbf{m})$, the original polynomial p must contain (not only linear but also) quadratic factor:

$$p \mid_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} = \lambda_{\mathbf{m}}^2 \cdots$$

(times an unknown factor, possibly zero).

9.3.3 Continuity Across a Side

Look now at the face $\triangle(\mathbf{k},\mathbf{l},\mathbf{m})$, and the edges in it. Assume now that p has vanishing degrees of freedom not only in $(\mathbf{k},\mathbf{l})$ but also in the two other edges: $(\mathbf{l},\mathbf{m})$ and $(\mathbf{m},\mathbf{k})$. This way, p has a total of 36 vanishing degrees of freedom: ten vanishing partial derivatives (of orders 0, 1, and 2) at $\mathbf{k}$, $\mathbf{l}$, and $\mathbf{m}$, and two vanishing nontangential derivatives at $(\mathbf{k}+\mathbf{l})/2$, $(\mathbf{l}+\mathbf{m})/2$, and $(\mathbf{m}+\mathbf{k})/2$.

Thus, the discussion in Sect. 9.3.2 applies to all three edges: not only $(\mathbf{k},\mathbf{l})$ but also $(\mathbf{l},\mathbf{m})$ and $(\mathbf{m},\mathbf{k})$. Therefore, p must contain a factor as big as

$$p \mid_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} = \lambda_{\mathbf{k}}^2 \lambda_{\mathbf{l}}^2 \lambda_{\mathbf{m}}^2 \cdots .$$

But p is of degree five only, so it must vanish throughout the entire face:

$$p \mid_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} \equiv 0.$$

As a bonus, p also has two vanishing tangential derivatives throughout the entire face. This will be useful later.

Unfortunately, the gradient of p, although vanishes in the edges, not necessarily vanishes throughout the entire face. For example, ∇p may still take there the nonzero form

$$(\nabla p) \mid_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} \equiv \gamma \lambda_{\mathbf{k}} \lambda_{\mathbf{l}} \lambda_{\mathbf{m}},$$

where γ is a nonzero three-dimensional column vector.

As a matter of fact, even if ∇p happened to vanish at the side midpoint $(\mathbf{k}+\mathbf{l}+\mathbf{m})/3$ as well, it might not vanish throughout the entire side. After all, as a polynomial of degree four, it might still take the nonzero form

$$(\nabla p)\,|_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} \equiv \gamma\lambda_{\mathbf{k}}\lambda_{\mathbf{l}}\lambda_{\mathbf{m}}\left(\frac{1}{3}-\lambda_{\mathbf{k}}\right)$$

$$\text{or } (\nabla p)\,|_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} \equiv \gamma\lambda_{\mathbf{k}}\lambda_{\mathbf{l}}\lambda_{\mathbf{m}}\left(\frac{1}{3}-\lambda_{\mathbf{l}}\right)$$

$$\text{or } (\nabla p)\,|_{\triangle(\mathbf{k},\mathbf{l},\mathbf{m})} \equiv \gamma\lambda_{\mathbf{k}}\lambda_{\mathbf{l}}\lambda_{\mathbf{m}}\left(\frac{1}{3}-\lambda_{\mathbf{m}}\right).$$

So, we are stuck: we can say no more about ∇p. Fortunately, we can still say more about p itself. In fact, since p vanishes throughout $\triangle(\mathbf{k},\mathbf{l},\mathbf{m})$, it must contain a linear factor of the form

$$p = \lambda_{\mathbf{n}}\cdots.$$

Assume now that p has vanishing degrees of freedom in the three other faces as well. This way, in total, p has 52 vanishing degrees of freedom: ten at each corner, and two at each edge midpoint. We can then do the same in each face. As a result, p must contain the factor

$$p = \lambda_{\mathbf{k}}\lambda_{\mathbf{l}}\lambda_{\mathbf{m}}\lambda_{\mathbf{n}}\cdots.$$

More precisely, since p is of degree five at most, it must be of the form

$$p = \lambda_{\mathbf{k}}\lambda_{\mathbf{l}}\lambda_{\mathbf{m}}\lambda_{\mathbf{n}}\left(\alpha_{\mathbf{k}}\lambda_{\mathbf{k}}+\alpha_{\mathbf{l}}\lambda_{\mathbf{l}}+\alpha_{\mathbf{m}}\lambda_{\mathbf{m}}+\alpha_{\mathbf{n}}\lambda_{\mathbf{n}}\right),$$

where $\alpha_{\mathbf{k}}$, $\alpha_{\mathbf{l}}$, $\alpha_{\mathbf{m}}$, and $\alpha_{\mathbf{n}}$ are some scalars. What could they be?

9.3.4 Independent Degrees of Freedom

Assume now that p also has four more vanishing degrees of freedom: at each side midpoint, it has a vanishing nontangential derivative. (Thanks to the bonus in Sect. 9.3.3, at each side midpoint, p actually has a zero gradient.) So, in total, p indeed has 56 vanishing degrees of freedom in t. Is p identically zero?

To find out, calculate its gradient at $(\mathbf{k}+\mathbf{l}+\mathbf{m})/3$. At this point, the barycentric coordinates are

$$\lambda_{\mathbf{k}} = \lambda_{\mathbf{l}} = \lambda_{\mathbf{m}} = \frac{1}{3} \quad \text{and} \quad \lambda_{\mathbf{n}} = 0.$$

Thanks to the chain rule (Sect. 9.2.4), we can calculate the gradient indirectly: apply $\nabla_{\boldsymbol{\lambda}}$ rather than ∇. This way, we differentiate with respect to $\lambda_{\mathbf{k}}$, $\lambda_{\mathbf{l}}$, $\lambda_{\mathbf{m}}$, and $\lambda_{\mathbf{n}}$ rather than x, y, or z.

What happens when such a differentiation is carried out? Well, upon evaluating at $\lambda_{\mathbf{n}} = 0$, the term that contains $\lambda_{\mathbf{n}}^2$ drops. The three other terms, on the other hand, must be differentiated with respect to $\lambda_{\mathbf{n}}$, or they would drop as well. In summary, at $(\mathbf{k}+\mathbf{l}+\mathbf{m})/3$, we have

$$\begin{aligned}
(0,0,0) &= \nabla^t p \\
&= \nabla_{\boldsymbol{\lambda}}^t p \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \nabla_{\boldsymbol{\lambda}}^t \left(\lambda_{\mathbf{k}}\lambda_{\mathbf{l}}\lambda_{\mathbf{m}}\lambda_{\mathbf{n}} \left(\alpha_{\mathbf{k}}\lambda_{\mathbf{k}} + \alpha_{\mathbf{l}}\lambda_{\mathbf{l}} + \alpha_{\mathbf{m}}\lambda_{\mathbf{m}} + \alpha_{\mathbf{n}}\lambda_{\mathbf{n}}\right)\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \lambda_{\mathbf{k}}\lambda_{\mathbf{l}}\lambda_{\mathbf{m}} \nabla_{\boldsymbol{\lambda}}^t \lambda_{\mathbf{n}} \left(\alpha_{\mathbf{k}}\lambda_{\mathbf{k}} + \alpha_{\mathbf{l}}\lambda_{\mathbf{l}} + \alpha_{\mathbf{m}}\lambda_{\mathbf{m}}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \lambda_{\mathbf{k}}\lambda_{\mathbf{l}}\lambda_{\mathbf{m}}(0,0,0,1) \left(\alpha_{\mathbf{k}}\lambda_{\mathbf{k}} + \alpha_{\mathbf{l}}\lambda_{\mathbf{l}} + \alpha_{\mathbf{m}}\lambda_{\mathbf{m}}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= 3^{-4}(0,0,0,1) \left(\alpha_{\mathbf{k}} + \alpha_{\mathbf{l}} + \alpha_{\mathbf{m}}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right).
\end{aligned}$$

Now, the fourth row in $\partial\boldsymbol{\lambda}/\partial\mathbf{d}$ cannot be $(0,0,0)$, or $\lambda_{\mathbf{n}}$ would be just a constant, which is impossible. Therefore, we must have

$$\alpha_{\mathbf{k}} + \alpha_{\mathbf{l}} + \alpha_{\mathbf{m}} = 0.$$

The same could be done at the three other side midpoints as well. In summary, we have four linear equations:

$$\begin{aligned}
\alpha_{\mathbf{l}} + \alpha_{\mathbf{m}} + \alpha_{\mathbf{n}} &= 0 \\
\alpha_{\mathbf{k}} + \alpha_{\mathbf{m}} + \alpha_{\mathbf{n}} &= 0 \\
\alpha_{\mathbf{k}} + \alpha_{\mathbf{l}} + \alpha_{\mathbf{n}} &= 0 \\
\alpha_{\mathbf{k}} + \alpha_{\mathbf{l}} + \alpha_{\mathbf{m}} &= 0.
\end{aligned}$$

More compactly, this could be written as a four-dimensional linear system:

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha_{\mathbf{l}} \\ \alpha_{\mathbf{m}} \\ \alpha_{\mathbf{n}} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Look at this 4×4 matrix. Is it singular? Fortunately not. Indeed, it has no zero eigenvalue. In fact, it has only two eigenvalues: 3 and -1. To see this, just write it as

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} - I,$$

where I is the 4×4 identity matrix.

On the right-hand side, look at the first matrix. It can be written as a column vector times a row vector:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} (1, 1, 1, 1).$$

What are the eigenvectors? Well, $(1, 1, 1, 1)^t$ is an eigenvector, with the eigenvalue 4. Every vector orthogonal to $(1, 1, 1, 1)^t$, on the other hand, has the eigenvalue 0. Thus, once I is subtracted, we get the eigenvalues 3 and -1, as asserted.

Thus, the only solution to the above linear system is

$$\alpha_{\mathbf{k}} = \alpha_{\mathbf{l}} = \alpha_{\mathbf{m}} = \alpha_{\mathbf{n}} = 0.$$

Thus, our original polynomial is identically zero:

$$p \equiv 0.$$

This means that our original 56 degrees of freedom are indeed independent of each other, as asserted all along. This is the key to designing basis functions.

9.4 Piecewise-Polynomial Functions

9.4.1 Smooth Piecewise-Polynomial Function

Later on, we will extend a function from t to the entire mesh. Could this be done smoothly? In other words, could the extended function remain continuous across the boundary of t, and perhaps even have a continuous gradient?

To study this, let t_1 and t_2 be two neighbor tetrahedra that share a joint edge: $t_1 \cap t_2$. Let p_1 and p_2 be two different polynomials of degree five (or less), defined in t_1 and t_2, respectively. Assume also that, in their joint edge, both p_1 and p_2 share the same 22 degrees of freedom: ten partial derivatives at each endpoint, and two nontangential derivatives at the midpoint. As discussed above, $p_1 - p_2$ and $\nabla p_1 - \nabla p_2$ must then vanish throughout the entire edge.

Now, in the union $t_1 \cup t_2$, define a new piecewise-polynomial function:

$$u(x, y, z) \equiv \begin{cases} p_1(x, y, z) & \text{if } (x, y, z) \in t_1 \\ p_2(x, y, z) & \text{if } (x, y, z) \in t_2. \end{cases}$$

In what sense is u smooth? Well, in $t_1 \cap t_2$, u is continuous: it is the same from both sides. Furthermore, ∇u is continuous as well: it is the same from both sides as well. These properties will be useful later.

9.4.2 Continuous Piecewise-Polynomial Function

Assume now that t_1 and t_2 share not only a mere edge but also a complete face: $t_1 \cap t_2$ is now a face, not just an edge. Moreover, assume now that both p_1 and p_2 share the same 36 degrees of freedom in $t_1 \cap t_2$: ten at each vertex, and two more at each edge midpoint. As discussed above, $p_1 - p_2$ must then vanish throughout the entire face. Thus, u is continuous throughout the entire face $t_1 \cap t_2$: it is the same from both sides. This property will be useful later: it will help design a piecewise-polynomial basis function.

9.5 Basis Functions

9.5.1 Side Midpoint Basis Function

We are now ready to design our first basis function. For a start, we define it in t as a polynomial of degree five. Later on, we will extend it outside t as well.

Consider, for example, the side midpoint

$$\mathbf{w} \equiv \frac{\mathbf{k} + \mathbf{l} + \mathbf{m}}{3}.$$

Recall that, at $\mathbf{w}$, the barycentric coordinates are

$$\lambda_{\mathbf{k}} = \lambda_{\mathbf{l}} = \lambda_{\mathbf{m}} = \frac{1}{3} \quad \text{and} \quad \lambda_{\mathbf{n}} = 0.$$

Assume that

$$\frac{\partial \lambda_{\mathbf{n}}}{\partial z} \neq 0.$$

This is not new: we have already seen this in Sect. 9.1.4 in a (yet stronger) geometrical version, which makes sure that this partial derivative is far away from

zero. (This is why we picked the z-partial derivative as our chosen degree of freedom at $\mathbf{w}$.) Now, let us define the corresponding basis function $\psi_{\mathbf{w}}$ in t:

$$\psi_{\mathbf{w}} \equiv \alpha \lambda_{\mathbf{k}} \lambda_{\mathbf{l}} \lambda_{\mathbf{m}} \lambda_{\mathbf{n}} \left(\frac{1}{3} - \lambda_{\mathbf{n}} \right),$$

where α is a constant scalar, to be specified later.

Why is this a good candidate for a basis function? Because it has just one nonzero degree of freedom. Indeed,

- at each corner in t, there is a triple product of vanishing barycentric coordinates. For this reason, the partial derivatives of order 0, 1, and 2 vanish there, as required.
- At each edge midpoint, there are two vanishing factors, so the nontangential derivatives vanish there as well.
- At every side midpoint but $\mathbf{w}$, there are two vanishing factors, so the nontangential derivative vanishes there as well.
- So, our only task is to pick α cleverly, to make sure that the final degree of freedom (the nontangential derivative at $\mathbf{w}$) is correct as well.

So, what should α be? Pick α to make the z-partial derivative equal to 1 at $\mathbf{w}$. How to do this? As discussed in Sect. 9.3.4, use the chain rule. This way, at $\mathbf{w}$,

$$\begin{aligned}
(\cdot, \cdot, 1) &= \nabla^t \psi_{\mathbf{w}} \\
&= \nabla^t_{\boldsymbol{\lambda}} \psi_{\mathbf{w}} \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \\
&= \nabla^t_{\boldsymbol{\lambda}} \left(\alpha \lambda_{\mathbf{k}} \lambda_{\mathbf{l}} \lambda_{\mathbf{m}} \lambda_{\mathbf{n}} \left(\frac{1}{3} - \lambda_{\mathbf{n}} \right) \right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \\
&= \alpha \lambda_{\mathbf{k}} \lambda_{\mathbf{l}} \lambda_{\mathbf{m}} \nabla^t_{\boldsymbol{\lambda}} \lambda_{\mathbf{n}} \left(\frac{1}{3} - \lambda_{\mathbf{n}} \right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \\
&= 3^{-4} \alpha (0, 0, 0, 1) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right).
\end{aligned}$$

More explicitly, define

$$\alpha \equiv \frac{3^4}{\frac{\partial \lambda_{\mathbf{n}}}{\partial z}}.$$

This way, only at $\mathbf{w}$ is the degree of freedom equal to 1. Elsewhere, on the other hand, the degrees of freedom vanish, as required. Thus, $\psi_{\mathbf{w}}$ is our first basis function.

In practice, we may have not only one tetrahedron t but also a complete mesh. In particular, t may have a neighbor tetrahedron from the other side of $\triangle(\mathbf{k}, \mathbf{l}, \mathbf{m})$. How to define $\psi_{\mathbf{w}}$ there? Just use the same approach, and define $\psi_{\mathbf{w}}$ as a different

polynomial there. Still, at the joint side, both definitions agree with each other (Sect. 9.4.2). Furthermore, thanks to symmetry, their gradients must agree as well. Thus, $\psi_{\mathbf{w}}$ is indeed a differentiable piecewise-polynomial function, as required.

In the rest of the mesh, on the other hand, $\psi_{\mathbf{w}}$ is defined as zero. This way, $\psi_{\mathbf{w}}$ is indeed a proper basis function: differentiable, piecewise-polynomial, and with just one nonzero degree of freedom: at $\mathbf{w}$ only.

The above could be done not only at $\mathbf{w}$ but also at any other side midpoint. Next, we define yet another kind of basis function.

9.5.2 *Edge-Midpoint Basis Function*

Consider now the edge midpoint

$$\mathbf{h} \equiv \frac{\mathbf{k}+\mathbf{l}}{2}.$$

Clearly, at $\mathbf{h}$, the barycentric coordinates are

$$\lambda_{\mathbf{k}} = \lambda_{\mathbf{l}} = \frac{1}{2} \quad \text{and} \quad \lambda_{\mathbf{m}} = \lambda_{\mathbf{n}} = 0.$$

Assume that, in the Jacobian $\partial\boldsymbol{\lambda}/\partial\mathbf{d}$, the 2×2 lower-left block is nonsingular:

$$\det\left(\frac{\partial\left(\lambda_{\mathbf{m}}, \lambda_{\mathbf{n}}\right)}{\partial(x, y)}\right) \neq 0.$$

This is not new: we have already seen this in Sect. 9.1.4 in a (yet stronger) geometrical version, which makes sure that this determinant is far away from zero. This is why we picked the x- and y-partial derivatives as our chosen degrees of freedom at $\mathbf{h}$.

Let us start with the x-partial derivative. Let us introduce the corresponding basis function $\psi_{\mathbf{h},1}$:

$$\psi_{\mathbf{h},1} \equiv \lambda_{\mathbf{k}}^2 \lambda_{\mathbf{l}}^2 \left(\alpha_{\mathbf{m}} \lambda_{\mathbf{m}} + \alpha_{\mathbf{n}} \lambda_{\mathbf{n}}\right),$$

where $\alpha_{\mathbf{m}}$ and $\alpha_{\mathbf{n}}$ are constant scalars, to be specified later.

Why is this a good candidate for a basis function? Because it has just one nonzero degree of freedom. Indeed,

- at each corner of t, there is here a product of at least three vanishing barycentric coordinates. For this reason, the partial derivatives of order 0, 1, and 2 vanish there, as required.
- At every edge midpoint but $\mathbf{h}$, the quadratic factor $\lambda_{\mathbf{k}}^2$ or $\lambda_{\mathbf{l}}^2$ vanishes, so the nontangential derivatives vanish there as well.

- At each side midpoint that lies across from $\mathbf{h}$, the quadratic factor $\lambda_{\mathbf{k}}^2$ or $\lambda_{\mathbf{l}}^2$ vanishes, so the nontangential derivative vanishes there as well.
- Later on, we will also make sure that this is also true at those side midpoints that lie nearby $\mathbf{h}$.
- Before doing this, however, we have more urgent business: to pick the above α's cleverly, and make sure that the degrees of freedom are correct at $\mathbf{h}$ itself.

So, what should $\alpha_{\mathbf{m}}$ and $\alpha_{\mathbf{n}}$ be? Pick them to make the x-partial derivative equal to 1, and the y-partial derivative equal to 0 at $\mathbf{h}$. In other words, at $\mathbf{h}$,

$$\begin{aligned}
(1, 0, \cdot) &= \nabla^t \psi_{\mathbf{h},1} \\
&= \nabla_{\boldsymbol{\lambda}}^t \psi_{\mathbf{h},1} \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \\
&= \nabla_{\boldsymbol{\lambda}}^t \left(\lambda_{\mathbf{k}}^2 \lambda_{\mathbf{l}}^2 \left(\alpha_{\mathbf{m}} \lambda_{\mathbf{m}} + \alpha_{\mathbf{n}} \lambda_{\mathbf{n}} \right) \right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \\
&= \lambda_{\mathbf{k}}^2 \lambda_{\mathbf{l}}^2 \nabla_{\boldsymbol{\lambda}}^t \left(\alpha_{\mathbf{m}} \lambda_{\mathbf{m}} + \alpha_{\mathbf{n}} \lambda_{\mathbf{n}} \right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \\
&= \lambda_{\mathbf{k}}^2 \lambda_{\mathbf{l}}^2 \left(\alpha_{\mathbf{m}} \nabla_{\boldsymbol{\lambda}}^t \lambda_{\mathbf{m}} + \alpha_{\mathbf{n}} \nabla_{\boldsymbol{\lambda}}^t \lambda_{\mathbf{n}} \right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \\
&= 2^{-4} \left(\alpha_{\mathbf{m}} (0, 0, 1, 0) + \alpha_{\mathbf{n}} (0, 0, 0, 1) \right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right) \\
&= 2^{-4} \left(0, 0, \alpha_{\mathbf{m}}, \alpha_{\mathbf{n}} \right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}} \right).
\end{aligned}$$

These are two linear equations in two unknowns: $\alpha_{\mathbf{m}}$ and $\alpha_{\mathbf{n}}$. What is the coefficient matrix? It is a familiar block: the 2×2 lower-left block in the original Jacobian $\partial \boldsymbol{\lambda} / \partial \mathbf{d}$. More precisely, we actually look at the transpose system, so we actually look at the transpose block. Anyway, it has the same determinant: nonzero. So, it is nonsingular, as required. Therefore, $\alpha_{\mathbf{m}}$ and $\alpha_{\mathbf{n}}$ can be solved for uniquely.

So, at $\mathbf{h}$, the degrees of freedom are correct. What about the rest of the degrees of freedom in t? Well, most of them already vanish, as required. Only at nearby side midpoints may they still be nonzero. How to fix this?

Consider, for instance, the side midpoint $\mathbf{w}$, discussed in the previous subsection. Fortunately, we already have a basis function: $\psi_{\mathbf{w}}$ in t. Let us go ahead and subtract a multiple of it from $\psi_{\mathbf{h},1}$ in t:

$$\psi_{\mathbf{h},1} \leftarrow \psi_{\mathbf{h},1} - \left(\psi_{\mathbf{h},1} \right)_z (\mathbf{w}) \psi_{\mathbf{w}}.$$

This is indeed a fix: after this substitution, we have

$$\left(\psi_{\mathbf{h},1} \right)_z (\mathbf{w}) = 0,$$

as required. Fortunately, this substitution does not spoil the rest of the degrees of freedom, which remain correct.

By now, we took care of $\mathbf{w}$. Still, $\mathbf{w}$ is mirrored by yet another side midpoint: $(\mathbf{k}+\mathbf{l}+\mathbf{n})/3$. To fix it too, subtract

$$\psi_{\mathbf{h},1} \leftarrow \psi_{\mathbf{h},1} - \left(\psi_{\mathbf{h},1}\right)_z \left(\frac{\mathbf{k}+\mathbf{l}+\mathbf{n}}{3}\right) \psi_{(\mathbf{k}+\mathbf{l}+\mathbf{n})/3}.$$

This is indeed a fix: after this substitution is carried out in t, we have

$$\left(\psi_{\mathbf{h},1}\right)_z \left(\frac{\mathbf{k}+\mathbf{l}+\mathbf{n}}{3}\right) = 0,$$

as required. Fortunately, this substitution does not spoil any other degree of freedom. Thus, in its final form, $\psi_{\mathbf{h},1}$ makes a proper basis function in t.

So far, we have defined $\psi_{\mathbf{h},1}$ in t only. It is now time to extend it to the entire mesh as well. First, let us work in those neighbor tetrahedra that share $(\mathbf{k},\mathbf{l})$ as their joint edge: repeat the same procedure there as well. This way, in each edge-sharing tetrahedron, $\psi_{\mathbf{h},1}$ is defined as a different polynomial of degree five. Still, across $(\mathbf{k},\mathbf{l})$, $\psi_{\mathbf{h},1}$ remains differentiable (Sect. 9.4.1).

In the rest of the tetrahedra in the mesh, which do not use $(\mathbf{k},\mathbf{l})$ as an edge, $\psi_{\mathbf{h},1}$ is defined as zero. This way, $\psi_{\mathbf{h},1}$ is indeed a proper basis function throughout the entire mesh.

So far, we have focused on the x-partial derivative at $\mathbf{h}$. Now, let us consider the y-partial derivative. For this purpose, in the beginning of the above development, just replace $(1,0,\cdot)$ by $(0,1,\cdot)$. This produces a new function $\psi_{\mathbf{h},2}$, corresponding to the y-partial derivative at $\mathbf{h}$, as required.

The same could be done not only at $\mathbf{h}$ but also at any other edge midpoint. Next, let us move on to yet another kind of basis function: corner basis function.

9.5.3 Hessian-Related Corner Basis Function

Consider the corner $\mathbf{n} \in t$. Clearly, at $\mathbf{n}$, the barycentric coordinates are

$$\lambda_{\mathbf{k}} = \lambda_{\mathbf{l}} = \lambda_{\mathbf{m}} = 0 \quad \text{and} \quad \lambda_{\mathbf{n}} = 1.$$

Let us define the basis function corresponding to the xx-partial derivative at $\mathbf{n}$:

$$\psi_{\mathbf{n},5} \equiv \lambda_{\mathbf{n}}^3 \sum_{\mathbf{i},\mathbf{j}\in\{\mathbf{k},\mathbf{l},\mathbf{m}\}} \alpha_{\mathbf{i},\mathbf{j}} \lambda_{\mathbf{i}} \lambda_{\mathbf{j}}.$$

Why is this a good candidate for a basis function? Because it has just one nonzero degree of freedom. Indeed,

- thanks to the cubic factor $\lambda_{\mathbf{n}}^3$, $\psi_{\mathbf{n},5}$ has vanishing degrees of freedom at $\mathbf{k}$, $\mathbf{l}$, and $\mathbf{m}$, as required.
- This is also true at the edge- and side midpoints, at least those that lie on the face $\triangle(\mathbf{k}, \mathbf{l}, \mathbf{m})$ (across from $\mathbf{n}$).
- Later on, we will make sure that this is also true at those that lie nearby $\mathbf{n}$.
- Before doing this, however, we have more urgent business: to pick the above α's cleverly, and make sure that the degrees of freedom are correct at $\mathbf{n}$ itself.

Thanks to symmetry, we may assume here that

$$\alpha_{\mathbf{i},\mathbf{j}} = \alpha_{\mathbf{j},\mathbf{i}},$$

so we are actually looking for six unknown α's. To find them, solve the following six equations at $\mathbf{n}$:

$$\begin{aligned}
\begin{pmatrix} 1\,0\,0 \\ 0\,0\,0 \\ 0\,0\,0 \end{pmatrix} &= \nabla\nabla^t \psi_{\mathbf{n},5} \\
&= \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right)^t \left(\nabla_{\boldsymbol{\lambda}} \nabla_{\boldsymbol{\lambda}}^t \psi_{\mathbf{n},5}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right)^t \nabla_{\boldsymbol{\lambda}} \nabla_{\boldsymbol{\lambda}}^t \left(\lambda_{\mathbf{n}}^3 \sum_{\mathbf{i},\mathbf{j} \in \{\mathbf{k},\mathbf{l},\mathbf{m}\}} \alpha_{\mathbf{i},\mathbf{j}} \lambda_{\mathbf{i}} \lambda_{\mathbf{j}}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right)^t \left(2\lambda_{\mathbf{n}}^3 \sum_{\mathbf{i},\mathbf{j} \in \{\mathbf{k},\mathbf{l},\mathbf{m}\}} \alpha_{\mathbf{i},\mathbf{j}} \nabla_{\boldsymbol{\lambda}} \lambda_{\mathbf{i}} \nabla_{\boldsymbol{\lambda}}^t \lambda_{\mathbf{j}}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= 2 \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right)^t \begin{pmatrix} \alpha_{\mathbf{k},\mathbf{k}} & \alpha_{\mathbf{k},\mathbf{l}} & \alpha_{\mathbf{k},\mathbf{m}} & 0 \\ \alpha_{\mathbf{l},\mathbf{k}} & \alpha_{\mathbf{l},\mathbf{l}} & \alpha_{\mathbf{l},\mathbf{m}} & 0 \\ \alpha_{\mathbf{m},\mathbf{k}} & \alpha_{\mathbf{m},\mathbf{l}} & \alpha_{\mathbf{m},\mathbf{m}} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= 2 \left(\frac{\partial\,(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}})}{\partial(x, y, z)}\right)^t \begin{pmatrix} \alpha_{\mathbf{k},\mathbf{k}} & \alpha_{\mathbf{k},\mathbf{l}} & \alpha_{\mathbf{k},\mathbf{m}} \\ \alpha_{\mathbf{l},\mathbf{k}} & \alpha_{\mathbf{l},\mathbf{l}} & \alpha_{\mathbf{l},\mathbf{m}} \\ \alpha_{\mathbf{m},\mathbf{k}} & \alpha_{\mathbf{m},\mathbf{l}} & \alpha_{\mathbf{m},\mathbf{m}} \end{pmatrix} \left(\frac{\partial\,(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}})}{\partial(x, y, z)}\right).
\end{aligned}$$

More explicitly, define

$$\begin{pmatrix} \alpha_{\mathbf{k},\mathbf{k}} & \alpha_{\mathbf{k},\mathbf{l}} & \alpha_{\mathbf{k},\mathbf{m}} \\ \alpha_{\mathbf{l},\mathbf{k}} & \alpha_{\mathbf{l},\mathbf{l}} & \alpha_{\mathbf{l},\mathbf{m}} \\ \alpha_{\mathbf{m},\mathbf{k}} & \alpha_{\mathbf{m},\mathbf{l}} & \alpha_{\mathbf{m},\mathbf{m}} \end{pmatrix} \equiv \frac{1}{2} \left(\frac{\partial\,(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}})}{\partial(x, y, z)}\right)^{-t} \begin{pmatrix} 1\,0\,0 \\ 0\,0\,0 \\ 0\,0\,0 \end{pmatrix} \left(\frac{\partial\,(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}})}{\partial(x, y, z)}\right)^{-1}.$$

This way, $\psi_{\mathbf{n},5}$ has the correct degrees of freedom at $\mathbf{n}$ as well.

Still, this is not good enough. To become a proper basis function, this function must now be modified. For each nearby side midpoint, subtract a multiple of a basis function like that defined in Sect. 9.5.1 in t. Furthermore, for each nearby edge midpoint, subtract a multiple of two basis functions like those defined in Sect. 9.5.2 in t.

This defines our new basis function in t. How to extend it to those neighbor tetrahedra that share $\mathbf{n}$ as their joint corner? In each of them, just repeat the above procedure, and define $\psi_{\mathbf{n},5}$ as a different polynomial of degree five. In the rest of the mesh, on the other hand, define it as zero. This extends $\psi_{\mathbf{n},5}$ into a continuous piecewise-polynomial basis function in the entire mesh.

The same method could also be used to design five more basis functions, corresponding to the xy-, yy-, xz-, yz-, and zz-partial derivative at $\mathbf{n}$. In fact, to define $\psi_{\mathbf{n},6}$, $\psi_{\mathbf{n},7}$, $\psi_{\mathbf{n},8}$, $\psi_{\mathbf{n},9}$, and $\psi_{\mathbf{n},10}$, make just a small change in the above: just replace

$$\begin{pmatrix} 1&0&0\\0&0&0\\0&0&0 \end{pmatrix} \text{ by } \begin{pmatrix} 0&1&0\\1&0&0\\0&0&0 \end{pmatrix}, \begin{pmatrix} 0&0&0\\0&1&0\\0&0&0 \end{pmatrix},$$

$$\begin{pmatrix} 0&0&1\\0&0&0\\1&0&0 \end{pmatrix}, \begin{pmatrix} 0&0&0\\0&0&1\\0&1&0 \end{pmatrix}, \text{ or } \begin{pmatrix} 0&0&0\\0&0&0\\0&0&1 \end{pmatrix},$$

respectively.

9.5.4 Gradient-Related Corner Basis Function

Likewise, let us go ahead and design a new basis function, corresponding to the x-partial derivative at $\mathbf{n}$:

$$\psi_{\mathbf{n},2} \equiv \lambda_{\mathbf{n}}^3 \left(\alpha_{\mathbf{k}}\lambda_{\mathbf{k}} + \alpha_{\mathbf{l}}\lambda_{\mathbf{l}} + \alpha_{\mathbf{m}}\lambda_{\mathbf{m}}\right).$$

How to find the unknowns $\alpha_{\mathbf{k}}$, $\alpha_{\mathbf{l}}$, and $\alpha_{\mathbf{m}}$? Solve three linear equations at $\mathbf{n}$:

$$\begin{aligned}
(1,0,0) &= \nabla^t \psi_{\mathbf{n},2} \\
&= \left(\nabla_{\boldsymbol{\lambda}}^t \psi_{\mathbf{n},2}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \nabla_{\boldsymbol{\lambda}}^t \left(\lambda_{\mathbf{n}}^3 \left(\alpha_{\mathbf{k}}\lambda_{\mathbf{k}} + \alpha_{\mathbf{l}}\lambda_{\mathbf{l}} + \alpha_{\mathbf{m}}\lambda_{\mathbf{m}}\right)\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \lambda_{\mathbf{n}}^3 \left(\alpha_{\mathbf{k}} \nabla_{\boldsymbol{\lambda}}^t \lambda_{\mathbf{k}} + \alpha_{\mathbf{l}} \nabla_{\boldsymbol{\lambda}}^t \lambda_{\mathbf{l}} + \alpha_{\mathbf{m}} \nabla_{\boldsymbol{\lambda}}^t \lambda_{\mathbf{m}}\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right)
\end{aligned}$$

$$
\begin{aligned}
&= \lambda_{\mathbf{n}}^3 \left(\alpha_{\mathbf{k}}(1,0,0,0) + \alpha_{\mathbf{l}}(0,1,0,0) + \alpha_{\mathbf{m}}(0,0,1,0)\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \left(\alpha_{\mathbf{k}}, \alpha_{\mathbf{l}}, \alpha_{\mathbf{m}}, 0\right) \left(\frac{\partial \boldsymbol{\lambda}}{\partial \mathbf{d}}\right) \\
&= \left(\alpha_{\mathbf{k}}, \alpha_{\mathbf{l}}, \alpha_{\mathbf{m}}\right) \left(\frac{\partial \left(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}}\right)}{\partial (x, y, z)}\right).
\end{aligned}
$$

More explicitly, define

$$
\begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha_{\mathbf{l}} \\ \alpha_{\mathbf{m}} \end{pmatrix} \equiv \left(\frac{\partial \left(\lambda_{\mathbf{k}}, \lambda_{\mathbf{l}}, \lambda_{\mathbf{m}}\right)}{\partial (x, y, z)}\right)^{-t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.
$$

Still, this is not good enough. To become a proper basis function, this function must now be modified. For each nearby side midpoint, subtract a multiple of the basis function defined in Sect. 9.5.1 in t. Furthermore, for each nearby edge midpoint, subtract a multiple of those two basis functions defined in Sect. 9.5.2 in t. Finally, subtract a multiple of those six basis functions defined in Sect. 9.5.3:

$$
\begin{aligned}
\psi_{\mathbf{n},2} &\leftarrow \psi_{\mathbf{n},2} - \left(\psi_{\mathbf{n},2}\right)_{xx}(\mathbf{n})\psi_{\mathbf{n},5} \\
\psi_{\mathbf{n},2} &\leftarrow \psi_{\mathbf{n},2} - \left(\psi_{\mathbf{n},2}\right)_{xy}(\mathbf{n})\psi_{\mathbf{n},6} \\
\psi_{\mathbf{n},2} &\leftarrow \psi_{\mathbf{n},2} - \left(\psi_{\mathbf{n},2}\right)_{yy}(\mathbf{n})\psi_{\mathbf{n},7} \\
\psi_{\mathbf{n},2} &\leftarrow \psi_{\mathbf{n},2} - \left(\psi_{\mathbf{n},2}\right)_{xz}(\mathbf{n})\psi_{\mathbf{n},8} \\
\psi_{\mathbf{n},2} &\leftarrow \psi_{\mathbf{n},2} - \left(\psi_{\mathbf{n},2}\right)_{yz}(\mathbf{n})\psi_{\mathbf{n},9} \\
\text{and finally } \psi_{\mathbf{n},2} &\leftarrow \psi_{\mathbf{n},2} - \left(\psi_{\mathbf{n},2}\right)_{zz}(\mathbf{n})\psi_{\mathbf{n},10}
\end{aligned}
$$

in t. As before, $\psi_{\mathbf{n},2}$ is now extended into a continuous piecewise-polynomial basis function in the entire mesh.

The same approach can also be used to design two more basis functions, corresponding to the y- and z-partial derivative at $\mathbf{n}$. In fact, to define $\psi_{\mathbf{n},3}$ and $\psi_{\mathbf{n},4}$, just replace $(1, 0, 0)$ above by $(0, 1, 0)$ or $(0, 0, 1)$, respectively.

9.5.5 Corner Basis Function

Finally, let us define the basis function corresponding to the function itself at $\mathbf{n}$:

$$
\psi_{\mathbf{n},1} \equiv \lambda_{\mathbf{n}}^3.
$$

Still, this is not good enough. To become a proper basis function, this function must now be modified. For each nearby side midpoint, subtract a multiple of a side midpoint basis function in t. Furthermore, for each nearby edge midpoint, subtract a multiple of two edge-midpoint basis functions in t. Moreover, as in Sect. 9.5.4, subtract a multiple of six Hessian-related basis functions in t. Finally, subtract a multiple of three gradient-related basis functions:

$$\begin{aligned} \psi_{\mathbf{n},1} &\leftarrow \psi_{\mathbf{n},1} - \big(\psi_{\mathbf{n},1}\big)_x(\mathbf{n})\psi_{\mathbf{n},2} \\ \psi_{\mathbf{n},1} &\leftarrow \psi_{\mathbf{n},1} - \big(\psi_{\mathbf{n},1}\big)_y(\mathbf{n})\psi_{\mathbf{n},3} \\ \text{and finally } \psi_{\mathbf{n},1} &\leftarrow \psi_{\mathbf{n},1} - \big(\psi_{\mathbf{n},1}\big)_z(\mathbf{n})\psi_{\mathbf{n},4} \end{aligned}$$

in t. As before, $\psi_{\mathbf{n},1}$ should now be extended into a continuous piecewise-polynomial basis function in the entire mesh.

This completes the design of ten new basis functions for $\mathbf{n}$. The same can now be done for the other corners as well.

9.6 Numerical Experiment: Electromagnetic Waves

9.6.1 Frequency and Wave Number

How to test our high-order finite elements? For this purpose, consider the three-dimensional unit cube

$$[0,1]^3 \equiv [0,1] \times [0,1] \times [0,1]$$

(Fig. 9.2). In it, draw a mesh of tetrahedra. This could be done adaptively, level by level [75].

On this mesh, solve the Helmholtz equation that models an electromagnetic wave. For this equation, the typical solution is a pure wave of the form

$$\sin(2\pi\omega_1 x)\sin(2\pi\omega_2 y)\sin(2\pi\omega_3 z),$$

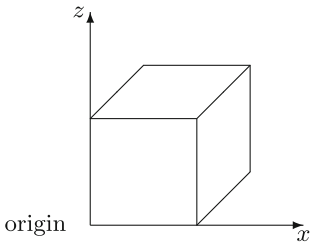


Fig. 9.2 How to model an electromagnetic wave in a three-dimensional cube?

where ω_1, ω_2, and ω_3 are the wave numbers. For example, $\omega_1 = 1/2$ stands for a rather smooth wave that hardly oscillates with x at all: it only makes half a cycle in x: from $\sin(0) = 0$ to $\sin(\pi) = 0$. $\omega_1 = 1$, on the other hand, is already far more interesting: it makes one complete cycle in x: from $\sin(0) = 0$ to $\sin(2\pi) = 0$.

Still, this kind of pure wave is too ideal: it only characterizes a simple model case. In the more general case, on the other hand, many different waves could mix or interfere with each other and produce all sorts of nonstandard waves with just any shape.

Thanks to the wave numbers, we can also define the frequency:

$$\text{frequency} \equiv \sqrt{\omega_1^2 + \omega_2^2 + \omega_3^2}.$$

Furthermore, the wave-length is defined as

$$\text{wave-length} \equiv \frac{2\pi}{\text{frequency}}.$$

These parameters are more general: they apply not only to a pure wave but also to the general solution.

The general case is quite difficult: there is no general method to solve for the true analytic solution. This is where finite elements come useful: they help approximate the true solution numerically.

In our experiments, we design the boundary conditions cleverly, so that the true solution happens to be known. This way, we can also calculate the numerical error: the difference between the numerical solution and the true solution. This may help examine the finite-element scheme and evaluate its accuracy and quality.

9.6.2 Adaptive Mesh Refinement

How to design the mesh? By iterative refinement: start from a coarse mesh, and refine it more and more, level by level [75]. Best do this adaptively: refine only those tetrahedra that are worth it. This may be quite economic and save a lot of computational resources.

Moreover, the high-order finite elements are particularly efficient and accurate. Although there is a lot of preparation work, it is still well-parallelizable.

How to refine the coarse mesh? Once we have decided which tetrahedra to refine, in what order to do this? Better use a trick: in the coarse mesh, better split long edges before short ones. This may improve regularity: the tetrahedra become "fat" and thick, and not too thin. This will be discussed widely in Chap. 11.

With this trick, adaptive refinement wins: it gets as accurate as ever. Why? Since it uses less edges, each individual tetrahedron is rather big and thick. In such a

tetrahedron, our polynomial of degree five has a good chance to approximate the true solution well. This is indeed demonstrated in the numerical results below.

9.7 Numerical Results

9.7.1 High-Order Finite Elements

Global refinement is naive: in the coarse mesh, all tetrahedra split. This way, the fine mesh is too fine: it uses too many degrees of freedom, for no reason.

Adaptive refinement, on the other hand, is cleverer: it refines only where absolutely necessary. This is more economic and efficient.

In both approaches, there is yet another trick: long edges split before short ones. This improves regularity: the fine tetrahedra are still thick, and not too thin.

In each level of refinement, adaptive refinement introduces some new edges, but not too many. This way, the fine tetrahedra are not too small. This may improve accuracy: after all, in a big tetrahedron, a polynomial of degree five has a good chance to approximate the true solution well.

To evaluate accuracy, we calculate the numerical errors: the difference between the numerical solution and the true solution. This is done in two different ways:

- The node error: this is the average numerical error over the nodes in the mesh.
- Still, this is not the whole story. After all, our polynomials approximate the true solution not only at the nodes but also in between. To evaluate this, draw an oblique line in the unit cube, from corner to corner: from $(0, 0, 0)$ to $(1, 1, 1)$. This gives the line error: the average numerical error on this line.

From Tables 9.1 and 9.2, it is apparent that adaptive refinement wins! It is better than naive global refinement. Next, let us see that our polynomials of degree five win as well: they are better than naive polynomials of degree one.

Table 9.1 Numerical results with high-order (globally refined) finite elements

Level	Nodes	Tetrahedra	Degrees of freedom	Frequency	Node error	Line error
1	8	6	136	1	0.033	0.020
2	27	48	586	1	0.044	0.020
3	125	384	3322	1	0.012	0.013
1	8	6	136	$\sqrt{2}$	0.265	0.170
2	27	48	586	$\sqrt{2}$	0.141	0.099
3	125	384	3322	$\sqrt{2}$	0.061	0.083

Table 9.2 Numerical results with high-order (adaptive) finite elements

Level	Nodes	Tetrahedra	Degrees of freedom	Frequency	Node error	Line error
1	8	6	136	1	0.033	0.020
2	20	34	428	1	0.020	0.012
3	60	172	1544	1	0.018	0.008
1	8	6	136	$\sqrt{2}$	0.265	0.170
2	17	28	356	$\sqrt{2}$	0.049	0.031
3	49	128	1212	$\sqrt{2}$	0.004	0.004

Table 9.3 Numerical results with linear (adaptive) finite elements (frequency: 1)

Level	Nodes	Average node error
3	125	0.85
4	168	0.68
5	394	0.50
6	518	0.40
7	1131	0.28
8	2162	0.15
9	2352	0.08

9.7.2 Linear Adaptive Finite Elements

So far, we have seen that adaptive refinement wins: it is better than naive global refinement. Why? Because it uses more brain power: it looks at each coarse tetrahedron, thinks, and decides whether to split it or not.

Do polynomials of degree fine win as well? Are they better than naive polynomials of degree one? They certainly are! After all, we have invested a lot of brain power to design and analyze them in the first place, so we must have some gain.

Indeed, in Tables 9.3 and 9.4, one can see the results with linear polynomials of degree one, not five. Even with our best adaptive refinement, they are clearly inferior to high-order finite elements (Table 9.2). So, it was indeed worthwhile to invest time and effort, and design them in the first place.

9.8 Exercises

1. Consider a polynomial $p(x, y)$ of two independent variables, and of degree m. How many different monomials could it contain? Hint:

$$\binom{m+2}{2} = \frac{(m+2)!}{m! \cdot 2!} = \frac{(m+1)(m+2)}{2}$$

   (Chap. 8, Sect. 8.10.4).
2. Set $m = 5$. What is the concrete answer now?

Table 9.4 Numerical results with linear (adaptive) finite elements (frequency: $\sqrt{2}$)

Level	Nodes	Average node error
3	125	1.18
4	222	0.96
5	341	0.82
6	559	0.77
7	1007	0.69
8	2065	0.59
9	2983	0.49
10	3076	0.44
11	3401	0.35
12	3966	0.27
13	4271	0.20
14	4741	0.15

$$\binom{5+2}{2} = \frac{6 \cdot 7}{2} = 21.$$

3. How many independent degrees of freedom are required to specify p uniquely in the Cartesian plane? Hint: 21.
4. What could they be? Hint: the 21 coefficients in p.
5. This is a bit too explicit. Could they be more implicit? Hint: see below.
6. In the Cartesian x-y plane, draw a triangle. Look at one of its corners. At this corner, how many partial derivatives of order 0, 1, or 2 are there? Hint:

$$\binom{2+2}{2} = \frac{4!}{2! \cdot 2!} = \frac{4 \cdot 3}{2} = 6$$

(Chap. 8, Sect. 8.10.4).

7. List them one by one in a row. Hint:

$$p, \ p_x, \ p_y, \ p_{xx}, \ p_{xy}, \quad \text{and} \quad p_{yy}.$$

8. Specify them, not only in this corner but also in the two others. This gives a total of 18 degrees of freedom.
9. Now, look at the edges of the triangle. At each edge midpoint, specify a nontangential derivative of p. This gives a total of 21 degrees of freedom.
10. Are they independent of each other? Hint: yes! See Sect. 9.3.3.
11. Do they specify p uniquely? Hint: yes!
12. Consider now polynomials of three variables: $p(x, y, z)$, defined in a tetrahedron t in the Cartesian space.
13. What algebraic condition must t satisfy? Hint:

$$\det(S_t) \neq 0.$$

14. What does this mean geometrically? Hint: the corners of t are not on the same plane. As a result, t is nondegenerate: it has a positive volume.
15. Prove algebraically that the barycentric coordinates satisfy

$$\lambda_{\mathbf{i}}(\mathbf{j}) = \begin{cases} 1 & \text{if } \mathbf{i} = \mathbf{j} \\ 0 & \text{if } \mathbf{i} \neq \mathbf{j} \end{cases} \qquad \mathbf{i}, \mathbf{j} \in \{\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n}\}.$$

    Hint: in Cramer's formula, used in Sect. 9.2.3, the matrix in the numerator has duplicate columns.
16. Prove this geometrically as well. Hint: see the end of Sect. 9.2.3.
17. Show that the side midpoint basis function defined in Sect. 9.5.1 has only one nonzero degree of freedom in t.
18. Show that, once extended to the entire mesh, it makes a proper basis function: piecewise-polynomial, differentiable, and with just one nonzero degree of freedom in the entire mesh,.
19. Show that the edge-midpoint basis function defined in Sect. 9.5.2 has only one nonzero degree of freedom in t.
20. Show that, once extended to the entire mesh, it makes a proper basis function: piecewise-polynomial, continuous, and with only one nonzero degree of freedom in the entire mesh.
21. Show that the Hessian-related corner basis function defined in Sect. 9.5.3 has only one nonzero degree of freedom in t.
22. Show that, once extended to the entire mesh, it makes a proper basis function: piecewise-polynomial, continuous, and with just one nonzero degree of freedom in the entire mesh.
23. Show that the gradient-related corner basis function defined in Sect. 9.5.4 has only one nonzero degree of freedom in t.
24. Show that, once extended to the entire mesh, it makes a proper basis function: piecewise-polynomial, continuous, and with just one nonzero degree of freedom in the entire mesh.
25. Show that the corner basis function defined in Sect. 9.5.5 has only one nonzero degree of freedom in t.
26. Show that, once extended to the entire mesh, it makes a proper basis function: piecewise-polynomial, continuous, and with just one nonzero degree of freedom in the entire mesh.

Part IV
Finite Elements in 3-D

To define useful basis functions, one must first have a proper mesh. Consider a three-dimensional domain, convex or nonconvex. To approximate it well, design a mesh of disjoint (nonoverlapping) tetrahedra. In numerical analysis, these are called finite elements [9, 84].

In the mesh, we have nodes: the corners of the tetrahedra. The node is the most elementary ingredient in the mesh. An individual node may serve as a corner in a few tetrahedra.

If they belong to the same tetrahedron, then the nodes must be connected to each other by an edge. An edge is often shared by a few adjacent tetrahedra. There are two kinds of edges: a boundary edge could be shared by two tetrahedra. An inner edge, on the other hand, must be shared by more tetrahedra.

Each tetrahedron is bounded by four triangles: its sides or faces. In the mesh, there are two kinds of faces: an inner face is shared by two adjacent tetrahedra. A boundary face, on the other hand, belongs to one tetrahedron only.

How to construct the mesh? Start from a coarse mesh that approximates the domain poorly, and improve it step by step. For this purpose, refine: split coarse tetrahedra. At the same time, introduce new (small) tetrahedra next to the convex parts of the boundary to improve the approximation from the inside. This procedure may then repeat time and again iteratively, producing finer and finer meshes at higher and higher levels.

This makes a multilevel hierarchy of finer and finer meshes, approximating the original domain better and better. In the end, at the top level, those tetrahedra that exceed the domain may drop from the final mesh. This completes the automatic algorithm to approximate the original domain well.

The mesh should be as regular as possible: the tetrahedra should be thick and nondegenerate. Furthermore, the mesh should be as convex as possible. Only at the top level may the mesh become concave again. A few tricks are introduced to have these properties.

To verify accuracy, numerical integration is then carried out on the fine mesh. For this purpose, we use a simple example, for which the analytic integral is well-known in advance. The numerical integral is then subtracted from the analytic integral. This

is the error: it turns out to be very small in magnitude. Furthermore, our regularity estimates show that the mesh is rather regular, as required.

Once the basis functions are well-defined in the fine mesh, they can be used to approximate a given function, defined in the original domain. This is indeed the spline problem [5, 13, 19, 33, 47, 60, 71, 96, 97]: design a (rather smooth) piecewise-polynomial function to "tie" (or match) the original values of the function at the mesh nodes.

The spline problem could also be formulated as follows: consider a discrete grid function, defined at the mesh nodes only. Extend it into a complete spline: a (rather smooth) piecewise-polynomial function, defined not only at the nodes but also in between. The solution must be optimal in terms of minimum "energy." This is indeed the smoothest solution possible.

Our (regular) finite-element mesh could be used not only in the spline problem but also in many other practical problem. Later on, we will see interesting applications in modern physics and chemistry.

Chapter 10
Automatic Mesh Generation



Consider a complicated domain in three spatial dimensions. How to store it on the computer? For this purpose, it must be discretized: approximated by a discrete mesh, ready to be used in practical algorithms.

To approximate the domain well, best use a mesh of tetrahedra. This way, the tetrahedra may take different shapes and sizes. Next to the curved boundary, many small tetrahedra should be used, to approximate the boundary well. Next to the flat part of the boundary, on the other hand, a few big tetrahedra may be sufficient. This is indeed local refinement: small tetrahedra should be used only where absolutely necessary.

The automatic refinement algorithm starts from a coarse mesh that approximates the domain rather poorly. Then, this mesh refines time and again, producing finer and finer meshes that approximate the domain better and better. Indeed, in a finer mesh, small tetrahedra may be added next to the curved boundary, to approximate it better from the inside. This produces a multilevel hierarchy of more and more accurate meshes.

At the intermediate levels, the mesh is often still convex. This is good enough for a convex domain, but not for a more complicated, nonconvex domain. Fortunately, at the top level, this gets fixed: those tetrahedra that exceed the domain too much drop. This way, the finest mesh at the top level gets nonconvex, and ready to approximate the original nonconvex domain.

10.1 The Refinement Step

10.1.1 Iterative Multilevel Refinement

What is multilevel refinement? It uses a hierarchy of finer and finer meshes, to approximate the original domain better and better [51, 56, 75].

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At the bottom level, one may place a rather poor mesh, containing only a few big tetrahedra. Do not worry: the initial coarse mesh will soon refine and improve. This is indeed the refinement step, producing the next finer mesh in the next higher level.

How does the refinement step work? Well, each coarse tetrahedron splits into two subtetrahedra. How is this done? Well, a coarse edge is divided into two subedges. For this purpose, its original midpoint is connected to those two corners that lie across from it. This produces two new subtetrahedra. In the finer mesh, they are going to replace the original coarse tetrahedron, providing a better resolution.

10.1.2 Conformity

What happens to those neighbor tetrahedra that share the same coarse edge? To preserve conformity, they must split in the same way as well. This way, the above midpoint is connected to those corners that lie across from it not only in the original tetrahedron but also in each adjacent (edge-sharing) tetrahedron.

So far, we have been busy splitting tetrahedra. This was done in two possible ways. A coarse tetrahedron may split out of its own initiative, to refine. Still, this is not the only way: a tetrahedron may also be forced to split not to refine but only to preserve conformity and fit to a neighbor tetrahedron that has already refined.

Still, the refinement step may not only split existing tetrahedra but also introduce new ones, to improve the approximation next to the convex part of the boundary. This is called boundary refinement. This completes the refinement step, producing the next finer mesh at the next higher level. This mesh is now ready for the next refinement step, and so on.

The boundary of the original domain may contain two parts: the convex part and the concave (or nonconvex) part. The refinement step distinguishes between the two. At the convex part, a few new (small) tetrahedra are introduced, to approximate the curved boundary better from the inside. At the concave part, on the other hand, no new tetrahedron is introduced. On the contrary: those tetrahedra that exceed the domain too much even drop in the end. This way, the finest mesh at the top level will get accurate again, even for a complicated nonconvex domain.

10.1.3 Regular Mesh

What is a regular mesh? Well, in a regular mesh, the tetrahedra are thick and nondegenerate [7, 45]. As discussed in Chap. 9, Sect. 9.1.1, this is a most important property.

Fortunately, at the bottom level, one can often pick a rather regular initial mesh. How to make sure that the finer mesh in the next higher level remains regular as well?

10.1.4 How to Preserve Regularity?

Well, here is the trick. Just before the refinement step, order the coarse tetrahedra one by one by maximal edge: a tetrahedron with a longer maximal edge before a tetrahedron with a shorter maximal edge. In their new order, scan the tetrahedra one by one: refine a tetrahedron only if it is indeed a coarse tetrahedron. Otherwise, leave it. This way, each coarse tetrahedron refines only once: only its maximal edge splits into two subedges. This way, the original coarse tetrahedron splits into two new subtetrahedra, which can no longer split any more in this refinement step.

Or can't they? Well, there may still be one exception. Once a coarse tetrahedron splits (at its maximal edge, as above), all those neighbor tetrahedra that share this edge must split as well, to preserve conformity (Sect. 10.1.2). In such a neighbor tetrahedron, however, this edge is not necessarily maximal: it could be submaximal.

Could it really? After all, if the neighbor tetrahedron contained a longer edge, then it would have already been listed before and would have already split long ago. Thus, it cannot be a coarse tetrahedron: it could only be a subtetrahedron, obtained from an earlier split.

This is not so good: splitting a subtetrahedron at its submaximal edge may produce a rather thin (irregular) subtetrahedron. As a result, regularity may decrease a little. Fortunately, this will soon get fixed: just before the next refinement step, the tetrahedra are going to be reordered once again in terms of maximal edge, so this problem should probably get fixed in the next refinement step, increasing regularity again.

10.2 Approximating a 3-D Domain

10.2.1 Implicit Domain

The original three-dimensional domain $\Omega \subset \mathbb{R}^3$ may be rather complicated: its boundary $\partial\Omega$ may be curved, nonstandard, and irregular. In fact, $\partial\Omega$ may be available only implicitly, in terms of a given real function $F(x, y, z)$:

$$\partial\Omega \equiv \left\{(x, y, z) \in \mathbb{R}^3 \;\mid\; F(x, y, z) = 0\right\}.$$

This defines $\partial\Omega$ implicitly as the zero level set of F: the set of points at which F vanishes.

Assume, for example, that we want $\partial\Omega$ to be a sphere centered at $(1/2, 1/2, 1/2)$. Assume also that we want this sphere to confine the unit cube:

$$[0, 1]^3 \subset \Omega.$$

In this case, F could be defined as

$$F(x, y, z) \equiv \left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 + \left(z - \frac{1}{2}\right)^2 - \frac{3}{4}.$$

Indeed, with this definition, $\partial\Omega$ is a sphere of radius $\sqrt{3}/2$ around $(1/2, 1/2, 1/2)$. In fact, F is negative inside the sphere, positive outside it, and zero on the sphere itself.

In practice, however, F is rarely available in its closed analytic form. More often, it is only available as a computer function: for every given x, y, and z, $F(x, y, z)$ can be calculated on the computer. Let us see a more interesting example.

10.2.2 Example: A Nonconvex Domain

In the above example, Ω is convex: the interior of the sphere that confines the unit cube. Here, on the other hand, we consider a more complicated example, in which Ω is no longer convex.

Let $\mathbb{R}^+$ be the nonnegative part of the real axis:

$$\mathbb{R}^+ \equiv \{x \in \mathbb{R} \mid x \geq 0\} \subset \mathbb{R}.$$

Assume that Ω lies in the nonnegative "octant" of the three-dimensional Cartesian space:

$$\Omega \subset \left(\mathbb{R}^+\right)^3 \equiv \left\{(x, y, z) \in \mathbb{R}^3 \mid x \geq 0,\ y \geq 0,\ z \geq 0\right\}.$$

This way, every point in Ω must have three nonnegative coordinates: $x \geq 0$, $y \geq 0$, and $z \geq 0$.

Still, we are not done yet: we want Ω to be much smaller than $(\mathbb{R}^+)^3$. For this purpose, let R_1 and R_2 be some positive parameters, to be specified later ($0 < R_1 < R_2$). Assume also that Ω lies in the sphere of radius R_2, centered at $(0, 0, 0)$. This way, every point in Ω must have a magnitude of R_2 or less.

Finally, assume also that Ω lies outside of the sphere of radius R_1. This way, every point in Ω must have a magnitude of R_1 or more.

In summary, Ω contains those points with magnitude between R_1 and R_2, and with three nonnegative coordinates:

$$\Omega \equiv \left\{(x, y, z) \in \mathbb{R}^3 \mid x \geq 0,\ y \geq 0,\ z \geq 0,\ R_1^2 \leq x^2 + y^2 + z^2 \leq R_2^2\right\}.$$

This defines Ω in a closed analytic form: a nonconvex domain.

It would be more interesting, though, to pretend that we do not have this closed form available. After all, this is usually the case in practice. Thus, we should better train in using F, not Ω. For this purpose, how to define F?

Well, from the original definition, the origin is not in Ω. Thus, one could issue a ray from the origin toward Ω. Clearly, the ray meets $\partial\Omega$ at two points: it enters Ω through a point of magnitude R_1 and leaves through a point of magnitude R_2. On the ray, it makes sense to define F as a parabola that vanishes at these two points and has its unique minimum in between. This would indeed define F in $(\mathbb{R}^+)^3$.

So far, we have "defined" F in $(\mathbb{R}^+)^3$ only. Still, this is not the end of it. After all, F must be defined outside Ω as well. In fact, it must be positive there: it must increase monotonically away from Ω.

To extend F to the rest of the Cartesian space, one must consider negative coordinates as well. Each negative coordinate should contribute its absolute value to F, to increase the value of F linearly as the point leaves $(R^+)^3$. This way, F indeed increases monotonically away from Ω, as required.

So, we only need to specify the above parabolas explicitly. This will indeed make one factor in F: $a \equiv a(x, y, z)$. This way, both a and F will indeed vanish on the round parts of $\partial\Omega$, where the magnitude is either R_1 or R_2. Still, this is not good enough. After all, F must vanish on the flat sides of Ω as well. For this purpose, F must contain yet another factor: $b \equiv b(x, y, z)$. So, we can already view F as a product of two functions: $F = ab$.

How should b look like? Well, each nonzero three-dimensional vector $(x, y, z)^t$ makes three cosines with the x-, y-, and z-axes:

$$\frac{x}{\sqrt{x^2+y^2+z^2}}, \quad \frac{y}{\sqrt{x^2+y^2+z^2}}, \quad \text{and} \quad \frac{z}{\sqrt{x^2+y^2+z^2}}$$

(Chap. 2, Sect. 2.3.3). Consider the minimal cosine (in absolute value):

$$b \equiv b(x, y, z) \equiv \frac{\min(|x|, |y|, |z|)}{\sqrt{x^2+y^2+z^2}}.$$

This way, b is positive throughout $(\mathbb{R}^+)^3$, except at the planes $x \equiv 0$, $y \equiv 0$, and $z \equiv 0$, where it vanishes. Thus, these planes are contained in the zero level set of b:

$$\begin{aligned}\partial\left((\mathbb{R}^+)^3\right) &= \left\{(x, y, z) \in \mathbb{R}^3 \;\mid\; x \geq 0,\ y \geq 0,\ z \geq 0,\ xyz = 0\right\} \\ &\subset \left\{(x, y, z) \in \mathbb{R}^3 \;\mid\; xyz = 0\right\} \\ &= \left\{(x, y, z) \in \mathbb{R}^3 \;\mid\; b(x, y, z) = 0\right\} \cup \{(0, 0, 0)\}.\end{aligned}$$

In particular, the zero level set of b contains the flat sides of Ω, as required.

Let us now go ahead and define a explicitly as well:

$$a \equiv a(x, y, z) \equiv \left(R_1 - \sqrt{x^2+y^2+z^2}\right)\left(R_2 - \sqrt{x^2+y^2+z^2}\right).$$

This way, in each ray issuing from the origin toward Ω, a makes a parabola, as required. Each parabola vanishes at two points only: the point of magnitude R_1 and the point of magnitude R_2.

Fortunately, b is constant in the ray, so the product ab still makes a parabola in the ray, as required. Moreover, ab still has a unique minimum in between these points. For this reason, ab must also have a unique minimum in the entire domain Ω. As a matter of fact, this minimum is obtained at

$$\frac{1}{\sqrt{3}} \cdot \frac{R_1 + R_2}{2}(1, 1, 1).$$

We are now ready to define F in the entire Cartesian space:

$$F(x, y, z) \equiv \begin{cases} ab & \text{if } (x, y, z) \in \Omega \\ |a| & \text{if } (x, y, z) \in \left(\mathbb{R}^+\right)^3 \setminus \Omega \\ F(\max(x, 0), \max(y, 0), \max(z, 0)) & \\ \quad - \min(x, 0) - \min(y, 0) - \min(z, 0) & \text{if } (x, y, z) \notin \left(\mathbb{R}^+\right)^3. \end{cases}$$

This way, outside $(\mathbb{R}^+)^3$, F is defined recursively from its value at the nearest point in $(\mathbb{R}^+)^3$. With this definition, F indeed increases linearly as either x or y or z gets more and more negative.

This completes the definition of F. Indeed, F is negative in the interior of Ω, zero on the boundary $\partial\Omega$, and positive outside of Ω. Furthermore, F increases monotonically away from Ω, as required.

We thus have a good example to model a realistic case. In fact, we can just pretend that Ω has never been disclosed to us explicitly, but only implicitly, in terms of F. As a matter of fact, we can even pretend that F is available not analytically but only computationally: given x, y, and z, we have a computer program to calculate $F(x, y, z)$ for us. As we will see below, this is enough to design a proper mesh to approximate Ω.

10.2.3 How to Find a Boundary Point?

As discussed above, to model a realistic case, we pretend that $\partial\Omega$ is not available explicitly. What is available is the computer function F: for every given x, y, and z, $F(x, y, z)$ can be calculated on the computer. Thus, $\partial\Omega$ is only available implicitly, as the zero level set of F:

$$(x, y, z) \in \partial\Omega \quad \text{if} \quad F(x, y, z) = 0.$$

Fortunately, this is good enough to find a new boundary point on $\partial\Omega$. How to do this? Well, for this purpose, one must find some concrete coordinates x, y, and z,

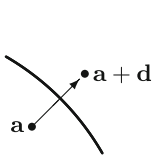


Fig. 10.1 The good case: the arrow leading from $\mathbf{a}$ to $\mathbf{a}+\mathbf{d}$ indeed contains a boundary point in $\partial\Omega$. In this case, F indeed changes sign over the arrow: $F(\mathbf{a}) < 0 < F(\mathbf{a}+\mathbf{d})$ or $F(\mathbf{a}) > 0 > F(\mathbf{a}+\mathbf{d})$

for which

$$F(x, y, z) = 0.$$

Let $\mathbf{a} \in \mathbb{R}^3$ be some initial point. Furthermore, let $\mathbf{d} \in \mathbb{R}^3$ be some nonzero direction vector. The task is to find a boundary point of the form $\mathbf{a} + \alpha\mathbf{d}$, for some unknown (nonnegative) scalar α.

For this purpose, consider the arrow leading from $\mathbf{a}$ to $\mathbf{a}+\mathbf{d}$ (Fig. 10.1). If F changes sign over the arrow:

$$F(\mathbf{a})F(\mathbf{a}+\mathbf{d}) < 0,$$

then the required boundary point lies in between $\mathbf{a}$ and $\mathbf{a}+\mathbf{d}$ and can be found by iterative bisection: split the original arrow into two subarrows. If F changes sign over the first subarrow:

$$F(\mathbf{a})F\left(\mathbf{a}+\frac{\mathbf{d}}{2}\right) < 0,$$

then the boundary point must lie on the first subarrow, in between $\mathbf{a}$ and $\mathbf{a}+\mathbf{d}/2$. In this case, the first subarrow is picked to substitute the original arrow:

$$\mathbf{d} \leftarrow \frac{\mathbf{d}}{2}.$$

If, on the other hand, F changes sign over the second subarrow:

$$F\left(\mathbf{a}+\frac{\mathbf{d}}{2}\right)F(\mathbf{a}+\mathbf{d}) < 0,$$

then the boundary point must lie on the second subarrow, in between $\mathbf{a}+\mathbf{d}/2$ and $\mathbf{a}+\mathbf{d}$. In this case, the second subarrow is picked:

$$\mathbf{a} \leftarrow \mathbf{a}+\frac{\mathbf{d}}{2} \text{ and } \mathbf{d} \leftarrow \frac{\mathbf{d}}{2}.$$

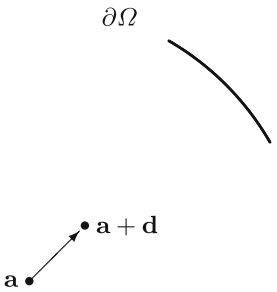


Fig. 10.2 The bad case: the arrow leading from $\mathbf{a}$ to $\mathbf{a}+\mathbf{d}$ is too short, so the boundary $\partial\Omega$ remains ahead of it. The arrow must first shift or stretch forward, until its head passes $\partial\Omega$, as in the previous figure

This procedure repeats time and again, until $\mathbf{d}$ gets sufficiently short, so the required boundary point is found with a sufficient accuracy.

Unfortunately, the situation is not always so benign. Some preparation work may be needed before iterative bisection can start. To study such a case, let us return to the original $\mathbf{a}$ and $\mathbf{d}$.

In Fig. 10.2, for example, the original arrow is too short. It must first shift (or stretch) ahead, until its head passes $\partial\Omega$. Only then can iterative bisection start, as above.

10.3 Approximating a Convex Boundary

10.3.1 Boundary Refinement

Thanks to the above method, we can now find a new boundary point. This can now be used in the refinement step. This way, the mesh refines not only in the interior of the domain but also next to the curved boundary. For this purpose, a few new (small) tetrahedra are introduced next to the convex part of the boundary, to improve the approximation there from the inside.

For instance, consider a boundary edge that lies next to the convex part of the boundary, with both its endpoints on the boundary. Such an edge must then be shared by two boundary triangles that lie next to the boundary as well, with all vertices on the boundary. In fact, each boundary triangle may serve as the face in one tetrahedron only.

Since the boundary is locally convex there, the boundary edge (and the boundary triangles) should lie (mostly) inside the domain. In the refinement step, the coarse boundary edge may split. In this case, a normal vector issues from its midpoint toward the boundary. This normal vector may make a good direction vector that points toward a new boundary point, as in Sect. 10.2.3.

This new boundary point is then connected to five points: three on the original boundary edge (two endpoints and one midpoint), and two off it (one vertex in each boundary triangle). This adds four new tetrahedra, to improve the approximation next to the locally convex boundary from the inside.

10.3.2 Boundary Edge and Triangle

What is a boundary edge? So far, we have assumed that a boundary edge must have both endpoints on $\partial\Omega$. But what happens when Ω is nonconvex? In this case, the initial coarse mesh might be quite different from Ω: it might exceed it quite a bit (Fig. 10.3).

In this case, a standard boundary edge (with both endpoints on $\partial\Omega$) may not do: its midpoint may lie outside Ω. After all, Ω may be locally concave there.

In this case, once this edge splits into two subedges, there is no boundary refinement. But what about the subedge, split in the next refinement step? Well, it is no longer a boundary edge: it has just one endpoint on $\partial\Omega$, and one endpoint off $\partial\Omega$. For this reason, in the next refinement step too, the subedge may split, but with no boundary refinement. This is indeed unfortunate: the domain will never be approximated accurately there.

To fix this, we must redefine a boundary edge in terms of the current mesh M, not the original domain Ω. A boundary edge must have both endpoints on ∂M, not necessarily on $\partial\Omega$.

How could this help? Well, consider the above situation once again (Fig. 10.4). In the first refinement step, there is no change: the original boundary edge is so coarse that its midpoint lies outside the nonconvex domain, so there is no boundary

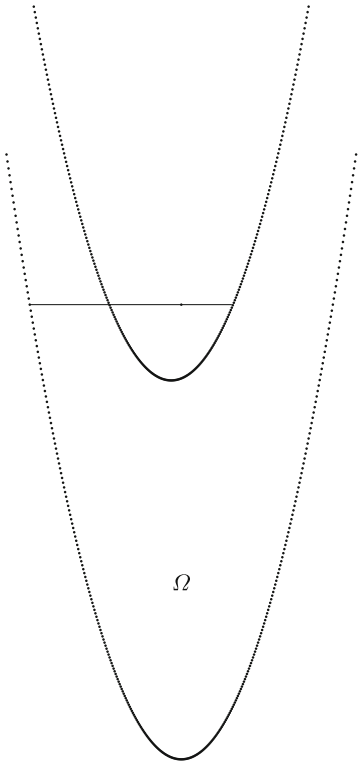


Fig. 10.3 A boundary edge with both its endpoints on $\partial\Omega$. Unfortunately, its midpoint may still lie outside the nonconvex domain, so no boundary refinement is carried out there. Furthermore, the subedge is no longer a boundary edge: it has an endpoint off $\partial\Omega$. Therefore, no boundary refinement will take place ever

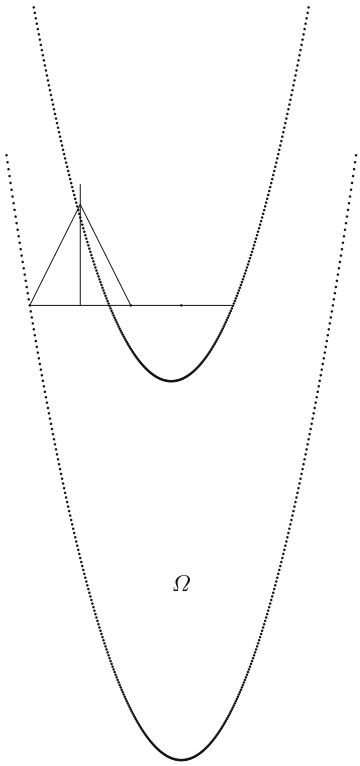


Fig. 10.4 A boundary edge of M, although not of Ω. In this sense, the left subedge is a legitimate boundary edge, with both endpoints on ∂M, although not on $\partial\Omega$. Its own midpoint lies well inside Ω, so boundary refinement will indeed take place there in the next refinement step. Although the mesh gets slightly nonconvex, this should produce no overlapping tetrahedra

refinement there. Fortunately, its subedge is a boundary edge of M, although not of Ω. As such, since its own midpoint lies well inside Ω, it may indeed split there in the next refinement step, this time with boundary refinement, as required.

This way, the fine mesh will no longer be convex. Later on, we will see that this may be risky: in yet finer meshes, tetrahedra may overlap, making a complete mess. Fortunately, here M is nonconvex outside the domain only, so there should be no risk.

Thus, from now on, a boundary edge will mean a boundary edge of M, not Ω, and a boundary triangle will mean a boundary triangle of M as well. For this purpose, we will define a new mechanism to detect such an edge, based on M only, and independent of Ω or F.

10.3.3 How to Fill a Valley?

In Sect. 10.1.4, we have seen that only the maximal edge should split. Or should it? Well, sometimes a submaximal edge should split instead, to make the fine mesh more convex. After all, in Fig. 10.4, we only see a two-dimensional projection.

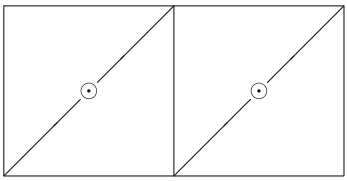


Fig. 10.5 The coarse mesh: a view from above. In the refinement step, the oblique edge splits, and a normal vector issues from its midpoint toward your eyes, to hit the boundary above it

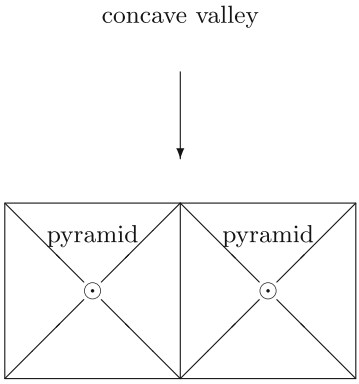


Fig. 10.6 The next finer mesh: two pyramids, with a concave valley in between

In reality, on the other hand, the three-dimensional mesh may suffer from a local concavity: a "valley."

Consider, for example, the coarse mesh M, as viewed from above (Fig. 10.5). Assume that M is flat from above, so this is indeed a real view. Which is the maximal edge? Clearly, the oblique one. This is indeed the edge that should split in the refinement step.

Assume that $\partial\Omega$ lies above M, closer to your eyes. In this case, there is also a boundary refinement: from the edge midpoint, a normal vector issues toward your eyes, to meet the boundary above it, next to your eye. This is indicated by the "$\odot$" in Fig. 10.5. This forms a little pyramid, made of four new tetrahedra, to approximate the boundary above it better.

Unfortunately, this also produces a concave valley in between the new pyramids (Fig. 10.6). How to fill it? Well, in the next refinement step, in each pyramid, split not the maximal edge but rather the vertical edge along the valley, in between the pyramids (Fig. 10.7). This way, a normal vector will issue from the middle of the valley toward your eyes, to help fill the valley with four new tetrahedra, next to the boundary above it.

Thus, the original order in Sect. 10.1.4 must change. First, list those tetrahedra near a valley: those near a "deep" valley before others. This way, a tetrahedron near a deep valley splits earlier, as required, and the valley gets filled sooner, even though the edge along it is submaximal. Later on, we will state more clearly what "deep" means.

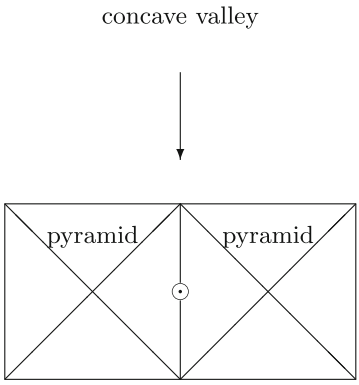


Fig. 10.7 The next refinement step: the vertical edge along the valley, although submaximal, splits, and a normal vector issues from its midpoint "⊙" toward your eyes, to hit the boundary above it, and fill the valley with four new tetrahedra

After these, list those tetrahedra that are near no valley. These are ordered as before: those with a long edge before others. After all, they should refine at their maximal edge, as before.

Finally, list those tetrahedra that exceed the (nonconvex) domain too much, regardless of the length of their edges. After all, they should have actually been dropped, so they have little business to refine. Later on, we will state more clearly what "too much" mean.

10.3.4 How to Find a Boundary Edge?

In Sect. 10.3.2, we have explained why we are interested in a boundary edge of M, not of Ω. How to find such an edge? More precisely, given an edge, how to check whether it is indeed a boundary edge, and find the boundary triangles that share it? After all, the endpoints of the edge must now lie on ∂M, which is not available numerically!

Well, for this purpose, let $t_1 \subset M$ be some tetrahedron in the mesh, and let $e \subset t_1$ be an edge in it. Then, e is shared by two sides $s_1, s_2 \subset t_1$:

$$e = s_1 \cap s_2.$$

To check whether e is a boundary edge or not, let us check whether it belongs to any boundary triangle. For this purpose, let us search for a neighbor tetrahedron t_2 that shares s_2 as its joint face:

$$s_2 = t_1 \cap t_2.$$

Now, t_2 must have another face, s_3, that also shares e as its joint edge:

$$e = s_1 \cap s_2 \cap s_3.$$

We can now apply the same procedure iteratively to t_2 rather than t_1, to find yet another triangle, s_4 that shares e as well, and so on. This produces a list of triangles

$$s_1, s_2, s_3, \ldots, s_n$$

that share e as their joint edge:

$$e = s_1 \cap s_2 \cap s_3 \cap \cdots \cap s_n = \cap_{i=1}^{n} s_i .$$

Now, if

$$s_n = s_1,$$

then e cannot be a boundary edge: it is surrounded by tetrahedra from all directions, so it must be away from ∂M. If, on the other hand,

$$s_n \neq s_1,$$

then s_n must be a boundary triangle, and e must indeed be a boundary edge.

In this case, how to find the second boundary triangle? Apply the same procedure once again, only this time interchange the roles of s_1 and s_2.

10.3.5 Locally Convex Boundary: Gram–Schmidt Process

Let

$$t \equiv (\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n}) \subset M$$

be some tetrahedron in the mesh, vertexed at corners **k**, **l**, **m**, and **n** (Fig. 9.1). Let

$$e \equiv (\mathbf{m}, \mathbf{n}) \subset t$$

be an edge in t. Assume that, in the refinement step, e should split at its midpoint:

$$\mathbf{a} \equiv \frac{\mathbf{m} + \mathbf{n}}{2}.$$

We must then know: is e a boundary edge? After all, if it is, then boundary refinement may be required at **a** (Sect. 10.3.1). Fortunately, we already know how to find out the answer (Sect. 10.3.4).

Thus, assume that e is indeed a boundary edge:

$$e \subset \partial M.$$

In this case, we also get a bonus: two boundary triangles:

$$\triangle(\mathbf{m}, \mathbf{n}, \mathbf{u}) \text{ and } \triangle(\mathbf{m}, \mathbf{n}, \mathbf{v}) \subset \partial\Omega.$$

These will be useful below.

Furthermore, assume also that the domain is locally convex at **a**:

$$F(\mathbf{a}) < 0, \text{ so } \mathbf{a} \in \Omega \setminus \partial\Omega.$$

In this case, it makes sense to fill the gap between e and $\partial\Omega$ with four new tetrahedra. For this purpose, issue a normal vector **d** from **a** toward $\partial\Omega$, to hit $\partial\Omega$ at the new boundary point $\mathbf{w} \in \partial\Omega$ (Sect. 10.2.3). Then, connect **w** to five points: **m**, **n**, **a**, **u**, and **v** (Fig. 10.8). This indeed produces four new tetrahedra, to approximate the boundary better at **a** from the inside, as required.

Now, how to define the direction vector **d**? Well, it should point in between **u** and **v**, toward $\mathbf{w} \in \partial\Omega$. For this purpose, let

$$\mathbf{e} \equiv \frac{\mathbf{n} - \mathbf{m}}{\|\mathbf{n} - \mathbf{m}\|_2}$$

be the unit vector parallel to e. Furthermore, define the differences

$$\mathbf{o} \equiv \mathbf{u} - \mathbf{a} \text{ and } \mathbf{p} \equiv \mathbf{v} - \mathbf{a}.$$

Now, project both **o** and **p** onto the plane perpendicular to e:

$$\mathbf{o} \leftarrow \mathbf{o} - (\mathbf{o}, \mathbf{e})\mathbf{e}$$

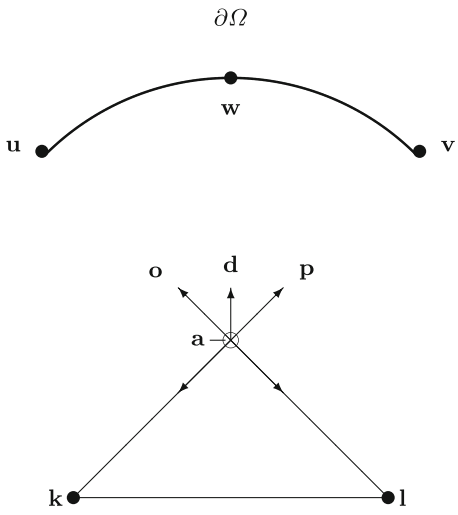


Fig. 10.8 Projection onto the plane perpendicular to the boundary edge $e \equiv (\mathbf{m}, \mathbf{n})$. The direction vector **d** points from $\mathbf{a} \equiv (\mathbf{m} + \mathbf{n})/2$ toward $\partial\Omega$, as required

$$\mathbf{p} \leftarrow \mathbf{p} - (\mathbf{p}, \mathbf{e})\mathbf{e}.$$

Indeed, after these substitutions,

$$(\mathbf{o}, \mathbf{e}) = (\mathbf{p}, \mathbf{e}) = 0.$$

This is actually a Gram–Schmidt process (Chap. 2, Sect. 2.3.2). Next, normalize both $\mathbf{o}$ and $\mathbf{p}$:

$$\mathbf{o} \leftarrow \frac{\mathbf{o}}{\|\mathbf{o}\|_2}$$
$$\mathbf{p} \leftarrow \frac{\mathbf{p}}{\|\mathbf{p}\|_2}.$$

This produces the picture in Fig. 10.8. Now, let $\mathbf{d}$ be the vector product

$$\mathbf{d} \equiv (\mathbf{p} - \mathbf{o}) \times \mathbf{e}.$$

Next, normalize $\mathbf{d}$ as well:

$$\mathbf{d} \leftarrow \frac{\mathbf{d}}{\|\mathbf{d}\|_2}.$$

Is $\mathbf{d}$ a good direction vector? Well, in Fig. 10.8, assume that $\mathbf{e}$ points into the page, away from you. Thanks to the right-hand rule, $\mathbf{d}$ indeed points away from M, as required (See exercises at the end of Chap. 2).

But what if the orientation is the other way around, and

$$\det((\mathbf{k} - \mathbf{a} \mid \mathbf{e} \mid \mathbf{l} - \mathbf{a})) < 0?$$

In this case, $\mathbf{d}$ must reverse:

$$\mathbf{d} \leftarrow -\mathbf{d}.$$

(See exercises at the end of Chap. 2.)

By now, in either case, $\mathbf{d}$ is a good direction vector: it indeed points away from M, toward $\partial\Omega$, as required.

We can now also tell what a "deep" valley is: if

$$(\mathbf{o} + \mathbf{p}, \mathbf{d})$$

is large, then the valley along e is indeed deep. In this case, t should be listed early, and split early at $\mathbf{a}$, filling the valley with four new tetrahedra:

$$(\mathbf{a}, \mathbf{w}, \mathbf{m}, \mathbf{u}),\ (\mathbf{a}, \mathbf{w}, \mathbf{m}, \mathbf{v}),\ (\mathbf{a}, \mathbf{w}, \mathbf{n}, \mathbf{u}),\quad \text{and}\ (\mathbf{a}, \mathbf{w}, \mathbf{n}, \mathbf{v}),$$

as required. This helps approximate Ω better from the inside.

Still, there is a condition. To use these four new tetrahedra, $\mathbf{w}$ must be away from $\mathbf{a}$:

$$\|\mathbf{w} - \mathbf{a}\|_2 \geq 10^{-2}\|\mathbf{n} - \mathbf{m}\|_2.$$

Otherwise, these new tetrahedra are too thin and degenerate and should better be left out, and not added to M.

10.4 Approximating a Nonconvex Domain

10.4.1 Locally Concave Boundary

Still, there is one more problem. So far, the domain has been approximated from the inside, at its (locally) convex part only. At its locally concave (nonconvex) part, on the other hand, no boundary refinement has been carried out. After all, the midpoint of the boundary edge often lies outside the domain. So, the approximation is still poor.

How to fix this? Well, a few strategies have been tested. One method is as follows. Recall that, in the refinement step, the original coarse tetrahedron

$$t \equiv (\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n})$$

is replaced by two subtetrahedra:

$$(\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{a}) \quad \text{and} \quad (\mathbf{k}, \mathbf{l}, \mathbf{a}, \mathbf{n}),$$

where

$$\mathbf{a} \equiv \frac{\mathbf{m} + \mathbf{n}}{2}$$

is the edge midpoint. Here, on the other hand, since $\mathbf{a} \notin \Omega$, it might make sense to replace it by the nearest point $\mathbf{c} \in \partial\Omega$. This way, in the refinement step, t is replaced by two new fine tetrahedra:

$$(\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{c}) \quad \text{and} \quad (\mathbf{k}, \mathbf{l}, \mathbf{c}, \mathbf{n}).$$

Is this a good fix? Unfortunately not: it might produce overlapping tetrahedra in the next refinement step, and a complete mess.

A better approach is as follows. Wait until the entire multilevel hierarchy is complete. Then, fix the top level only: from the finest mesh, drop those tetrahedra that exceed Ω too much in the sense of having 3–4 corners outside Ω. A tetrahedron

with only 1–2 corners outside Ω, on the other hand, must not drop. This way, at the top level, Ω gets approximated at its concave part as well: not from the inside, but from the outside.

10.4.2 Convex Meshes

Why fix only the top level, not the intermediate ones? Well, dropping tetrahedra from an intermediate mesh might spoil the special structure required to carry out the next refinement step.

In fact, even in a nonconvex domain, the mesh should better remain as convex as possible for as long as possible. Dropping a tetrahedron, on the other hand, may produce a "hole" in the mesh, making it highly nonconvex too early.

What would then happen in the next refinement step? Well, consider a boundary edge in such a hole. Recall that this is a boundary edge of the mesh, not necessarily of the domain (Sect. 10.3.2). The normal vector issuing from its midpoint (Sect. 10.3.5) may then hit the other bank of the hole, producing overlapping tetrahedra, and a complete mess.

This is why no tetrahedron should drop from any intermediate mesh. This way, in most of the multilevel hierarchy, the meshes remain as convex as possible. Only from the finest mesh may some tetrahedra drop, leaving it as nonconvex as the original domain, as required.

10.5 Exercises

1. Consider the closed unit cube
$$\Omega \equiv [0,1]^3 \equiv \left\{(x,y,z) \in \mathbb{R}^3 \;\middle|\; 0 \le x,y,z \le 1\right\}.$$
2. Show that Ω is convex.
3. Define the function
$$F(x,y,z) \equiv \max\left(\left|x-\frac{1}{2}\right|, \left|y-\frac{1}{2}\right|, \left|z-\frac{1}{2}\right|\right) - \frac{1}{2}.$$
4. Show that F is negative in the interior of Ω, zero on its boundary, positive outside it, and monotonically increasing away from it.
5. Conclude that $\partial\Omega$ is indeed the zero level set of F, as required.
6. Show that F is monotonically increasing on each ray issuing from the middle of Ω at $(1/2, 1/2, 1/2)$.
7. Write the unit cube as the union of six disjoint tetrahedra.
8. Make sure that this mesh is conformal.

9. Apply a refinement step to this mesh.
10. Make sure that the fine mesh is conformal as well.
11. Assume now that Ω is the interior of the sphere that confines the unit cube.
12. Show that Ω is convex.
13. Define F as in Sect. 10.2.1.
14. Show that F is negative in the interior of Ω, zero on its boundary, positive outside it, and monotonically increasing away from it.
15. Conclude that $\partial\Omega$ is indeed the zero level set of F, as required.
16. Show that F is monotonically increasing on each ray issuing from the middle of Ω at $(1/2, 1/2, 1/2)$.
17. Now, let Ω be the nonconvex domain in Sect. 10.2.2. Show that Ω is indeed the intersection of the ball of radius R_2, the outside of the ball of radius R_1, and $(\mathbb{R}^+)^3$.
18. Show that Ω is indeed nonconvex.
19. What is the convex part of the boundary of Ω?
20. What is the concave (nonconvex) part of the boundary of Ω?
21. What is the flat part of the boundary of Ω?
22. Show that the function F defined in Sect. 10.2.2 is indeed negative in the interior of Ω, zero on $\partial\Omega$, positive outside of Ω, and monotonically increasing away from Ω.
23. Conclude that $\partial\Omega$ is indeed the zero level set of F, as required.
24. Show that F has a unique minimum on each ray issuing from the origin toward Ω.
25. Show that F increases monotonically on each ray issuing from $\partial\Omega$ away from Ω.
26. Show that F has a unique minimum in Ω.
27. Find this minimum explicitly.
28. Consider now some mesh of tetrahedra. Show that a boundary triangle serves as a face in exactly one tetrahedron.
29. Show that a boundary edge is shared by exactly two faces: the boundary triangles.
30. Show that, if these boundary triangles serve as faces in one and the same tetrahedron, then this is the only tetrahedron that uses the above boundary edge.
31. Show that the algorithm in Sect. 10.3.4 indeed tells whether a given edge is a boundary edge or not.
32. Show that, if this is indeed a boundary edge, then this algorithm also finds the boundary triangles that share it.
33. In this case, how can the second boundary triangle be found as well?
34. Show that the direction vector $\mathbf{d}$ defined in Sect. 10.3.5 (in its final form) indeed points from the midpoint $\mathbf{a}$ toward $\partial\Omega$, in between the boundary triangles.
35. Show that the four new tetrahedra that are added to the mesh at the end of Sect. 10.3.5 indeed improve the approximation at the convex part of the boundary from the inside.
36. Show that the refinement step preserves conformity.

37. Show that the dropping technique in Sect. 10.4.1 indeed improves the approximation at the concave part of the boundary from the outside.
38. Why should this dropping technique be applied to the finest mesh only?
39. Use the present mesh to linearize and assemble the Maxwell system. The solution can be found in Chapter 21 in [77].

Chapter 11
Mesh Regularity



In Chap. 10, Sects. 10.1.3–10.1.4, we have already met the important concept of mesh regularity and took it into account in the refinement step. Here, we continue to discuss it and introduce a few reliable tests to estimate it. This way, we can make sure that our multilevel refinement is indeed robust: the meshes are not only more and more accurate but also fairly regular.

Some regularity tests could be rather misleading and inadequate. Here, we highlight this problem and avoid it. We are careful to use regularity tests that are genuine and adequate.

Why are tetrahedra so suitable to serve as finite elements in our mesh? Because they are flexible: can take all sorts of shapes and sizes. This is the key for an efficient local refinement: use small tetrahedra only where absolutely necessary. Still, there is a price to pay: regularity must be compromised. After all, to approximate the curved boundary well, the tetrahedra must be a little thin. Still, thanks to our tricks, regularity decreases only moderately and linearly from level to level. This is not too bad: it is unavoidable and indeed worthwhile to compromise some regularity for the sake of high accuracy.

11.1 Angle and Sine in 3-D

11.1.1 Sine in a Tetrahedron

How to measure mesh regularity? For this purpose, we must first ask: how to measure the regularity of one individual tetrahedron? Or, how to measure how thick it is?

Consider the general tetrahedron

$$t \equiv (\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n}),$$

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Y. Shapira, *Linear Algebra and Group Theory for Physicists and Engineers*,
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vertexed at its distinct corners $\mathbf{k}$, $\mathbf{l}$, $\mathbf{m}$, and $\mathbf{n}$ (Fig. 9.1). For example, if

$$\begin{aligned}\mathbf{k} &= (0, 0, 0)^t \\ \mathbf{l} &= (1, 0, 0)^t \\ \mathbf{m} &= (0, 1, 0)^t \\ \mathbf{n} &= (0, 0, 1)^t,\end{aligned}$$

then t is just the unit tetrahedron T in Fig. 8.5.

Recall the 3×3 matrix

$$S_t \equiv (\mathbf{l} - \mathbf{k} \mid \mathbf{m} - \mathbf{k} \mid \mathbf{n} - \mathbf{k}),$$

whose columns are the vectors leading from $\mathbf{k}$ to the three other corners. As discussed in Chap. 9, Sect. 9.1.1, S_t is the Jacobian of the mapping that maps T onto t. For example, if $t = T$, then S_t is just the 3×3 identity matrix: the Jacobian of the identity mapping.

In a two-dimensional triangle, we already have the sine function, to help estimate the individual angles. Could the sine function be extended to a three-dimensional tetrahedron as well? After all, unlike the triangle, the tetrahedron has no angles in the usual sense!

For this purpose, let us define the "sine" of t at some corner, say $\mathbf{k}$. This new sine will have a value between 0 and 1, to tell us how straight (or right-angled) t is at $\mathbf{k}$:

$$\sin(t, \mathbf{k}) \equiv \frac{|\det(S_t)|}{\|\mathbf{l} - \mathbf{k}\| \cdot \|\mathbf{m} - \mathbf{k}\| \cdot \|\mathbf{n} - \mathbf{k}\|}.$$

In other words, take S_t, normalize its columns, calculate the determinant, and take the absolute value.

In the extreme case in which t is degenerate, its sine is as small as 0. In the more optimistic case in which t is as straight as the unit tetrahedron T, on the other hand, its sine is as large as 1. Thus, the sine indeed tells us how straight and right-angled t is at $\mathbf{k}$.

11.1.2 Minimal Angle

To estimate $\sin(t, \mathbf{k})$, let us write it in terms of angles between edges or faces in t. This may help bound $\sin(t, \mathbf{k})$ from below, indicating regularity.

Consider the face $\triangle(\mathbf{k}, \mathbf{l}, \mathbf{m}) \subset t$. In this face, let α be the angle vertexed at $\mathbf{k}$. In Fig. 11.1, for example, $\triangle(\mathbf{k}, \mathbf{l}, \mathbf{m})$ is horizontal, so α would be a horizontal angle as well.

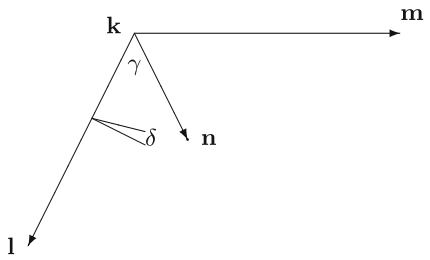


Fig. 11.1 The tetrahedron t: a view from above. It sits on its horizontal base: $\triangle(\mathbf{k}, \mathbf{l}, \mathbf{m})$. Its left edge $(\mathbf{k}, \mathbf{l})$ and its top corner $\mathbf{n}$ make a nonhorizontal face: $\triangle(\mathbf{k}, \mathbf{l}, \mathbf{n})$. Between these two faces, there is a vertical angle: δ

Furthermore, consider the edge $(\mathbf{k}, \mathbf{n})$. Consider its orthogonal projection onto the above face. This produces a new angle between $(\mathbf{k}, \mathbf{n})$ and this projection: β. In Fig. 11.1, for example, β would be the vertical angle between $(\mathbf{k}, \mathbf{n})$ and the x-y plane.

Consider now another face: $\triangle(\mathbf{k}, \mathbf{l}, \mathbf{n}) \subset t$. In this face, let γ be the angle vertexed at $\mathbf{k}$. In Fig. 11.1, for example, γ is a nonhorizontal angle.

Finally, let δ be the angle between the above faces (or between their normal vectors). Then we have

$$\sin(t, \mathbf{k}) = \sin(\alpha)\sin(\beta) = \sin(\alpha)\sin(\gamma)\sin(\delta).$$

Now, how to make sure that t is nondegenerate? Well, require that these angles are nonzero (have a positive sine). Furthermore, how to make sure that t is quite thick? Well, require that these angles are far from zero: their sine is bounded from below by a positive constant. This is indeed the minimal-angle criterion.

Still, this is a rather geometrical criterion. Is there a more algebraic, easily calculated criterion?

Well, let us try. Assume that t is regular in the sense that

$$\sin(t, \mathbf{k}) \geq C$$

for some positive constant $0 < C \leq 1$, independent of $\mathbf{k}$. Then we also have

$$\sin(\alpha) \geq \sin(\alpha)\sin(\gamma)\sin(\delta) = \sin(t, \mathbf{k}) \geq C$$
$$\sin(\delta) \geq \sin(\alpha)\sin(\gamma)\sin(\delta) = \sin(t, \mathbf{k}) \geq C,$$

or

$$\alpha \geq \arcsin(C)$$
$$\delta \geq \arcsin(C),$$

where $0 < \arcsin(C) \leq \pi/2$. In fact, even as $C \to 0$, we still have

$$\alpha \geq \arcsin(C) \sim C$$
$$\delta \geq \arcsin(C) \sim C.$$

Furthermore, since C is independent of $\mathbf{k}$, the same could be done for every angle in t, either between two distinct edges or between two distinct faces. All such angles are indeed bounded from below by $\arcsin(C) > 0$. Thus, t is indeed thick enough.

Is the reverse also true? Well, let us try the other way around: assume now that all angles like α and δ are bounded from below by a positive constant $0 < C_4 \leq \pi/2$. Then we have

$$\sin(t, \mathbf{k}) = \sin(\alpha)\sin(\gamma)\sin(\delta) \geq \sin^3(C_4) > 0.$$

Furthermore, since this is true for any α, γ, and δ, $\mathbf{k}$ in the above estimate could also be replaced by $\mathbf{l}$, $\mathbf{m}$, or $\mathbf{n}$.

Do we have here two equivalent criteria to estimate regularity? Unfortunately not, as $C_4 \to 0$, the latter estimate is too weak and gives us little information:

$$\sin(t, \mathbf{k}) \geq \sin^3(C_4) \sim C_4^3 \ll C_4 \ll 1.$$

As discussed below, this kind of "equivalence" may be misleading and inadequate.

11.1.3 Proportional Sine

Unfortunately, $\sin(t, \mathbf{k})$ does not tell us the whole story. After all, even a straight and right-angled tetrahedron may still be disproportionate and nonsymmetric: the edges issuing from $\mathbf{k}$ may still be different from each other in length. To account for this, let us introduce the so-called proportional sine:

$$\mathrm{Psine}(t, \mathbf{k}) \equiv \sin(t, \mathbf{k}) \frac{\min(\|\mathbf{l} - \mathbf{k}\|, \|\mathbf{m} - \mathbf{k}\|, \|\mathbf{n} - \mathbf{k}\|)}{\max(\|\mathbf{l} - \mathbf{k}\|, \|\mathbf{m} - \mathbf{k}\|, \|\mathbf{n} - \mathbf{k}\|)}.$$

Still, this is not the end of it. To tell how straight and symmetric t is, we might want to look at it from the best point of view. So far, we have looked at it only from $\mathbf{k}$. There might, however, be a better direction. This motivates the definition of the maximal proportional sine:

$$\mathrm{maxPsine}(t) \equiv \max_{\mathbf{q} \in \{\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n}\}} \mathrm{Psine}(t, \mathbf{q}).$$

For example, in terms of maximal proportional sine, the unit tetrahedron T has the maximal possible regularity: 1.

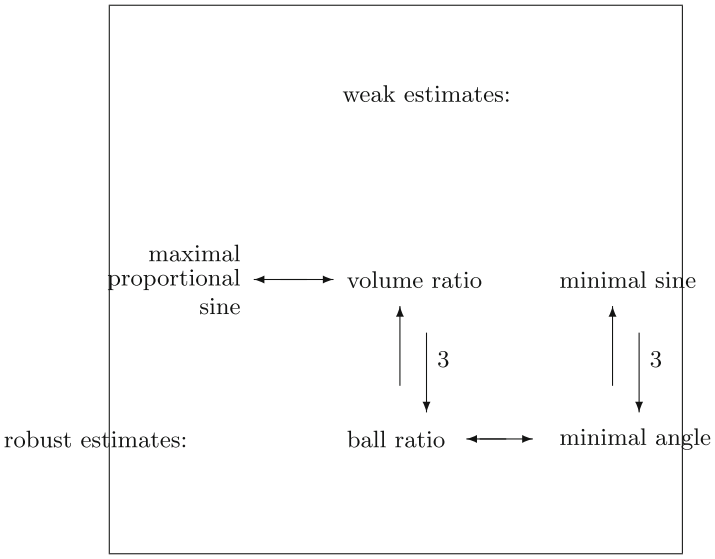


Fig. 11.2 Strong vs. weak regularity estimates. The weak estimates at the top are equivalent to each other, but inferior to the robust estimates at the bottom

11.1.4 Minimal Sine

The maximal proportional sine is an algebraic criterion, easy to calculate on the computer. Later on, we will see that it is actually equivalent to the minimal sine:

$$\text{minSine}(t) \equiv \min_{\mathbf{q}\in\{\mathbf{k},\mathbf{l},\mathbf{m},\mathbf{n}\}} \sin(t, \mathbf{q}).$$

In terms of minimal sine, the most regular tetrahedron is no longer T, but rather the even equilateral tetrahedron, whose edges have the same length.

In Sect. 11.1.2, we have already seen that minimal sine is "equivalent" to minimal angle: a tetrahedron regular in one sense is also regular in the other sense. Still, this "equivalence" is inadequate and misleading. The minimal-angle criterion is much more robust and reliable (Fig. 11.2). Unfortunately, it is geometrical in nature and not easy to calculate. Later on, we will introduce a new regularity estimate that is not only robust but also easy to calculate: ball ratio.

11.2 Adequate Equivalence

11.2.1 Equivalent Regularity Estimates

How thick is t? In the above, we already gave three possible estimates: its minimal angle, minimal sine, or maximal proportional sine. Below, we will see that these

estimates are not completely independent of each other. On the contrary: they may be related, or even equivalent to each other. In fact, minimal sine and maximal proportional sine are both weak, whereas minimal angle is strong and robust (Fig. 11.2).

To see this, let us first introduce yet another (weak) regularity estimate—volume ratio:

$$\frac{|\det(S_t)/6|}{\text{maxEdge}^3(t)},$$

where $\text{maxEdge}(t)$ is the maximal edge length in t. Let us show that this estimate is not really new: it is actually equivalent to the maximal proportional sine:

$$\text{maximal proportional sine} \;\geq\; \text{constant} \cdot \text{(volume ratio)}$$

(which is obvious), and

$$\text{volume ratio} \;\geq\; \text{constant} \cdot \text{(maximal proportional sine)}$$

(which is not so obvious).

Let us prove the "not so obvious" bit. For this purpose, consider the maximal edge in t. It is shared by two faces in t. Each such face contains at least one more edge that is also as long as $\text{maxEdge}(t)/2$.

Fortunately, every corner in t belongs to at least one of these faces. Thus, from every corner in t, there issues at least one edge that is as long as $\text{maxEdge}(t)/2$.

Now, in the maximal edge, look at that endpoint from which two long edges issue: the maximal edge itself, and another edge that is also as long as $\text{maxEdge}(2)/2$. Look at the proportional sine at this corner. This completes the proof.

This equivalence is indeed genuine and adequate. The volume ratio is thus not really a new estimate: it gives us the same information as does the maximal proportional sine. Indeed, if t is thick, then both tell us this. If, on the other hand, t is too thin, then both tell us this in the same way.

Below, on the other hand, we will see that this is not always the case: two different regularity estimate may seem equivalent but are not.

11.2.2 Inadequate Equivalence

Unfortunately, the "equivalence" introduced in [7] is not good enough. It uses an inequality like above, but only for a thick tetrahedron, whose regularity estimate is bounded from below by a positive constant, not for a thin tetrahedron, whose regularity estimate approaches zero. This kind of "equivalence" may be rather misleading and inadequate.

In Sect. 11.1.2, we have already seen an example of an inadequate equivalence: as $C_4 \to 0$, the minimal angle is bounded from below much better than the minimal sine.

This is also the case with the ball ratio [the radius of the ball inscribed in t, divided by maxEdge(t)]. The ball ratio and the volume ratio may seem equivalent to each other but are not. After all, in the proof in [7], while the ball ratio is well bounded from below by $C_2 > 0$, the volume ratio may get as small as $C_2^3 \ll C_2 \ll 1$.

All this is still too theoretical. To establish that an "equivalence" is inadequate, it is not enough to study its proof: there is a need to design a concrete counter example of a limit process in which the tetrahedron t gets less and less regular, yet its regularity estimates disagree with each other. This is done next.

Consider, for example, the flat tetrahedron vertexed at

$$(-1, 0, 0),\ (1, 0, 0),\ (0, -1, \varepsilon),\quad \text{and}\quad (0, 1, \varepsilon),$$

where $\varepsilon > 0$ is a small parameter (Fig. 11.3). Is this tetrahedron regular? Well, for $\varepsilon \ll 1$, it certainly is not: its volume is as small as ε, but all its edges are as long as 1. Therefore, all regularity estimates agree with each other: they are as small as $\varepsilon \ll 1$.

So, by now, we have no evidence of any inadequacy. The above is no counter example: the regularity estimates still agree with each other.

Let us try and design yet another example: a thin tetrahedron, vertexed at

$$(-1, 0, 0),\ (1, 0, 0),\ (0, -\varepsilon, \varepsilon),\quad \text{and}\quad (0, \varepsilon, \varepsilon)$$

(Fig. 11.4). Is this tetrahedron regular? No, it is most certainly not: its volume is now as small as ε^2. Still, its regularity estimates disagree with each other. Indeed, it has only one edge as short as ε, and five edges that are as long as 1. For this reason, its weak regularity estimates lie: its volume ratio, minimal sine, and maximal proportional sine are as small as ε^2, which is too harsh. Its strong regularity estimates, on the other hand, are more realistic: its minimal angle and ball ratio are only as small as ε, not ε^2.

What might happen if a weak regularity estimate was used in a stopping criterion in multilevel refinement? Well, we might then believe that a particular tetrahedron is too thin, even when it is not. This might be too pedantic, leading to stopping too early, and rejecting good legitimate fine meshes.

Picking a smaller stopping threshold is no cure: this might be too loose, leading to stopping too late, and accepting flat tetrahedra, for which all regularity tests are as good (Fig. 11.3).

A robust regularity test is thus clearly necessary. To be practical, it must also be easy to calculate. This is done next.

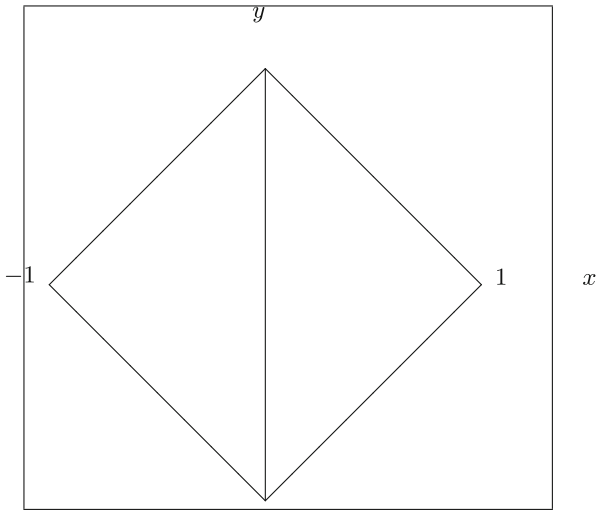


Fig. 11.3 A flat tetrahedron: a view from above. All regularity estimates (weak and strong alike) are as good

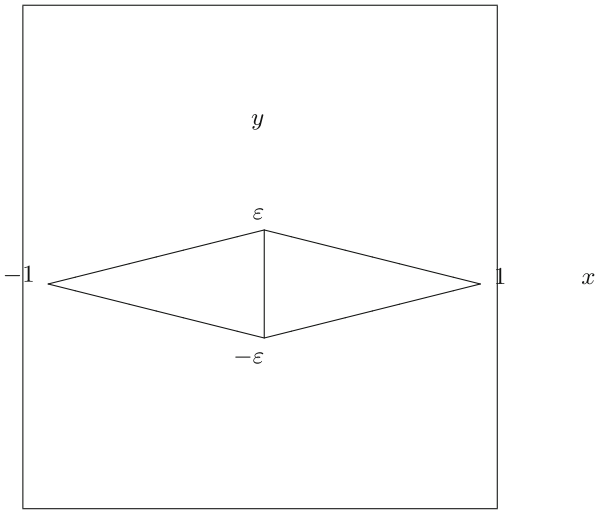


Fig. 11.4 A thin tetrahedron: a view from above. The weak estimates lie: they are as small as ε^2, but the true estimate is only as small as ε

11.2.3 Ball Ratio

Like the minimal angle, the ball ratio is a robust regularity estimate. This is the radius of the ball inscribed in t, divided by $\text{maxEdge}(t)$.

In other words, look at the largest ball that can be contained in t. Clearly, this ball is tangent to the faces of t from the inside. Denote its center by $\mathbf{o}$, and its radius by r. Now, take r, and divide by the length of the maximal edge in t.

Unfortunately, this is still a geometrical definition, not easy to calculate. Is there an algebraic formula, easy to calculate on the computer?

Fortunately, there is. For this purpose, connect **o** to the four corners of t: **k**, **l**, **m**, and **n**. This splits t into four disjoint subtetrahedra.

Clearly, the volume of t is the sum of the volumes of these subtetrahedra. Furthermore, in each subtetrahedron, the radius issuing from **o** makes a right angle with the face that lies across from **o**. Thus,

$$\begin{aligned} |\det(S_t)| &= \left|\det(S_{(\mathbf{o},\mathbf{l},\mathbf{m},\mathbf{n})})\right| + \left|\det(S_{(\mathbf{k},\mathbf{o},\mathbf{m},\mathbf{n})})\right| + \left|\det(S_{(\mathbf{k},\mathbf{l},\mathbf{o},\mathbf{n})})\right| + \left|\det(S_{(\mathbf{k},\mathbf{l},\mathbf{m},\mathbf{o})})\right| \\ &= r\left(\|(\mathbf{m}-\mathbf{l})\times(\mathbf{n}-\mathbf{l})\| + \|(\mathbf{m}-\mathbf{k})\times(\mathbf{n}-\mathbf{k})\|\right. \\ &\quad \left. + \|(\mathbf{l}-\mathbf{k})\times(\mathbf{n}-\mathbf{k})\| + \|(\mathbf{l}-\mathbf{k})\times(\mathbf{m}-\mathbf{k})\|\right), \end{aligned}$$

where "$\times$" stands for vector product.

This formula can now be used to calculate r. The ball ratio is then obtained immediately as $r/\text{maxEdge}(t)$.

11.3 Numerical Experiment

11.3.1 Mesh Regularity

What is the regularity of the entire mesh M? Naturally, it is just the minimum regularity of the individual tetrahedra in M. Still, not all tetrahedra should be considered. After all, a tetrahedron with 3–4 corners outside Ω should be disregarded (Sect. 10.4.1):

$$\text{minimal sine}(M) \equiv \min_{t\subset M, t \text{ has } 2\text{--}4 \text{ corners in } \Omega} \text{minSine}(t),$$

or

$$\text{minimal maxPsine}(M) \equiv \min_{t\subset M, t \text{ has } 2\text{–}4 \text{ corners in } \Omega} \text{maxPsine}(t),$$

or

$$\text{minimal ballR}(M) \equiv \min_{t\subset M, t \text{ has } 2\text{–}4 \text{ corners in } \Omega} \text{ballR}(t).$$

11.3.2 Numerical Results

To test the quality of our multilevel refinement, we consider the nonconvex domain in Chap. 10, Sect. 10.2.2, with $R_2 = 1$ and $R_1 = 0.75$. The initial (coarse) mesh is just a hexahedron: three edge-sharing tetrahedra, each two sharing a face as well.

In our first (dummy) test, only inner splitting is used: no boundary refinement is carried out at all. This way, all meshes at all levels remain confined to the initial hexahedron. As a result, Ω remains poorly approximated.

Why is this test important? Well, it may help filter out the effect of inner splitting only: may this affect regularity?

Fortunately, not much. Indeed, from Table 11.1, it turns out that, in 11 levels, regularity decreases by 50% only. Furthermore, different regularity estimates have nearly the same value: minimal sine is as large as ball ratio. This tells us that no tetrahedron is probably as thin as in Fig. 11.4.

Since no boundary refinement is used, no new valley is produced. Therefore, before each refinement step, the tetrahedra are ordered in terms of maximal edge only, as in Chap. 10, Sect. 10.1.4. After all, no valley is filled, so no deep-valley criterion is relevant. This approach is used in the next test as well.

In the second test, we move on to a more interesting implementation: use boundary refinement as well, to help approximate Ω better. Still, we do not use the trick in Chap. 10, Sect. 10.3.3, as yet: no valley is filled as yet. After all, $\partial\Omega$ is rather smooth, so the mesh in Fig. 10.6 is only slightly concave, producing no overlapping tetrahedra in the next refinement steps.

To estimate the accuracy of the mesh, we also report the volume error:

$$\left| \int\int\int_{\Omega} dxdydz - \sum_{t \subset M, t \text{ has 2–4 corners in } \Omega} \int\int\int_{t} dxdydz. \right|.$$

This is discussed in detail in Chap. 12 below.

From Tables 11.2 and 11.3, it turns out that it is indeed a good idea to order the tetrahedra by maximal edge before each refinement step. Although this may require more nodes, this is a price worth paying for the sake of better accuracy and regularity. Indeed, regularity decreases as slowly as linearly, whereas accuracy improves as fast as exponentially.

In the third and final test, on the other hand, we let the deep-valley criterion dominate the maximal-edge criterion. Before each refinement step, the tetrahedra are now ordered as in Chap. 10, Sect. 10.3.3. Then, the tetrahedra split in this order as well, at the edge of deeper valley, if any. This may help fill the valleys in Fig. 10.7.

Unfortunately, in the first three levels, regularity drops (Table 11.4). After all, the original hexahedron is slightly concave from below, so a submaximal edge may split there, leaving a maximal edge coarse and long. Fortunately, in the next higher levels, things get better: the regularity remains nearly constant, with a very good accuracy in a moderate number of nodes. Thus, in practice, it may make sense to ignore those old valleys that already exist in the initial mesh.

Table 11.1 The dummy test: no boundary refinement is carried out, so all meshes are confined to the original hexahedron, with inner refinement only. Three regularity estimates are reported at the 11th level

Ordering strategy	Minimal ballR	Minimal sine	Minimal maxPsine
Leave disordered	0.012	0.007	0.023
Order by maximal edge	0.033	0.056	0.129

Table 11.2 The nonconvex domain: $R_2 = 1$, $R_1 = 0.75$. The tetrahedra are left disordered No valley is filled

Level	Nodes	Tetrahedra	Minimal ballR	Minimal sine	Minimal maxPsine	Volume error
1	5	3	0.0548	0.1301	0.24839	0.2305
2	8	6	0.0758	0.1330	0.24569	0.2305
3	14	24	0.0465	0.1045	0.18739	0.1942
4	32	84	0.0307	0.0371	0.06409	0.1358
5	74	276	0.0164	0.0328	0.07749	0.0579
6	173	720	0.0119	0.0328	0.04239	0.0234
7	447	1938	0.0088	0.0146	0.02999	0.0153
8	1080	5187	0.0090	0.0091	0.01649	0.0084
9	2770	13,740	0.0060	0.0071	0.0081	0.0037
10	7058	36,651	0.0035	0.0036	0.0093	0.0017
11	18,116	92,792	0.0018	0.0021	0.0059	0.0003

Table 11.3 Before each refinement step, order the tetrahedra by maximal edge: those with a longer edge before others. No valley is filled

Level	Nodes	Tetrahedra	Minimal ballR	Minimal sine	Minimal maxPsine	Volume error
1	5	3	0.0547	0.1301	0.2483	0.2305
2	8	6	0.0758	0.1330	0.2456	0.2305
3	14	24	0.0465	0.1045	0.1873	0.1942
4	32	84	0.0307	0.0371	0.0640	0.1358
5	83	282	0.0170	0.0371	0.0525	0.0588
6	212	858	0.0233	0.0269	0.0524	0.0279
7	560	2472	0.0103	0.0173	0.0299	0.0145
8	1530	7386	0.0102	0.0148	0.0285	0.0073
9	4297	21,516	0.0048	0.0077	0.0134	0.0027
10	11,897	61,446	0.0048	0.0058	0.0140	0.0006
11	32,976	168,602	0.0048	0.0062	0.0135	0.00001

11.4 Exercises

1. Show that, in terms of maximal proportional sine, the unit tetrahedron T has the maximal possible regularity: 1.
2. Write the even equilateral tetrahedron explicitly, and calculate its minimal sine and its maximal proportional sine.

Table 11.4 The deep-valley criterion dominates the maximal-edge criterion. This way, an edge along a valley splits early, even though it is submaximal

Level	Nodes	Tetrahedra	Minimal ballR	Minimal sine	Minimal maxPsine	Volume error
1	5	3	0.0547	0.1301	0.2483	0.2305
2	9	16	0.0094	0.0055	0.0130	0.2199
3	19	50	0.0045	0.0026	0.0039	0.2100
4	48	176	0.0045	0.0026	0.0039	0.1522
5	128	523	0.0049	0.0020	0.0053	0.0726
6	324	1479	0.0048	0.0020	0.0053	0.0252
7	879	4263	0.0049	0.0020	0.0053	0.0132
8	2484	12,674	0.0040	0.0020	0.0053	0.0030
9	7197	37,603	0.0035	0.0013	0.0053	0.00013

3. Show that, in terms of minimal sine, the even equilateral tetrahedron has the maximal possible regularity.
4. Why are both minimal sine and maximal proportional sine not as robust as minimal angle or ball ratio? Hint: see Fig. 11.4.
5. Write (and prove) an explicit formula to calculate the ball ratio. Hint: see Sect. 11.2.3.

Chapter 12
Numerical Integration



Does our multilevel refinement work well? Does it approximate well the original domain? To check on this, we use numerical integration.

Fortunately, our numerical results are encouraging: as the mesh refines, the numerical integral gets more and more accurate. This indicates that our original algorithm is indeed robust and could be used in even more complicated domains.

Of course, there is a price to pay: regularity must decrease. After all, this is why tetrahedra are so suitable to serve as finite elements in our mesh: they are flexible and may come in all sorts of shapes and sizes. This is indeed the key for an efficient local refinement: use small tetrahedra only where absolutely necessary. Still, to approximate the curved boundary well, some tetrahedra must also be a little thin. Fortunately, in our numerical experiments, regularity decreases only moderately and linearly from level to level. This is good enough: after all, it is unavoidable and indeed worthwhile to compromise some regularity for the sake of high accuracy.

12.1 Integration in 3-D

12.1.1 Volume of a Tetrahedron

Consider again a tetrahedron of the form

$$t \equiv (\mathbf{k}, \mathbf{l}, \mathbf{m}, \mathbf{n}),$$

vertexed at $\mathbf{k}$, $\mathbf{l}$, $\mathbf{m}$, and $\mathbf{n}$ (Fig. 9.1). In Chap. 9, Sect. 9.1.2, we have already seen how to integrate in t, using an easier calculation in the unit tetrahedron T:

$$\int\int\int_t F(x, y, z)dxdydz = |\det(S_t)| \int\int\int_T (F \circ E_t)(x, y, z)dxdydz,$$

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Y. Shapira, *Linear Algebra and Group Theory for Physicists and Engineers*,
https://doi.org/10.1007/978-3-031-22422-5_12

where E_t maps T onto t, and S_t is its Jacobian. In this chapter, we explain this formula in some more detail, and in a wider context.

The standard coordinates x, y, and z used in t could also be written in terms of reference coordinates: $\hat{x}$, $\hat{y}$, and $\hat{z}$ in T (Fig. 8.5). These new coordinates could be defined implicitly in T by

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = E_t \left(\begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} \right) = \mathbf{k} + S_t \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix},$$

where E_t and S_t are as in Chap. 9, Sect. 9.1.1. This way, every point $(x, y, z) \in t$ is given uniquely in terms of its own reference point $(\hat{x}, \hat{y}, \hat{z}) \in T$.

The reference coordinates can now be used to integrate in t. In particular, the volume of t is

$$\begin{aligned} \int\int\int_t dxdydz &= \int\int\int_T \left| \det \left(\frac{\partial(x, y, z)}{\partial(\hat{x}, \hat{y}, \hat{z})} \right) \right| d\hat{x}d\hat{y}d\hat{z} \\ &= \int\int\int_T |\det(S_t)| \, d\hat{x}d\hat{y}d\hat{z} \\ &= |\det(S_t)| \int\int\int_T d\hat{x}d\hat{y}d\hat{z} \\ &= \frac{|\det(S_t)|}{6}. \end{aligned}$$

(See Chap. 8, Sect. 8.12.5, and verify that the volume of T is indeed $1/6$.) This result is particularly useful in the numerical integration below.

12.1.2 Integral in 3-D

We have already seen integration in two spatial dimensions (Chap. 8, Sect. 8.10.3). Let us extend this to three spatial dimensions as well. For this purpose, we can use the mesh designed in Chap. 10. All that is left to do is to let the mesh size approach zero:

$$\text{meshsize}(M) \equiv \max_{t \subset M, t \text{ has 2–4 corners in } \Omega} \text{maxEdge}(t) \to 0.$$

Consider a domain $\Omega \subset \mathbb{R}^3$, and assume that a function $f \equiv f(x, y, z)$ is defined in it. Assume also that a multilevel hierarchy of meshes is available to approximate Ω better and better. Let M be some mesh in this hierarchy.

The following limit, if indeed exists, defines the integral of f in Ω:

$$\int\int\int_{\Omega} f(x,y,z)dxdydz \equiv \lim_{\text{meshsize}(M)\to 0} \sum_{t\equiv(\mathbf{k},\mathbf{l},\mathbf{m},\mathbf{n})\subset M, t \text{ has 2–4 corners in } \Omega} \frac{|\det(S_t)|}{6} \cdot \frac{f(\mathbf{k})+f(\mathbf{l})+f(\mathbf{m})+f(\mathbf{n})}{4}.$$

12.1.3 Singularity

But what if f had a singularity? Well, if f is not well-defined at some corner (say $\mathbf{k}$) in some tetrahedron t, then, in the contribution from t to the above sum, substitute

$$f(\mathbf{k}) \leftarrow \frac{f(\mathbf{l})+f(\mathbf{m})+f(\mathbf{n})}{3}.$$

If the singularity is not too sharp, then this might help. For example, assume that, at a distance r from $\mathbf{k}$, f is as small as

$$|f| \leq \text{ constant } \cdot r^{-2}.$$

Assume also that the mesh is fairly regular, so t is rather thick. In this case, f is as small as

$$\max\left(|f(\mathbf{l})|, |f(\mathbf{m})|, |f(\mathbf{n})|\right) \leq \text{ constant } \cdot \text{ maxEdge}^{-2}(t).$$

Thus, the above substitution indeed helps. After all, the contribution from t is also multiplied by the volume of t, which dominates: it is as small as

$$\frac{|\det(S_t)|}{6} \leq \text{maxEdge}^3(t).$$

Let us see some interesting examples. For this purpose, let us introduce spherical coordinates in three spatial dimensions.

12.2 Changing Variables

12.2.1 Spherical Coordinates

We have already used spherical coordinates implicitly to construct the unit sphere in the first place (Chap. 6, Sect. 6.1.4). Here, however, we introduce them fully and explicitly and use them more widely.

A nonzero vector $(x, y, z) \in \mathbb{R}^3$ could be written in terms of its unique spherical coordinates:

- $r \geq 0$: the magnitude of the vector.
- $-\pi/2 \leq \phi \leq \pi/2$: the angle between the original vector and its orthogonal projection onto the x-y plane.
- and $0 \leq \theta < 2\pi$: the angle between this projection and the positive part of the x-axis.

Our new independent variables are now r, θ, and ϕ. θ is known as the azimuthal angle: it is confined to the horizontal x-y plane. ϕ, on the other hand, measures the elevation from the x-y plane upwards. Its complementary angle, $\pi/2 - \phi$, is known as the polar angle between the original vector and the positive part of the z-axis (Fig. 12.1).

The original Cartesian coordinates x, y, and z can now be viewed as dependent variables. After all, they now depend on our new independent variables r, θ, and ϕ:

$$\begin{aligned} x &= r \cdot \cos(\phi) \cos(\theta) \\ y &= r \cdot \cos(\phi) \sin(\theta) \\ z &= r \cdot \sin(\phi). \end{aligned}$$

Let us use the new spherical coordinates in integration.

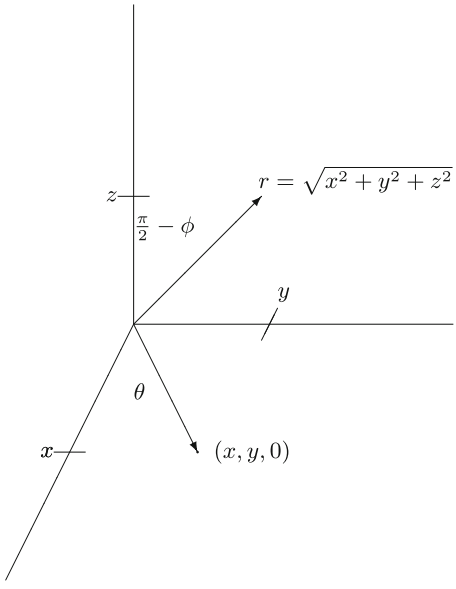


Fig. 12.1 The vector (x, y, z) in its spherical coordinates: r (the magnitude of the vector), ϕ (the angle between the original vector and its projection onto the x-y plane), and θ (the angle between this projection and the x-axis)

12.2.2 Partial Derivatives

Since x, y, and z are functions of r, θ, and ϕ, they also have partial derivatives with respect to them (Chap. 8, Sects. 8.12.1–8.12.4). For example, the partial derivative of x with respect to θ, denoted by $\partial x/\partial\theta$, is obtained by keeping r and ϕ fixed, and differentiating x as a function of θ only. These partial derivatives form the Jacobian.

12.2.3 The Jacobian

As discussed in Chap. 8, Sect. 8.12.4, the Jacobian is the matrix of partial derivatives:

$$\begin{aligned}
&\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} \\
&\equiv \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{pmatrix} \\
&= \begin{pmatrix} \cos(\phi)\cos(\theta) & -r\cdot\cos(\phi)\sin(\theta) & -r\cdot\sin(\phi)\cos(\theta) \\ \cos(\phi)\sin(\theta) & r\cdot\cos(\phi)\cos(\theta) & -r\cdot\sin(\phi)\sin(\theta) \\ \sin(\phi) & 0 & r\cdot\cos(\phi) \end{pmatrix}.
\end{aligned}$$

12.2.4 Determinant of Jacobian

As in Chap. 2, Sect. 2.1.1, the determinant of the above Jacobian is

$$\begin{aligned}
&\det\left(\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}\right) \\
&= \det\left(\begin{pmatrix} \cos(\phi)\cos(\theta) & -r\cdot\cos(\phi)\sin(\theta) & -r\cdot\sin(\phi)\cos(\theta) \\ \cos(\phi)\sin(\theta) & r\cdot\cos(\phi)\cos(\theta) & -r\cdot\sin(\phi)\sin(\theta) \\ \sin(\phi) & 0 & r\cdot\cos(\phi) \end{pmatrix}\right) \\
&= r^2\cos(\phi)\det\left(\begin{pmatrix} \cos(\phi)\cos(\theta) & -\sin(\theta) & -\sin(\phi)\cos(\theta) \\ \cos(\phi)\sin(\theta) & \cos(\theta) & -\sin(\phi)\sin(\theta) \\ \sin(\phi) & 0 & \cos(\phi) \end{pmatrix}\right) \\
&= r^2\cos(\phi)\left(\sin^2(\phi)\left(\sin^2(\theta)+\cos^2(\theta)\right)+\cos^2(\phi)\left(\cos^2(\theta)+\sin^2(\theta)\right)\right) \\
&= r^2\cos(\phi)\left(\sin^2(\phi)+\cos^2(\phi)\right)
\end{aligned}$$

$$= r^2 \cos(\phi).$$

Thanks to this determinant, we can now go ahead and integrate in spherical coordinates.

12.2.5 Integrating a Composite Function

Let us write f as a composite function of the spherical coordinates r, θ, and ϕ:

$$f(x, y, z) \equiv f(x(r, \theta, \phi), y(r, \theta, \phi), z(r, \theta, \phi)) \equiv f(r, \theta, \phi).$$

This way, we can now go ahead and integrate in spherical (rather than Cartesian) coordinates:

$$\begin{aligned}
&\int\int\int_{\Omega} f(x, y, z) dx dy dz \\
&= \int\int\int_{\Omega} f(r, \theta, \phi) \left| \det \left(\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} \right) \right| dr d\theta d\phi \\
&= \int\int\int_{\Omega} f(r, \theta, \phi) r^2 \cos(\phi) dr d\theta d\phi.
\end{aligned}$$

For this purpose, however, Ω must be written in spherical coordinates as well. In some symmetric cases, this is easy to do.

12.3 Integration in the Meshes

12.3.1 Integrating in a Ball

Assume that Ω is a ball of radius $R > 0$, centered at the origin:

$$\begin{aligned}
\Omega &\equiv \left\{ (x, y, z) \mid x^2 + y^2 + z^2 \leq R^2 \right\} \\
&= \left\{ (r, \theta, \phi) \mid 0 < r \leq R,\ 0 \leq \theta < 2\pi,\ -\frac{\pi}{2} < \phi < \frac{\pi}{2} \right\}.
\end{aligned}$$

Note that, in spherical coordinates, the vertical line $x = y = 0$ (or $\phi = \pm\pi/2$) must be excluded from Ω, because θ is not defined uniquely there, and the Jacobian is singular there. Fortunately, this line has no effect on the integral: it has zero volume, or zero three-dimensional measure.

Assume also that the integrand f depends on r only:

$$f \equiv f(r).$$

In this case, the integral of f in Ω is

$$\begin{aligned}\int\int\int_{\Omega} f dxdydz &= \int\int\int_{\Omega} f(r)r^2\cos(\phi)drd\theta d\phi \\ &= \int_0^{2\pi} d\theta \int_{-\pi/2}^{\pi/2} \cos(\phi)d\phi \int_0^R f(r)r^2 dr \\ &= 2\pi\left(\sin\left(\frac{\pi}{2}\right) - \sin\left(\frac{-\pi}{2}\right)\right)\int_0^R f(r)r^2 dr \\ &= 4\pi\int_0^R f(r)r^2 dr.\end{aligned}$$

Here are some straightforward examples. If $f(r) = 1/r$, then we have

$$\int\int\int_{\Omega} \frac{dxdydz}{\sqrt{x^2+y^2+z^2}} = 4\pi\int_0^R r \cdot dr = 2\pi R^2$$

(Chap. 8, Sect. 8.7.2). If, on the other hand, $f \equiv 1$ is just the constant integrand, then we obtain the volume of the ball of radius R:

$$\int\int\int_{\Omega} dxdydz = 4\pi\int_0^R r^2 dr = \frac{4\pi}{3}R^3.$$

12.3.2 Stopping Criterion

The above formula may help calculate the volume error in Chap. 11, Sect. 11.3.2. This way, our multilevel refinement proves robust: the volume error indeed decreases rapidly as the mesh refines.

In more general cases, on the other hand, Ω is not as simple, and its volume is not available in a closed analytic form. How then could we make sure that our multilevel refinement works well? Well, on each mesh, calculate the numerical integral. Then, calculate the difference between each two consecutive numerical integrals: the numerical integral on this level, minus the numerical integral on the previous level. Does this difference approach zero (in absolute value) as the mesh refines? If it does, then we must be on the right track. If, on some level, it fails to decrease, then some tetrahedra might overlap, so we should better stop refining.

12.3.3 Richardson Extrapolation

To have a yet better numerical integral, one might also want to use Richardson extrapolation: take the numerical integral on this level, and multiply it by $4/3$. Then, take the numerical integral on the previous level, and multiply it by $1/3$. Then, subtract the latter from the former. The result may improve on the standard numerical integral: it is often twice as accurate. Furthermore, like the standard numerical integral in Sect. 12.3.2, it could be used in a stopping criterion as well.

12.4 Exercises

1. As in Sect. 12.3.1, use spherical coordinates to compute the integral

$$\int\int\int_{\Omega} \frac{dxdydz}{\sqrt{x^2+y^2+z^2}},$$

   where Ω is the ball of radius R_2 (centered at the origin), minus a smaller ball of radius R_1.
2. Let $R_1 \to 0$. What is the limit?
3. Use the above exercises to define and compute the integral

$$\int\int\int_{\Omega} \frac{dxdydz}{\sqrt{x^2+y^2+z^2}},$$

   where Ω is the ball of radius R_2, centered at the origin.
4. Use spherical coordinates to compute the integral

$$\int\int\int_{\Omega} \frac{dxdydz}{x^2+y^2+z^2},$$

   where Ω is the ball of radius R_2 (centered at the origin), minus a smaller ball of radius R_1.
5. Let $R_1 \to 0$. What is the limit?
6. Use the above exercises to define and compute the integral

$$\int\int\int_{\Omega} \frac{dxdydz}{x^2+y^2+z^2},$$

   where Ω is the ball of radius R_2, centered at the origin.
7. Explain why the numerical integral in Sect. 12.1.2 sums contributions only from those tetrahedra with at least two corners in Ω.

8. Explain why each such tetrahedron contributes the average of f over its four corners, times its volume.
9. Explain why, at a corner of singularity, the missing value of f should be replaced by the average of f over the three other corners (in Sect. 12.1.3).

Chapter 13
Spline: Variational Model in 3-D



By now, our finite-element mesh is quite regular and accurate. It is now time to define basis functions (B-splines) on it. What is a basis function? It has the following properties:

- Piecewise-polynomial.
- Continuous throughout the entire mesh.
- It has only one nonzero degree of freedom in the mesh.
- Its gradient is continuous across all edges and at all side midpoints.
- Its Hessian is continuous at all mesh nodes.

Why are they called basis functions? Because they span a wide space of functions. In other words, they can combine to form a general function, with fewer properties:

- Piecewise-polynomial.
- Its gradient is continuous at all mesh nodes and all edge- and side midpoints.
- Its Hessian is continuous at all mesh nodes.

There are fewer requirements here. Therefore, a function that meets them is not necessarily a basis function. Still, it can be written as a (unique) linear combination of the basis functions. This is indeed a basis for a linear space of functions.

What is a spline? It is a function of the above type, splined (nailed) at the mesh nodes: at each individual node, it must have some value, prescribed in advance. This fixes a few degrees of freedom: those corresponding to the corner basis functions. Still, there are many degrees of freedom that are still free. How to determine them best? This is called the spline problem.

What does "best" mean? Well, in theory, there are many possible splines. After all, the total number of degrees of freedom is very big: far bigger than the total number of nodes. Still, we are looking for one special spline: the minimum-energy spline.

Let us formulate the problem in a slightly different way. Look at the nodes in the mesh. Together, they form a discrete (nonuniform) grid. On it, define a new grid

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function: at each node, this function gives the value prescribed at this node. This value could also be called Dirichlet data.

The grid function is discrete: defined on the nodes only, not in between. How to extend it as smoothly as possible and define it not only at the individual nodes but also in between? This is indeed the spline problem. Here, "smooth" means having as little "energy" as possible.

Fortunately, this problem can be formulated algebraically as a system of linear equations. The solution to this system gives us the long vector that contains the unknown degrees of freedom in the entire mesh, as required.

13.1 Expansion in Basis Functions

13.1.1 Degrees of Freedom in the Mesh

Thanks to multilevel refinement (Chaps. 10, 11), we now have a fairly regular mesh M, which approximates the original three-dimensional domain well. Let N be the set of nodes, E the set of edges, and L the set of sides in M. This way, M contains $|N|$ nodes, $|E|$ edges, and $|L|$ sides.

What are the degrees of freedom in M? Well, each node has ten degrees of freedom (Chap. 9, Sects. 9.5.3–9.5.5), making a total of $10|N|$ degrees of freedom in M. Furthermore, each edge midpoint has two degrees of freedom (Chap. 9, Sect. 9.5.2), making $2|E|$ more degrees of freedom in M. Finally, each side midpoint has one degree of freedom (Chap. 9, Sect. 9.5.1), making $|L|$ more degrees of freedom in M. Thus, in total, M has

$$K \equiv 10|N| + 2|E| + |L|$$

degrees of freedom.

For each of these degrees of freedom, we have one basis function: a continuous piecewise-polynomial function, with only one nonzero degree of freedom in M. Let us go ahead and use them.

13.1.2 The Function Space and Its Basis

Why are these functions called basis functions? Because they form a basis! Indeed, they make a unique linear combination to help write every given function.

More precisely, not quite every function, but only a function f with the following properties:

1. f is piecewise-polynomial: in each individual tetrahedron $t \subset M$, f is a (different) polynomial of degree five (or less).

2. In each individual node in N, f has continuous gradient and Hessian.
3. In every edge- and side midpoint in M, f has continuous nontangential partial derivatives.

As we have seen in Chap. 9, Sect. 9.4.1, f must then be rather smooth: it must have a continuous gradient throughout the entire edge (for all edges in E). Furthermore, f must also be continuous throughout M (Chap. 9, Sect. 9.4.2).

Thus, f can be written as a unique linear combination of the basis functions:

$$\begin{aligned} f = &\sum_{\mathbf{j}\in N} c_{\mathbf{j},1}\psi_{\mathbf{j},1} + c_{\mathbf{j},2}\psi_{\mathbf{j},2} + \cdots + c_{\mathbf{j},10}\psi_{\mathbf{j},10} \\ &+ \sum_{\mathbf{h}=(\mathbf{j}+\mathbf{q})/2,\ (\mathbf{j},\mathbf{q})\in E} c_{\mathbf{h},1}\psi_{\mathbf{h},1} + c_{\mathbf{h},2}\psi_{\mathbf{h},2} \\ &+ \sum_{\mathbf{w}=(\mathbf{j}+\mathbf{q}+\mathbf{u})/3,\ \triangle(\mathbf{j},\mathbf{q},\mathbf{u})\in L} c_{\mathbf{w}}\psi_{\mathbf{w}}. \end{aligned}$$

What are the coefficients in this expansion? Well, they are just the corresponding degrees of freedom of f. For example, $c_{\mathbf{w}}$ is the nontangential derivative of f at the side midpoint $\mathbf{w}$, say

$$c_{\mathbf{w}} \equiv f_z(\mathbf{w})$$

(Chap. 9, Sects. 9.1.4 and 9.5.2).

Why is the above expansion true? Well, consider some tetrahedron $t \subset M$. Now, in t, both sides of the above expansion are polynomials of degree five with the same 56 degrees of freedom. Therefore, they must coincide throughout t (Chap. 9, Sect. 9.3.4). This is true for every $t \subset M$. Thus, they must also coincide throughout M, as required.

This means that the basis functions indeed form a basis. After all, every function f with the above properties can be written as a unique linear combination of them. In other words, the basis functions span a new function space, containing all these fs.

To represent f yet more easily, let us reindex the basis functions one by one in a row:

$$\psi_1,\ \psi_2,\ \psi_3,\ \ldots,\ \psi_K.$$

(Later on, we will discuss the inner order within this list.) This way, f could be written simply as

$$f = \sum_{j=1}^{K} c_j\psi_j.$$

This form will be most useful below.

13.2 The Stiffness Matrix

13.2.1 Assemble the Stiffness Matrix

To design the optimal spline, we need a new $K \times K$ matrix. This is the coefficient matrix (or the stiffness matrix), denoted by A. Let us specify its elements:

$$\begin{aligned} a_{i,j} &\equiv \int\int\int_M \nabla^t \psi_j \nabla \psi_i dxdydz \\ &= \sum_{t \subset M} \int\int\int_t \nabla^t \psi_j \nabla \psi_i dxdydz \\ &= \sum_{t \subset M} |\det(\S_t)| \int\int\int_T \nabla^t \left(\psi_j \circ E_t\right) S_t^{-1} S_t^{-t} \nabla \left(\psi_i \circ E_t\right) dxdydz \end{aligned}$$

($1 \leq i, j \leq K$, Chap. 9, Sect. 9.1.3). This way, to calculate A, we can now integrate in T rather than t.

How to assemble A efficiently? Fortunately, this could be done iteratively. Initially, A is just the zero $K \times K$ matrix. Then, scan the tetrahedra one by one. Each tetrahedron $t \subset M$ may then contribute to a few matrix element:

$$a_{i,j} \leftarrow a_{i,j} + |\det(\S_t)| \int\int\int_T \nabla^t \left(\psi_j \circ E_t\right) S_t^{-1} S_t^{-t} \nabla \left(\psi_i \circ E_t\right) dxdydz.$$

Clearly, this contribution is often zero. For instance, if ψ_i (or ψ_j) vanishes throughout t, then there is no contribution, and $a_{i,j}$ remains the same. Only if both ψ_i and ψ_j are nonzero in t may the contribution be nonzero. In other words, only if both ψ_i and ψ_j have a nonzero degree of freedom at a corner or an edge- or side midpoint in t may the contribution be nonzero.

Alternatively, one could also use the chain rule in Chap. 9, Sect. 9.2.5, to integrate in barycentric coordinates. After all, in practice, this integration is in T as well. This is particularly relevant if the ψ_i's are still available in barycentric coordinates.

Once all nonzero contributions have been assembled from all tetrahedra, the coefficient matrix A is ready. Below, we explain why this matrix is indeed relevant to design the best spline.

13.2.2 How to Order the Basis Functions?

How to order the basis functions

$$\psi_1,\ \psi_2,\ \psi_3,\ \ldots,\ \psi_K$$

one by one in a row? So far, this was not specified, because it did not matter. Now, it is time to order them more explicitly, at least block by block.

In the latter block, place the corner basis functions introduced in Chap. 9, Sect. 9.5.5. For this purpose, split our indices into two disjoint subsets:

$$\{1, 2, 3, \ldots, K\} = R \cup G,$$

where

$$\begin{aligned} R &\equiv \{1, 2, 3, \ldots, K - |N|\} \\ G &\equiv \{K - |N| + 1, K - |N| + 2, K - |N| + 3, \ldots, K\}. \end{aligned}$$

G is now ready to index the corner basis functions defined in Chap. 9, Sect. 9.5.5, and place them in the latter block:

$$\psi_G \equiv \left\{\psi_{K-|N|+1}, \psi_{K-|N|+2}, \psi_{K-|N|+3}, \ldots, \psi_K\right\} \equiv \left\{\psi_{\mathbf{j},1}\right\}_{\mathbf{j}\in N}.$$

R, on the other hand, will index the other basis functions, and place them in the former block:

$$\psi_R \equiv \left\{\psi_1, \psi_2, \psi_3, \ldots, \psi_{K-|N|}\right\}.$$

This also splits the coefficients in Sect. 13.1.2 into two subvectors:

$$\mathbf{c} \equiv (c_1, c_2, c_3, \ldots, c_K)^t = \begin{pmatrix} \mathbf{c}_R \\ \mathbf{c}_G \end{pmatrix}.$$

Assume now that f in Sect. 13.1.2 is just a discrete grid function: it is only given at the nodes, not in between. This means that only those $|N|$ degrees of freedom in $\mathbf{c}_G$ are available:

$$\mathbf{c}_{\mathbf{j},1} = f(\mathbf{j}),\quad \mathbf{j} \in N$$

(Chap. 9, Sect. 9.5.5). This splines (or nails) f at the nodes only, leaving it still free and unspecified in between. To specify f uniquely in between the nodes as well, we must also specify the other degrees of freedom in $\mathbf{c}_R$. How to do this best? In other words, how to extend f to the optimal spline?

13.3 Finding the Optimal Spline

13.3.1 Minimum Energy

What does "best" or "optimal" mean? To formalize this, define the energy of f. (In physics, this is often called kinetic energy, as in Chap. 15, Sect. 15.6.3):

$$\begin{aligned}
\text{energy}(f) &\equiv \int\int\int_M \left(f_x^2 + f_y^2 + f_z^2\right) dxdydz \\
&= \int\int\int_M \|\nabla f\|_2^2 dxdydz \\
&= \int\int\int_M \nabla^t f \nabla f dxdydz \\
&= \int\int\int_M \sum_{j=1}^{K} c_j \nabla^t \psi_j \sum_{i=1}^{K} c_i \nabla \psi_i dxdydz \\
&= \sum_{j=1}^{K}\sum_{i=1}^{K} c_j c_i \int\int\int_M \nabla^t \psi_j \nabla \psi_i dxdydz \\
&= \sum_{j=1}^{K}\sum_{i=1}^{K} c_j a_{i,j} c_i \\
&= \mathbf{c}^t A \mathbf{c}.
\end{aligned}$$

To minimize this kind of energy, let us decompose A cleverly.

13.3.2 The Schur Complement

Thanks to the above splitting, A has a new block form:

$$A = \begin{pmatrix} A_{RR} & A_{RG} \\ A_{GR} & A_{GG} \end{pmatrix}.$$

In practice, A never changes: it remains in its original form. Its block form remains implicit and theoretical only: it helps minimize the energy. For this purpose, let us see some algebraic properties.

Fortunately, A is symmetric. Therefore,

$$A_{RG} = A_{GR}^t.$$

Now, let us focus on the upper-left block: A_{RR}. Clearly, it is symmetric as well. Furthermore, it is also positive definite: its eigenvalues are positive (see exercises below). Therefore, A_{RR} is nonsingular: it has a unique inverse matrix A_{RR}^{-1}, which is symmetric and positive definite as well. Thus, A can be written as a triple product:

$$A = \begin{pmatrix} A_{RR} & A_{RG} \\ A_{GR} & A_{GG} \end{pmatrix} = \begin{pmatrix} A_{RR} & 0 \\ A_{GR} & I \end{pmatrix} \begin{pmatrix} A_{RR}^{-1} & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} A_{RR} & A_{RG} \\ 0 & I \end{pmatrix},$$

where I is the identity matrix of order $|N|$, and S is the Schur-complement matrix:

$$S \equiv A_{GG} - A_{GR} A_{RR}^{-1} A_{RG}.$$

Thanks to this decomposition, the energy can now be written as

$$\begin{aligned} \mathbf{c}^t A \mathbf{c} &= \mathbf{c}^t \begin{pmatrix} A_{RR} & 0 \\ A_{GR} & I \end{pmatrix} \begin{pmatrix} A_{RR}^{-1} & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} A_{RR} & A_{RG} \\ 0 & I \end{pmatrix} \mathbf{c} \\ &= \left(\begin{pmatrix} A_{RR} & A_{RG} \\ 0 & I \end{pmatrix} \mathbf{c} \right)^t \begin{pmatrix} A_{RR}^{-1} & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} A_{RR} & A_{RG} \\ 0 & I \end{pmatrix} \mathbf{c} \\ &= \begin{pmatrix} A_{RR}\mathbf{c}_R + A_{RG}\mathbf{c}_G \\ \mathbf{c}_G \end{pmatrix}^t \begin{pmatrix} A_{RR}^{-1} & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} A_{RR}\mathbf{c}_R + A_{RG}\mathbf{c}_G \\ \mathbf{c}_G \end{pmatrix} \\ &= (A_{RR}\mathbf{c}_R + A_{RG}\mathbf{c}_G)^t A_{RR}^{-1} (A_{RR}\mathbf{c}_R + A_{RG}\mathbf{c}_G) + \mathbf{c}_G^t S \mathbf{c}_G. \end{aligned}$$

How to minimize this? Clearly, the latter term is not in our hands to minimize. Indeed, it is fixed: $\mathbf{c}_G$ is given in advance. So, we can only "play" with the former term. How to minimize it?

Well, the former term is nonnegative (because A_{RR}^{-1} is symmetric and positive definite). So, best make it zero! For this purpose, pick $\mathbf{c}_R$ such that

$$A_{RR}\mathbf{c}_R + A_{RG}\mathbf{c}_G = \mathbf{0},$$

or

$$A_{RR}\mathbf{c}_R = -A_{RG}\mathbf{c}_G.$$

Since A_{RR} is nonsingular, this system indeed has a unique solution $\mathbf{c}_R$. (To find it numerically, use Chapter 17 in [75].) This will produce the entire vector $\mathbf{c}$, including the new degrees of freedom in $\mathbf{c}_R$, required to extend f to the entire mesh. This is indeed the desired spline.

13.4 Exercises

1. Look at the coefficient (stiffness) matrix A. Is it symmetric?
2. Conclude that its eigenvalues are real. Hint: see Chap. 1, Sect. 1.9.4.
3. Conclude that its eigenvectors are orthogonal to each other. Hint: see Chap. 1, Sect. 1.9.5.
4. Let v be some K-dimensional vector. Show that

$$v^t A v \geq 0.$$

   Hint: this is just energy.
5. Conclude that the eigenvalues of A are nonnegative. Hint: pick v to be an eigenvector.
6. Now, look at A_{RR} (the upper-left block in A). Is it symmetric as well?
7. Conclude that its eigenvalues are real.
8. Conclude also that its eigenvectors are orthogonal to each other.
9. Let v be a $(K - |N|)$-dimensional vector. Show that

$$v^t A_{RR} v \geq 0.$$

   Hint: add dummy zeroes to v. For this purpose, let $\mathbf{0}$ be the $|N|$-dimensional zero vector. This way,

$$v^t A_{RR} v = \left(v^t, \mathbf{0}^t\right) A \begin{pmatrix} v \\ \mathbf{0} \end{pmatrix} \geq 0.$$

10. Conclude that the eigenvalues of A_{RR} are nonnegative.
11. Show that, if

$$v^t A_{RR} v = 0,$$

   then v must be the zero vector. Hint: use the components of v as degrees of freedom in a new function:

$$g \equiv \sum_{j \in R} v_j \psi_j.$$

   Clearly, g vanishes at the nodes. Furthermore, g has zero energy. We need to show that g vanishes identically. For this purpose, pick some tetrahedron $t \subset M$. In it, pick some corner $\mathbf{q} \in t$. Now, for each point $(x, y, z) \in t$, use the line integral:

$$g(x, y, z) = g(\mathbf{q}) + \int_{\mathbf{q}}^{(x,y,z)} \nabla g \cdot dl = 0 + \int_{\mathbf{q}}^{(x,y,z)} \mathbf{0} \cdot dl = 0,$$

as required.

12. Conclude that, if v is a nonzero vector, then

$$v^t A_{RR} v > 0.$$

13. Conclude that the eigenvalues of A_{RR} can never vanish.
14. Conclude that the eigenvalues of A_{RR} are not only nonnegative but also positive.
15. Conclude that A_{RR} is nonsingular: it has a unique inverse matrix A_{RR}^{-1}.
16. Show that A_{RR}^{-1} is symmetric as well.
17. Show that A_{RR}^{-1} has the same eigenvectors as A_{RR}, with the reciprocal eigenvalue.
18. Let v be a nonzero $(K - |N|)$-dimensional vector. Show that

$$v^t A_{RR}^{-1} v > 0.$$

Hint: write v as

$$v = A_{RR} A_{RR}^{-1} v.$$

19. Conclude that, to minimize the energy of the spline, $\mathbf{c}_R$ should better solve the linear system

$$A_{RR}\mathbf{c}_R = -A_{RG}\mathbf{c}_G.$$

Part V
Permutation Group and the Determinant in Quantum Chemistry

So far, we used linear algebra to pave the way to group theory: we used matrices to represent groups. In this part, on the other hand, we work the other way around: we use group theory to pave the way to linear algebra. This leads to a practical application in quantum chemistry.

To this end, we introduce the permutation group. This helps study the determinant of a matrix, and its algebraic properties.

Why is the determinant so important? Because it helps design our quantum-mechanical model. In this model, the energy (and other physical quantities) is no longer known for certain. On the contrary, it is nondeterministic: a random variable (as in Chaps. 3 and 7). As such, we can still study its expectation and variance.

Thanks to the determinant, we can now write the expected energy and obtain the Hartree–Fock system: a pseudo-eigenvalue problem. Thanks to linear algebra, the (pseudo) eigenvectors have an attractive property: orthogonality. This is how linear algebra combines with group theory to form a complete theory, with practical applications.

Chapter 14
Permutation Group and the Determinant



Let us design a new group: the group of permutations. It will help define the determinant in a new way. This will give us a few attractive properties. Later on, in quantum chemistry, this will help analyze the electronic structure in the atom.

14.1 Permutation

14.1.1 Permutation

We have already seen how group theory benefits from linear algebra. Let us see the other way around: linear algebra can also benefit from group theory. In fact, permutations and their group will help redefine the determinant from scratch.

How to do this? Consider the set of n natural numbers:

$$\{1,\ 2,\ 3,\ \ldots, n\}$$

(for some natural number n). The braces tell us that this is just a set, with no specific order. The increasing order used here is just a writing style: a convention.

A permutation is a one-to-one mapping from this set onto itself. This means that each number $1 \leq i \leq n$ is mapped to a distinct number $1 \leq p(i) \leq n$. The entire mapping is denoted by

$$p\left(\{1,\ 2,\ 3,\ \ldots, n\}\right).$$

To denote a list (with a specific order), on the other hand, we use triangular brackets. In the beginning, we always assume an increasing order:

$$< 1, 2, 3, \ldots, n > .$$

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After the permutation, on the other hand, the numbers reorder:

$$< p^{-1}(1),\ p^{-1}(2),\ p^{-1}(3),\ \ldots,\ p^{-1}(n) > .$$

Now, if you want to apply yet another permutation on top, then you need to rename the numbers in the list, so they take their increasing order again, as in the beginning:

$$< 1,\ 2,\ 3,\ \ldots,\ n > .$$

Only now are you ready to design the next list, and so on.

14.1.2 Switch

For example, the switch

$$(1 \to 3)$$

switches 1 with 3: at the same time, 1 maps to 3, 3 maps back to 1, and the rest remain the same: 2 maps to 2, 4 to 4, and so on. For this reason, the switch is symmetric: it could also be written as

$$(1 \to 3) = (3 \to 1).$$

And what happens to the list? Before the switch, we assume an increasing order, as always:

$$< 1,\ 2,\ 3,\ \ldots,\ n > .$$

After the switch, on the other hand, the numbers reorder:

$$< 3,\ 2,\ 1,\ 4,\ 5,\ \ldots, n > .$$

We also say that the switch is odd: it picks a minus sign. This is denoted by

$$e((1 \to 3)) = -1.$$

14.1.3 Cycle

A cycle, on the other hand, is more complicated. For example,

$$(1 \to 3 \to 2)$$

maps 1 to 3, 3 to 2, and 2 back to 1 (at the same time). This is why the cycle is indeed cyclic: it could also be written as

$$(1 \to 3 \to 2) = (2 \to 1 \to 3) = (3 \to 2 \to 1).$$

And what happens to the list? Before the cycle, we assume an increasing order, as always:

$$< 1,\ 2,\ 3,\ \ldots,\ n > .$$

After the cycle, on the other hand, the new list is

$$< 2,\ 3,\ 1,\ 4,\ 5,\ \ldots, n > .$$

Why? Because the cycle works like this: it maps 1 to 3. To make room, both 3 and 2 must shift leftwards.

14.2 Decomposition

14.2.1 Composition (Product)

The above cycle can also be written as a composition (or product) of two switches:

$$(3 \to 2 \to 1) = (3 \to 2)(2 \to 1).$$

This is read leftwards: first, 1 switches with 2, producing

$$< 2,\ 1,\ 3,\ 4,\ 5,\ \ldots, n > .$$

Then, it also switches with 3, producing

$$< 2,\ 3,\ 1,\ 4,\ 5,\ \ldots, n >,$$

as required.

Why is this decomposition useful? Because it tells us that the cycle is even, not odd. Indeed, each switch picks a minus sign, which cancel each other:

$$e\left((3 \to 2 \to 1)\right) = e\left((3 \to 2)\right) e\left((2 \to 1)\right) = (-1)(-1) = 1.$$

14.2.2 3-Cycle

Here is a short notation for the above cycle:

$$[3 \to 1] \equiv (3 \to 2 \to 1).$$

Let us use this style to say once again that the cycle is even:

$$e([3 \to 1]) = 1.$$

This is also called 3-cycle. The switch, on the other hand, is also called 2-cycle.

14.2.3 4-Cycle

Likewise, we can also write a yet longer cycle—a 4-cycle:

$$[4 \to 1] \equiv (4 \to 3 \to 2 \to 1).$$

This cycle can be decomposed as a composition of three switches:

$$[4 \to 1] = (4 \to 3)(3 \to 2)(2 \to 1).$$

Again, this is read leftwards: 1 switches with 2, then with 3, then with 4. The resulting list is

$$< 2,\ 3,\ 4,\ 1,\ 5,\ 6,\ \ldots,\ n >,$$

as required. This is why this cycle is odd, not even:

$$e([4 \to 1]) = -1,$$

and so on.

14.2.4 How to Decompose a Permutation?

Let us decompose a general permutation p. For this purpose, start from 1. How to map it? Map it to some $1 \leq k \leq n$. This occupies $k = p(1)$. Next, look at the rest: from 2 to n. How to map them? They must map from

$$\{2, 3, 4, \ldots, n\} \to \{1, 2, 3, \ldots, k-1, k+1, k+2, \ldots, n\}.$$

(The braces tell us that these are only sets: the order is not specified here.) How to do this? In two stages:

- First, mix them (using a smaller permutation q):

$$q : \{2, 3, 4, \ldots, n\} \to \{2, 3, 4, \ldots, n\} .$$

  (1 is left unchanged.)
- If $2 \leq q(i) \leq k$, then subtract 1 from it:

$$q(i) \to q(i) - 1.$$

This way, k is not used, as required. In summary, the original permutation has been decomposed as

$$p(\{1,\ 2,\ 3,\ \ldots, n\}) = [k \to 1]q(\{2,\ 3,\ 4,\ \ldots, n\}) ,$$

for a unique (smaller) permutation q that mirrors p. As a matter of fact, q acts on $\{1, 2, 3, \ldots, n\}$, but always leaves 1 unchanged, so this can be disregarded.

14.3 Permutations and Their Group

14.3.1 Group of Permutations

Look at all permutations on $\{1, 2, 3, \ldots, n\}$. Place them in a new group:

$$P(\{1,\ 2,\ 3,\ \ldots, n\}) .$$

Recall that a group is a set of mathematical objects, closed under some algebraic operation. It must be associative, but not necessarily commutative. It must also contain an identity (or unit) object. Finally, each object must have its own inverse.

In our case, the objects are the permutations, and the operation is composition. This way, the group is indeed closed: the composition of two permutations is a permutation as well. Clearly, composition is associative, as required. Furthermore, we have the identity permutation that changes nothing. Finally, each individual permutation has its (unique) inverse.

Thanks to Sect. 14.2.4, the entire group can now be written as a union of smaller groups:

$$P(\{1,\ 2,\ 3,\ \ldots, n\}) = \cup_{k=1}^{n} [k \to 1]P(\{2,\ 3,\ 4,\ \ldots, n\}) .$$

Note that, in this union, the k-cycles $[k \to 1]$ have alternating signs: even, odd, even, odd, and so on:

$$e([k \to 1]) = (-1)^{k-1}.$$

This will be useful later.

14.3.2 How Many Permutations?

Let us denote our group by

$$P \equiv P(\{1, 2, 3, \ldots, n\})$$

for short. How many permutations are there in P? Well, in the above union, k could take n possible values: from 1 to n. Thanks to mathematical induction, P must contain $n!$ different permutations:

$$|P| = n!.$$

Half of them are odd, and half are even. This could be proved by mathematical induction on $n \geq 2$.

14.4 Determinant

14.4.1 Determinant: A New Definition

Let us use the permutation group to redefine the determinant. For this purpose, consider an $n \times n$ (complex) matrix:

$$A \equiv \left(a_{i,j}\right)_{1 \leq i,j \leq n}.$$

Its determinant is a (complex) number: sum of products of elements. Each product multiplies n elements: one from each row, and one from each column:

$$\det(A) \equiv \sum_{p \in P} e(p) a_{1,p(1)} a_{2,p(2)} a_{3,p(3)} \cdots a_{n,p(n)}.$$

Why is this the same as the old definition in Chap. 2, Sect. 2.1.1? To see this, use mathematical induction on n, and use the union in Sect. 14.3.1 to mirror the minors $A^{(1,k)}$:

$$\begin{aligned}
\det(A) &\equiv \sum_{p\in P} e(p)a_{1,p(1)}a_{2,p(2)}a_{3,p(3)}\cdots a_{n,p(n)} \\
&= \sum_{k=1}^{n} a_{1,k}e([k\to 1]) \sum_{q\in P(\{2,3,4,\ldots,n\})} e(q)a_{2,[k\to 1]q(2)}a_{3,[k\to 1]q(3)}\cdots a_{n,[k\to 1]q(n)} \\
&= \sum_{k=1}^{n}(-1)^{k-1}a_{1,k} \sum_{q\in P(\{1,2,3,\ldots,n-1\})} e(q)A^{(1,k)}_{1,q(1)}A^{(1,k)}_{2,q(2)}\cdots A^{(1,k)}_{n-1,q(n-1)} \\
&= \sum_{k=1}^{n}(-1)^{k-1}a_{1,k}\det\left(A^{(1,k)}\right)
\end{aligned}$$

(thanks to the induction hypothesis for the minors).

14.4.2 Determinant of the Transpose

Look at the transpose matrix. What is its determinant? To find out, mirror each permutation by its unique inverse. After all, if the permutation is even (odd), then its inverse is even (odd) as well. Indeed,

$$1 = e\left(pp^{-1}\right) = e(p)e\left(p^{-1}\right).$$

Thus, instead of scanning all permutations, scan all their inverses, one by one:

$$\begin{aligned}
\det(A) &\equiv \sum_{p\in P} e(p)a_{1,p(1)}a_{2,p(2)}a_{3,p(3)}\cdots a_{n,p(n)} \\
&= \sum_{p^{-1}\in P} e(p)a_{1,p(1)}a_{2,p(2)}a_{3,p(3)}\cdots a_{n,p(n)} \\
&= \sum_{p^{-1}\in P} e\left(p^{-1}\right)a_{p^{-1}(1),1}a_{p^{-1}(2),2}a_{p^{-1}(3),3}\cdots a_{p^{-1}(n),n} \\
&= \sum_{p\in P} e\left(p\right)a_{p(1),1}a_{p(2),2}a_{p(3),3}\cdots a_{p(n),n} \\
&= \det\left(A^t\right).
\end{aligned}$$

Thus, the transpose has the same determinant. Let us use this to calculate the determinant of a product of two matrices.

14.4.3 Determinant of a Product

Consider now two (complex) matrices of order n:

$$A \equiv \left(a_{i,j}\right)_{1\le i,j\le n} \quad \text{and} \quad B \equiv \left(b_{i,j}\right)_{1\le i,j\le n}.$$

What is the determinant of AB? Thanks to the new definition, this is

$$\det(AB) \equiv \sum_{p\in P} e(p)(AB)_{1,p(1)}(AB)_{2,p(2)}\cdots(AB)_{n,p(n)}.$$

Inside the sum, we have a product of n factors of the form

$$(AB)_{i,p(i)} \equiv \left(\sum_{j=1}^{n} a_{i,j}b_{j,p(i)}\right).$$

The above product scans over $i = 1, 2, 3, \ldots, n$ and multiplies these factors one by one. Upon opening parentheses, one must pick one particular j from each such factor. Which j to pick? There is no point to pick the same j from two different factors, say the kth and lth factors. After all, the resulting product will soon cancel with a similar product, obtained from a permutation of the form

$$p(k \to l),$$

which mirrors p: it is nearly the same as p, but also switches l and k beforehand, picking a minus sign. Apart from sign, the rest is the same: since j is the same, multiply the four matrix elements like this: first times fourth times third times second:

$$\ldots\cdot a_{k,j}b_{j,p(k)}a_{l,j}b_{j,p(l)}\cdot\ldots - \ldots\cdot a_{k,j}b_{j,p(l)}a_{l,j}b_{j,p(k)}\cdot\ldots = 0.$$

So, we better focus on a more relevant option: pick a different j from each factor, say

$$j = q(i),$$

for some permutation $q \in P$. This way,

$$\begin{aligned}&\det(AB)\\ &\equiv \sum_{p,q\in P} e(p)a_{1,q(1)}b_{q(1),p(1)}a_{2,q(2)}b_{q(2),p(2)}\cdots a_{n,q(n)}b_{q(n),p(n)}\end{aligned}$$

$$
\begin{aligned}
&= \sum_{p,q^{-1}\in P} e(p) a_{q^{-1}(1),1} b_{1,pq^{-1}(1)} a_{q^{-1}(2),2} b_{2,pq^{-1}(2)} \cdots a_{q^{-1}(n),n} b_{n,pq^{-1}(n)} \\
&= \sum_{p,q\in P} e(p) a_{q(1),1} b_{1,pq(1)} a_{q(2),2} b_{2,pq(2)} \cdots a_{q(n),n} b_{n,pq(n)} \\
&= \sum_{r,q\in P} e(r)e(q) a_{q(1),1} b_{1,r(1)} a_{q(2),2} b_{2,r(2)} \cdots a_{q(n),n} b_{n,r(n)} \\
&= \det\left(A^t\right) \det(B) \\
&= \det(A) \det(B)
\end{aligned}
$$

(where $r = pq$ helps change the summation). In summary, the determinant of the product is the product of determinants:

$$\det(AB) = \det(A) \det(B).$$

Let us use this result further.

14.4.4 Orthogonal Matrix

In Chap. 3, we used a Markov matrix to store probabilities. For this purpose, the matrix had to satisfy an algebraic condition: its columns had to sum to 1.

In quantum mechanics, on the other hand, the model is a bit different: the probabilities are not the matrix elements, but their square. For this reason, the columns should be a bit different: their sum of squares should be 1. This leads to a new kind of matrix: a (real) orthogonal matrix, or a (complex) unitary matrix. What about their determinant?

Thanks to the above, if O is an orthogonal matrix, then its determinant is either 1 or -1:

$$1 = \det\left(O^t O\right) = \det\left(O^t\right) \det(O) = (\det(O))^2 .$$

14.4.5 Unitary Matrix

Let A be a general complex matrix. Look at its Hermitian adjoint: A^t. What is its determinant? It is the complex conjugate of the original determinant:

$$\det\left(A^h\right) = \det\left(\bar{A}^t\right) = \det\bar{(}A^t) = \det\bar{(}A).$$

Let us use this: let U be a unitary matrix. What is its determinant? It is a complex number of absolute value 1:

$$1 = \det\left(U^h U\right) = \det\left(U^h\right)\det(U) = |\det(U)|^2.$$

This will be useful later.

14.5 The Characteristic Polynomial

14.5.1 The Characteristic Polynomial

Let us use the determinant to study the characteristic polynomial, introduced in Chap. 3, Sect. 3.1.1. This is a polynomial in the independent variable λ:

$$\det(A - \lambda I),$$

which vanishes if (and only if) λ happens to be an eigenvalue. Thus, it could be written as

$$\det(A - \lambda I) = q_0 + q_1\lambda + q_2\lambda^2 + \cdots + q_{n-1}\lambda^{n-1} + q_n\lambda^n.$$

What are these coefficients? Well, let us start with the leading term: $q_n\lambda^n$. What is the coefficient q_n? To tell this, look at $\det(A - \lambda I)$ in terms of the new definition. Only one permutation contributes to λ^n: the identity permutation. Indeed, it produces the product

$$(a_{1,1} - \lambda)(a_{2,2} - \lambda)(a_{3,3} - \lambda)\cdots(a_{n,n} - \lambda).$$

Upon opening parentheses, pick $-\lambda$ from each factor. This produces the leading term:

$$q_n\lambda^n = (-1)^n\lambda^n.$$

Thus,

$$q_n = (-1)^n.$$

Next, what is q_{n-1}? Again, only the identity permutation contributes to λ^{n-1}. (All others contribute to λ^{n-2} at most.) As discussed above, it produces the product

$$(a_{1,1} - \lambda)(a_{2,2} - \lambda)(a_{3,3} - \lambda)\cdots(a_{n,n} - \lambda).$$

Upon opening parentheses, pick $-\lambda$ from most factors. Only from one factor don't pick $-\lambda$. This can be done in n different ways, producing

$$q_{n-1}\lambda^{n-1} = (-1)^{n-1}\lambda^{n-1}\sum_{i=1}^{n} a_{i,i}.$$

Thus,

$$q_{n-1} = (-1)^{n-1}\sum_{i=1}^{n} a_{i,i} = (-1)^{n-1}\text{trace}(A),$$

where the trace of a matrix is the sum of its main-diagonal elements.

By now, we have already uncovered two coefficients in the characteristic polynomial. Finally, what is q_0? Again, look at $\det(A - \lambda I)$ in its new definition. This time, however, look at all permutations. Each one produces a product of n factors. From these factors, never pick $-\lambda$. After all, we are now interested in the free term, which contains no power of λ at all. Thus, its coefficient is

$$q_0 = \det(A).$$

Let us use these results in practice.

14.5.2 Trace—Sum of Eigenvalues

How to use the above in practice? For this purpose, we must write the characteristic polynomial in a new form: not as a sum, but as a product. Fortunately, it has degree n, so it has n (complex) roots:

$$\lambda_1,\ \lambda_2,\ \lambda_3,\ \ldots,\ \lambda_n.$$

(Some of these λ_i's may be the same, but this does not matter.) At these λ_i's, the characteristic polynomial vanishes. These are its roots. They are also the eigenvalues of A. Thanks to them, the characteristic polynomial can also be written as

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)(\lambda_3 - \lambda)\cdots(\lambda_n - \lambda).$$

This way, by setting $\lambda = \lambda_i$ $(1 \le i \le n)$, we indeed obtain zero, as required. Furthermore, the leading term is indeed $(-1)^n\lambda^n$, as required.

In the latter form, let us open parentheses, and pick $-\lambda$ from $n-1$ factors only. This way, we obtain $q_{n-1}\lambda^{n-1}$ in a new form:

$$q_{n-1}\lambda^{n-1} = (-1)^{n-1}\lambda^{n-1}\sum_{i=1}^{n} \lambda_i.$$

Thus, the trace is also the sum of eigenvalues:

$$\text{trace}(A) = \sum_{i=1}^{n} \lambda_i.$$

14.5.3 Determinant—Product of Eigenvalues

Finally, upon opening parentheses above, never pick $-\lambda$ from any factor. This way, we obtain q_0 in a new form:

$$q_0 = \lambda_1 \lambda_2 \lambda_3 \cdots \lambda_n.$$

Thus, the determinant is also the product of eigenvalues:

$$\det(A) = \lambda_1 \lambda_2 \lambda_3 \cdots \lambda_n.$$

Let us use this in quantum chemistry.

14.6 Exercises: Permutation and Its Structure

14.6.1 Decompose as a Product of Switches

1. For some $1 \leq i < k \leq n$, consider a switch of the form $(i \to k)$. What does it do? Hint: at the same time, i maps to k, and k maps back to i.
2. Show that it is symmetric:

$$(i \to k) = (k \to i).$$

3. What is its inverse? Hint: itself.
4. Consider a cycle of the form

$$[k \to i] \equiv (k \to k-1 \to k-2 \to \cdots \to i+1 \to i).$$

5. What does it do? Hint: at the same time, k maps to $k-1$, $k-1$ maps to $k-2$, …, $i+1$ maps to i, and i maps back to k.
6. Is it symmetric? Hint: only if $k = i+1$.
7. Write it as a product (composition) of switches:

$$[k \to i] = (k \to k-1)(k-1 \to k-2) \cdots (i+1 \to i).$$

8. What is its inverse? Hint:

$$[k \to i]^{-1} = (i \to i+1)(i+1 \to i+2)\cdots(k-1 \to k) = [i \to k].$$

9. Write the original switch $(i \to k)$ as a composition of two such cycles. Hint:

$$(i \to k) = [i \to k-1] \circ [k \to i].$$

10. Conclude once again that the original switch is odd. Hint:

$$\begin{aligned} e((i \to k)) &= e([i \to k-1] \circ [k \to i]) \\ &= e([i \to k-1])\, e([k \to i]) \\ &= (-1)^{k-1-i+k-i} \\ &= -1. \end{aligned}$$

11. Consider now a more general 3-cycle:

$$(l \to k \to i).$$

12. What does it do? Hint: at the same time, l maps to k, k maps to i, and i maps back to l.
13. Write it as a product of two switches. Hint:

$$(l \to k \to i) = (l \to k)(k \to i).$$

14. Conclude that it is even.
15. Consider now a general permutation:

$$p \in P.$$

16. Write it as a product of general cycles. Hint: start from 1. It must map to some number, which must map to some other number, and so on, until returning back to 1. This completes one general cycle. The rest is a disjoint (smaller) permutation, which can benefit from an induction hypothesis.
17. Conclude that every permutation can be written as a product of general cycles, each written as a product of switches, each written as a product of two more elementary cycles, each written as a product of most elementary switches, as above.
18. In the language of group theory, the elementary switches (that switch two neighbors with each other) *generate* the entire group of permutations.
19. What can you say about the determinant of the transpose? Hint: it is the same.
20. What can you say about the determinant of the Hermitian adjoint? Hint: it is the complex conjugate.

21. What can you say about the determinant of a Hermitian matrix? Hint: it is real.
22. What can you say about the determinant of a product of two matrices? Hint: it is the product of determinants.
23. What can you say about the determinant of an orthogonal matrix? Hint: it is ± 1.
24. What can you say about the determinant of a unitary matrix? Hint: it is a complex number of magnitude 1.

Chapter 15
Electronic Structure in the Atom: The Hartree–Fock System



Linear algebra and group theory go hand in hand. Indeed, the matrices introduced so far help represent groups. This makes groups much more concrete, transparent, and easy to implement on the computer.

It is time for groups to return the favor and pay their debt back. This was done just now: the permutation group helped redefine the determinant and study its algebraic properties. Let us use this in quantum chemistry to uncover the inner structure of the electrons in the atom.

Look at an individual electron, placed at some point in space. It has an electric charge: minus one. Why does not it repel itself? This paradox is answered in quantum mechanics: the electron is not really at one individual point, but at many points at the same time, each with a certain probability. This is called superposition. This way, the position of the electron is nondeterministic: a random variable.

The probability makes a distribution: a nonnegative function. What is the probability to find the electron in between two points on the real axis? To find out, integrate the area between these points, underneath the graph of the distribution. On the other hand, what is the probability to find the electron at a specific point on the real axis? Zero! After all, the vertical issuing from the point is too narrow: it has no width or area at all. Fortunately, in practice, we only need a finite precision: it is good enough to locate the electron in a tiny interval, within a tiny error.

How to uncover the electronic structure in the atom? For this purpose, let us combine group theory with linear algebra. How? Well, the position of each electron is a random variable: we cannot tell it for sure, but only at some probability. Likewise, energy and momentum are nondeterministic as well: we do not know what they really are, but only what they *could* be, with some uncertainty. This is not because we are ignorant, but because, at a very small scale, nature is stochastic.

Fortunately, we can still tell where each electron *might* be, and how likely this is. For this purpose, we need its wave function: its orbital.

In quantum mechanics, electrons are often indistinguishable from each other. To define their orbitals properly, we must use group theory. Thanks to the permutation

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group, the wave function takes the form of a determinant, leading to a simple formula for the (expected) energy. This way, we can model indistinguishable electrons of the same spin.

How to find the correct orbital? For this purpose, look at the total energy of the electron: not only kinetic but also potential, due to electrostatic attraction to the nucleus, and repulsion from other electrons. Again, this is a random variable: it must never be calculated explicitly, but only in terms of expectation. Still, this is good enough for us: thanks to linear algebra, the orbital can be solved for as an eigenvector, with a physical eigenvalue: its energy level.

Each electron may have a different orbital, with a different energy level. These orbitals are the (pseudo) eigenvectors of the same Hermitian matrix. Thanks to linear algebra, they are indeed orthonormal (with respect to some inner product). This is indeed their canonical form.

This leads to the Hartree–Fock system [1, 22, 26]. This is indeed how group theory and linear algebra join forces to solve a practical problem.

15.1 Wave Function

15.1.1 Particle and Its Wave Function

Consider a particle in 3-D. In classical mechanics, it has a deterministic position: (x, y, z). In quantum mechanics, on the other hand, its position is nondeterministic: a random variable, known at some probability only.

In Chap. 7, Sect. 7.11.5, we already met the state v: a grid function, defined on a uniform $m \times m \times m$ grid:

$$v \equiv \left(v_{i,j,k}\right)_{1 \le i,j,k \le m} \in \mathbb{C}^{m \times m \times m}.$$

This tells us the (nondeterministic) position of the particle in 3-D. How likely is it to be at $(X_{k,k}, X_{j,j}, X_{i,i})$? The probability for this is $|v_{i,j,k}|^2$.

Now, let us extend this and make it not only discrete but also continuous. Instead of v, let us talk about a wave function $w(x, y, z)$, defined on the entire three-dimensional Cartesian space. This way, the particle could now be everywhere: not only on a discrete grid but also in any point in 3-D. How likely is it to be at $(x, y, z) \in \mathbb{R}^3$? The probability for this is $|w(x, y, z)|^2$ (not strongly but only weakly—in terms of integration on a tiny volume, as discussed above).

This makes sense: after all, the position should better be a continuous random variable, which may take any value, not necessarily in a discrete grid. For this purpose, however, the sums and inner products used in Chap. 7 should be replaced by integrals. In particular, to make a legitimate probability, w must be normalized to satisfy

$$\int\int\int |w(x, y, z)|^2 dxdydz = 1,$$

where each integral sign integrates over one spatial coordinate, from $-\infty$ to ∞. This can be viewed as an extension of the vector norm defined in Chap. 1, Sect. 1.7.4. In this sense, w has norm 1. Later on, we will make sure that this normalization is carried out in advance.

So far, we talked mainly about one observable: position. Still, there is yet another important observable, which can never be continuous: energy. Indeed, only some energy levels are allowed, and the rest remain nonphysical. This is indeed quantum mechanics: energy comes in discrete quanta.

How to uncover the wave function w? This is the subject of this chapter. For simplicity, we use atomic units, in which the particle has mass 1, and Planck constant is 1 as well.

15.1.2 Entangled Particles

Consider now two particles that may interact and even collide with each other. (In the language of quantum mechanics, they are *entangled* to one another, as in the exercises at the end of Chap. 7.) In this case, the particles do not have independent wave functions, but only one joint wave function: $w(r_1, r_2)$, where $r_1 = (x_1, y_1, z_1)$ and $r_2 = (x_2, y_2, z_2)$ are their possible positions. Once $w(r_1, r_2)$ is solved for, it must be normalized to satisfy

$$\int\int\int\int\int\int |w(r_1, r_2)|^2 dx_1 dy_1 dz_1 dx_2 dy_2 dz_2 = 1.$$

This way, $|w(r_1, r_2)|^2$ may indeed serve as a legitimate probability function (or distribution), to tell us how likely the particles are to be at r_1 and r_2 at the same time (not pointwise but only in a tiny volume, as discussed above).

15.1.3 Disentangled Particles

Unfortunately, there are still a few problems with this model. First, what about three or four or more particles? The dimension gets too high, and the model gets impractical. Furthermore, even with just two particles, what is the meaning of the joint wave function? It tells us nothing about the individual particle! Thus, it makes more sense to assume that the particles do not interact at all, so their wave function could be factored as

$$w(r_1, r_2) = v^{(1)}(r_1)v^{(2)}(r_2),$$

where $v^{(1)}(r_1)$ and $v^{(2)}(r_2)$ are the wave functions of the individual particles. (In the language of quantum mechanics, they are now *disentangled* from each other.) Later on, we will improve on this model yet more, to handle indistinguishable particles as well.

15.2 Electrons in Their Orbitals

15.2.1 Atom: Electrons in Orbitals

Consider now a special kind of particle: an electron. More precisely, consider an atom with M (disentangled) electrons. In particular, look at the nth electron ($1 \leq n \leq M$). It has a (nondeterministic) position $r_n \equiv (x_n, y_n, z_n)$ in 3-D.

Where is the electron? We will never know for sure! After all, measuring the position is not a good idea: it would spoil the original wave function. Without doing this, the best we can tell is that the electron *might* be at r_n. The probability for this is $|v^{(n)}(r_n)|^2$, where $v^{(n)}$ is the wave function of the nth electron: its orbital.

In general, $v^{(n)}$ is a complex function, defined in the entire three-dimensional Cartesian space. Like every complex number, $v^{(n)}$ has a polar decomposition. In it, what matters is the absolute value. The phase, on the other hand, has no effect on the probability $|v^{(n)}|^2$. Still, it does play an important role in the dynamics of the system: it tells us the (linear and angular) momentum of the electron, at least nondeterministically.

The function $v^{(n)}(r_n)$ is also known as the nth orbital: it tells us where the nth electron could be in the atom. Unfortunately, $v^{(n)}$ is not yet known. To uncover it, we must solve a generalized eigenvalue problem. This is the subject of this chapter.

15.2.2 Potential Energy and Its Expectation

What is the potential energy in the atom? This is a random variable too: we can never tell it for sure. Fortunately, we can still tell its expectation. For this purpose, assume again that the electrons are entangled to each other, so

$$w(r_1, r_2, \ldots, r_M)$$

is their joint wave function. This way, $|w(r_1, r_2, \ldots, r_M)|^2$ is the probability to find them at $r_1, r_2, \ldots, r_n$ at the same time. Later on, we will write w more explicitly.

The potential energy has a few terms, coming from electrostatics. The first term comes from attraction to the nucleus, placed at the origin. To have its expectation, take the probability $|w|^2$, multiply it by the potential $1/\|r\|$, sum over all the electrons, and integrate:

$$-\int\int\int\cdots\int\int\int\sum_{n=1}^{M}\frac{|w|^2}{\|r_n\|}dx_1dy_1dz_1dx_2dy_2dz_2\cdots dx_Mdy_Mdz_M.$$

This is a $3M$-dimensional integral: each integral sign integrates over one individual coordinate, from $-\infty$ to ∞. Later on, we will simplify this considerably.

On top of this, there is yet more potential energy, due to the electrostatic repulsion of every two electrons from each other:

$$\int\int\int\cdots\int\int\int\sum_{i=1}^{M}\sum_{n=i+1}^{M}\frac{|w|^2}{\|r_i-r_n\|}dx_1dy_1dz_1dx_2dy_2dz_2\cdots dx_Mdy_Mdz_M.$$

This sums all pairs of electrons, indexed by $1\le i<n\le M$. This way, each pair appears only once, not twice.

These are the Coulomb integrals. Let us simplify them a little.

15.3 Distinguishable Electrons

15.3.1 Hartree Product

As discussed above, this w is not so useful: it mixes different orbitals with each other. Better separate variables: assume again that the electrons are disentangled from each other, so w can be factored as

$$w(r_1,r_2,\ldots,r_M)=v^{(1)}(r_1)v^{(2)}(r_2)\cdots v^{(M)}(r_M).$$

This is the Hartree product. Thanks to it, we will have more information about the nth individual electron, and how likely it is to be at r_n. In fact, the probability for this will be $|v^{(n)}(r_n)|^2$.

To have a Hartree product, the electrons must be not only disentangled but also *distinguishable* from each other. For example, two electrons can be distinguished by spin: one has spin-up, and the other has spin-down, so they cannot be one and the same. This leaves no ambiguity about who is who: each has its own identity, with no identity crisis. This, however, is not always the case. Electrons may have the same spin and be completely indistinguishable from one another. Such electrons must have a more complicated wave function. Still, for the time being, assume that the Hartree product is valid.

15.3.2 Potential Energy of the Hartree Product

In the Hartree product, each individual orbital is a legitimate probability function:

$$\int\int\int |v^{(i)}(r)|^2 dxdydz = 1, \quad 1 \le i \le M.$$

This way, the expected potential energy simplifies to read

$$-\sum_{n=1}^{M} \int\int\int \frac{|v^{(n)}|^2}{\|r\|} dxdydz$$
$$+\sum_{i=1}^{M}\sum_{n=i+1}^{M} \int\int\int\int\int\int |v^{(i)}(r)|^2 \frac{1}{\|r-\tilde{r}\|} |v^{(n)}(\tilde{r})|^2 dxdydzd\tilde{x}d\tilde{y}d\tilde{z}.$$

Here, both $r \equiv (x, y, z)$ and $\tilde{r} \equiv (\tilde{x}, \tilde{y}, \tilde{z})$ are dummy variables in this six-dimensional integral. This is why there is no need to index them by i or n any more.

15.4 Indistinguishable Electrons

15.4.1 Indistinguishable Electrons

Still, two electrons can be distinguished by spin only: one has spin-up, and the other has spin-down. If, on the other hand, they have the same spin, then one can never tell who is who: they have mixed identities and can be considered as one and the same.

Thus, it makes sense to split our electrons into two disjoint subsets. For this purpose, let $0 \le L \le M$ be a new integer number. Now, assume that the L former electrons have spin-up, and the $M - L$ latter electrons have spin-down.

Let us focus on the L former electrons. What is their joint wave function? It can no longer be a simple Hartree product, which assumes distinguishability.

15.4.2 Pauli's Exclusion Principle: Slater Determinant

What is the wave function of the L former electrons? This is a Slater determinant: the determinant of a new $L \times L$ matrix:

$$\frac{1}{\sqrt{L!}} \det\left(\left(v^{(n)}(r_i)\right)_{1\le i,n\le L}\right).$$

This supports indistinguishability. Indeed, interchanging rows (or columns) only picks a minus sign, with no effect on the absolute value. Furthermore, the electrons satisfy Pauli's exclusion principle: two electrons can never have the same state (same spin and also same orbital). Indeed, if they did, then the above matrix would be singular: it would have two identical columns, and zero determinant.

Thanks to the Slater determinant, the expected potential energy will be simplified yet more. To see this, we need some more algebra.

15.5 Orbitals and Their Canonical Form

15.5.1 The Overlap Matrix and Its Diagonal Form

Consider functions that are defined in the three-dimensional Cartesian space. (More precisely, consider only those functions that are square-integrable.) They form a new linear space: each function is like a "vector." As such, they also have their own inner product (or overlap): take one function, multiply it by the complex conjugate of the other function, and integrate. This extends the standard inner product of vectors, defined in Chap. 1, Sect. 1.7.1. This way, we can now talk about orthogonality, and even orthonormality.

In particular, every two orbitals have their overlap: their new inner product. This makes the new $L \times L$ overlap matrix:

$$O \equiv \left(O_{i,n}\right)_{1\le i,n\le L} \equiv \left(\int\int\int \bar{v}^{(i)}(r)v^{(n)}(r)dxdydz\right)_{1\le i,n\le L},$$

where $r = (x, y, z)$ is the dummy variable. Note that O is Hermitian, but not necessarily orthogonal.

A proper orbital should have norm 1: overlap 1 with itself. This way, it makes a legitimate wave function. Here, however, we do not have this as yet: the main-diagonal elements $O_{i,i}$ may still differ from 1.

Fortunately, O is Hermitian and positive semidefinite (Chap. 1, Sect. 1.12.1). As such, it can be diagonalized by a unitary matrix U:

$$O = U^h D U,$$

where

$$D \equiv \mathrm{diag}\left(D_{1,1}, D_{2,2}, \ldots, D_{L,L}\right)$$

is a real diagonal matrix, with the eigenvalues of O on its main diagonal:

$$D_{n,n} \geq 0, \quad 1 \leq n \leq L.$$

Often, the orbitals are linearly independent: they have no linear combination that vanishes (almost) everywhere. In this case, O is even positive definite—its eigenvalues are strictly positive:

$$D_{n,n} > 0, \quad 1 \leq n \leq L.$$

In this case, the overlap matrix has a positive determinant:

$$\det(O) = \det(D) = D_{1,1} D_{2,2} \cdots D_{L,L} > 0.$$

15.5.2 Unitary Transformation

Since U is unitary, its complex conjugate $\bar{U}$ is unitary as well. Let us use it to transform our orbitals. For this purpose, let us place them in an L-dimensional column vector:

$$\mathbf{v} \equiv \mathbf{v}(r) \equiv \begin{pmatrix} v^{(1)}(r) \\ v^{(2)}(r) \\ \vdots \\ v^{(L)}(r) \end{pmatrix}.$$

Thanks to this, our overlap matrix can now be written simply as

$$O = \int\int\int \bar{\mathbf{v}}(r)\mathbf{v}^t(r)dxdydz$$

(where $\mathbf{v}^t$ is the transpose of $\mathbf{v}$—the row vector). Thanks to our unitary matrix U, we now have

$$\begin{aligned} D &= UOU^h \\ &= UO\bar{U}^t \\ &= U\int\int\int \bar{\mathbf{v}}(r)\mathbf{v}^t(r)dxdydz\bar{U}^t \\ &= \int\int\int U\bar{\mathbf{v}}(r)\mathbf{v}^t(r)\bar{U}^t dxdydz \\ &= \int\int\int \bar{U}\bar{\mathbf{v}}(r)\left(\bar{U}\mathbf{v}\right)^t(r)dxdydz. \end{aligned}$$

15.5.3 Orthogonal Orbitals

So, we got a new column vector:

$$\bar{U}\mathbf{v},$$

containing L new orbitals. They are easier to work with. Why? Because they have a diagonal overlap matrix. This means that they have zero overlap with one another. This means that they are orthogonal (in the present sense, using integration).

15.5.4 Slater Determinant and Unitary Transformation

And what about the Slater determinant? How does it change? Not much. Indeed, in the old orbitals, it was written as

$$\frac{1}{\sqrt{L!}}\det\left(\begin{pmatrix}\mathbf{v}^t\,(r_1)\\ \mathbf{v}^t\,(r_2)\\ \vdots\\ \mathbf{v}^t\,(r_L)\end{pmatrix}\right).$$

In the new orbitals, on the other hand, it now takes the form:

$$\frac{1}{\sqrt{L!}}\det\left(\begin{pmatrix}\mathbf{v}^t\,(r_1)\\ \mathbf{v}^t\,(r_2)\\ \vdots\\ \mathbf{v}^t\,(r_L)\end{pmatrix}\bar{U}^t\right)=\frac{1}{\sqrt{L!}}\det\left(\bar{U}\right)\frac{1}{\sqrt{L!}}\det\left(\begin{pmatrix}\mathbf{v}^t\,(r_1)\\ \mathbf{v}^t\,(r_2)\\ \vdots\\ \mathbf{v}^t\,(r_L)\end{pmatrix}\right),$$

which is just a phase shift, with no effect on the probabilities (Chap. 14, Sects. 14.4.2–14.4.5).

15.5.5 Orthonormal Orbitals: The Canonical Form

By now, our new orbitals are orthogonal. How to normalize them? Like this:

$$D^{-1/2}\bar{U}\mathbf{v}.$$

For these new orbitals, the overlap matrix is even simpler: the $L \times L$ identity matrix. This means that they are orthonormal (in the present sense). This is called the canonical form.

Fortunately, there is no need to calculate it explicitly: it is good enough to know that it exists. Thanks to this, we can now substitute

$$\mathbf{v} \leftarrow D^{-1/2}\bar{U}\mathbf{v}.$$

This means that we can assume that our original orbitals are already in their canonical form from the start: orthonormal with respect to the present inner product (involving integration).

How does this affect the Slater determinant? We have already seen that the unitary transformation does not really affect it. The normalization, on the other hand, has a good effect: it normalizes the Slater determinant too, getting it ready to serve as a legitimate wave function. To see this, our canonical form will prove most useful.

15.5.6 Slater Determinant and Its Overlap

What is the Slater determinant? Like every determinant, it is a sum of $L!$ different products, using $L!$ different permutations of $\{1, 2, \ldots, L\}$. Let us calculate its overlap with itself. For this purpose, recall that we now assume that the orbitals are already in their canonical form. In other words, they are orthonormal with respect to the present inner product: overlap. We are now ready to combine orthonormality with the new definition of the determinant (Chap. 14, Sect. 14.4.1):

$$\begin{aligned}
&\frac{1}{L!}\int\int\int\cdots\int\int\int\left|\det\left(\left(v^{(n)}(r_i)\right)_{1\le i,n\le L}\right)\right|^2 dx_1dy_1dz_1\cdots dx_Ldy_Ldz_L\\
&=\frac{1}{L!}\int\int\int\cdots\int\int\int\sum_{p\in P}\left|v^{(p(1))}(r_1)\right|^2\cdots\left|v^{(p(L))}(r_L)\right|^2 dx_1dy_1dz_1\cdots dx_Ldy_Ldz_L\\
&=\int\int\int\left|v^{(1)}(r)\right|^2 dxdydz\int\int\int\left|v^{(2)}(r)\right|^2 dxdydz\cdots\int\int\int\left|v^{(L)}(r)\right|^2 dxdydz\\
&=1.
\end{aligned}$$

Indeed, in terms of the new orbitals, what is the integrand here? It is the determinant times its complex conjugate. In both, better pick the same permutation

$$p \in P \equiv P(\{1, 2, 3, \ldots, L\}).$$

Otherwise, there would be no contribution at all (thanks to orthogonality). Once the same p is picked from both the determinant and its complex conjugate, it makes the same integral for all p. To see this, each dummy variable is renamed as $r \equiv (x, y, z)$, independent of p. Thus, we get the same integral $L!$ times. This is balanced by the coefficient $1/L!$.

Thanks to the canonical form, we proved that the Slater determinant is indeed normalized. Thus, it is indeed a legitimate wave function. We are now ready to calculate the expected energy.

15.6 Expected Energy

15.6.1 Coulomb and Exchange Integrals

There is one more Slater determinant, defined for the $M - L$ latter orbitals of the remaining spin-down electrons. Assume that they are in their canonical form as well. So, our up-to-date wave function is a product of two Slater determinants:

$$w(r_1, r_2, \ldots, r_M) \equiv \frac{1}{\sqrt{L!}} \det\left(\left(v^{(n)}(r_i)\right)_{1\le i,n\le L}\right) \frac{1}{\sqrt{(M-L)!}} \det\left(\left(v^{(n)}(r_i)\right)_{L<i,n\le M}\right).$$

With this new wave function, what is the expected potential energy? Fortunately, it is much simpler than the general form in Sect. 15.2.2. In fact, it is nearly as simple as in Sect. 15.3.2: it contains just one more double sum of new integrals: exchange integrals.

Recall that $v^{(1)}, v^{(2)}, \ldots, v^{(L)}$ are assumed orthonormal in terms of overlap. So are also $v^{(L+1)}, v^{(L+2)}, \ldots, v^{(M)}$. Later on, we will make this plausible.

In the expected energy, we often integrate on the probability function $|w|^2$. It is sometimes more convenient to write it as

$$|w|^2 = \bar{w} \cdot w.$$

This way, the potential due to attraction to the nucleus simplifies to read

$$\begin{aligned} &-\int\int\int\cdots\int\int\int \sum_{n=1}^{M} \frac{|w|^2}{\|r_n\|} dx_1 dy_1 dz_1 dx_2 dy_2 dz_2 \cdots dx_M dy_M dz_M \\ &= -\int\int\int\cdots\int\int\int \bar{w} \sum_{n=1}^{M} \frac{1}{\|r_n\|} w dx_1 dy_1 dz_1 dx_2 dy_2 dz_2 \cdots dx_M dy_M dz_M \\ &= -\sum_{n=1}^{M} \int\int\int \bar{v}^{(n)} \frac{1}{\|r\|} v^{(n)} dx dy dz. \end{aligned}$$

Why? Well, let us focus on the former Slater determinant in w. Like every determinant, it is the sum of products, each uses a different permutation of $\{1, 2, \ldots, L\}$. Now, to have a nonzero integral, better pick the same permutation

in $\bar{w}$ as in w. (Otherwise, thanks to orthogonality, the integral would vanish.) These $L!$ permutations produce the same integral $L!$ times. (Indeed, the dummy variable is renamed as $r \equiv (x, y, z)$.) This is balanced by the coefficient $1/\sqrt{L!}$.

In the Coulomb integrals, on the other hand, things get a little more complicated. To have a nonzero contribution, we have two options. In the main option, pick the same permutation in $\bar{w}$ as in w. This will produce new Coulomb integrals of the form

$$\sum_{i=1}^{M} \sum_{n=i+1}^{M} \int\int\int\int\int\int |v^{(i)}(r)|^2 \frac{1}{\|r - \tilde{r}\|} |v^{(n)}(\tilde{r})|^2 dx dy dz d\tilde{x} d\tilde{y} d\tilde{z}.$$

In the secondary option, on the other hand, pick a slightly different permutation in $\bar{w}$: before the above permutation, switch i and n ($1 \le i < n \le L$, or $L < i < n \le M$). For instance, pick p in w, and

$$p(n \to i)$$

in $\bar{w}$. This will pick a minus sign:

$$e\left((n \to i)\, p\right) = e\left((n \to i)\right) e(p) = -e(p).$$

This will produce the so-called exchange integrals:

$$-\sum_{i=1}^{M} \sum_{n>i \text{ same spin}} \int\int\int\int\int\int \bar{v}^{(i)}(r) v^{(i)}(\tilde{r}) \frac{1}{\|r - \tilde{r}\|} \bar{v}^{(n)}(\tilde{r}) v^{(n)}(r) dx dy dz d\tilde{x} d\tilde{y} d\tilde{z}.$$

Let us see what this means for each individual orbital.

15.6.2 *Effective Potential Energy*

Now, let us focus on the nth electron only. What is the effective potential that it feels? Well, it feels attraction to the nucleus:

$$-\int\int\int \bar{v}^{(n)} \frac{1}{\|r\|} v^{(n)} dx dy dz.$$

On top of this, it also feels repulsion from all other electrons:

$$+\sum_{i=1}^{M} \int\int\int\int\int\int |v^{(i)}(r)|^2 \frac{1}{\|r - \tilde{r}\|} |v^{(n)}(\tilde{r})|^2 dx dy dz d\tilde{x} d\tilde{y} d\tilde{z}.$$

Here one may ask: does it feel any repulsion from itself? No, it does not. There is one fictitious term here: the term for which $i = n$. Do not worry: it will drop soon.

On top of this, it also feels the exchange force from all other electrons of the same spin:

$$-\sum_{i,\text{ same spin as } n} \int\int\int\int\int\int \bar{v}^{(i)}(r)v^{(i)}(\tilde{r})\frac{1}{\|r-\tilde{r}\|}\bar{v}^{(n)}(\tilde{r})v^{(n)}(r)dxdydzd\tilde{x}d\tilde{y}d\tilde{z}.$$

Here one may ask: does it feel any exchange force from itself? No, it does not. There is one fictitious term here: the term for which $i = n$. Fortunately, it is the same as the previous fictitious term, except for a minus sign. Therefore, both drop.

Thanks to these fictitious terms, we now have uniformity: the sums go over $i = 1, 2, \ldots, M$, including $i = n$. Thanks to this, the nth orbital will soon solve the same equation as the other ones.

15.6.3 Kinetic Energy

On top of this, the nth electron also has its own kinetic energy:

$$\frac{1}{2}\int\int\int \nabla^t \bar{v}^{(n)} \cdot \nabla v^{(n)} dxdydz,$$

where "∇" is the gradient: the vector of partial derivatives (and "∇^t" is its transpose: the row vector).

15.6.4 The Schrodinger Equation in Its Integral Form

Together, all these terms must sum to the expected (total) energy of the nth electron:

$$E\int\int\int |v^{(n)}|^2 dxdydz,$$

where E is a constant energy level: an eigenvalue of the Hamiltonian. After all, in quantum mechanics, energy comes in discrete quanta. Only these energy levels are allowed.

This is indeed Schrodinger's equation for the nth electron: its kinetic and effective potential energy (including the exchange terms) sum to its entire expected energy.

15.7 The Hartree–Fock System

15.7.1 Basis Functions and the Coefficient Matrix

So far, our orbital was a function in the three-dimensional Cartesian space. This is too general. To help uncover the orbital, we better approximate it by a piecewise-polynomial function. For this purpose, let us expand it as a linear combination of basis functions:

$$v^{(n)} \doteq \sum_{j=1}^{K} c_j \psi_j,$$

where the ψ_j's are the basis functions, and the c_j's are their (unknown) complex coefficients. (As a matter of fact, c_j depends on n as well, but we can disregard this, since we only focus on one particular n.) Let us plug this into the effective energy in Sect. 15.6.2, term by term. For this purpose, in each term, replace $\bar{v}^{(n)}$ by $\bar{\psi}_l$, and $v^{(n)}$ by ψ_j. (Although ψ_l is often real, we still take its complex conjugate, to be on the safe side.) This will assemble the (l, j)th element in the coefficient matrix A:

$$\begin{aligned}
&a_{l,j} \\
&\equiv -\int\int\int \bar{\psi}_l \frac{1}{\|r\|} \psi_j dx dy dz \\
&+ \sum_{i=1}^{M} \int\int\int\int\int\int |v^{(i)}(r)|^2 \frac{1}{\|r-\tilde{r}\|} \bar{\psi}_l(\tilde{r}) \psi_j(\tilde{r}) dx dy dz d\tilde{x} d\tilde{y} d\tilde{z} \\
&- \sum_{i, \text{ same spin as } n} \int\int\int\int\int\int \bar{v}^{(i)}(r) v^{(i)}(\tilde{r}) \frac{1}{\|r-\tilde{r}\|} \bar{\psi}_l(\tilde{r}) \psi_j(r) dx dy dz d\tilde{x} d\tilde{y} d\tilde{z} \\
&+ \frac{1}{2} \int\int\int \nabla^t \bar{\psi}_l \cdot \nabla \psi_j dx dy dz
\end{aligned}$$

$(1 \le l, j \le K)$. This way, A is no longer a constant matrix: it depends on the unknown orbitals: the $v^{(i)}$'s in the above sums. Still, for fixed orbitals, A is Hermitian, as required.

15.7.2 The Mass Matrix

Now, define also the mass matrix B. Its (l, j)th element is just the overlap of ψ_l with ψ_j:

$$b_{l,j} \equiv \int\int\int \bar{\psi}_l \psi_j dx dy dz$$

$(1 \leq l, j \leq K)$.

How to solve for the unknown c_j's? For this purpose, place them in a new K-dimensional column vector:

$$\mathbf{c} \equiv (c_1,\ c_2,\ c_3,\ \ldots,\ c_K)^t\ .$$

This way, we can now plug our discrete approximation in. The effective energy in Sects. 15.6.2–15.6.3 will then take a discrete form:

$$\mathbf{c}^h A\mathbf{c} = E\mathbf{c}^h B\mathbf{c},$$

where E is the (unknown) energy level. Thus, this is a nonlinear equation, with two types of unknowns: the vector $\mathbf{c}$ and the scalar E.

15.7.3 The Pseudo-Eigenvalue Problem

What should the energy level E be? It should be minimal. For this purpose, we need to solve a pseudo-eigenvalue problem:

$$A\mathbf{c} = E\,B\mathbf{c}.$$

The term "pseudo" tells us that this is actually a nonlinear system: A depends on the (unknown) orbitals themselves. Still, for fixed orbitals (say the solutions), A is Hermitian, as required.

The mass matrix B is Hermitian and is the same for all orbitals. The coefficient matrix A is Hermitian too, but *not* always the same: it depends on spin. For $1 \leq n \leq L$ (spin-up), A takes one form. For $L < n \leq M$ (spin-down), on the other hand, A takes another form.

Furthermore, on the right-hand side, we have a symmetric (and positive definite) matrix: B. This is a generalized eigenvalue problem (as in the exercises at the end of Chap. 1).

Thus, same-spin orbitals are expanded by vectors that solve the same pseudo-eigenvalue problem, with the same A and B. Are these vectors orthogonal (in the generalized sense)? There are two options: if they share the same (generalized) eigenvalue (energy level), then assume that they were already orthogonalized by a (generalized) Gram–Schmidt process. If, on the other hand, they have different (generalized) eigenvalues, then they are already orthogonal (in the sense in the exercises at the end of Chap. 1). In either case, they are orthogonal (in the generalized sense). Once normalized properly, they are also orthonormal (in the generalized sense).

15.7.4 Is the Canonical Form Plausible?

All along, we assumed that the orbitals $v^{(n)}$ are in their canonical form. Is this plausible? To see this, plug two distinct (orthogonal) eigenvectors in the right-hand side, with B in between. You get zero. After all, this is a generalized orthogonality.

Once used to expand a linear combination of basis functions, these two vectors make two orthogonal orbitals (of zero overlap). This is what we wanted: orthogonal orbitals. Once normalized, they are indeed in their canonical form, as asserted all along.

15.8 Exercises: Electrostatic Potential

15.8.1 Potential: Divergence of Flux

1. Consider a ball, centered at the origin: (0, 0, 0). (See Fig. 15.1.)
2. Assume that the ball has charge -1.
3. Outside the ball, consider a point r. What is the electrostatic field at r? Hint: it is radial and outgoing, and its magnitude is $1/\|r\|^2$. Thus, it must be $r/\|r\|^3$.
4. Around the ball, draw a sphere of radius $\|r\|$.
5. What is its surface area? Hint: $4\pi\|r\|^2/3$.
6. On the sphere, draw a little circle around r.
7. What is its surface area? Hint: $\|r\|^2$ (times a constant, which we disregard).
8. What is the flux of the field through the circle? Hint: up to a constant,

$$\text{flux} = \text{field} \cdot \text{surface area} \sim \frac{r}{\|r\|^3}\|r\|^2 = \frac{r}{\|r\|}.$$

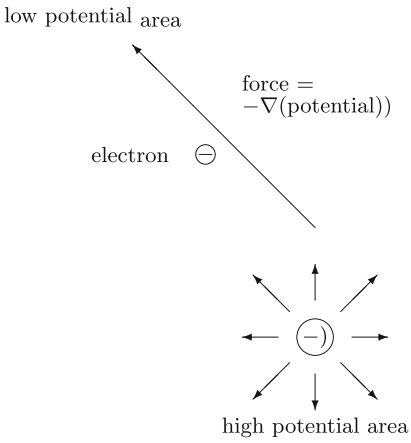


Fig. 15.1 The large (negative) charge at the center introduces a radial electrostatic field. The potential varies: high at the middle, and low away from it. This is why the electron tends to fly away, to minimize its (positive) energy

9. What is the gradient of a scalar function? Hint: vector of partial derivatives.
10. How is it denoted? Hint: by ∇, followed by the scalar function.
11. For instance, what is the gradient of $1/\|r\|$? Hint:

$$\begin{aligned}\nabla\left(\frac{1}{\|r\|}\right) &= \nabla\left(\left(x^2+y^2+z^2\right)^{-1/2}\right)\\ &= -\frac{1}{2}\left(x^2+y^2+z^2\right)^{-3/2}(2x,2y,2z)^t\\ &= -\frac{r}{\|r\|^3}.\end{aligned}$$

12. Is this familiar? Hint: this is the good old electrostatic field (with a minus sign).
13. What is the physical meaning of this? Hint: the field is minus the gradient of the potential (Fig. 15.1).
14. What is the divergence of a vector function? Hint: sum of partial derivatives.
15. How is it denoted? Hint: by ∇, followed by the vector function.
16. For instance, what is the divergence of r? Hint:

$$\nabla r = \nabla\left((x,y,z)^t\right) = \frac{\partial x}{\partial x}+\frac{\partial y}{\partial y}+\frac{\partial z}{\partial z} = 1+1+1 = 3.$$

17. What is the divergence of the flux? Hint: up to a constant,

$$\begin{aligned}\nabla\,(\text{flux}) &= \nabla\,(\text{field}\ \cdot\ \text{surface area})\\ &\sim \nabla\left(\frac{r}{\|r\|}\right)\\ &= \frac{\nabla r}{\|r\|}+r^t\nabla\left(\frac{1}{\|r\|}\right)\\ &= \frac{3}{\|r\|}-r^t\frac{r}{\|r\|^3}\\ &= \frac{3}{\|r\|}-\frac{1}{\|r\|}\\ &= \frac{2}{\|r\|}.\end{aligned}$$

18. Is this familiar?
19. What is this? Hint: this is the electrostatic potential.
20. From a physical point of view, what is the divergence of the flux? Hint: it tells us how much new flux is produced at r, and to what extend the arrows in Fig. 15.1 get more and more spread out at r.

Part VI
The Jordan Form

In this part, we look at a general matrix and design its Jordan form. For this purpose, we introduce new vectors: generalized eigenvectors. Thanks to them, the matrix looks nearly diagonal: it may have nonzero elements on two diagonals only. On the main diagonal, we have the eigenvalues. On the superdiagonal just above it, we have 1s or 0s. All the rest is zero.

To design this, we also use a fundamental theorem in number theory: the Chinese remainder theorem. This will help design the Jordan decomposition. To show how useful this is, we will also introduce a new algebraic structure: an algebra. In it, we will design a subalgebra, in which every derivation has its own Jordan decomposition. This will get us ready for more advanced material: Lie algebras and Cartan's criterion [38, 41, 76].

Chapter 16
The Jordan Form



In this chapter, we look at a general matrix and design its Jordan form. For this purpose, we introduce new vectors: generalized eigenvectors. Thanks to these vectors, the matrix looks nearly diagonal: it may have nonzero elements on two diagonals only. On the main diagonal, we have the eigenvalues. On the superdiagonal above it, we have 1s or 0s. All the rest is zero.

This has a lot of applications in applied science and engineering. For a Hermitian matrix, for example, this will help design a new orthonormal eigenbasis. This will be useful in the Fourier transform, and its real versions: the sine and cosine transforms.

16.1 Nilpotent Matrix and Generalized Eigenvectors

16.1.1 Nilpotent Matrix

Let us start from a special kind of matrix: a nilpotent matrix. Let B be a square (complex) matrix. Assume that there is a natural number $m \geq 1$ for which

$$B^m = (0)$$

(the zero matrix, whose elements vanish). Then we say that B is nilpotent. Clearly, m is not unique: it could increase and still satisfy the above equation. (Later on, we will pick the minimal possible m.)

Next, let us apply B to a vector. Let u be a nonzero vector:

$$u \neq \mathbf{0}.$$

Clearly,

$$B^m u = (0)u = \mathbf{0}.$$

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Still, there is no need to go to such a high power. There could be another number

$$1 \leq k \leq m$$

for which

$$B^k u = \mathbf{0}$$

as well. Unlike m, which depends on B only, k depends on u too. Still, like m, k is not unique. Let us pick k as small as possible.

16.1.2 Cycle and Invariant Subspace

Assume that k is indeed minimal. This means that

$$B^{k-1} u \neq \mathbf{0}.$$

Therefore, this is an eigenvector of B, with eigenvalue 0:

$$B\left(B^{k-1} u\right) = B^k u = \mathbf{0}.$$

We can now design an eigenvector: start from u, and apply B time and again. This will make a cycle: the k new vectors

$$u,\ Bu,\ B^2 u,\ \ldots,\ B^{k-1} u.$$

These are called generalized eigenvectors. Only the latter is a legitimate eigenvector. The former, on the other hand, are not.

Clearly, the cycle is invariant under B: each vector is mapped to the next one. Only the latter is mapped to zero. For this reason, the cycle spans an invariant subspace.

In the simple case of $k = 1$, the cycle is very short. It has length 1 and contains one vector only: u alone. Only in this case is u a legitimate eigenvector.

16.1.3 Generalized Eigenvectors and Their Linear Independence

There could be many cycles. How to order them? Let us order them in terms of decreasing length. For example, the longest cycle could by of length 4 and contain four vectors:

$$u,\ Bu,\ B^2u,\ B^3u.$$

Here, only B^3u is a legitimate eigenvector: its eigenvalue is 0. The rest, on the other hand, are only *generalized* eigenvectors, not eigenvectors.

The next cycle could be of length 3 and contain three vectors:

$$v,\ Bv,\ B^2v$$

(where v is some other nonzero vector). Here, only B^2v is an eigenvector: its eigenvalue is 0. The rest, on the other hand, are only *generalized* eigenvectors but not eigenvectors.

Still, all seven vectors are generalized eigenvectors. After all, a generalized eigenvector could be an eigenvector or not.

Are they linearly independent? To make sure, we need yet another assumption. Assume that the eigenvectors B^3u and B^2v are linearly independent of one another. Are all seven vectors linearly independent as well? To check on this, assume that

$$c_0u + c_1Bu + c_2B^2u + c_3B^3u + d_0v + d_1Bv + d_2B^2v = \mathbf{0}.$$

We need to prove that

$$c_0 = c_1 = c_2 = c_3 = d_0 = d_1 = d_2 = 0.$$

To prove this, place these coefficients in two rows: the longer cycle at the top, and the shorter at the bottom (Fig. 16.1, top diagram). Now, apply B to the entire linear combination:

$$B\left(c_0u + c_1Bu + c_2B^2u + c_3B^3u + d_0v + d_1Bv + d_2B^2v\right) = \mathbf{0}.$$

Since $B^4u = B^3v = \mathbf{0}$, this simplifies to read

$$c_0Bu + c_1B^2u + c_2B^3u + d_0Bv + d_1B^2v = \mathbf{0}.$$

This is illustrated geometrically in Fig. 16.1: the entire diagram shifts leftwards, and the leftmost cells drop. Next, apply B once again. This shifts the diagram even more, dropping two more cells from the left. Finally, apply B one last time. This shifts the diagram one last time, leaving one last cell, containing c_0. This means that

$$c_0 = 0.$$

This "shaves" the rightmost cell off. The process can now restart all over again, as in Fig. 16.2, to conclude that

$$c_1 = d_0 = 0.$$

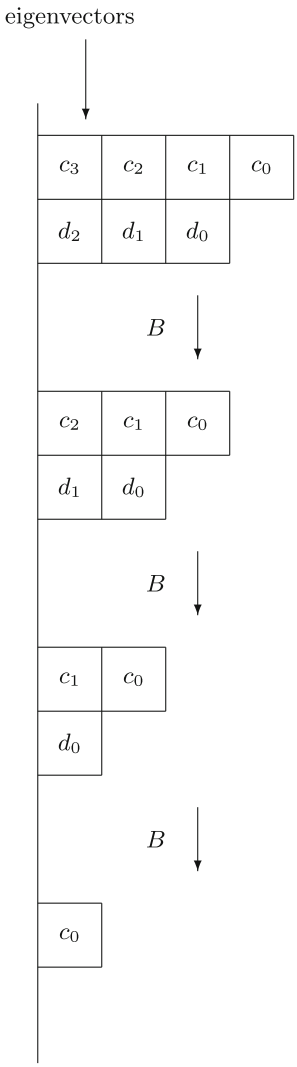


Fig. 16.1 If the eigenvectors B^3u and B^2v are linearly independent, then so are all seven generalized eigenvectors. Indeed, apply B to the entire linear combination time and again, shifting the diagram leftwards, more and more. Those cells that pass the left edge drop, leaving the rightmost cell only. It must therefore vanish: $c_0 = 0$

This "shaves" off two more cells from the right, and so on. In the end, we will conclude that all seven coefficients vanish, as asserted.

16.1.4 More General Cases

This is not limited to cycles of length three or four only. It could be extended to longer cycles as well, using a nested mathematical induction. Furthermore, this is not limited to two cycles only. It could be extended to more cycles as well (ordered

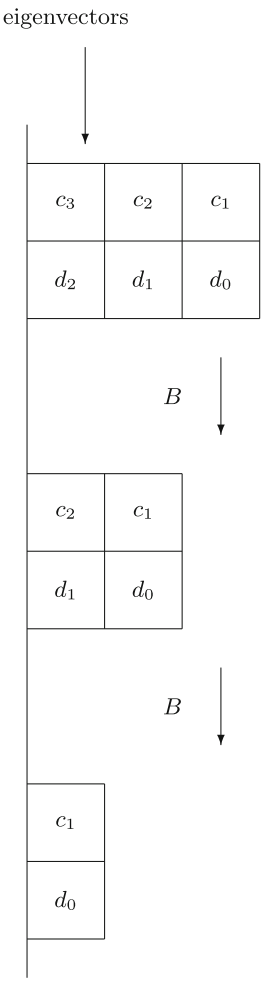


Fig. 16.2 After dropping c_0, repeat the same process from the start. As before, shift leftwards, more and more. In the end, only two cells survive. Since the eigenvectors are linearly independent, we must have $d_0 = c_1 = 0$. Drop these coefficients, and restart all over again, and so on

in terms of decreasing length, from the longest to the shortest). So, this is our conclusion: if a few eigenvectors are linearly independent, then so are also all the generalized eigenvectors in their cycles.

In particular, this also holds for one isolated cycle: the generalized vectors in it are linearly independent. Therefore, the length of a cycle can never exceed the order of B.

16.1.5 Linear Dependence

So far, we proved linear independence. Next, let us work the other way around. This time, we will prove linear dependence.

For this purpose, assume now that there is no new eigenvector any more. Is there no new *generalized* eigenvector either? To check on this, let w be some vector, with a very short cycle:

$$w,\ Bw.$$

This way, Bw is an eigenvector, with eigenvalue 0. This short cycle could be placed at the bottom row, in agreement with our decreasing-length convention.

Now, assume that Bw is not really a new eigenvector: it depends linearly on the former eigenvectors:

$$Bw = \alpha B^3 u + \beta B^2 v$$

(for some α and β, not both zero). What about w? Is it really new, or does it depend linearly on the former vectors in the former cycles?

In the above equation, B maps w to a linear combination of eigenvectors. But w is not the only one: there is another vector mapped in the same way! To design it, just shift the coefficients α and β rightwards (decrease the power of B, as in the following parentheses):

$$B\left(\alpha B^2 u + \beta B v\right) = \alpha B^3 u + \beta B^2 v = Bw.$$

So, we have two vectors that are mapped to the same vector: Bw. Thus, their difference must map to zero and must therefore be an eigenvector in its own right:

$$w - \alpha B^2 u - \beta B v = \gamma B^3 u + \delta B^2 v.$$

So, w is not really new, as asserted.

16.1.6 More General Cases

Let us see another example, a little more complicated. For this purpose, assume now that w makes a little longer cycle:

$$w,\ Bw,\ B^2 w$$

(where $B^2 w$ is now an eigenvector, with eigenvalue 0). This is still in agreement with our decreasing-length rule (Fig. 16.3).

Assume now that the "new" eigenvector $B^2 w$ is not really new: it depends linearly on the former eigenvectors:

$$B^2 w = \alpha B^3 u + \beta B^2 v.$$

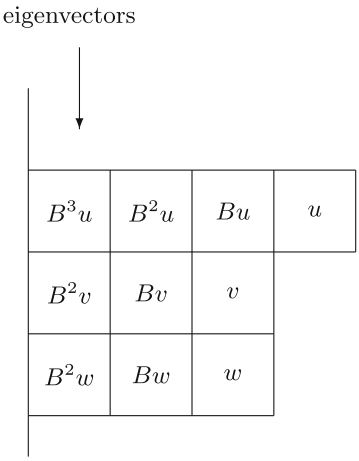


Fig. 16.3 If there is no new eigenvector, then there is no new generalized eigenvector either. For example, if B^2w depends linearly on B^3u and B^2v, then both w and Bw depend linearly on the two former cycles (the first and second rows)

Now, do to Bw what you did before to w. This way, Bw is now written as a linear combination:

$$Bw = \alpha B^2u + \beta Bv + \gamma B^3u + \delta B^2v.$$

So, B maps w to this linear combination. But w is not the only one: there is yet another vector mapped in the same way. To design it, just shift the coefficients α, β, γ, and δ rightwards (in the following parentheses, the power of B decreases):

$$B\left(\alpha Bu + \beta v + \gamma B^2u + \delta Bv\right) = \alpha B^2u + \beta Bv + \gamma B^3u + \delta B^2v = Bw.$$

So, we have two vectors mapped to the same vector: Bw. What about their difference? It must therefore map to zero, so it is an eigenvector in its own right:

$$w - \alpha Bu - \beta v - \gamma B^2u - \delta Bv = \zeta B^3u + \eta B^2v$$

(for some new coefficients ζ and η). So, w is not really new: it depends linearly on the previous generalized eigenvectors.

This is not limited to these examples only: it could be extended to many more rows and columns (using a nested mathematical induction). In summary, if there is a really new w, then there is also a really new eigenvector. The cycle leading to it cannot be too long, or it would be used earlier (as in our convention). Thus, the entire cycle must be really new. After all, we already know that all generalized eigenvectors (in all these cycles) are linearly independent (Sect. 16.1.3). Together, they form a new basis: the Jordan basis.

16.2 Nilpotent Matrix and Its Jordan Form

16.2.1 How to Design a Jordan Basis?

Recall that B is nilpotent:

$$B^m = (0).$$

Recall also that m is not unique: it could increase, and the same equation still holds. How to pick a suitable m?

Let us pick a big m, which is always available: the order of B. This m is not minimal, but is still good: no cycle could be longer:

$$k \leq m.$$

Indeed, by contradiction: if $k > m$, then the vectors in the cycle would depend linearly on each other, in violation of Sect. 16.1.4.

We can now design a Jordan basis. Nothing could stop us: pick a vector, generate its cycle, then pick a really new vector (linearly independent of the former cycle), generate its cycle, and so on, until you have m linearly independent generalized eigenvectors. (In this process, if a cycle is longer than a former one, then drop the former one.) This is a Jordan basis.

So far, our m is too big. Is there a smaller m? Yes, let m be as big as the length of the longest cycle. This is indeed the minimal m.

16.2.2 The Reverse Ordering

So far, we ordered our cycles in terms of decreasing length. Next, let us work the other way around: order them in terms of increasing length: from the shortest to the longest.

Moreover, in each cycle, reverse the ordering as well. In the above example, this give the following new ordering:

$$\left\{\ldots,\ B^2v,\ Bv,\ v,\ B^3u,\ B^2u,\ Bu,\ u\right\}.$$

16.2.3 Jordan Blocks

In each cycle, how does B act? Each generalized eigenvector goes to the previous one, until the first one: the eigenvector, mapped to zero.

Now, in the above list, represent each vector by a standard unit vector: only one component is 1, and the rest are zero. This way, B takes a new form:

$$B = \begin{pmatrix} \ddots & & & & & & & \\ & \ddots & & & & & & \\ & & 0 & 1 & & & & \\ & & & 0 & 1 & & & \\ & & & & 0 & & & \\ & & & & & 0 & 1 & \\ & & & & & & 0 & 1 \\ & & & & & & & 0 \end{pmatrix}.$$

This is the Jordan form. Most elements are zero. Only one diagonal is nonzero: the superdiagonal, just above the main diagonal. In our example, there are two Jordan blocks: at the lower-right corner, there is a 4×4 block, with three 1s in the superdiagonal. Before it, there is a 3×3 block, with just two 1s in the superdiagonal. In general, there could be many more blocks.

16.2.4 Jordan Blocks and Their Powers

What is the order of B? It is the total number of generalized eigenvectors, in all cycles together.

Next, look at higher powers of B: B^2, B^3, ... In these powers, what happens to the Jordan blocks? Well, the 1s shift rightwards, to the next higher superdiagonal, until they exceed the Jordan block at the upper-right corner and drop out. This is why, in our example,

$$B^4 = (0), \quad \text{so} \quad m = 4.$$

So far, we discussed the nilpotent matrix B. What happens in a more general matrix, nilpotent or not? There is one change only: in the Jordan form, the main diagonal could be nonzero. Let us design this.

16.3 General Matrix

16.3.1 Characteristic Polynomial: Eigenvalues and Their Multiplicity

So far, we have studied the nilpotent matrix B. Next, let us discuss a more general matrix A (of order n), nilpotent or not. Its Jordan form is slightly different: the main-diagonal elements are zero or not.

The characteristic polynomial of A could be written as

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)(\lambda_3 - \lambda) \cdots (\lambda_n - \lambda),$$

where λ is the independent variable, and the λ_is (the eigenvalues of A) are not necessarily distinct: some of them could be the same. To avoid this, better rewrite this as

$$\det(A - \lambda I) = (\lambda_1 - \lambda)^{n_1}(\lambda_2 - \lambda)^{n_2}(\lambda_3 - \lambda)^{n_3} \cdots (\lambda_k - \lambda)^{n_k},$$

for some new $k \geq 1$, and new λ_is that are now *distinct*, each with its own multiplicity: n_i. Clearly, the multiplicities sum to n:

$$n_1 + n_2 + n_3 + \cdots + n_k = n.$$

16.3.2 Block and Its Invariant Subspace

Let us focus on the first eigenvalue: λ_1. Let us subtract λ_1 from the main diagonal of A, and write the difference in a new block form:

$$A - \lambda_1 I = \begin{pmatrix} C - \lambda_1 I & \\ E & J - \lambda_1 I \end{pmatrix}$$

(for some square blocks C and J, and some rectangular block E). The upper-right block vanishes. In a moment, we will see why. Here, I is an identity matrix of a suitable order.

More specifically, what is J? It restricts A to the following subspace:

$$V \equiv \left\{ v \mid (A - \lambda_1 I)^{n_1} v = \mathbf{0} \right\}.$$

In fact, V is the null space of $(A - \lambda_1 I)^{n_1}$. Clearly, V is invariant under $A - \lambda_1 I$. This is why $A - \lambda_1 I$ has no upper-right block. Indeed, let J be the restriction of A to V:

$$J \equiv A\,|_V\ .$$

This is why J is in the lower-right corner: it acts on the latter standard unit vectors, which span V.

16.3.3 Block and Its Jordan Form

As a matter of fact, we already know these vectors. Indeed, thanks to the above definition, $J - \lambda_1 I$ is nilpotent:

$$(J - \lambda_1 I)^{n_1} = (A - \lambda_1 I)^{n_1}\,|_V = (0).$$

Thus, $J - \lambda_1 I$ (and indeed J itself) has a Jordan form, as in Sect. 16.2.3:

$$J = \begin{pmatrix} \ddots & & & & & & & \\ & \ddots & & & & & & \\ & & \lambda_1 & 1 & & & & \\ & & & \lambda_1 & 1 & & & \\ & & & & \lambda_1 & & & \\ & & & & & \lambda_1 & 1 & \\ & & & & & & \lambda_1 & 1 \\ & & & & & & & \lambda_1 \end{pmatrix}.$$

This is a more general Jordan form: there are nonzero elements not only on the superdiagonal but also on the main diagonal.

16.3.4 Block and Its Characteristic Polynomial

How big is J? To see this, decompose the characteristic polynomial of A:

$$\det(A - \lambda I) = \det(C - \lambda I)\det(J - \lambda I),$$

where I is an identity matrix of a suitable order.

So, how could the characteristic polynomial of J look like? It could only contain factors from the characteristic polynomial of A. Could it contain a factor like $\lambda_2 - \lambda$? No! After all, no eigenvector corresponding to λ_2 could ever belong to V.

Thus, the characteristic polynomial of J must look like

$$\det(J - \lambda I) = (\lambda_1 - \lambda)^l$$

for some natural number l. This also follows from the Jordan form of J, illustrated above.

What could l be? Could

$$l > n_1?$$

No! After all, the characteristic polynomial of J could only contain a factor from the characteristic polynomial of A. On the other hand, could

$$l < n_1?$$

If this were true, then C would have the factor $\lambda_1 - \lambda$ in its own characteristic polynomial. This way, C would also have a new eigenvector $w \neq \mathbf{0}$, corresponding to the eigenvalue λ_1. But then we could design a new eigenvector for A as well: just add a few dummy components to w:

$$(A - \lambda_1 I) \begin{pmatrix} w \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ Ew \end{pmatrix},$$

so

$$(A - \lambda_1 I)^{l+1} \begin{pmatrix} w \\ \mathbf{0} \end{pmatrix} = (A - \lambda_1)^l \begin{pmatrix} \mathbf{0} \\ Ew \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ (J - \lambda_1 I)^l \, Ew \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ (0)Ew \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix},$$

so

$$(A - \lambda_1)^{n_1} \begin{pmatrix} w \\ \mathbf{0} \end{pmatrix} = \mathbf{0}$$

as well, in violation of the very definition of V (since $w \neq \mathbf{0}$ is outside V).

So, we must have

$$l = n_1.$$

So, J is an $n_1 \times n_1$ block. Its Jordan form is already illustrated above.

Now, there is nothing special about λ_1: the same could be done for all other eigenvalues as well: $\lambda_2, \lambda_3, \ldots, \lambda_k$. Thus, in its final Jordan form, A will be block-diagonal, with k such blocks along its main diagonal. To design this more explicitly, we need some background from number theory.

16.4 Exercises: Hermitian Matrix and Its Eigenbasis

16.4.1 Nilpotent Hermitian Matrix

1. Let B be a nilpotent matrix of order l.
2. Could B have a nonzero eigenvalue? Hint: by contradiction: assume that B had a nonzero eigenvalue $\lambda \neq 0$, with an eigenvector $v \neq \mathbf{0}$. Then we would have

$$B^i v = \lambda^i v \neq \mathbf{0}, \quad i = 1, 2, 3, \ldots,$$

   so B could never be nilpotent.
3. Conclude that B has one eigenvalue only: 0.
4. Look at a short cycle of B, of length 2 only:

$$w, \ Bw.$$

   What can you say about Bw? Hint: it is an eigenvector of B, corresponding to eigenvalue 0:

$$B(Bw) = \mathbf{0}.$$

5. Assume now that B is also Hermitian.
6. Could such a cycle exist? Hint: no! Indeed, this would lead to a contradiction:

$$0 = (\mathbf{0}, w) = (B(Bw), w) = (Bw, Bw) > 0.$$

7. So, B has no cycle of length 2.
8. Still, could B have a longer cycle of length 3, like

$$w, \ Bw, \ B^2 w?$$

   Hint: no! Indeed, since B is Hermitian, this would lead to a contradiction again:

$$0 = (\mathbf{0}, Bw) = \Big(B(B^2 w), Bw\Big) = \Big(B^2 w, B^2 w\Big) > 0.$$

9. Could B have a yet longer cycle, of length 4 or more? Hint: no! To prove this, use the same technique.
10. Conclude that B has only very short "cycles," of length 1 each.
11. Conclude that B has no generalized eigenvectors, but only genuine eigenvectors.
12. Conclude that B has l linearly independent eigenvectors.
13. Orthonormalize them. Hint: use a Gram–Schmidt process.
14. What is their joint eigenvalue? Hint: 0.
15. Conclude that B is the zero matrix.

16.4.2 Hermitian Matrix and Its Orthonormal Eigenbasis

1. Let A be a Hermitian matrix (nilpotent or not).
2. Let λ_1 be an eigenvalue of A (of multiplicity n_1).
3. What can you say about λ_1? Hint: it is real (because A is Hermitian).
4. Pick two different eigenvectors from two different Jordan blocks of A. What can you say about them? Hint: they are orthogonal to one another (because they have different eigenvalues).
5. Define the subspace V, as in Sect. 16.3.2.
6. What is V? Hint: V is the null space of $(A - \lambda_1 I)^{n_1}$.
7. Define the matrix J, as in Sect. 16.3.2.
8. What is J? Hint: J is the restriction of A to V.
9. What is the order of J? Hint: n_1.
10. Is J Hermitian? Hint: yes. Indeed, for every two vectors $p, q \in V$, J behaves just like A:

$$(Jp, q) = (Ap, q) = (p, Aq) = (p, Jq).$$

11. Let I be the identity matrix of order n_1.
12. Look at $J - \lambda_1 I$.
13. Is it Hermitian? Hint: yes (because λ_1 is real).
14. Is it nilpotent? Hint: yes. Indeed, look at its n_1st power. This is just the zero matrix. Indeed, for every vector $q \in V$,

$$(J - \lambda_1 I)^{n_1} q = (A - \lambda_1 I)^{n_1} q = \mathbf{0}.$$

15. Conclude that $J - \lambda_1 I$ is nilpotent and Hermitian at the same time.
16. Conclude that it is just the zero matrix:

$$J - \lambda_1 I = (0).$$

17. Conclude that

$$J = \lambda_1 I.$$

18. Do the same for the other Jordan blocks of A as well.
19. Conclude that A has a diagonal Jordan form, corresponding to an orthonormal eigenbasis.

Chapter 17
Jordan Decomposition of a Matrix



So far, we designed the Jordan form of a matrix. It contains two parts: the main diagonal (containing the eigenvalues), plus the superdiagonal (containing 1's and 0's).

The main diagonal is more interesting. After all, it contains the eigenvalues. How to obtain it more directly from the original matrix? For this, we need the Jordan decomposition. In it, we will see quite clearly how the main diagonal is obtained as a (unique) polynomial: a linear combination of powers of the original matrix. This will get you ready for more advanced material: Lie algebras and Cartan's criterion [38, 41, 76].

17.1 Greatest Common Divisor

17.1.1 Integer Division with Remainder

For a start, we need some background in number theory. Let $n > m > 0$ be two natural numbers. What is the ratio between n and m? By "ratio," we do not mean the simple fraction n/m, but only its integer part: the maximal integer number that does not exceed it:

$$\left\lfloor \frac{n}{m} \right\rfloor = \max \left\{ j \in \mathbb{Z} \;\middle|\; j \leq \frac{n}{m} \right\},$$

where $\mathbb{Z}$ is the set of integer numbers. This way, we can now divide n by m with remainder (residual). This means to decompose n as

$$n = km + l,$$

where

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Y. Shapira, *Linear Algebra and Group Theory for Physicists and Engineers*,
https://doi.org/10.1007/978-3-031-22422-5_17

$$k = \left\lfloor \frac{n}{m} \right\rfloor,$$

and l is the remainder (or residual) that is too small to be divided by m:

$$0 \leq l < m.$$

This l has a new name: n modulus m. It is often placed in parentheses:

$$l = (n \bmod m).$$

17.1.2 Congruence: Same Remainder

The above equation could also be written as

$$n \equiv l \pmod{m}.$$

These parentheses are often dropped:

$$n \equiv l \bmod m.$$

What does this mean? It means that n and l are related: they are *congruent* (modulus m). Indeed, when decomposed, they have the same remainder:

$$n = km + l$$
$$l = 0 \cdot m + l.$$

This means that both n and l are indistinguishable modulus m, because they have the same remainder: l:

$$(n \bmod m) = l = (l \bmod m).$$

In other words, their difference $n - l$ could be divided by m (evenly, with no remainder at all):

$$m \mid n - l.$$

This is indeed a mathematical equivalence relation. (Check!)

17.1.3 Common Divisor

Next, consider a special case, in which there is no remainder at all:

$$(n \bmod m) = 0.$$

This means that m divides n:

$$m \mid n.$$

This is very special. In general, m does not divide n. Still, m may share a common divisor with n: a third number that divides both n and m. For example, if both n and m are even, then 2 is their common divisor. But 2 is not the only one: there could be others. What is the maximal one?

17.1.4 The Euclidean Algorithm

How to calculate the greatest common divisor (GCD) of n and m? Well, there are two possibilities: if m divides n, then m itself is the greatest common divisor. If not, then

$$\begin{aligned} \mathrm{GCD}(n, m) &= \mathrm{GCD}(km + l, m) \\ &= \mathrm{GCD}(l, m) \\ &= \mathrm{GCD}(m, l) \\ &= \mathrm{GCD}(m, n \bmod m). \end{aligned}$$

So, instead of the original numbers

$$n > m,$$

we can now work with smaller numbers:

$$m > l = (n \bmod m).$$

This leads to the (recursive) Euclidean algorithm:

$$\mathrm{GCD}(n, m) = \begin{cases} m & \text{if } (n \bmod m) = 0 \\ \mathrm{GCD}(m, n \bmod m) & \text{if } (n \bmod m) > 0. \end{cases}$$

How to prove this? By mathematical induction on $n = 2, 3, 4, \ldots$ (see below).

17.1.5 The Extended Euclidean Algorithm

Thanks to mathematical induction, we will have more than that. We will be able to write the GCD even more explicitly, as a linear combination:

$$\mathrm{GCD}(n, m) = an + bm,$$

for some integer numbers a and b (positive or negative or zero). Indeed, if m divides n, then this is easy:

$$\mathrm{GCD}(n, m) = m = 0 \cdot n + 1 \cdot m,$$

as required. If not, then the induction hypothesis tells us that

$$\mathrm{GCD}(n, m) = \mathrm{GCD}(m, l) = \tilde{a} \cdot m + \tilde{b} \cdot l,$$

for some new integer numbers $\tilde{a}$ and $\tilde{b}$. Now, in this formula, substitute

$$l = n - km.$$

This gives

$$\begin{aligned} \mathrm{GCD}(n, m) &= \mathrm{GCD}(m, l) \\ &= \tilde{a} \cdot m + \tilde{b} \cdot l \\ &= \tilde{a} \cdot m + \tilde{b}\,(n - km) \\ &= \tilde{b} \cdot n + \left(\tilde{a} - k \cdot \tilde{b}\right) m. \end{aligned}$$

To have the required linear combination, define

$$\begin{aligned} a &= \tilde{b} \\ b &= \tilde{a} - k \cdot \tilde{b} = \tilde{a} - \left\lfloor \frac{n}{m} \right\rfloor \tilde{b}. \end{aligned}$$

This is the extended Euclidean algorithm.

17.1.6 Confining the Coefficients

But there is still a problem: the new coefficients a and b could be negative. Worse, they could be too big, and even exceed n. How to avoid this? Easy: throughout the above algorithm (including the recursion), introduce just one change: take modulus n. This will force the new coefficients to remain moderate: between 0 and $n - 1$:

$$0 \leq a, b < n.$$

In the induction hypothesis, do the same: confine its own coefficients to the same interval: $[0, n-1]$:

$$0 \leq \tilde{a}, \tilde{b} < n.$$

17.1.7 The Modular Extended Euclidean Algorithm

How to do this? First, let us rewrite the algorithm as follows:

$$\begin{aligned} a &= \tilde{b} \\ s &= \left\lfloor \frac{n}{m} \right\rfloor \tilde{b} \\ b &= \tilde{a} - s. \end{aligned}$$

This is still the same as before. But now, we are going to change it. To make sure that

$$0 \leq a, b < n,$$

take modulus n:

$$\begin{aligned} a &= \tilde{b} \\ s &= \left(\left\lfloor \frac{n}{m} \right\rfloor \tilde{b} \right) \bmod n \\ b &= \begin{cases} \tilde{a} - s & \text{if } \tilde{a} \geq s \\ \tilde{a} + n - s & \text{if } \tilde{a} < s. \end{cases} \end{aligned}$$

Now, in the induction hypothesis, assume also that

$$0 \leq \tilde{a}, \tilde{b} < n.$$

Thanks to this, we also have

$$0 \leq a, s, b < n,$$

as required. Furthermore, modulus n, nothing has changed: the algorithm still does the same job. Thus, we got a new linear combination modulus n:

$$\mathrm{GCD}(n, m) \equiv (an + bm) \bmod n.$$

Let us go ahead and use this in practice.

17.2 Modular Arithmetic

17.2.1 Coprime

Consider now a natural number n. What is a coprime of n? This is another number p that shares no common divisor with n:

$$\mathrm{GCD}(n, p) = 1.$$

(p could be prime or not.) There could be many legitimate coprimes. Which one to pick? Better pick a moderate one. For this purpose, start from an initial guess, say a small prime number:

$$p \leftarrow 5.$$

Does it divide n? If not, then pick it as our desired coprime. If, on the other hand, p divides n, then we must keep looking:

$$p \leftarrow \text{ the next prime number,}$$

and so on. In at most $(\log_2 n)$ guesses, we will eventually find a new prime number p that does not divide n any more. This will be our final p.

17.2.2 Modular Multiplication

Let us consider yet another task. Let n and j be two natural numbers. How to calculate the product

$$nj \bmod m?$$

This is not so easy: both n and j could be very long and contain many digits. Their product nj could be even longer and not easy to store on the computer and work with. How to avoid this? Let us decompose them (as in Sect. 17.1.2):

$$n = \left\lfloor \frac{n}{m} \right\rfloor m + (n \bmod m)$$
$$j = \left\lfloor \frac{j}{m} \right\rfloor m + (j \bmod m).$$

In the right-hand side, only the latter term is interesting. The former term, on the other hand, is a multiple of m, which is going to drop anyway. Therefore, instead of nj, better calculate

$$(n \bmod m)(j \bmod m) \bmod m.$$

What did we do here? We took the modulus *before* starting to multiply. This way, we only multiply moderate numbers, which never exceed $m - 1$.

17.2.3 Modular Power

Here is a more difficult job: calculate a power like

$$n^k \bmod m.$$

We have the same problem as before: n^k could be too long to store or use. How to avoid this? In the algorithm in Chap. 8, Sect. 8.5.2, before using n^2, takes modulus m:

$$\text{power}(n, k, m) = \begin{cases} 1 & \text{if } k = 0 \\ n \bmod m & \text{if } k = 1 \\ n \cdot \text{power}(n^2 \bmod m, (k-1)/2, m) \bmod m & \text{if } k > 1 \text{ and } k \text{ is odd} \\ \text{power}(n^2 \bmod m, k/2, m) & \text{if } k > 1 \text{ and } k \text{ is even.} \end{cases}$$

Better yet, before starting the calculation, substitute

$$n \leftarrow (n \bmod m).$$

This avoids the big number n^2 in the first place. Instead, the recursion is applied to the moderate number

$$(n \bmod m)^2 \leq (m-1)^2.$$

All intermediate products are now moderate and never exceed $(m-1)^2$. To prove this, use mathematical induction on k (see exercises below).

Still, isn't this a little too pedantic? After all, taking the modulus is not for free. Why apply it so often? Better apply it only when absolutely necessary: when detecting an inner number too long to store or use efficiently.

17.2.4 Modular Inverse

Let p and q be some given coprime numbers (prime or not):

$$\mathrm{GCD}(p, q) = 1.$$

Here is an important task: find the inverse of q modulus p, denoted by

$$q^{-1} \bmod p.$$

What is this? This is the unique solution x of the equation

$$qx \equiv 1 \bmod p.$$

Does x exist? Is it unique? To prove this, consider the set S, containing the integer numbers from 0 to $p - 1$:

$$S = \{0, 1, 2, \ldots, p - 1\}.$$

Define the new mapping

$$M : S \to S$$

by

$$M(s) = (qs \bmod p), \quad s \in S.$$

What number is mapped to 1? This will be our x. To find it, let us look at M, and study it.

Is M one-to-one? To find out, consider two numbers $a, b \in S$ (say, $a \geq b$). Assume that

$$M(a) = M(b).$$

This means that

$$q(a - b) \equiv 0 \bmod p,$$

or

$$p \mid q(a - b).$$

But p shares no common divisor with q. Therefore, it must divide $a - b$:

$$p \mid a - b,$$

or

$$a \equiv b \bmod p.$$

Thus, M is indeed one-to-one. As such, it preserves the total number of numbers in S:

$$|M(S)| = |S| = p.$$

Thus, M is a one-to-one mapping from S onto S. Thus, it has an inverse mapping:

$$M^{-1} : S \to S.$$

In particular, we can now define

$$x = M^{-1}(1).$$

In summary, the inverse of q modulus p exists uniquely:

$$\left(q^{-1} \bmod p\right) = M^{-1}(1).$$

17.2.5 How to Find the Modular Inverse?

How to solve for x in practice? For this purpose, assume that

$$q < p.$$

(Otherwise, just substitute

$$q \leftarrow q - p$$

time and again, until q becomes small enough. After all, this makes no difference to x.) Now, use the modular extended Euclidean algorithm (Sect. 17.1.7). This way, we can now write

$$1 = \text{GCD}(p, q) \equiv (ap + bq) \equiv bq \bmod p,$$

for some integer numbers

$$0 \leq a, b < p.$$

Finally, define

$$x = b.$$

So, we have got our inverse:

$$\left(q^{-1} \bmod p\right) = b.$$

This will be useful below.

17.3 The Chinese Remainder Theorem

17.3.1 Modular Equation

We are now ready to prove the Chinese remainder theorem. This is useful: it will also give us a practical algorithm.

Consider k natural numbers (prime or not), greater than 1:

$$p_1,\ p_2,\ p_3,\ \ldots,\ p_k > 1.$$

Assume that they are mutually coprime: if you pick any two of them, then they share no common divisor:

$$\mathrm{GCD}(p_i, p_j) = 1 \quad (1 \leq i < j \leq k).$$

Consider now the following equation:

$$x \equiv d_1 \bmod p_1,$$

where $0 \leq d_1 < p_1$ is a given (integer) number, and x is the (integer) unknown. What could x be? Easy:

$$x = d_1.$$

But this is not what we want. We want x to have a different form.

17.3.2 How to Use the Coprime?

Let P be the product of all the p_i's:

$$P = p_1 p_2 p_3 \cdots p_k.$$

Likewise, let P_1 be the product of all the p_i's but the first one:

$$P_1 = p_2 p_3 p_4 \cdots p_k = \frac{P}{p_1}.$$

We want x to have the special form

$$x = P_1 x_1.$$

(Later on, we will see why.) To have such an x, we must first have x_1. What could x_1 be? Well, we already know what it is. After all, p_2, p_3, p_4, ..., p_k are coprime to p_1. Therefore, their product P_1 is coprime to p_1 as well:

$$\mathrm{GCD}(P_1, p_1) = 1.$$

Let $\tilde{x}_1$ be the inverse of P_1 modulus p_1:

$$P_1 \tilde{x}_1 \equiv 1 \bmod p_1.$$

($\tilde{x}_1$ could be calculated by the extended Euclidean algorithm, as in Sects. 17.2.4–17.2.5). Now, define

$$x_1 = d_1 \tilde{x}_1.$$

This way,

$$P_1 x_1 \equiv P_1 d_1 \tilde{x}_1 \equiv d_1 \bmod p_1,$$

as required. So, $P_1 x_1$ is indeed a legitimate solution to our original equation.

Better yet, let us design a more general solution: to $P_1 x_1$, add a multiple of p_1. For example, add P_2: the product of all the p_i's but p_2:

$$P_2 = p_1 p_3 p_4 p_5 \cdots p_k = \frac{P}{p_2}.$$

This designs a new solution:

$$P_1 x_1 + P_2.$$

Indeed, this addition makes no harm: since P_2 is a multiple of p_1, our original equation still holds.

So far, we had just one equation. Next, let us introduce more equations.

17.3.3 Modular System of Equations

Let us move on to a complete system of k modular equations:

$$\begin{aligned} x &\equiv d_1 \bmod p_1 \\ x &\equiv d_2 \bmod p_2 \\ x &\equiv d_3 \bmod p_3 \\ &\vdots \\ x &\equiv d_k \bmod p_k, \end{aligned}$$

where

$$\begin{aligned} 0 \le d_1 &< p_1 \\ 0 \le d_2 &< p_2 \\ 0 \le d_3 &< p_3 \\ &\vdots \\ 0 \le d_k &< p_k \end{aligned}$$

are given (integer) numbers. For this new system, our old x is not good enough any more: it only solves the first equation but not the others. How to improve it?

So far, we solved the first equation alone. Next, let us do the same for each individual equation as well. For this purpose, define new products of the form

$$\begin{aligned} P_1 &= \frac{P}{p_1} \\ P_2 &= \frac{P}{p_2} \\ P_3 &= \frac{P}{p_3} \\ &\vdots \\ P_k &= \frac{P}{p_k}. \end{aligned}$$

Each product contains all of the p_i's but one. Let us use this to define more equations:

$$P_1 x_1 \equiv d_1 \bmod p_1$$

$$\begin{aligned} P_2 x_2 &\equiv d_2 \bmod p_2 \\ P_3 x_3 &\equiv d_3 \bmod p_3 \\ &\vdots \\ P_k x_k &\equiv d_k \bmod p_k. \end{aligned}$$

We already know how to solve each individual equation on its own. We are now ready to define our new x:

$$x = P_1 x_1 + P_2 x_2 + P_3 x_3 + \cdots + P_k x_k.$$

This x indeed solves all the equations at the same time. For example, it solves the first equation. Indeed, it is not much different from $P_1 x_1$: it only contains a few more multiples of p_1 that make no harm. The same is true for the other equations as well, as required.

17.3.4 Uniqueness

In the above, there is still a problem: x could get too big. How to make x moderate? For this purpose, subtract P time and again. After all, as discussed above, this makes no harm: since P is the product of all the p_i's, all equations still hold. Thus, we can substitute

$$x \leftarrow x - P$$

time and again, until x gets as small as

$$0 \leq x < P.$$

In this range, x is unique. Indeed, let

$$0 \leq y < P$$

be yet "another" solution of the same system. Let us show that y is not really different from x. Indeed, assume that

$$x \geq y.$$

(Otherwise, just interchange them.) Now, look at the difference:

$$0 \leq x - y < P.$$

It solves the homogeneous system:

$$\begin{aligned} x - y &\equiv 0 \bmod p_1 \\ x - y &\equiv 0 \bmod p_2 \\ x - y &\equiv 0 \bmod p_3 \\ &\vdots \\ x - y &\equiv 0 \bmod p_k. \end{aligned}$$

In other words, $x - y$ is divided by all the p_i's:

$$p_1,\ p_2,\ p_3, \ldots,\ p_k \ \mid\ x - y.$$

Thus, their product divides $x - y$ as well:

$$P \ \mid\ x - y.$$

Therefore,

$$x - y = 0,$$

or

$$x = y.$$

Thus, in the above range, x is indeed unique. This is the Chinese remainder theorem.

17.4 How to Use the Remainder?

17.4.1 Integer Decomposition

Let us use the remainder in practice. As in Sect. 17.1.1, let us decompose n once again, with different notations:

$$n = lm + r,$$

where $l \geq 0$ is a nonnegative integer number, and

$$0 \leq r < m$$

is the remainder. This will be useful below.

17.4.2 Binary Number

But this was not the first time we used the remainder. In fact, we have already seen the remainder earlier (but only implicitly). In Chap. 8, Sect. 8.4.2, we wrote a natural number k in its binary form:

$$k = b_m b_{m-1} b_{m-2} \cdots b_1 b_0,$$

where b_m, b_{m-1}, …, b_1, b_0 are the binary digits (0 or 1), ordered rightwards: from the most significant to the least significant. This is actually a binary polynomial:

$$k = b_0 + b_1 \cdot 2 + b_2 \cdot 2^2 + \cdots + b_m \cdot 2^m = \sum_{j=0}^{m} b_j 2^j = q(2),$$

where

$$q(x) = \sum_{i=0}^{m} b_i x^i,$$

and x is the independent variable.

Given a natural number k, how to uncover its binary form? For this purpose, just divide by 2 with remainder:

$$k = 2l + b_0.$$

Here, the remainder is b_0: the least significant digit. What are the other digits? To find out, use recursion on the above l. The least significant binary digit of l will then serve as b_1, and so on. Let us extend this to a more general polynomial as well.

17.4.3 Horner's Algorithm

The same could be done in a more general polynomial as well. In fact, in Chap. 8, Sect. 8.2.4, we looked at a polynomial of degree n:

$$p(x) = \sum_{i=0}^{n} a_i x^i,$$

where x is the (independent) variable, and the a_i's are given (real or even complex) coefficients.

Now, the monomial x is a polynomial in its own right (although quite short). How to divide by x with remainder? Easy: as before, decompose p as

$$p(x) = a_0 + x\,\hat{p}(x),$$

where $\hat{p}(x)$ is a shorter polynomial:

$$\hat{p}(x) = a_1 + a_2 x + a_3 x^2 + \cdots + a_n x^{n-1} = \sum_{i=0}^{n-1} a_{i+1} x^i.$$

In Chap. 8, Sect. 8.2.4, we have used this to help calculate the value of p at x. Here, on the other hand, we want more: to actually divide p with remainder.

But we have already got our remainder: the constant a_0. After all, this is a "polynomial" in its own right, even shorter than x. In fact, x is of degree 1, whereas a_0 is of degree 0 only. The same could now be repeated recursively for $\hat{p}$ as well. This will recover all coefficients again, one by one. (this has a theoretical value only, because we already know them.)

In summary, most of what we have done in number theory could now be extended to polynomials as well. In particular, this applies to the Chinese remainder theorem as well. This will have a nice application in linear algebra.

17.5 Polynomials and the Chinese Remainder Theorem

17.5.1 Characteristic Polynomial: Root and its Multiplicity

Let us go back to our $n \times n$ matrix A. Let us write its characteristic polynomial in a slightly different form:

$$P(x) = (x - \lambda_1)^{n_1} (x - \lambda_2)^{n_2} (x - \lambda_3)^{n_3} \cdots (x - \lambda_k)^{n_k},$$

where x is the independent variable. As before, the λ_i's are the *distinct* eigenvalues of A. These are also the roots of P. In fact, at $x = \lambda_i$, P vanishes:

$$P(\lambda_i) = 0, \quad 1 \le i \le k.$$

Each λ_i comes with its own multiplicity: n_i. Together, the multiplicities sum to n:

$$n_1 + n_2 + n_3 + \cdots + n_k = n.$$

17.5.2 Multiplicity and Jordan Subspace

But the multiplicity n_i has yet another role: it is also the dimension of V_i, the null space of $(A - \lambda_i I)^{n_i}$:

$$V_i = \left\{ v \mid (A - \lambda_i I)^{n_i} v = \mathbf{0} \right\}$$

(Chap. 16, Sect. 16.3.4).

The new subspace V_i is the ith Jordan subspace. This will be useful in the Jordan decomposition below.

We are now ready to rewrite the Chinese remainder theorem: not with numbers, but with polynomials. Instead of the p_i's, we will have new polynomials: factors from the characteristic polynomial P.

17.5.3 The Chinese Remainder Theorem with Polynomials

In the Chinese remainder theorem, the p_i's are mutually coprime numbers. Here, on the other hand, we replace numbers with polynomials. After all, we have already seen that numbers are just a special kind of polynomials. So, consider now a new system:

$$\begin{aligned} Q &\equiv \lambda_1 \bmod (x - \lambda_1)^{n_1} \\ Q &\equiv \lambda_2 \bmod (x - \lambda_2)^{n_2} \\ Q &\equiv \lambda_3 \bmod (x - \lambda_3)^{n_3} \\ &\vdots \\ Q &\equiv \lambda_k \bmod (x - \lambda_k)^{n_k} . \end{aligned}$$

If A is nonsingular, then add one more equation:

$$Q \equiv 0 \bmod x .$$

(In a moment, we will see why.) If, on the other hand, A is singular, then there is no need, because 0 is already an eigenvalue. For example, if $\lambda_1 = 0$, then we already have a stronger equation:

$$Q \equiv 0 \bmod x^{n_1} .$$

Instead of adding an extra equation, one could just raise the power and obtain a yet stronger equation:

$$Q \equiv 0 \bmod x^{n_1+1} .$$

This is not a must: we do it only for uniformity. Our system is now ready.

In this system, what is Q? This is the unknown polynomial: $Q(x)$. In each equation, Q is congruent to a "polynomial" of degree zero: the constant λ_i. This congruence is modulus $(x - \lambda_i)^{n_i}$.

17.6 The Jordan Decomposition

17.6.1 Properties of Q

Thanks to the Chinese remainder theorem, Q indeed exists. After all, in our system, on the right-hand side, we have mutually coprime polynomials. Furthermore, Q is unique modulus xP. In other words, Q is the only polynomial of degree n or less that solves the system: no other polynomial of degree n or less could be added to it.

How does Q look like? Thanks to the extra equation in Sect. 17.5.3, Q has the factor x in it. Thus, once written as a sum, Q has no constant (free) term at all. It starts from the linear monomial x^1, x^2, ... So, 0 is a root as well:

$$Q(0) = 0.$$

We can now plug A for x and obtain $Q(A)$: a linear combination of A and its powers. This is a legitimate $n \times n$ matrix in its own right.

17.6.2 The Diagonal Part

Look at the difference $Q(A) - \lambda_i I$. Thanks to the ith equation, it is the same as $(A - \lambda_i I)^{n_i}$ times some other polynomial. On V_i, this vanishes. So, on V_i, $Q(A)$ acts like a diagonal matrix: $\lambda_i I$. This is true for every $1 \leq i \leq k$.

We therefore say that $Q(A)$ is diagonalizable (or semisimple): in some basis (the Jordan basis), it is diagonal: on its main diagonal, it contains the eigenvalues of A (as in the Jordan form).

17.6.3 The Nilpotent Part

Finally, $Q(A)$ has another attractive property: it approximates A well. In what sense? Look at the difference $A - Q(A)$. Thanks to the ith equation, it satisfies

$$A - Q(A) \equiv A - \lambda_i I \bmod (A - \lambda_i I)^{n_i}.$$

Once restricted to V_i, this becomes even simpler:

$$(A - Q(A)) \mid_{V_i} = (A - \lambda_i I) \mid_{V_i} .$$

What is the n_ith power of this? Since V_i is invariant under $A - \lambda_i I$, we simply have

$$(A - Q(A))^{n_i} \mid_{V_i} = (A - \lambda_i I)^{n_i} \mid_{V_i} = (0).$$

Thus, every greater power is zero as well:

$$(A - Q(A))^{n} \mid_{V_i} = (0).$$

This is true for all $1 \leq i \leq k$. In summary, the difference is nilpotent:

$$(A - Q(A))^{n} = (0).$$

(In the Jordan basis, the difference is just the superdiagonal, which is indeed nilpotent.) In this sense, $Q(A)$ indeed approximates A well.

17.6.4 The Jordan Decomposition

In summary, thanks to the Chinese remainder theorem, we have decomposed A as the sum of two parts: the diagonal part, plus a less important part—the nilpotent part:

$$A = Q(A) + (A - Q(A)) .$$

Let us use this in a few examples.

17.7 Example: Space of Polynomials

17.7.1 Polynomial and its Differentiation

Consider a polynomial of degree n (or less):

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = \sum_{i=0}^{n} a_i x^i .$$

The coefficients could be placed in an $(n + 1)$-dimensional vector:

$$(a_0, a_1, a_2, \ldots, a_n) .$$

On polynomials like p, define the differentiation operator:

$$Dp = p'.$$

This is a linear operator, so it has a Jordan form. In fact, its Jordan decomposition is very simple:

$$D = (0) + D.$$

Why? Because D is nilpotent. Indeed, its nth power is the nth derivative of p, which is just a constant:

$$D^n p = p^{(n)} = n! a_n$$

(Chap. 8, Sect. 8.6.3). For this reason, the next derivative vanishes:

$$D^{n+1} p = p^{(n+1)} = 0.$$

This is true for every polynomial p. Thus, D is indeed nilpotent: its $(n+1)$st power vanishes.

17.7.2 The Jordan Basis

D has just one Jordan block. To see this, let us design the Jordan basis: n linearly independent polynomials of degree n or less. For a start, pick a "polynomial" of degree zero: the constant 1. Is this an eigenvector of D? It sure is:

$$D1 = 1' = 0.$$

So, the eigenvalue is just zero. Next, pick an antiderivative: the linear monomial x. Indeed,

$$Dx = x' = 1.$$

Next, pick an antiderivative of x: $x^2/2$, and so on. This produces the entire Jordan basis:

$$\left\{1, x, \frac{x^2}{2}, \frac{x^3}{3!}, \ldots, \frac{x^n}{n!}\right\}.$$

17.7.3 The Jordan Block

In this basis, how does D look like? Well, D maps each generalized eigenvector to the previous one, until the first one, mapped to zero. This is what we wanted: D indeed takes its Jordan form:

$$D = \begin{pmatrix} 0 & 1 & & & & \\ & \ddots & \ddots & & & \\ & & \ddots & \ddots & & \\ & & & 0 & 1 & \\ & & & & 0 & 1 \\ & & & & & 0 \end{pmatrix}.$$

This is indeed an $(n+1) \times (n+1)$ matrix, as required. It has n nonzero elements: on the superdiagonal.

17.7.4 The Jordan Block and its Powers

D^2, on the other hand, has just $n-1$ nonzero elements—on the next higher superdiagonal:

$$D^2 = \begin{pmatrix} 0 & & 1 & & & \\ & \ddots & & \ddots & & \\ & & \ddots & & \ddots & \\ & & & 0 & & 1 \\ & & & & 0 & \\ & & & & & 0 \end{pmatrix}.$$

The higher powers of D, on the other hand, have yet fewer nonzero elements:

$$D^3 = \begin{pmatrix} 0 & & & 1 & & \\ & \ddots & & & \ddots & \\ & & \ddots & & & \ddots \\ & & & 0 & & \\ & & & & 0 & \\ & & & & & 0 \end{pmatrix}$$

$$\vdots$$

$$D^n = \begin{pmatrix} 0 & & & & & 1 \\ & \ddots & & & & \\ & & \ddots & & & \\ & & & 0 & & \\ & & & & 0 & \\ & & & & & 0 \\ & & & & & & 0 \end{pmatrix}.$$

Finally, the next power vanishes:

$$D^{n+1} = (0).$$

After all, we have already seen that D is nilpotent.

17.8 Exercises: Numbers—Polynomials

17.8.1 Natural Numbers: Binary Form

1. Consider a natural number n.
2. Write it in its binary form.
3. What is the rightmost (least significant) binary digit?
4. Assume that n is even. What is its least significant binary digit? Hint: 0.
5. Assume that n is odd. What is its least significant binary digit? Hint: 1.
6. No matter whether n is odd or even, what is its least significant binary digit? Hint: n mod 2.
7. Divide n by 2 with remainder. What is the remainder? Hint: n mod 2.
8. How is the remainder related to the least significant binary digit? Hint: they are the same.
9. What is the next (more significant) binary digit? Hint: divide n by 2 with remainder. Disregard the remainder. Look at the result: $\lfloor n/2 \rfloor$. Use the same trick to find its own least significant binary digit: divide $\lfloor n/2 \rfloor$ by 2, and look at the new remainder.
10. What are the rest of the binary digits in n? Hint: repeat the above recursively (Sect. 17.4.2).
11. How is this related to Horner's algorithm? Hint: both use the same recursion: divide by x (or 2) with remainder, to shift a list of numbers time and again.
12. What is the difference? Hint: in the above algorithm, we shift the (unknown) binary digits time and again, to uncover them, one by one. In Horner's algorithm, on the other hand, we work the other way around: shift the (well-known) coefficients time and again, and use them one by one, to find the (unknown) value.

13. How to add two (arbitrarily long) natural numbers? Hint: write the longer number in the top line. Write the shorter number just below it. Start from the rightmost (least significant) digit. Add digit by digit, all the way leftwards. (Do not forget the carry!)
14. How to add two polynomials to each other? Hint: likewise: monomial by monomial. (But there is no carry.)
15. How to subtract two natural numbers from each other? Hint: assume that the first one is longer. (Otherwise, interchange them, subtract as below, and attach a minus sign to the result.) Write it in the top line. Write the shorter number just below it. Start from the rightmost digit. Work vertically, digit by digit, all the way leftwards.
16. How to subtract two polynomials from each other? Hint: likewise, monomial by monomial.
17. How to multiply two natural numbers by each other? Hint: vertically, leftwards, digit by digit.
18. Is there a better way? Hint: try FFT (Chap. 5 in [75], and Chap. 14 in [77]).
19. How to multiply two polynomials by each other? Hint: likewise, monomial by monomial.
20. Is there a better way?
21. How to divide two natural numbers by each other (with remainder)? Hint: vertically, rightwards, digit by digit (assuming that the second one is nonzero).
22. How to divide two polynomials by each other (with remainder)? Hint: likewise, monomial by monomial.
23. After the division, look at the remainder. What is its meaning? Hint: this is the result of the modulus operation: the first number (or polynomial) modulus the second one.
24. Use the above to write the Euclidean algorithm for polynomials as well. This will find their greatest common divisor: a new polynomial that divides them both (evenly, with no remainder).
25. Write the extended Euclidean algorithm (Sect. 17.1.5) for polynomials as well. This will write their greatest common divisor as their "sum," with some "coefficients," which are now polynomials in their own right.
26. Apply the Chinese remainder theorem, with polynomials instead of numbers.
27. Why is this legitimate?
28. If $\lambda_1 \neq \lambda_2$, are the polynomials $(x - \lambda_1)^{n_1}$ and $(x - \lambda_2)^{n_2}$ coprime?
29. Look at the solution Q. What is its degree? Hint: n at most.
30. In what sense is Q unique? Hint: no other polynomial of degree n (or less) could be added to it.
31. Show that, if A is an $n \times n$ matrix, then $Q(A)$ is an $n \times n$ matrix as well.
32. Show that, in its own Jordan basis, $Q(A)$ is diagonal.
33. This means that $Q(A)$ is diagonalizable (semisimple).
34. Show that the difference $A - Q(A)$ is nilpotent.

Chapter 18
Algebras and Their Derivations and Their Jordan Form



So far, we have designed the Jordan decomposition of a matrix. Let us extend this to a more general algebraic structure: an algebra. On the algebra, one can define (linear) operators. In particular, we will define a special kind of operator: a derivation.

In the algebra, we'll then specify a subalgebra. In it, the derivation has an interesting Jordan decomposition. This will get you ready for more advanced material: Lie algebras and Cartan's criterion [38, 41, 76].

18.1 Eigenfunctions

18.1.1 Polynomials of Any Degree

So far, we considered polynomials of a fixed degree: n. Still, n was never specified: it could be as large as you like. Thus, one may now ask: why stop at n at all? Why not consider the infinite-dimensional space of all polynomials? This way, our differentiation operator D will be an infinite matrix (or operator), with the same Jordan decomposition as before:

$$D = (0) + D,$$

where D is now "nilpotent" in a nonstandard sense: a given polynomial p, if differentiated sufficiently many times, gives zero. For instance, if p is of degree n, then

$$D^{n+1} p = 0.$$

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Still, $n + 1$ is no longer global: it depends on p. This is no longer nilpotency in its standard sense.

As before, there is just one Jordan subspace, which is now very big: it contains all polynomials, of any degree. This new space is infinite-dimensional. Let us go ahead and extend it yet more.

18.1.2 Eigenfunction

Consider now the exponent function $\exp(\lambda x)$, where x is the independent variable, and λ is a (real or even complex) parameter. How to differentiate? Well, this is a composite function:

$$D\exp(\lambda x) = \lambda \exp'(\lambda x) = \lambda \exp(\lambda x).$$

This is an eigenfunction of D, with eigenvalue λ. Thus, it is in the null space of $D - \lambda$:

$$(D - \lambda)\exp(\lambda x) = 0.$$

Thus, it belongs to a new Jordan subspace. How to design it?

18.1.3 The Leibniz (Product) Rule

To design the entire Jordan subspace, let us consider a product: an exponent times a polynomial. How to differentiate this? As in the Leibniz rule (or the product rule):

$$D(p\exp(\lambda x)) = (Dp)\exp(\lambda x) + pD\exp(\lambda x).$$

18.1.4 Mathematical Induction

From both sides of this equation, subtract the same thing:

$$(D - \lambda)(p\exp(\lambda x)) = (Dp)\exp(\lambda x).$$

Could this be extended to a yet higher power:

$$(D - \lambda)^k(p\exp(\lambda x)) = (D^k p)\exp(\lambda x)?$$

To prove this, use mathematical induction on $k = 0, 1, 2, \ldots$:

$$\begin{aligned}(D-\lambda)^{k+1}(p\exp(\lambda x)) &= (D-\lambda)^k(D-\lambda)(p\exp(\lambda x)) \\ &= (D-\lambda)^k((Dp)\exp(\lambda x)) \\ &= ((D^{k+1}p)\exp(\lambda x))\end{aligned}$$

(because Dp is a polynomial in its own right). So, we proved that

$$(D-\lambda)^k(p\exp(\lambda x)) = (D^k p)\exp(\lambda x), \quad k \geq 0.$$

(In particular, if k is bigger than the degree of p, then this is zero.) So, we got our (infinite-dimensional) Jordan subspace: all polynomials times $\exp(\lambda x)$. What Jordan basis spans this? Take all monomials, times $\exp(\lambda x)$:

$$\operatorname{span}_{k=0}^{\infty}\left\{x^k\exp(\lambda x)\right\}.$$

This is one Jordan subspace. Now, the same could be done for each and every (real or even complex) λ. In summary, we have infinitely many Jordan subspaces. This way, D acts on a rather big function space, containing all polynomials times all exponents. This is spanned by its basis: all monomials, times all exponents:

$$\operatorname{span}_{k\geq 0,\lambda\in\mathbb{C}}\left\{x^k\exp(\lambda x)\right\}.$$

This has a new name: subalgebra.

18.2 Algebra and Its Derivation

18.2.1 Leibniz Rule

What is the Leibniz rule? So far, it was the same as the product rule, telling us how to differentiate a product of two functions:

$$(fg)' = f'g + fg'.$$

Let us extend this: f and g can now be not only functions but also members of a new structure: an algebra. What is this? This is a vector space (or linear space) with one more operation: multiplication (product).

18.2.2 Product (Multiplication)

In a linear (vector) space, pick two vectors. You can always add them and obtain a new vector. So, the linear space is closed under addition. But how about multiplication? Inner product is not good enough: it makes a scalar, not a vector. Vector product is also not good enough: it is limited to 3-D only. So, in general, we cannot multiply.

In an algebra, on the other hand, we can: if f and g are in the algebra, then their product fg is in the algebra as well. In other words, the algebra is closed under multiplication. This is a vector-times-vector (not scalar-times-vector) multiplication. It could be rather strange: it must be distributive (bilinear), but not necessarily commutative or even associative. We have already seen one example: vector product in 3-D.

18.2.3 Derivation

In our algebra, consider now a new linear operator D that maps each member to some other member. Here, D is an extended differentiation: a derivation. To deserve this name, D must obey a new Leibniz rule:

$$D(fg) = (Df)g + fDg.$$

This way, D indeed behaves like a legitimate derivation: to act on a product, first act on the first member, then on the second member, then sum up. This is indeed a natural extension of the good old differentiation.

18.3 Product and Its Derivation

18.3.1 Two-Level (Virtual) Binary Tree

Next, let us use two (real or even complex) parameters: τ and λ. In our algebra, consider the operator $D - \tau - \lambda$. When applied to a product, it obeys the new Leibniz rule:

$$(D - \tau - \lambda)(fg) = ((D - \tau)f)g + f(D - \lambda)g.$$

This way, the original operator $D - \tau - \lambda$ splits into two parts: $D - \tau$ acts on f, and $D - \lambda$ acts on g. This makes a two-level virtual binary tree (Fig. 18.1).

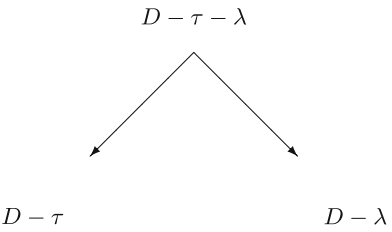


Fig. 18.1 At the head, we have the original operator $D-\tau-\lambda$. At the bottom, it splits: $D-\tau$ on the left, and $D-\lambda$ on the right

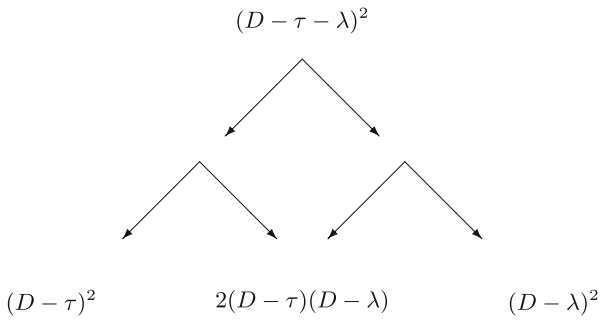


Fig. 18.2 At the head, we have the original operator $(D-\tau-\lambda)^2$. At the bottom, it splits: $(D-\tau)^2$ on the left, $(D-\lambda)^2$ on the right, and $2(D-\tau)(D-\lambda)$ at the middle, as in Pascal's triangle

What is so good about this? Well, if f and g are eigenvectors of D with the eigenvalues τ and λ (respectively), then fg is an eigenvector as well, with the new eigenvalue $\tau+\lambda$ (provided that $fg \neq \mathbf{0}$). Let us extend this to a higher power.

18.3.2 Multilevel Tree

How does $(D-\tau-\lambda)^2$ split? It makes a three-level tree (Fig. 18.2). More precisely, at the middle, both branches meet and add up, introducing the new coefficient 2, as in the second row in Pascal's triangle:

$$\begin{aligned}(D-\tau-\lambda)^2(fg) &= (D-\tau-\lambda)(D-\tau-\lambda)(fg)\\ &= (D-\tau-\lambda)\left(((D-\tau)f)\,g + f(D-\lambda)g\right)\\ &= \Big((D-\tau)^2 f\Big)\,g + 2(D-\tau)f(D-\lambda)g + f(D-\lambda)^2 g.\end{aligned}$$

18.3.3 Pascal's Triangle and the Binomial Formula

This is how the first and second powers split. What about the kth power? It splits as in the binomial formula—as in the kth row in Pascal's triangle:

$$(D-\tau-\lambda)^k(fg)=\sum_{i=0}^{k}\binom{k}{i}(D-\tau)^i f(D-\lambda)^{k-i}g.$$

18.4 Product and Its Jordan Subspace

18.4.1 Two Members from Two Jordan Subspaces

Does D have a Jordan decomposition? We do not know! After all, the Jordan decomposition was designed only in finite dimension (Chap. 17, Sect. 17.6.4). Here, on the other hand, we could have infinitely many dimensions.

Still, we could work the other way around: start from two Jordan subspaces, and merge them into a new Jordan subspace, and so on. In the end, we design a complete subalgebra, in which D indeed has a Jordan form.

For this purpose, assume that D has (at least) two Jordan subspaces, corresponding to two eigenvalues: τ and λ. Assume that f belongs to the former, and g to the latter. What about their product fg? Does it belong to any Jordan subspace?

18.4.2 Product and Its New Jordan Subspace

To check on this, look at the sum in Sect. 18.3.3. Only a few terms survive. Most terms, on the other hand, contain a high power and vanish.

More precisely, in each term, there is a high power: either of $D-\tau$ (applied to f) or of $D-\lambda$ (applied to g). Let us make k so big that all terms vanish. This way, fg indeed belongs to a new Jordan subspace, corresponding to a new eigenvalue: $\tau+\lambda$.

18.4.3 Example: Polynomials Times Exponents

As a matter of fact, we have already seen an example: polynomial times an exponent. Each belongs to its own Jordan subspace: the polynomial is associated with eigenvalue 0, and the exponent with eigenvalue λ.

What about their product? Does it belong to any Jordan subspace? It sure does: the Jordan subspace corresponding to the sum: the "new" eigenvalue

$$0+\lambda=\lambda.$$

Thanks to our new multilevel tree (Fig. 18.2), we can now extend this to $\tau\neq 0$ as well. For this purpose, consider two polynomials: q and p. Consider also two

functions: $q \cdot \exp(\tau x)$ and $p \cdot \exp(\lambda x)$. The former is in the Jordan subspace associated with τ. The latter, on the other hand, is in the Jordan subspace associated with λ. What about their product

$$q \ \exp(\tau x) \cdot p \ \exp(\lambda x) = (qp) \exp((\tau + \lambda)x)?$$

It is indeed in a new Jordan subspace, associated with $\tau + \lambda$.

18.5 Derivation on a Subalgebra

18.5.1 Restriction to a Subalgebra

Still, our original algebra may be too big: it may contain members that belong to no Jordan subspace at all. To fix this, let us restrict our discussion to a subalgebra, containing only the generalized eigenvectors of D (and their linear combinations).

Why is this a legitimate subalgebra? Because it is closed under multiplication (as we have seen above).

18.6 More Examples

18.6.1 Smooth Functions

Look at an interesting algebra: C^∞. This contains all smooth functions, differentiable as many times as you like.

But this is too big. Better focus on a subalgebra: polynomials times exponents (and their linear combinations). In it, differentiation has a Jordan form. To see this, split it into many (disjoint) Jordan subspaces, each of which has a unique exponent, times just any polynomial.

In C^∞ differentiation has no Jordan form. Only in the subalgebra does differentiation have a Jordan form. Let us see yet another example.

18.6.2 Finite Dimension

Consider some finite-dimensional algebra. In Chap. 17, Sect. 17.6.4, we have already seen that every linear mapping (including our D) has a Jordan form.

But our D is also a derivation: it obeys the new Leibniz rule. Thus, there is no need to design any subalgebra: just use the original algebra. Once f and g are

picked from two Jordan subspaces, fg must belong to a Jordan subspace as well, corresponding to the sum of their eigenvalues.

Finally, let us see an interesting property. (This will be relevant in finite or infinite dimension.)

18.7 The Diagonal Part

18.7.1 Is It a Derivation?

In our subalgebra, we can safely decompose D in its Jordan decomposition:

$$D = \Lambda + (D - \Lambda),$$

where Λ is diagonal, and the difference $D - \Lambda$ is nilpotent. If our subalgebra is infinite-dimensional, then this "nilpotency" is only nonstandard (Chap. 17, Sect. 18.1.1).

D is a derivation: it obeys the new Leibniz rule. What about Λ? Is it a derivation in its own right? Does Λ obey a Leibniz rule as well? To check on this, pick two members of the subalgebra: f and g. Assume that f belongs to the Jordan subspace associated with τ. Because Λ is diagonal,

$$\Lambda f = \tau f.$$

Likewise, assume that g belongs to the Jordan subspace associated with λ. Because Λ is diagonal,

$$\Lambda g = \lambda g.$$

Now, what about their product? As discussed above, it must belong to the Jordan subspace associated with $\tau + \lambda$. Therefore,

$$\begin{aligned}\Lambda(fg) &= (\tau + \lambda)(fg)\\ &= \tau(fg) + \lambda(fg)\\ &= (\tau f)g + f\lambda g\\ &= (\Lambda f)g + f\Lambda g.\end{aligned}$$

This is nothing but Leibniz rule for Λ. This could be extended linearly. In summary, we have an interesting property: Λ is a legitimate derivation in its own right.

18.8 Exercises: Derivation and Its Exponent

18.8.1 Exponent—Product Preserving

1. What is an algebra? Hint: a vector space with a new operation: product or multiplication (Sect. 18.2.1).
2. This is a vector-times-vector multiplication: do not confuse it with the usual scalar-times-vector multiplication.
3. Is the algebra closed under multiplication? In what sense? Hint: the product is in the algebra as well.
4. What are the properties of this product? Hint: distributive (bilinear), but not necessarily commutative or even associative (see below).
5. Let D be a derivation in the algebra (Sect. 18.2.3). Let f and g be some members of the algebra. Write a Leibniz rule for D. Hint:

$$D(fg) = Df \cdot g + f\,Dg.$$

6. Assume that D is nilpotent. What does this mean? Hint: for some (sufficiently large) m, D^m is the zero mapping: it maps everything to $\mathbf{0}$.
7. Show that, for every $l \geq m$, D^l is the zero mapping as well. Hint: as a linear mapping, $D\mathbf{0} = \mathbf{0}$.
8. Define its exponent:

$$\exp(D) = \sum_{i=0}^{\infty} \frac{D^i}{i!}.$$

9. Is this really infinite? Hint: only the m leading terms survive.
10. Show that this is already in the Jordan form. Hint: the first term is just the identity mapping. This is the diagonal part. The rest is the "tail"

$$D + \frac{D^2}{2} + \frac{D^3}{3!} + \cdots + \frac{D^{m-1}}{(m-1)!} = \sum_{i=1}^{m-1} \frac{D^i}{i!}.$$

    This is nilpotent. To see this, look at its mth power: it contains only high powers of D that vanish.
11. What is the inverse of $\exp(D)$? Hint: $\exp(-D)$.
12. Write it in terms of the above tail. Hint: take the tail, and look at its (finite) power series, with alternating signs.
13. How does the derivation D and its powers act on a product like fg? Hint: mirror the binomial formula:

$$D^n(fg) = \sum_{i=0}^{n} \binom{n}{i} D^i f \cdot D^{n-i} g$$

(Figs. 18.1, 18.2).

14. From this formula, is D^2 a derivation in its own right? Hint: no! It obeys no Leibniz rule:

$$D^2(fg) = D^2 f \cdot g + 2Df \cdot Dg + f D^2 g \neq D^2 f \cdot g + f D^2 g$$

(in general).

15. Likewise, for any $1 < l < m$, is D^l a derivation?
16. Consider the real exponent function: $\exp(x)$. Recall that

$$\exp(x + y) = \exp(x) \exp(y).$$

17. So far, we have seen how D and its powers act on a product like fg. How does $\exp(D)$? Hint: mirror the above formulas, and obtain

$$\exp(D)(fg) = \exp(D) f \cdot \exp(D) g.$$

18. In summary, thanks to the above exercises, what can you say about $\exp(D)$? Hint: it is an invertible mapping that preserves multiplication (Fig. 18.3).
19. Is it a derivation? Hint: no! It obeys no Leibniz rule:

$$\exp(D)(fg) = \exp(D) f \cdot \exp(D) g \neq \exp(D) f \cdot g + f \cdot \exp(D) g$$

(in general).

20. Let E be a mapping of the same kind: linear, invertible, product preserving, but not necessarily a derivation.
21. Let E^{-1} be its inverse.

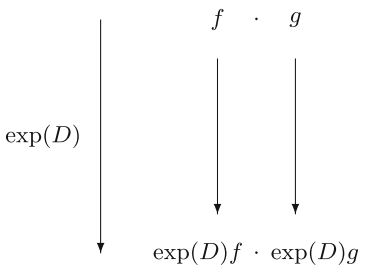


Fig. 18.3 Thanks to the binomial formula, $\exp(D)$ preserves multiplication. In other words, order does not matter: either multiply in the upper level and then apply $\exp(D)$ to fg or apply $\exp(D)$ to f and g separately, and then multiply in the lower level

22. Show that it is of the same kind: linear, invertible, product preserving, but not necessarily a derivation.
23. Look at the composition of E with its inverse:

$$EE^{-1} = E^{-1}E.$$

Show that this is just the identity mapping that changes nothing.
24. Show that the identity mapping is of the same kind: linear, invertible, product preserving, but not a derivation.
25. Look at a triple composition:

$$EDE^{-1}.$$

How does it act? Hint: leftwards, in three stages: first, apply E^{-1}. Then, apply D. Finally, apply E.
26. Is it a derivation in its own right? Hint: yes. To see this, apply it to a product of the form Ef times Eg:

$$\begin{aligned} EDE^{-1}(Ef \cdot Eg) &= ED(fg) \\ &= E(Df \cdot g + fDg) \\ &= EDfEg + EfEDg \\ &= EDE^{-1}EfEg + EfEDE^{-1}Eg. \end{aligned}$$

This is indeed a legitimate Leibniz rule.
27. What are its powers? Hint:

$$\left(EDE^{-1}\right)^l = ED^lE^{-1}$$

($l \geq 0$). Indeed, the inner E and E^{-1} cancel out.
28. Is EDE^{-1} nilpotent?
29. Does it have an exponent?
30. How does it look like? Hint:

$$\exp\left(EDE^{-1}\right) = E\exp(D)E^{-1}.$$

31. Use the original algebra to design a new algebra. Hint: see below.
32. Look at the set of linear mappings from the original algebra into itself.
33. Is this a group? Hint: no, the zero mapping has no inverse.
34. Is this a legitimate linear space?
35. To multiply two such mappings F and G, use composition: $F \circ G$.
36. How does $F \circ G$ act on a member of the original algebra? Hint: leftwards. First, apply G. On the result, apply F.

37. Is the new algebra indeed closed under this kind of multiplication? Hint: $F \circ G$ is indeed a linear mapping as well. To see this, open parentheses in $F \circ G(\alpha f + \beta g)$, where α and β are some scalars.
38. Is this multiplication distributive (bilinear)? Hint: let H be a linear mapping too. Open parentheses in $(\alpha F + \beta G) \circ H$ and also in $H \circ (\alpha F + \beta G)$.
39. Is this multiplication associative? Hint: yes, composition is.
40. Is it commutative? Hint: no, composition is not (in general).
41. Give a few examples of such linear mappings. Hint: D and its powers and $\exp(D)$.
42. Linear mappings may commute or not. Does D commute with $\exp(D)$? Hint: yes, both are polynomials in D.
43. On our new algebra, define the new operator "$F\circ$" that converts the old mapping G into the new mapping $F \circ G$.
44. Is it linear? Hint: Open parentheses in $F \circ (\alpha G + \beta H)$.
45. Is it a derivation? Hint: no! It obeys no Leibniz rule:

$$F \circ (G \circ H) = (F \circ G) \circ H \neq (F \circ G) \circ H + G \circ (F \circ H)$$

(in general).

Part VII
Linearization in Numerical Relativity

So far, we have introduced special relativity from an algebraic point of view (Chap. 4). Here, on the other hand, we also talk about a geometrical concept: spacetime.

In special relativity, our world is ideal, with no gravity at all. Thanks to spacetime, we can also move on to a more realistic setting: general relativity, with gravity. Indeed, in spacetime, gravity takes a new geometrical face.

This leads to Einstein equations. To introduce them, we need some new features in linear algebra: symbols and tensors, and algebraic operations between them.

For this purpose, we must first introduce a new principle: Einstein's summation convention. It improves on the standard sums used in linear algebra. In fact, it tells us how to raise and lower indices, and sum up easily. Thanks to this, the system of nonlinear equations gets particularly easy to introduce.

How to solve it numerically? For this purpose, we also need to linearize it on (high-order) finite elements and march in time, step by step. This is described below.

Chapter 19
Einstein Equations and their Linearization



In many applications, we seek the unknown function. Fortunately, it is good enough to approximate it as a linear combination of basis functions. Still, in such an expansion, what should the coefficients be? They are still unknown! Fortunately, we still know something about them: how they couple with each other (nonlinearly). Indeed, they solve a system of nonlinear algebraic equations.

In general relativity, on the other hand, things get even more complicated. We have not only one unknown function but ten. They are stored in a 4×4 symmetric matrix: the metric g.

Since g is symmetric, it is sufficient to store its upper-triangular part only. Each entry of the form g_{mn} $(0 \leq m \leq n \leq 3)$ is a different function, with a different expansion, with different coefficients. How to uncover them?

Fortunately, the metric solves a system of partial differential equations (PDEs). How to discretize it? Well, we already did: once each unknown function is expanded as above, we have an algebraic (discrete) system to solve. This requires a few tricks: to have the familiar kinetic energy, one must integrate by parts in space. Furthermore, to discretize the time derivatives, one must also march in time, step by step (or grid-level by grid-level). In the end, we will have a discrete algebraic system for the unknown coefficients of each and every entry in the metric. Once this system is solved numerically, we will have our desired metric: each entry in it will be expanded in terms of basis functions. This will be available in each and every time-level, as required.

Still, this system is not only big but also nonlinear: it couples different unknown functions nonlinearly. How to solve it numerically? The best method is the Newton iteration. (See Chap. 9 in [75], and Chapter 21 in [77]). It avoids nonphysical solutions and converges rapidly to the desired physical solution.

How to use the Newton iteration? In each time-level, start from an initial guess, taken from the previous time-level. To improve on it, go ahead and iterate. For this purpose, you must first linearize: differentiate with respect to each coefficient of each basis function.

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In this chapter, we describe how to linearize the Einstein equations. This was not implemented in practice yet. Still, the algorithm could lead to a stable time-marching scheme. Moreover, it could be extended to many other (nonlinear) applications.

19.1 How to Discretize and Linearize?

19.1.1 How to Discretize in Time?

In general relativity, we have a *metric* PDE. In it, we seek a new metric g: a 4×4 matrix, with as many as sixteen unknown entries, each of which is a function in spacetime. Fortunately, g is symmetric, so it has just ten independent entries: say, those in the upper-triangular part.

In its general form, the original problem is too complicated to solve analytically: it must be approximated on a discrete grid. At each grid point, we seek ten unknowns: $g_{m,n}$ $(0 \le m \le n \le 3)$.

What is the domain that should be discretized? This is spacetime: a four-dimensional manifold. Each individual point in spacetime is called an event: a four-dimensional vector, specifying both place and time.

Ideally, we would like to discretize this original manifold. Still, we only have our lab coordinates: t, x, y, and z. For the sake of uniformity, these are often denoted by $x^0 \equiv t$, $x^1 \equiv x$, $x^2 \equiv y$, and $x^3 \equiv z$. These are often indexed by a small Greek letter, say $\alpha = 0, 1, 2, 3$.

Our time t is measured in our own clock, in our own lab. It is therefore not absolute but only relative: in your own lab, you may measure a different time.

Ideally, in spacetime, we should use the proper time. This is a local time: in each individual event, this is the time measured by a clock at this event. After all, why use our own time t, measured in our own lab? Is our lab better than any other lab? The proper time, on the other hand, is much more general and invariant: it could be calculated in a coherent way from just any other system as well. Indeed, in spacetime, the proper time s is measured by

$$c^2 ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2,$$

where c is the speed of light [2, 6, 15, 18, 52, 78]. Unfortunately, this is too ideal: in reality, we only have t to work with. After all, at each fixed node in space, $dx = dy = dz = 0$, so the best approximation to s is our local time t.

In spacetime, the proper time s is just another dimension, independent of the other three, and having the same status. To approximate spacetime fairly, we should ideally use four-dimensional finite elements, in the x^0-x^1-x^2-x^3 coordinates. This way, there is no favorite direction: time is treated just like any other dimension, as required.

But this is too ideal again. From our own (narrow) perspective, t is not just a coordinate: it is time, different from space. After all, nothing can move faster than light. In the language of general relativity, in spacetime, a free fall must follow a time-like geodesic, not space-like.

In this point of view, time is different from space. As in classical mechanics, time could be eliminated and remain implicit. This would give us a new three-dimensional manifold: the phase space. In it, the physical motion takes place along the spatial geodesic, with a private parameter to help advance from state to state. What is this parameter? This is the proper time s, measured in this geodesic, and telling us how fast the motion is.

For this reason, the mesh should better be uniform in time: march in a time grid, time-level by time-level, from $t = 0$ to $t = \triangle t,\ 2\triangle t,\ 3\triangle t, \ldots$. In each time-level, use a three-dimensional mesh of tetrahedra to assemble the spatial derivatives. The time derivative, on the other hand, is discretized by an (implicit) time differencing: the current time-level minus the previous one, divided by $\triangle t$. This could also be viewed as an integration against Dirac's delta function in time. Together, the time-levels make a complete four-dimensional grid in spacetime, as required.

This approach is more economic: although spacetime is four-dimensional, we only integrate on three-dimensional tetrahedra. For this purpose, we often use integration by parts. This way, the basis functions are differentiated only once, not twice.

Should we integrate by parts in time too? On first glance, this seems a good idea. After all, this could avoid the second time derivative. Still, for the sake of stability, better not. Better discretize the original tt-partial derivative implicitly: the current time-level, minus twice the previous one, plus the yet previous one, divided by $(\triangle t)^2$. As discussed above, linearization may help analyze stability.

19.1.2 How to Differentiate the Metric?

In our (high-order) finite elements, the metric is approximated as a linear combination of basis functions:

$$g(n\triangle t) = \sum_{j=1}^{O} c_j(n\triangle t)\psi_j,$$

where $\psi_j \equiv \psi_j(x, y, z)$ is a basis function: a polynomial of three variables, corresponding to the jth degree of freedom in the mesh.

Let us focus on a particular time-level. This way, the argument "$(n\triangle t)$" can now drop:

$$g = \sum_{j=1}^{O} c_j \psi_j$$

(where O is the total number of degrees of freedom in the mesh). Furthermore, the c_j's are no longer scalars. Like g, they are 4×4 upper-triangular matrices. Thus, the above expansion could be written more explicitly as

$$g_{mn} = \sum_{j=1}^{O} c_{j,mn} \psi_j, \quad 0 \le m \le n \le 3.$$

To uncover the c_j's, we will need to assemble the Einstein equations: multiply by ψ_i, integrate, and assemble in space. For this purpose, let i and j be fixed indices of two degrees of freedom in the mesh. What is the (i, j)th "element" in the coefficient matrix? This is no longer a scalar, but a new 10×10 block.

In this block, let us focus on just one column: the (k, l)th column ($0 \le k \le l \le 3$). To help assemble it, we have two useful polynomials: the basis functions ψ_i and ψ_j in the finite element that is being assembled (or integrated upon).

How to assemble this column in one go? Well, it contains ten entries, at rows indexed by $0 \le m \le n \le 3$. To assemble them, differentiate the Einstein equations with respect to $c_{j,kl}$. This will produce the Jacobian, to help linearize the system with respect to the true algebraic unknowns: the c_j's.

Since j, k, and l are already fixed, this can be denoted by a prime:

$$' \equiv \frac{\partial}{\partial c_{j,kl}}.$$

For example,

$$g' \equiv \frac{\partial g}{\partial c_{j,kl}} \equiv \frac{\partial g_{mn}}{\partial c_{j,kl}}.$$

This could be viewed as a variation: the metric is varied in terms of one coefficient only: $c_{j,kl}$.

Since the same is done for all $0 \le m \le n \le 3$ in one go, the indices m and n often drop as well. This way,

$$g' \equiv g'_{m,n} \equiv \begin{cases} \psi_j & \text{if} \quad m = k \quad \text{and} \quad n = l \\ 0 & \text{otherwise.} \end{cases}$$

Thus, g' is a 4×4 upper-triangular matrix (indexed by $0 \le m \le n \le 3$), with just one nonzero entry: the (k, l)th entry (where k and l have already been fixed). As always, this entry is actually a polynomial in x, y, and z, defined in the finite element under consideration.

These polynomials need to be stored, for further use. Likewise, the entries of g are some other polynomials, which need to be stored as well. In g^{-1}, on the other hand, the entries are not only polynomials but also rational functions: a polynomial in the numerator, divided by another polynomial in the denominator (as in Cramer's rule). Both need to be stored, for further use. Likewise, in each product of these matrices, the entries are new rational functions, which need to be stored as well.

Let us use these observations to differentiate composite functions of g as well. In this context, g is not necessarily the solution but can be just any metric. This general g (as well as g') is also a symmetric 4×4 matrix. Store all of it, not just its upper-triangular part. This will help multiply two such matrices later. In this implementation, g' contains two nonzero entries (because kl and lk refer to the same entry).

19.1.3 The Flat Minkowski Metric

What do we know in advance about the metric? Well, Einstein's equivalence principle tells us that, at each individual event in spacetime, the metric could be approximated locally by a constant (flat) metric: the Minkowski metric. Although this local approximation remains implicit and theoretical, it still tells us how smooth the metric really is.

The Minkowski metric has a simple form: for some coordinates (not necessarily our lab coordinates), it can be written as

$$-c^2dt^2 + dx^2 + dy^2 + dz^2.$$

19.1.4 Riemann's Normal coordinates

Thus, the (unknown) metric is locally hyperbolic: at each event in spacetime, there are some coordinates (known as Riemann's normal coordinates) for which the metric is (nearly) as simple as in special relativity. Unfortunately, Riemann's normal coordinates have just a theoretical value: they remain local and implicit and can never be "tied up" to form a useful (global) coordinate system.

19.1.5 Toward Gravity Waves

Still, there is one exception. To detect gravity waves, we do linearize around the Minkowski metric, assuming that the true metric is close enough to it. This way, the Riemann normal coordinates do take a more practical and global face. Indeed, to the original system, apply a suitable gauge transformation, to reduce it into an easier

(linear) system. No need to do this explicitly: just assume that the gauge conditions hold, and you get ten decoupled wave equations (each solved as in Tables 9.1, 9.2, 9.3, and 9.4).

19.1.6 Where to Linearize?

Why not do this here? Because here we seek the true metric, which could be completely different from the Minkowski metric. For this purpose, the linearization should be not around the Minkowski metric but around our best guess g (available from the previous iteration or time-level). For this purpose, we must know how to linearize around any (general) g. This is done next.

19.2 The Christoffel Symbol

19.2.1 The Gradient Symbol and its Variation

Let us calculate the gradient of g and store it in a new $4 \times 4 \times 4$ symbol, with three lower indices—m, n, and α:

$$\Upsilon_{mn,\alpha} \equiv \Upsilon_{mn,\alpha}(g) \equiv \frac{\partial g_{mn}}{\partial x^\alpha} \quad (0 \le m, n, \alpha \le 3).$$

Here, mn actually means "m, n". We often drop the comma. Only in the differentiation do we keep the comma: "$,\alpha$" means a partial derivative with respect to x^α ($0 \le \alpha \le 3$). For example, for $\alpha = 0$, this is the time derivative.

A symbol may be viewed as an extension of the concept of matrix: it may use more than two indices. In our context, it also depends on the event under consideration: it may change from event to event. Unlike the tensor introduced later, a symbol is not necessarily invariant under coordinate change. To denote a symbol, the convention is to use a capital Greek letter.

Later on, we will need to linearize in each iteration. For this, we will need to evaluate Υ at a given g.

When $\alpha = 0$, for example, $x^0 = t$ is the time variable. In this case, the t-partial derivative is finite-differenced: the current time-level minus the previous one, divided by $\triangle t$. Still, this entry in Υ must be stored in two versions.

Indeed, in Einstein's equations, Υ will be multiplied from the left by g^{-1} (for sour general guess g), then differentiated with respect to $c_{j,kl}$. Differentiating a product like $g^{-1}\Upsilon$ will produce two terms: in $(g^{-1})'\Upsilon$, only g^{-1} will be differentiated, but Υ just evaluated at g. In $g^{-1}\Upsilon'$, on the other hand, things will be the other way around: only Υ will be differentiated, but g^{-1} just evaluated at g. We must be able to deal with both terms. This is why we need two versions.

One version will be used for calculation and evaluation at the g. For this purpose, when $\alpha = 0$, the previous time-level must be calculated at its own g, and subtracted.

The second version, on the other hand, will be used for variation with respect to $c_{j,kl}$. For this purpose, when $\alpha = 0$, the previous time-level must not be subtracted any more. After all, it does not depend on $c_{j,kl}(n\triangle t)$ but only on $c_{j,kl}((n-1)\triangle t)$, which is just a fixed parameter.

Still, to avoid the latter version, differentiate Υ (with respect to $c_{j,kl}$) as a composite function of g. How to do this? Fortunately, we already know how to differentiate g and produce g'. Since Υ is linear in g,

$$\Upsilon'_{mn,\alpha} = \Upsilon_{mn,\alpha}(g') = \begin{cases} \psi_{j,\alpha} & \text{if } \ m = k,\ n = l, \text{ and } \ 0 \le \alpha \le 3 \\ 0 & \text{otherwise.} \end{cases}$$

This way, Υ' contains at most eight nonzero entries (because kl and lk refer to the same entry).

Let us reorder the indices, and sum up. This defines a new $4 \times 4 \times 4$ symbol:

$$\Theta_{\alpha mn} \equiv \Theta_{\alpha mn}(\Upsilon) \equiv \frac{1}{2}\left(\Upsilon_{\alpha m,n} + \Upsilon_{\alpha n,m} - \Upsilon_{mn,\alpha}\right).$$

Fortunately, Θ is linear in Υ, so it is easily differentiated:

$$\Theta'_{\alpha mn} = \Theta_{\alpha mn}(\Upsilon').$$

This way, Θ' contains at most 22 nonzero entries, each being a polynomial of three variables in the finite element under consideration.

19.2.2 The Inverse Matrix and its Variation

For a given g, look at its inverse: g^{-1}. How to differentiate it with respect to $c_{j,kl}$? For this purpose, look at the 4×4 identity matrix I. It is completely independent of $c_{j,kl}$:

$$I' = (0)$$

(the 4×4 zero matrix). Thanks to Leibniz rule,

$$(0) = I' = \left(gg^{-1}\right)' = g'g^{-1} + g\left(g^{-1}\right)'.$$

Thus,

$$g\left(g^{-1}\right)' = -g'g^{-1},$$

or

$$\left(g^{-1}\right)' = -g^{-1}g'g^{-1}.$$

This will be useful later.

19.2.3 Einstein Summation Convention

The inverse matrix is often denoted by two upper indices:

$$g^{\mu\nu} \equiv \left(g^{-1}\right)_{\mu\nu}.$$

Here, we use a small Greek letter as an index: $0 \leq \mu, \nu \leq 3$. Also, we often use Einstein summation convention:

> An index that appears twice (not only as an upper index but also as a lower index) is contracted (summed over).

This way, no "$\sum$" sign is needed any more. For example, in the identity matrix I, the individual elements are often denoted by

$$\delta^{\mu}_{\kappa} \equiv \begin{cases} 1 & \text{if} \quad \mu = \kappa \\ 0 & \text{if} \quad \mu \neq \kappa. \end{cases}$$

Thanks to Einstein summation convention, the trace (sum of main-diagonal elements) has a new short form, with no "$\sum$" at all:

$$\text{trace}(I) = \delta^{\mu}_{\mu} \equiv \sum_{\mu=0}^{3} \delta^{\mu}_{\mu} = \sum_{\mu=0}^{3} 1 = 1 + 1 + 1 + 1 = 4.$$

This way, the "$\sum$" sign is avoided and dropped, with no harm. We then say that the upper and lower indices were contracted with each other. We can now combine the above conventions and multiply g by its own inverse (from the left):

$$g^{\mu\nu}g_{\nu\kappa} \equiv \sum_{\nu=0}^{3} g^{\mu\nu}g_{\nu\kappa} \equiv \sum_{\nu=0}^{3} \left(g^{-1}\right)_{\mu\nu} g_{\nu\kappa} = \delta^{\mu}_{\kappa}.$$

This way, μ remains upper, κ remains lower, and ν (the dummy index) is gone, as required. In other words, the second index in $g^{\mu\nu}$ was contracted with the first index in $g_{\nu\kappa}$.

An upper index can be contracted with a lower index only, not with an upper index. This is why the above contraction is indeed legitimate.

Let us introduce a new convention, with which the above can also be written in a new style:

$$g^{-1}g_{\mu\kappa} \equiv g^{\mu\nu}g_{\nu\kappa}.$$

This way, applying g^{-1} from the left means contracting with the first lower index of the matrix (or tensor) on the right. Let us go ahead and use this.

19.2.4 The Christoffel Symbol and its Variation

Let us use this new convention to define

$$\Gamma^{\alpha}_{mn} \equiv \Gamma^{\alpha}_{mn}(g) \equiv g^{-1}\Theta_{\alpha mn} \equiv g^{\alpha\beta}\Theta_{\beta mn}.$$

This is the Christoffel symbol.

How to differentiate it? Use the Leibniz rule:

$$\Gamma'^{\alpha}_{mn} = \left(g^{-1}\right)' \Theta_{\alpha mn} + g^{-1}\Theta'_{\alpha mn} = -\left(g^{-1}g'g^{-1}\right)\Theta_{\alpha mn} + g^{-1}\Theta'_{\alpha mn}.$$

This way, Γ' is indeed a function of the given g, as required.

The Christoffel symbol is the key to the curvature of spacetime, stored in the Riemann tensor.

19.3 Einstein Equations in Vacuum

19.3.1 The Riemann Tensor and its Variation

Let us define a yet bigger $4 \times 4 \times 4 \times 4$ symbol, with four indices—one upper and three lower:

$$\Xi^{\rho}_{\mu\sigma\nu} \equiv \frac{\partial \Gamma^{\rho}_{\nu\mu}}{\partial x^{\sigma}} + \Gamma^{\rho}_{\sigma\lambda}\Gamma^{\lambda}_{\nu\mu}.$$

Later on, we will need to multiply this by ψ_i and integrate in the relevant finite element. To discuss this, let us focus on the first term in $\psi_i \Xi$. To integrate this, we might want to use integration by parts.

In high-order finite elements, this is not a must. After all, a polynomial of degree five has nonzero second-order partial derivatives: polynomials of degree three.

In linear finite elements, on the other hand, integration by parts is vital. After all, a linear polynomial has vanishing second-order partial derivatives, which contribute

nothing to the integral. To avoid this, integration by parts throws the x^σ-partial differentiation onto ψ_i (with the minus sign), plus a boundary term (if the finite element is next to the boundary).

This is good enough for the simple case of Einstein equations in vacuum, introduced below. In the more general case, on the other hand, this is not the whole story. Indeed, Ξ will be multiplied not just by ψ_i but actually by $\psi_i g^{-1}$ from the left. To integrate $\psi_i g^{-1}\Xi$ by parts, the x^σ-partial differentiation must be thrown not just on ψ_i but actually on the entire product $\psi_i g^{-1}$, producing not one but two new terms to integrate. Yet worse: later on, we will also need to multiply this by g', so integration by parts will actually produce not two but three new terms to integrate. To do this easily, we will see a new trick later (Sect. 19.6.1 below).

Only when $\sigma = 0$ there is no need to integrate by parts at all. After all, the t-partial derivative is finite-differenced: the current time-level minus the previous one, divided by $\triangle t$. Still, this entry in the first term in Ξ must be stored in two versions: one for calculation and evaluation at g, for which the previous time-level must be calculated at its own g and subtracted, and another for variation, from which the previous time-level must not be subtracted at all, because it depends on $c_{j,kl}((n-1)\triangle t)$, not $c_{j,kl}(n\triangle t)$ (Sect. 19.2.1). (To avoid treating t different, you might want to use four-dimensional finite elements in the first place.)

How to differentiate Ξ? As in Leibniz rule:

$$\Xi'^{\rho}_{\mu\sigma\nu} = \frac{\partial \Gamma'^{\rho}_{\nu\mu}}{\partial x^\sigma} + \Gamma'^{\rho}_{\sigma\lambda}\Gamma^{\lambda}_{\nu\mu} + \Gamma^{\rho}_{\sigma\lambda}\Gamma'^{\lambda}_{\nu\mu}.$$

To define the Riemann tensor, just antisymmetrize Ξ in terms of σ and ν. In other words, subtract a mirror term, with σ and ν interchanged:

$$R^{\rho}_{\mu\sigma\nu} \equiv \Xi^{\rho}_{\mu\sigma\nu} - \Xi^{\rho}_{\mu\nu\sigma}.$$

This is easy enough to differentiate:

$$R'^{\rho}_{\mu\sigma\nu} = \Xi'^{\rho}_{\mu\sigma\nu} - \Xi'^{\rho}_{\mu\nu\sigma}.$$

19.3.2 Vacuum and Curvature

The Riemann tensor tells us the curvature: how curved spacetime is at each individual event. This curvature comes from the mass (or energy) distribution. It actually defines the true (nonflat) geometry in spacetime, and the true metric in it. This produces a new system of (nonlinear) PDEs: Einstein equations, whose solution is the true metric.

For a start, let us model gravity in vacuum: an empty domain in space, with no matter (or any other source of curvature) at all. In this domain, the system is homogeneous: its right-hand side vanishes.

Outside the domain, on the other hand, there may be massive stars (like black holes) that may affect the domain from the outside, and curve spacetime globally. Still, they may affect our domain only indirectly: through boundary conditions. In the interior of the domain, on the other hand, there should be no right-hand side at all.

Does this mean that, in our domain, the Riemann tensor should vanish? No! If it did, then the metric would be constant, and spacetime would be completely flat (as in special relativity): every free fall would follow a straight line in spacetime, as in Minkowski space.

This may be good enough in special relativity, where no gravity is assumed at all. In general relativity, on the other hand, gravity curves spacetime around. As a result, a free fall follows not a straight but a curved geodesic: the shortest path in the underlying (nonflat) geometry.

This is indeed why the Earth orbits the Sun in the solar system: it "falls" not in a straight line but in an elliptic orbit. (To see how this looks like in spacetime, add a vertical time axis on top, and you would get a "spring," stretched up.) Although it looks elliptic to us, the orbit is considered as straight in the nonflat geometry.

In this example, what is our domain? Well, it does not contain the Sun itself, but only the space outside it, which is assumed to be completely empty.

So, to model gravity, the Riemann tensor must not vanish. This would be too pedant—it would throw us back to special relativity, losing all the interesting effects of gravity. To avoid this, better require something else to vanish: not the big Riemann tensor, but the smaller Ricci tensor.

19.3.3 The Ricci Tensor

What is the Ricci tensor? It is the "trace" of the Riemann tensor, obtained by contracting one upper index with one lower index:

$$R_{\mu\nu} \equiv R^{\rho}_{\mu\rho\nu}.$$

What do we have here? Recall that the original Riemann tensor uses one upper index, followed by three lower ones. Now, take the first index, and contract it with the third one. This is indeed legitimate: the first index is upper, whereas the third index is lower. Once they have been contracted with one another, they are gone (thanks to Einstein summation convention). This defines the new Ricci tensor. It is still denoted by a capital R. Still, it is smaller than the original Riemann tensor: it only uses two indices, not four.

19.3.4 Einstein Equations in Vacuum

In vacuum, the Ricci tensor must vanish:

$$R_{\mu\nu} = 0.$$

These are the Einstein equations in vacuum. Because $R_{\mu\nu}$ depends on g, this indeed makes a system of PDEs in the unknown metric g, as required.

How to solve for g numerically? Well, we already know how: march in time, level by level. In each time-level, pick an initial guess from the previous time-level. Then, use an inner Newton iteration to solve for g in this time-level.

In each iteration, we need to linearize. For this, we need to differentiate the Ricci tensor at a general g. How to do this? Easy:

$$R'_{\mu\nu} = R'^{\rho}_{\mu\rho\nu}.$$

19.3.5 Stable Time Marching

Linearization is important not only in the Newton iteration required in implicit time marching but also in the stability analysis carried out beforehand. After all, to pick the optimal $\triangle t$, one must estimate stability in advance. This cannot be done in the original nonlinear system, but only in a linearized system [54].

This way, one could make a fair comparison between explicit and (semi-)implicit schemes in terms of stability, accuracy, and efficiency: assemble the iteration matrix at some time-level, and calculate its spectral radius, using the inverse power method. For each scheme, pick its own optimal $\triangle t$, and estimate the total cost of the entire run.

Better yet, to avoid calculating the eigenvalues exactly, one could also estimate them locally: pick just one point in space, and assume that the stencil around it was used everywhere: not only locally but also globally. This way, in space, you have an infinite matrix with constant diagonals, ready for a discrete Fourier transform in space. This way, the eigenvalues of the iteration matrix can now be estimated. If they do not exceed 1 in magnitude, then the scheme must be stable.

So far, we considered the Einstein equations in vacuum. This system is good enough in the study of black holes and the solar system. After all, nobody is interested in the interior of a star but only in the empty space outside it.

But this is no longer good enough in cosmology. In this field, we can no longer assume that our domain is empty. On the contrary: we often use a very large scale: we average on the stars and the galaxies in the entire universe and consider them as a homogeneous dust, with a uniform density, and no pressure at all. Thus, we cannot assume vacuum any more: the universe is full of matter and energy. To model this,

the Einstein equations must take an inhomogeneous form: the right-hand side must be a nonzero tensor: the stress tensor.

19.4 The Trace-Subtracted Form

19.4.1 The Stress Tensor

In cosmology (and a few other topics), the Einstein equations are no longer homogeneous: the right-hand side is a new nonzero tensor: the stress (energy-momentum) tensor $T_{\mu\nu}$. Often, it takes the simple form

$$T_{\mu\nu} \equiv V_{\mu\nu} + pg_{\mu\nu} \quad (0 \le \mu, \nu \le 3),$$

where the scalar p and the tensor V may depend on g, g^{-1}, and the original field, in a rather simple way. What does "simple" mean? Let us see a few examples.

For simplicity, we often drop the indices μ and ν:

$$T \equiv pg + V.$$

(Later on, we will place T in Einstein's equations, in the right-hand side.) This form appears in a few important models:

1. In a scalar field, p is the Lagrangian: the kinetic energy minus the potential energy (as in a harmonic oscillator). This way, p depends linearly on g^{-1}.
2. In electromagnetics, on the other hand,

$$p = \frac{1}{4}\text{trace}\left(g^{-1}Fg^{-1}F\right) \quad \text{and} \quad V = -Fg^{-1}F,$$

where F is the anti-symmetric tensor that stores the electromagnetic field:

$$F = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & -B_3 & B_2 \\ -E_2 & B_3 & 0 & -B_1 \\ -E_3 & -B_2 & B_1 & 0 \end{pmatrix},$$

where (E_1, E_2E_3) is the electric field, and (B_1, B_2B_3) is the magnetic field. This leads to the Einstein–Maxwell equations.

3. Finally, in cosmology, in a very large scale, the stars and galaxies are averaged, and viewed as a perfect fluid, with pressure p. (In Friedmann's theory, in particular, they are considered as dust, with no pressure at all: $p = 0$.) In this model, we also assume spatial isotropy: at each individual point in space, every direction looks the same. This way, V has just one nonzero entry:

$$V = \begin{pmatrix} \rho + p & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

where ρ is the energy density. (Do not confuse it with the index ρ used in the Riemann tensor above!) Moreover, we also assume symmetry and homogeneity: in every point in space, everything is the same: $p \equiv p(t)$ and $\rho \equiv \rho(t)$. This way, the universe must be a highly symmetric three-dimensional manifold:

- Either a "closed" (compact) hypersphere, with a (constant) positive spatial curvature,
- Or an "open" (noncompact) hyperboloid, with a negative curvature,
- Or a flat (and noncompact) Euclidean space, with no curvature at all. (From observations, this option is most likely.)

As the universe expands, the matter gets more and more spread out, and the energy density decreases: $d\rho/dt < 0$. In reality, however, the universe may contain not only dust but also other kinds of radiation, with a nonzero pressure, proportional to their own energy density. For example, even if the universe were completely empty, it would still contain "vacuum energy," with $p = -\rho$, $dp/dt = d\rho/dt = 0$, and $V \equiv 0$. Because it has constant ρ and p, this "dark" energy is also called the cosmological constant. Still, the above models are too ideal: in a real cosmological model, the universe is a mix of dust and dark energy together.

19.4.2 The Stress Tensor and its Variation

Fortunately, in the above form, the stress tensor is easy to differentiate:

$$T' = p'g + pg' + V'.$$

Now, let us take T, and apply g^{-1} to it from the left:

$$g^{-1}T = pI + g^{-1}V.$$

Here, I is the 4×4 identity matrix, so $\mathrm{trace}(I) = 4$. Therefore,

$$\mathrm{trace}\left(g^{-1}T\right) = 4p + \mathrm{trace}\left(g^{-1}V\right).$$

This scalar can now be multiplied by g_{mn} from the right, with two new indices—m and n:

$$\text{trace}\left(g^{-1}T\right)g_{mn} = 4pg_{mn} + \text{trace}\left(g^{-1}V\right)g_{mn}.$$

To differentiate this, use Leibniz rule on both sides:

$$\begin{aligned}
&-\text{trace}\left(g^{-1}g'g^{-1}T\right)g_{mn} + \text{trace}\left(g^{-1}T'\right)g_{mn} + \text{trace}\left(g^{-1}T\right)g'_{mn} \\
&= 4p'g_{mn} + 4pg'_{mn} \\
&- \text{trace}\left(g^{-1}g'g^{-1}V\right)g_{mn} + \text{trace}\left(g^{-1}V'\right)g_{mn} + \text{trace}\left(g^{-1}V\right)g'_{mn}.
\end{aligned}$$

Still, this is true only if T has the above simple form. Otherwise, the left-hand side must be calculated explicitly, including T'.

19.4.3 The Trace-Subtracted Form

We are now ready to write Einstein equations in their inhomogeneous form, with a nonzero tensor on the right-hand side. This way, they are relevant not only in vacuum but also in matter or radiation, as required in cosmology. For this purpose, let G denote Newton's gravity constant.

The general Einstein equations have two possible forms. Let us start with the easier one: the trace-subtracted form. In this form, T is placed in the right-hand side, and its trace is placed in the left-hand side:

$$\frac{1}{8\pi G}R_{\mu\nu} + \frac{1}{2}\text{trace}\left(g^{-1}T\right)g_{\mu\nu} = T_{\mu\nu}.$$

Here, we better keep the indices μ and ν and not drop them. This indicates that we use the Ricci tensor $R_{\mu\nu}$, not the original Riemann tensor that takes four indices.

Fortunately, we already have an algorithm to solve this system numerically. Indeed, we already differentiated and linearized all the terms around any given g, as required in each Newton iteration in each time-level. Still, there could be some difficulties, which may convince us to switch to a more fundamental form.

19.5 Einstein Equations: General Form

19.5.1 The Ricci Scalar

The above system is good enough so long as T is available analytically as a function of the metric. Unfortunately, T may also depend on the original field, which is not yet known, and may depend on the metric as well. Furthermore, what if T was given

only experimentally or computationally, as a computer function? Moreover, what about more complicated models and field theories, arising in quantum mechanics, quantum field theory, and particle physics? In such cases, we better resort to a more fundamental form, which obeys a few conservation laws, and may therefore be more stable, accurate, and suitable for numerical discretization. This form uses not only the Ricci tensor but also a more fundamental tensor: the Einstein tensor.

So far, we considered a few kinds of tensors. The Riemann tensor is really big: it takes four indices. The Ricci tensor, on the other hand, is smaller: it takes two indices only.

The Ricci scalar, on the other hand, is yet smaller: it has no index at all. Still, it is denoted by the same capital R, followed by no index at all. Here is how it is defined:

$$R \equiv \text{trace}\left(g^{-1}R_{\mu\nu}\right).$$

To distinguish this from the Ricci and Riemann tensors, we keep their indices and do not drop them.

Since the trace is a linear operation,

$$\begin{aligned} R' &= \text{trace}'\left(g^{-1}R_{\mu\nu}\right) \\ &= \text{trace}\left(-g^{-1}g'g^{-1}R_{\mu\nu} + g^{-1}R'_{\mu\nu}\right) \\ &= -\text{trace}\left(g^{-1}g'g^{-1}R_{\mu\nu}\right) + \text{trace}\left(g^{-1}R'_{\mu\nu}\right). \end{aligned}$$

19.5.2 The Einstein Tensor

We are now ready to define the Einstein tensor: take the Ricci tensor, and subtract half the Ricci scalar times g:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}.$$

(Do not confuse $G_{\mu\nu}$ with Newton's constant G, which has no index at all.) How to differentiate this? Easy:

$$G'_{\mu\nu} = R'_{\mu\nu} - \frac{1}{2}\left(R'g_{\mu\nu} + Rg'_{\mu\nu}\right).$$

19.5.3 Einstein Equations—General Form

We are now ready to present Einstein equations in their new general form:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

(Distinguish between $G_{\mu\nu}$ on the left, a tensor with two indices, and G on the right, a constant with no index at all.) In this form, the stress tensor T appears in the right-hand side only. Its trace is never subtracted from the left-hand side any more. This is a more fundamental form, which may have a few advantages. To solve numerically for g, follow the guidelines at the end of Sect. 19.3.4.

19.6 How to Integrate?

19.6.1 Integration by Parts

In the Einstein tensor, look at the second term. To define it, we carried out four kinds of algebraic operations. To obtain the Riemann tensor, we antisymmetrized Ξ. Then, to obtain the Ricci tensor, we contracted the first and third indices. Finally, to obtain the Ricci scalar, we applied g^{-1} from the left and traced. In this process, we lost all four indices, including one important index: the original index σ in Ξ.

All this is good enough in high-order finite elements, where integration by parts is not a must and could be avoided. But what about linear finite elements, where integration by parts is vital, as discussed in Sect. 19.3.1? How can we do this, with no σ any more?

For this purpose, let us do things the other way around: wait with antisymmetrizing, contracting, and tracing until later. Before these operations, multiply Ξ by g^{-1} from the left (in terms of μ, as in our convention), and by g from the right (with no contraction at all, but with two new indices—m and n):

$$g^{-1}\Xi^{\rho}_{\mu\sigma\nu}g_{mn}.$$

Do this slowly and patiently. To apply g^{-1}, pick just one element from g^{-1}, say $g^{\mu\alpha}$, and multiply by it (without contracting as yet):

$$g^{\mu\alpha}\Xi^{\rho}_{\beta\sigma\nu}g_{mn}.$$

(Later on, we will sum over $\alpha = \beta = 0, 1, 2, 3$.) This tensor is really big: eight indices! Fortunately, no need to store all entries in it. After all, because α and β are going to be contracted soon, only those entries with $\alpha = \beta$ need to be stored. Furthermore, because we are going to antisymmetrize soon, we do not need those entries with $\sigma = \nu$, which will drop anyway. Moreover, because we are going to

contract soon, we only need those entries with $\rho = \sigma$ (or $\rho = \nu$, due to the above antisymmetrization). Finally, because we are going to trace soon, we only need those entries with $\mu = \nu$ (or $\mu = \sigma$, due to the above antisymmetrization).

But these entries are not the only ones. We also need their linearization, obtained from differentiation, before integrating by parts. This must be stored as well.

Once differentiated, the above term produces three new terms:

$$-\left(g^{-1}g'g^{-1}\right)\Xi^{\rho}_{\mu\sigma\nu}g_{mn} + g^{-1}\Xi'^{\rho}_{\mu\sigma\nu}g_{mn} + g^{-1}\Xi^{\rho}_{\mu\sigma\nu}g'_{mn}.$$

As before, to apply a matrix from the left, use a big tensor of eight indices. Again, no need to store all entries, but only those indicated above. Before storing, multiply by ψ_i, and integrate by parts: throw the x^{σ}-partial derivative from the first term in Ξ onto the rest, pick a minus sign, and store. This will produce three new polynomials to integrate (plus a boundary term, for those finite elements near the boundary). The same should also be done with σ and ν interchanged. Once this is done, we do not need σ any more: we can safely antisymmetrize, contract, and trace, to obtain the linearized second term in Einstein's tensor, as required.

19.6.2 Why to Linearize?

General relativity is a local field theory: to obtain the Einstein equations in their general form, one must minimize the Einstein–Hilbert action: an extension of the quadratic form minimized in the heat equation and other diffusion problems in classical mechanics.

What is this action? It is a functional of the metric: it integrates the Lagrange density: the kinetic energy minus the potential energy (as in a harmonic oscillator). In our case, this is the Ricci scalar. The integration is carried out over the path where the motion takes place in spacetime, with an absolute (coordinate-independent) measure.

The challenge is to design a new path and metric that minimize this action (at least locally). For this purpose, one must use calculus of variations: find a specific metric (and path) that could vary infinitesimally, leaving the action unchanged. At this specific metric (and path), the action must indeed be minimized (at least locally). This is an extension of function minimization in elementary calculus.

As discussed above, differentiation with respect to $c_{j,kl}$ is also a kind of variation: it varies just one coefficient of one basis function in one entry in the metric at a time. This helps linearize the Einstein tensor, as required in each Newton iteration, in each time-level.

What is the first term in the Einstein tensor? This is the Ricci tensor. This term is differentiated first.

Here one may ask: Why bother differentiate the Ricci tensor at all? After all, in spacetime, in a coordinate-invariant sense, this variation is a total derivative of

some other tensor, contributing nothing to the integral (but a boundary term at most). Furthermore, since we consider a motion from one fixed event to another, this boundary term has no effect whatsoever. Why then bother to linearize, as done all along?

Well, this may be true in coordinate-independent terms, but not necessarily in our own lab coordinates. Moreover, in each Newton iteration, we need to linearize not around the true metric but around some approximate metric, with an unknown error. Then, we actually solve for the error, which may be completely different from the true metric.

Still, these answers are rather technical. Here is a deeper answer: globally, the boundary term may be more important than it seems. After all, we seek not just any metric, but one specific metric that also satisfies the boundary conditions at the boundary of our domain.

Compare this with the Laplace equation: the steady state of the heat equation. As a matter of fact, this is the homogeneous version of the Poisson equation. In this case, the action is just the kinetic energy: the squared norm of the gradient, integrated over the domain. Its variation is indeed a total derivative: the divergence of the gradient, or the Laplacian. This is why it should vanish. Does this mean that the Laplace equation is trivial? No! Although the zero function is a solution, it is useless: it does not satisfy the boundary conditions.

It is true that, with homogeneous Neumann boundary conditions, the Laplace equation has infinitely many solutions. In fact, every constant function is a legitimate solution. This is indeed an ill-posed Sturm–Liouville problem. Why ill-posed? Because, when inhomogeneous boundary conditions are imposed, it may have no solution at all.

To make the problem well-posed, we must "nail" it with one extra integral equation, or with a Dirichlet condition at one boundary point. In this case, the above constant functions are no help any more. The new problem must be solved from scratch.

Likewise, in electromagnetics, the potential is not defined uniquely but only up to an additive constant. Still, "nailing" it at just one point (say, the infinity point) is good enough to make the problem well-posed.

What about a domain as big as the entire universe, with no boundary at all? In this case, there is no boundary condition any more. Instead, we impose a global integral constraint: we require that the solution be square-integrable in the entire universe. This is enough to make the problem well-posed, with a unique solution. In the Laplace equation discussed above, for example, the zero function is now the only legitimate solution.

In cosmology, our domain is indeed the entire universe. Fortunately, even with no boundary condition at all, the potential energy is still unique: it is "nailed" at a specific constant value. Indeed, even if the universe were completely empty, it would still contain an unchanging (ground-state) "vacuum energy": the cosmological constant. This is why the potential energy is unique: it is "nailed" at vacuum. This is what makes the problem well-posed, as required.

19.6.3 Back to the Trace-Subtracted Form

Look again at the Einstein equations, in their general form:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

Why is this fundamental? Because this is the original system, from which the trace-subtracted form is derived in the first place. To see this, apply g^{-1} to both sides:

$$g^{-1}G_{\mu\nu} = 8\pi G g^{-1}T_{\mu\nu}.$$

In this, plug the original definition

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu},$$

to obtain

$$g^{-1}\left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right) = g^{-1}R_{\mu\nu} - \frac{1}{2}RI_{\mu\nu} = 8\pi G g^{-1}T_{\mu\nu}.$$

Now, take the trace of both sides:

$$\text{trace}\left(g^{-1}R_{\mu\nu}\right) - \frac{1}{2}R \cdot 4 = R - 2R = -R = 8\pi G \cdot \text{trace}\left(g^{-1}T\right).$$

In summary, we now have the Ricci scalar in a more explicit form:

$$R = -8\pi G \cdot \text{trace}\left(g^{-1}T\right).$$

Let us plug this in Einstein equations, and divide by $8\pi G$:

$$\frac{1}{8\pi G}R_{\mu\nu} + \frac{1}{2}\text{trace}\left(g^{-1}T\right)g_{\mu\nu} = T_{\mu\nu}.$$

This is indeed the trace-subtracted form back again.

What do we have in this system? On the right-hand side, we have a well-known tensor: the stress tensor. On the left-hand side, on the other hand, we have two terms, both depending on the metric g, which is still unknown. How to solve for g numerically? Follow the above guidelines.

19.7 Exercises

1. What is a tensor?
2. What is a matrix? Is it a special kind of tensor?
3. How many indices does a matrix have?
4. Can a tensor have more? How many?
5. Implement a general tensor, with as many indices as you like.
6. Implement the arithmetic operations between tensors. The solution can be found in Chapter 20 in [77].
7. Let A be a 4×4 matrix:

$$A \equiv \left(a_{i,j}\right)_{0 \le i,j \le 3}.$$

   Apply g^{-1} to A from the left. For this purpose, contract with ν, which is the first lower index in A:

$$g^{-1}a_{\mu\kappa} = \left(g^{-1}\right)_{\mu\nu} a_{\nu\kappa} = g^{\mu\nu}a_{\nu\kappa}.$$

   Hint: use Einstein summation convention and our new convention.
8. Is this a legitimate contraction? Hint: ν is an upper index in $g^{\mu\nu}$, and a lower index in $a_{\nu\kappa}$, as required.
9. How could this new style be converted back to the naive linear-algebra style? Hint: introduce parentheses back again:

$$g^{-1}a_{\mu\kappa} = \left(g^{-1}A\right)_{\mu\kappa} = \sum_{\nu=0}^{3} \left(g^{-1}\right)_{\mu\nu} a_{\nu\kappa} = \sum_{\nu=0}^{3} g^{\mu\nu}a_{\nu\kappa} = g^{\mu\nu}a_{\nu\kappa}.$$

10. What is so good about this new style? Hint: thanks to it, g^{-1} can now be applied not only to a matrix like A but also to a bigger symbol like Θ, to help define the Christoffel symbol:

$$\Gamma^{\alpha}_{mn} \equiv g^{-1}\Theta_{\alpha mn} \equiv g^{\alpha\beta}\Theta_{\beta mn}$$

    (Sect. 19.2.4).
11. In the latter formula, what happened to the index β? Hint: it has been contracted upon, and gone. This is why it can no longer be found in Γ.
12. Is this a legitimate contraction? Hint: β is an upper index in $g^{\alpha\beta}$, and a lower index in Θ, as required.
13. Why is α an upper index in Γ? Hint: because it is an upper index in $g^{\alpha\beta}$ and should remain an upper index in Γ as well.
14. Give yet another example of applying g^{-1} in the new style. Hint: apply g^{-1} to the Ricci tensor, and take the trace:

$$R \equiv \text{trace}\left(g^{-1} R_{\mu\nu}\right).$$

This is the Ricci scalar.

15. What information is stored in the Riemann tensor? Hint: the entire curvature in spacetime.
16. Does it make sense to require that it vanishes? Hint: this is too pedantic—it would lead to a flat geometry, with no gravity at all, as in the Minkowski space in special relativity. This is too ideal: in reality, we do have gravity.
17. Does it make more sense to require that the Ricci tensor vanishes? Hint: only in vacuum: an empty domain, with no matter in it, but only outside it.
18. Give an example of an empty domain, with no matter in it, but only outside it. Hint: in the solar system, exclude the Sun itself, and consider the space outside it.
19. In this domain, does not the Earth itself violate the vacuum assumption? Hint: the Earth is the object in motion, upon which gravity acts. As such, it is considered as a single point, with no dimension at all.
20. In this domain, how could the Earth feel any gravity? Hint: it feels the Sun's gravity through the boundary conditions at the Sun's surface.
21. Why does the Earth orbit the Sun? Hint: in the nonflat geometry in spacetime, the Earth's orbit is as straight as ever. This is how the Earth actually "falls" freely along its orbit.
22. Look at a massive star in the sky. Does the time on it tick faster or slower than our own time here? Hint: slower. This is gravitational time dilation.
23. Is the speed of light the same here as there? Hint: yes. This is Einstein's axiom.
24. What color does the star have? Hint: due to gravitational redshift, it looks here redder than there.
25. What about a star as massive and dense as a black hole? Hint: the light gets so red that it cannot be seen here at all. In other words, it cannot escape from the powerful gravity of the black hole.

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